

Processes

Abhilash Jindal

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- Vocabulary (OSTEP Ch. 4)
 - What is a process? System calls, Scheduler, Address space, Process control block

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- Processes in action (xv6 Ch. 3: system calls, x86 protection, trap handlers)
 - User stack<>kernel stack, sys call handling

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- Memory isolation (OSTEP Ch. 13, 15, 16)
 - How to isolate memory into address spaces?
- Processes in action (xv6 Ch. 3: system calls, x86 protection, trap handlers)
 - User stack<>kernel stack, sys call handling
- Scheduling (xv6 Ch5: context switching, OSTEP Ch. 6-9)
 - Response time, throughput, fairness

Process is a running program

- Load program from disk to memory
 - Exactly how we loaded OS
- Give control to the process. Jump cs, eip

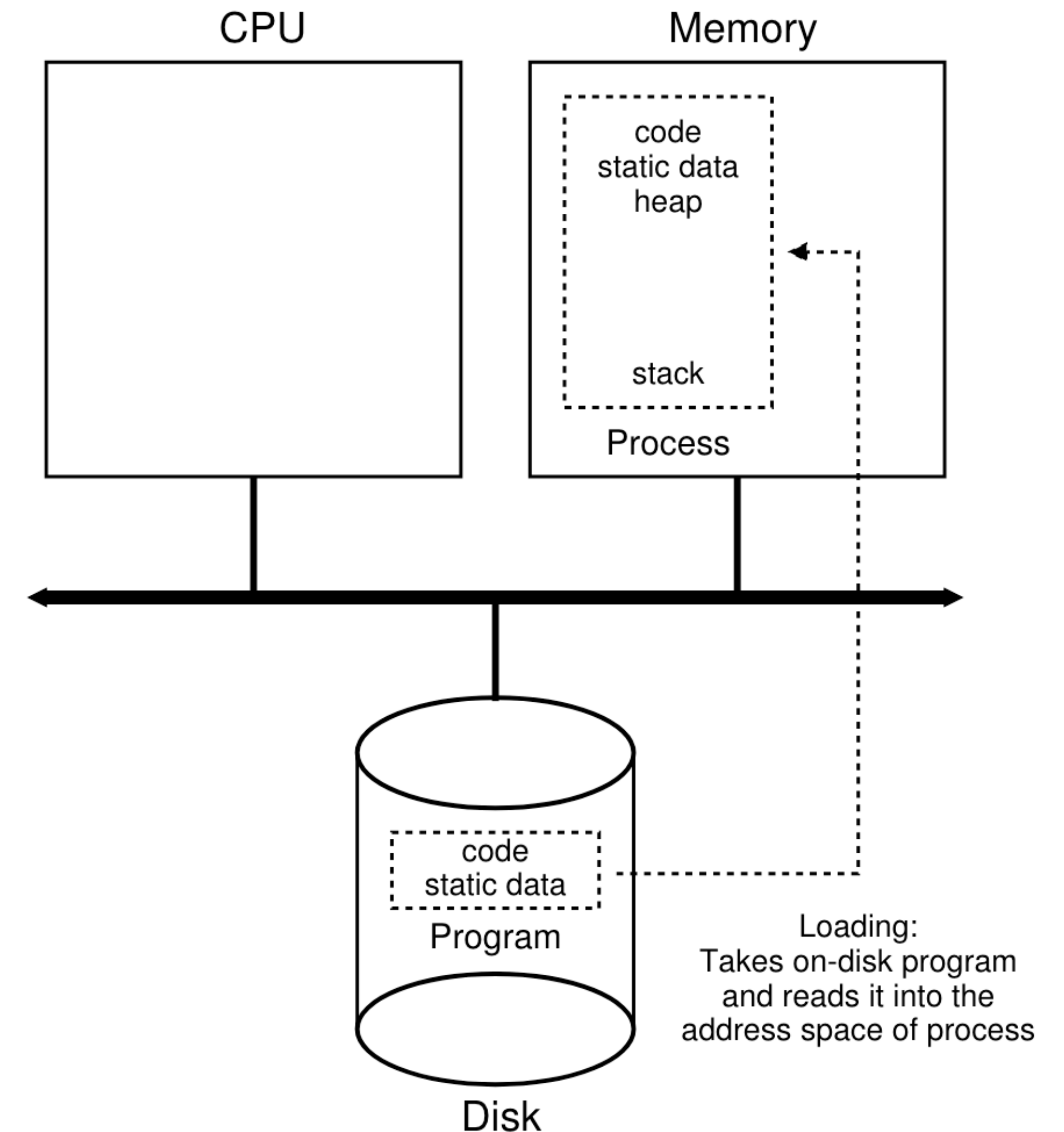


Figure 4.1: Loading: From Program To Process

Processes can ask OS to do work for them

System calls

```
$ strace cat /tmp/foo
...
openat(AT_FDCWD, "/tmp/foo", O_RDONLY) = 3
read(3, "hi\n", 131072) = 3
write(1, "hi\n", 3) = 3
...
```

OS maintains process states

Scheduler switches between processes

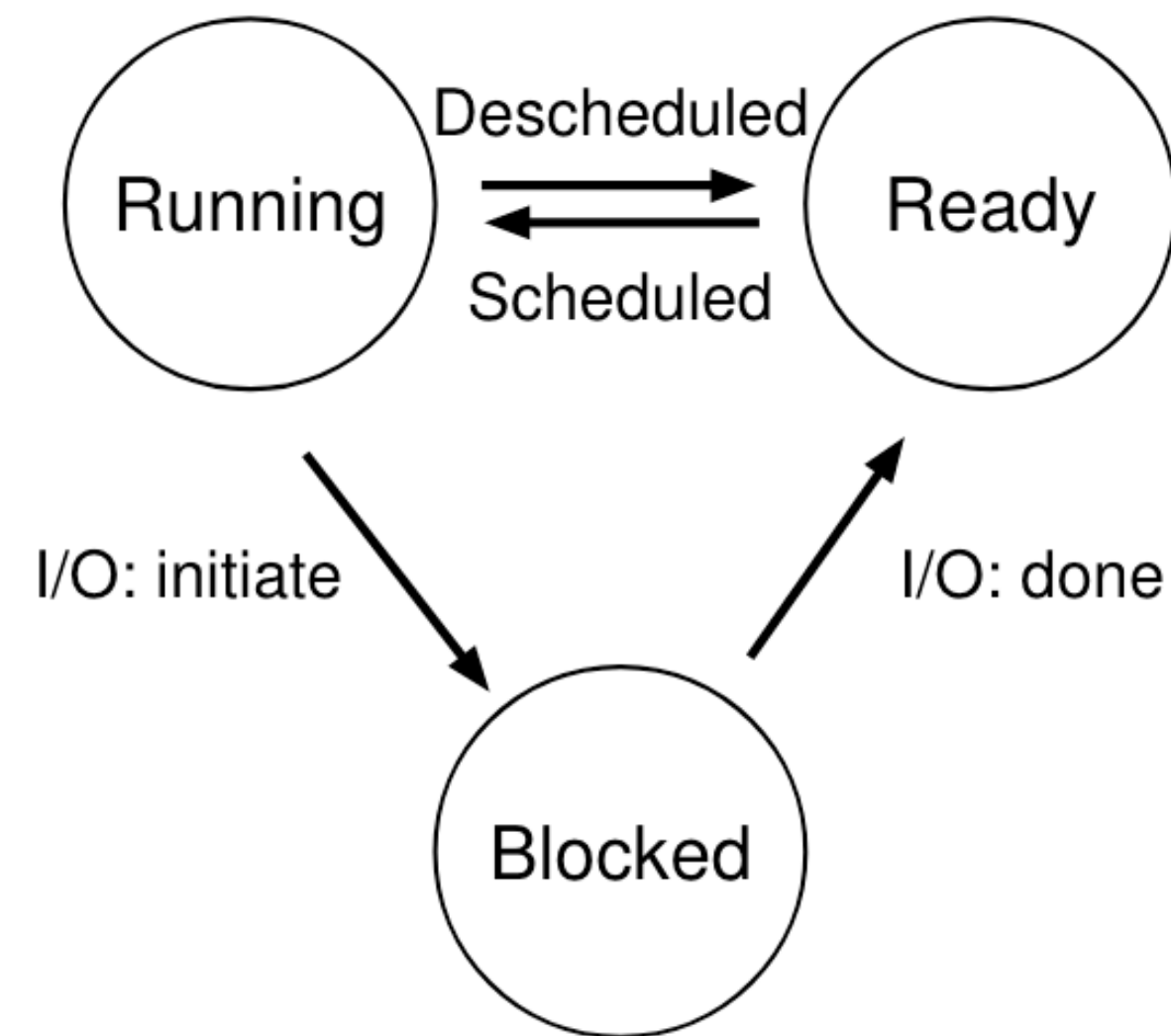


Figure 4.2: Process: State Transitions

OS maintains process states

Scheduler switches between processes

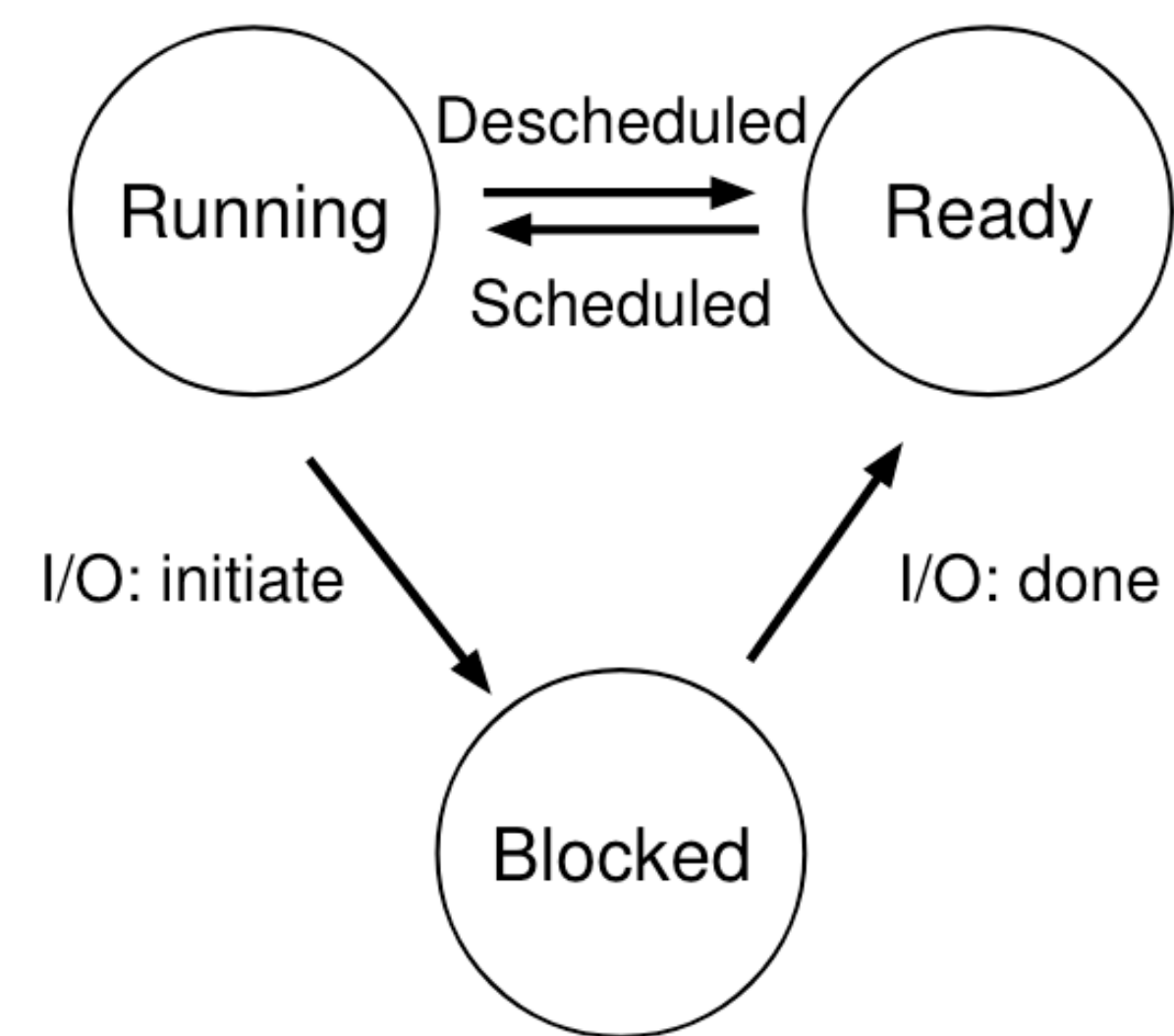


Figure 4.2: Process: State Transitions

Time	Process ₀	Process ₁	Notes
1	Running	Ready	
2	Running	Ready	
3	Running	Ready	Process ₀ initiates I/O
4	Blocked	Running	Process ₀ is blocked,
5	Blocked	Running	so Process ₁ runs
6	Blocked	Running	
7	Ready	Running	I/O done
8	Ready	Running	Process ₁ now done
9	Running	–	
10	Running	–	Process ₀ now done

Figure 4.4: Tracing Process State: CPU and I/O

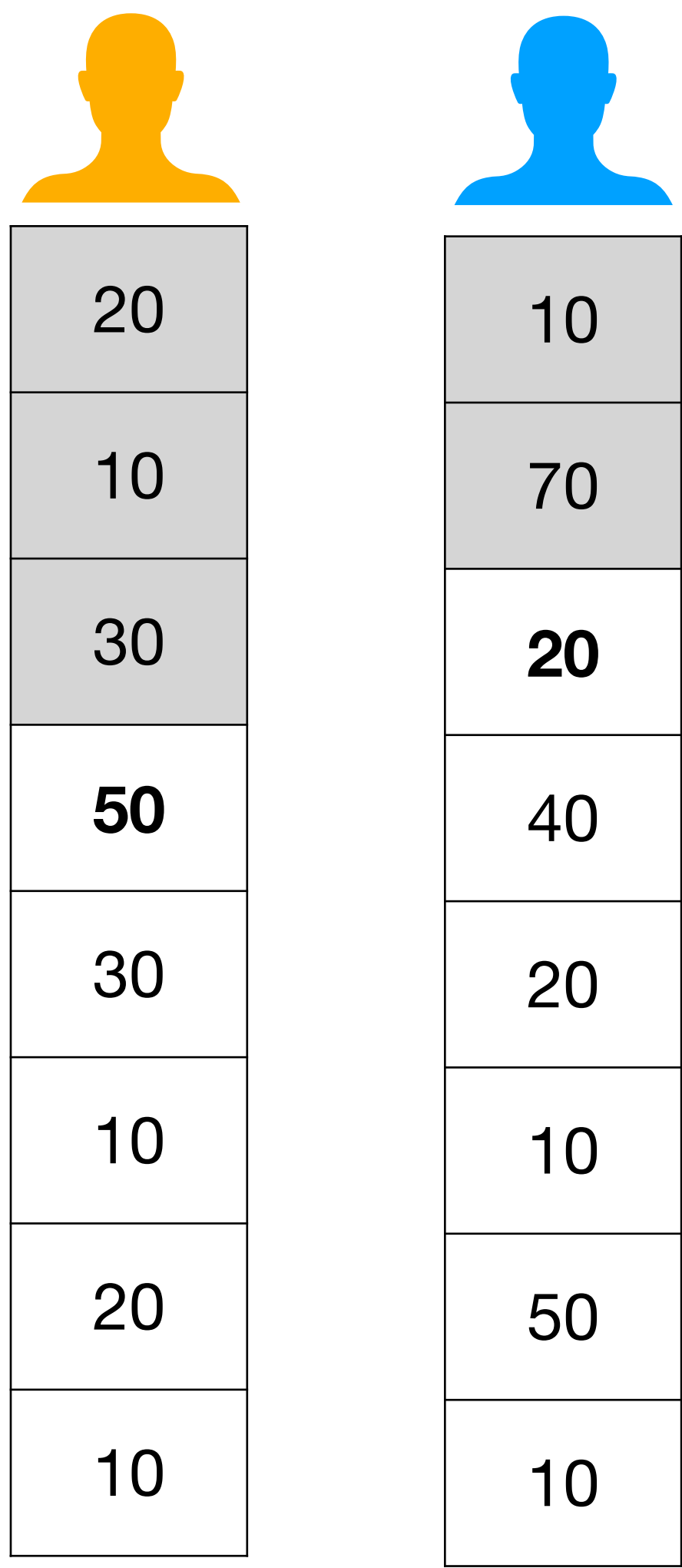
Calculator analogy: A process is computing long sum



20
10
30
50
30
10
20
10

- 2 0 = (move pointer to 10)
- + 1 0 = (move pointer to 30)
- + 3 0 = (move pointer to 50)
- + 5 0 = (move pointer to 30)
- + 3 0 = (move pointer to 10)
- + 1 0 = (move pointer to 20)
- + 2 0 = (move pointer to 10)

Context switch snatches the calculator



Context switch snatches the calculator



20
10
30
50
30
10
20
10



10
70
20
40
20
10
50
10

- Steps to share the calculator:
 - $20 + 10 = 30 + 30 = 60$
 - Write 60 in notebook, remember that we were done till 30, give calculator
 - $10 + 70 = 80$
 - Write 80 in notebook, remember that we were done till 70, give the calculator back

OS maintains list of Process Control Blocks

xv6 PCB

```
struct {
    struct proc proc[NPROC];
} ptable;

enum procstate { UNUSED, EMBRYO, SLEEPING, RUNNABLE, RUNNING, ZOMBIE };

struct proc {
    ...
    int pid;                // Process ID
    char name[16];          // Process name (debugging)
    struct proc *parent;    // Parent process
    enum procstate state;   // Process state
    struct file *ofile[NFILE]; // Open files
    struct inode *cwd;       // Current directory
    struct context *context; // Switch here to run process
};
```

Address space

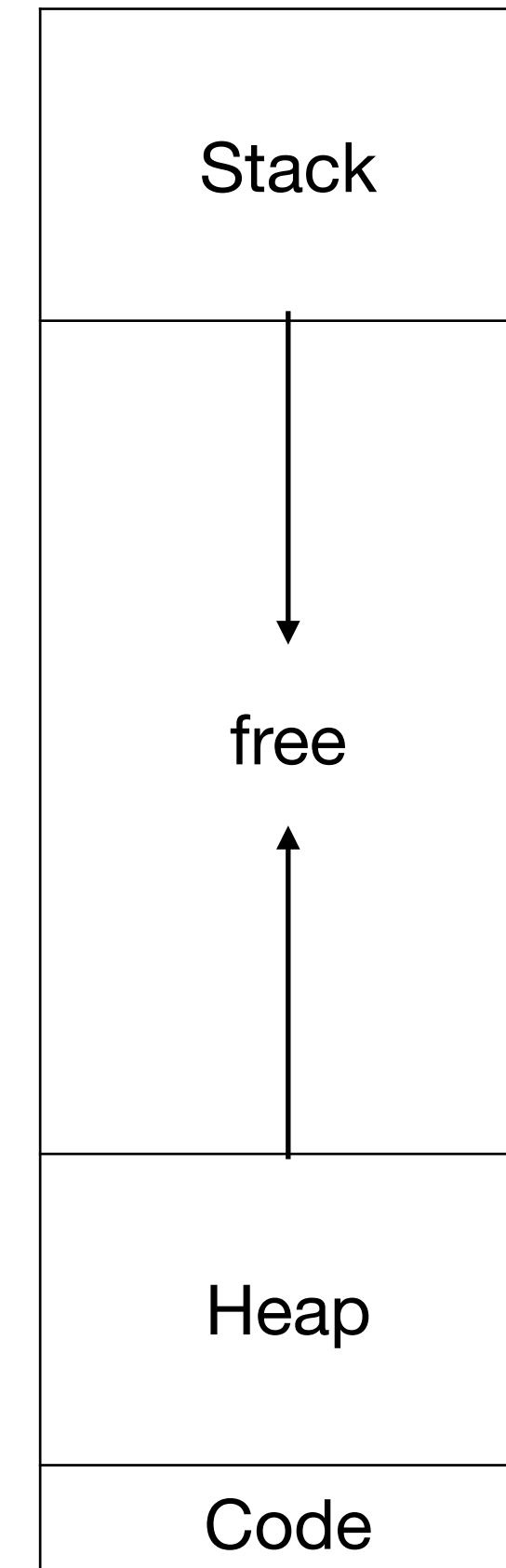
OSTEP Ch. 14,17

Abhilash Jindal

Process Address Space

Code, Heap, Stack

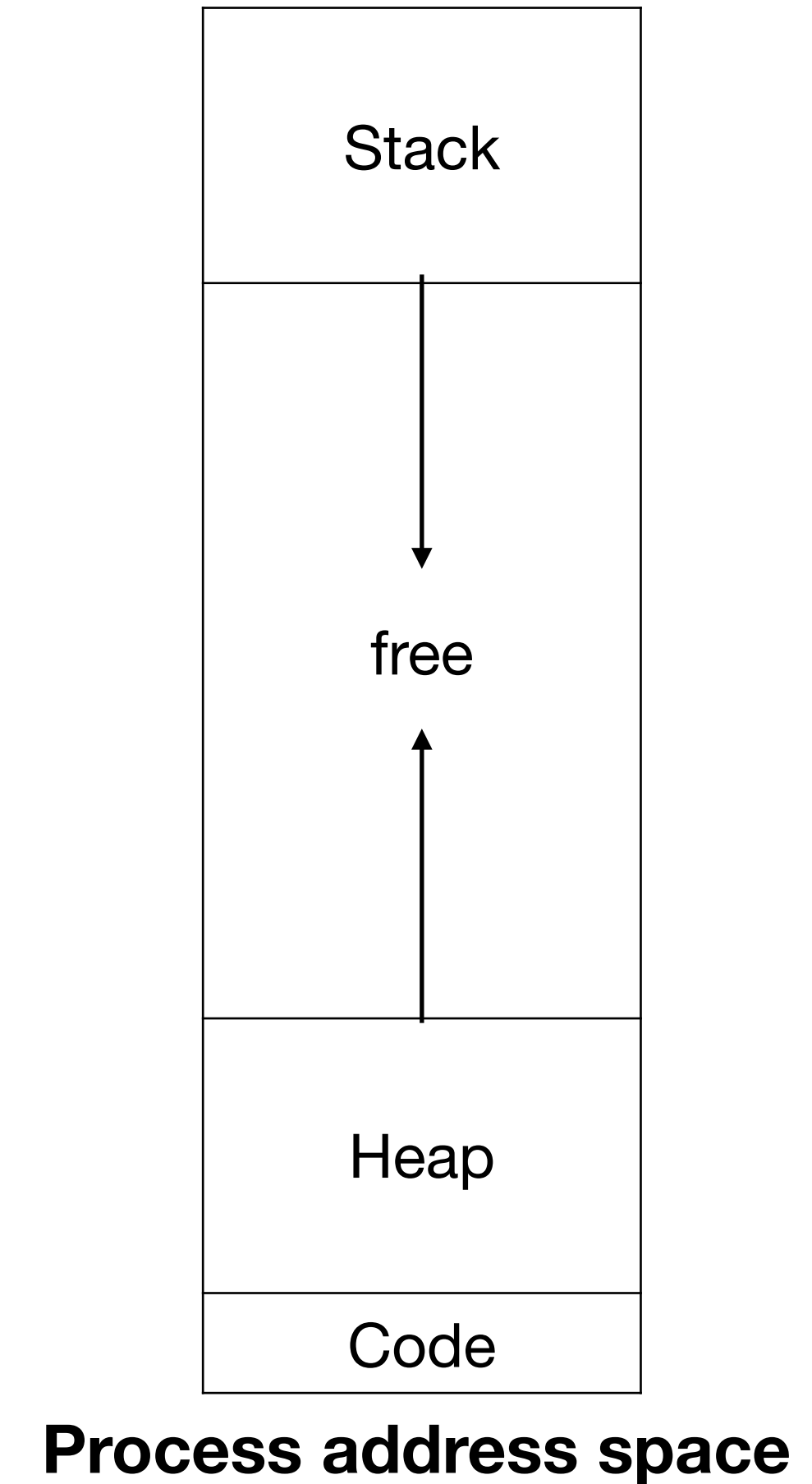
- Example: 01/mem.c
 - Multiple processes access the same pointer location but the OS and hardware isolated process to its own *address space*. Each process thinks they own the memory
- Code: Instructions, global variables. Loader reserved this space when loading from ELF.
- Stack: Local variables
- Heap: Dynamically allocated variables.
malloc/free



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Function calling in action

Stack

02.s

eip

→

_foo:

pushl %ebp

movl %esp, %ebp

movl 8(%ebp), %eax

addl 12(%ebp), %eax

popl %ebp

retl

.globl _main

.p2align 4, 0x90

_main:

pushl %ebp

movl %esp, %ebp

subl \$24, %esp

movl \$0, -4(%ebp)

movl \$41, (%esp)

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calll _foo

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retl

Save caller's base pointer

ebp = esp

eax = *(ebp + 8)

eax = eax + *(ebp + 12)

Restore caller's base pointer

change eip to return address

-- Begin function main

Save caller's base pointer

ebp = esp

esp = esp - 0x18

*(ebp-4)=0

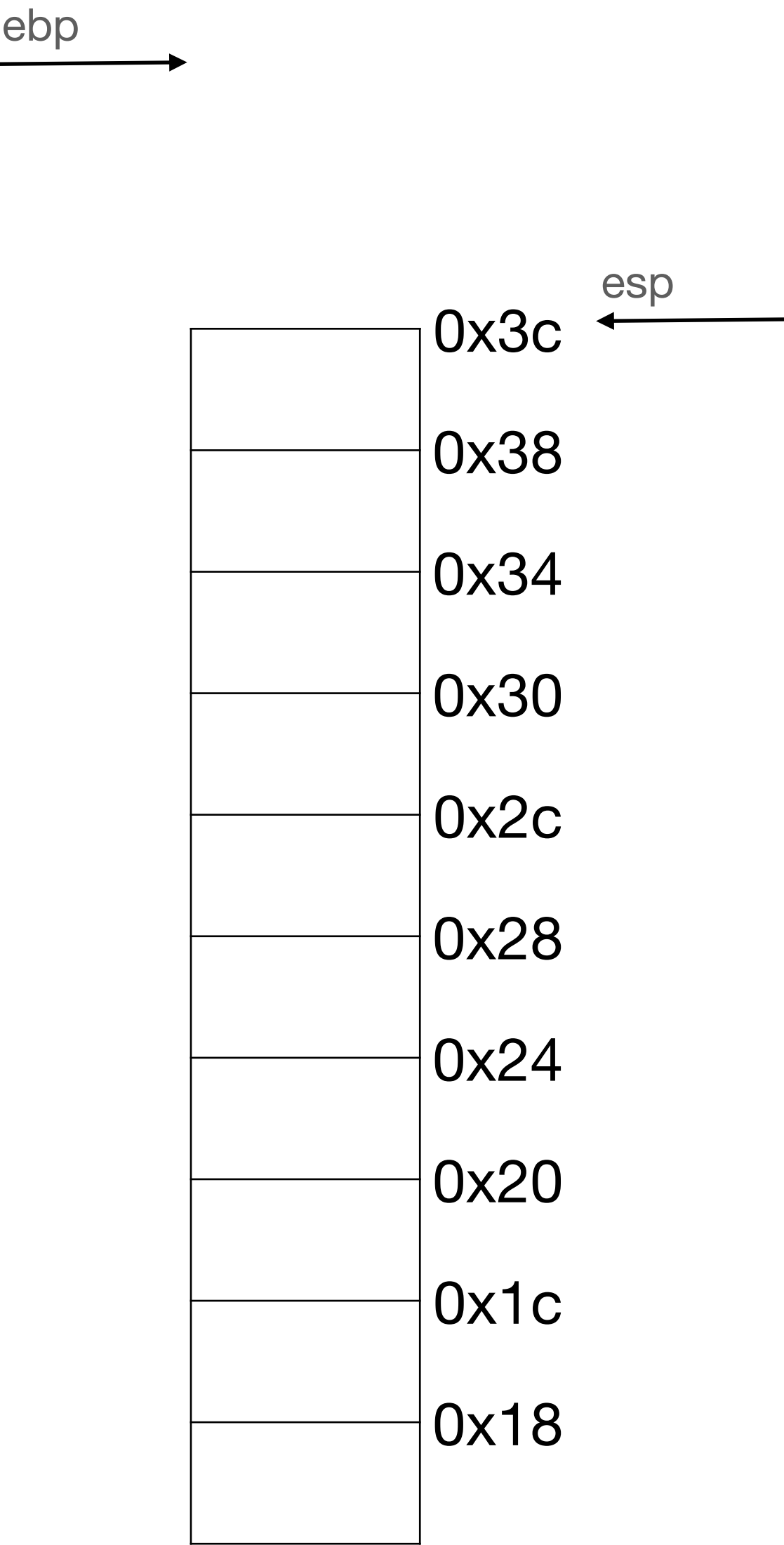
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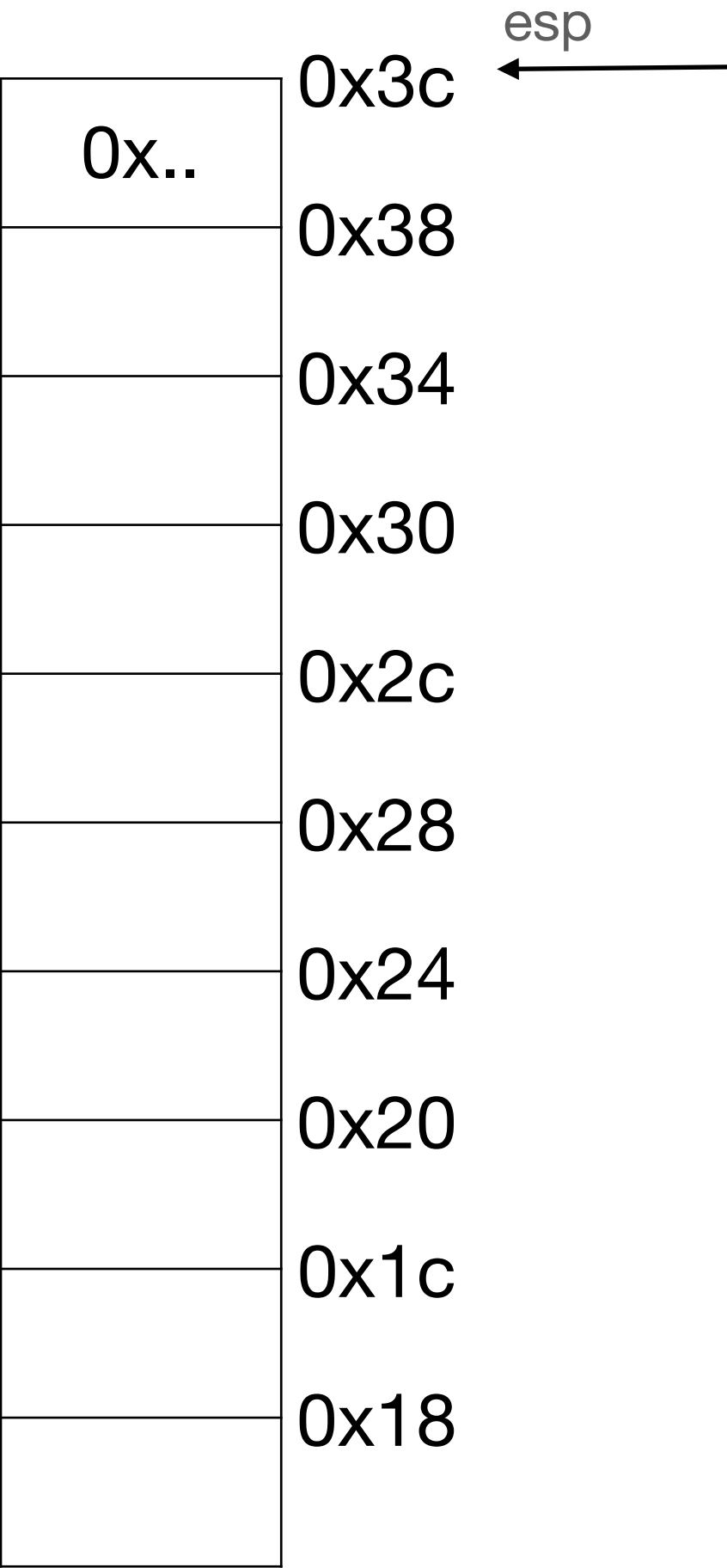
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02.s

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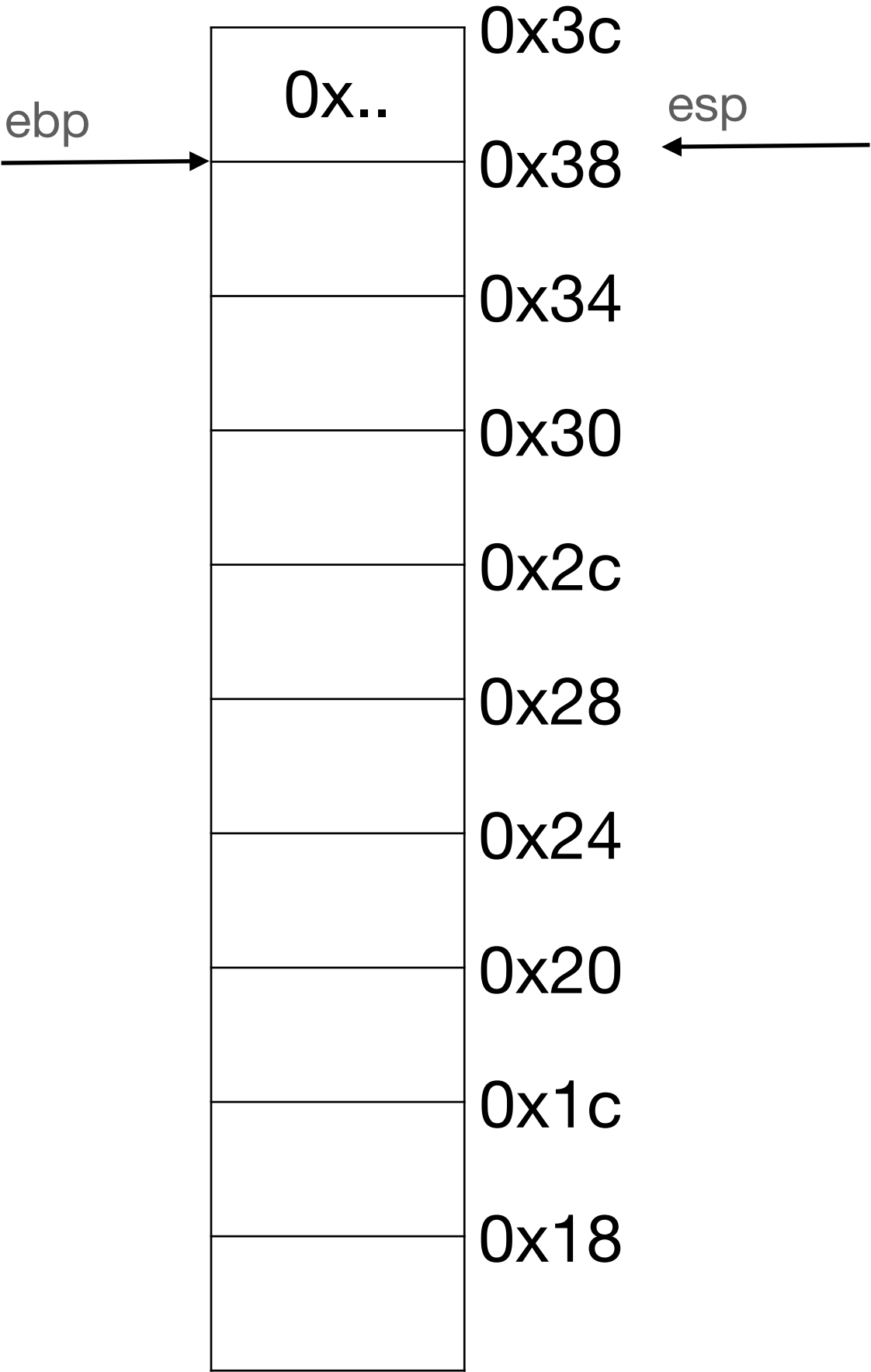
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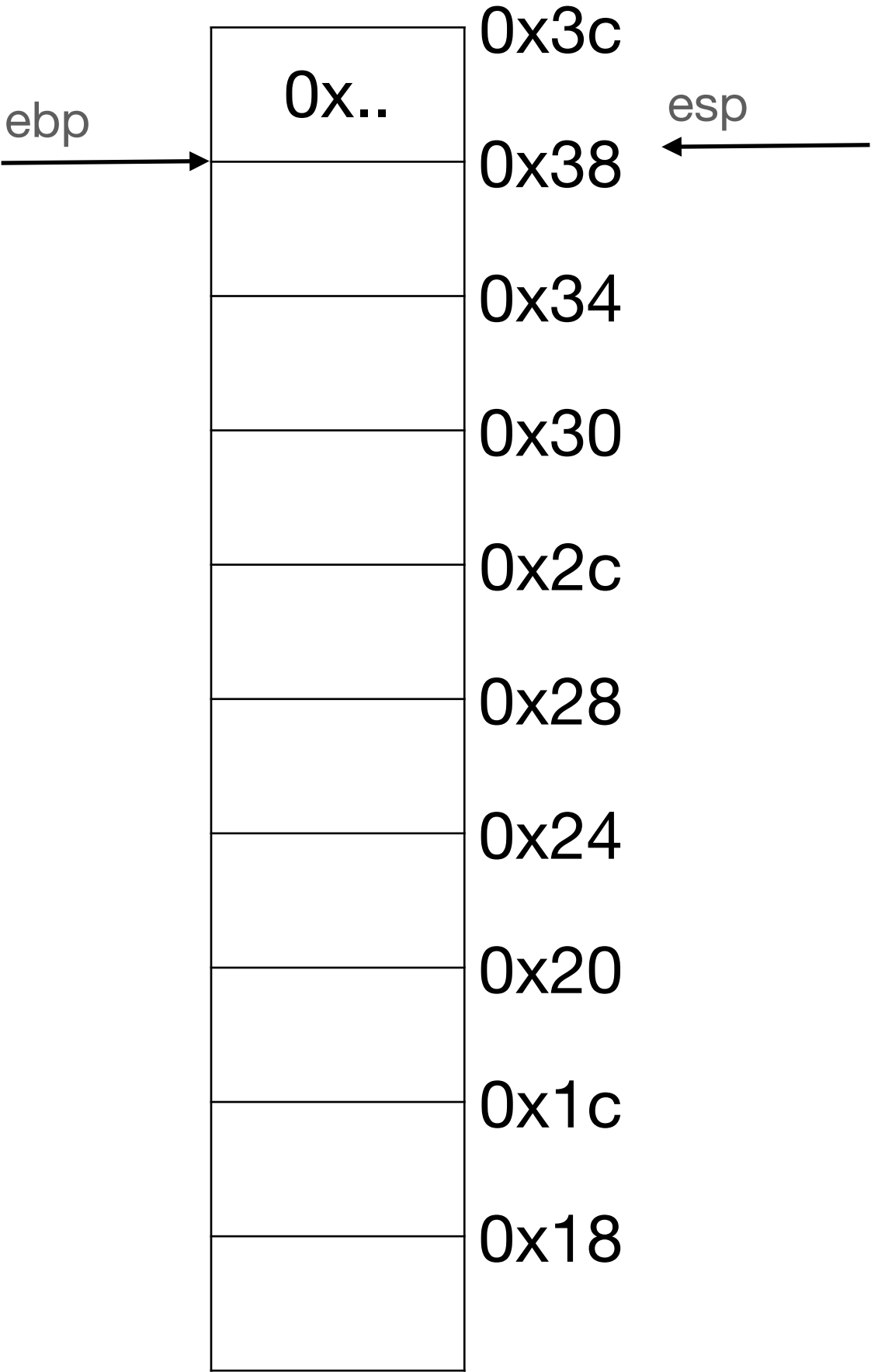
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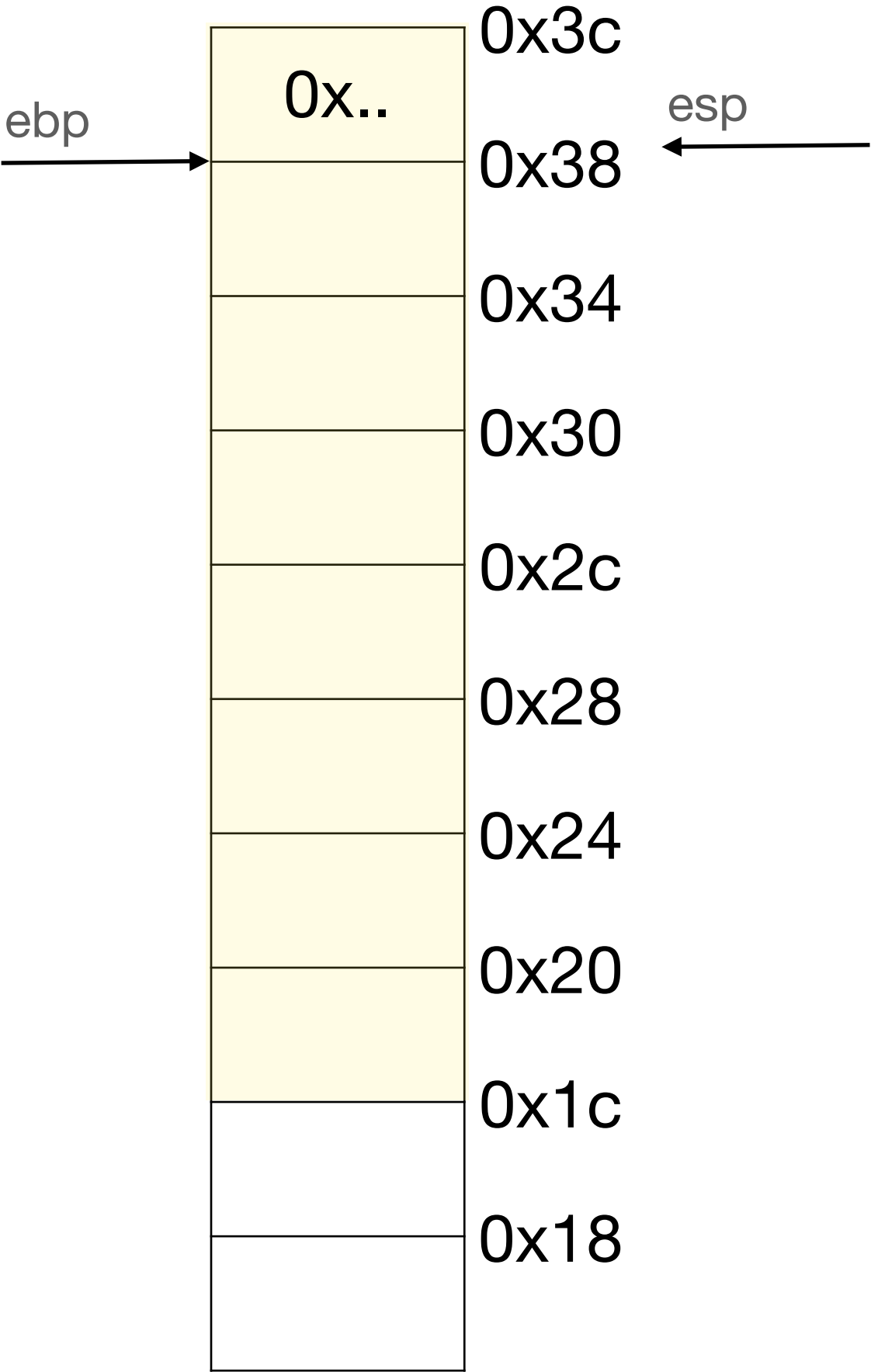
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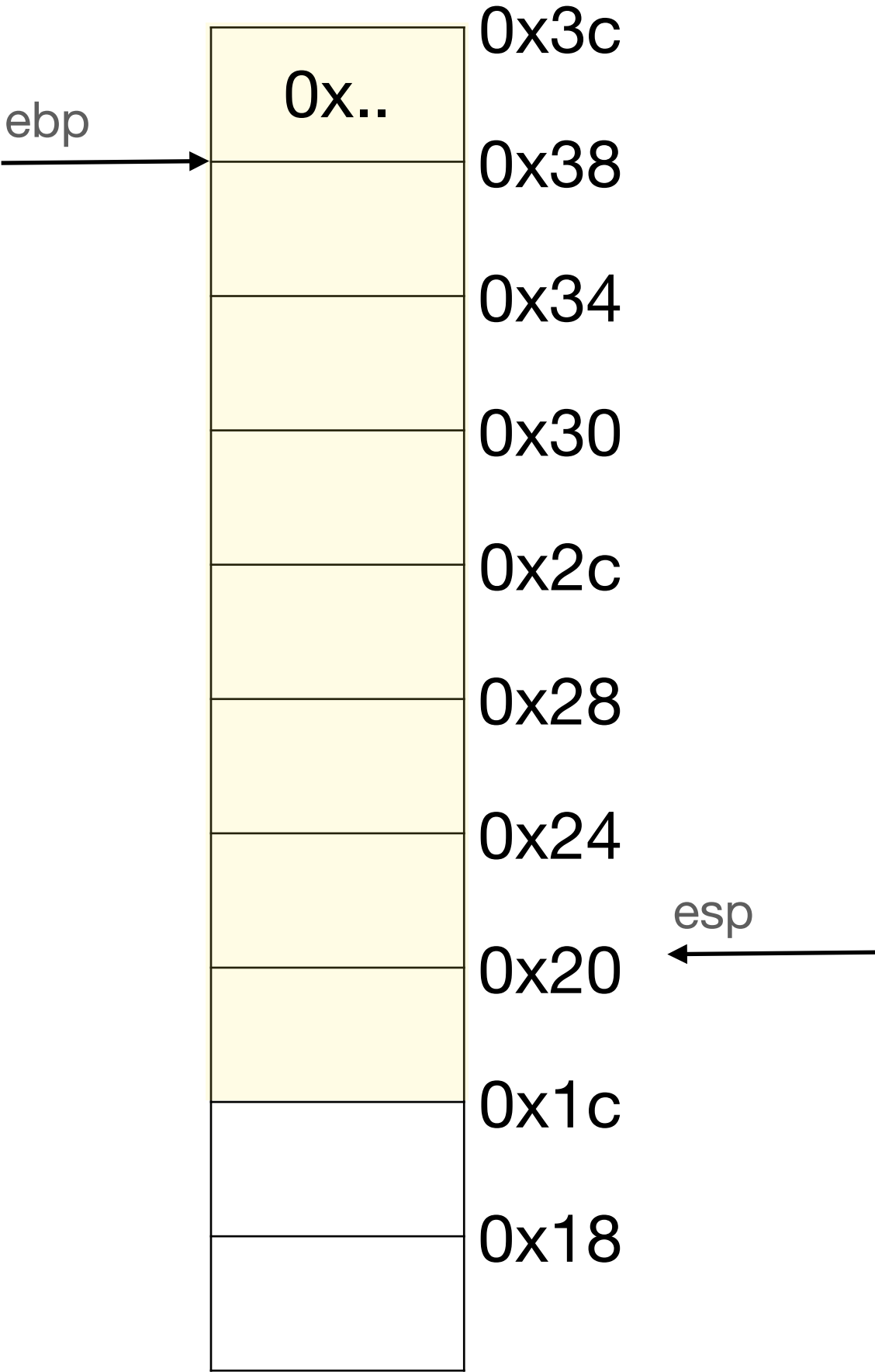
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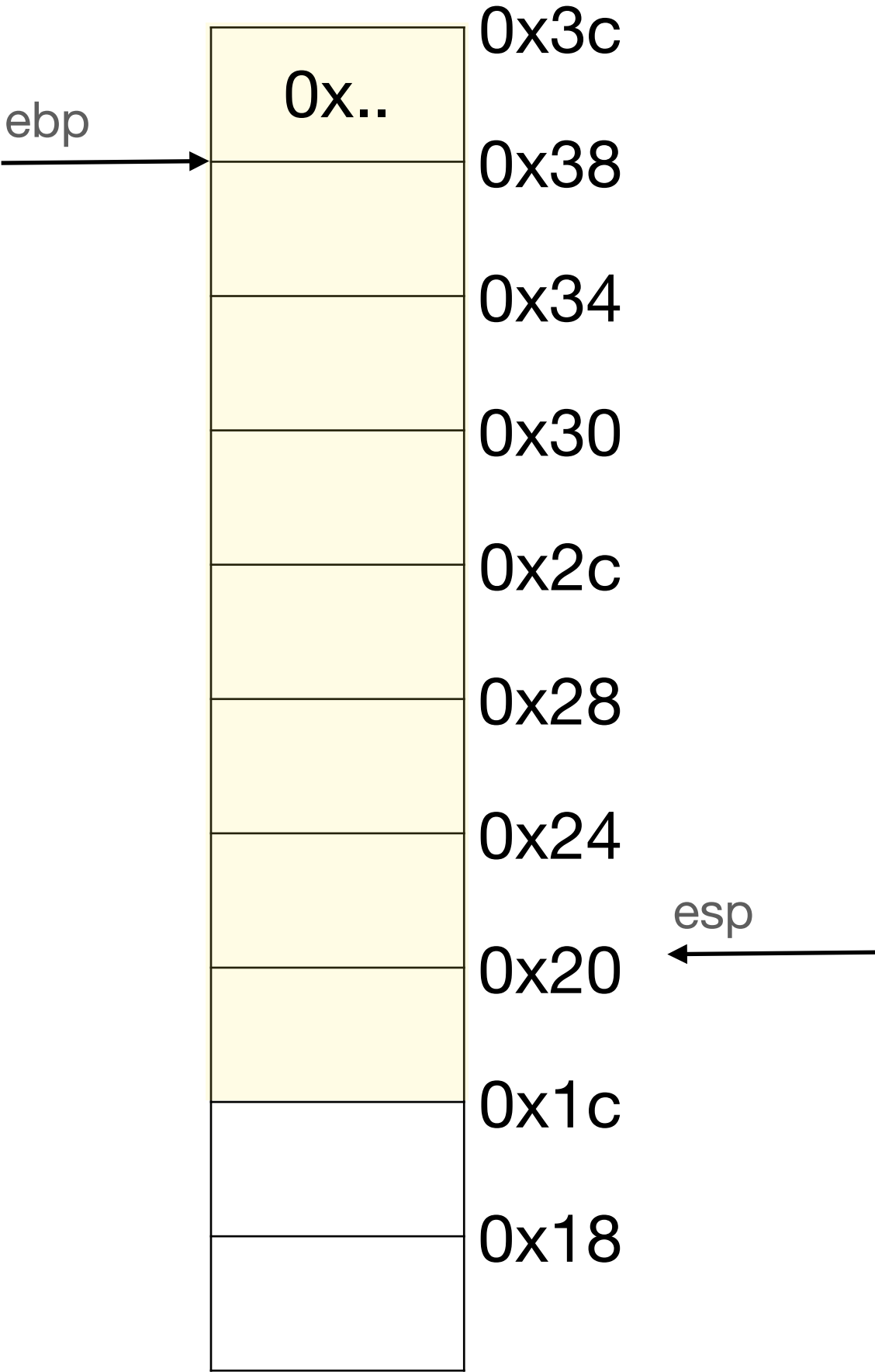
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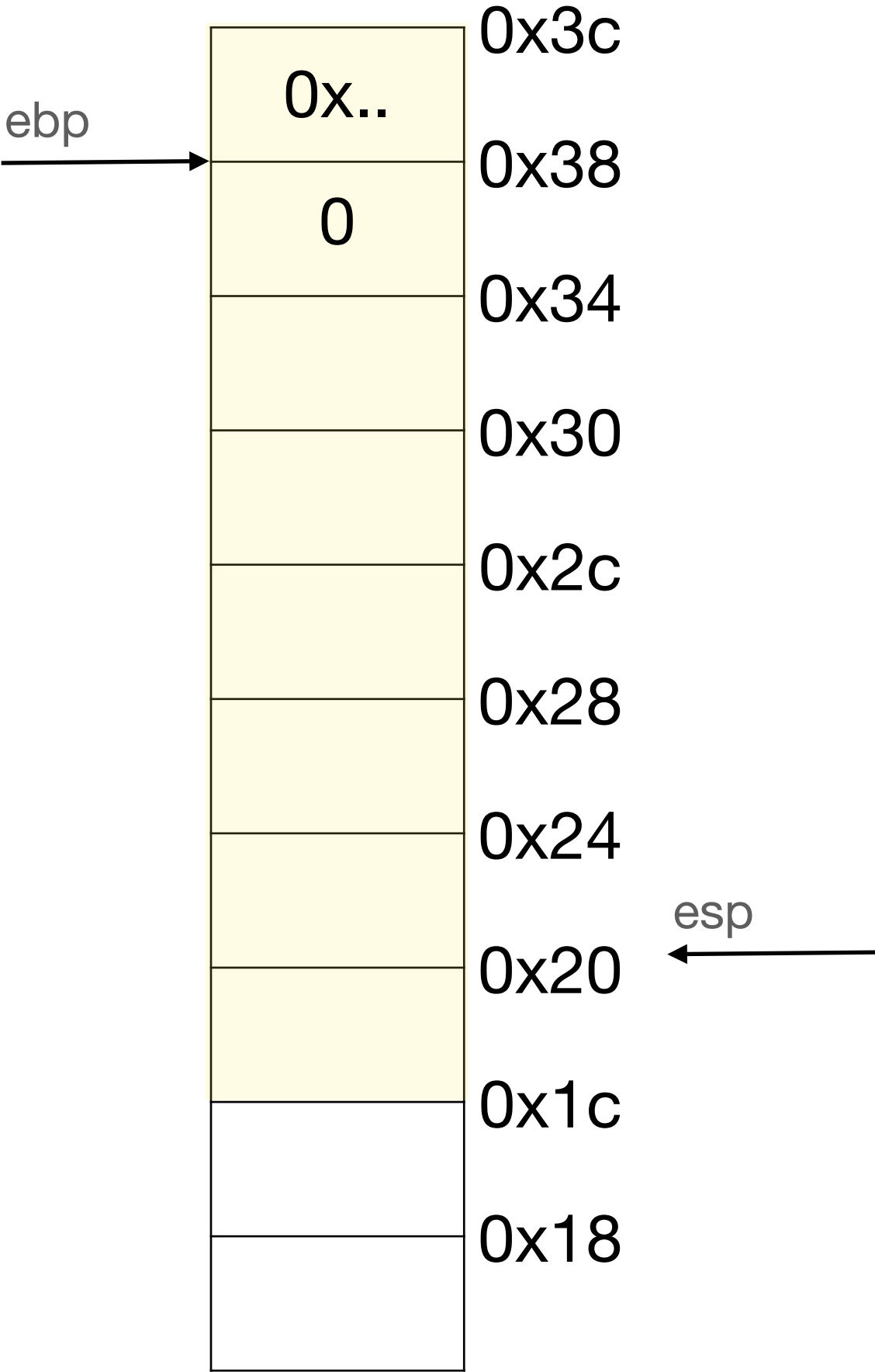
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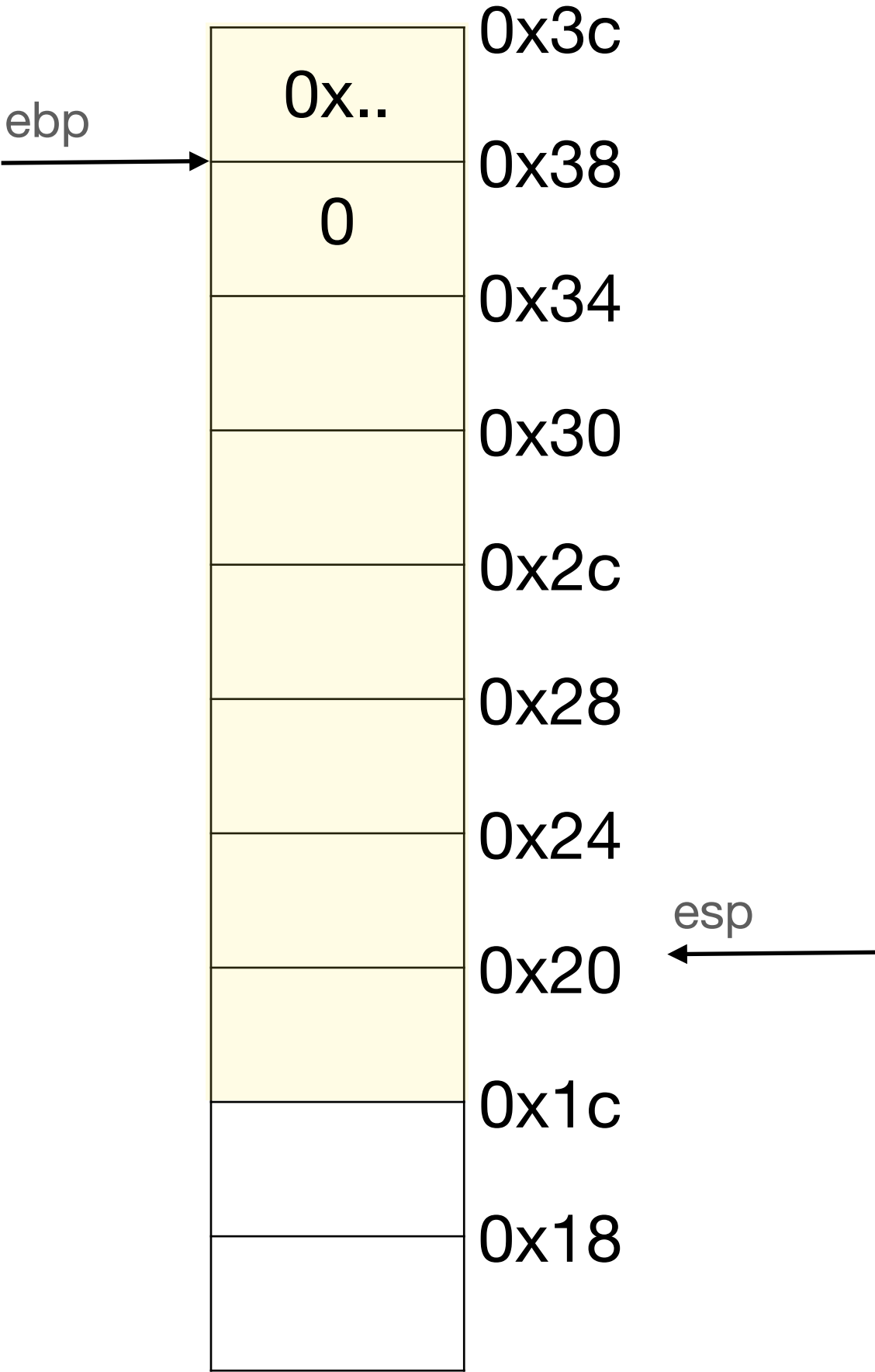
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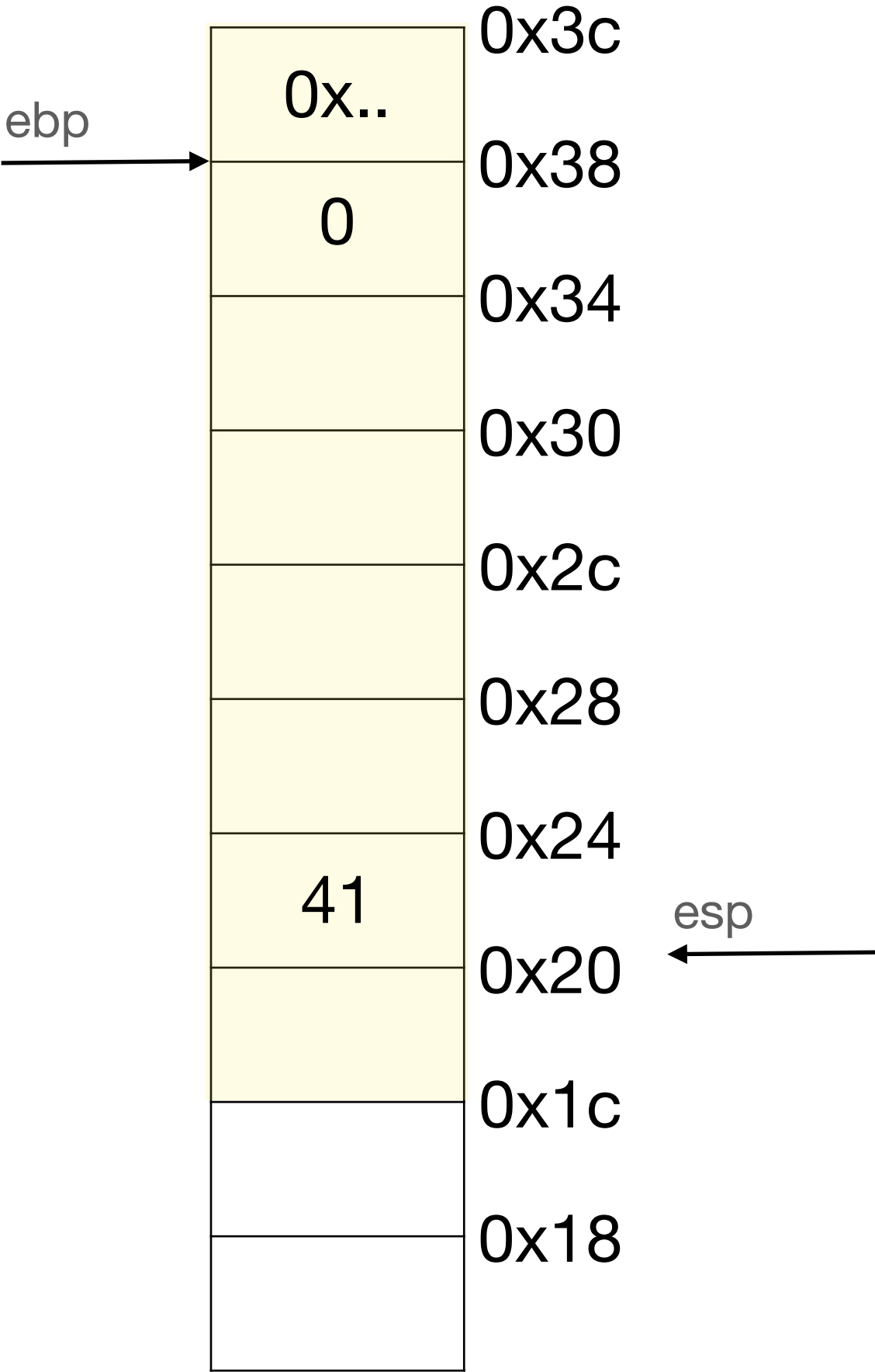
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ebp →

← esp

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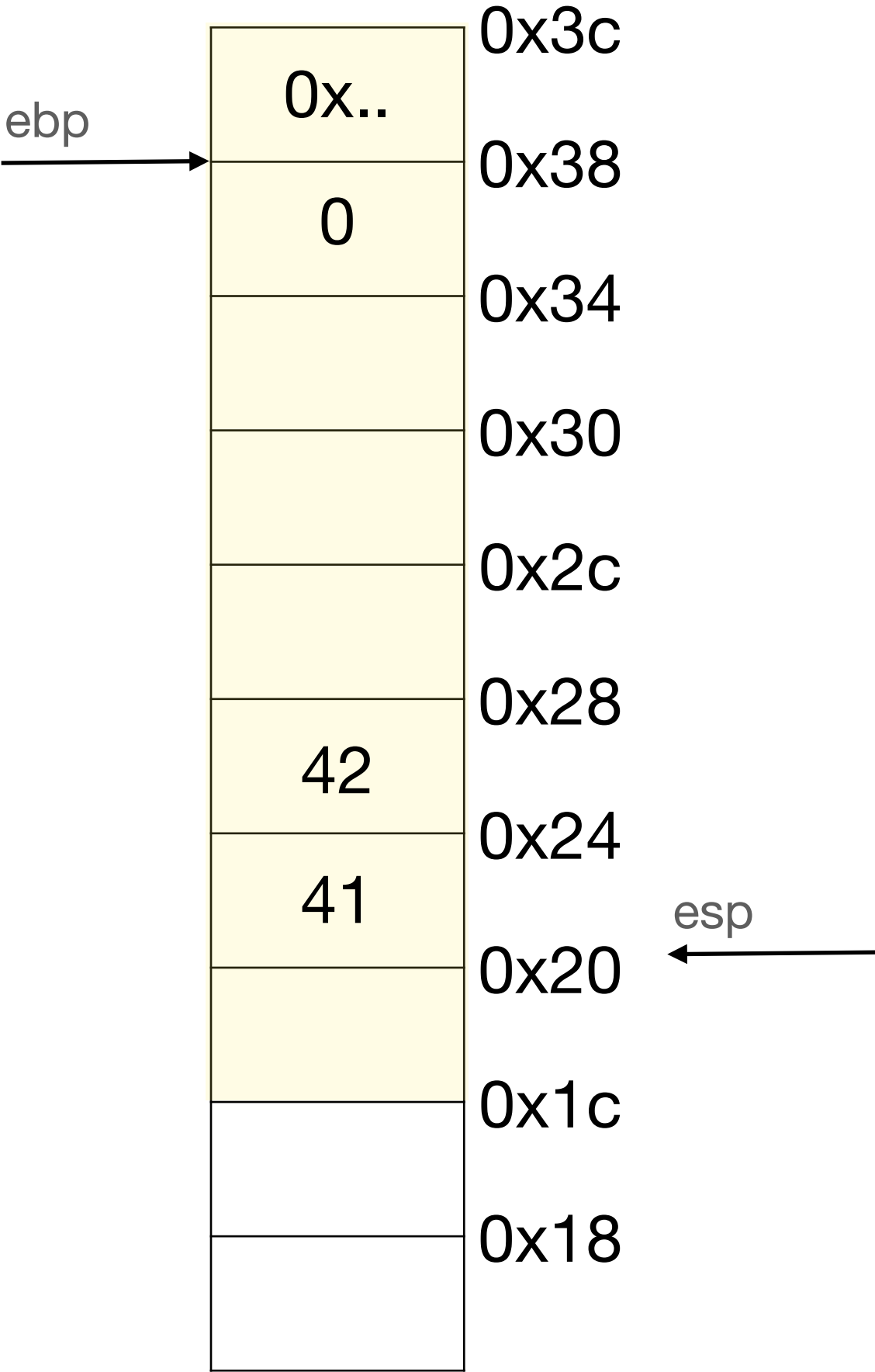
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	0x1c
	0x18

ebp →

← esp

Function calling in action

Stack

02.s

eip

→

_foo:

pushl %ebp

movl %esp, %ebp

movl 8(%ebp), %eax

addl 12(%ebp), %eax

popl %ebp

retl

.globl _main

.p2align 4, 0x90

_main:

pushl %ebp

movl %esp, %ebp

subl \$24, %esp

movl \$0, -4(%ebp)

movl \$41, (%esp)

movl \$42, 4(%esp)

calll _foo

addl \$24, %esp

popl %ebp

retl

Save caller's base pointer

ebp = esp

eax = *(ebp + 8)

eax = eax + *(ebp + 12)

Restore caller's base pointer

change eip to return address

-- Begin function main

Save caller's base pointer

ebp = esp

esp = esp - 0x18

*(ebp-4)=0

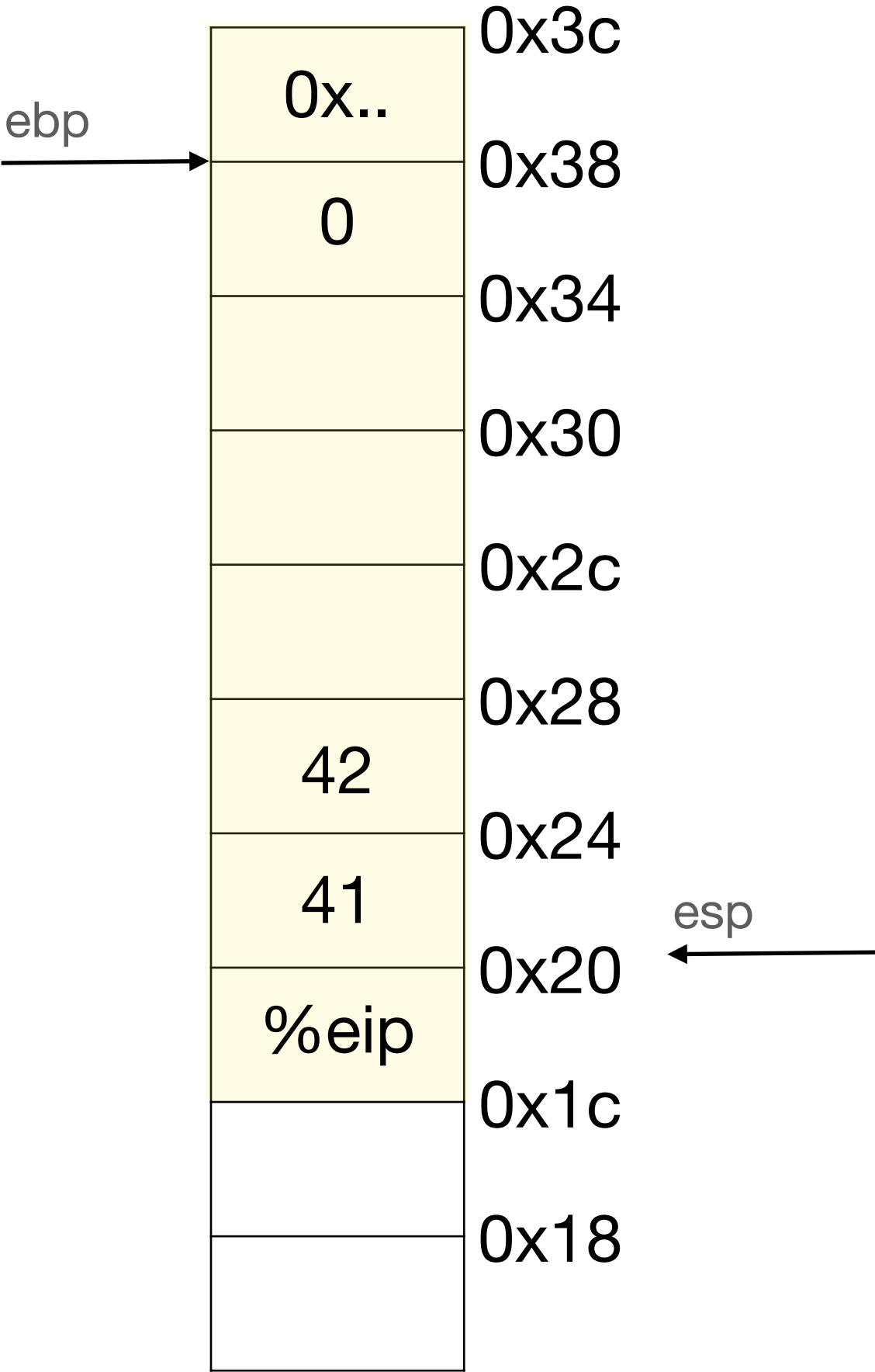
*(esp) = 41

*(esp+4) = 42

Push current eip on to stack, jump to foo

esp = esp + 24 (Restore caller's esp)

Restore caller's ebp



Function calling in action

Stack

02.s

eip

→

_foo:

pushl %ebp

movl %esp, %ebp

movl 8(%ebp), %eax

addl 12(%ebp), %eax

popl %ebp

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movl \$41, (%esp)

movl \$42, 4(%esp)

calll _foo

addl \$24, %esp

popl %ebp

retl

Save caller's base pointer

ebp = esp

eax = *(ebp + 8)

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Restore caller's base pointer

change eip to return address

-- Begin function main

Save caller's base pointer

ebp = esp

esp = esp - 0x18

*(ebp-4)=0

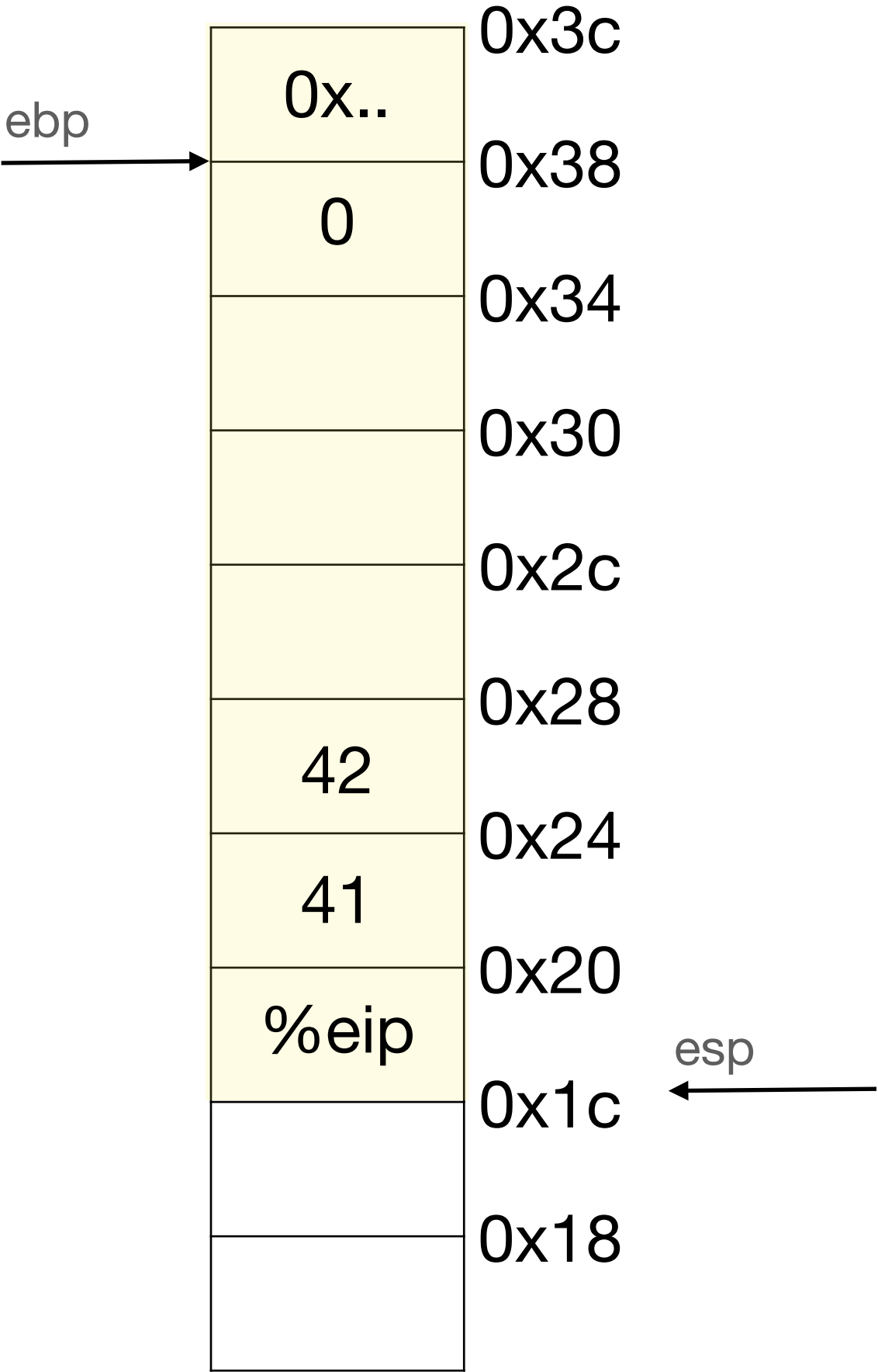
*(esp) = 41

*(esp+4) = 42

Push current eip on to stack, jump to foo

esp = esp + 24 (Restore caller's esp)

Restore caller's ebp



Function calling in action

Stack

02.s

eip

_foo:

pushl %ebp

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.p2align 4, 0x90

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change eip to return address

-- Begin function main

Save caller's base pointer

ebp = esp

esp = esp - 0x18

*(ebp-4)=0

*(esp) = 41

*(esp+4) = 42

Push current eip on to stack, jump to foo

esp = esp + 24 (Restore caller's esp)

Restore caller's ebp

	0x3c
ebp →	0x..
	0x38
	0
	0x34
	0x30
	0x2c
	0x28
	42
	0x24
	41
	0x20
	%eip
	0x1c
← esp	
	0x18

Function calling in action

Stack

02.s

eip

→

`_foo:`
`pushl %ebp`
`movl %esp, %ebp`
`movl 8(%ebp), %eax`
`addl 12(%ebp), %eax`
`popl %ebp`
`retl`

`.globl _main`
`.p2align 4, 0x90`
`_main:`
`pushl %ebp`
`movl %esp, %ebp`
`subl $24, %esp`
`movl $0, -4(%ebp)`
`movl $41, (%esp)`
`movl $42, 4(%esp)`
`calll _foo`
`addl $24, %esp`
`popl %ebp`
`retl`

Save caller's base pointer

`ebp = esp`

`eax = *(ebp + 8)`

`eax = eax + *(ebp + 12)`

Restore caller's base pointer

change eip to return address

`## -- Begin function main`

Save caller's base pointer

`ebp = esp`

`esp = esp - 0x18`

`*(ebp-4)=0`

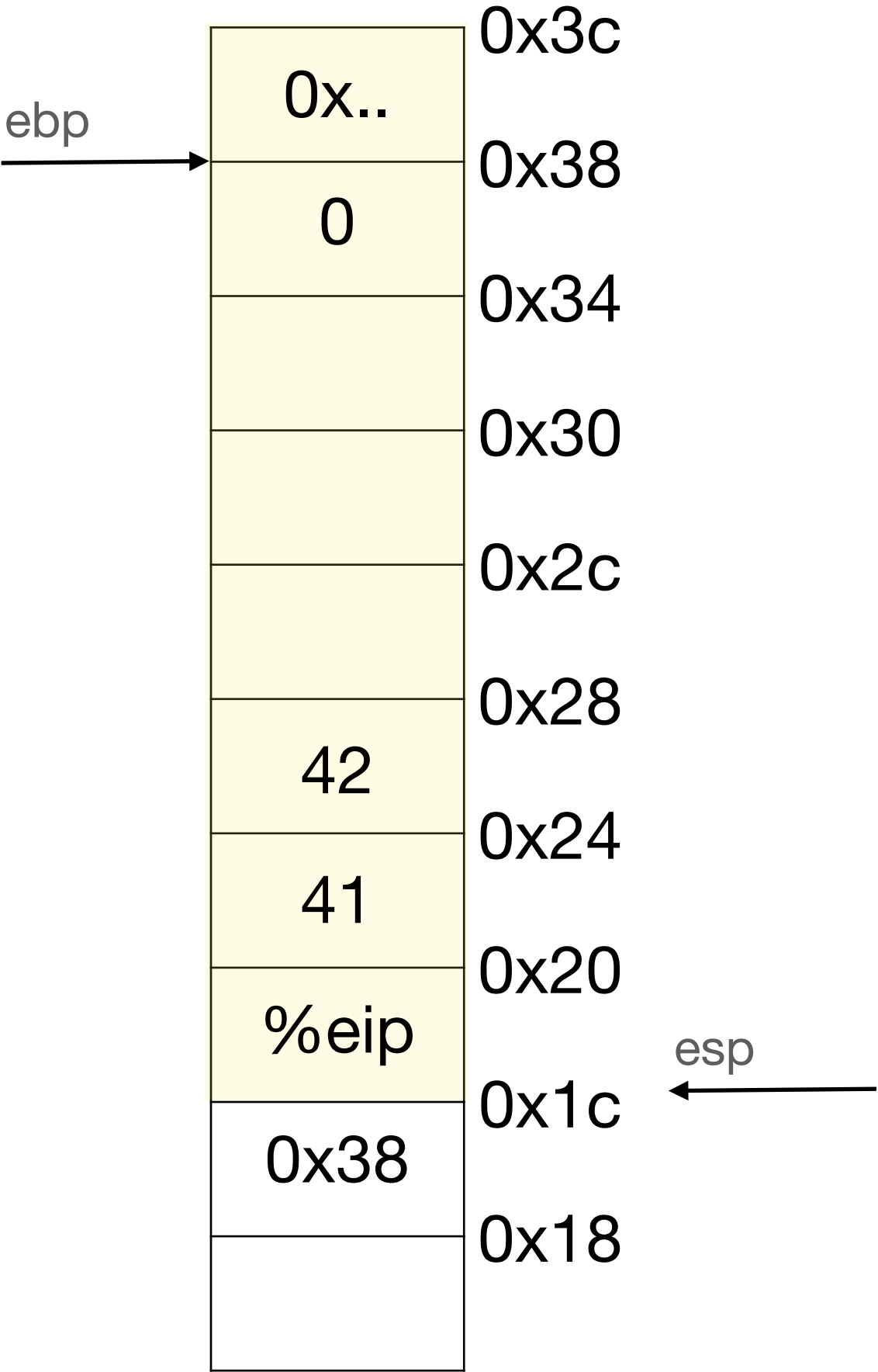
`*(esp) = 41`

`*(esp+4) = 42`

Push current eip on to stack, jump to foo

`esp = esp + 24 (Restore caller's esp)`

Restore caller's ebp



Function calling in action

Stack

02.s

eip

→

_foo:

pushl %ebp

movl %esp, %ebp

movl 8(%ebp), %eax

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.globl _main

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movl \$42, 4(%esp)

calll _foo

addl \$24, %esp

popl %ebp

retl

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change eip to return address

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ebp = esp

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*(ebp-4)=0

*(esp) = 41

*(esp+4) = 42

Push current eip on to stack, jump to foo

esp = esp + 24 (Restore caller's esp)

Restore caller's ebp

	0x3c
0x..	0x3c
0	0x38
	0x34
	0x30
	0x2c
42	0x28
41	0x24
%eip	0x20
0x38	0x1c
	0x18

ebp →

← esp

Function calling in action

Stack

02.s

eip

→

```
_foo:
    pushl %ebp
    movl  %esp, %ebp
    movl  8(%ebp), %eax
    addl  12(%ebp), %eax
    popl  %ebp
    retl

    .globl _main
    .p2align 4, 0x90
_main:
    pushl %ebp
    movl  %esp, %ebp
    subl  $24, %esp
    movl  $0, -4(%ebp)
    movl  $41, (%esp)
    movl  $42, 4(%esp)
    calll _foo
    addl  $24, %esp
    popl  %ebp
    retl
```

Save caller's base pointer

ebp = esp

eax = *(ebp + 8)

eax = eax + *(ebp + 12)

Restore caller's base pointer

change eip to return address

-- Begin function main

Save caller's base pointer

ebp = esp

esp = esp - 0x18

*(ebp-4)=0

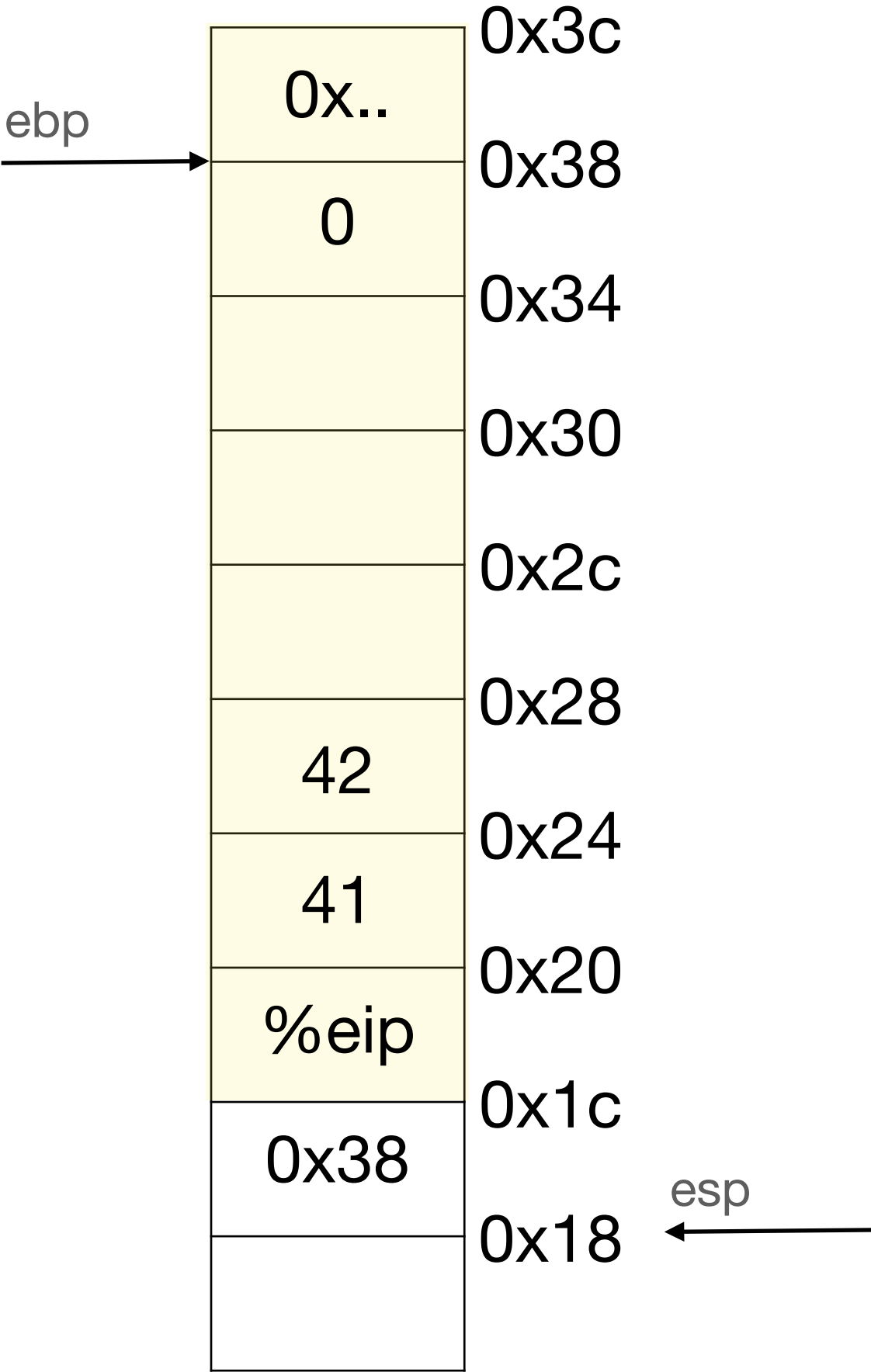
*(esp) = 41

*(esp+4) = 42

Push current eip on to stack, jump to foo

esp = esp + 24 (Restore caller's esp)

Restore caller's ebp



Function calling in action

Stack

02.s

eip

→

```
_foo:
    pushl %ebp
    movl  %esp, %ebp
    movl  8(%ebp), %eax
    addl  12(%ebp), %eax
    popl  %ebp
    retl

    .globl _main
    .p2align 4, 0x90
_main:
    pushl %ebp
    movl  %esp, %ebp
    subl  $24, %esp
    movl  $0, -4(%ebp)
    movl  $41, (%esp)
    movl  $42, 4(%esp)
    calll _foo
    addl  $24, %esp
    popl  %ebp
    retl
```

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ebp = esp

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Restore caller's base pointer

change eip to return address

-- Begin function main

Save caller's base pointer

ebp = esp

esp = esp - 0x18

*(ebp-4)=0

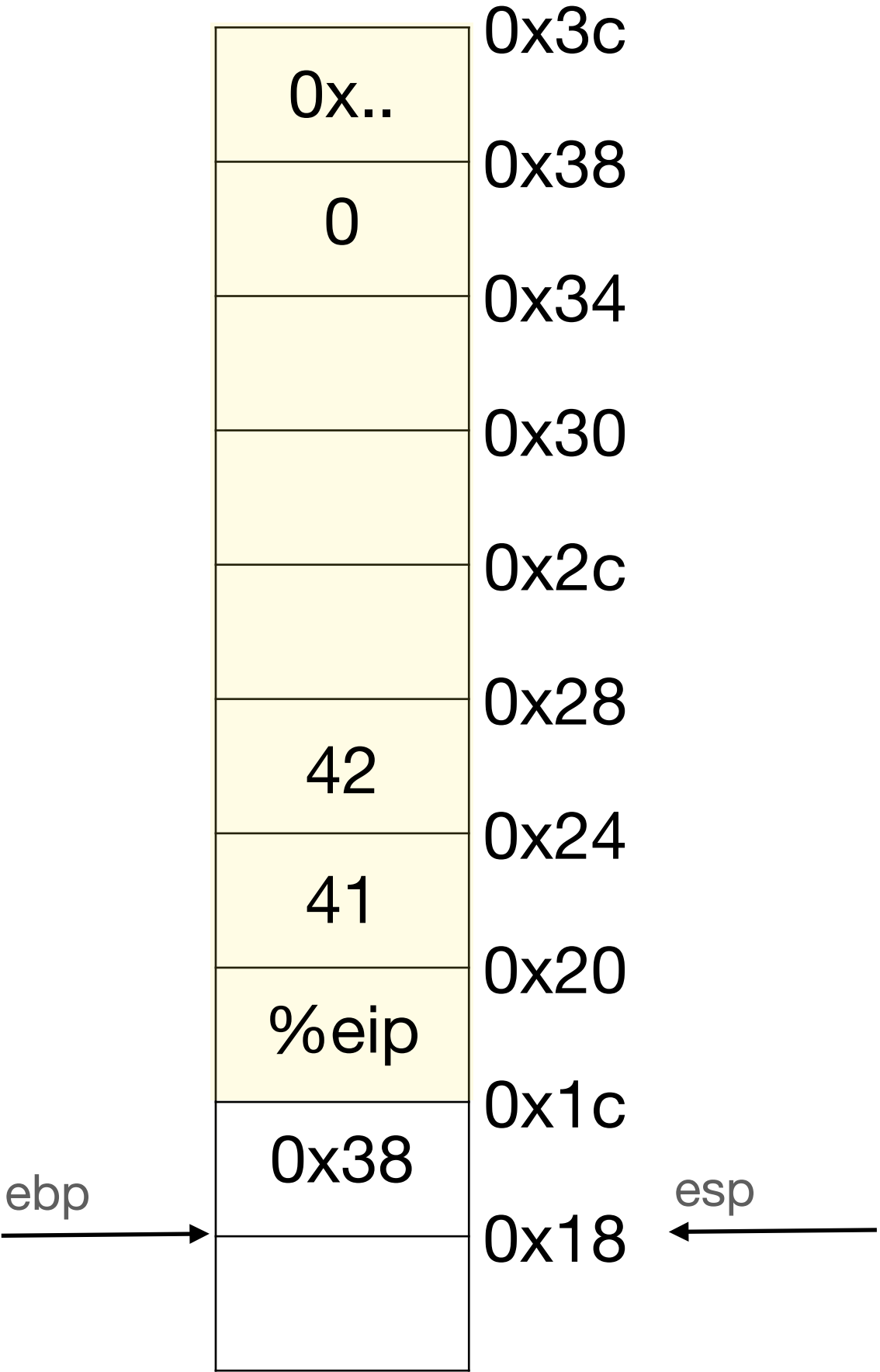
*(esp) = 41

*(esp+4) = 42

Push current eip on to stack, jump to foo

esp = esp + 24 (Restore caller's esp)

Restore caller's ebp



Function calling in action

Stack

02.s

eip

→

```
_foo:
    pushl %ebp
    movl  %esp, %ebp
    movl  8(%ebp), %eax
    addl  12(%ebp), %eax
    popl  %ebp
    retl

    .globl _main
    .p2align 4, 0x90
_main:
    pushl %ebp
    movl  %esp, %ebp
    subl  $24, %esp
    movl  $0, -4(%ebp)
    movl  $41, (%esp)
    movl  $42, 4(%esp)
    calll _foo
    addl  $24, %esp
    popl  %ebp
    retl
```

Save caller's base pointer

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change eip to return address

-- Begin function main

Save caller's base pointer

ebp = esp

esp = esp - 0x18

*(ebp-4)=0

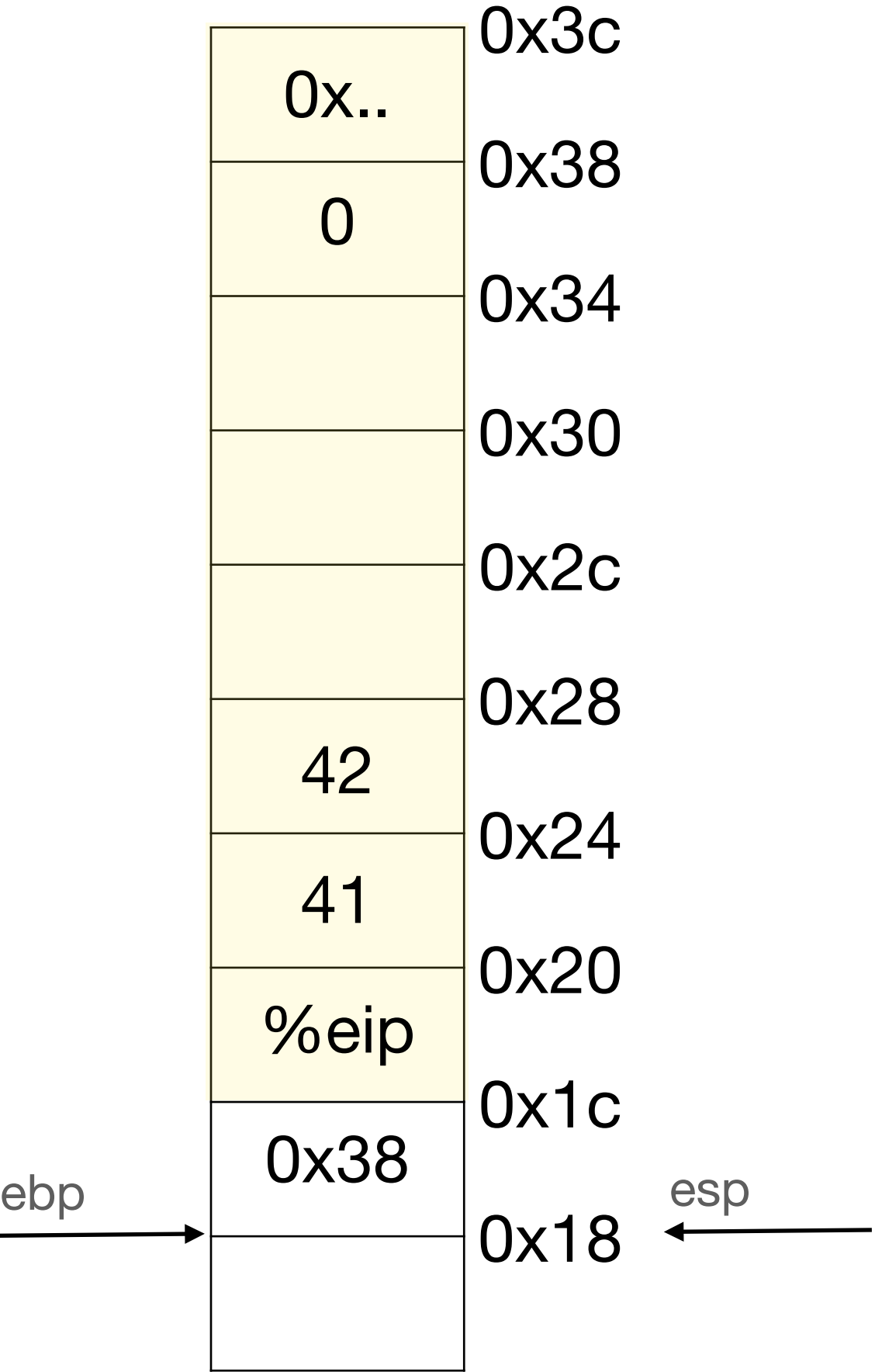
*(esp) = 41

*(esp+4) = 42

Push current eip on to stack, jump to foo

esp = esp + 24 (Restore caller's esp)

Restore caller's ebp



Function calling in action

Stack

02.s

eip

→

_foo:

pushl %ebp

movl %esp, %ebp

movl 8(%ebp), %eax

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popl %ebp

retl

.globl _main

.p2align 4, 0x90

_main:

pushl %ebp

movl %esp, %ebp

subl \$24, %esp

movl \$0, -4(%ebp)

movl \$41, (%esp)

movl \$42, 4(%esp)

calll _foo

addl \$24, %esp

popl %ebp

retl

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change eip to return address

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Save caller's base pointer

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esp = esp - 0x18

*(ebp-4)=0

*(esp) = 41

*(esp+4) = 42

Push current eip on to stack, jump to foo

esp = esp + 24 (Restore caller's esp)

Restore caller's ebp

0x3c	0x..
0x38	0
0x34	
0x30	
0x2c	
0x28	42
0x24	41
0x20	%eip
0x1c	0x38
0x18	

ebp → 0x38 ← esp

Function calling in action

Stack

02.s

eip

→

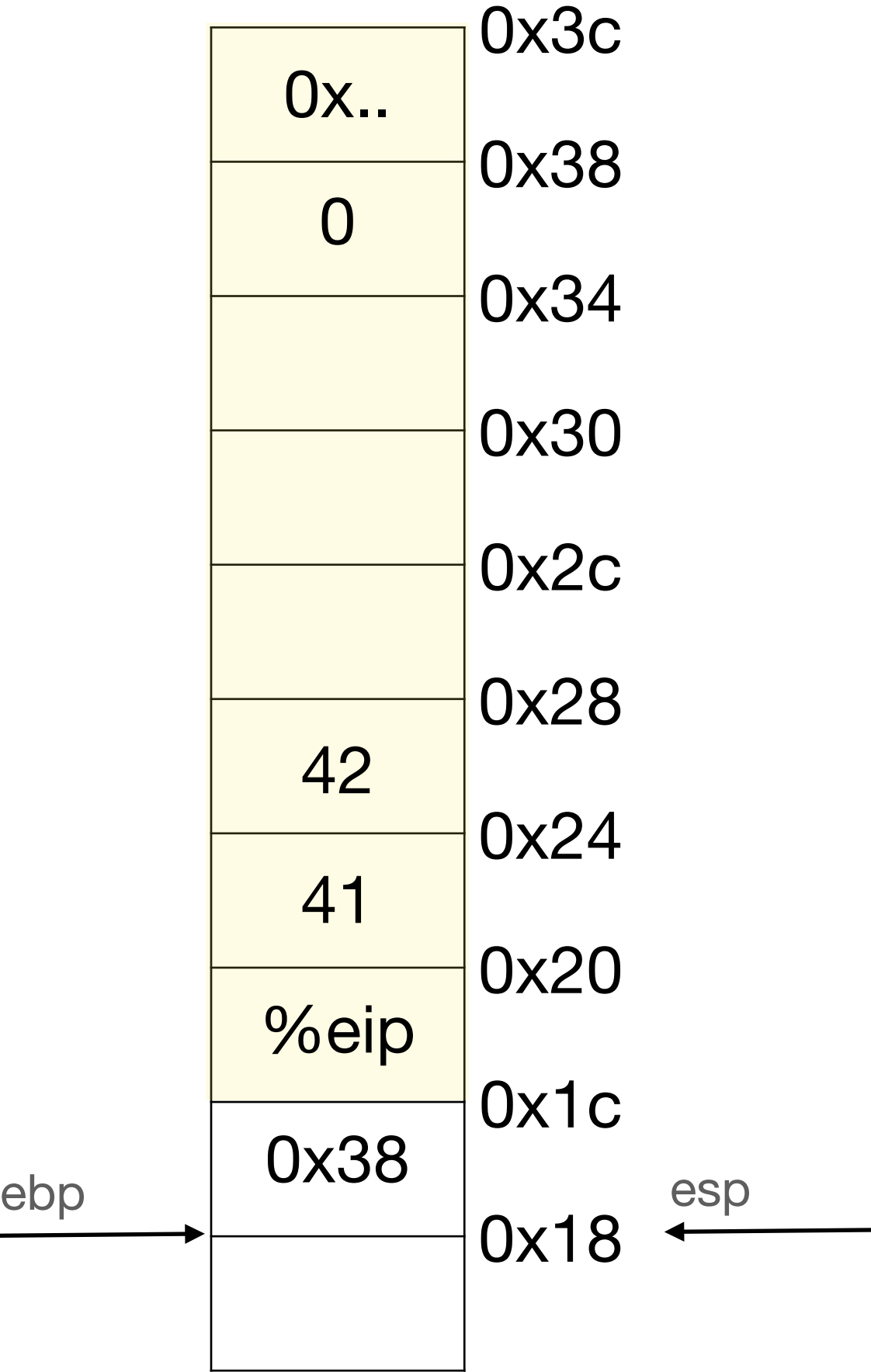
```
_foo:
    pushl %ebp
    movl  %esp, %ebp
    movl  8(%ebp), %eax
    addl  12(%ebp), %eax
    popl  %ebp
    retl

    .globl _main
    .p2align 4, 0x90
_main:
    pushl %ebp
    movl  %esp, %ebp
    subl  $24, %esp
    movl  $0, -4(%ebp)
    movl  $41, (%esp)
    movl  $42, 4(%esp)
    calll _foo
    addl  $24, %esp
    popl  %ebp
    retl
```

```
Save caller's base pointer
ebp = esp
eax = *(ebp + 8)
eax = eax + *(ebp + 12)
Restore caller's base pointer
change eip to return address

## -- Begin function main

Save caller's base pointer
ebp = esp
esp = esp - 0x18
*(ebp-4)=0
*(esp) = 41
*(esp+4) = 42
Push current eip on to stack, jump to foo
esp = esp + 24 (Restore caller's esp)
Restore caller's ebp
```



Function calling in action

Stack

02.s

eip

→

```
_foo:
    pushl %ebp
    movl  %esp, %ebp
    movl  8(%ebp), %eax
    addl  12(%ebp), %eax
    popl  %ebp
    retl

    .globl _main
    .p2align 4, 0x90
_main:
    pushl %ebp
    movl  %esp, %ebp
    subl  $24, %esp
    movl  $0, -4(%ebp)
    movl  $41, (%esp)
    movl  $42, 4(%esp)
    calll _foo
    addl  $24, %esp
    popl  %ebp
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```

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Save caller's base pointer

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esp = esp - 0x18

*(ebp-4)=0

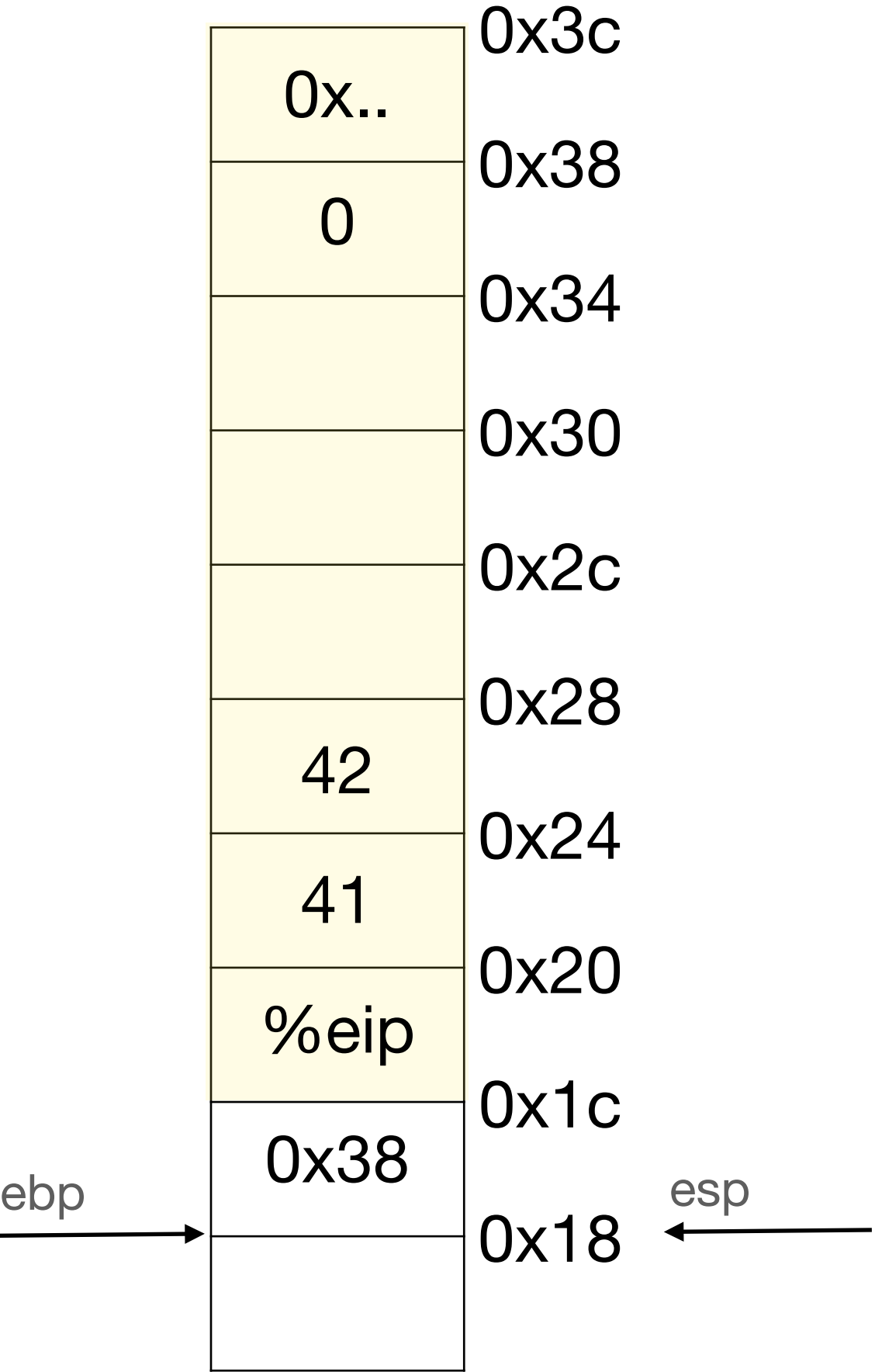
*(esp) = 41

*(esp+4) = 42

Push current eip on to stack, jump to foo

esp = esp + 24 (Restore caller's esp)

Restore caller's ebp



Function calling in action

Stack

02.s

eip

→

```
_foo:
    pushl %ebp
    movl  %esp, %ebp
    movl  8(%ebp), %eax
    addl  12(%ebp), %eax
    popl  %ebp
    retl

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    .p2align 4, 0x90
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    pushl %ebp
    movl  %esp, %ebp
    subl  $24, %esp
    movl  $0, -4(%ebp)
    movl  $41, (%esp)
    movl  $42, 4(%esp)
    calll _foo
    addl  $24, %esp
    popl  %ebp
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Restore caller's base pointer

change eip to return address

-- Begin function main

Save caller's base pointer

ebp = esp

esp = esp - 0x18

*(ebp-4)=0

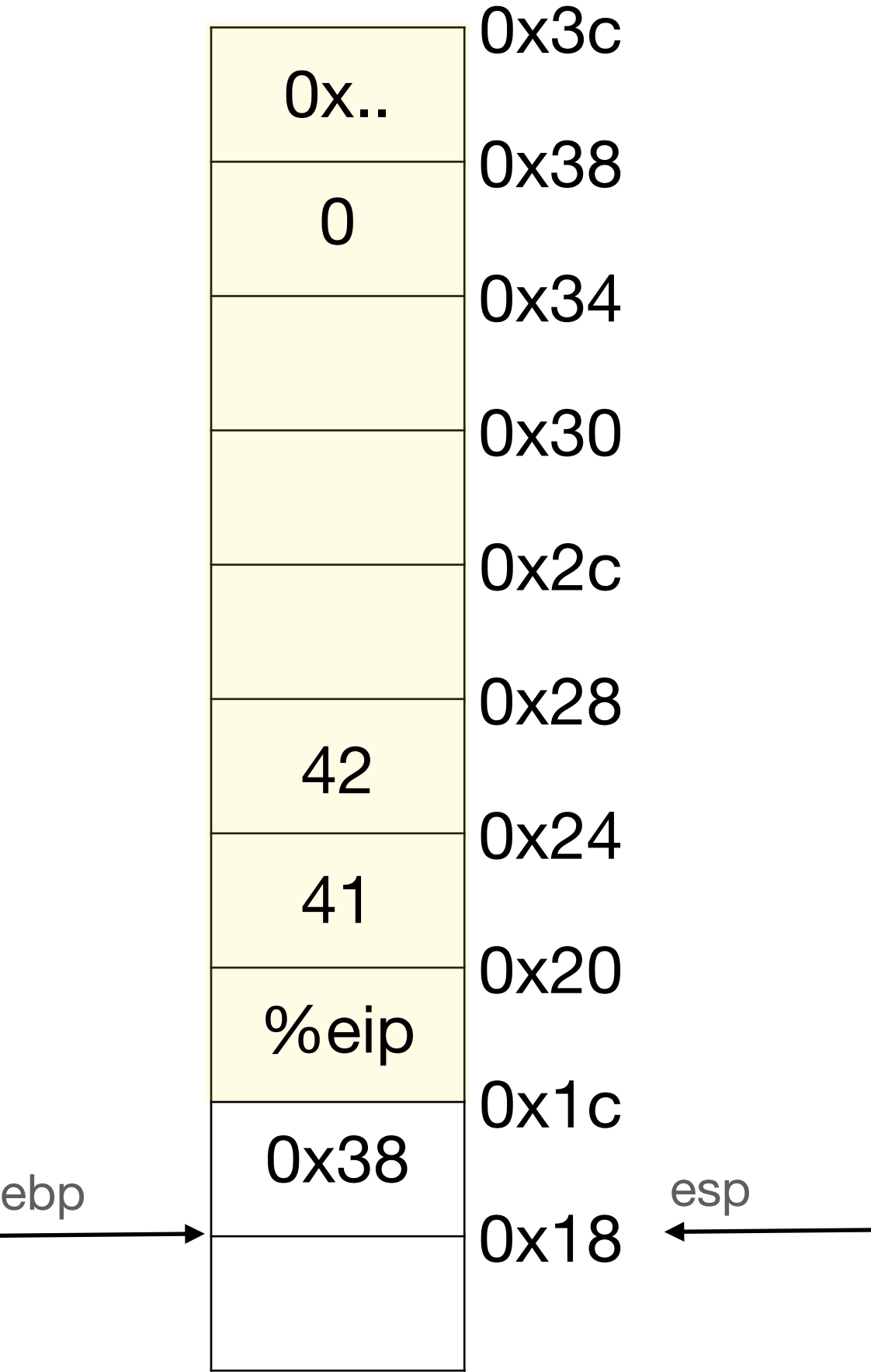
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Push current eip on to stack, jump to foo

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Restore caller's ebp



Function calling in action

Stack

02.s

eip

→

```
_foo:
    pushl %ebp
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    movl  8(%ebp), %eax
    addl  12(%ebp), %eax
    popl  %ebp
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    .globl _main
    .p2align 4, 0x90
_main:
    pushl %ebp
    movl  %esp, %ebp
    subl  $24, %esp
    movl  $0, -4(%ebp)
    movl  $41, (%esp)
    movl  $42, 4(%esp)
    calll _foo
    addl  $24, %esp
    popl  %ebp
    retl
```

Save caller's base pointer

ebp = esp

eax = *(ebp + 8)

eax = eax + *(ebp + 12)

Restore caller's base pointer

change eip to return address

-- Begin function main

Save caller's base pointer

ebp = esp

esp = esp - 0x18

*(ebp-4)=0

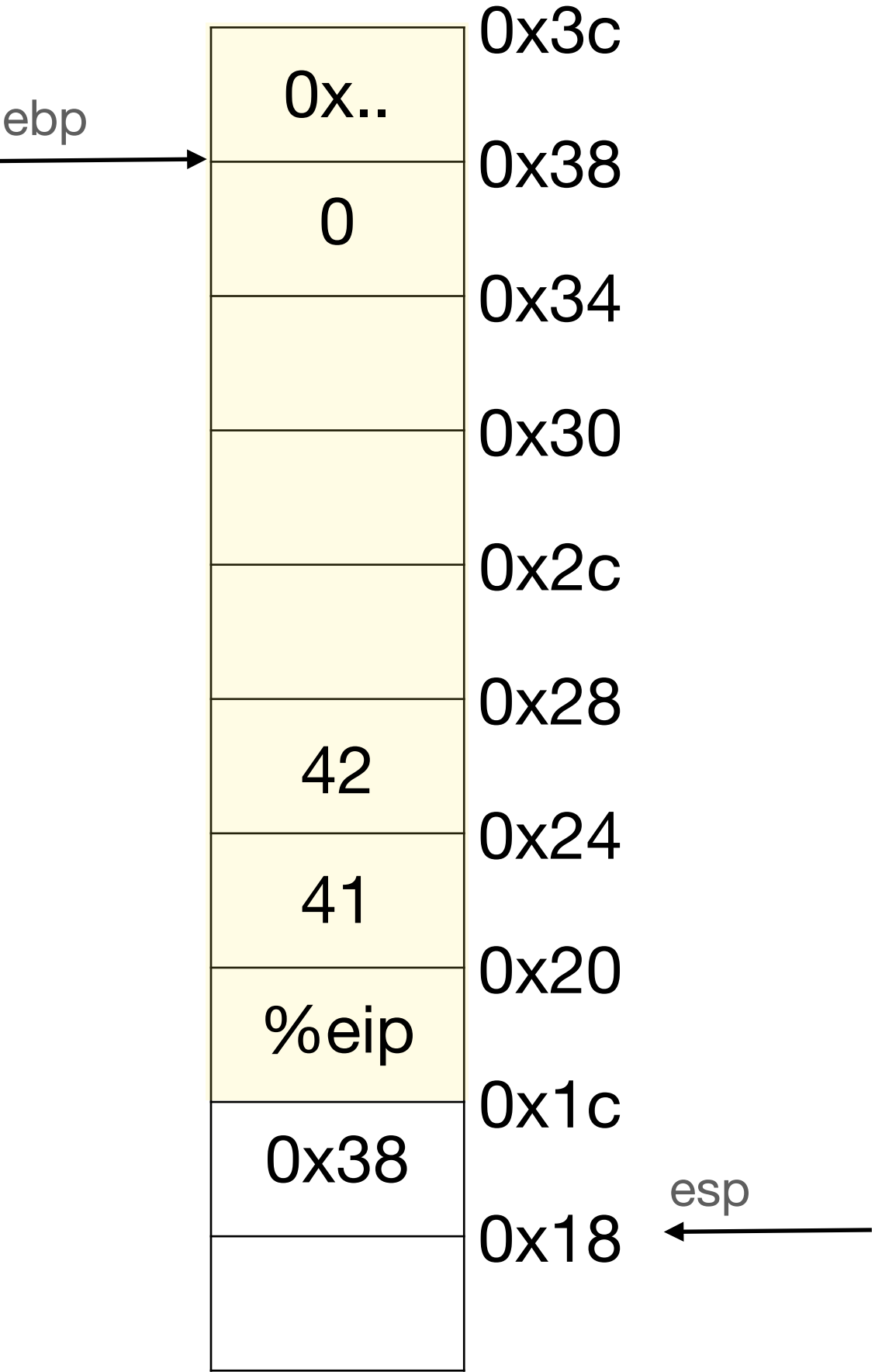
*(esp) = 41

*(esp+4) = 42

Push current eip on to stack, jump to foo

esp = esp + 24 (Restore caller's esp)

Restore caller's ebp



Function calling in action

Stack

02.s

eip

→

_foo:

pushl %ebp

movl %esp, %ebp

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popl %ebp

retl

.globl _main

.p2align 4, 0x90

_main:

pushl %ebp

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movl \$41, (%esp)

movl \$42, 4(%esp)

calll _foo

addl \$24, %esp

popl %ebp

retl

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change eip to return address

-- Begin function main

Save caller's base pointer

ebp = esp

esp = esp - 0x18

*(ebp-4)=0

*(esp) = 41

*(esp+4) = 42

Push current eip on to stack, jump to foo

esp = esp + 24 (Restore caller's esp)

Restore caller's ebp

	0x3c
0x..	
0	0x38
	0x34
	0x30
	0x2c
	0x28
42	0x24
41	0x20
%eip	0x1c ← esp
0x38	0x18

Function calling in action

Stack

02.s

eip

→

```
_foo:
    pushl %ebp
    movl  %esp, %ebp
    movl  8(%ebp), %eax
    addl  12(%ebp), %eax
    popl  %ebp
    retl

    .globl _main
    .p2align 4, 0x90
_main:
    pushl %ebp
    movl  %esp, %ebp
    subl  $24, %esp
    movl  $0, -4(%ebp)
    movl  $41, (%esp)
    movl  $42, 4(%esp)
    calll _foo
    addl  $24, %esp
    popl  %ebp
    retl
```

Save caller's base pointer

ebp = esp

eax = *(ebp + 8)

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Restore caller's base pointer

change eip to return address

-- Begin function main

Save caller's base pointer

ebp = esp

esp = esp - 0x18

*(ebp-4)=0

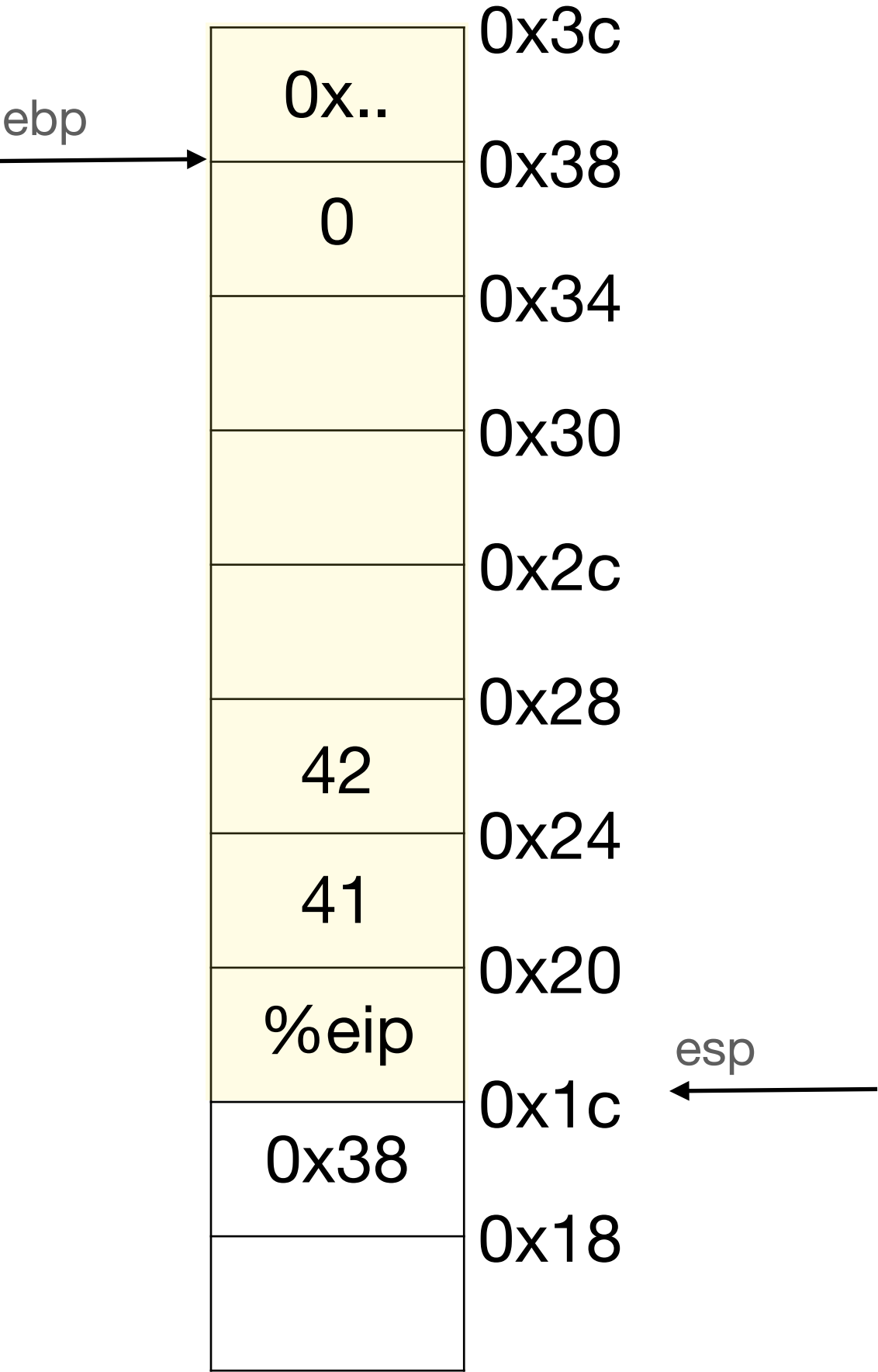
*(esp) = 41

*(esp+4) = 42

Push current eip on to stack, jump to foo

esp = esp + 24 (Restore caller's esp)

Restore caller's ebp



Function calling in action

Stack

02.s

eip

→

_foo:

pushl %ebp

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movl 8(%ebp), %eax

addl 12(%ebp), %eax

popl %ebp

retl

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.p2align 4, 0x90

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pushl %ebp

movl %esp, %ebp

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movl \$41, (%esp)

movl \$42, 4(%esp)

calll _foo

addl \$24, %esp

popl %ebp

retl

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Restore caller's base pointer

change eip to return address

-- Begin function main

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ebp = esp

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*(ebp-4)=0

*(esp) = 41

*(esp+4) = 42

Push current eip on to stack, jump to foo

esp = esp + 24 (Restore caller's esp)

Restore caller's ebp

	0x3c
0x..	0x38
0	0x34
	0x30
	0x2c
	0x28
42	0x24
41	0x20 ← esp
%eip	0x1c
0x38	0x18

Function calling in action

Stack

02.s

eip

→

_foo:

pushl %ebp

movl %esp, %ebp

movl 8(%ebp), %eax

addl 12(%ebp), %eax

popl %ebp

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pushl %ebp

movl %esp, %ebp

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movl \$0, -4(%ebp)

movl \$41, (%esp)

movl \$42, 4(%esp)

calll _foo

addl \$24, %esp

popl %ebp

retl

Save caller's base pointer

ebp = esp

eax = *(ebp + 8)

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Restore caller's base pointer

change eip to return address

-- Begin function main

Save caller's base pointer

ebp = esp

esp = esp - 0x18

*(ebp-4)=0

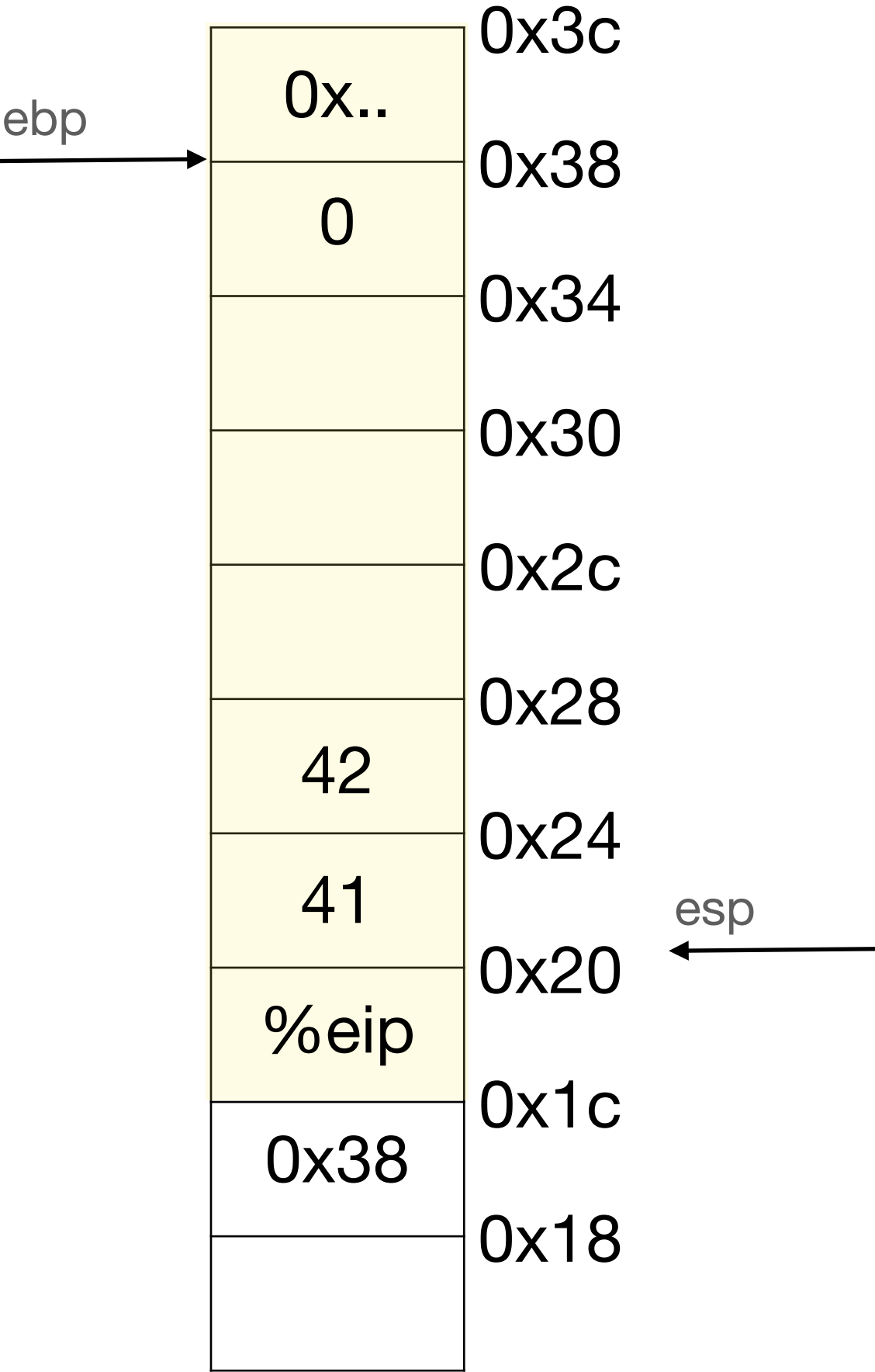
*(esp) = 41

*(esp+4) = 42

Push current eip on to stack, jump to foo

esp = esp + 24 (Restore caller's esp)

Restore caller's ebp



Function calling in action

Stack

02.s

eip

→

addl \$24, %esp

popl %ebp

retl

_foo:

pushl %ebp

movl %esp, %ebp

movl 8(%ebp), %eax

addl 12(%ebp), %eax

popl %ebp

retl

.globl _main

.p2align 4, 0x90

_main:

pushl %ebp

movl %esp, %ebp

subl \$24, %esp

movl \$0, -4(%ebp)

movl \$41, (%esp)

movl \$42, 4(%esp)

calll _foo

Save caller's base pointer

ebp = esp

eax = *(ebp + 8)

eax = eax + *(ebp + 12)

Restore caller's base pointer

change eip to return address

-- Begin function main

Push current eip on to stack, jump to foo

esp = esp + 24 (Restore caller's esp)

Restore caller's ebp

	0x..	0x3c
ebp →	0	0x38
		0x34
		0x30
		0x2c
	42	0x28
	41	0x24
	%eip	0x20 ← esp
	0x38	0x1c
		0x18

Function calling in action

Stack

02.s

eip

→

addl \$24, %esp

popl %ebp

retl

_foo:

pushl %ebp

movl %esp, %ebp

movl 8(%ebp), %eax

addl 12(%ebp), %eax

popl %ebp

retl

.globl _main

.p2align 4, 0x90

_main:

pushl %ebp

movl %esp, %ebp

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movl \$0, -4(%ebp)

movl \$41, (%esp)

movl \$42, 4(%esp)

calll _foo

Save caller's base pointer

ebp = esp

eax = *(ebp + 8)

eax = eax + *(ebp + 12)

Restore caller's base pointer

change eip to return address

-- Begin function main

Push current eip on to stack, jump to foo

esp = esp + 24 (Restore caller's esp)

Restore caller's ebp

	0x3c
0x..	0x38
0	0x34
	0x30
	0x2c
	0x28
42	0x24
41	0x20
%eip	0x1c
0x38	0x18
	0x14

Function calling in action

Stack

02.s

```
foo:
```

Save caller's base pointer

ebp = esp

```
eax = *(ebp + 8)
```

```
eax = eax + *(ebp + 12)
```

Restore caller's base pointer

change `eip` to return address

```
## -- Begin function main
```

```
## -- Begin function main
```

```
main:
```

Save caller's base pointer

ebp = esp

```
esp = esp - 0x18
```

$$*(ebp-4)=0$$

* (esp) = 41

```
* (esp+4) = 42
```

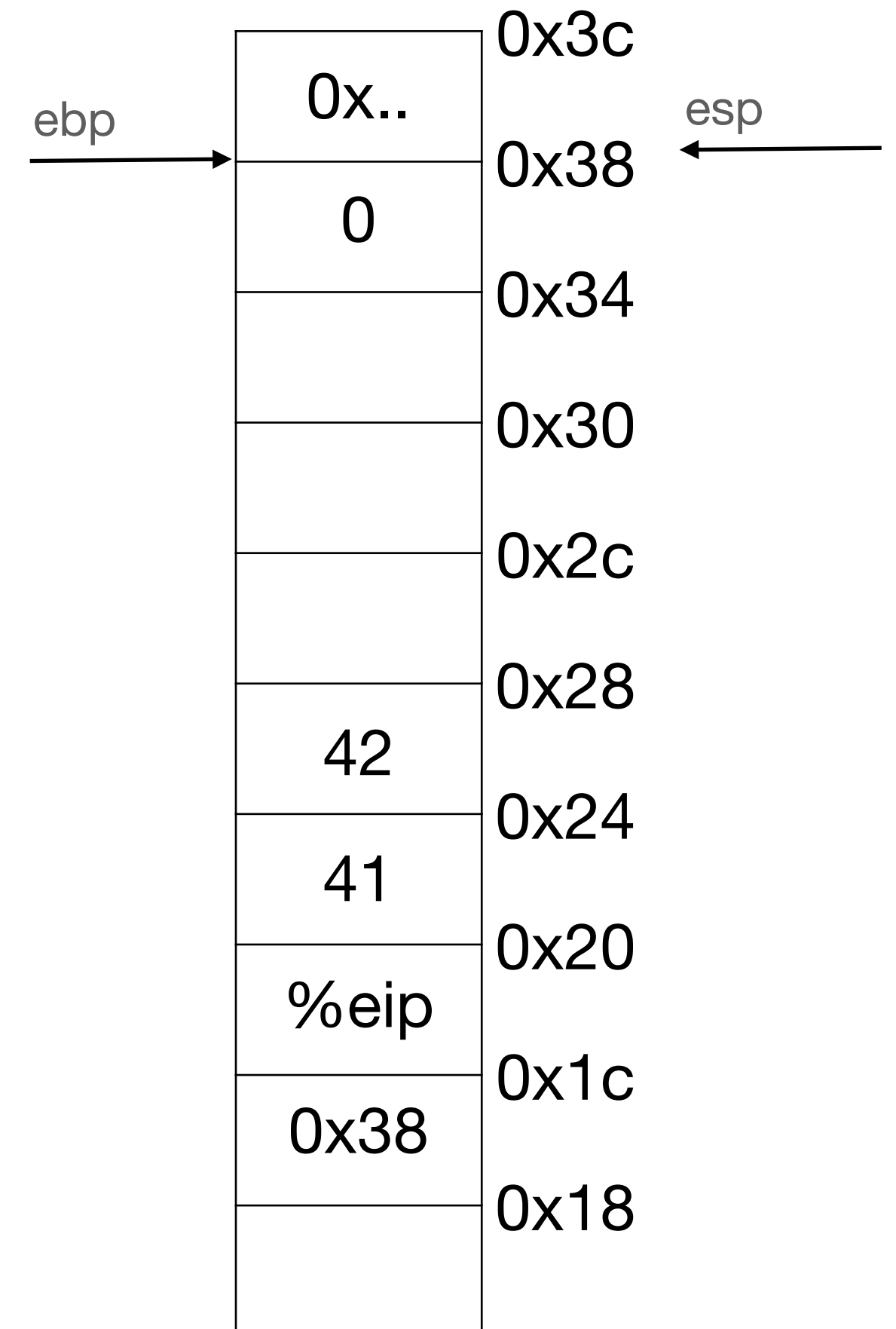
Push current eip on to stack, jump to foo

```
esp = esp + 24 (Restore caller's esp)
```

eip

Restore caller's ebp

Restore caller's ebp



Function calling in action

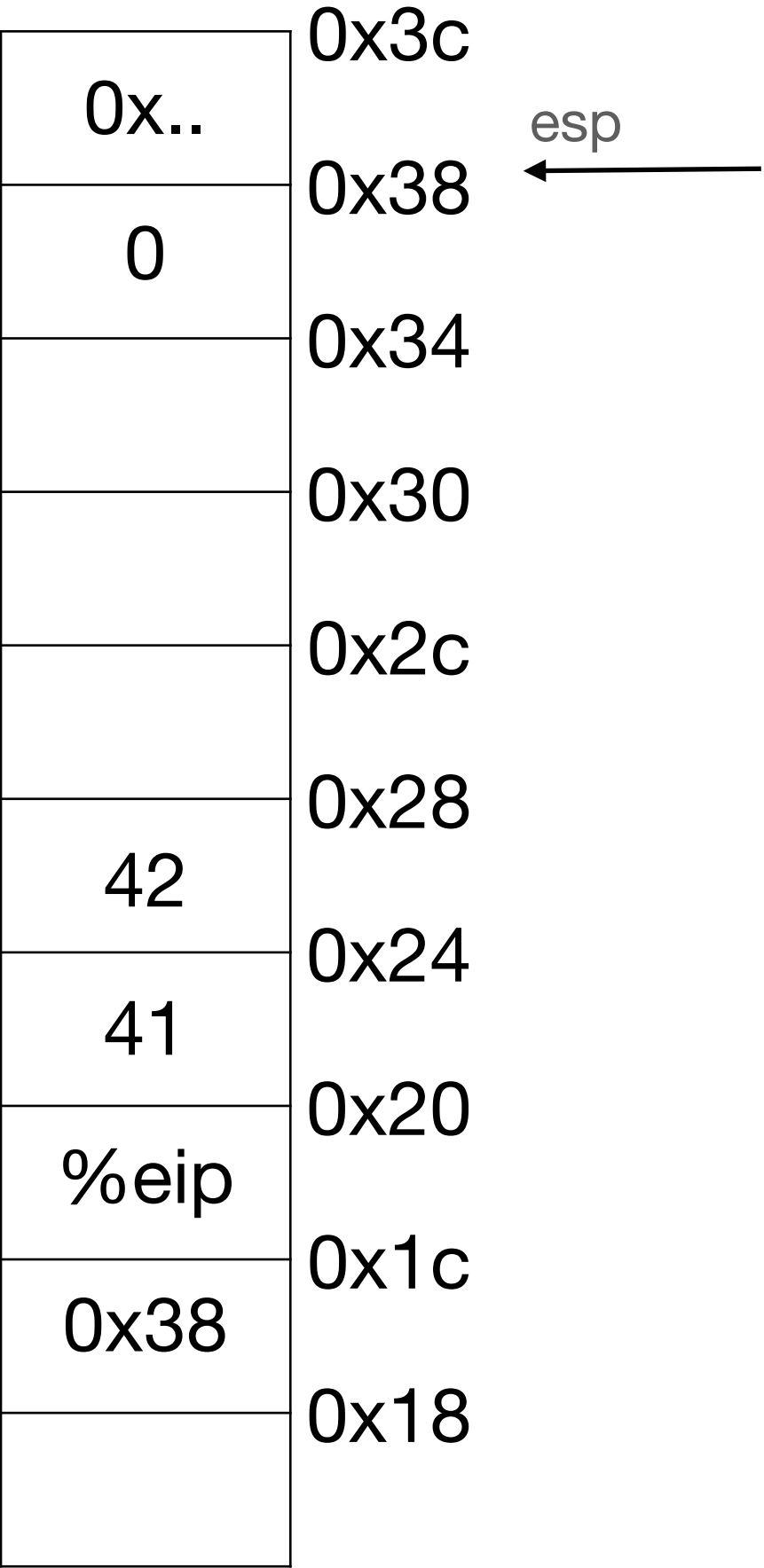
Stack

ebp
→

```
02.s
_foo:
    pushl %ebp                Save caller's base pointer
    movl  %esp, %ebp          ebp = esp
    movl  8(%ebp), %eax        eax = *(ebp + 8)
    addl  12(%ebp), %eax        eax = eax + *(ebp + 12)
    popl  %ebp                Restore caller's base pointer
    retl                      change eip to return address

    .globl _main               ## -- Begin function main
    .p2align 4, 0x90
_main:
    pushl %ebp                Save caller's base pointer
    movl  %esp, %ebp          ebp = esp
    subl  $24, %esp            esp = esp - 0x18
    movl  $0, -4(%ebp)         *(ebp-4)=0
    movl  $41, (%esp)          *(esp) = 41
    movl  $42, 4(%esp)         *(esp+4) = 42
    calll _foo                 Push current eip on to stack, jump to foo
    addl  $24, %esp            esp = esp + 24 (Restore caller's esp)
    popl  %ebp                Restore caller's ebp
    retl
```

eip
→



Function calling in action

Stack

02.s

```

|_foo:

```

```
pushl %ebp
```

```
movl    %esp, %ebp
```

```
movl    8(%ebp), %eax
```

```
addl    12(%ebp), %eax
```

```
popl    %ebp
```

```
retl
```

Save caller's base pointer

ebp = esp

```
eax = *(ebp + 8)
```

```
eax = eax + *(ebp + 12)
```

Restore caller's base pointer

change eip to return address

```
.globl _main
```

```
## -- Begin function main
```

```
.p2align 4, 0x90
```

```

main:

```

pushl %ebp

```
movl    %esp, %ebp
```

```
subl $24, %esp
```

```
movl    $0, -4(%ebp)
```

```
movl    $41, (%esp)
```

```
movl    $42, 4(%esp)
```

```
calll _foo
```

```
addl    $24, %esp
```

```
► popl %ebp
```

```
retl
```

Save caller's base pointer

ebp = esp

```
esp = esp - 0x18
```

```
* (ebp-4) = 0
```

```
* (esp) = 41
```

```
* (esp+4) = 42
```

Push current eip on to stack, jump to foo

```
esp = esp + 24 (Restore caller's esp)
```

Restore caller's ebp



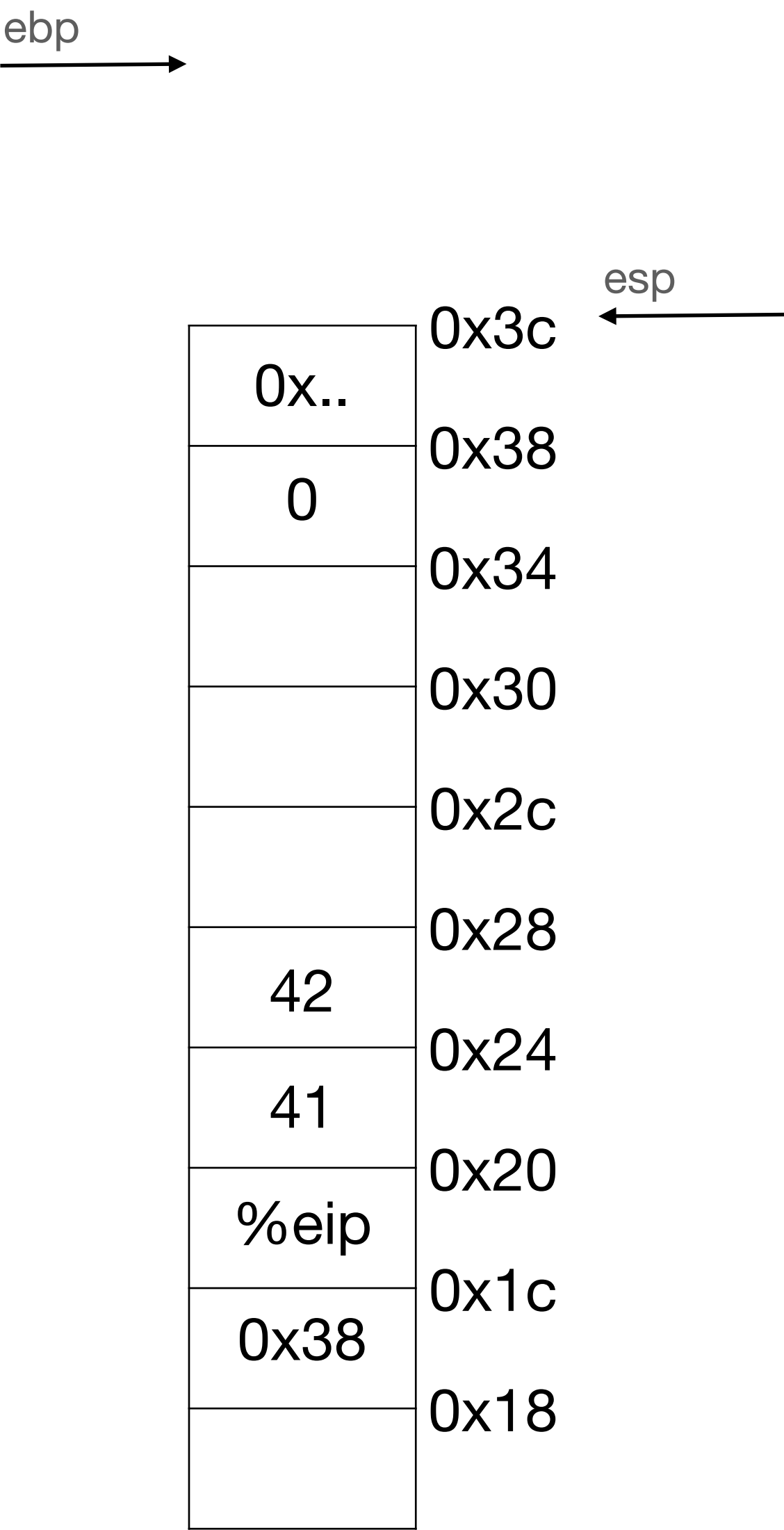
0x..	0x3c
0	0x38
	0x34
	0x30
	0x2c
	0x28
42	0x24
41	0x20
%eip	0x1c
0x38	0x18

Function calling in action

Stack

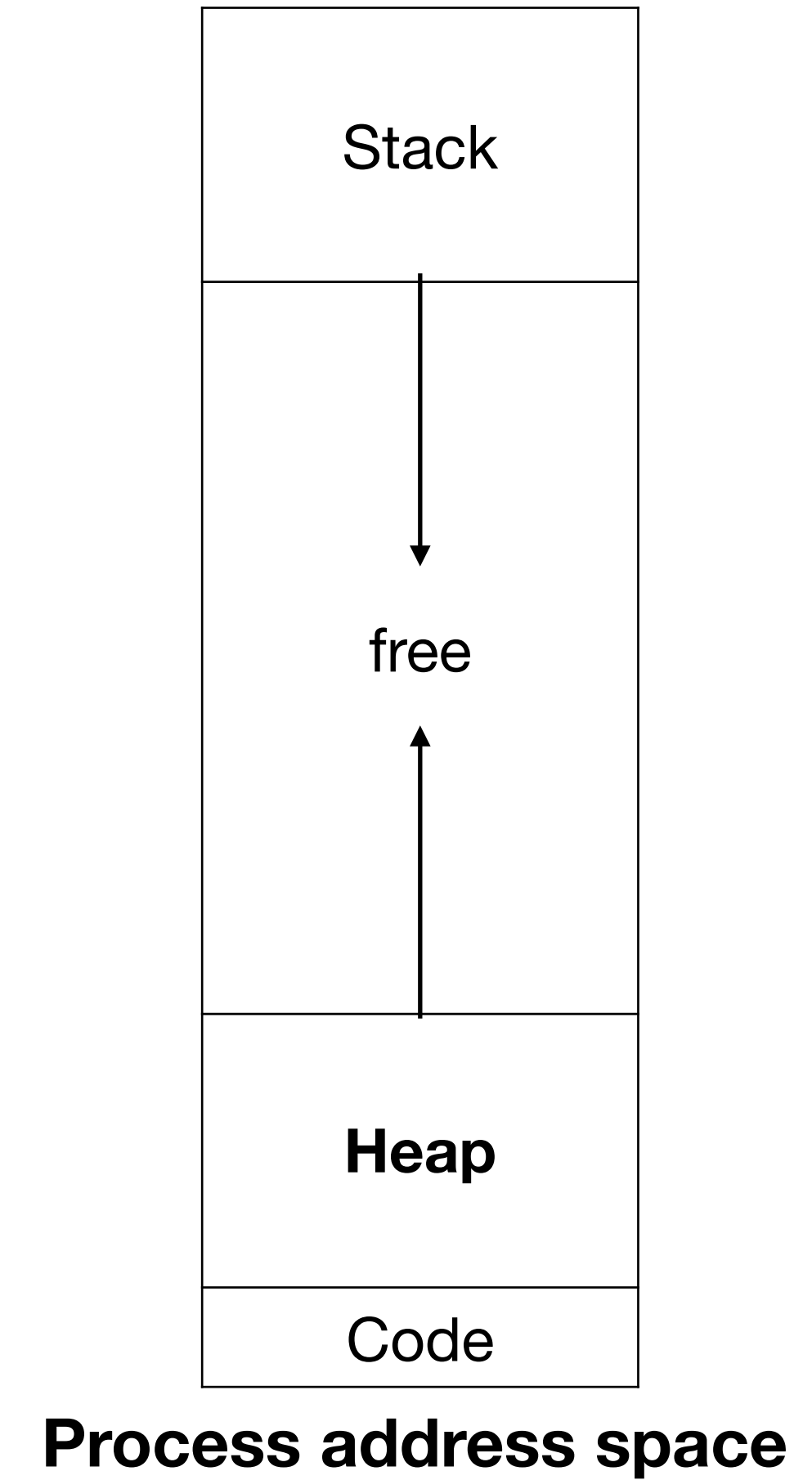
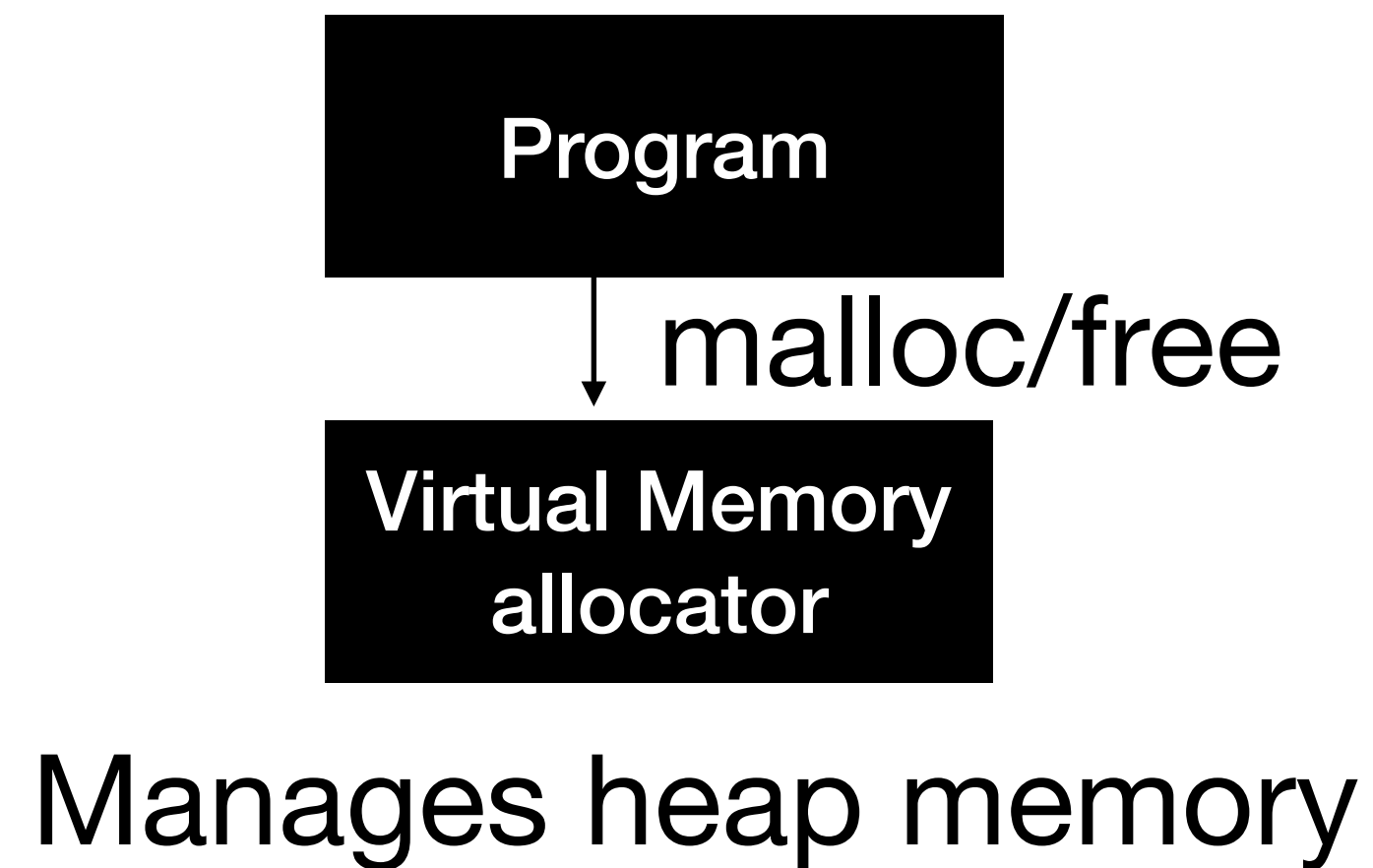
```
02.s
_foo:
    pushl %ebp                Save caller's base pointer
    movl  %esp, %ebp          ebp = esp
    movl  8(%ebp), %eax        eax = *(ebp + 8)
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    popl  %ebp                Restore caller's base pointer
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    .globl _main               ## -- Begin function main
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_main:
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    subl  $24, %esp            esp = esp - 0x18
    movl  $0, -4(%ebp)         *(ebp-4)=0
    movl  $41, (%esp)          *(esp) = 41
    movl  $42, 4(%esp)         *(esp+4) = 42
    calll _foo                 Push current eip on to stack, jump to foo
    addl  $24, %esp            esp = esp + 24 (Restore caller's esp)
    popl  %ebp                Restore caller's ebp
    retl
```

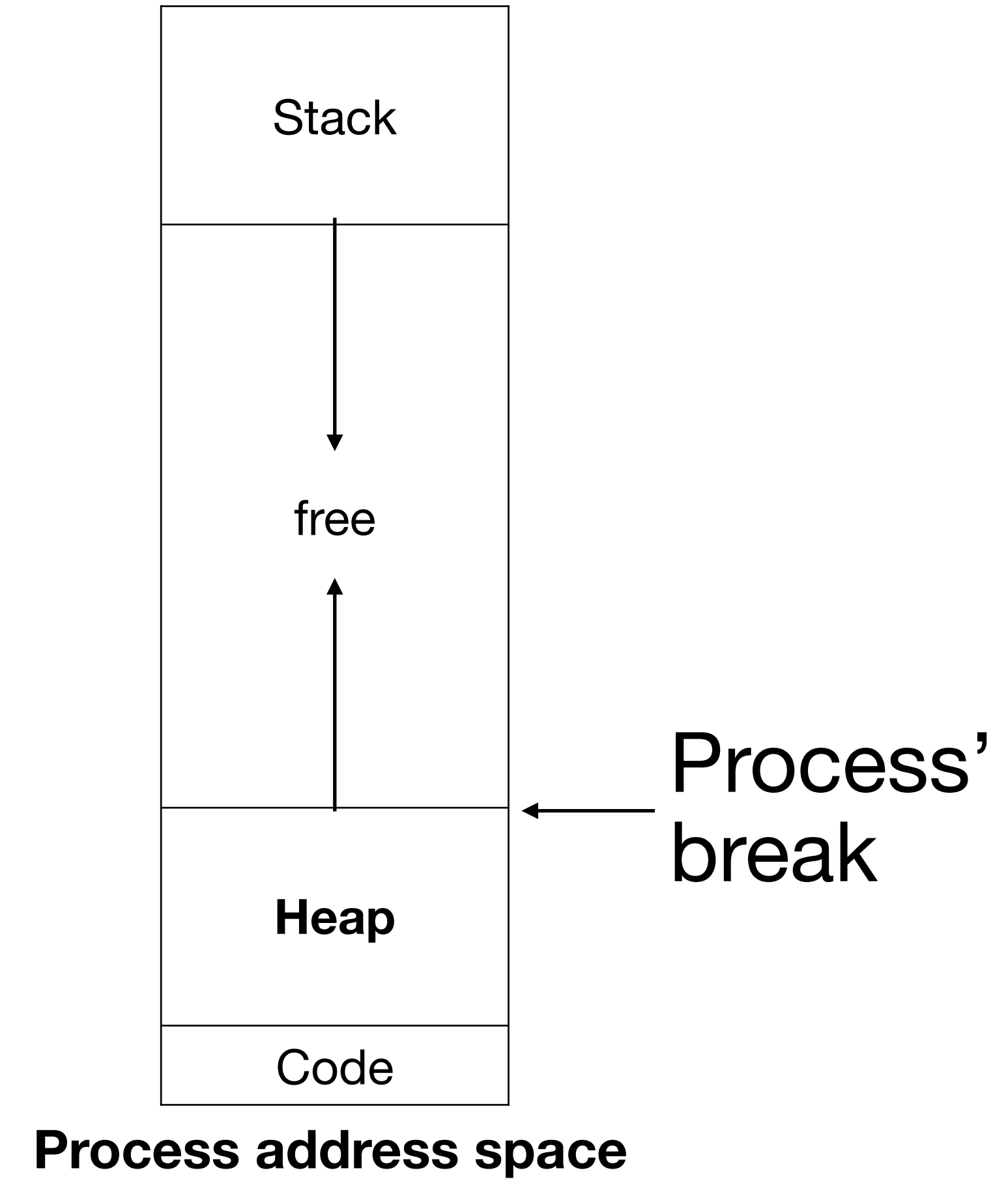
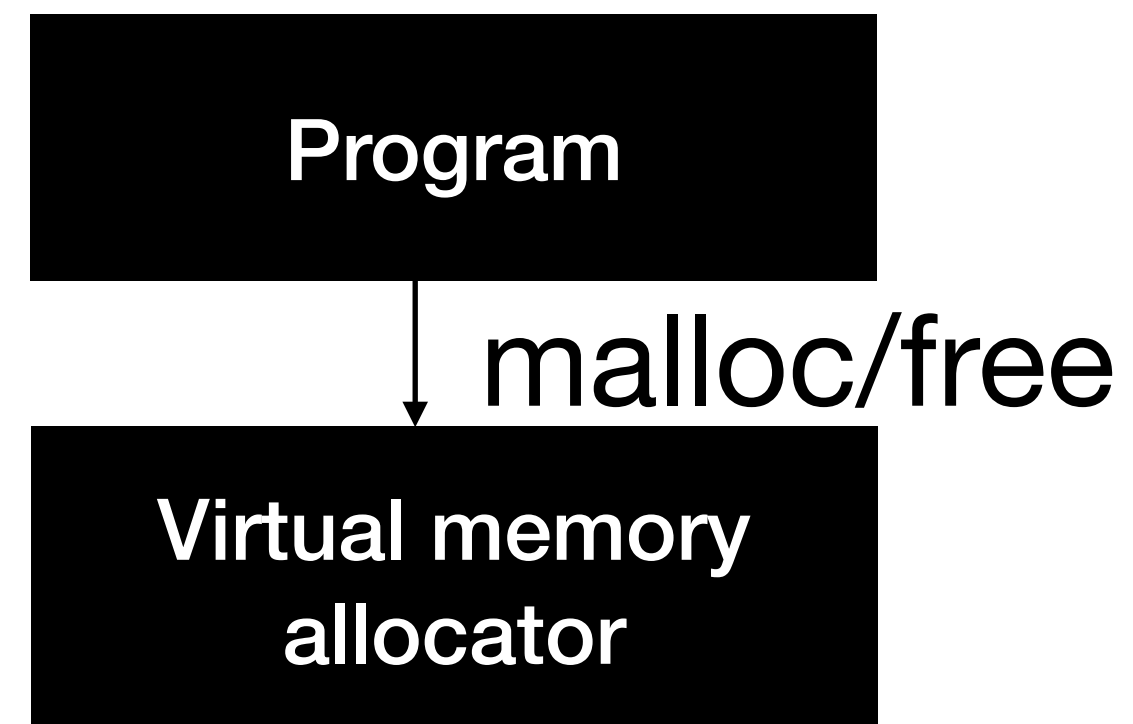


Memory allocator

Works with virtual memory

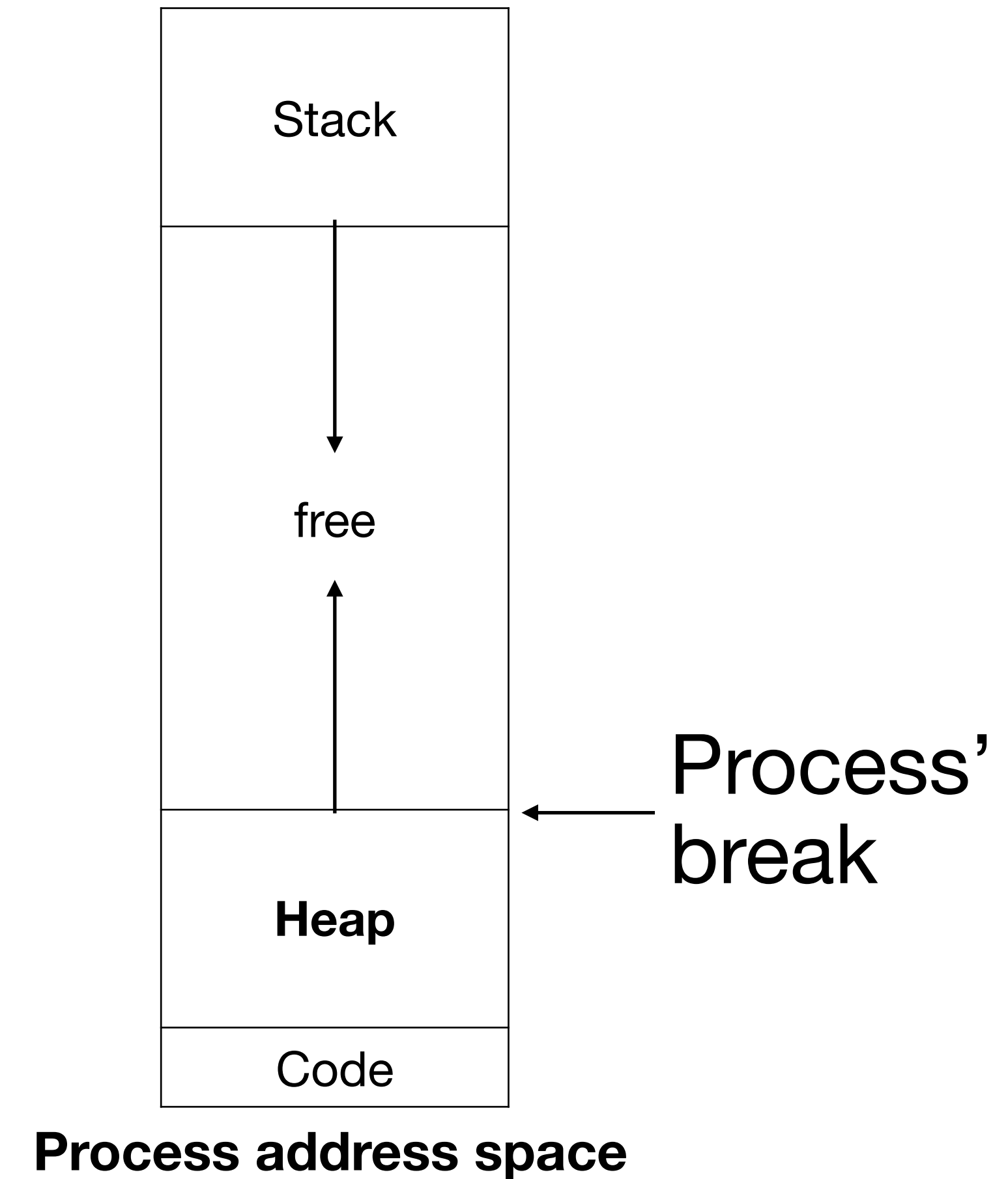
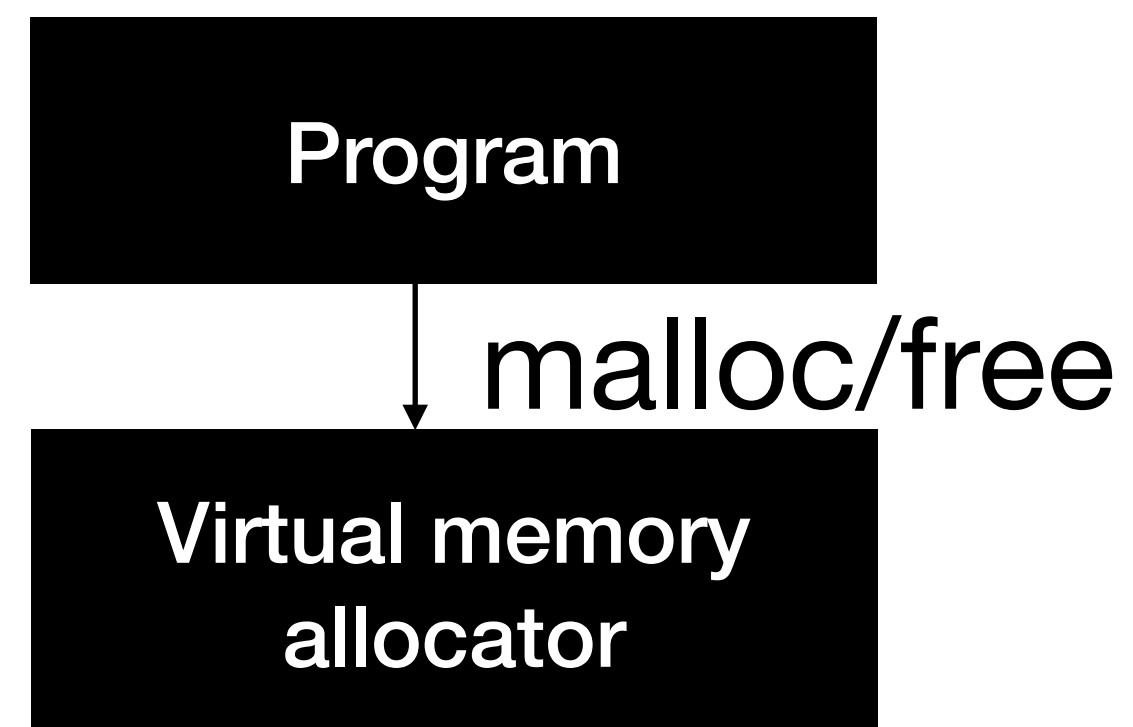


OS memory allocator



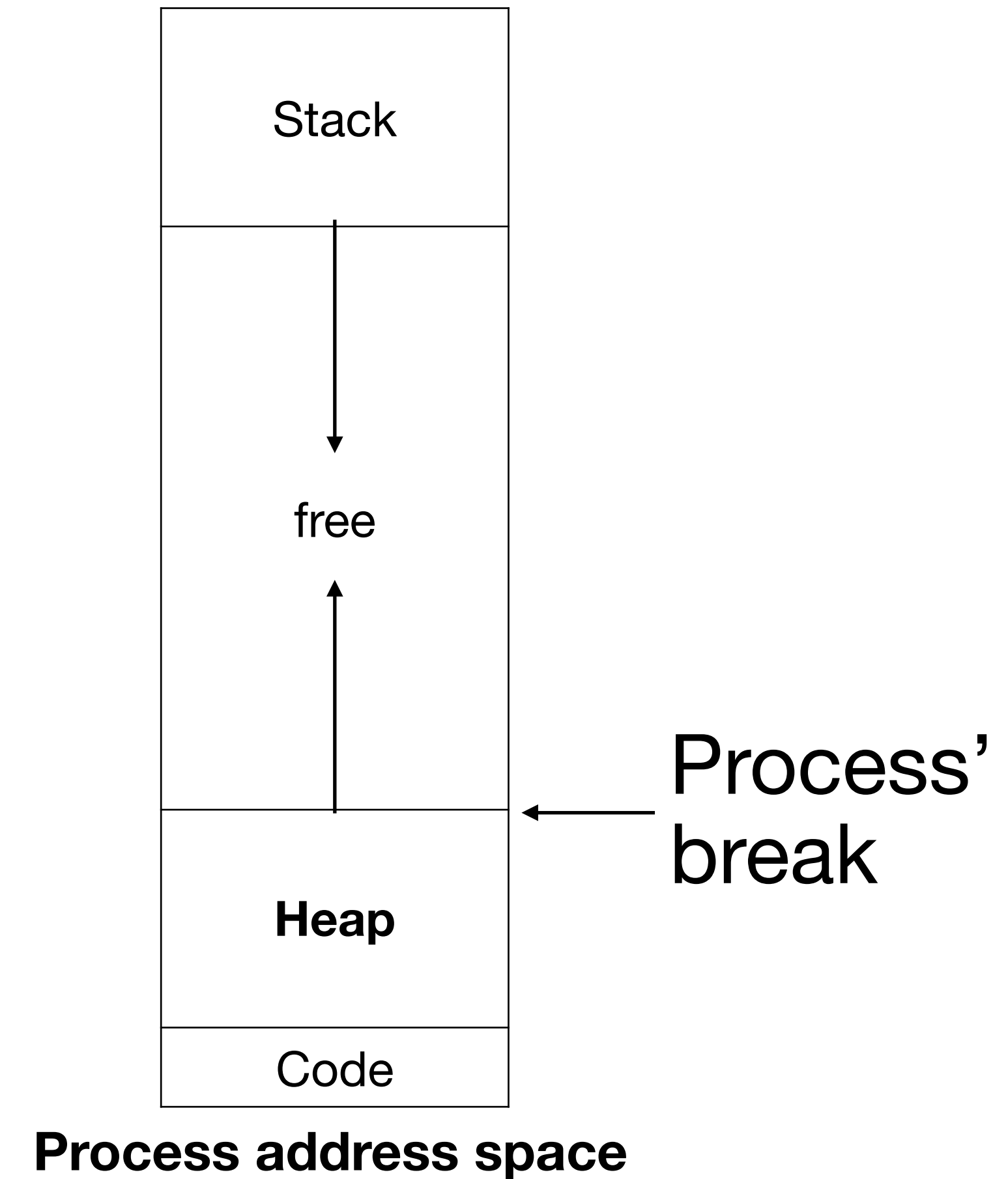
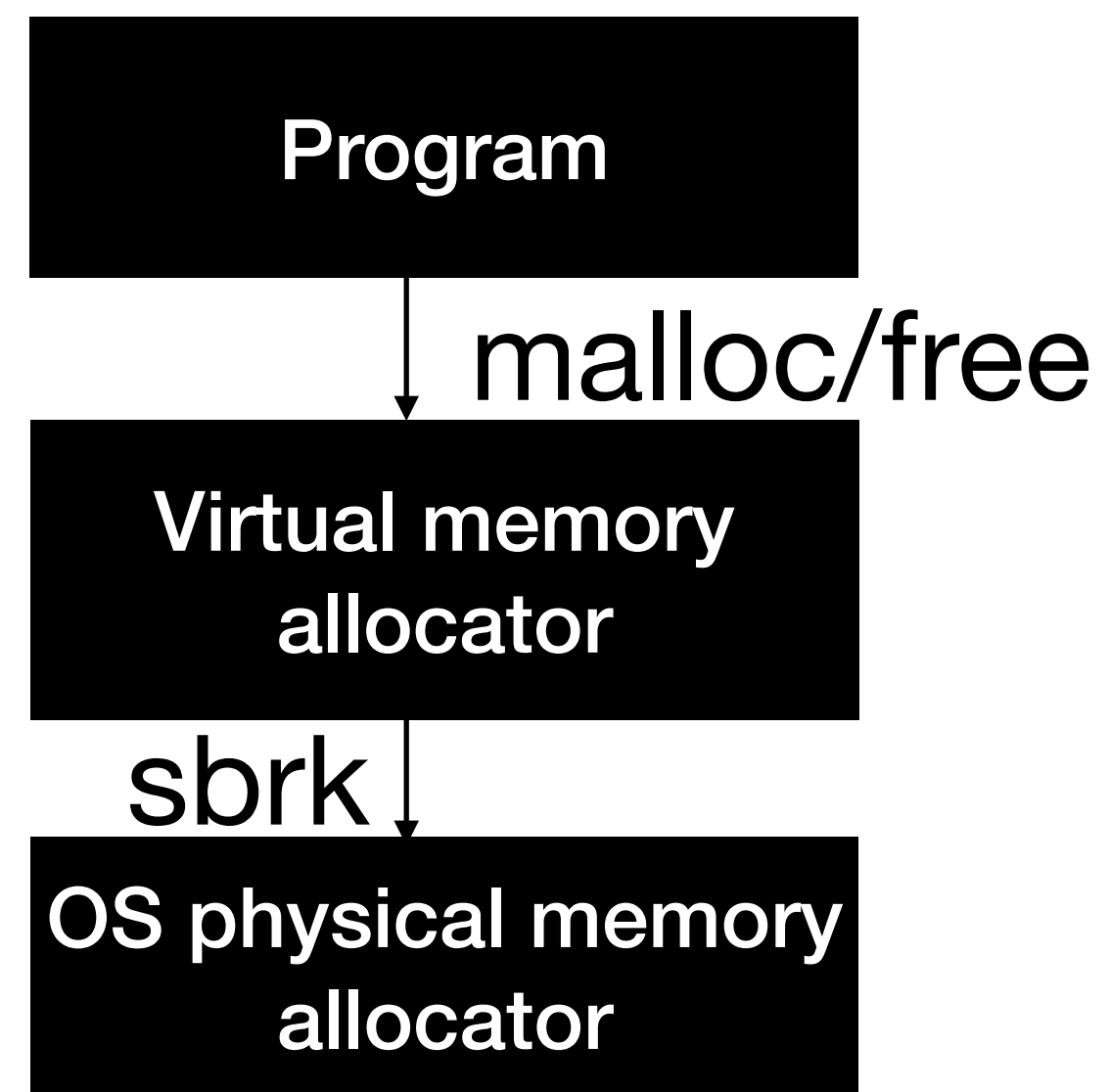
OS memory allocator

- `sbrk(int increment)` increments process' break. *increment* can be negative.



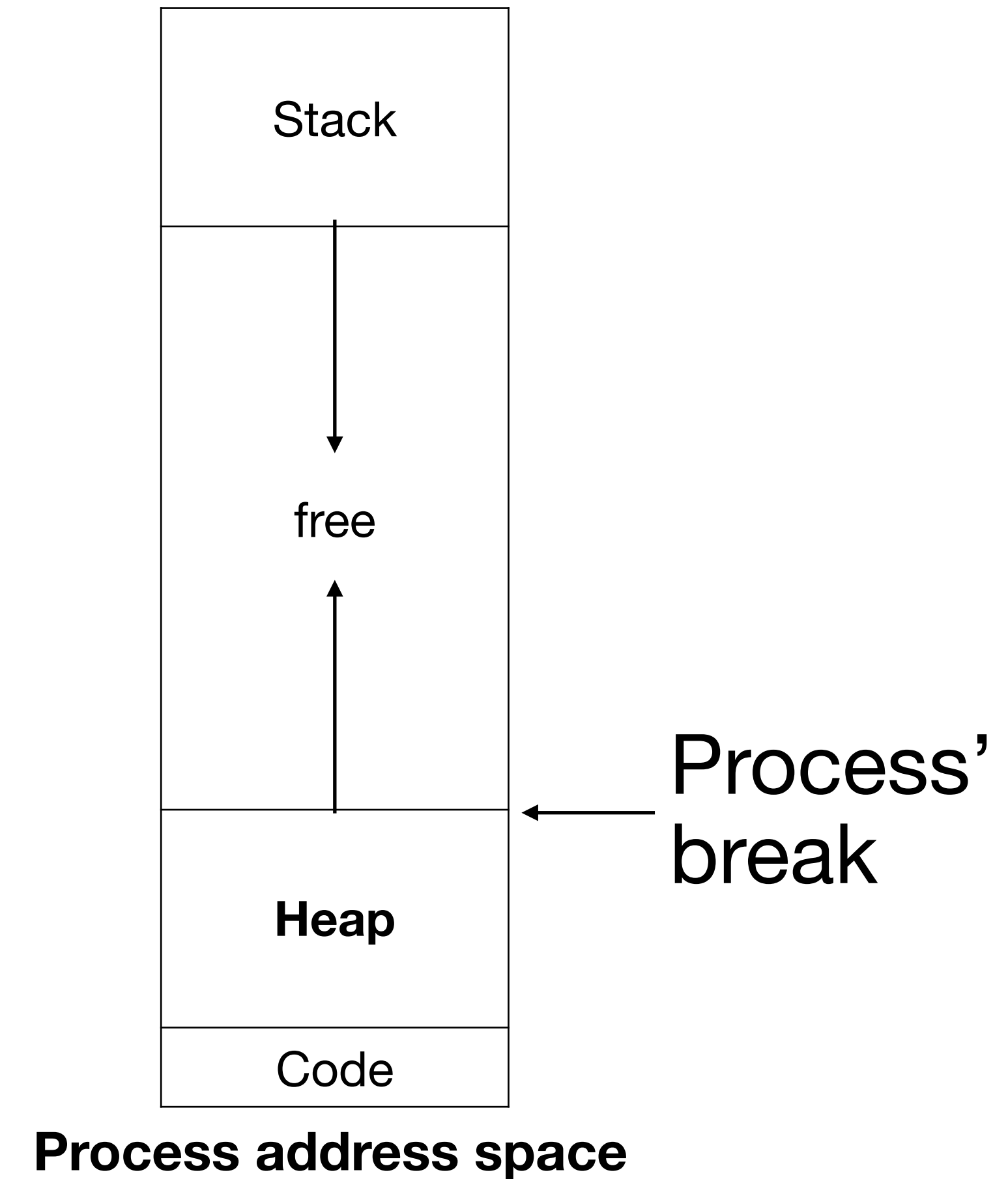
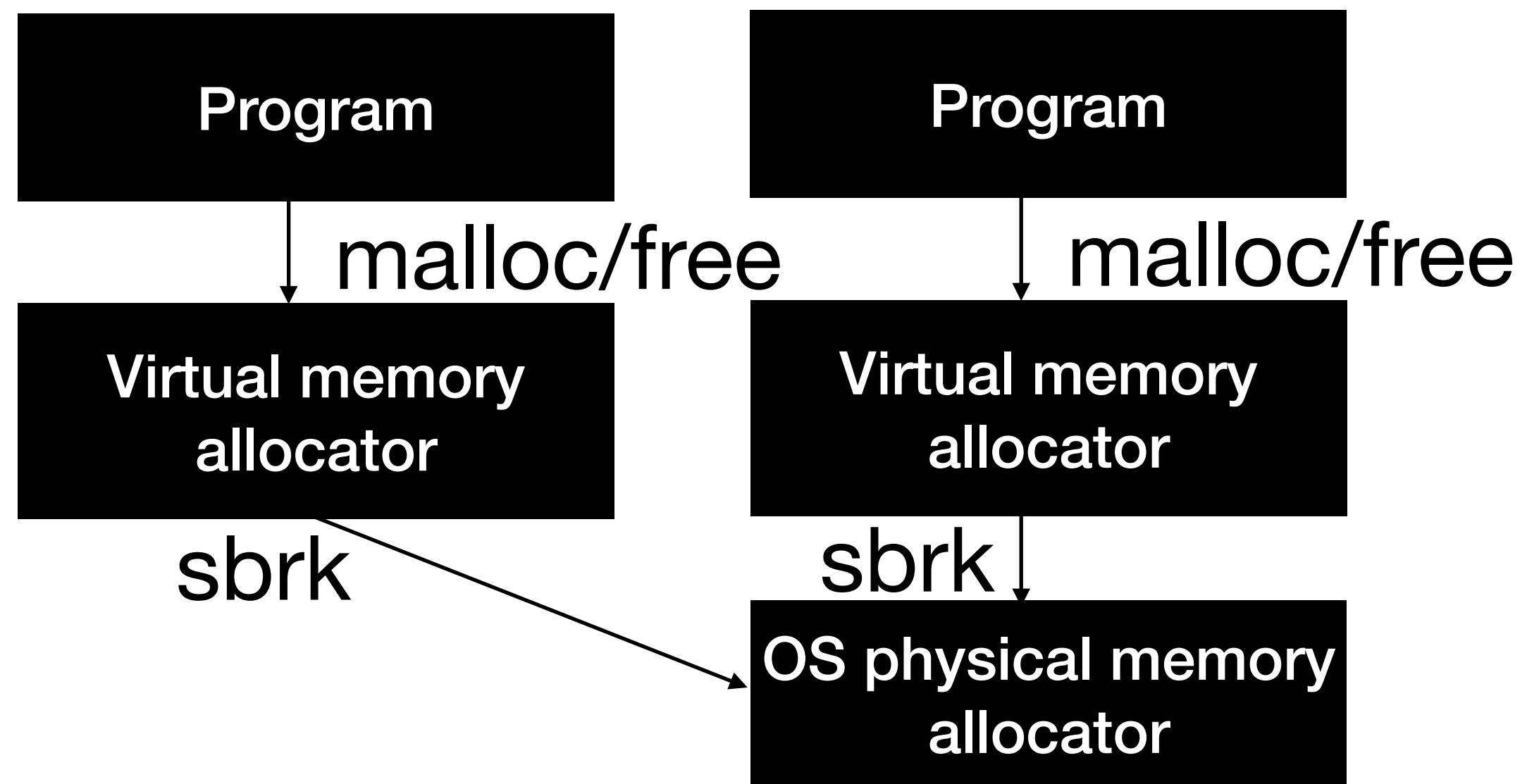
OS memory allocator

- `sbrk(int increment)` increments process' break. *increment* can be negative.



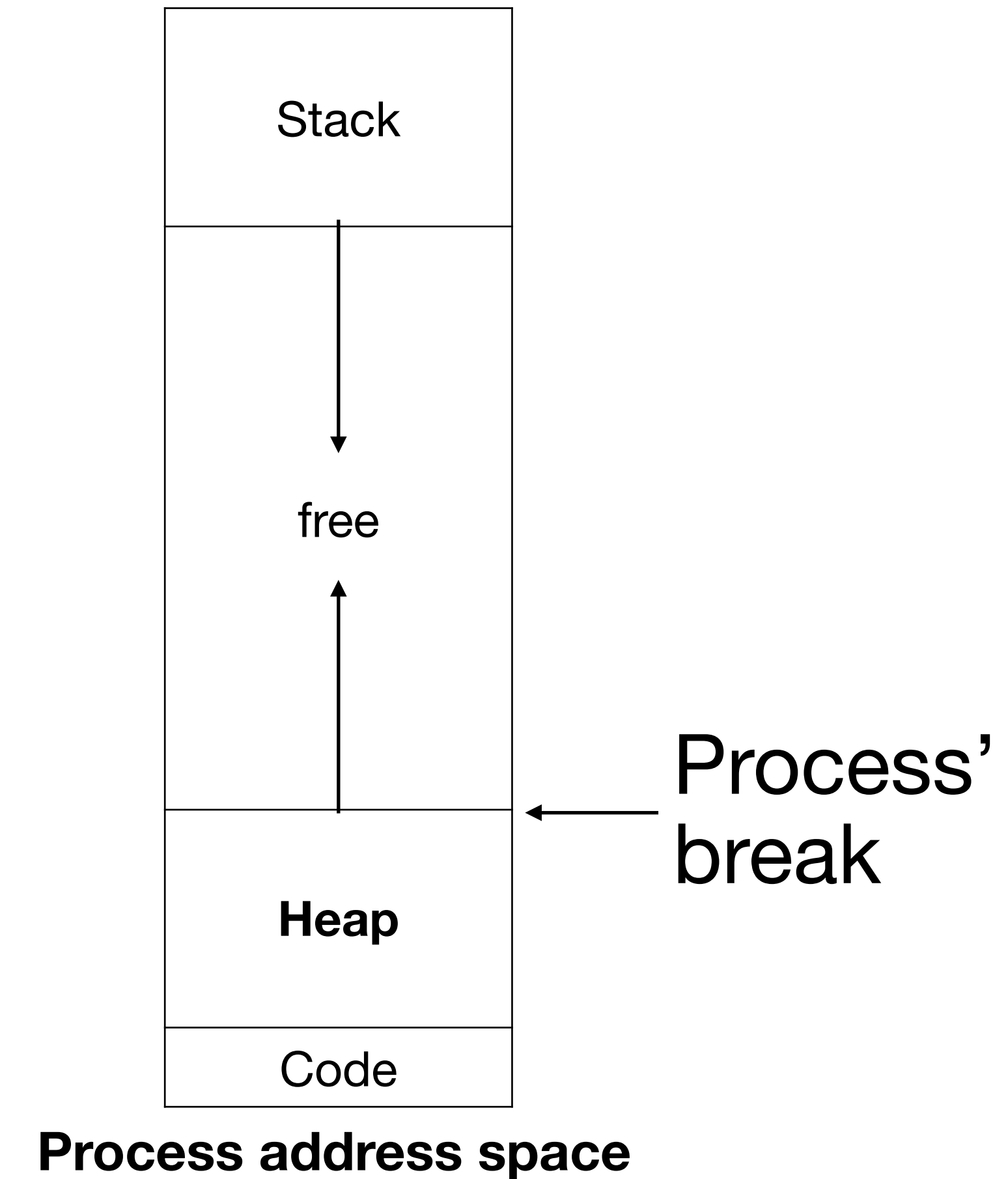
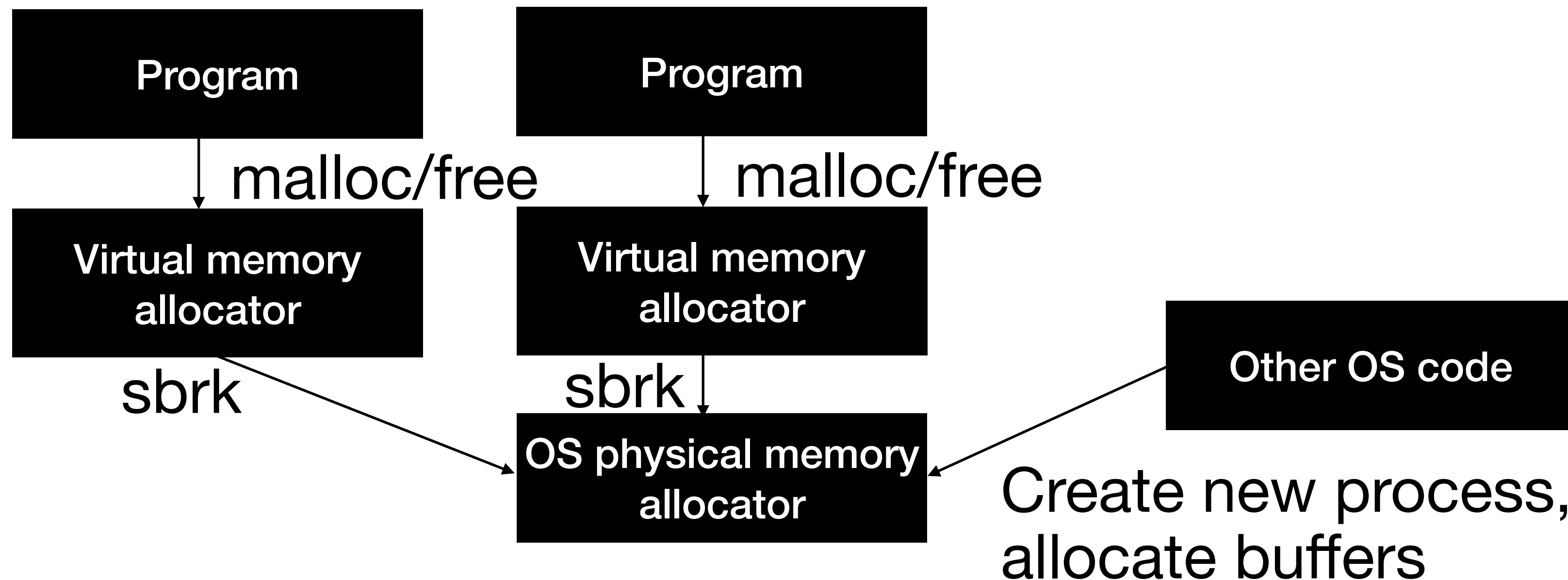
OS memory allocator

- `sbrk(int increment)` increments process' break. *increment* can be negative.



OS memory allocator

- `sbrk(int increment)` increments process' break. *increment* can be negative.



Heap memory APIs and bugs

- malloc, free. va.c.
 - malloc is for dynamic allocation. Allocation size may not be known at compile time. Slower than stack allocations. Memory allocator needs to find free space on heap.
 - Size of the allocated region is not passed in free. It must be tracked by the memory-allocation library itself.
- Null pointer dereference. null.c
- Memory leak. leak.c
- Buffer overflow. overflow.c
- Use after free. useafterfree.c
- Invalid free. invalidfree.c
- Double free. doublefree.c
- Uninitialised read. uninitread.c

Memory allocation

How to check for invalid frees? How much memory needs to be freed?

```
ptr=malloc(size_t size = 20);
```

```
free(ptr);
```


Memory allocation

How to check for invalid frees? How much memory needs to be freed?

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ptr=malloc(size_t size = 20);
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free(ptr);
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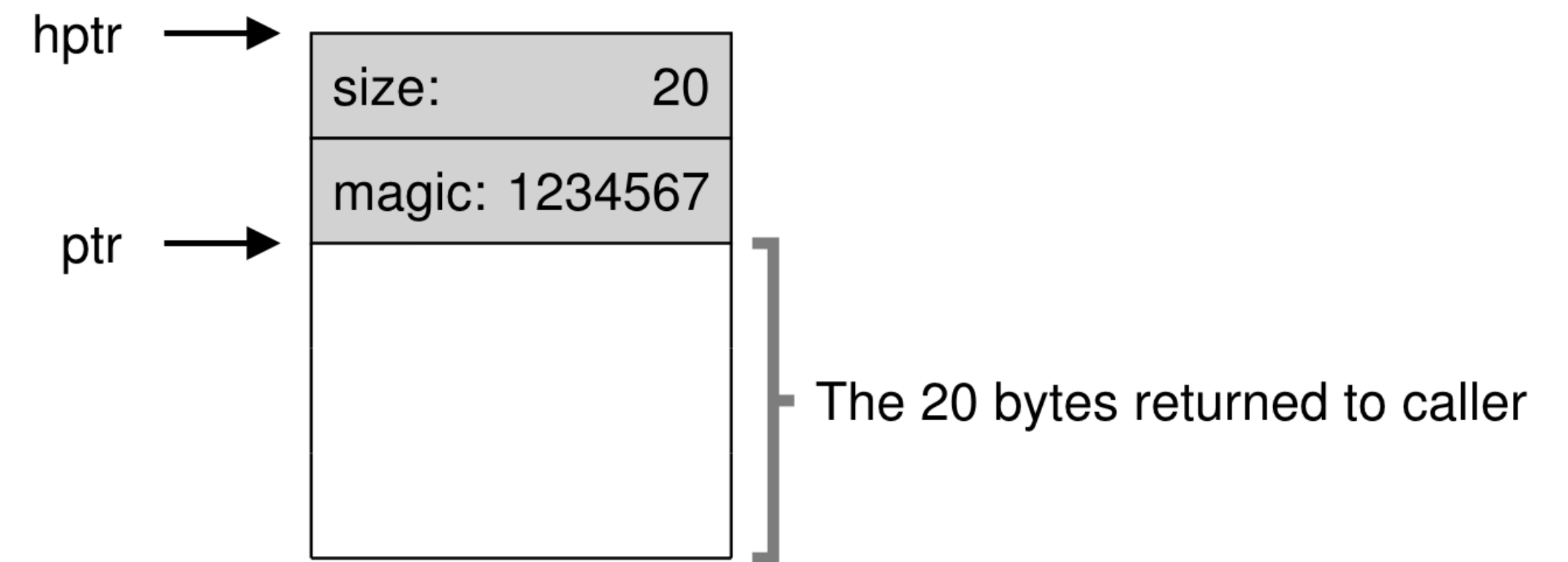


Figure 17.2: Specific Contents Of The Header

Memory allocation

How to check for invalid frees? How much memory needs to be freed?

```
ptr=malloc(size_t size = 20);  
free(ptr);
```

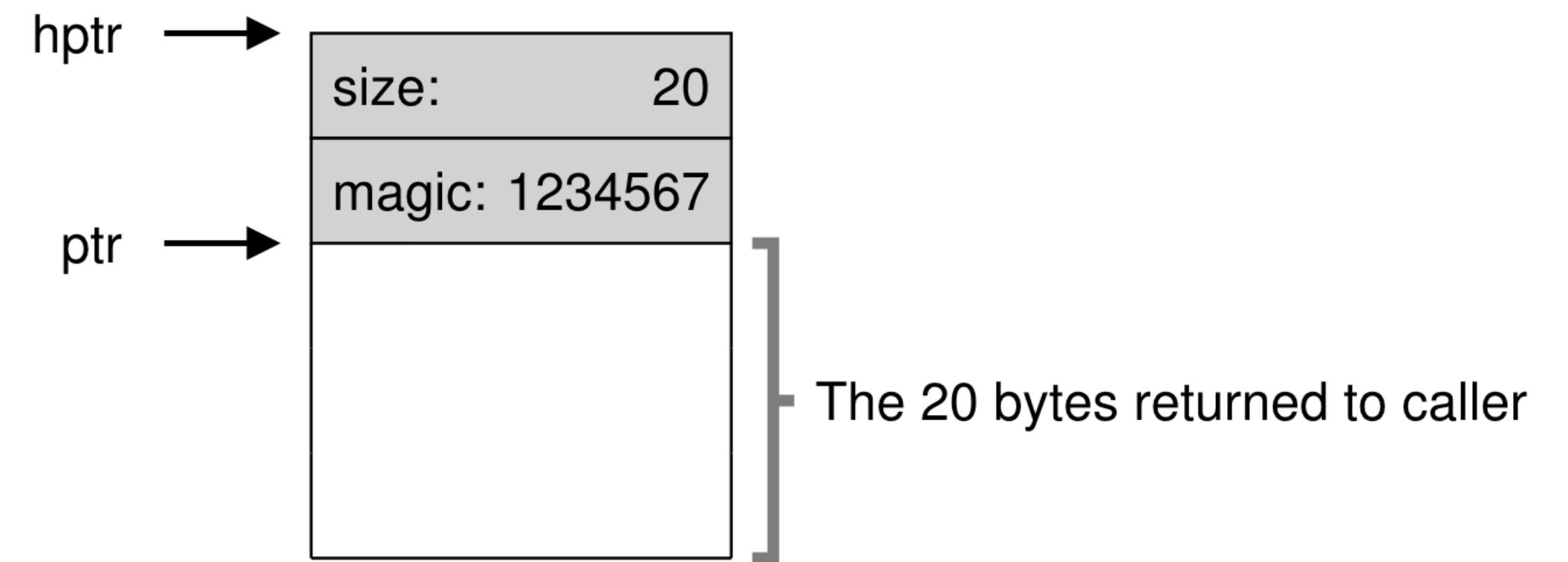
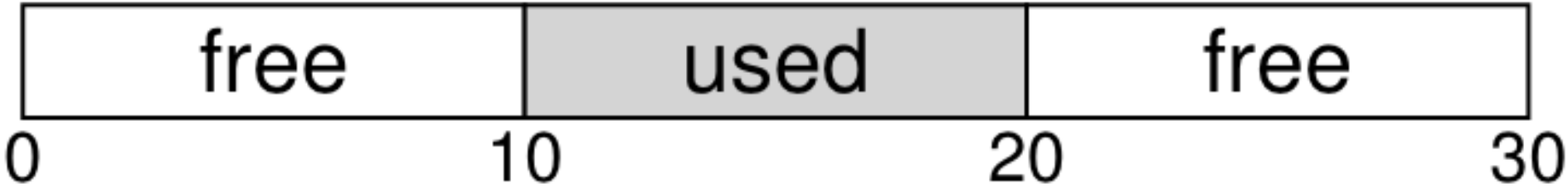


Figure 17.2: Specific Contents Of The Header

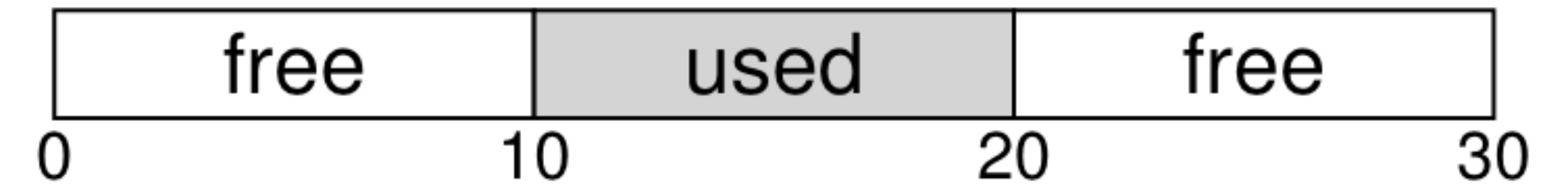
Stack allocations are faster than heap allocations. No need to find space.

Memory allocator



Fragmented heap over time

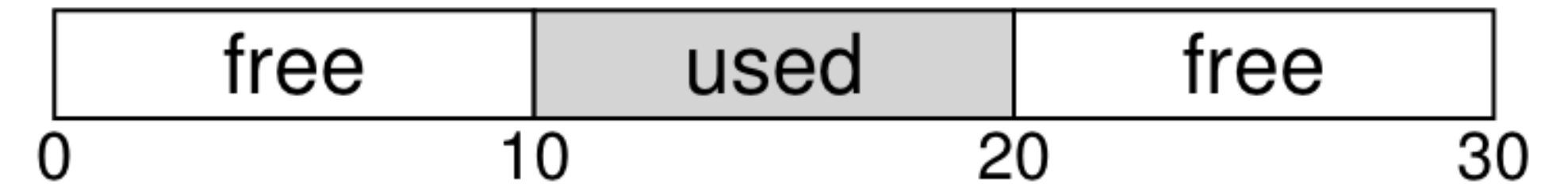
Memory allocator



Fragmented heap over time

- Assumptions
 - Do not apriori know allocation size and order
 - Cannot move memory once it is allocated. Program might have the pointer to it.

Memory allocator



Fragmented heap over time

- Assumptions
 - Do not apriori know allocation size and order
 - Cannot move memory once it is allocated. Program might have the pointer to it.
- Goals
 - Quickly satisfy variable-sized memory allocation requests. How to track free memory?
 - Minimize fragmentation
 - All allocators are *bad* for *some programs*. Understand pros and cons of allocator designs

Memory (de)allocation patterns

- Small mallocs can be frequent. Large mallocs are usually infrequent.
 - After malloc, program will initialise the memory area.
- “Clustered deaths”: Objects allocated together die together.

Free list splitting and coalescing

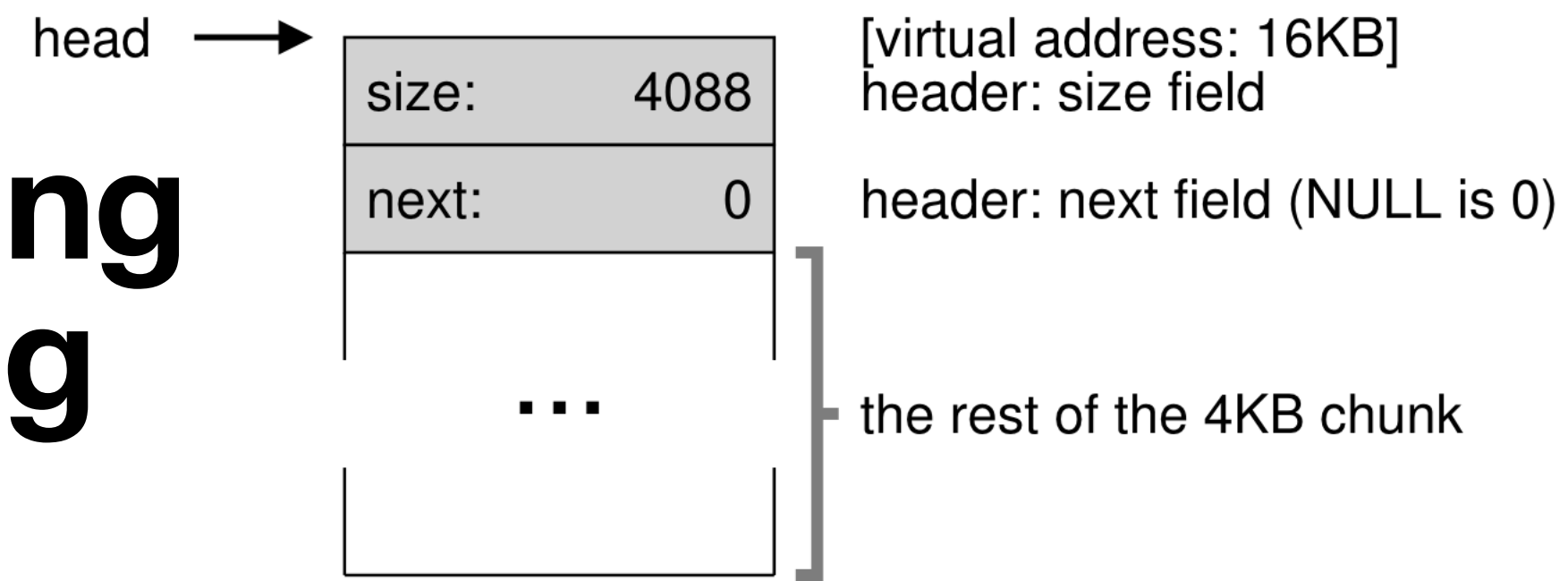


Figure 17.3: A Heap With One Free Chunk

```
→ ptr = malloc(100)
   sptr = malloc(100)
   optr = malloc(100)
   free(sptr)
   free(ptr)
   free(optr)
```

Free list splitting and coalescing

```
ptr = malloc(100)
```

```
→ sptr = malloc(100)
```

```
optr = malloc(100)
```

```
free(sptr)
```

```
free(ptr)
```

```
free(optr)
```

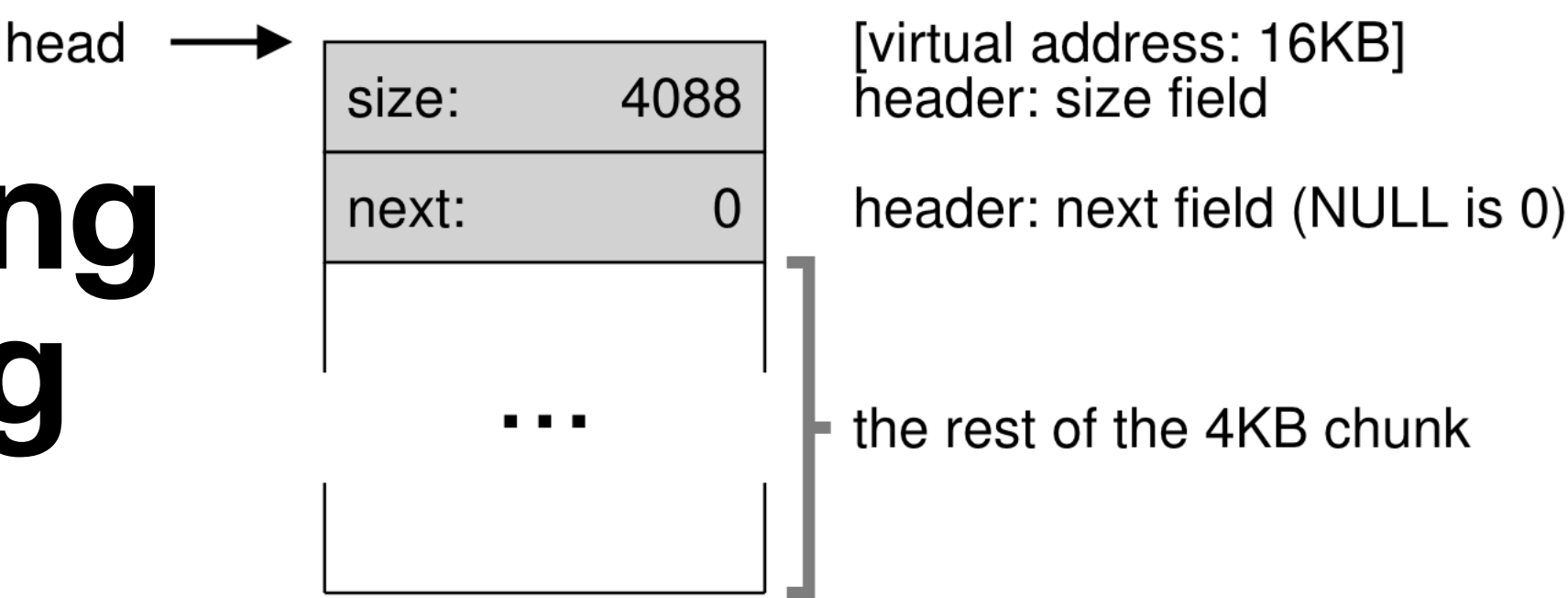


Figure 17.3: A Heap With One Free Chunk

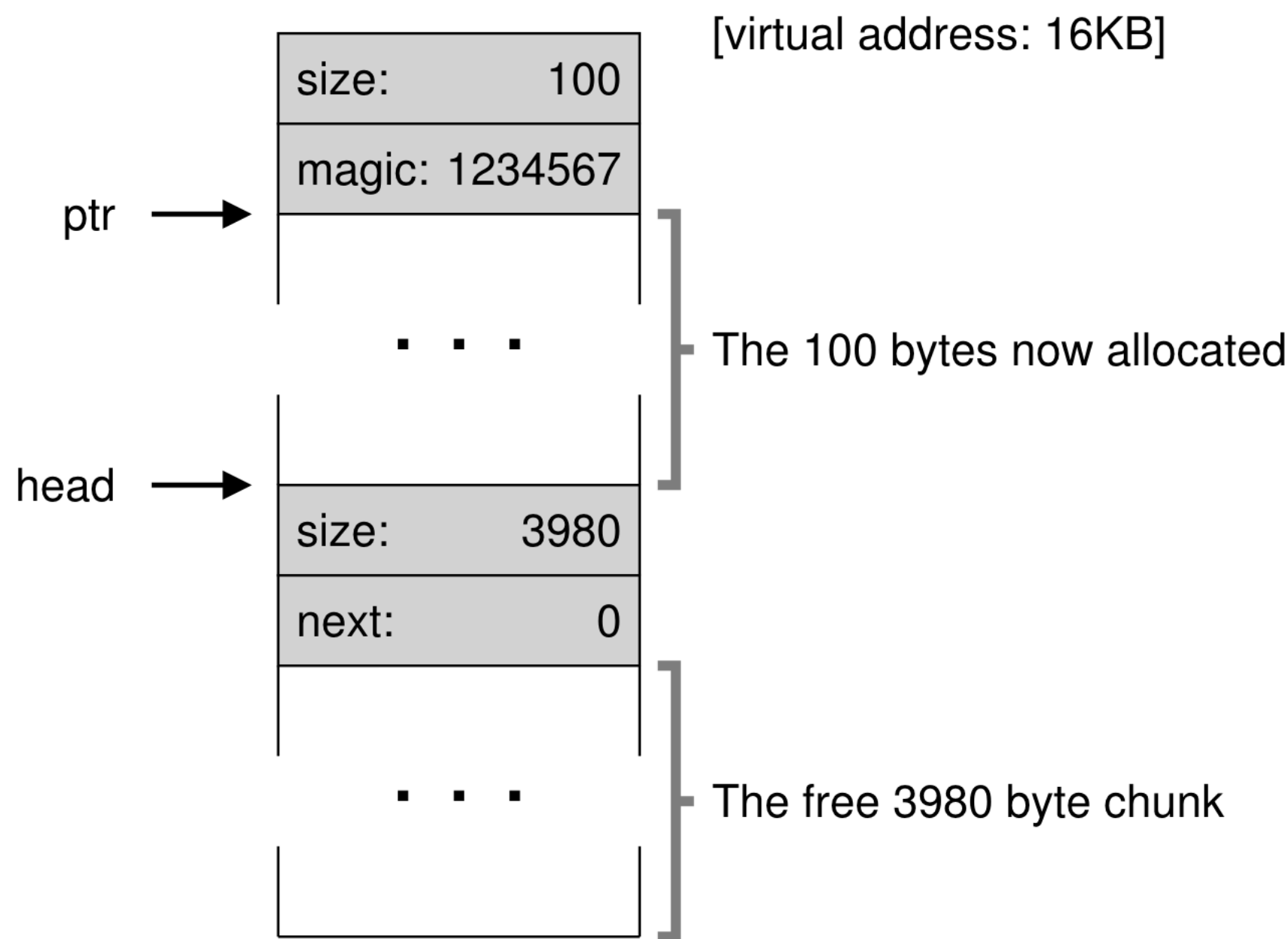


Figure 17.4: A Heap: After One Allocation

Free list splitting and coalescing

```
ptr = malloc(100)
sptr = malloc(100)
optr = malloc(100)
→ free(sptr)
free(ptr)
free(optr)
```

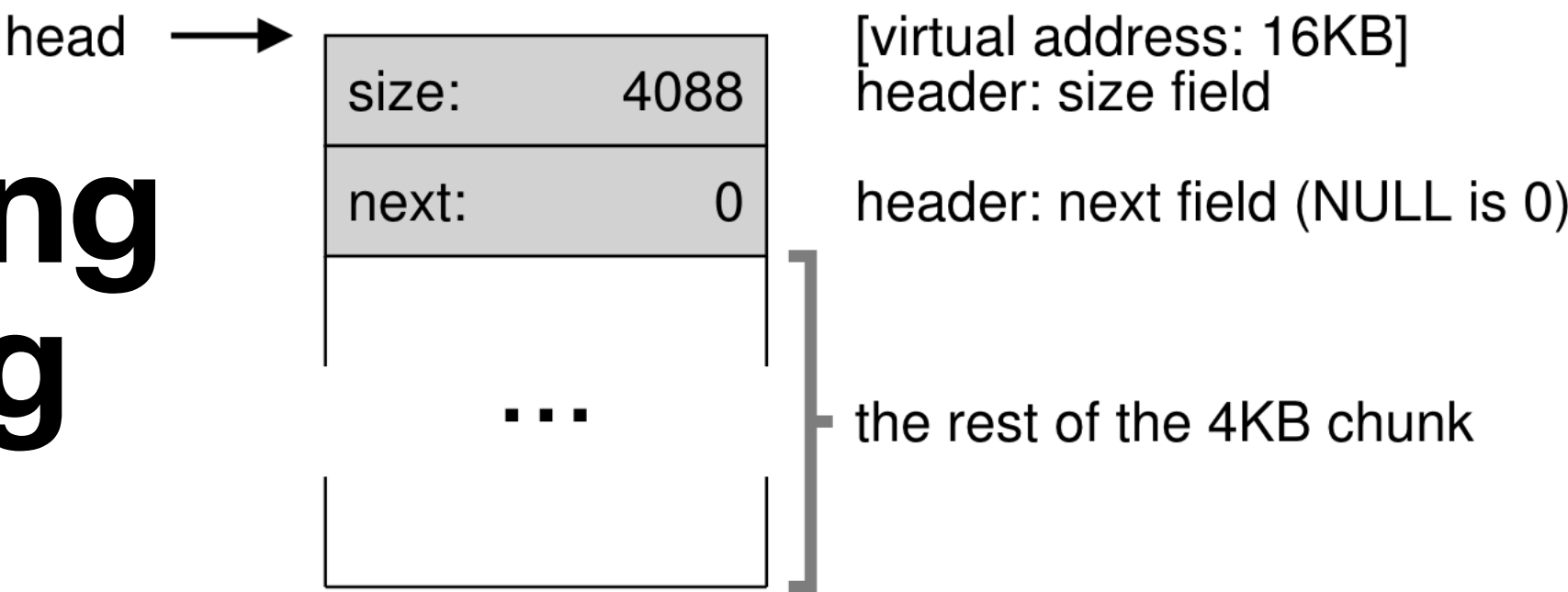


Figure 17.3: A Heap With One Free Chunk

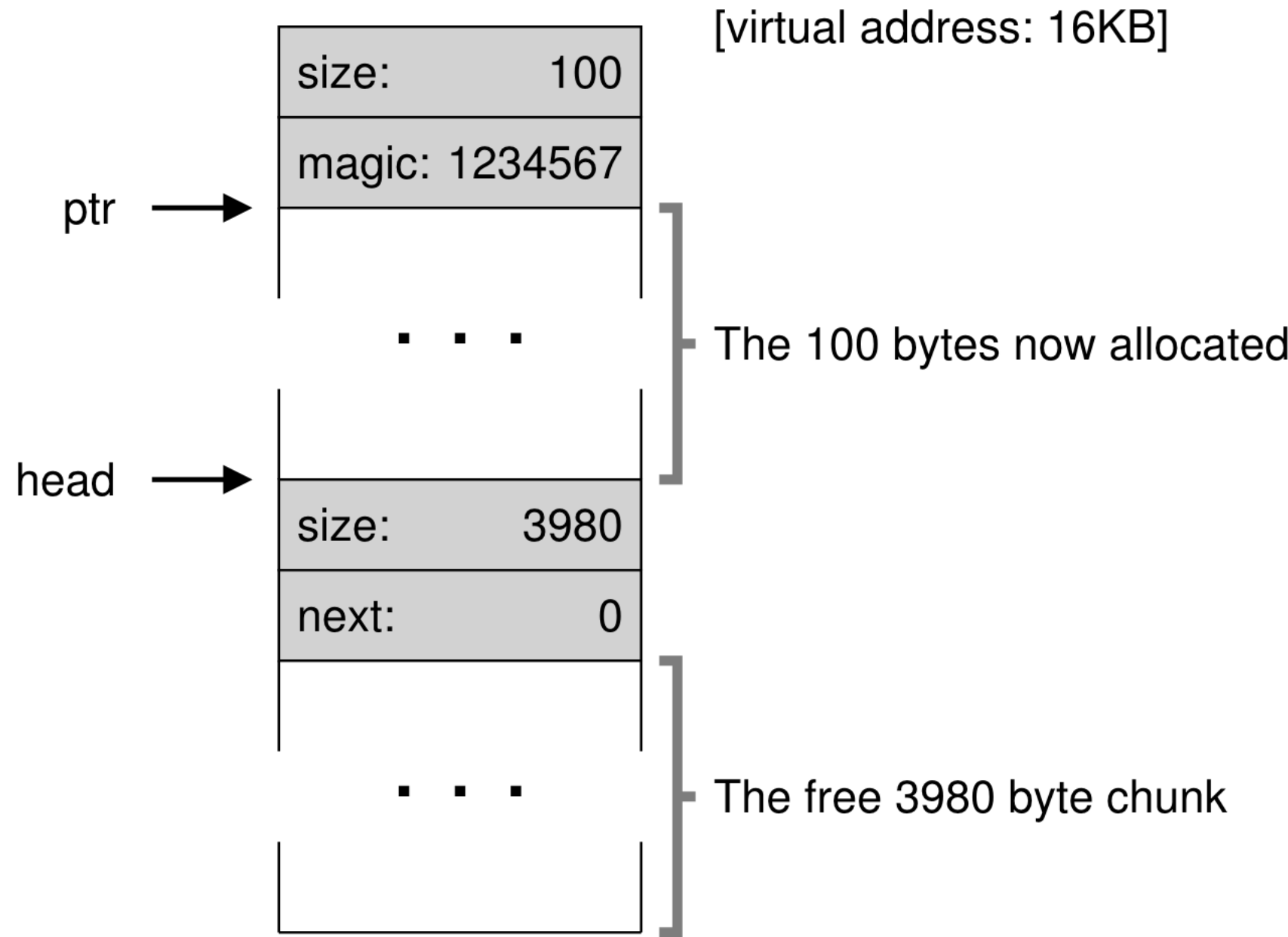


Figure 17.4: A Heap: After One Allocation

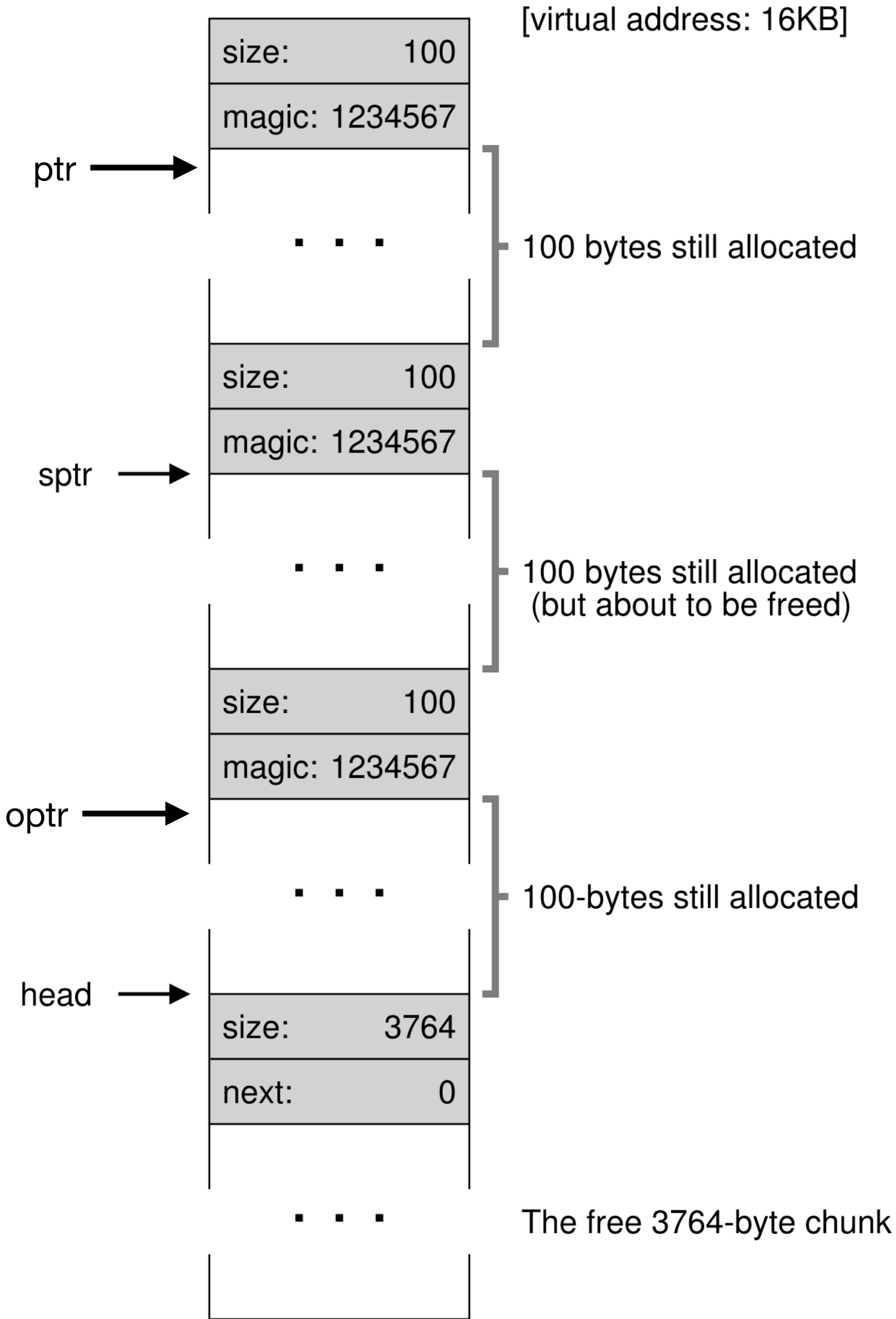


Figure 17.5: Free Space With Three Chunks Allocated

Free list splitting and coalescing

```
ptr = malloc(100)
sptr = malloc(100)
optr = malloc(100)
free(sptr)
→ free(ptr)
free(optr)
```

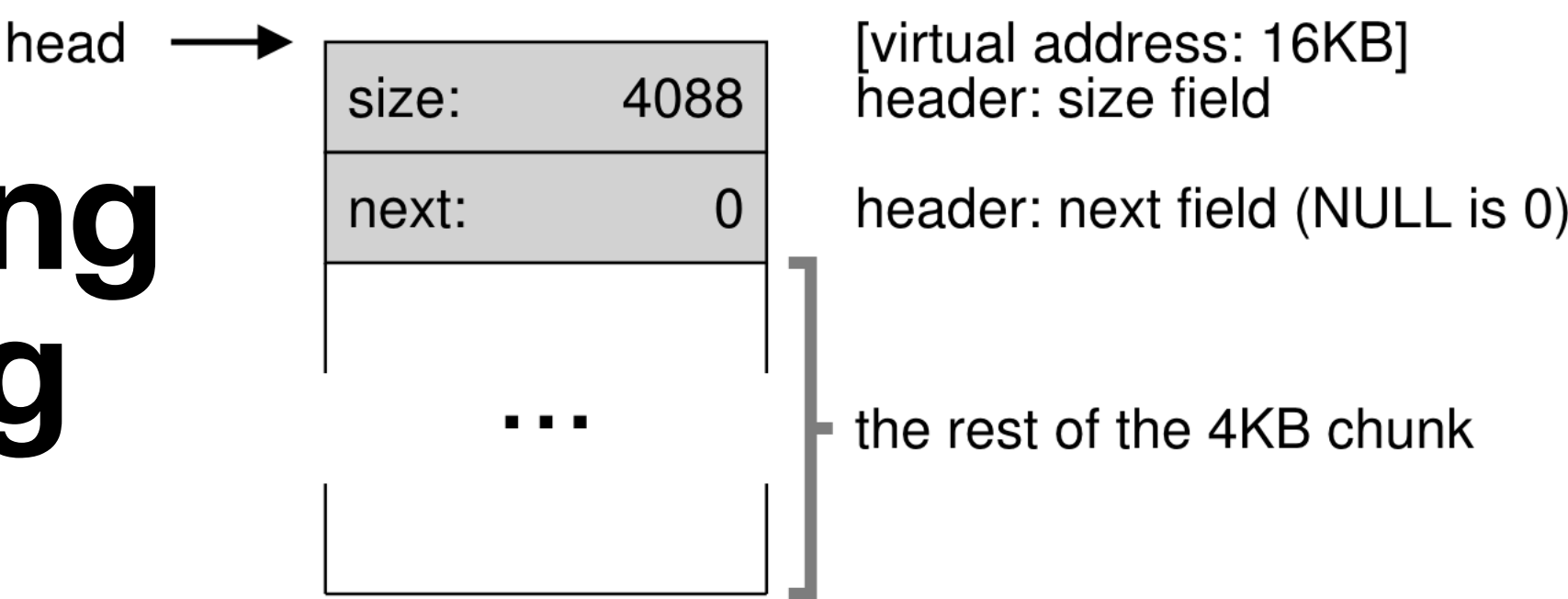


Figure 17.3: A Heap With One Free Chunk

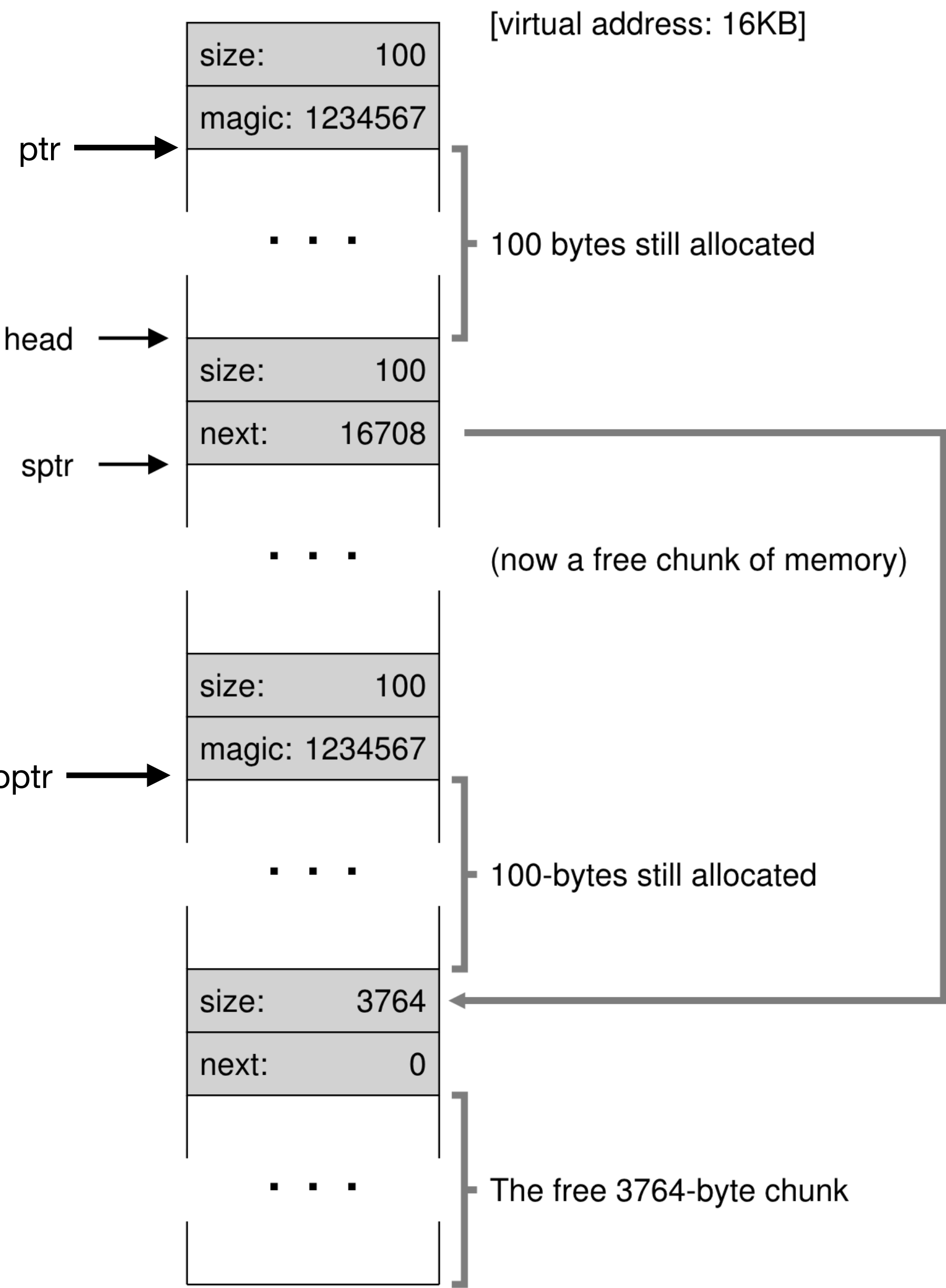


Figure 17.6: Free Space With Two Chunks Allocated

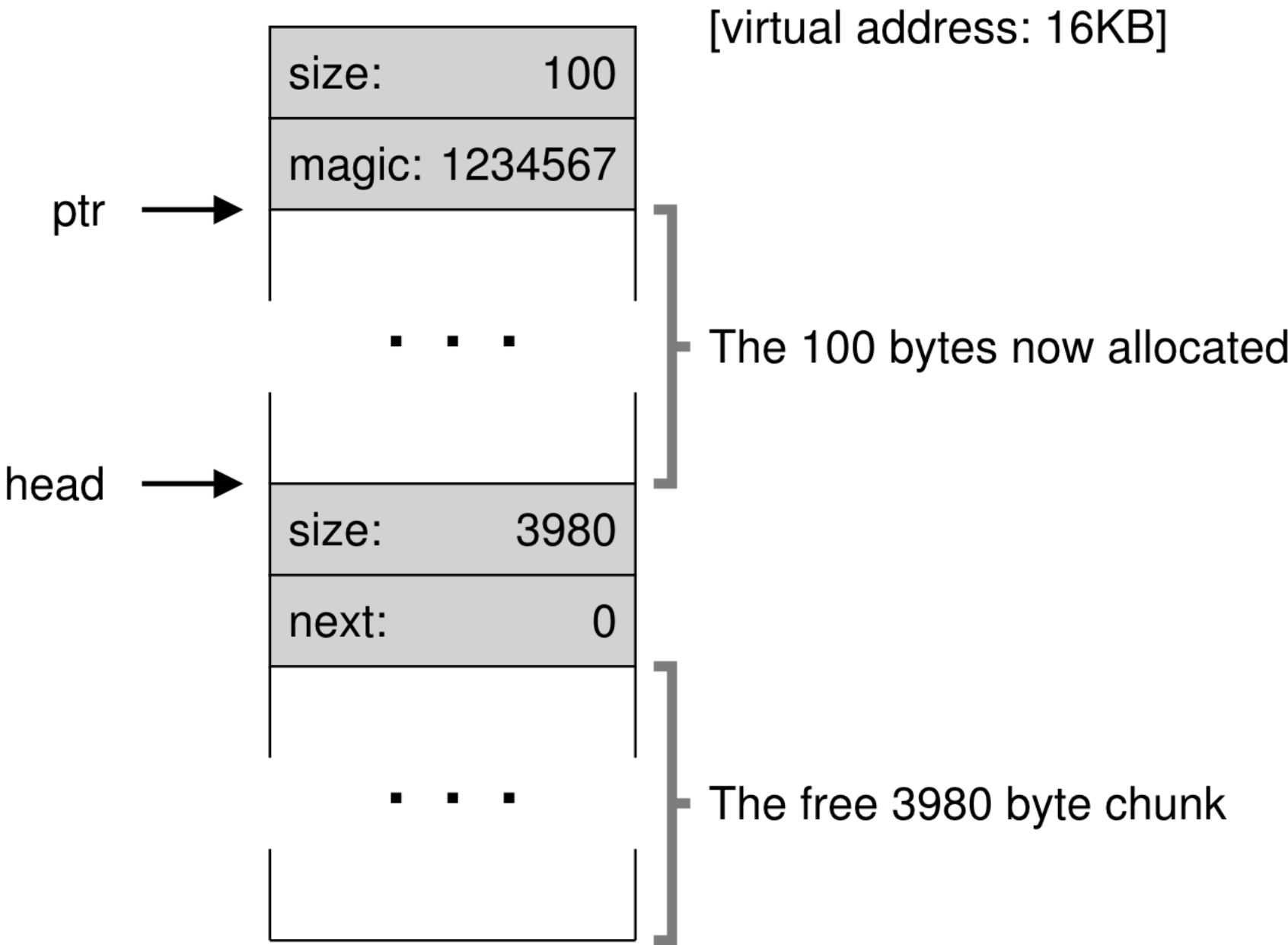


Figure 17.4: A Heap: After One Allocation

Free list splitting and coalescing

```
ptr = malloc(100)
sptr = malloc(100)
optr = malloc(100)
free(sptr)
free(ptr)
free(optr)
```

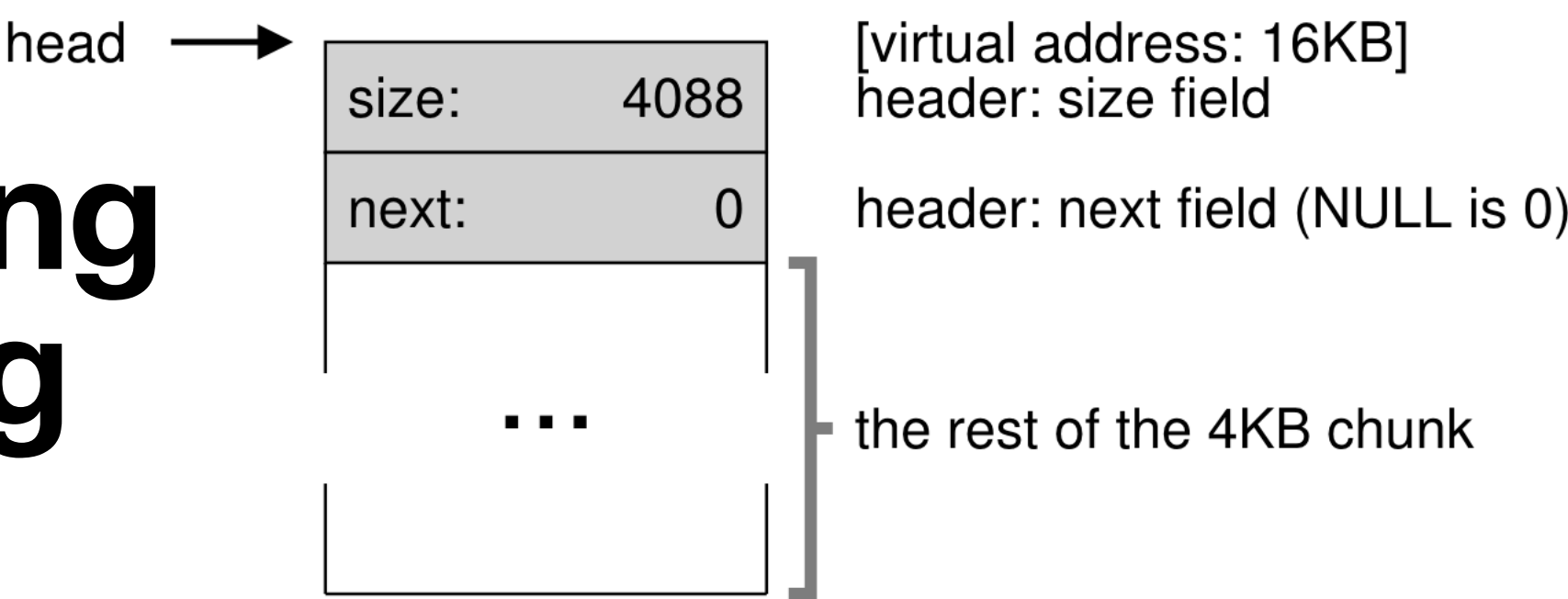


Figure 17.3: A Heap With One Free Chunk

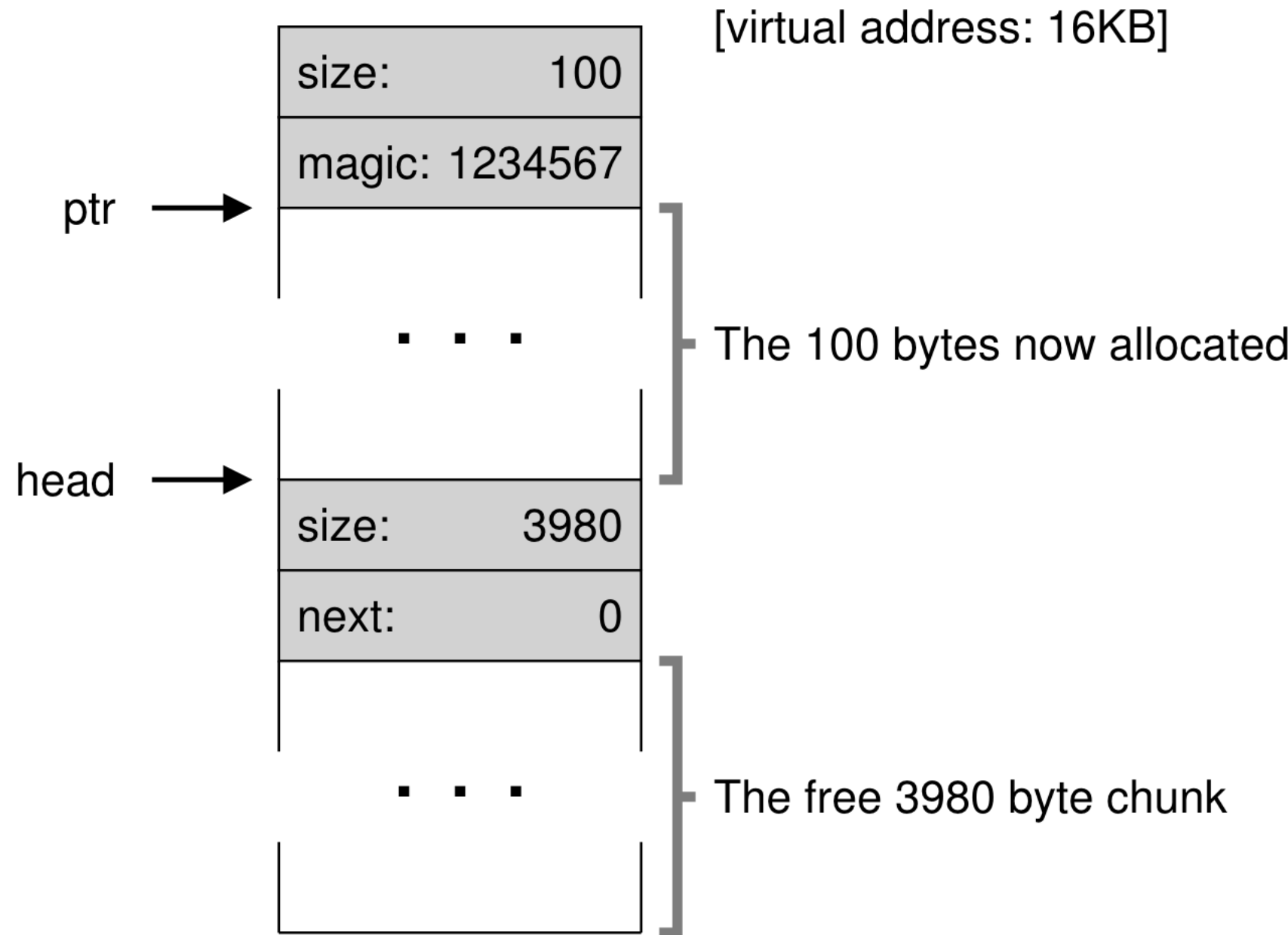


Figure 17.4: A Heap: After One Allocation

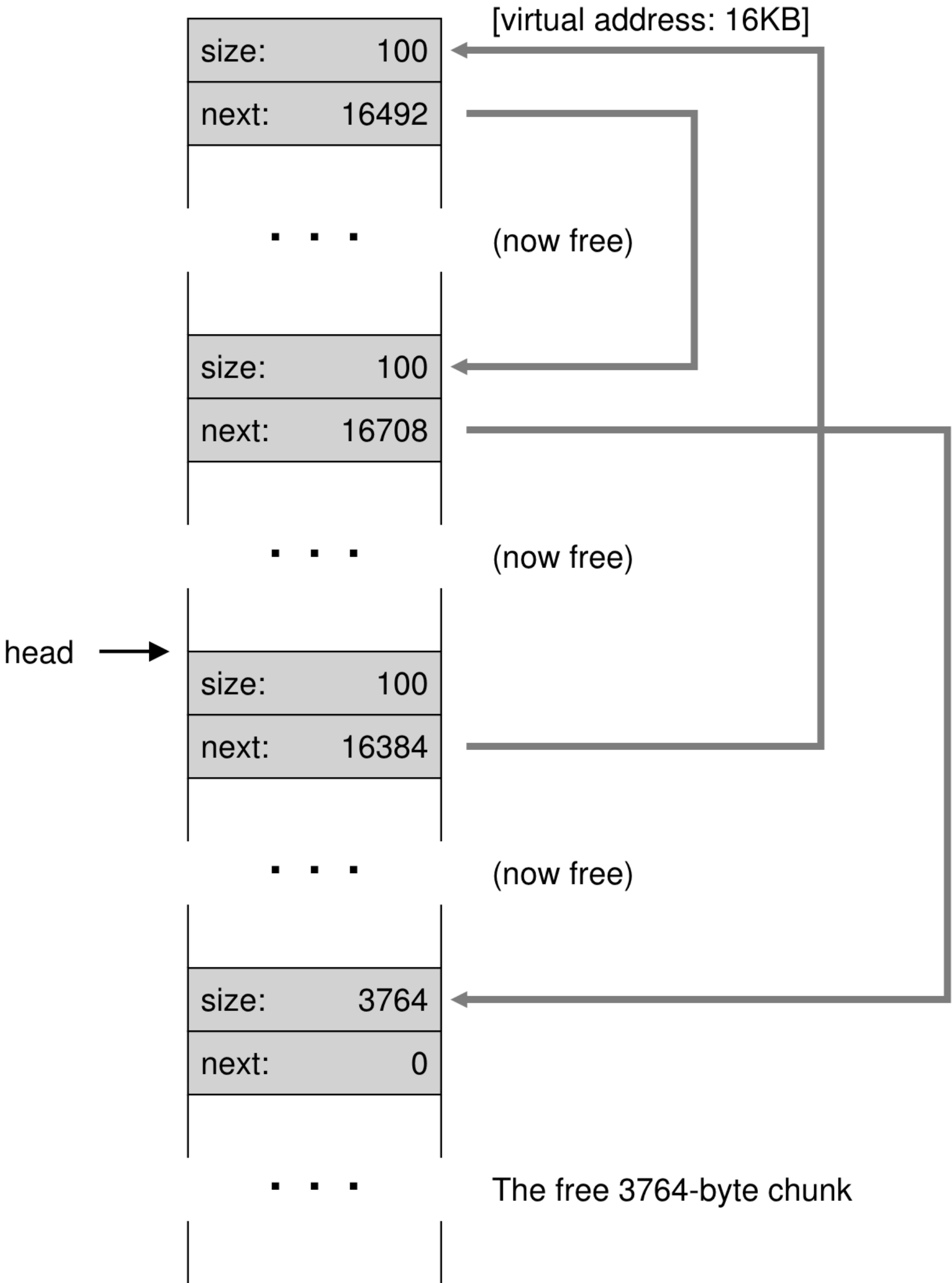


Figure 17.7: A Non-Coalesced Free List

Free list splitting and coalescing

```
ptr = malloc(100)
sptr = malloc(100)
optr = malloc(100)
free(sptr)
free(ptr)
free(optr)
```

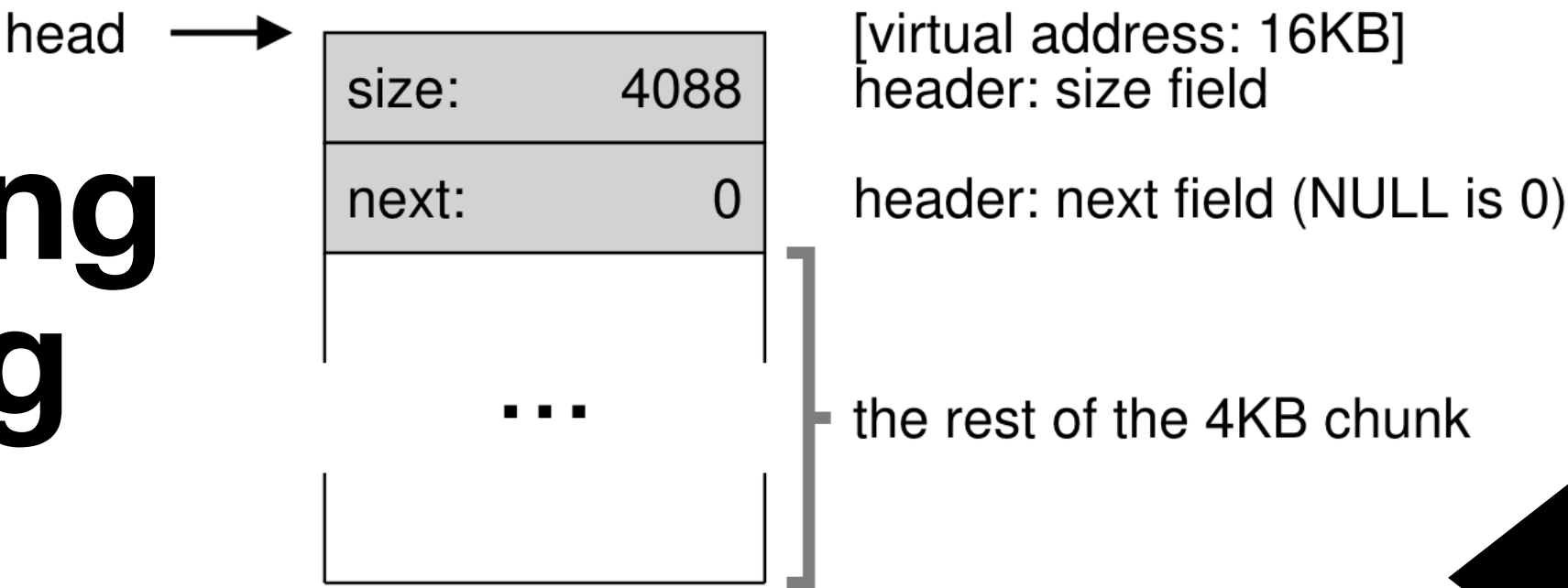


Figure 17.3: A Heap With One Free Chunk

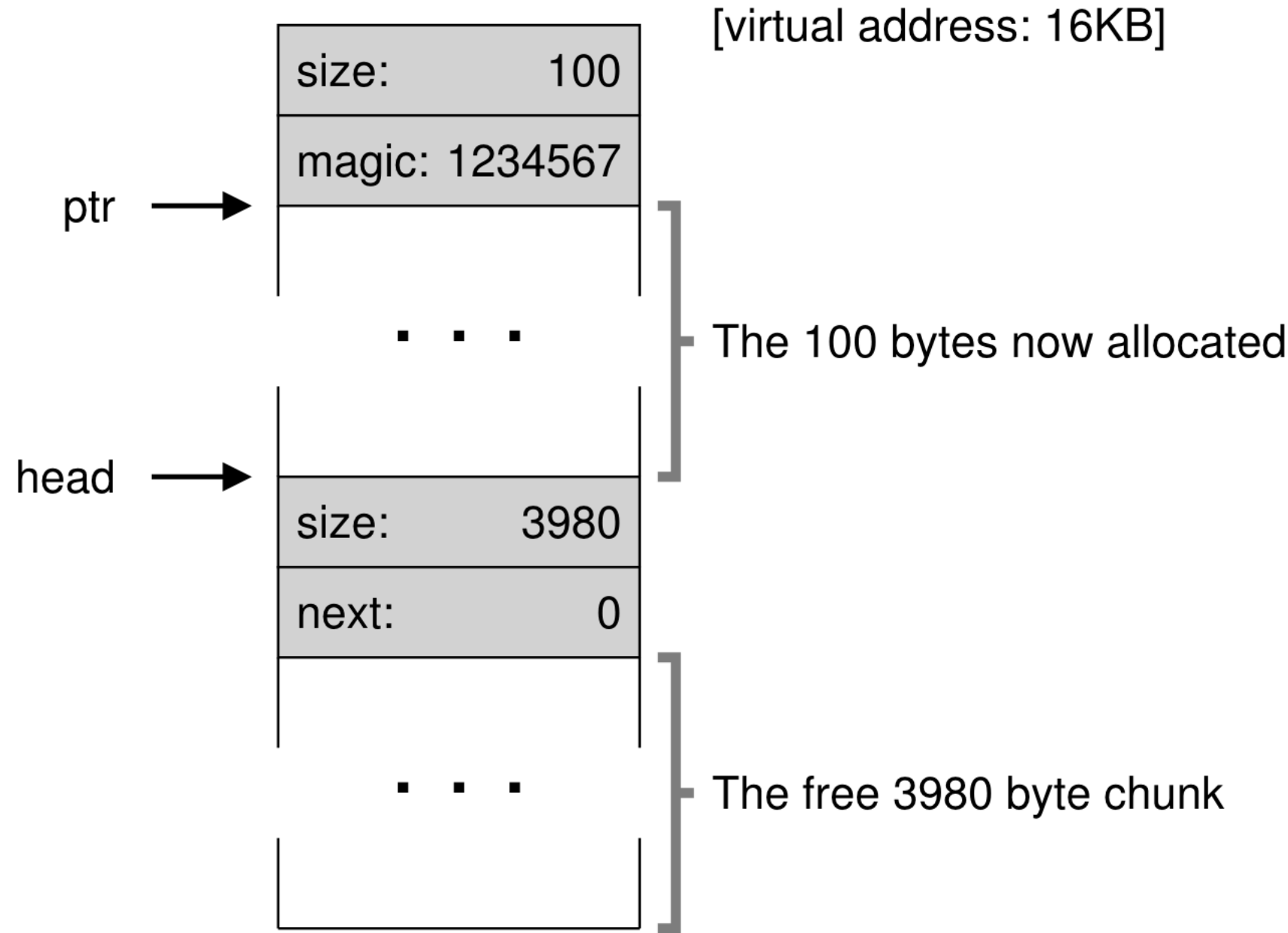


Figure 17.4: A Heap: After One Allocation

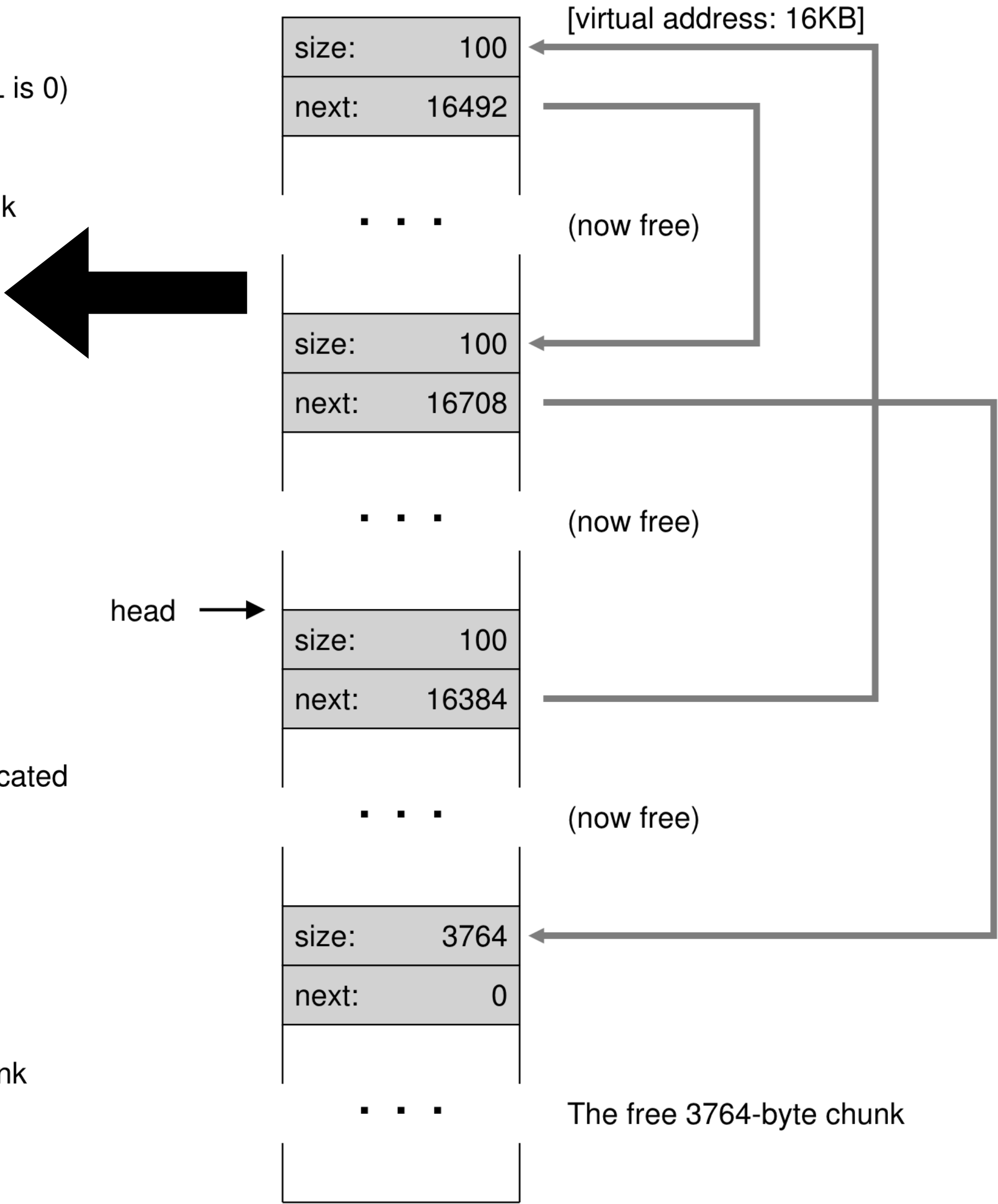


Figure 17.7: A Non-Coalesced Free List

Which block to allocate?

Example: malloc(15)



Which block to allocate?

Example: malloc(15)

- Best fit
 - Slow. need to search the whole list



Which block to allocate?

Example: malloc(15)

- Best fit
 - Slow. need to search the whole list



Which block to allocate?

Example: malloc(15)

- Best fit
 - Slow. need to search the whole list
- First fit
 - Faster. (xv6: umalloc.c)
 - May cause more fragmentation by breaking up larger free blocks
 - Example: malloc(25)



Which block to allocate?

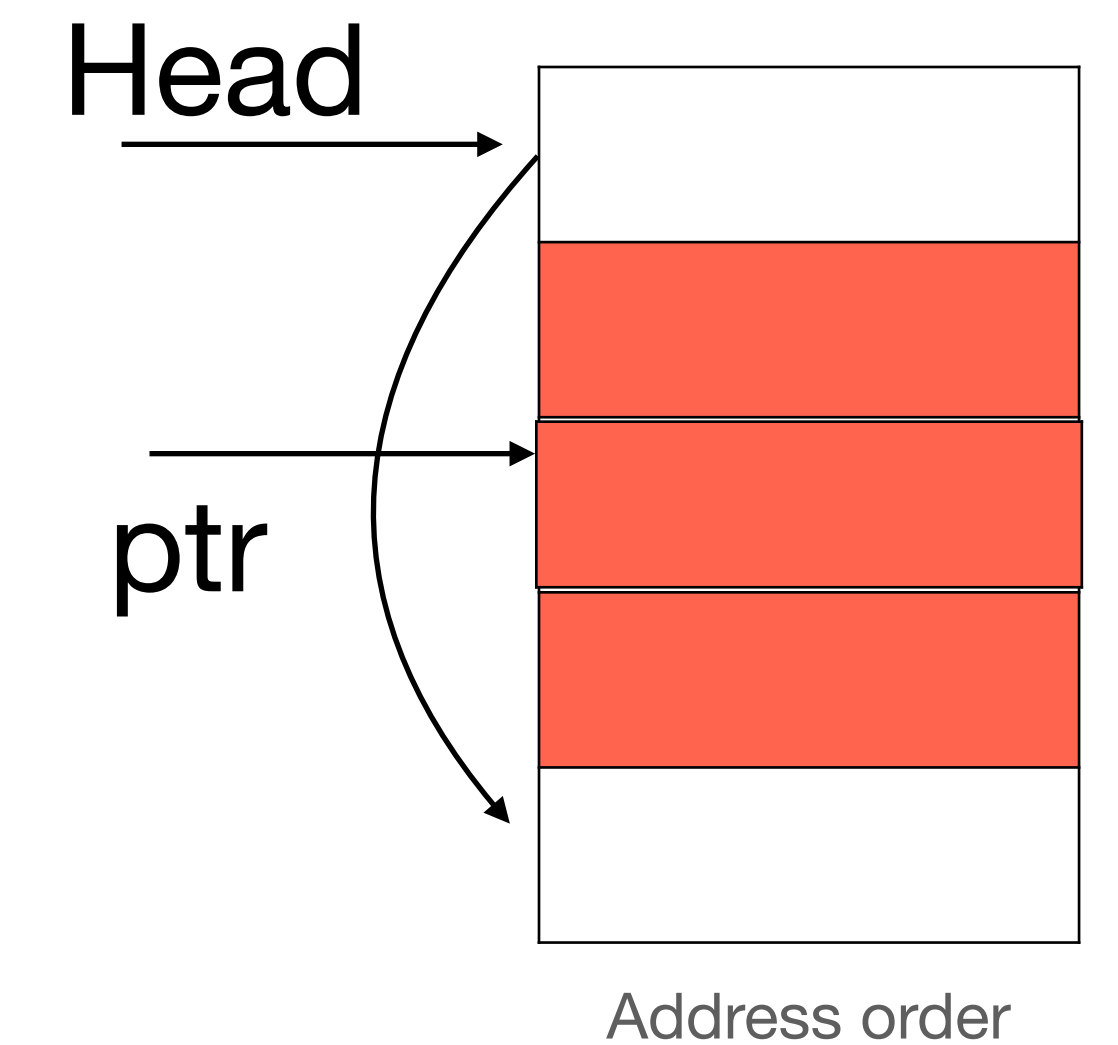
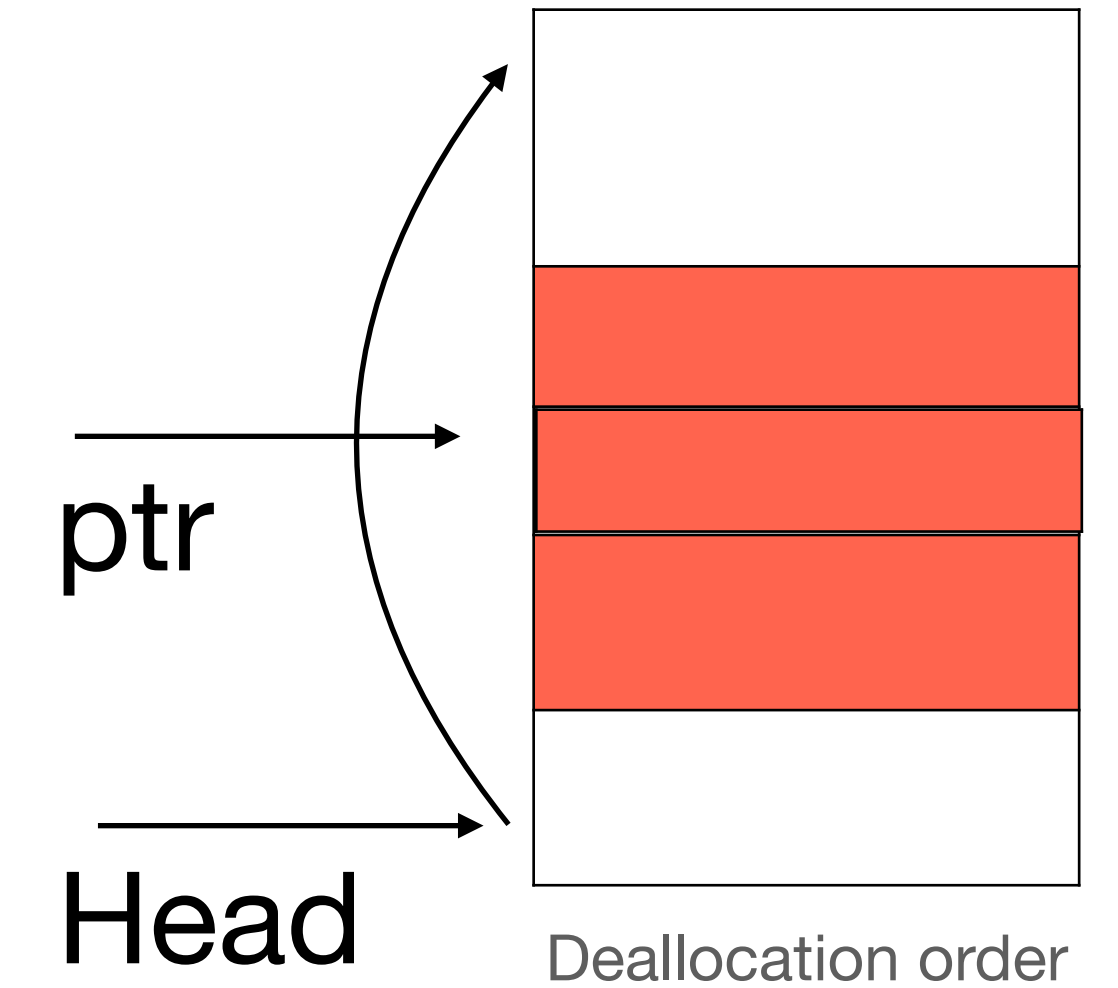
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- Best fit
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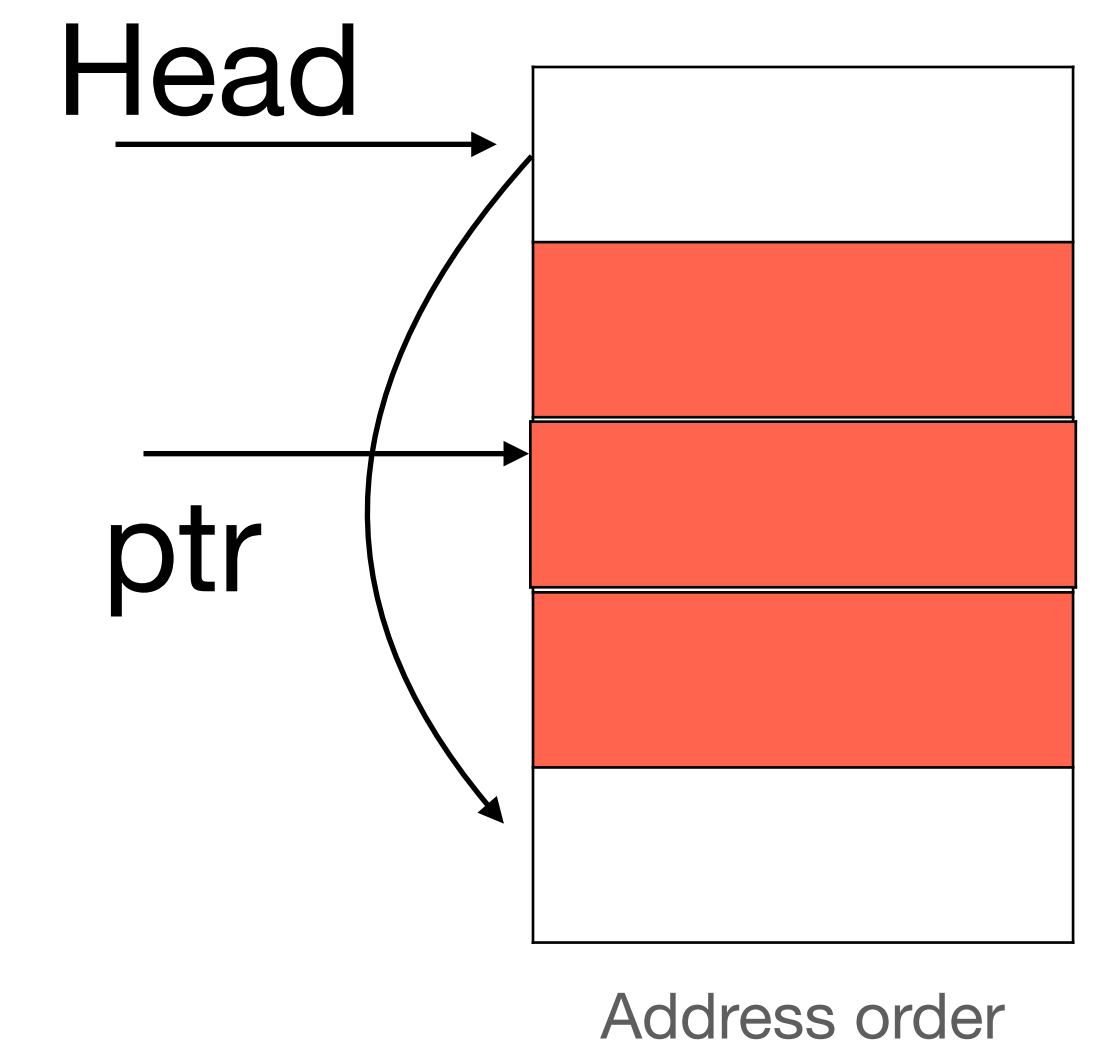
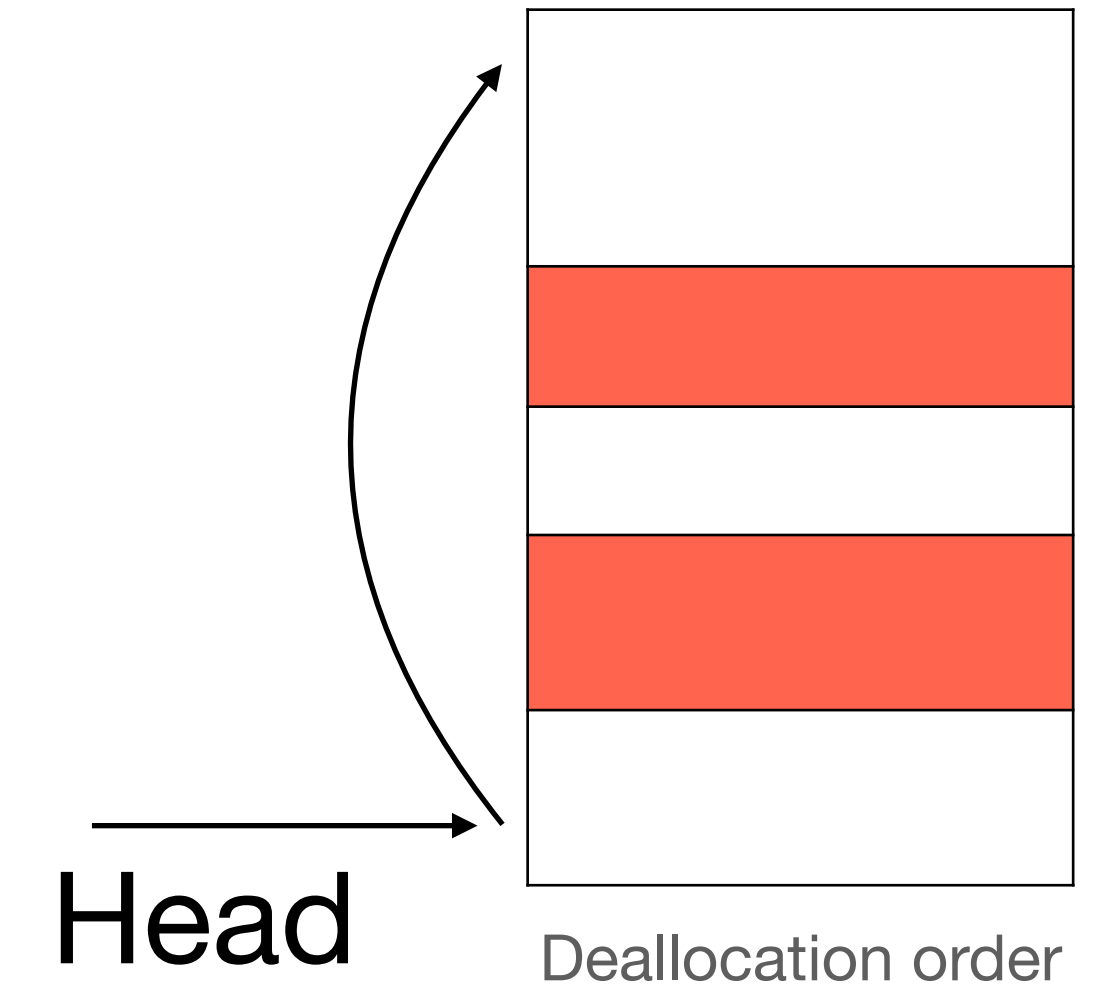
In which order should we maintain the free list?

- `free(ptr)`



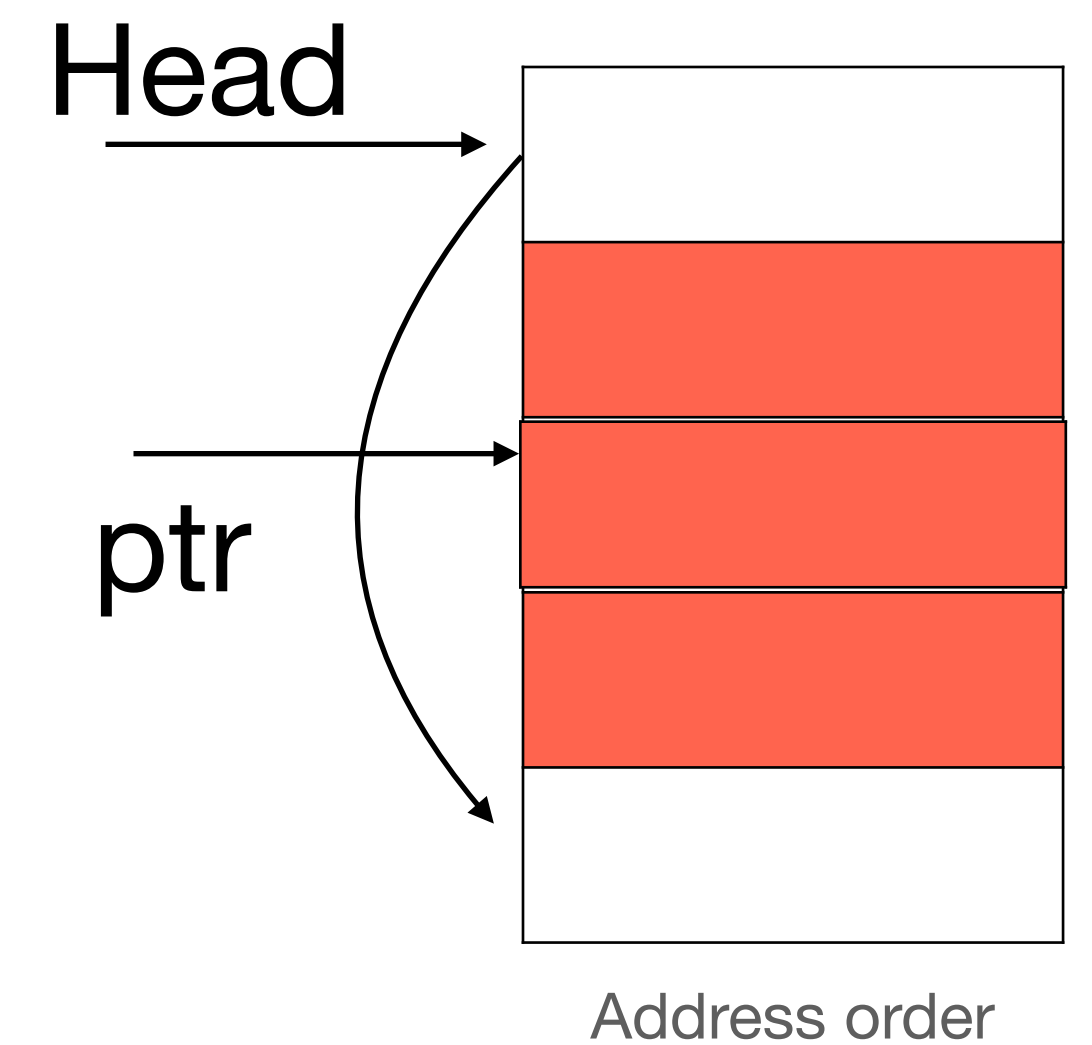
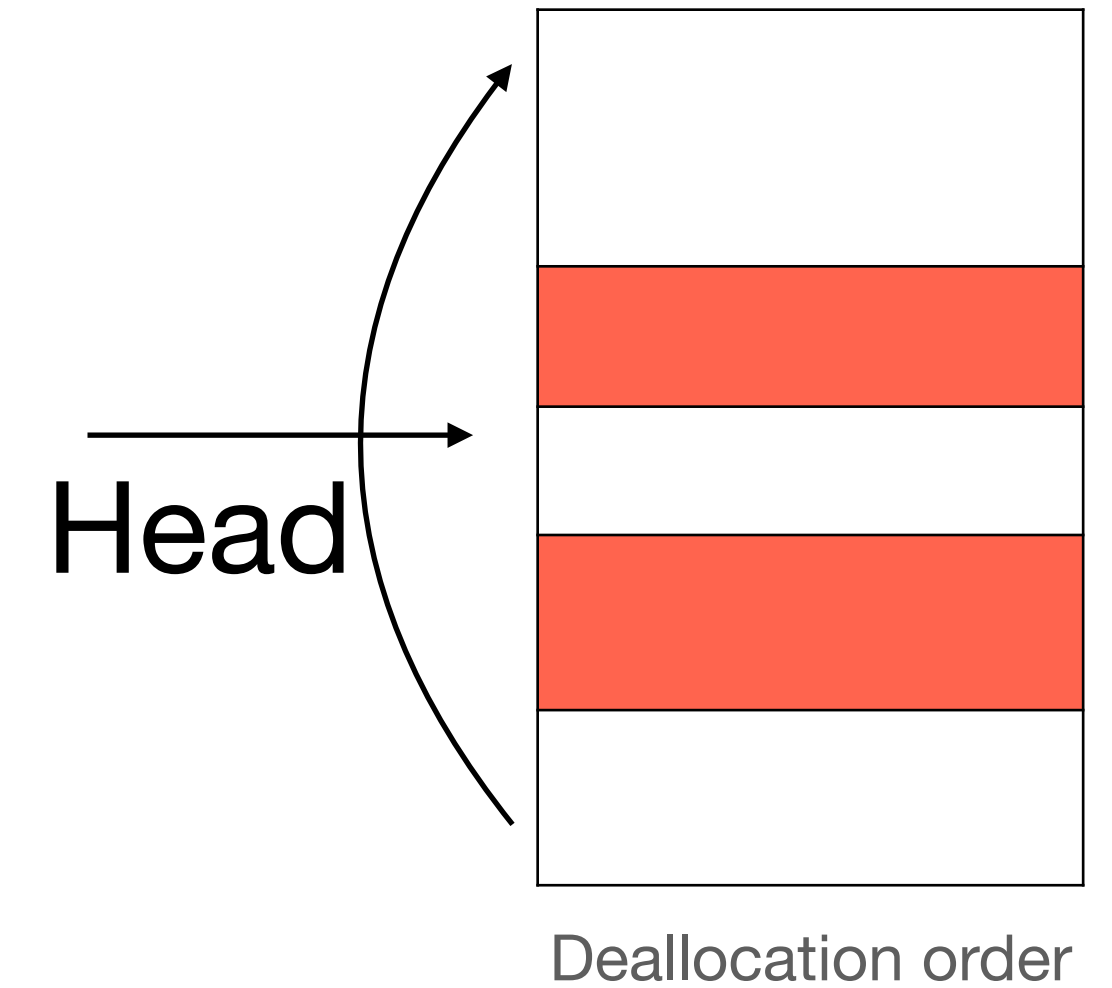
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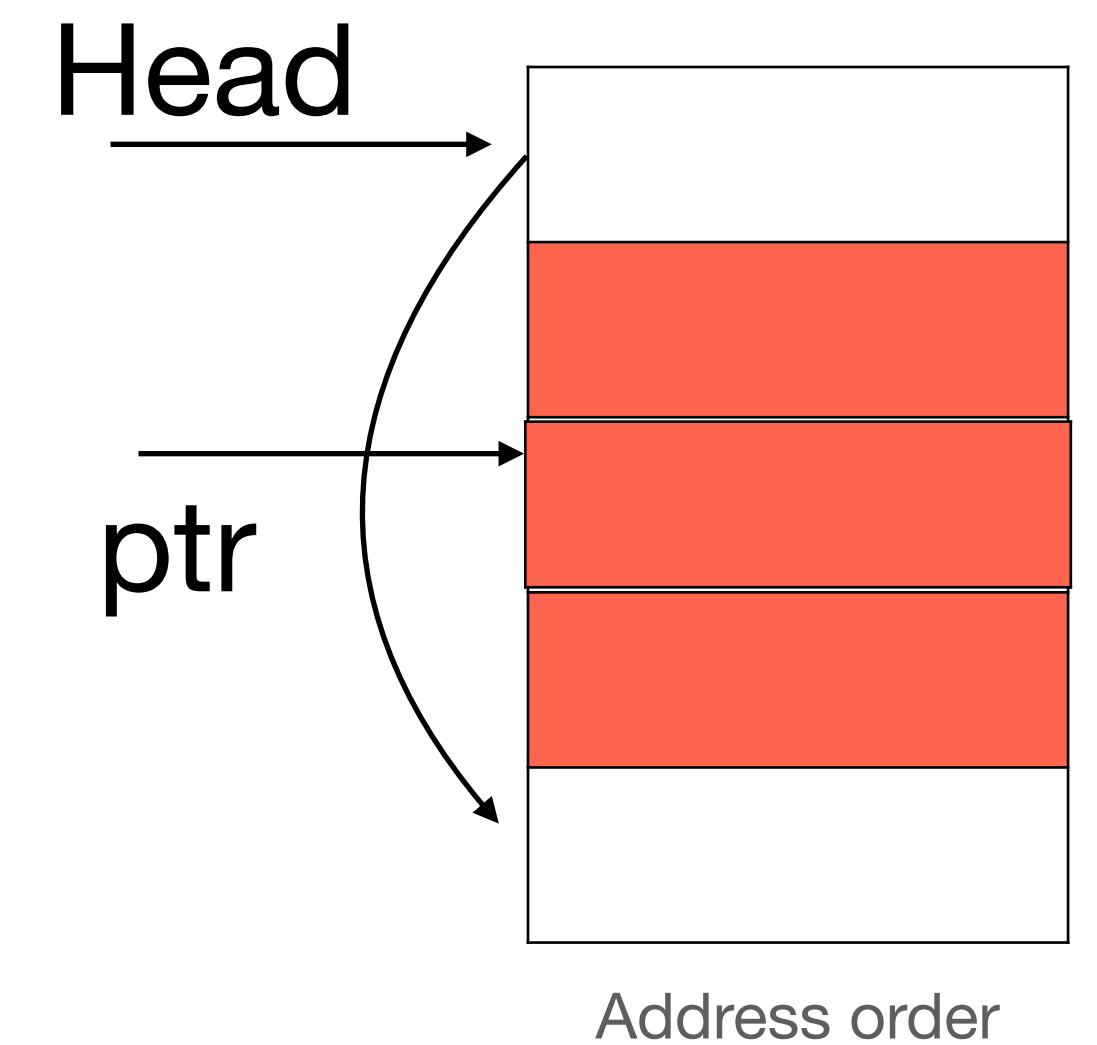
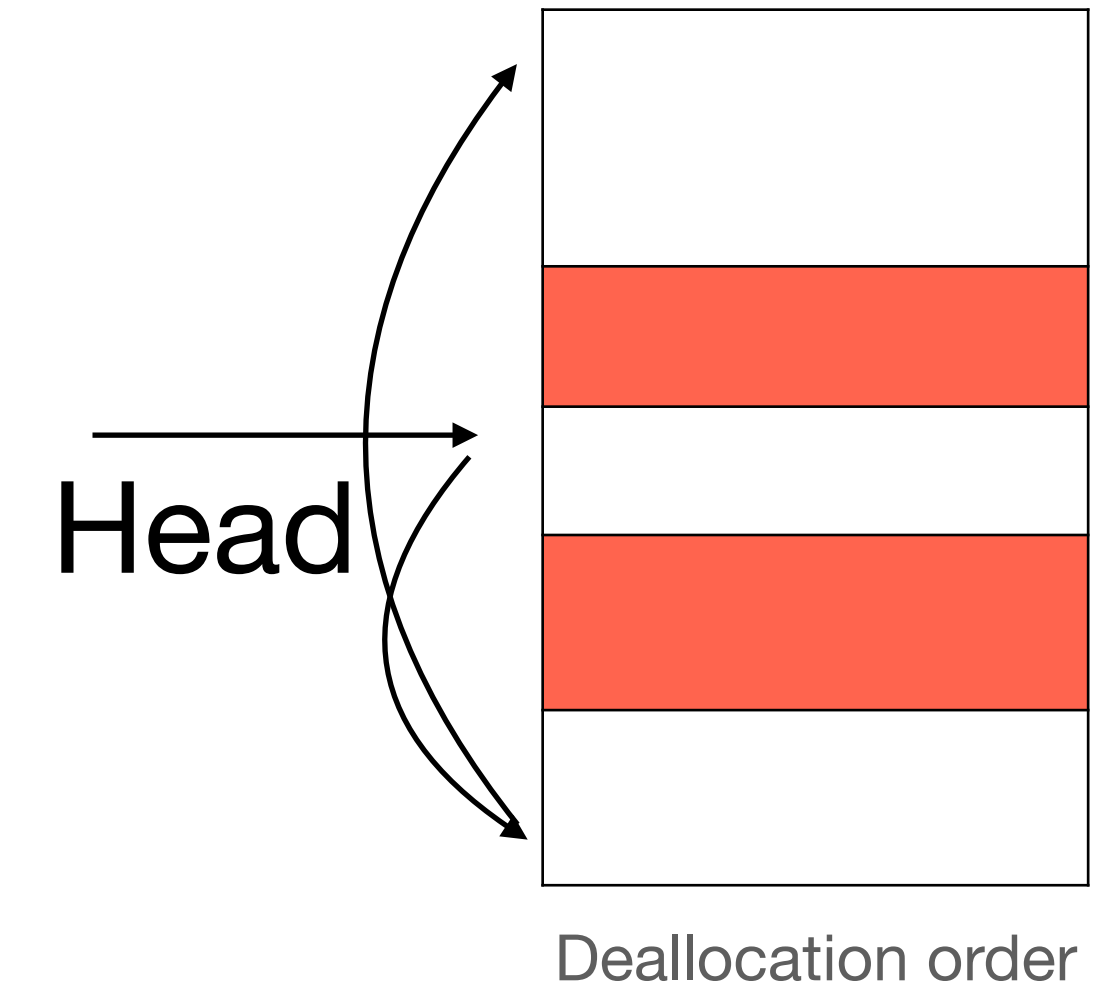
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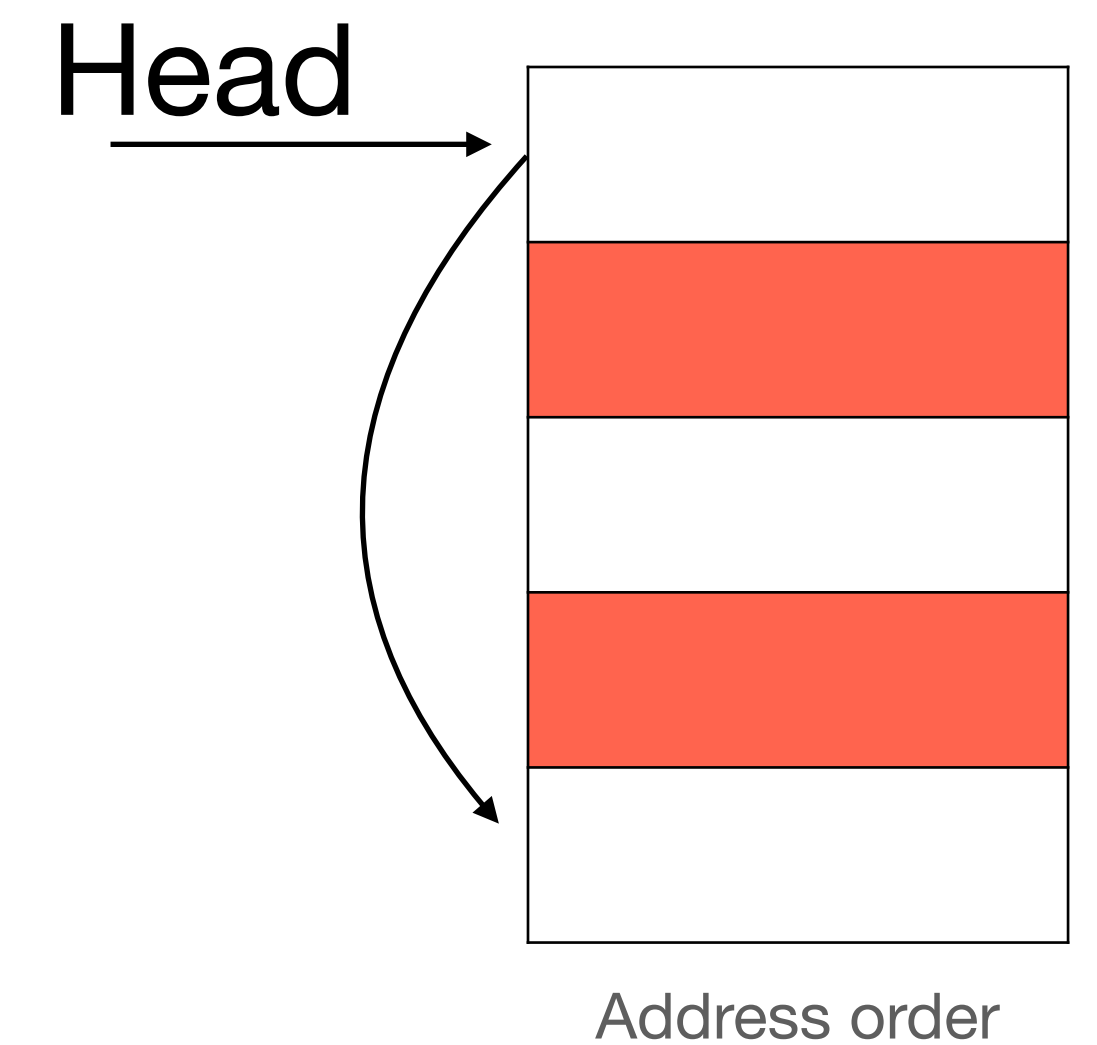
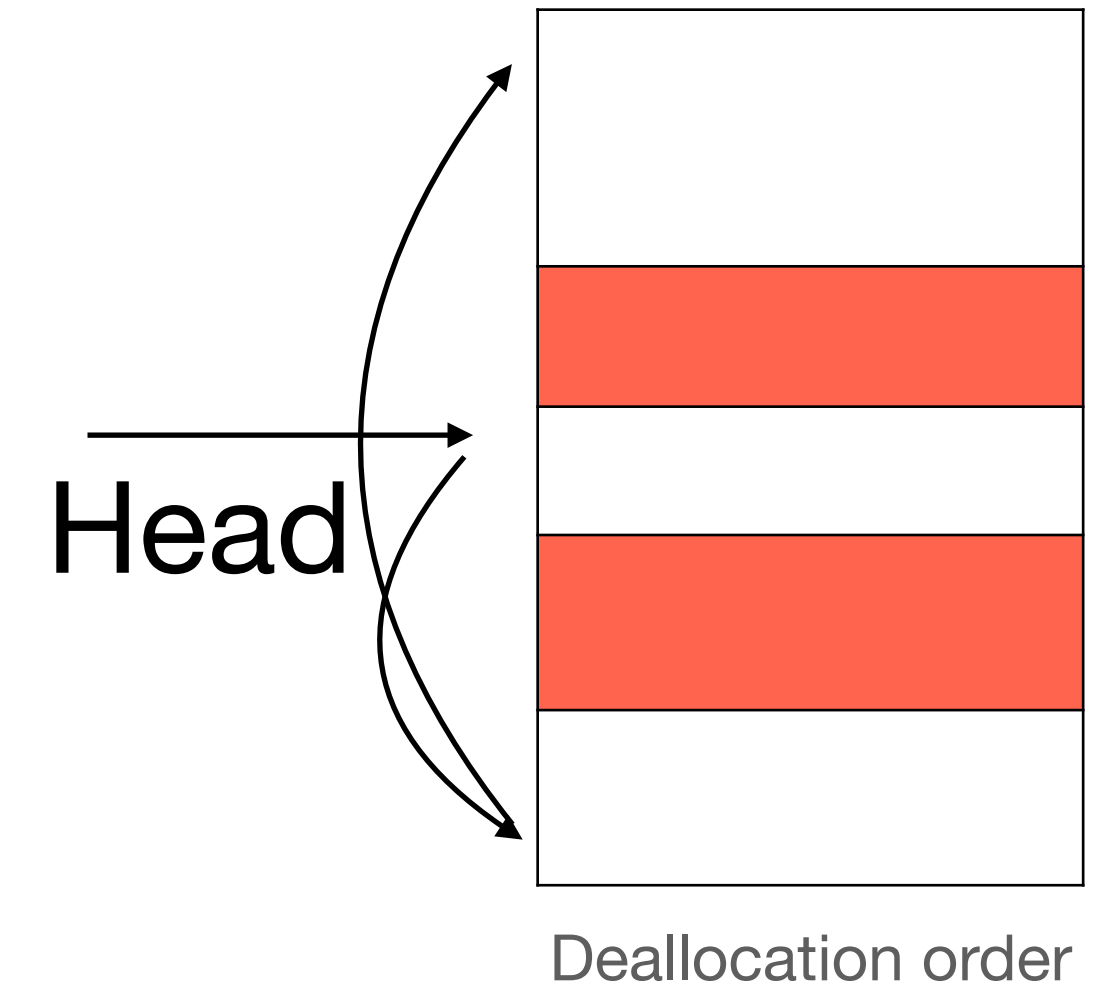
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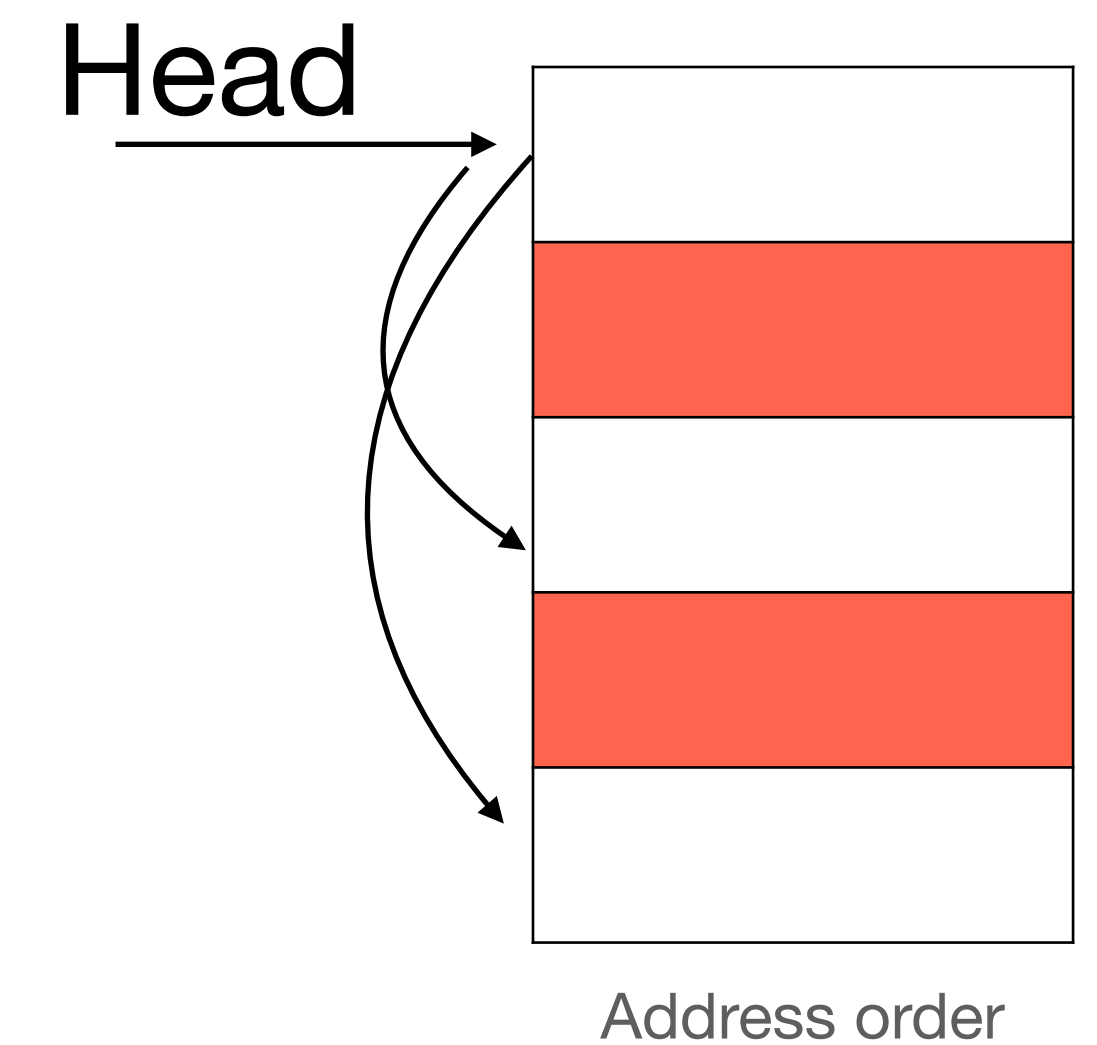
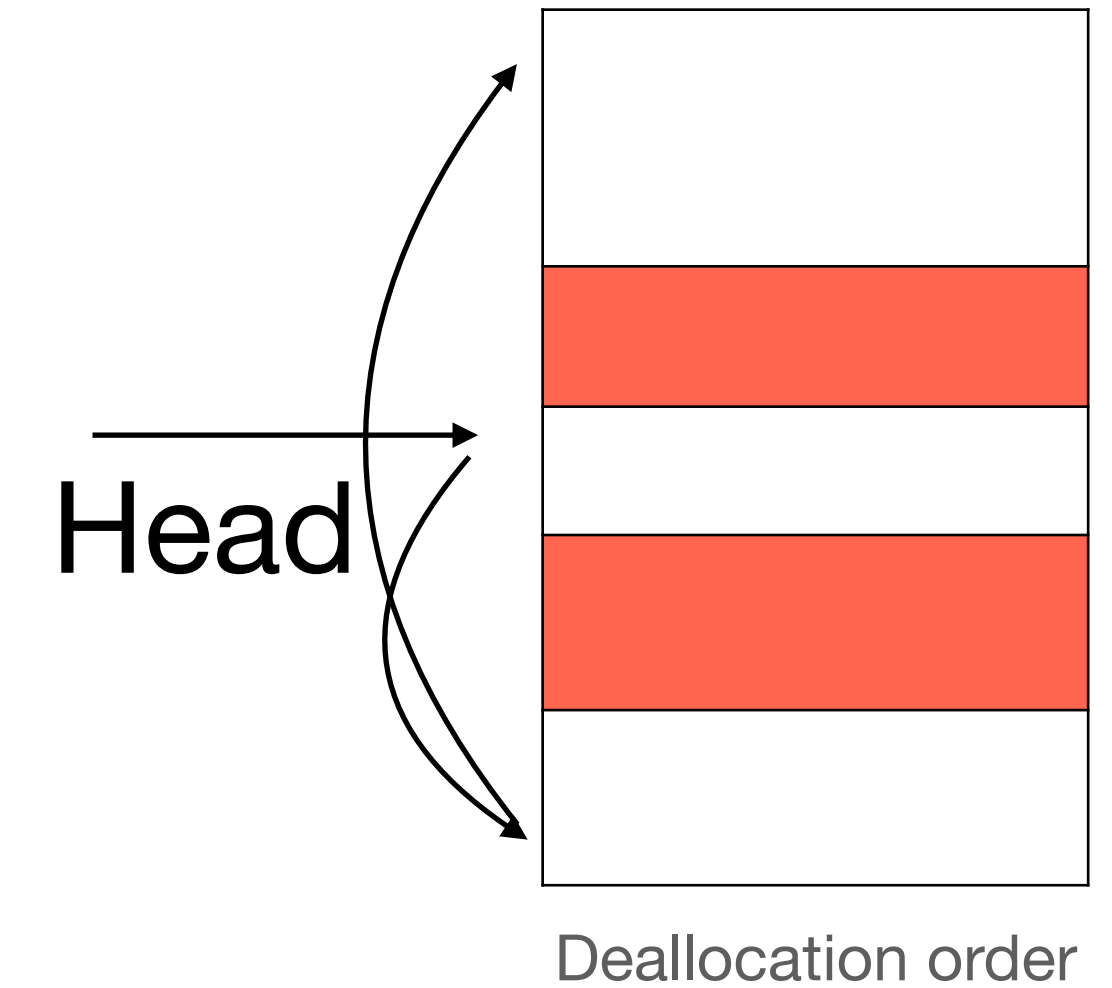
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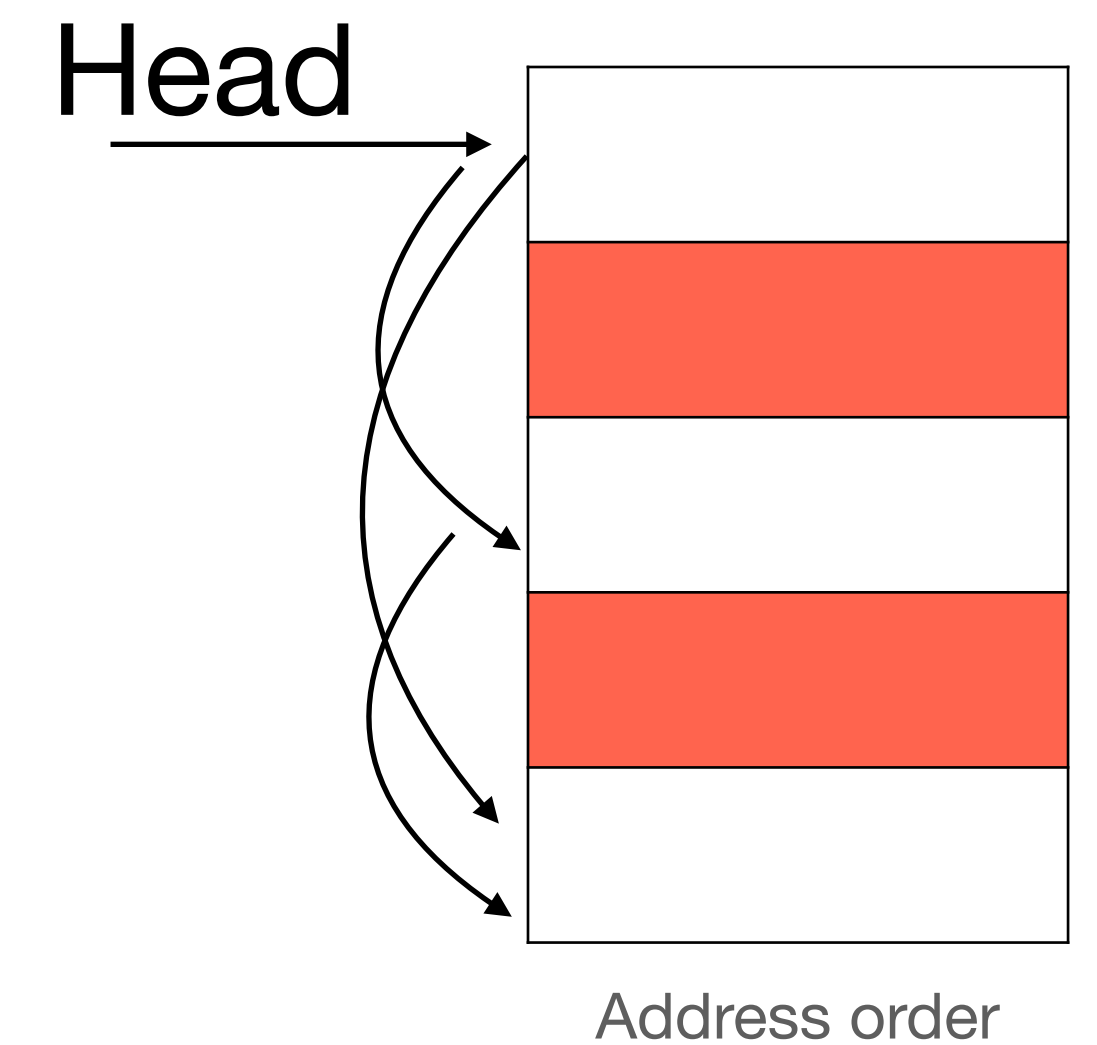
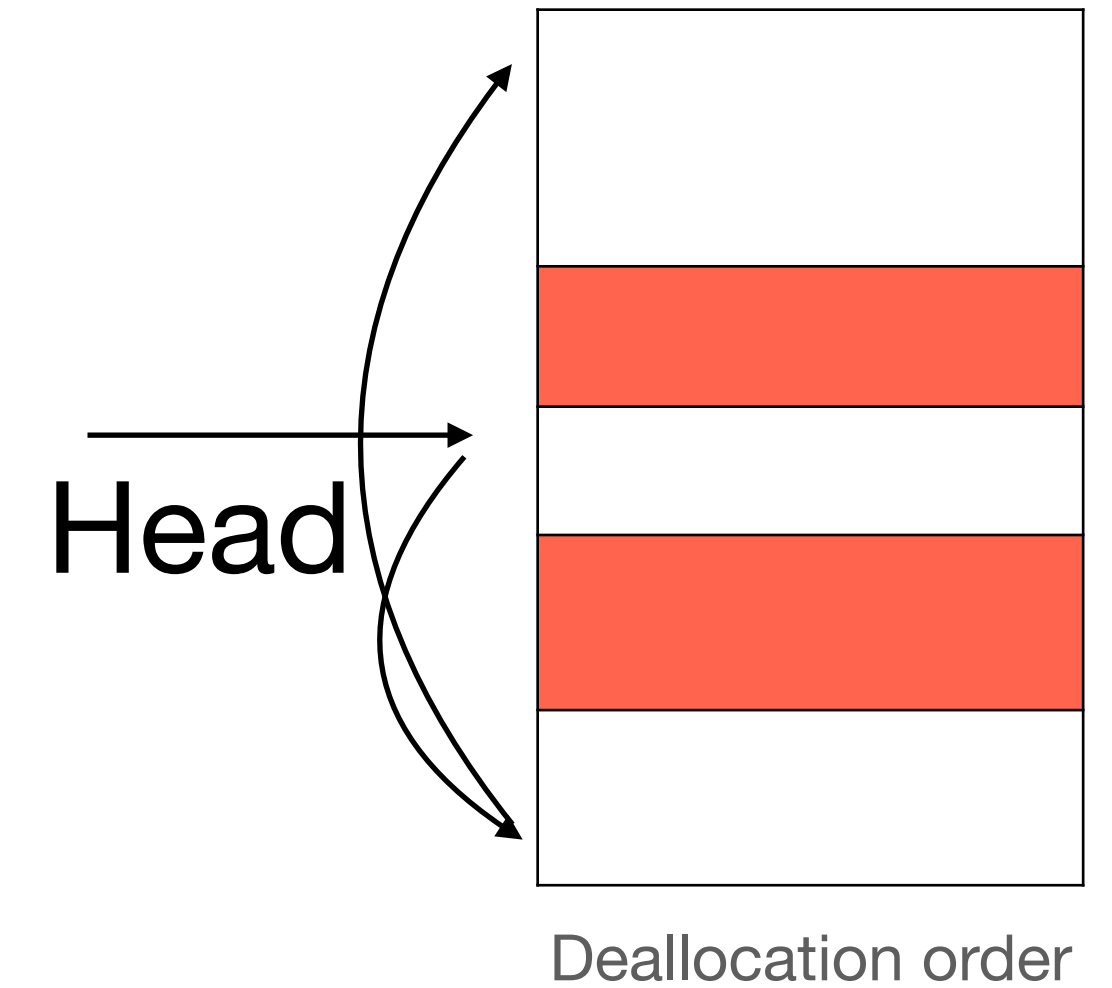
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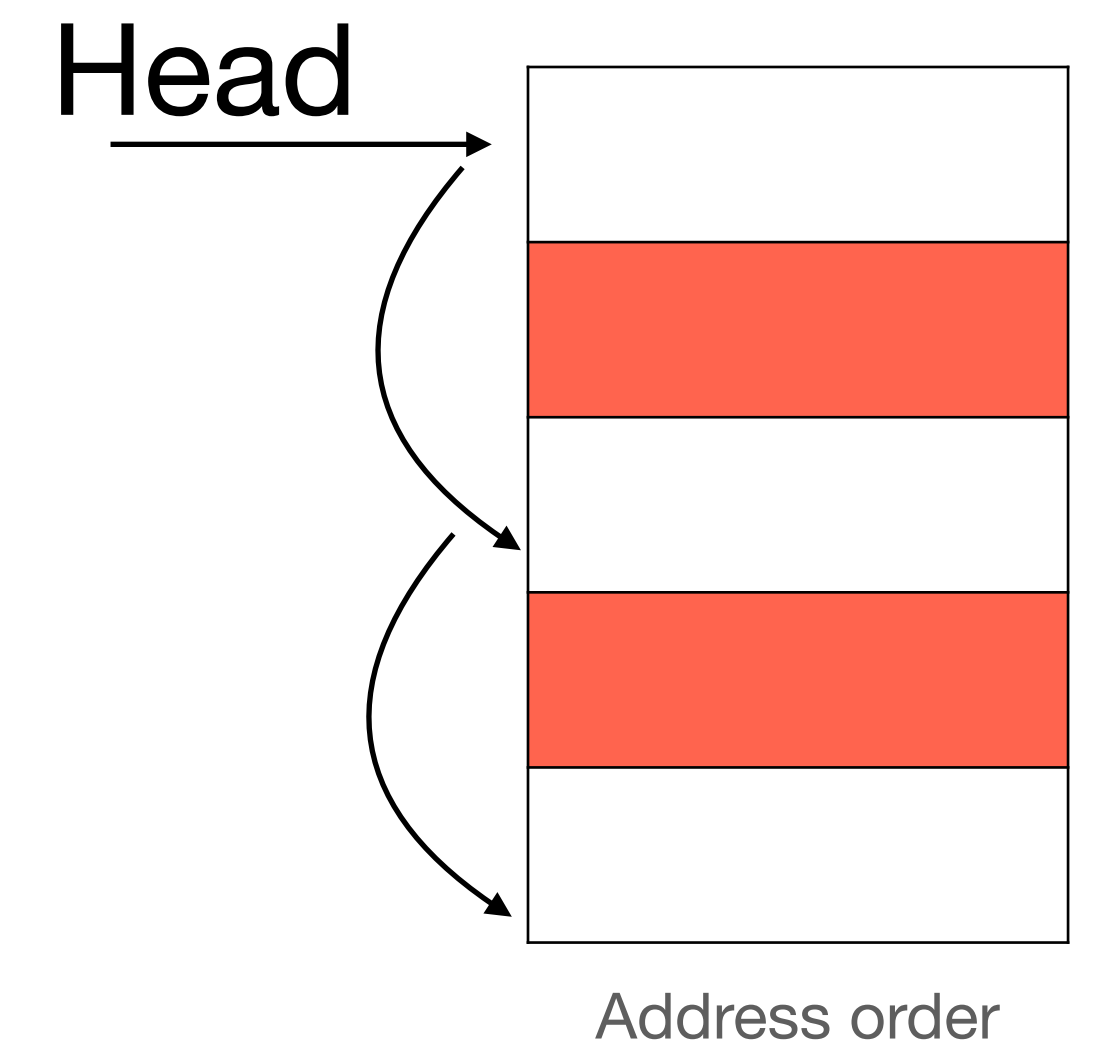
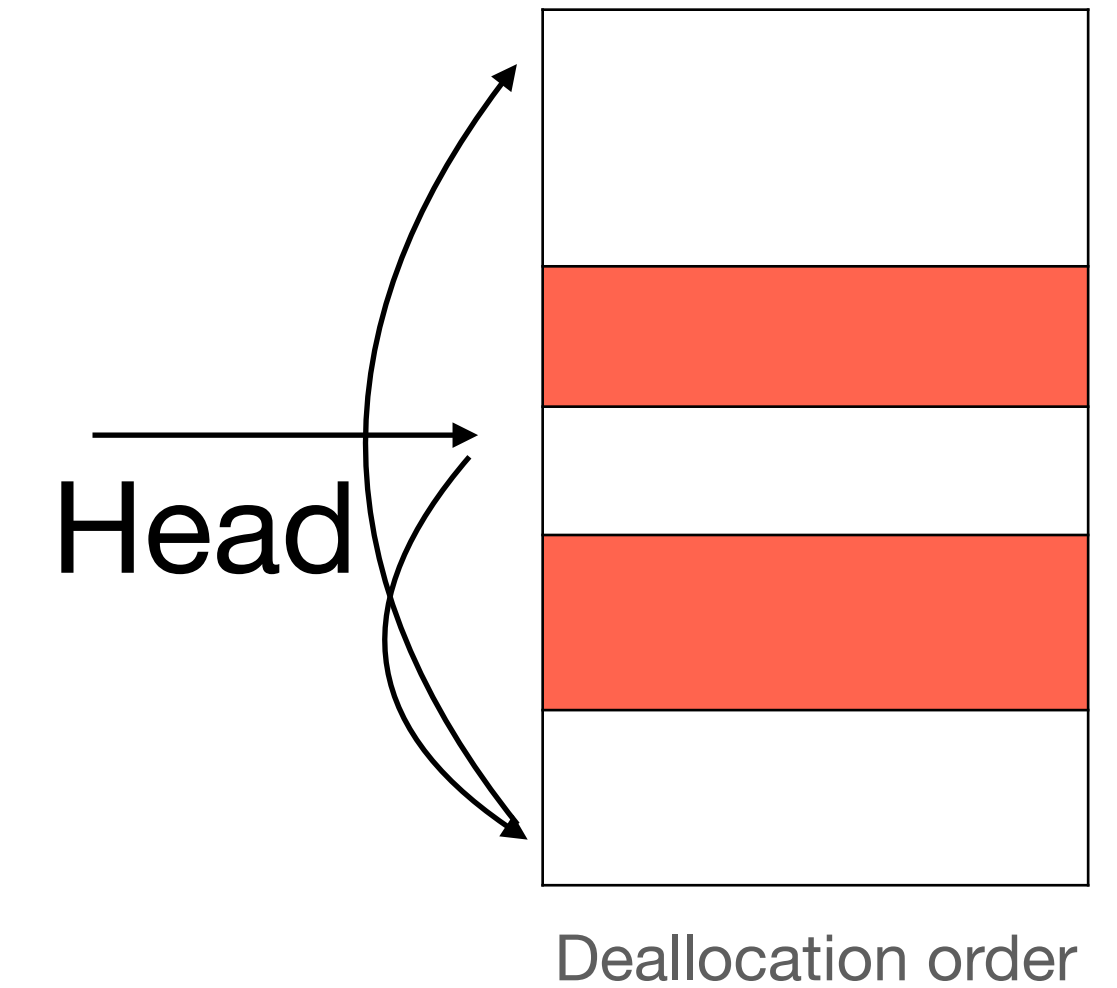
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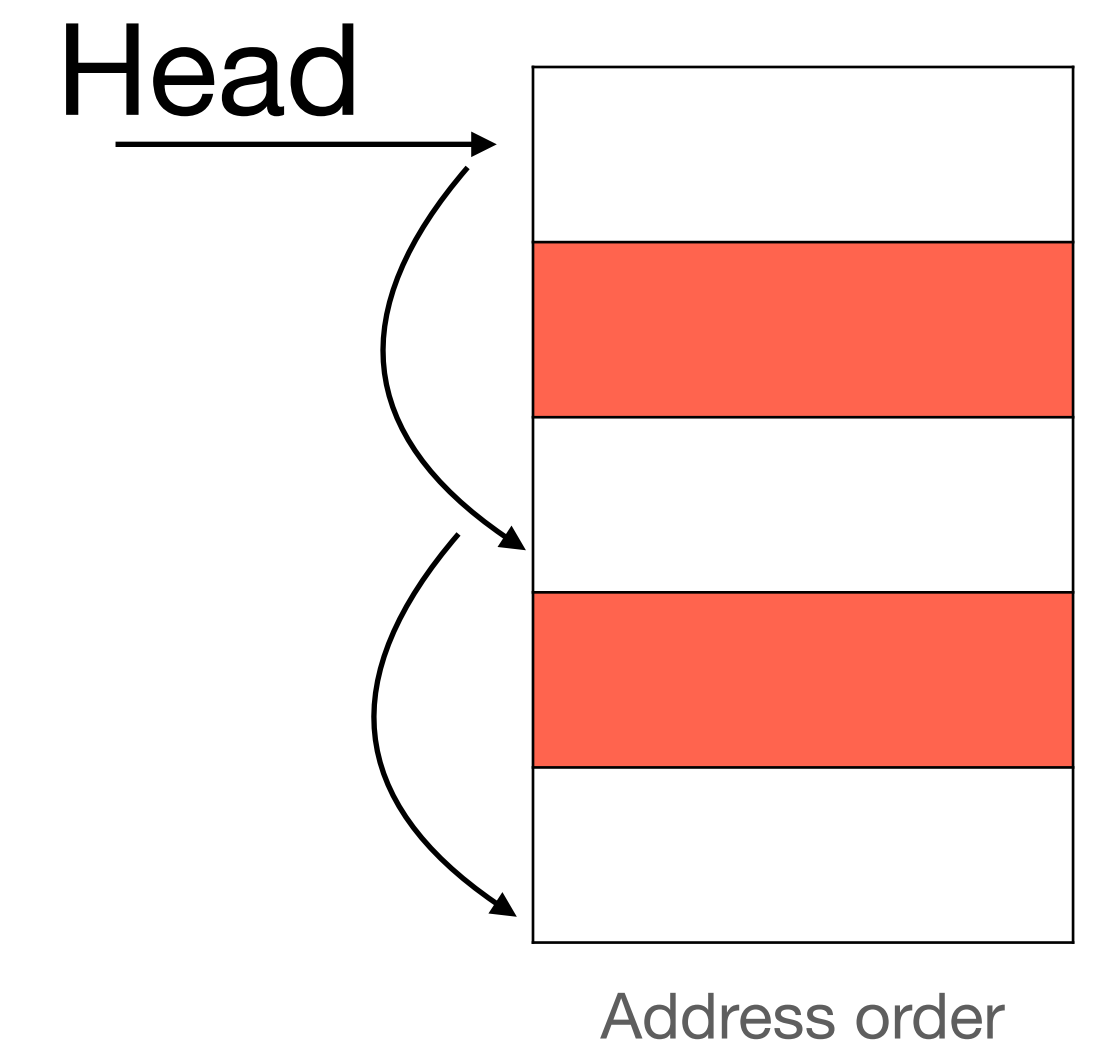
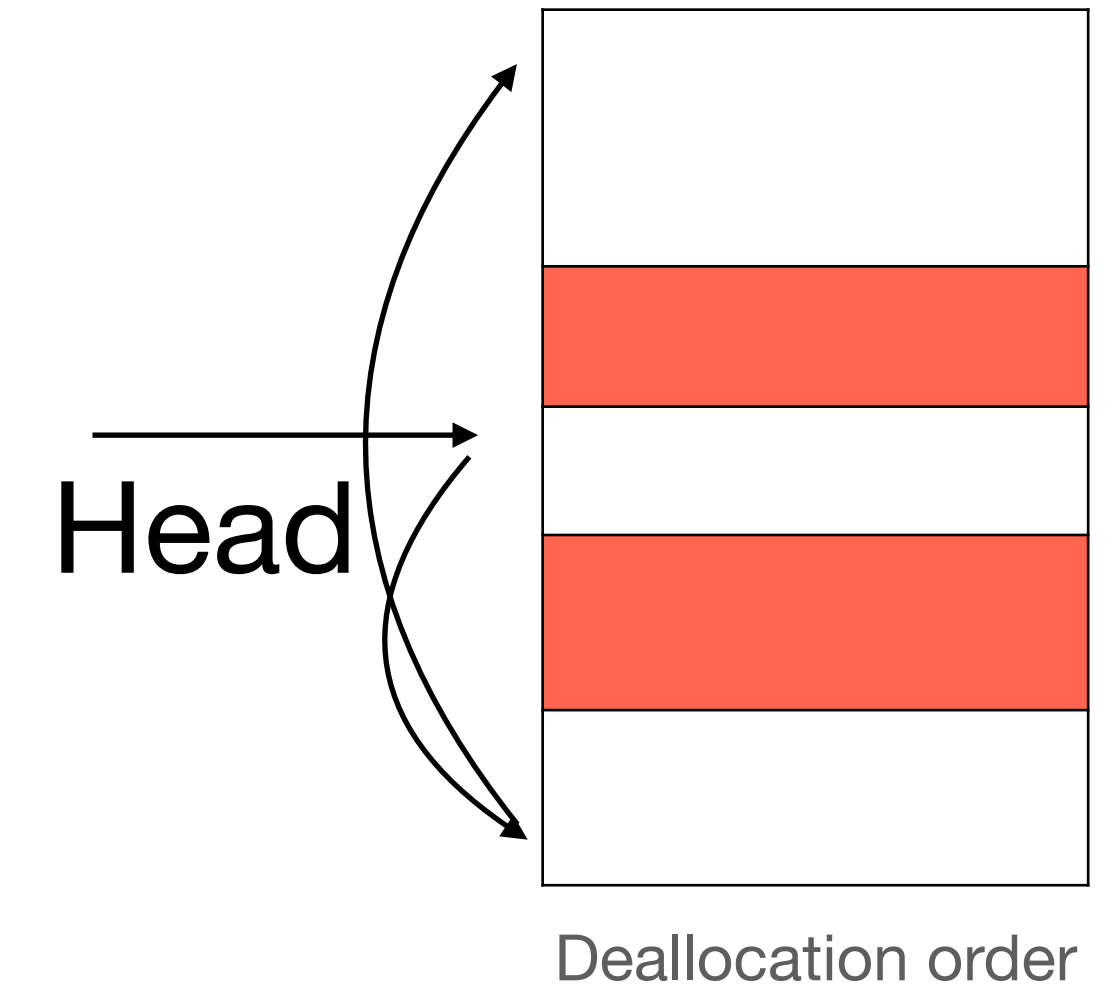
In which order should we maintain the free list?

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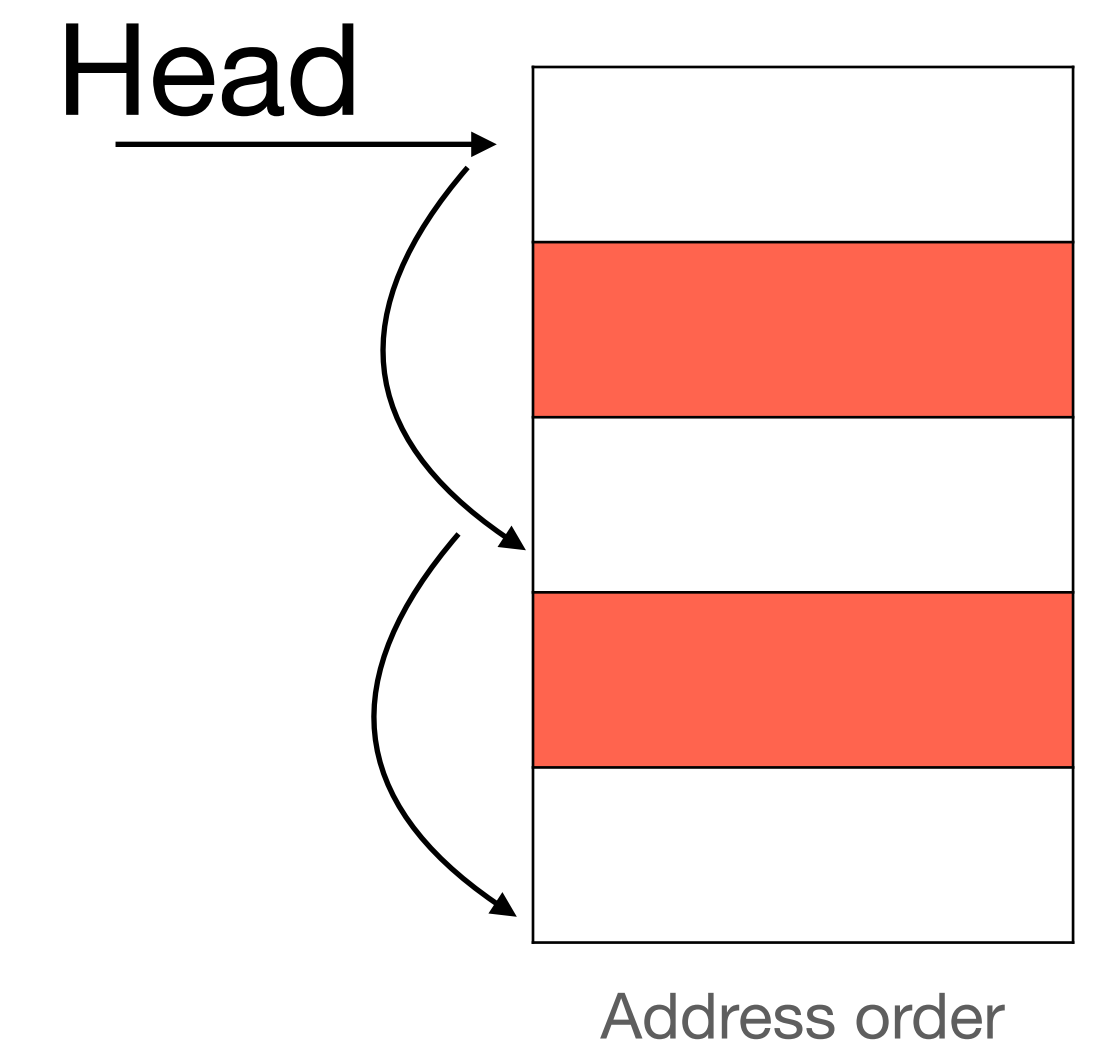
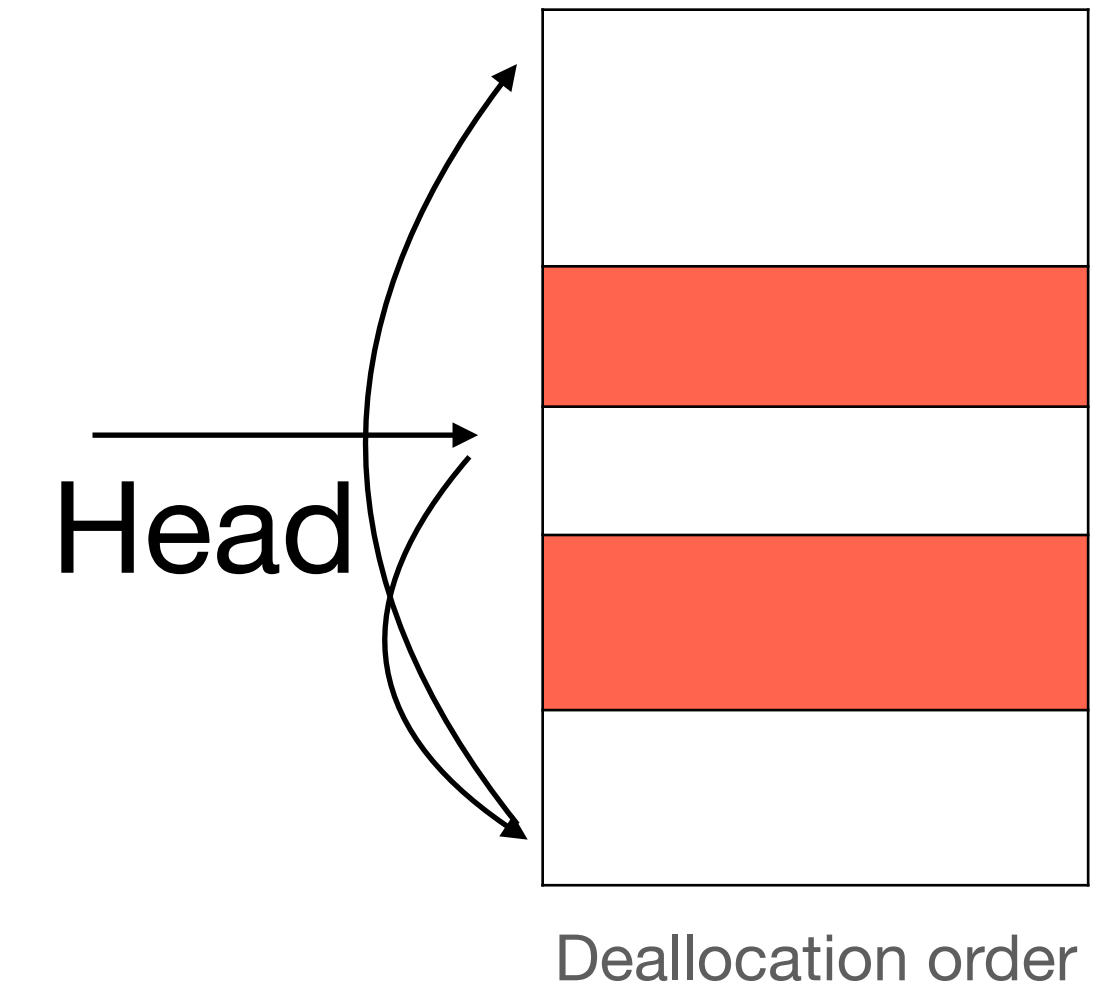
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- `free(ptr)`
- Slow frees in address order: need to traverse the free list



In which order should we maintain the free list?

- `free(ptr)`
- Slow frees in address order: need to traverse the free list
- `umalloc.c` in `xv6` implements address order

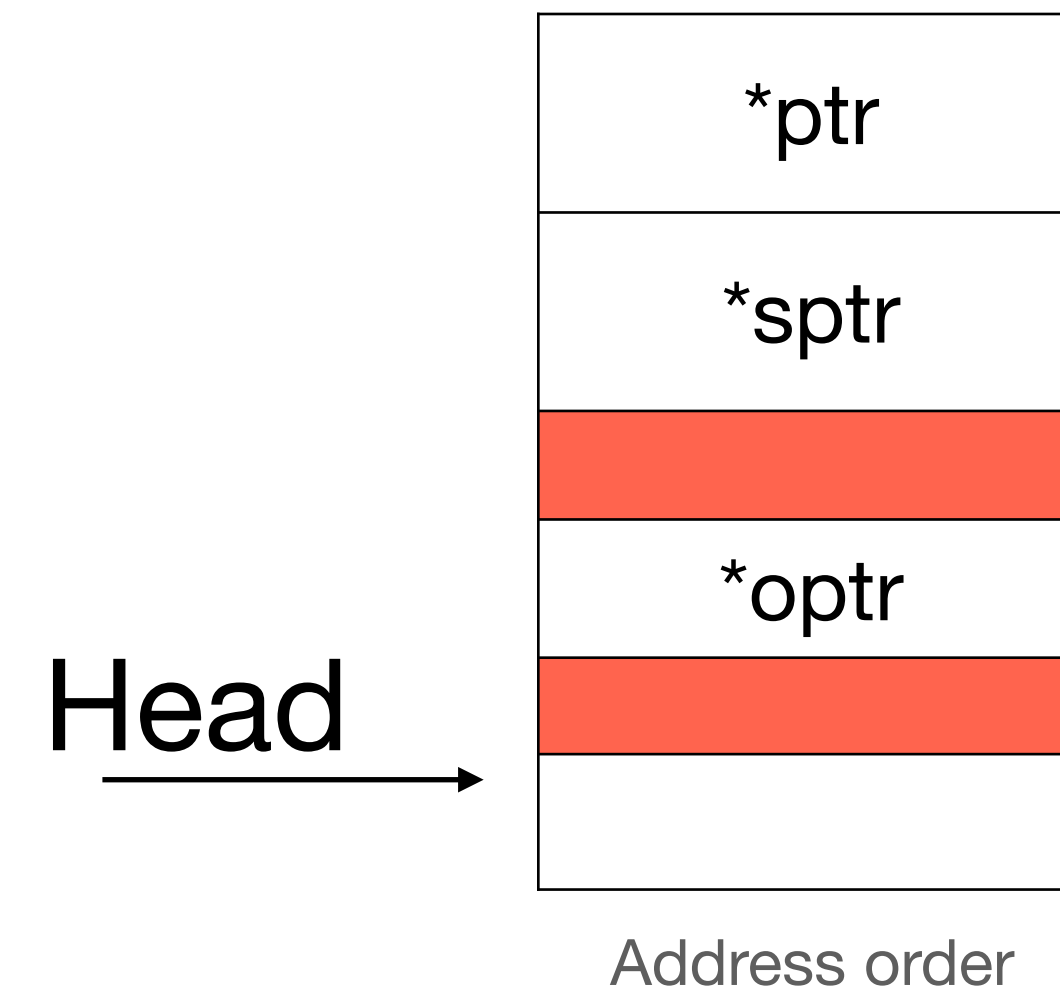
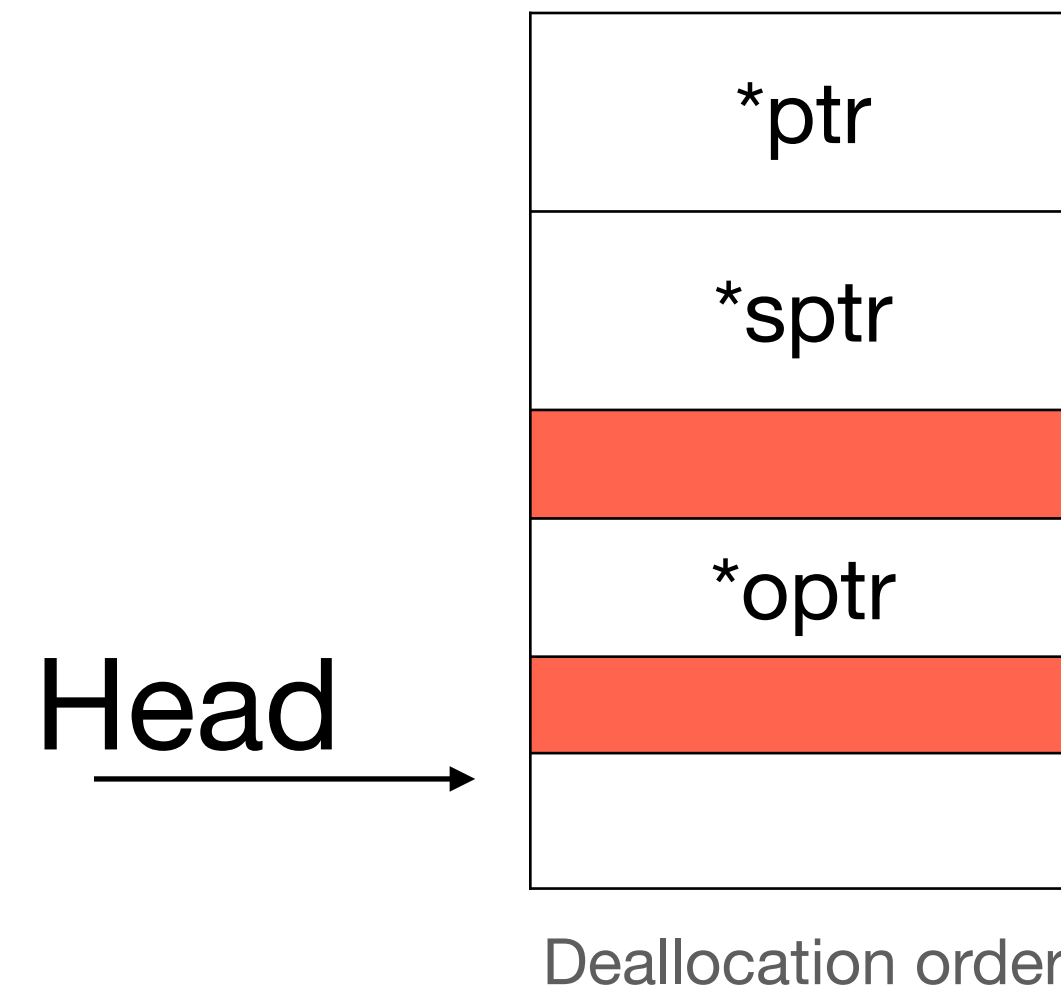


How to do coalescing?

`free(ptr)`

`free(optr)`

`free(sptr)`

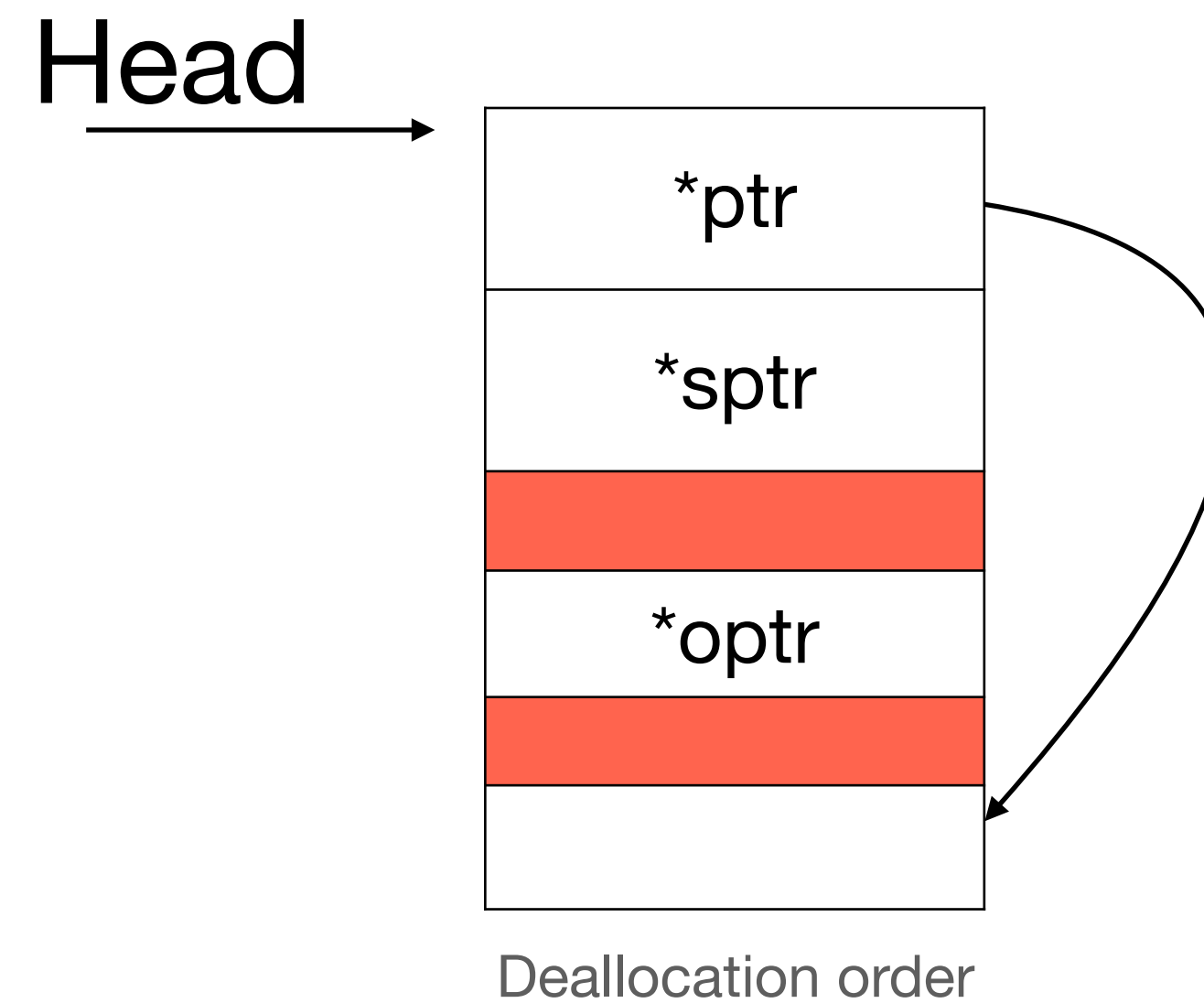


How to do coalescing?

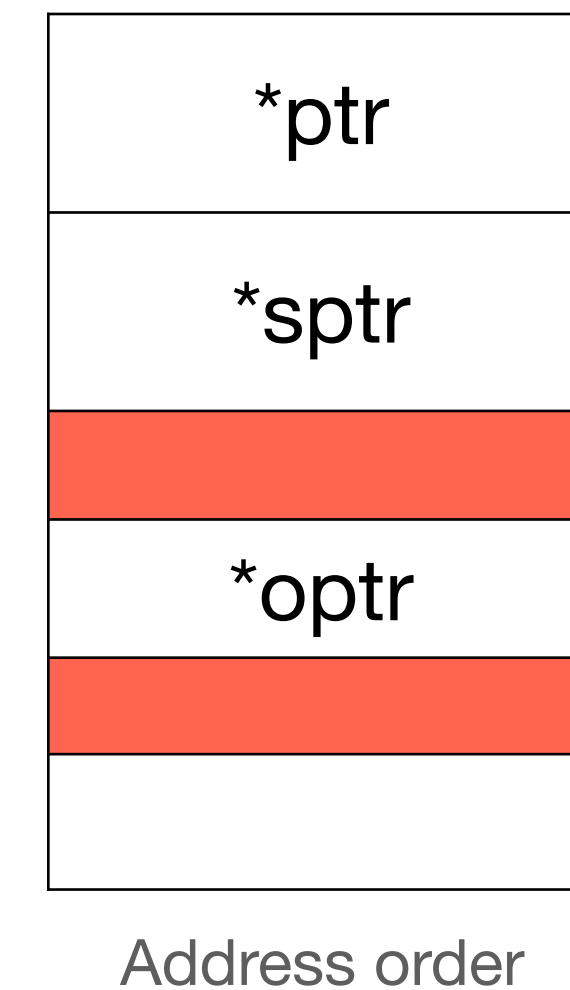
`free(ptr)`

`free(optr)`

`free(sptr)`



Head →

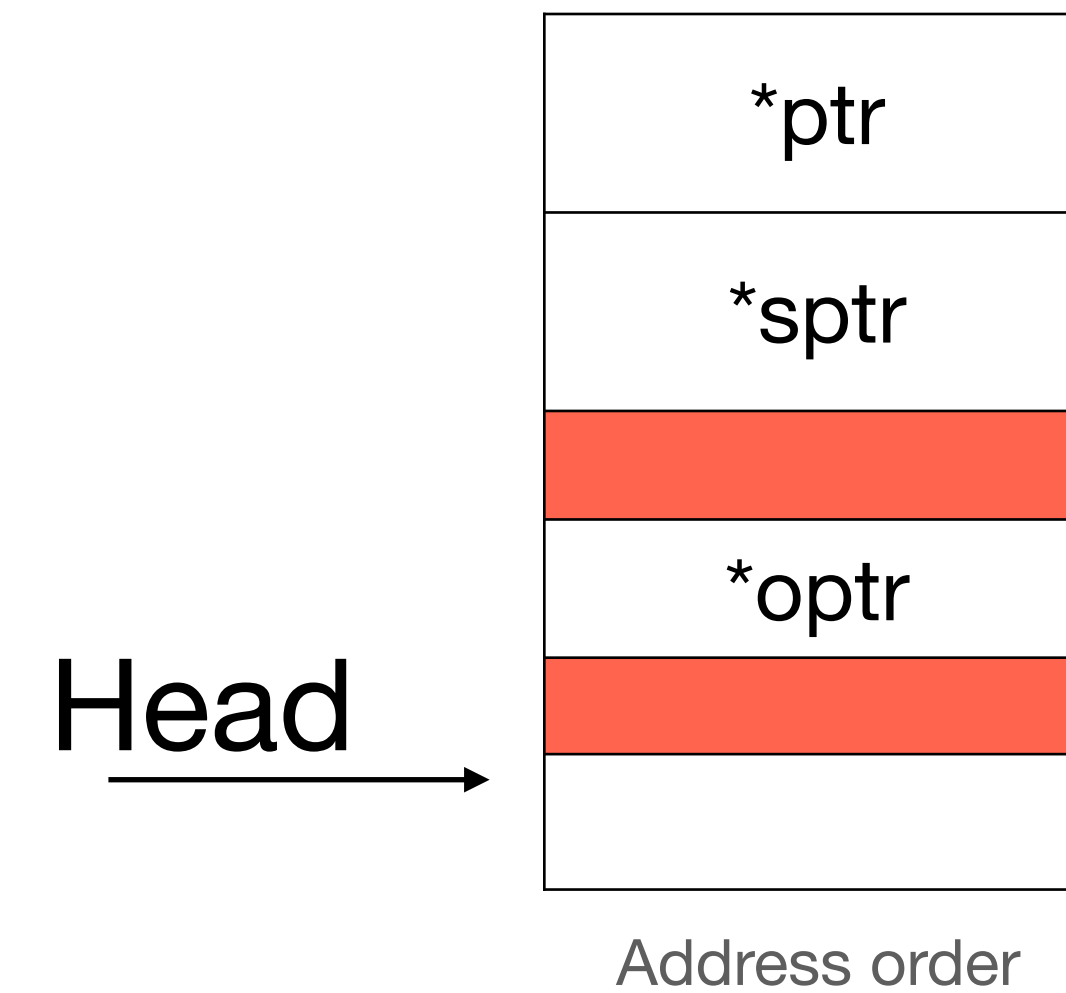
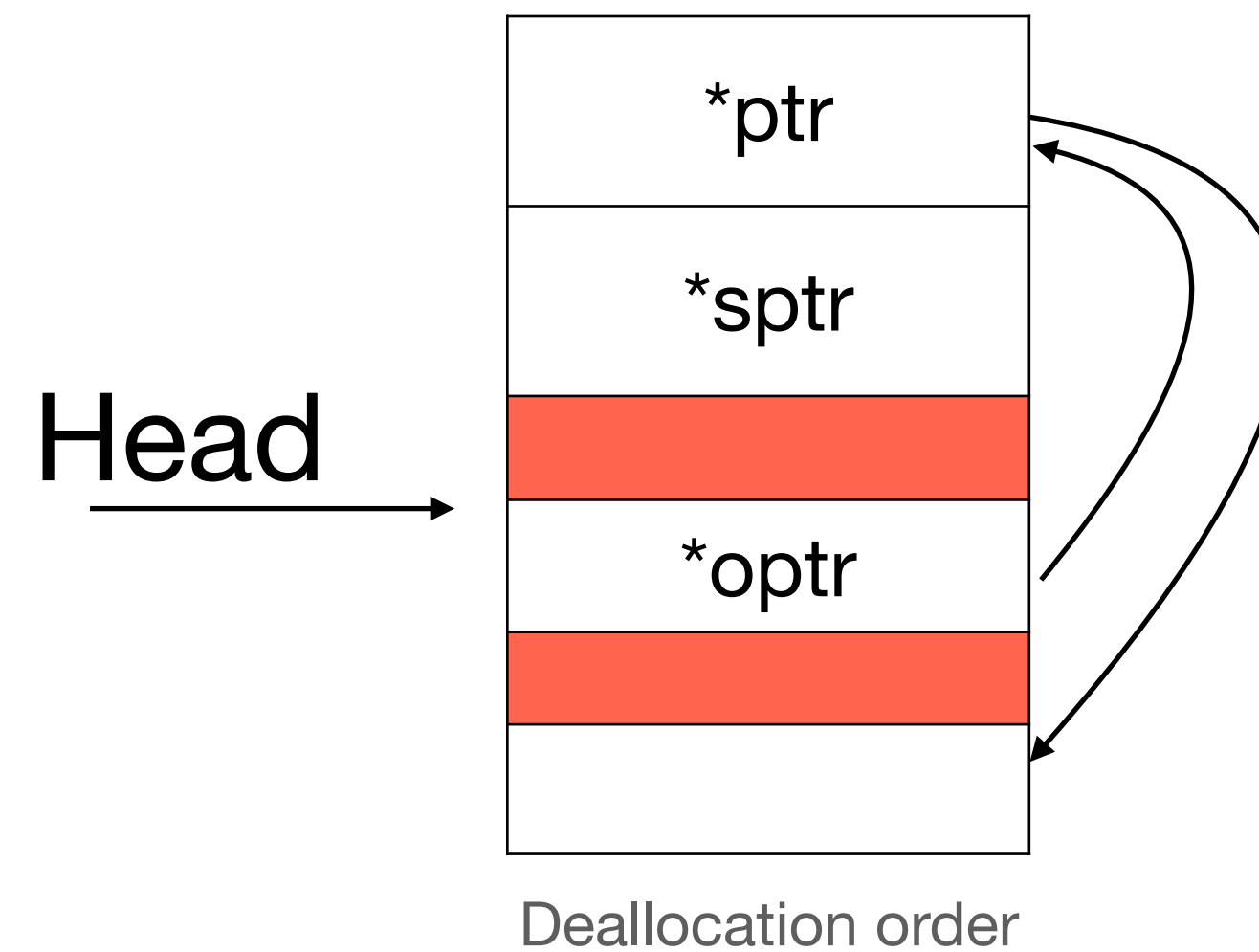


How to do coalescing?

`free(ptr)`

`free(optr)`

`free(sptr)`

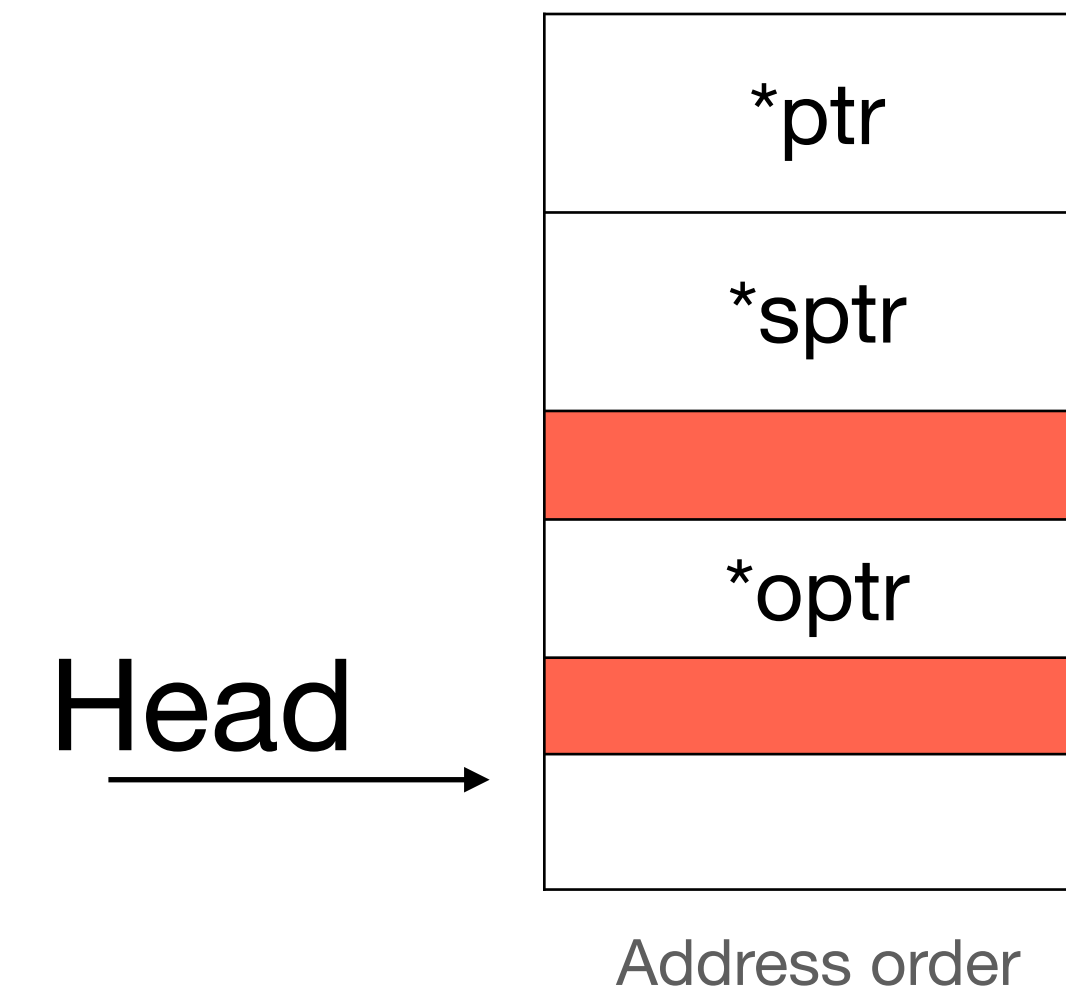
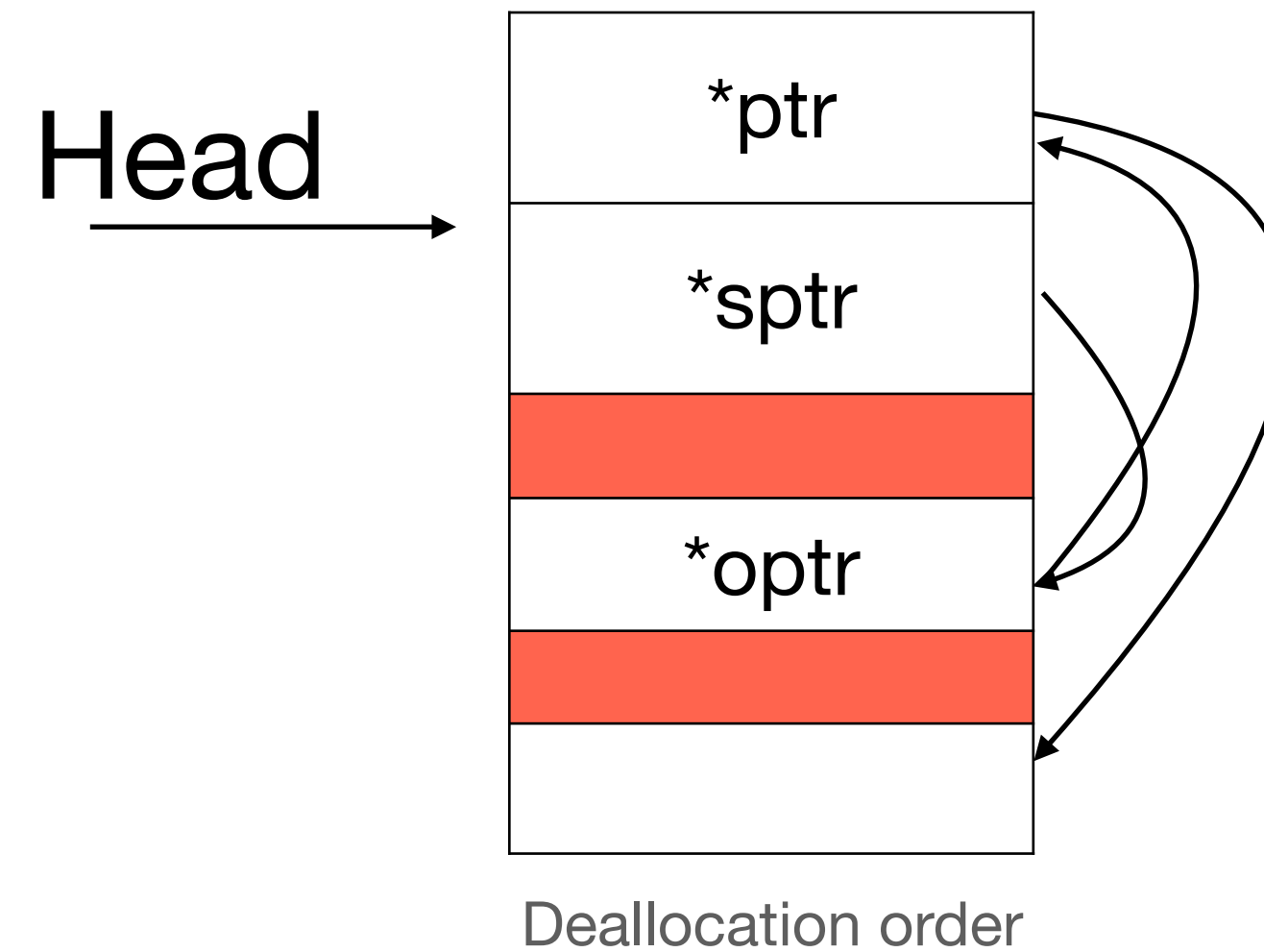


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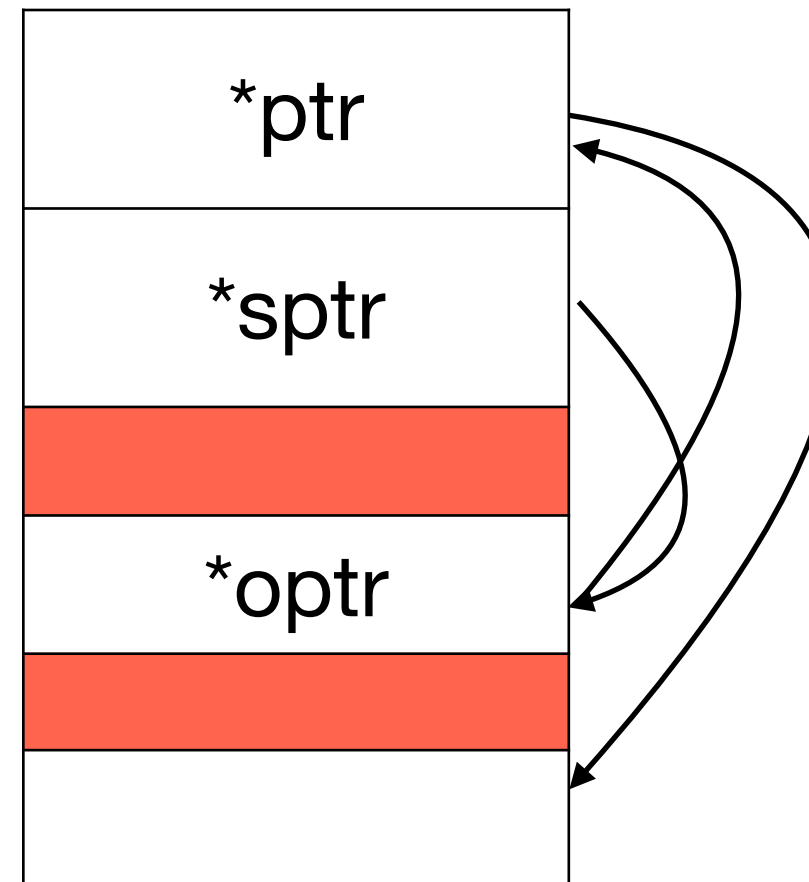
How to do coalescing?

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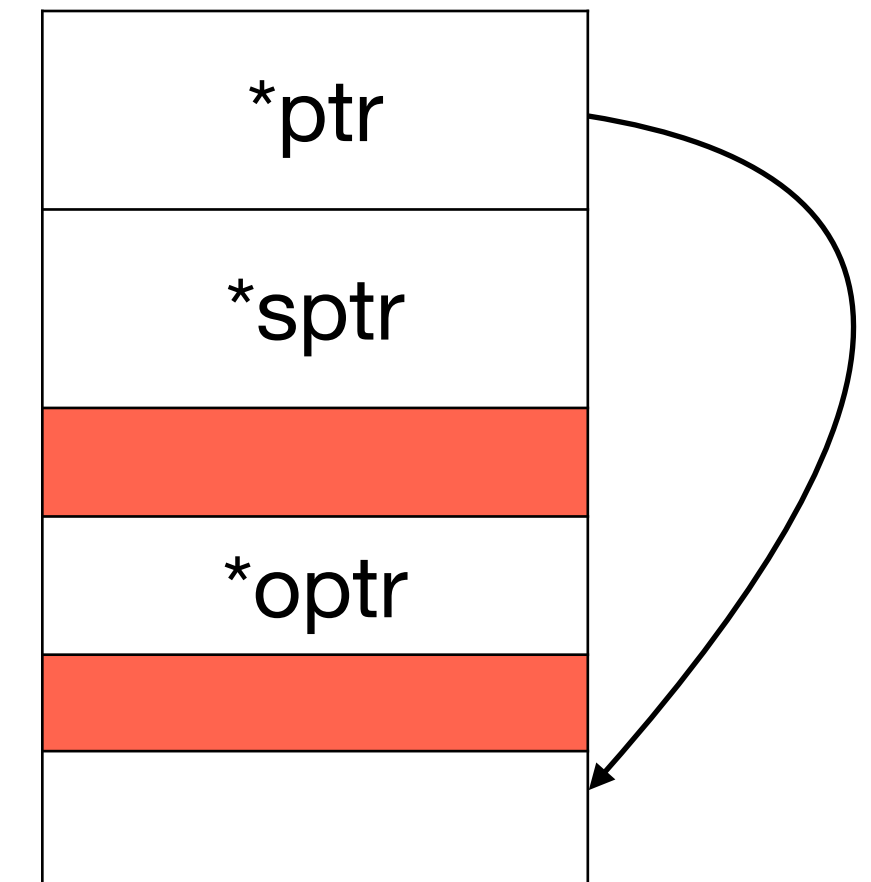
`free(sptr)`

Head



Deallocation order

Head



Address order

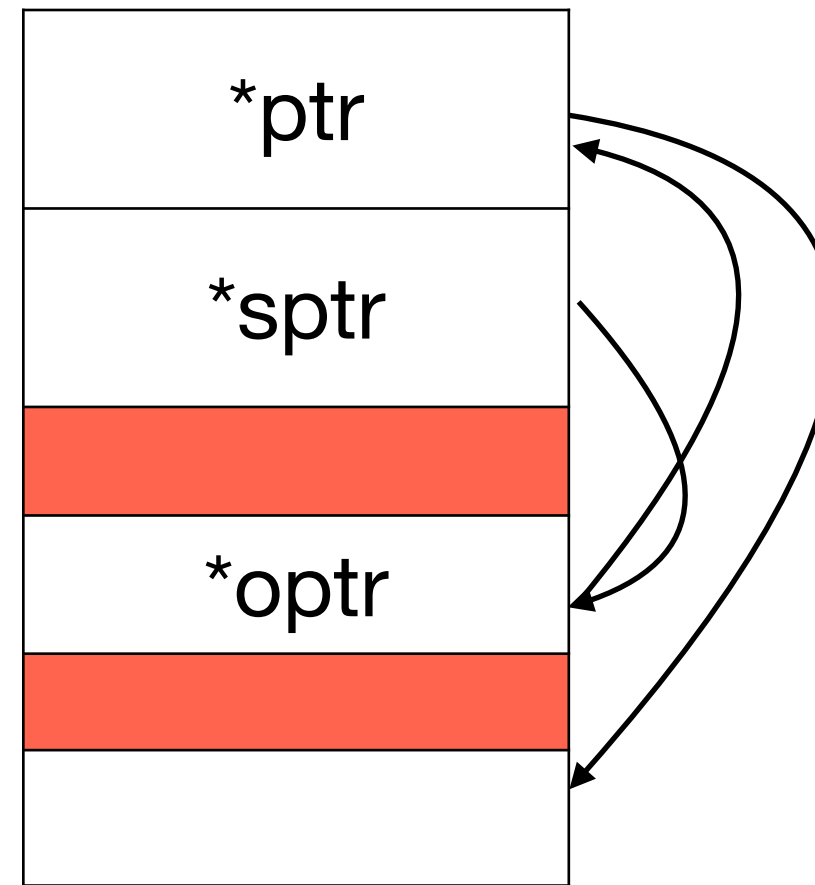
How to do coalescing?

`free(ptr)`

`free(optr)`

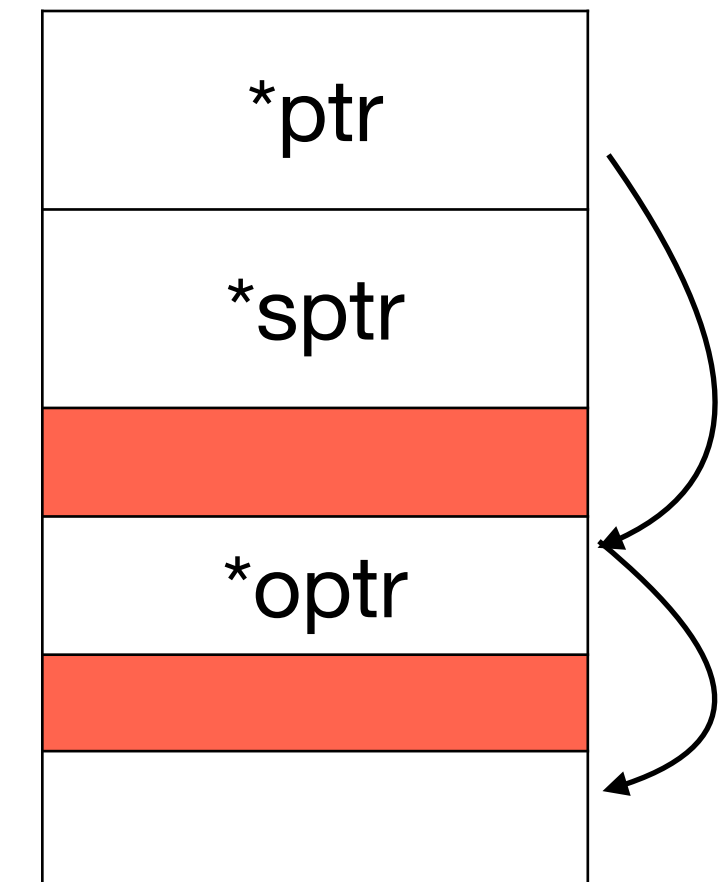
`free(sptr)`

Head



Deallocation order

Head

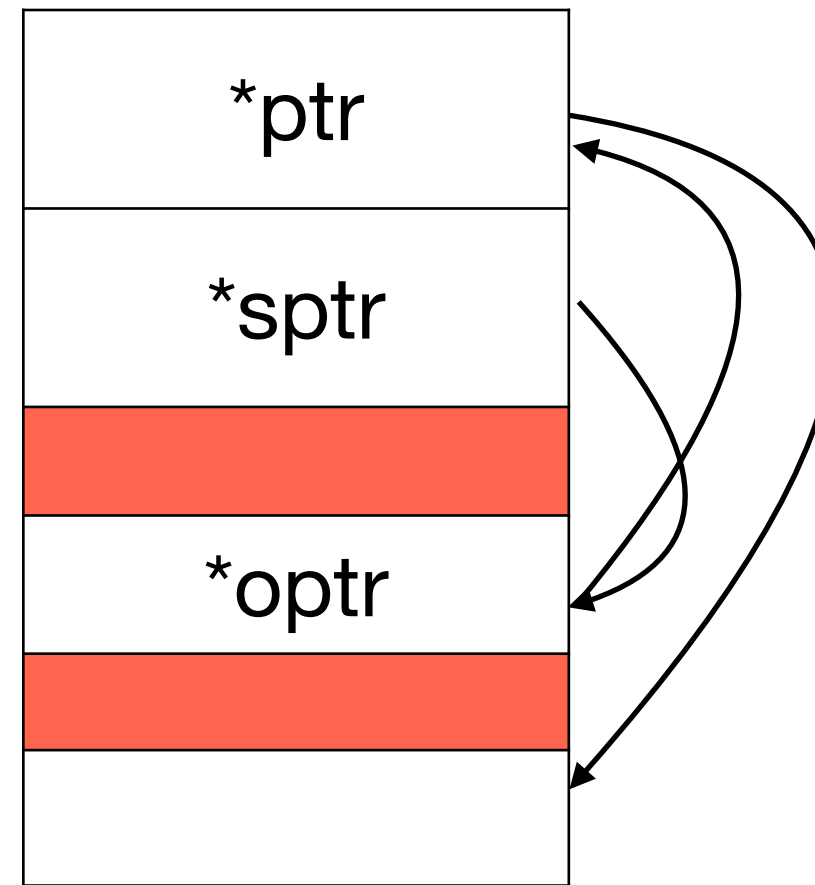


Address order

How to do coalescing?

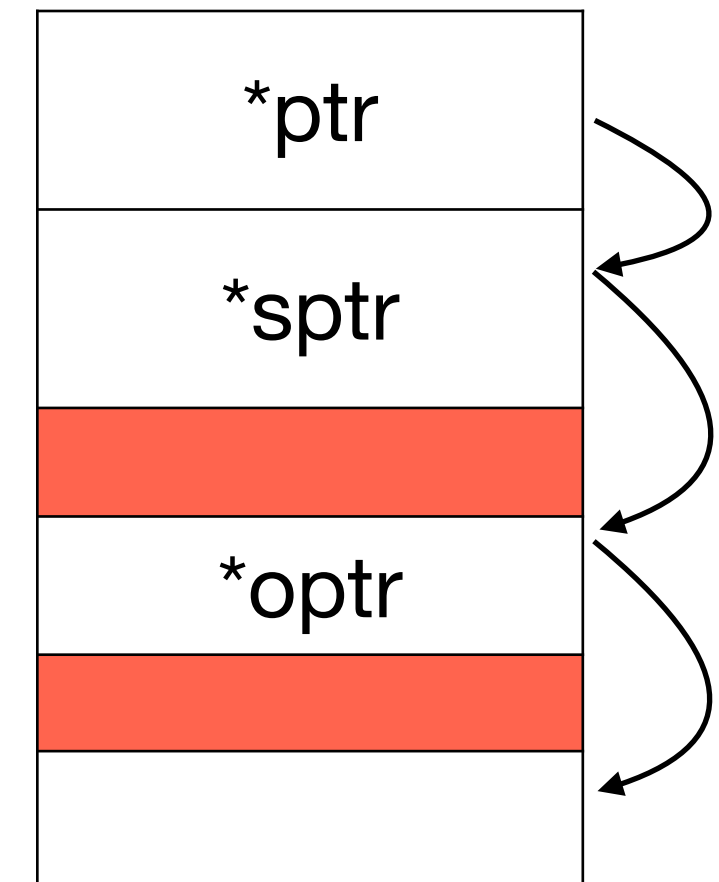
free(ptr)
free(optr)
free(sptr)

Head



Deallocation order

Head



Address order

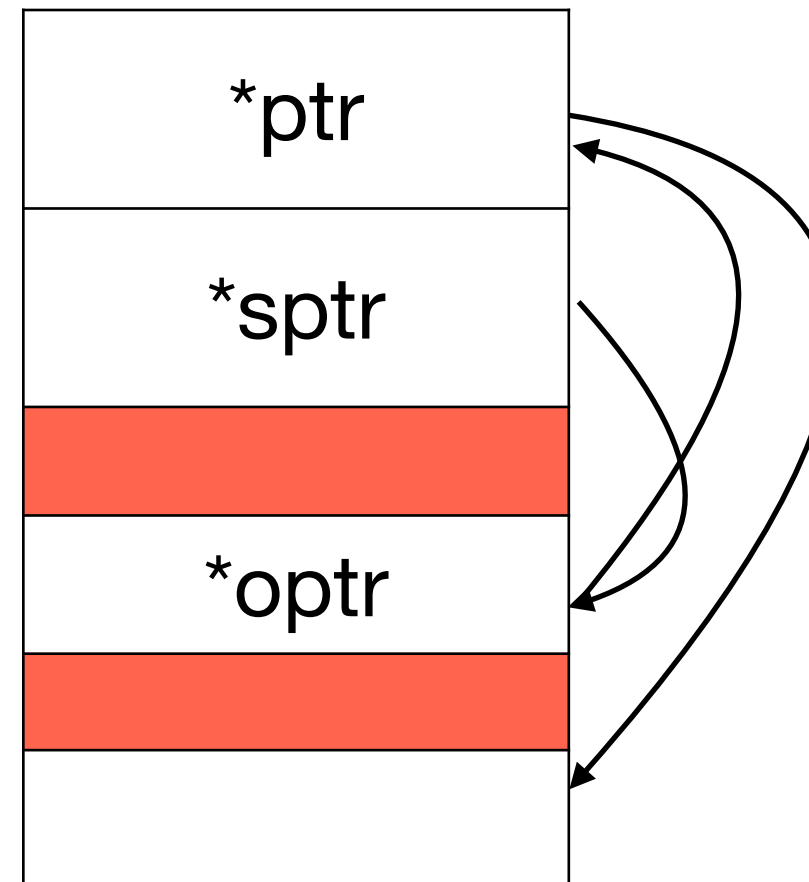
How to do coalescing?

`free(ptr)`

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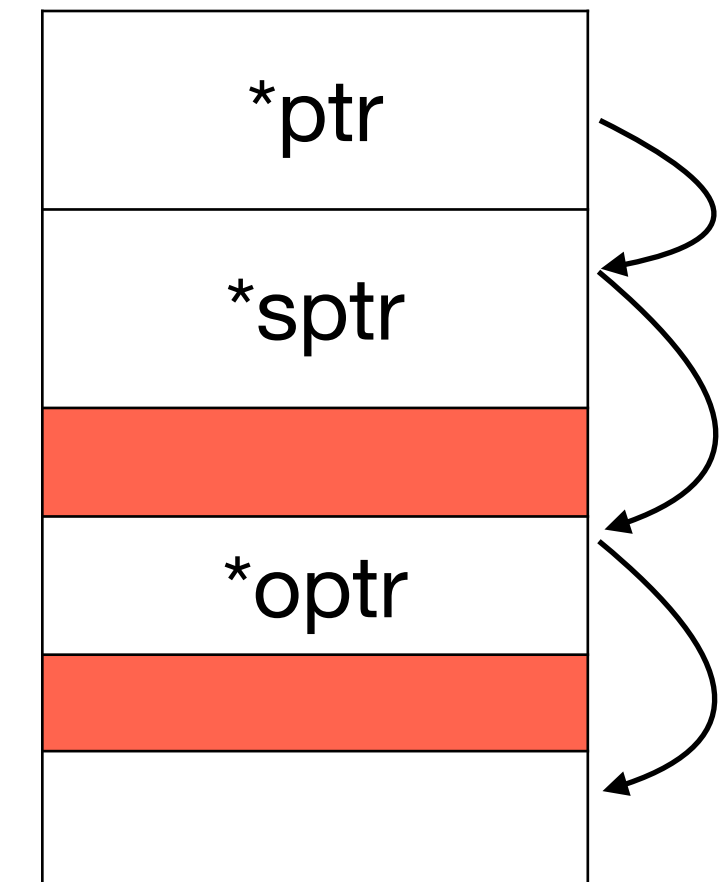
`free(sptr)`

Head →



Deallocation order

Head →



Address order

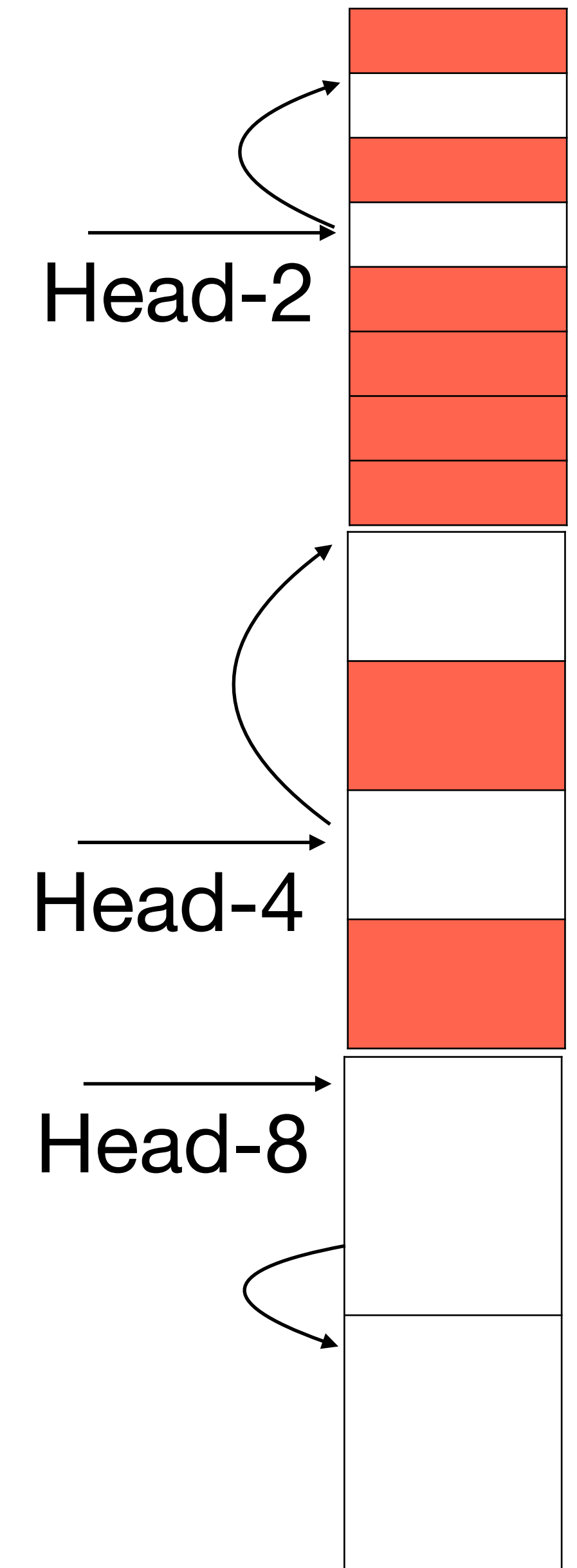
Easier to coalesce in an address-order free-list

Tradeoffs

- Allocation
 - First fit: faster search, leads to fragmentation
 - Best fit: slower search, less fragmentation
- Free list order
 - Address order: Slow frees, easier to coalesce
 - Deallocation order: Fast frees, difficult to coalesce

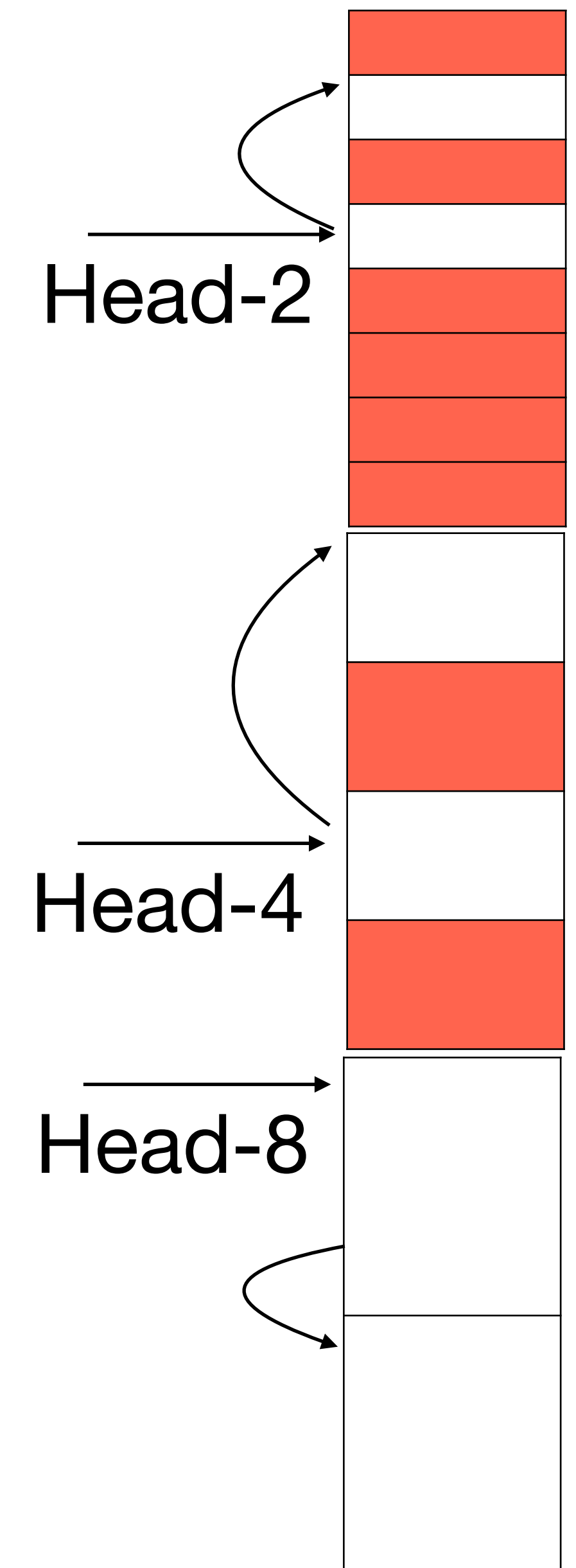
Segregated lists

- Separate lists for each size.



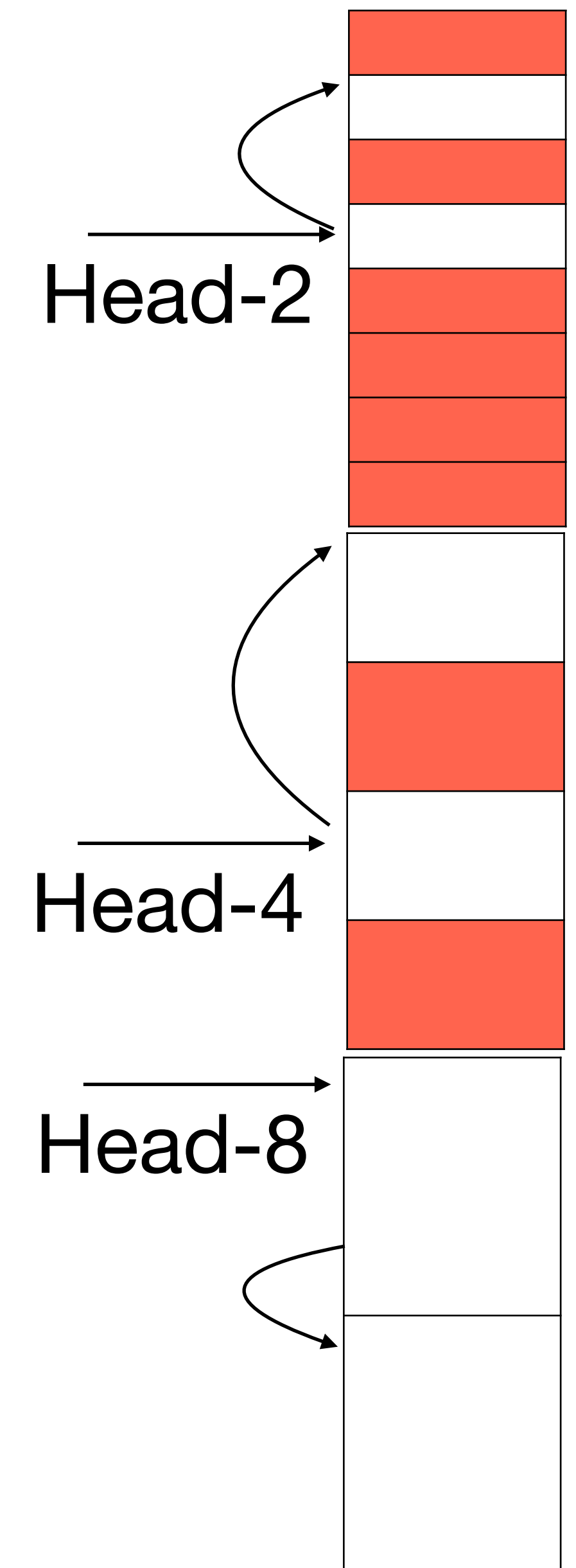
Segregated lists

- Separate lists for each size.
- “Segregated fit”: *First fit* in the smallest object list that can fit the object. Approximates *best fit*.



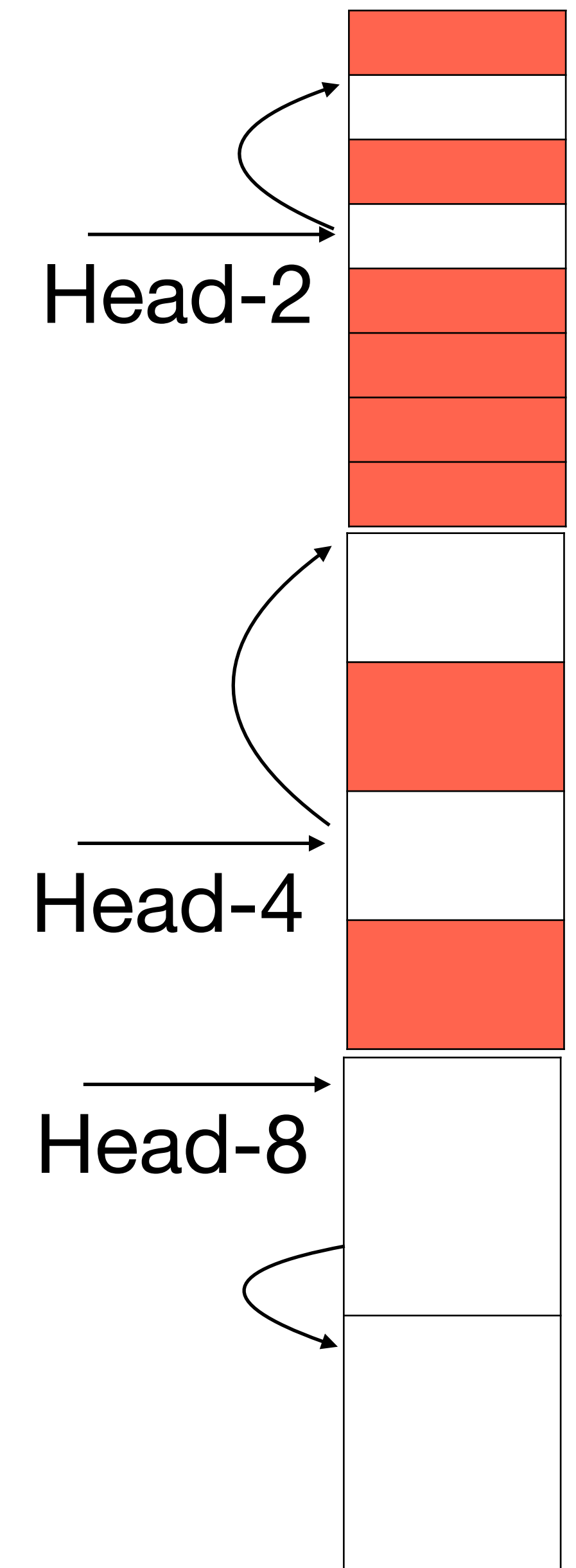
Segregated lists

- Separate lists for each size.
- “Segregated fit”: *First fit* in the smallest object list that can fit the object. Approximates *best fit*.
 - No splitting, coalescing



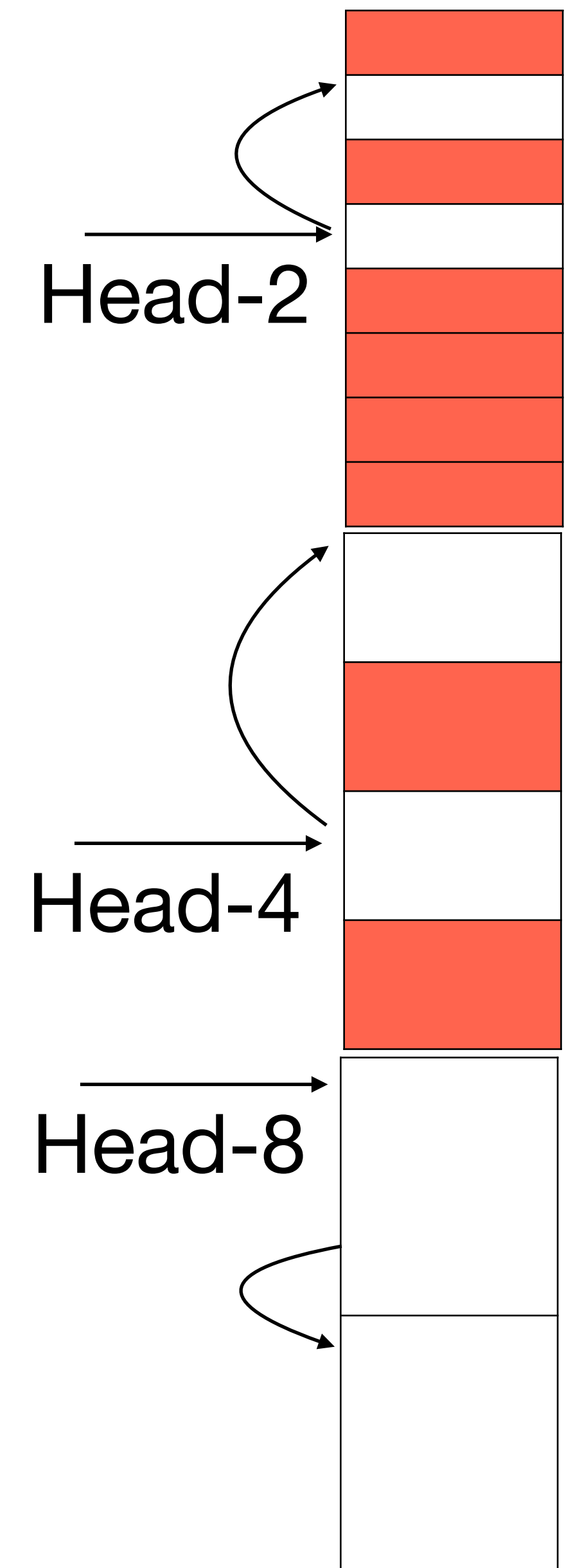
Segregated lists

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- Internal fragmentation: allocates more than asked



Segregated lists

- Separate lists for each size.
- “Segregated fit”: *First fit* in the smallest object list that can fit the object. Approximates *best fit*.
 - No splitting, coalescing
- Internal fragmentation: allocates more than asked
- Wastes memory: Reserving space in a segregated list is wasteful if allocations are not requested for that object size



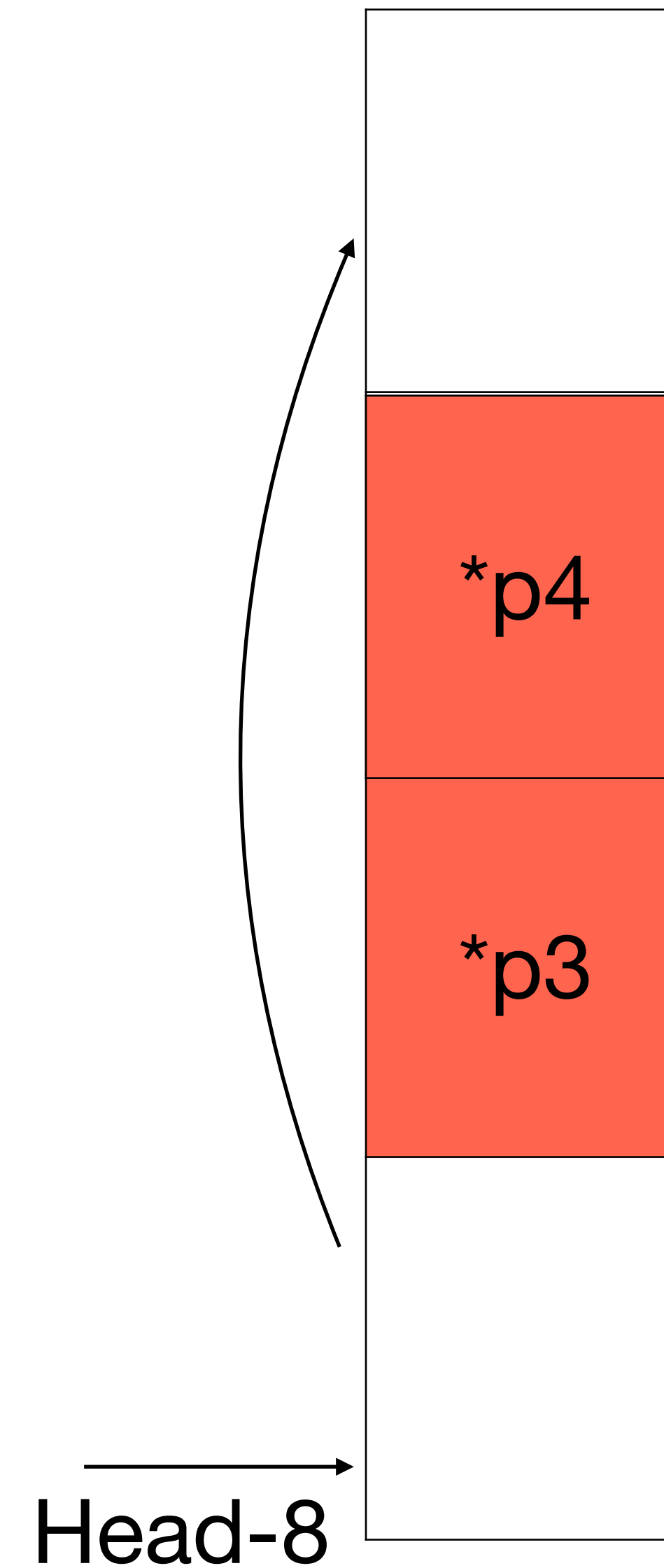
Slab allocator

Used in Linux Kernel

- Knows the size of allocations (process control blocks, buffer cache block, etc.) made by the Linux kernel. Keep segregated lists for each such struct.
 - No internal fragmentation: Exactly the size of the struct
 - Hierarchical allocator: Return unused lists back to global allocator

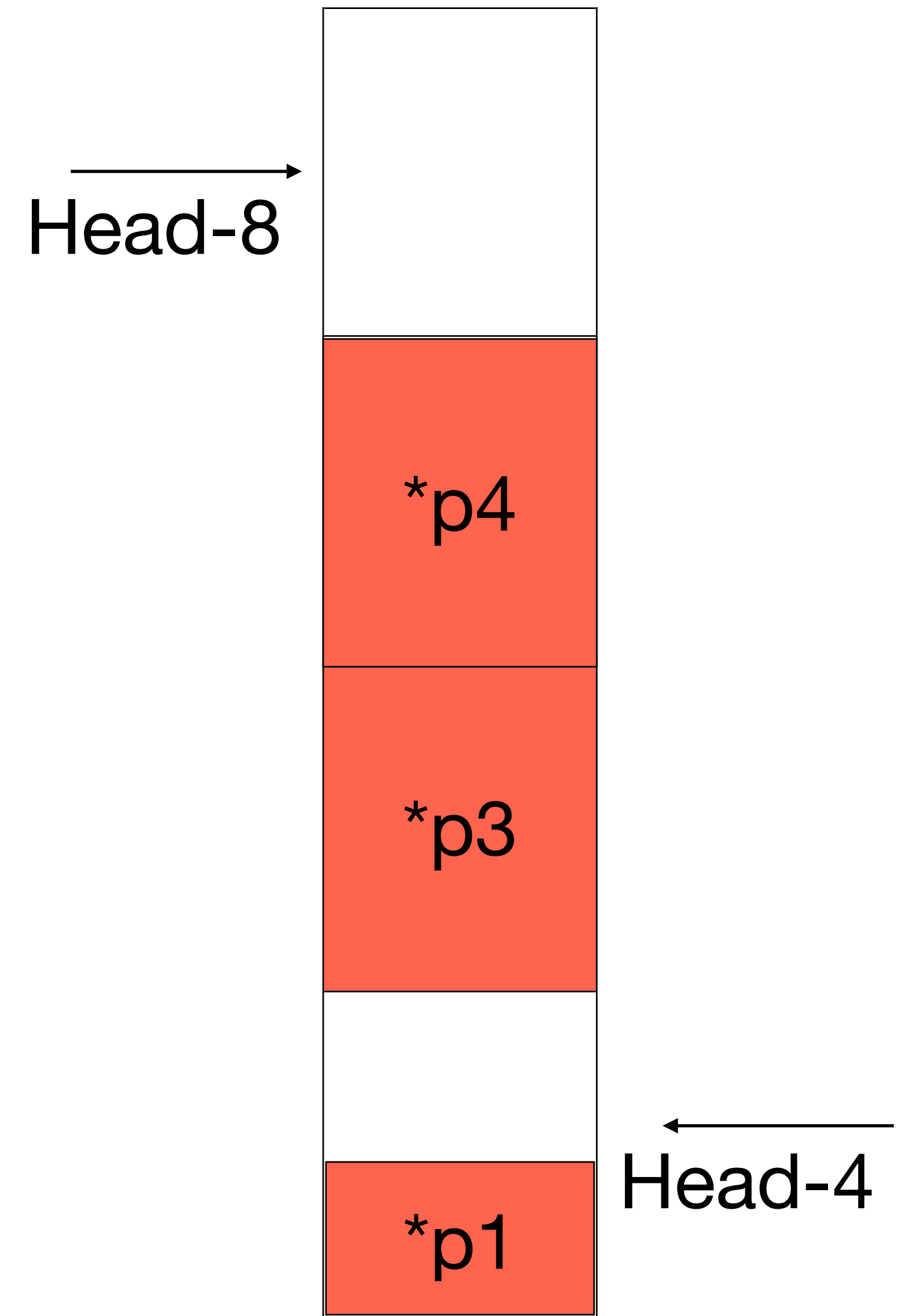
Buddy allocator

- Example:
 - `p1 = malloc(3)`
 - `p2 = malloc(2)`



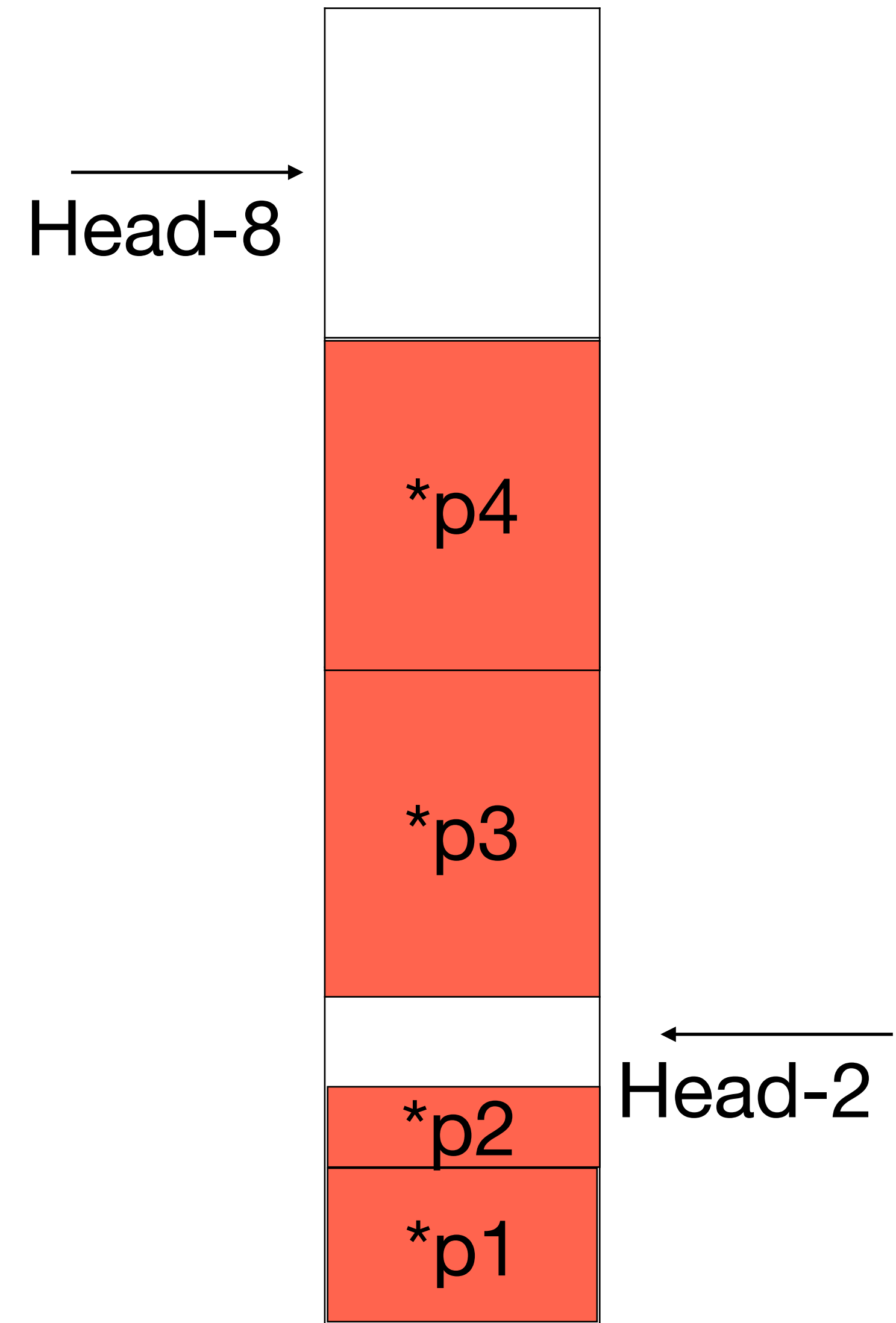
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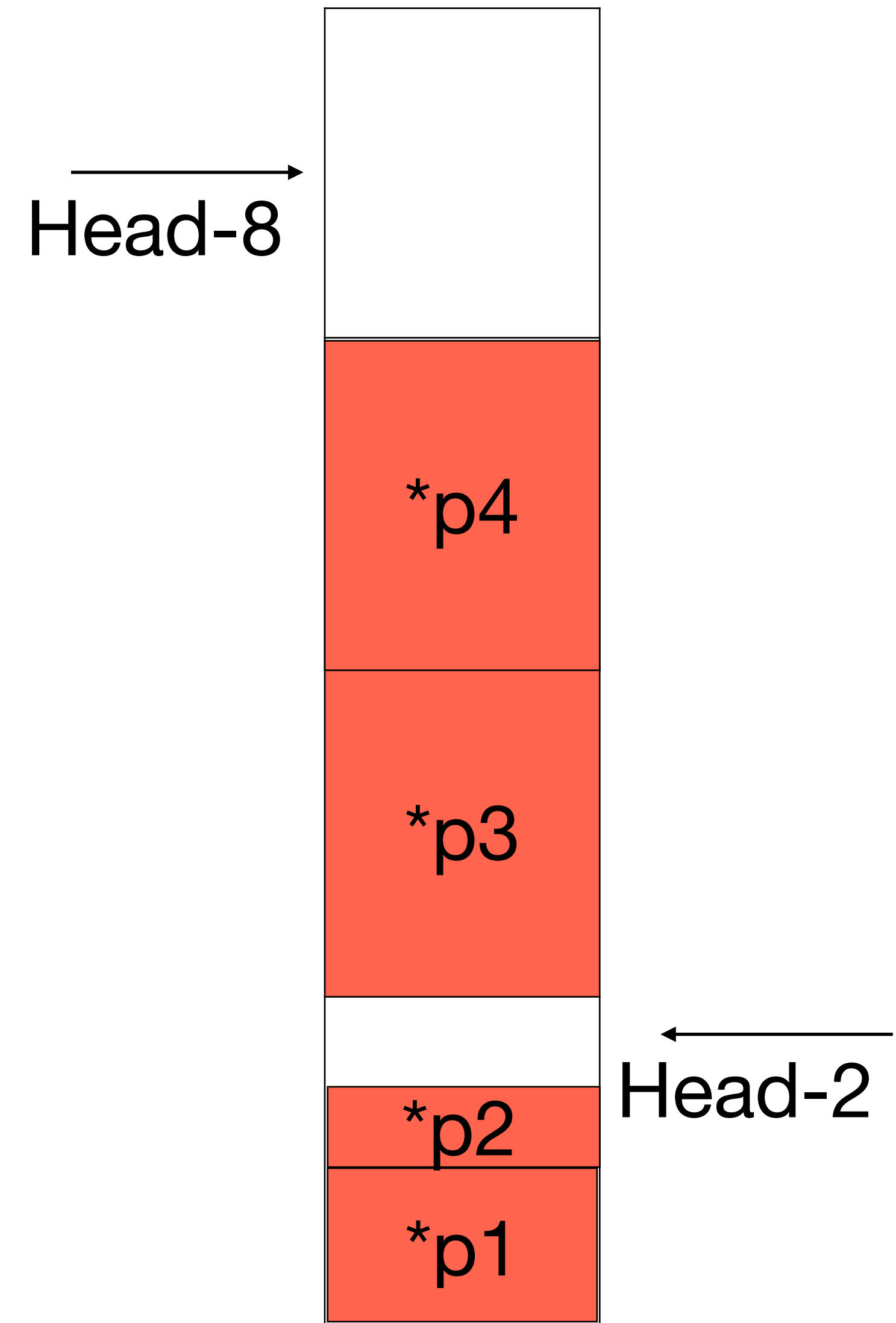
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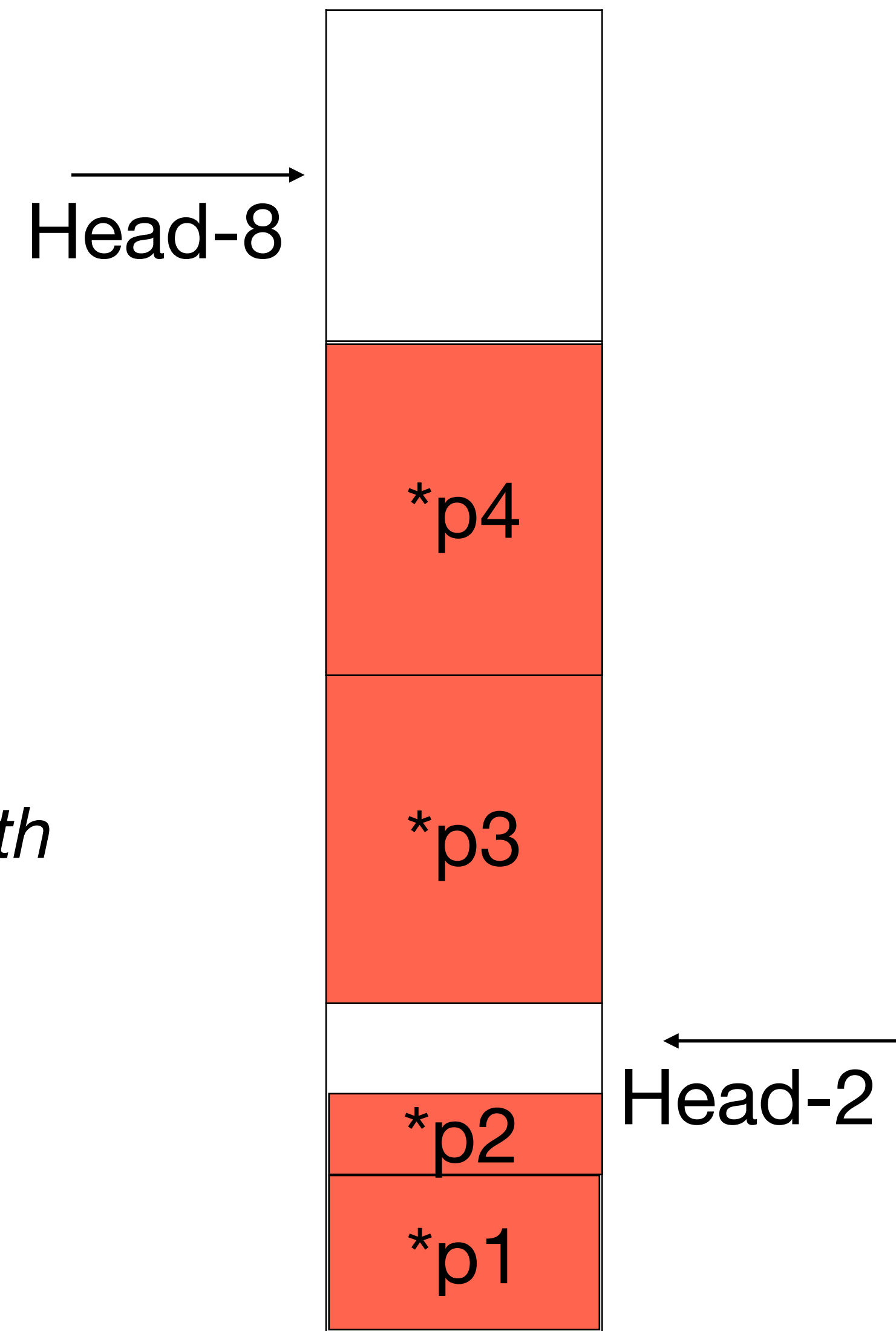
Buddy allocator

- Example:
 - $p1 = \text{malloc}(3)$
 - $p2 = \text{malloc}(2)$
- On-the-fly segregated lists



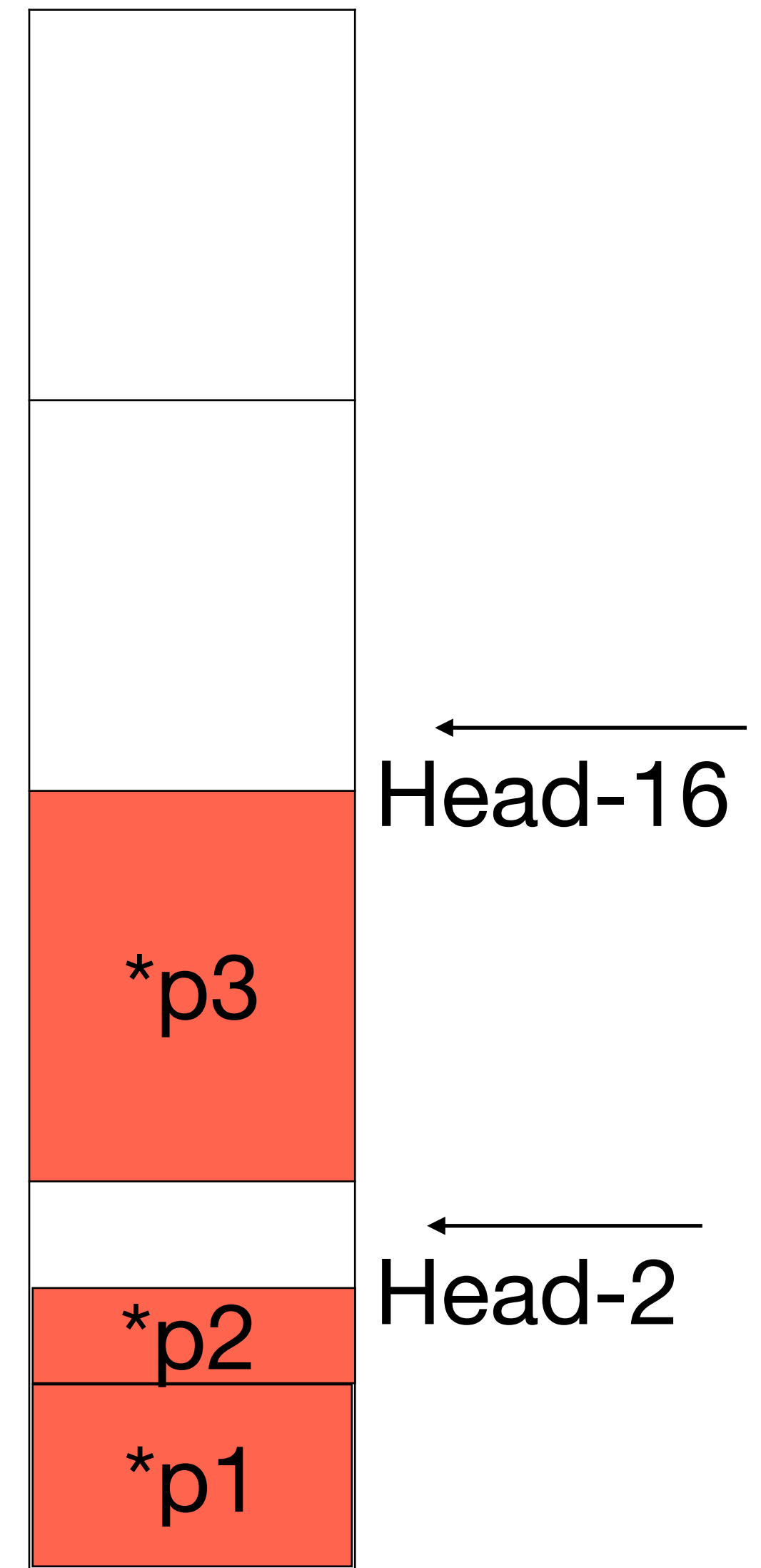
Buddy allocator

- Example:
 - `p1 = malloc(3)`
 - `p2 = malloc(2)`
- On-the-fly segregated lists
- Straightforward splitting and *coalescing* (*coalesce with my buddy*)
 - `free(p4)`



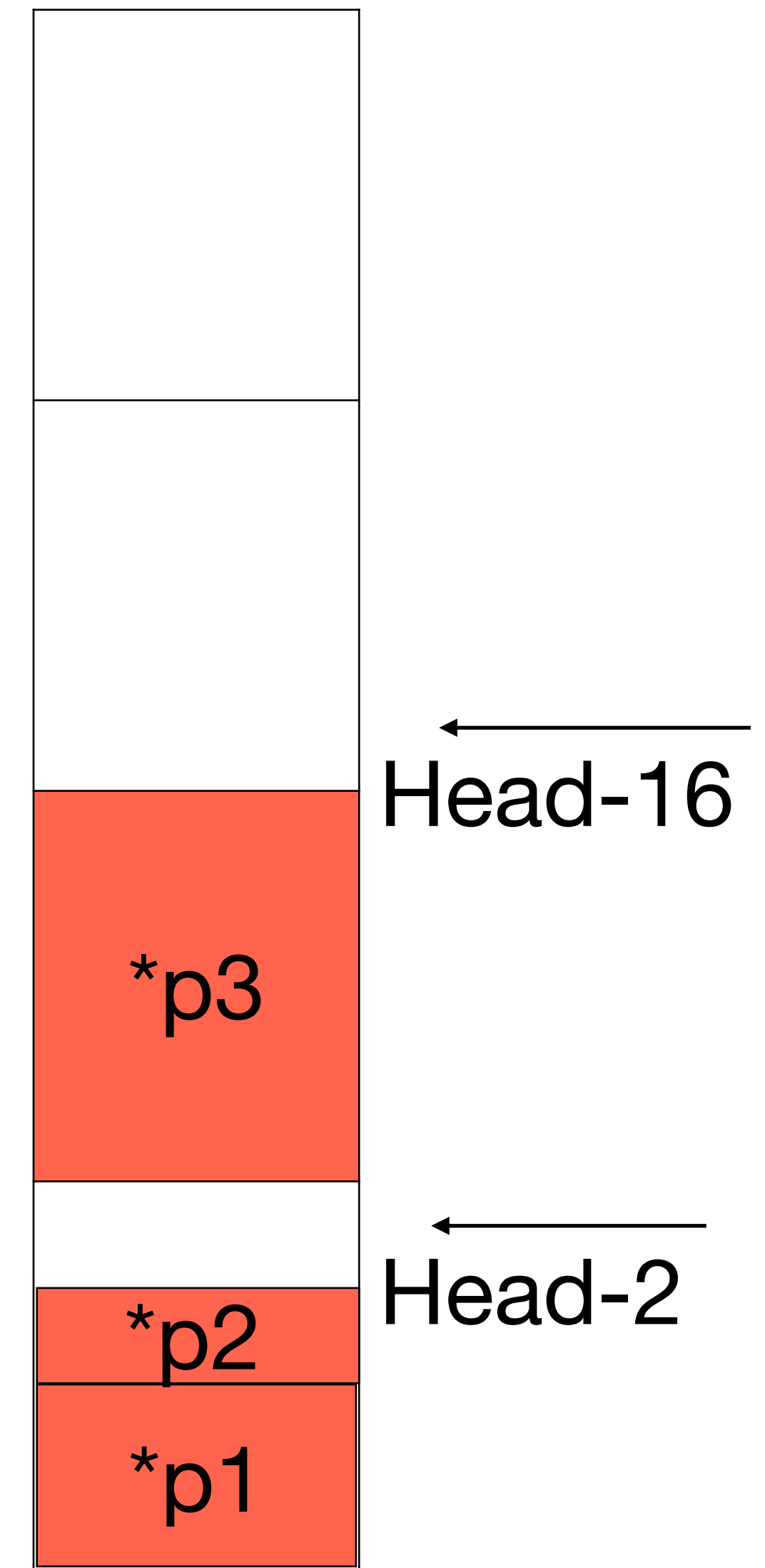
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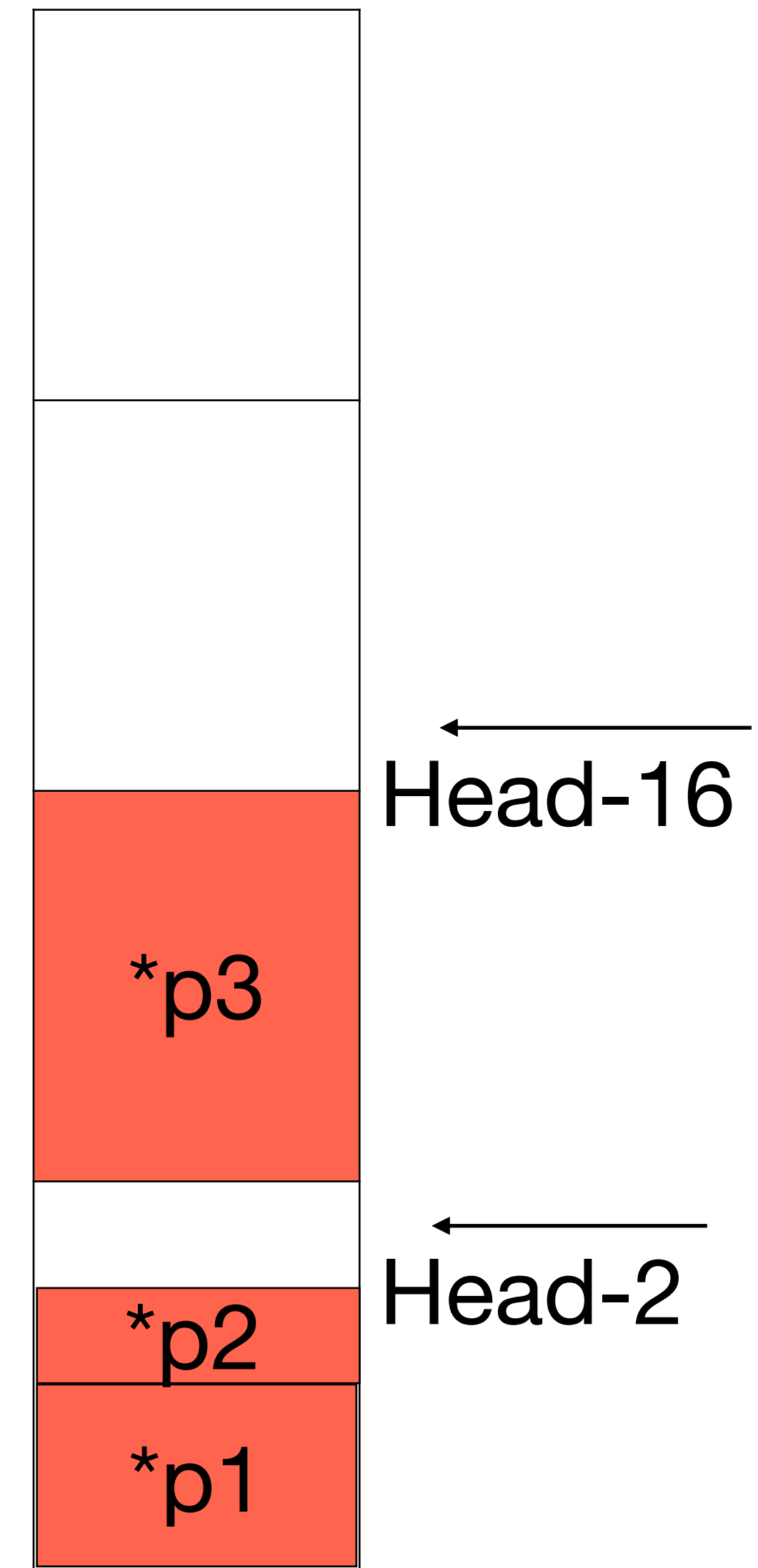
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- *Deallocation order*: fast frees



Buddy allocator

- Example:
 - `p1 = malloc(3)`
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- On-the-fly segregated lists
- Straightforward splitting and *coalescing* (*coalesce with my buddy*)
 - `free(p4)`
- *Deallocation order*: fast frees
- Used in Linux kernel



Aside: automatic memory management

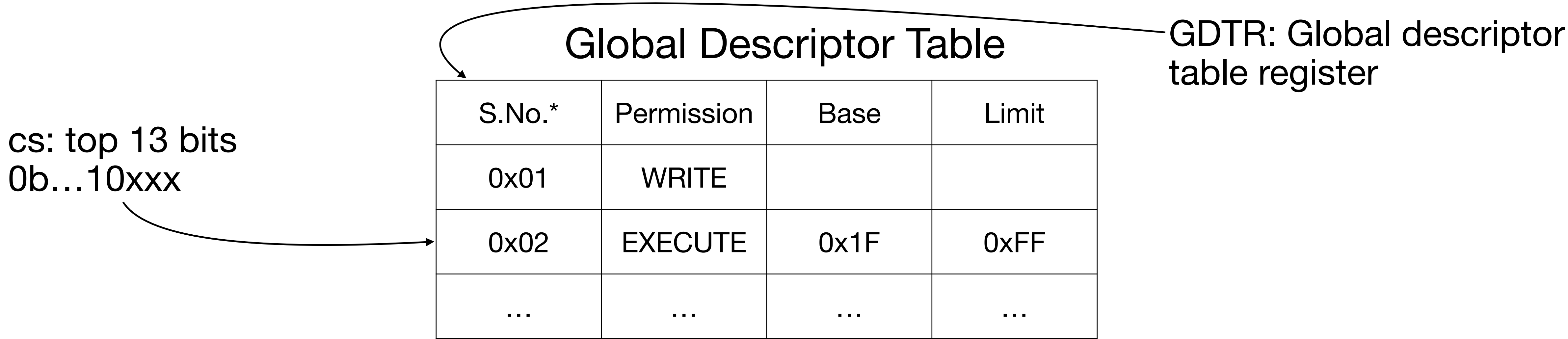
- “Higher level” languages do not expose raw pointers to programmers.
Example: Java, Python, Go
- The language runtime manages memory. Programmer need not call free.
- Largely prevents memory leaks, dangling references, double free, null pointer dereference
- Mark and sweep garbage collector, reference counting based garbage collector
- Copying GC can do compaction to defragment heap. Will rewrite pointers.
- Can incur heavy performance penalty

Memory isolation

OSTEP Ch. 13,15,16

Abhilash Jindal

Many segments can be initialised in GDT



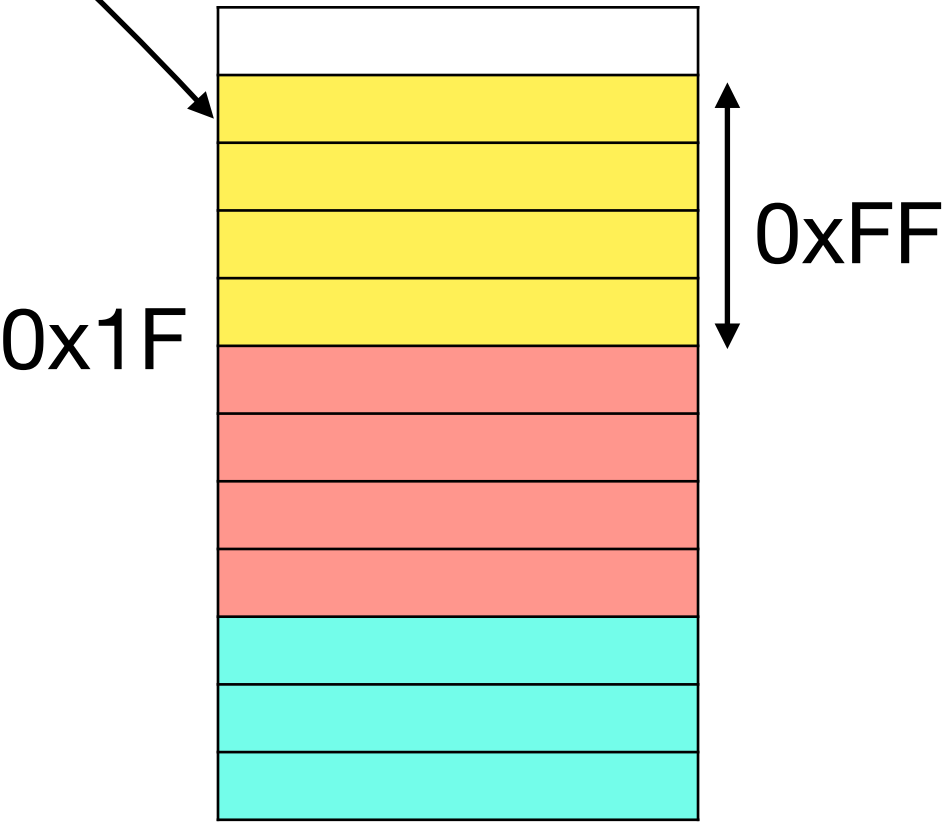
*: S.No. added only for illustration

Address translation

Global Descriptor Table

cs: top 13 bits
0b...10xxx

S.No.*	Permission	Base	Limit
0x01	WRITE		
0x02	EXECUTE	0x1F	0xFF
...	...	...	...

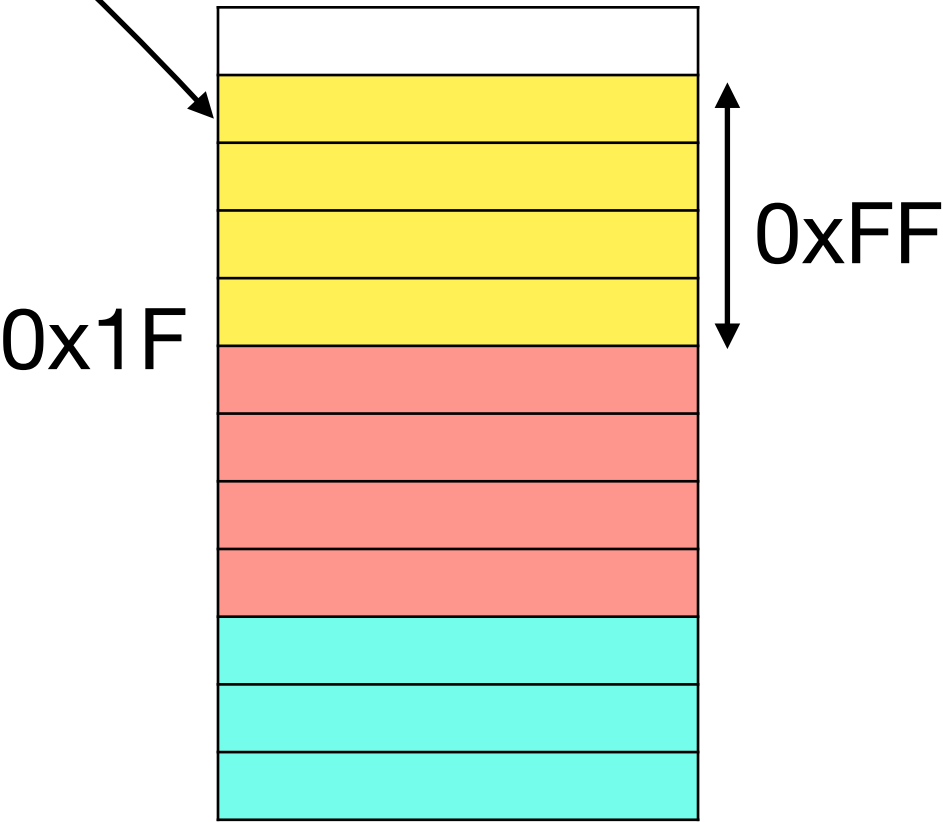
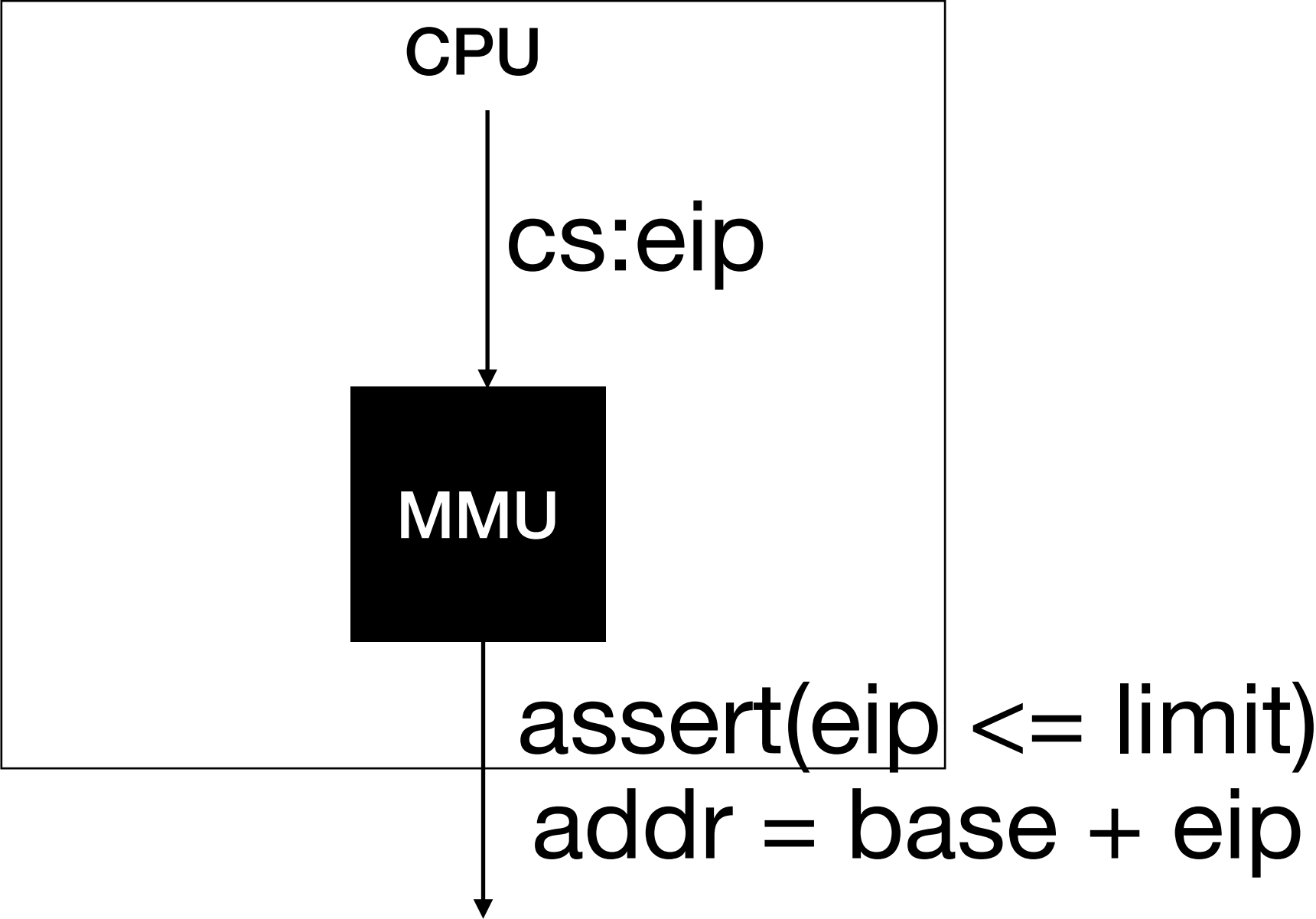


Address translation

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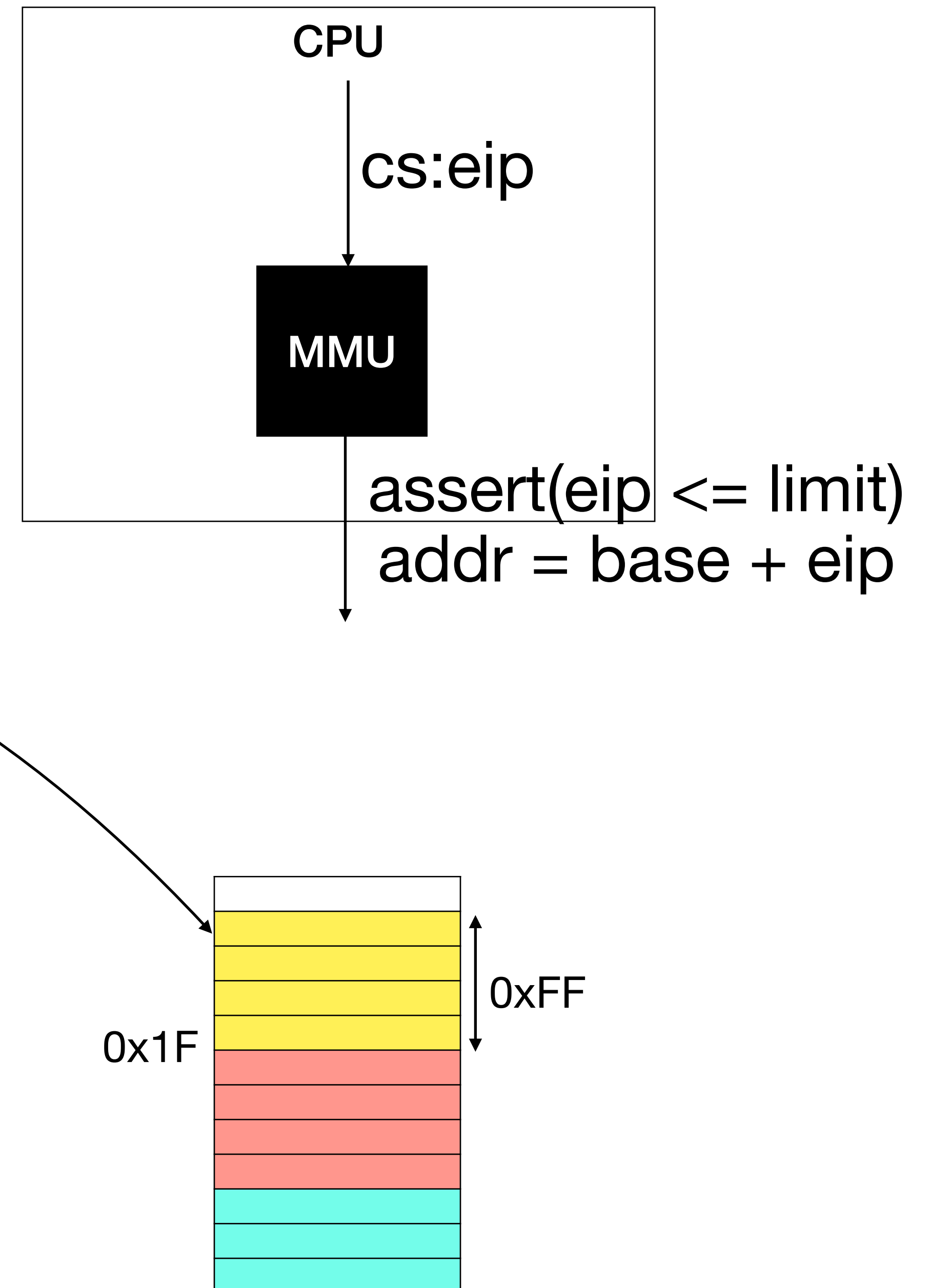
Address translation

Global Descriptor Table

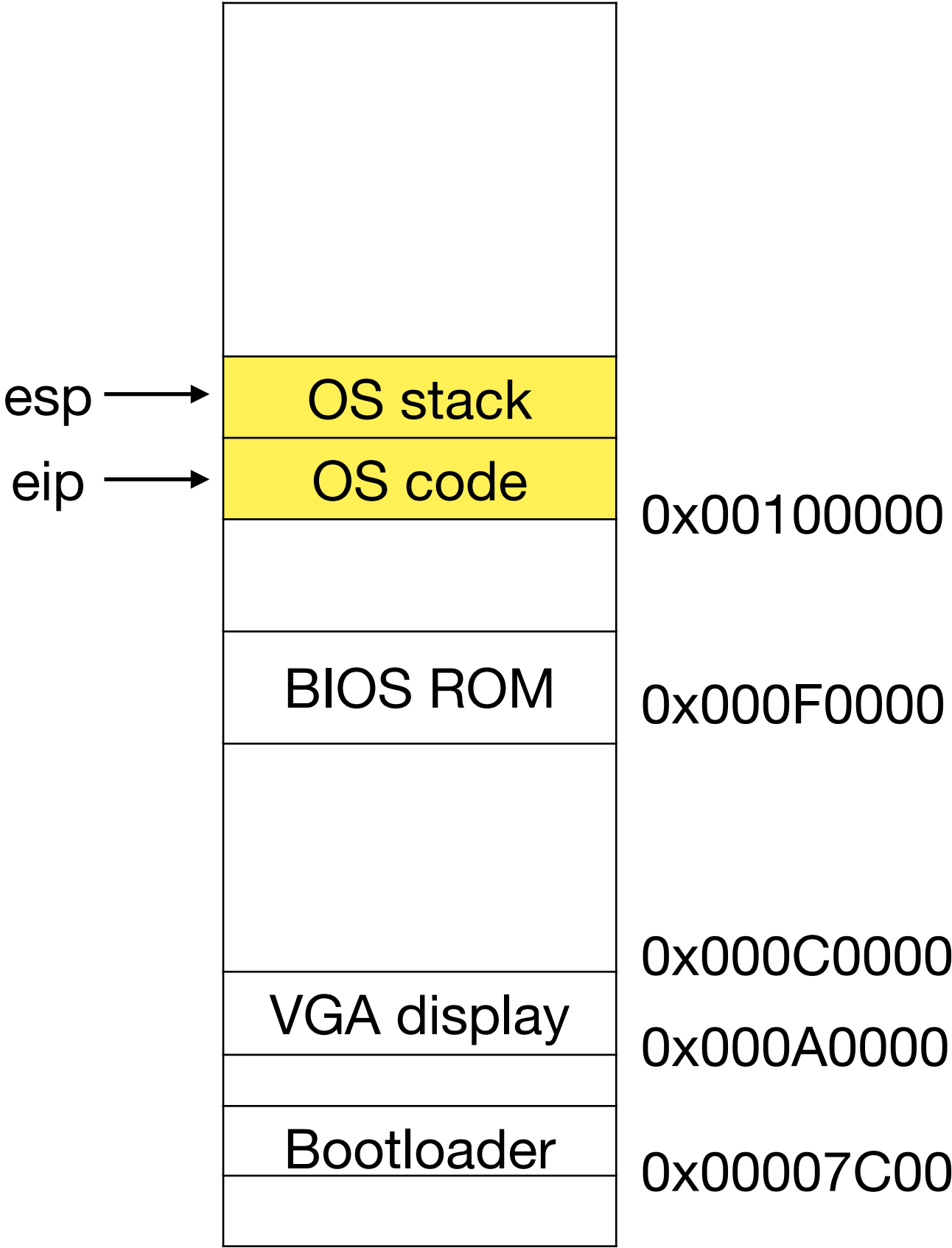
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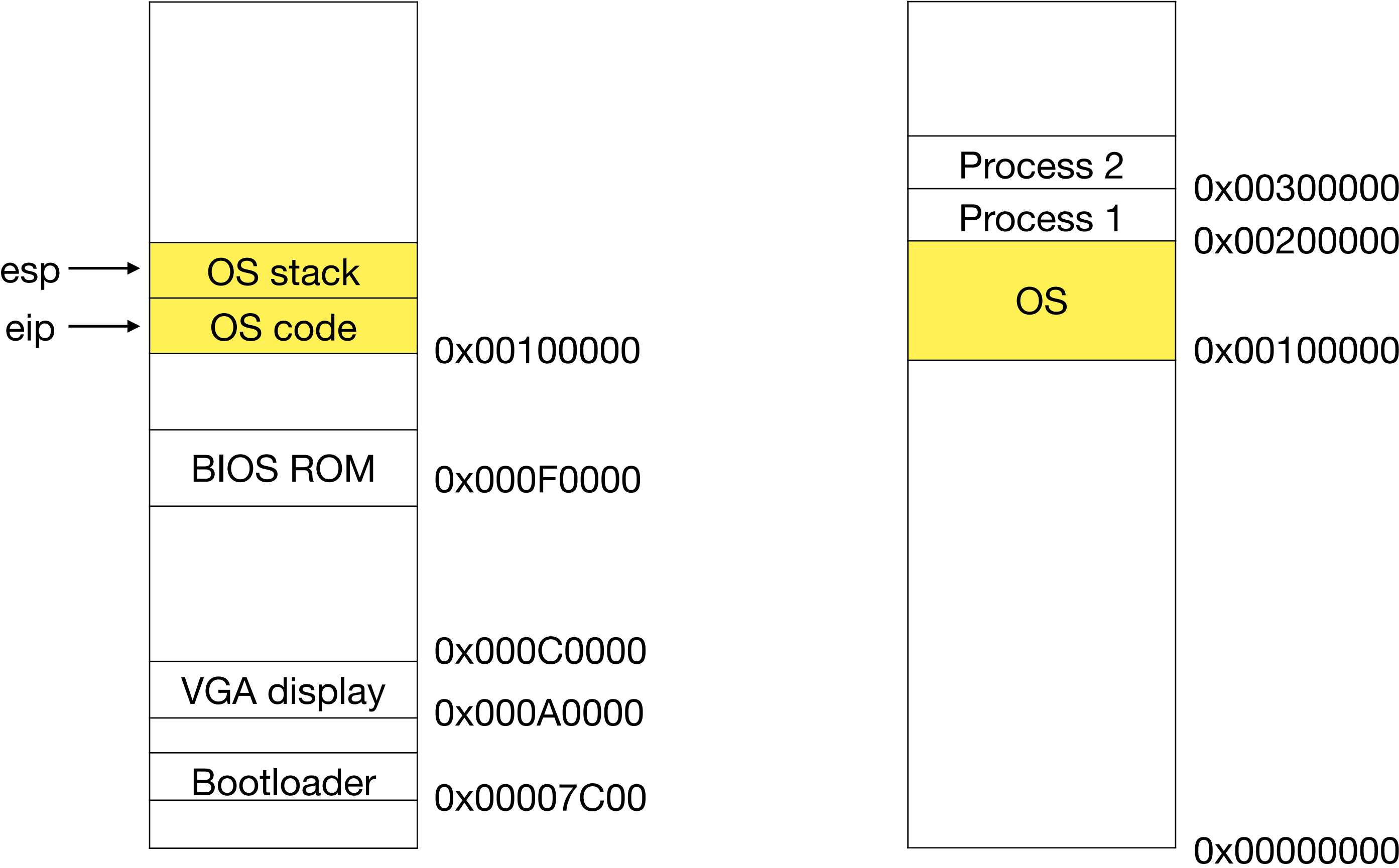
- Can “isolate” different segments from each other



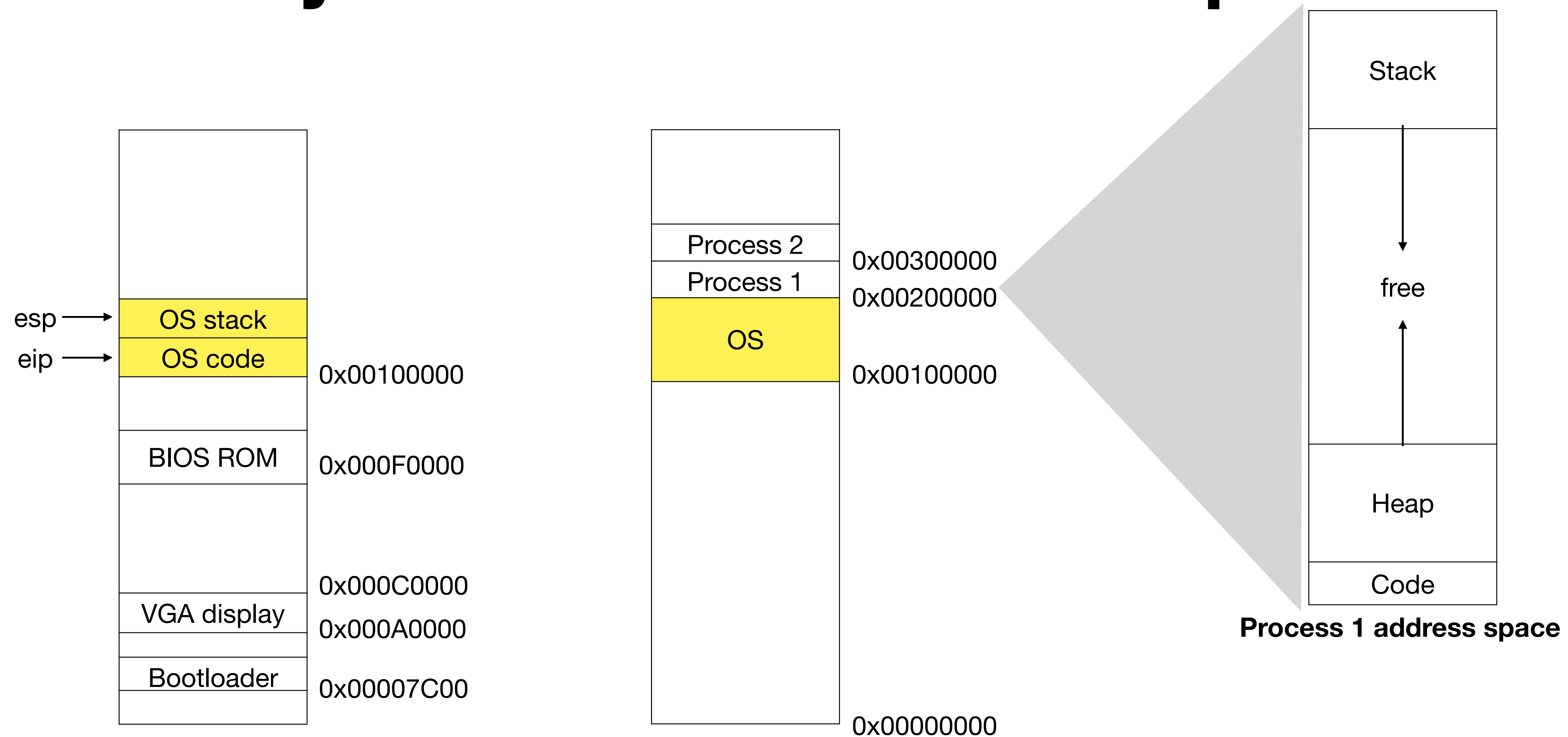
Memory isolation and address space



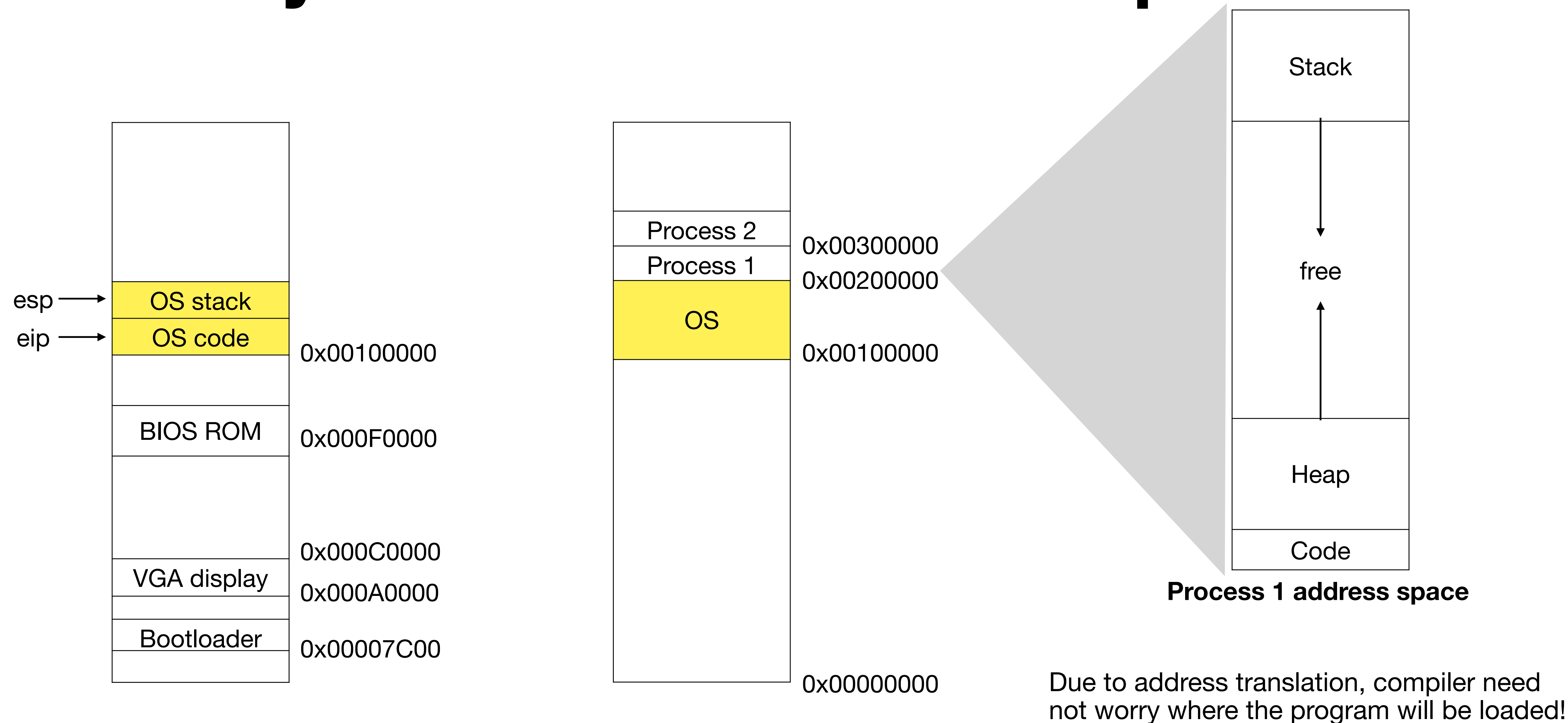
Memory isolation and address space



Memory isolation and address space

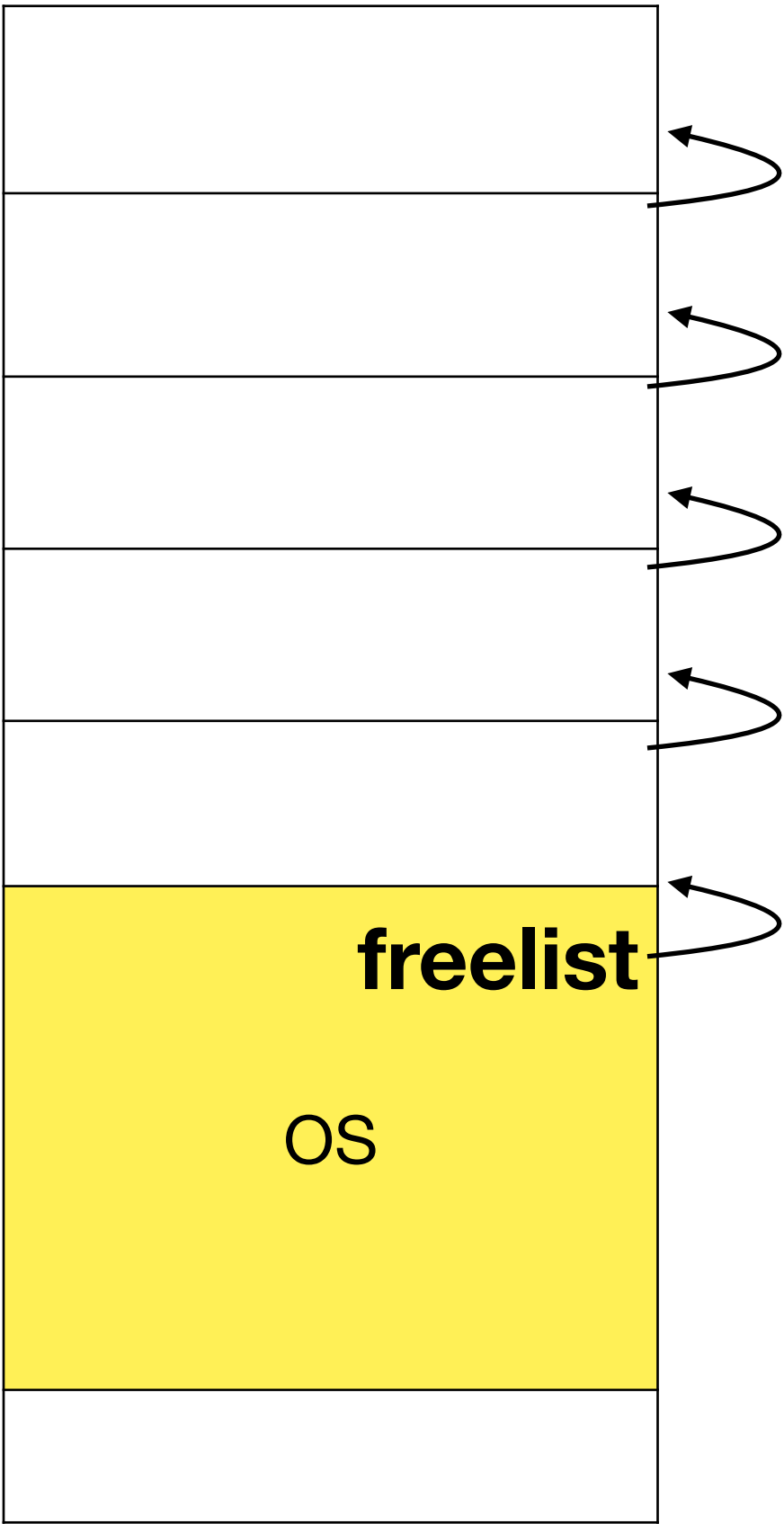


Memory isolation and address space



Allocating memory to processes

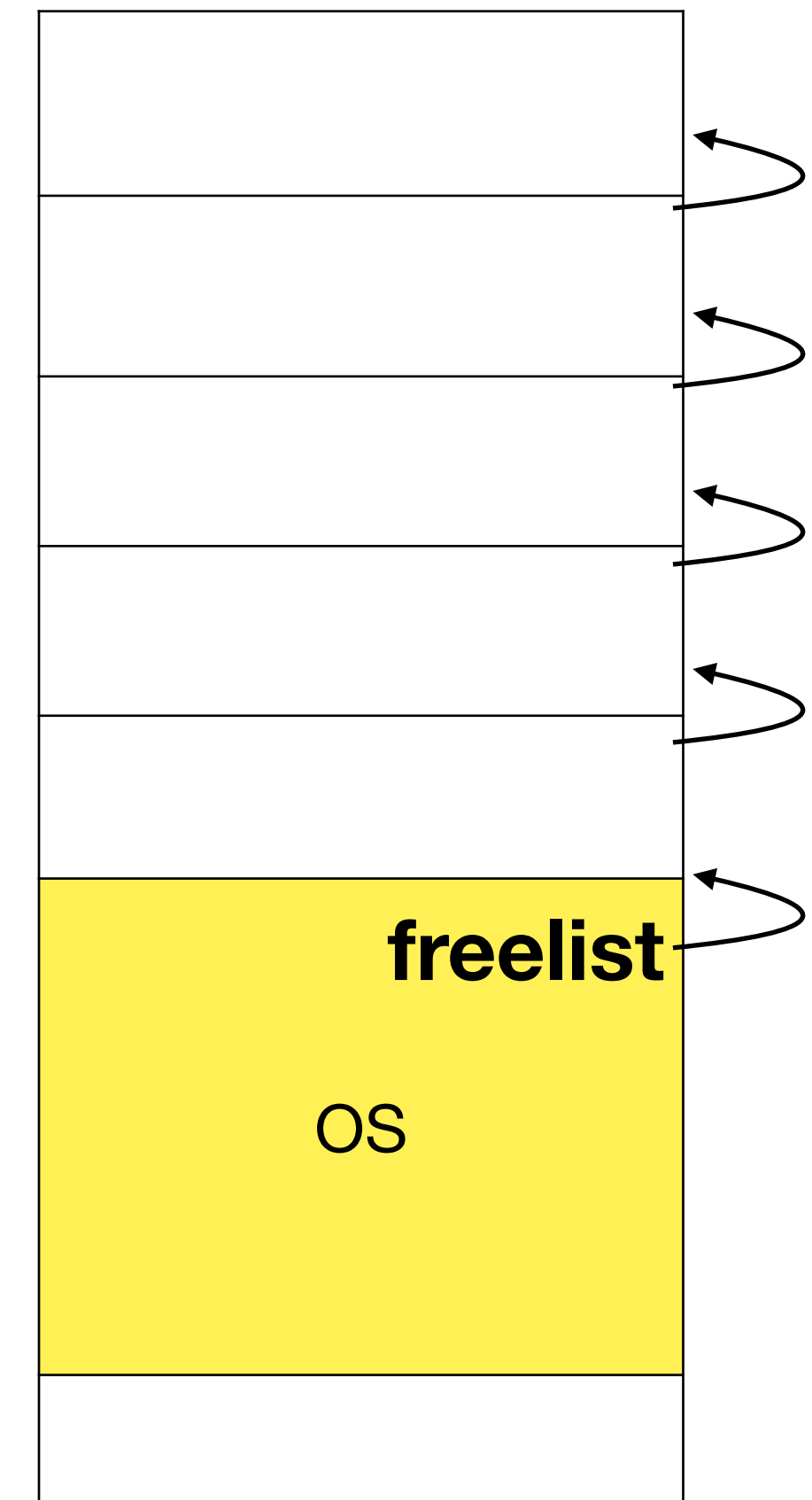
p17-free-space



Allocating memory to processes

p17-free-space

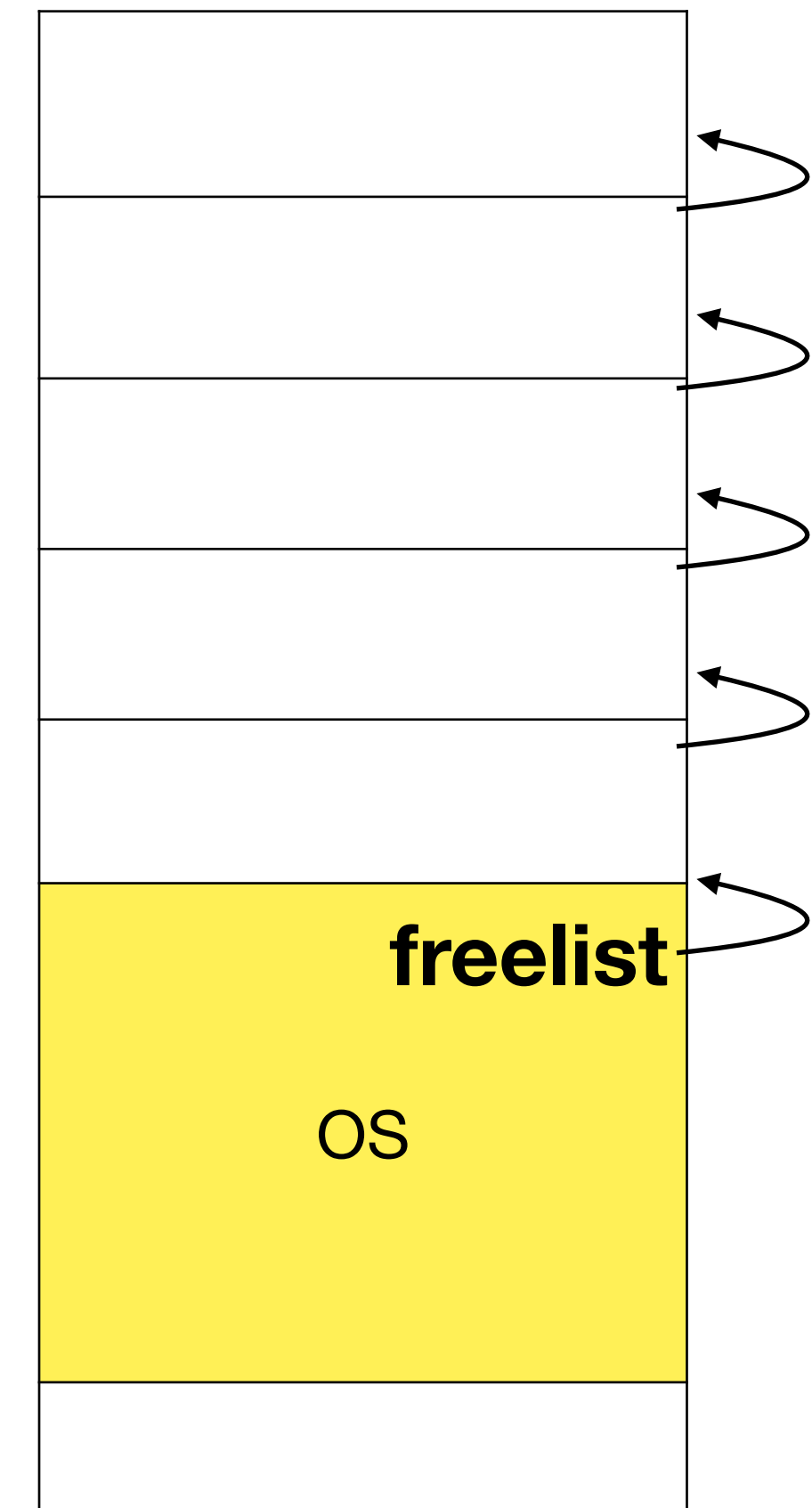
- Flat memory model: cs, ss, ds all point to the same memory segment



Allocating memory to processes

p17-free-space

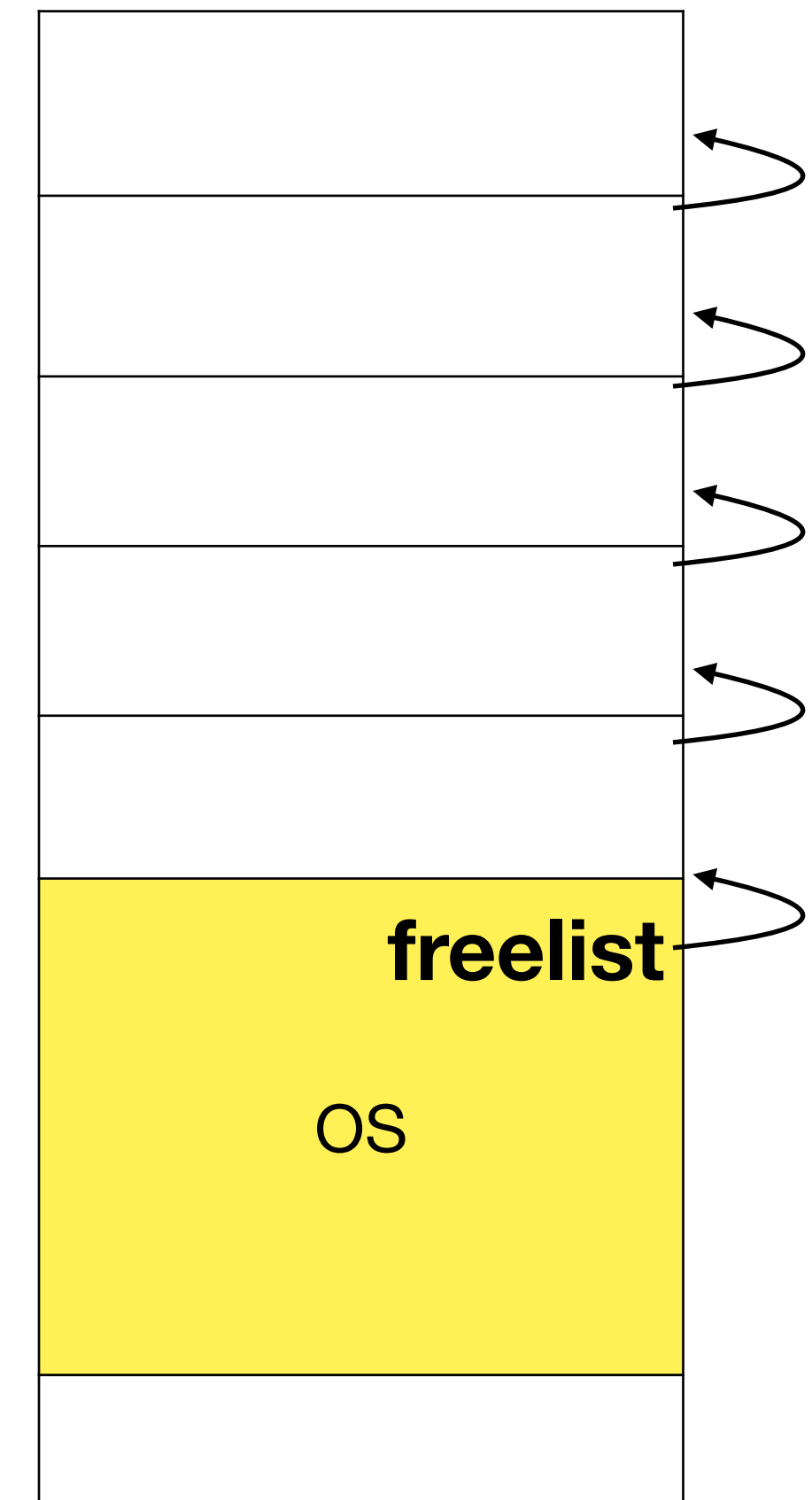
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- Break memory into equal sized parts (say 1MB). OS allocates/deallocates at this granularity by maintaining a free list



Allocating memory to processes

p17-free-space

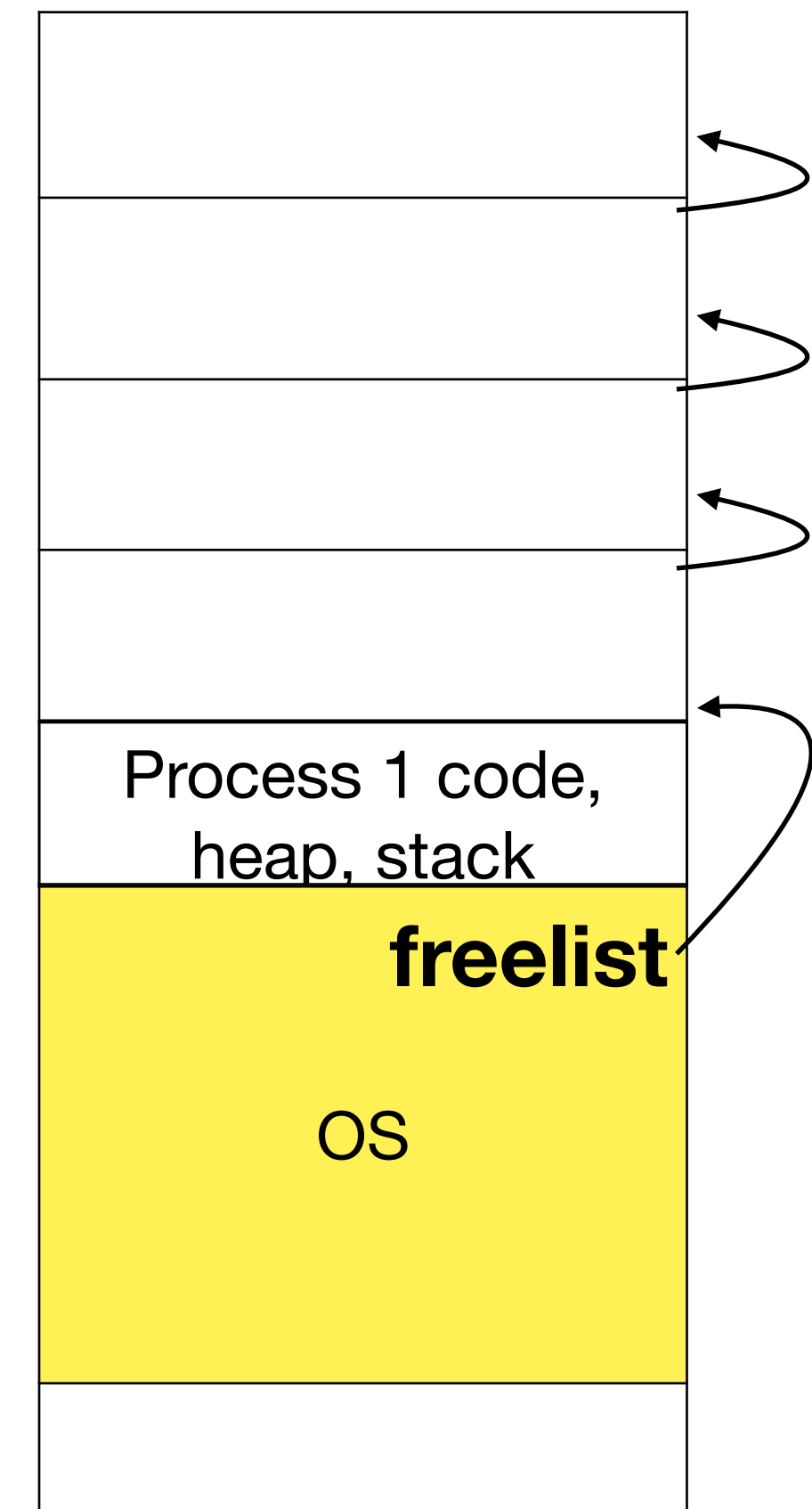
- Flat memory model: cs, ss, ds all point to the same memory segment
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- Create (delete) entries in GDT for the new process, finished process



Allocating memory to processes

p17-free-space

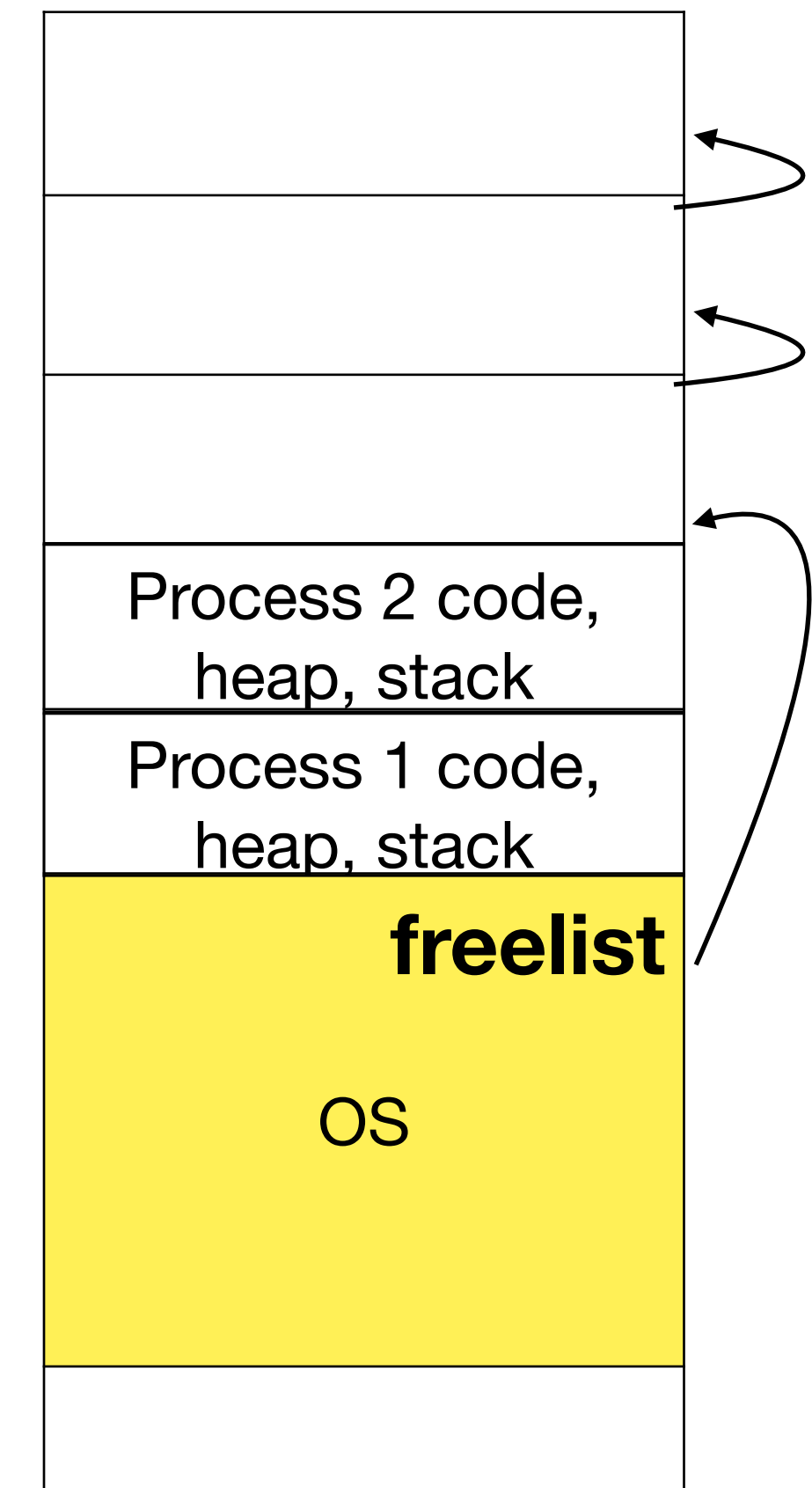
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Allocating memory to processes

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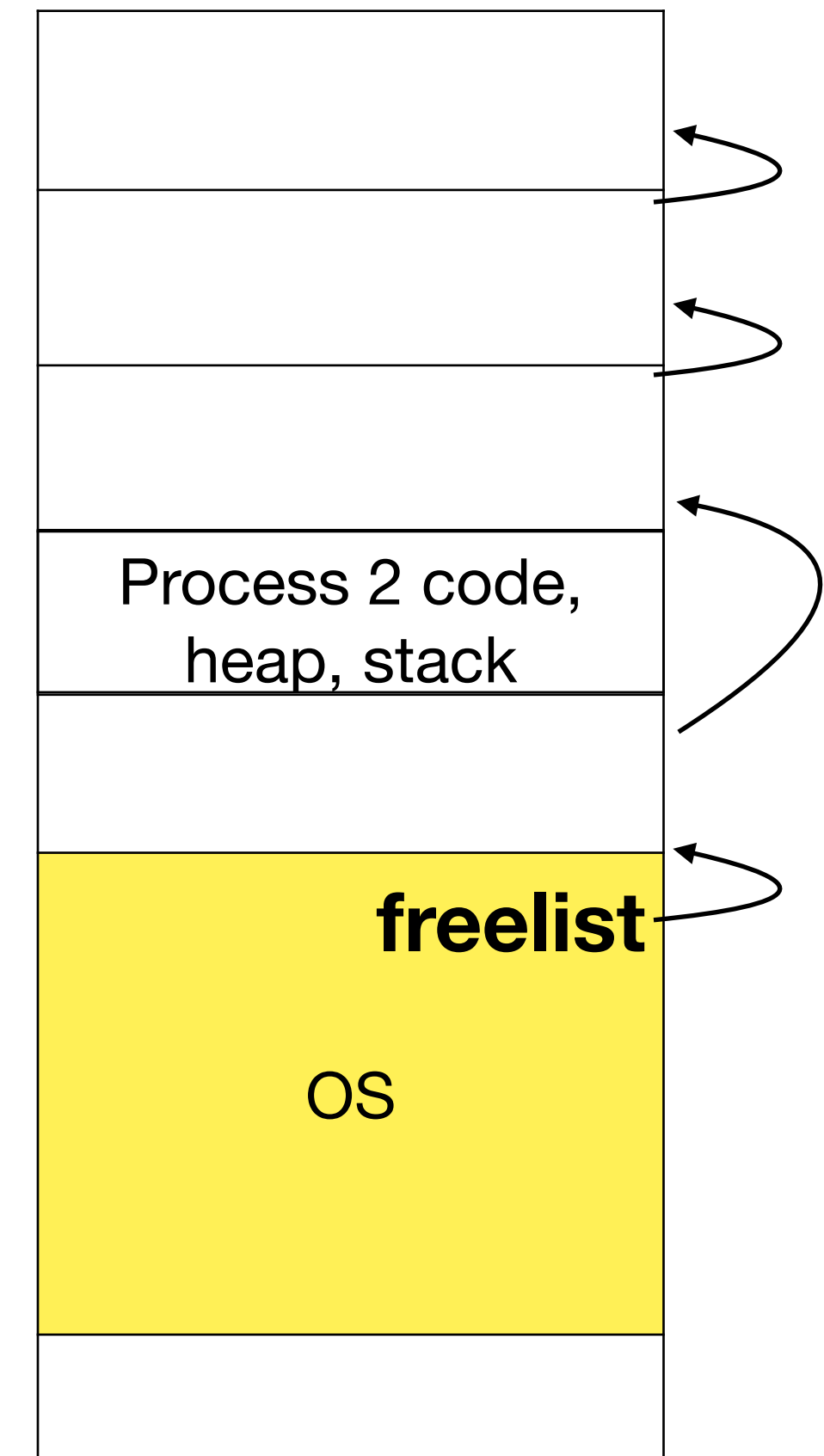
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Allocating memory to processes

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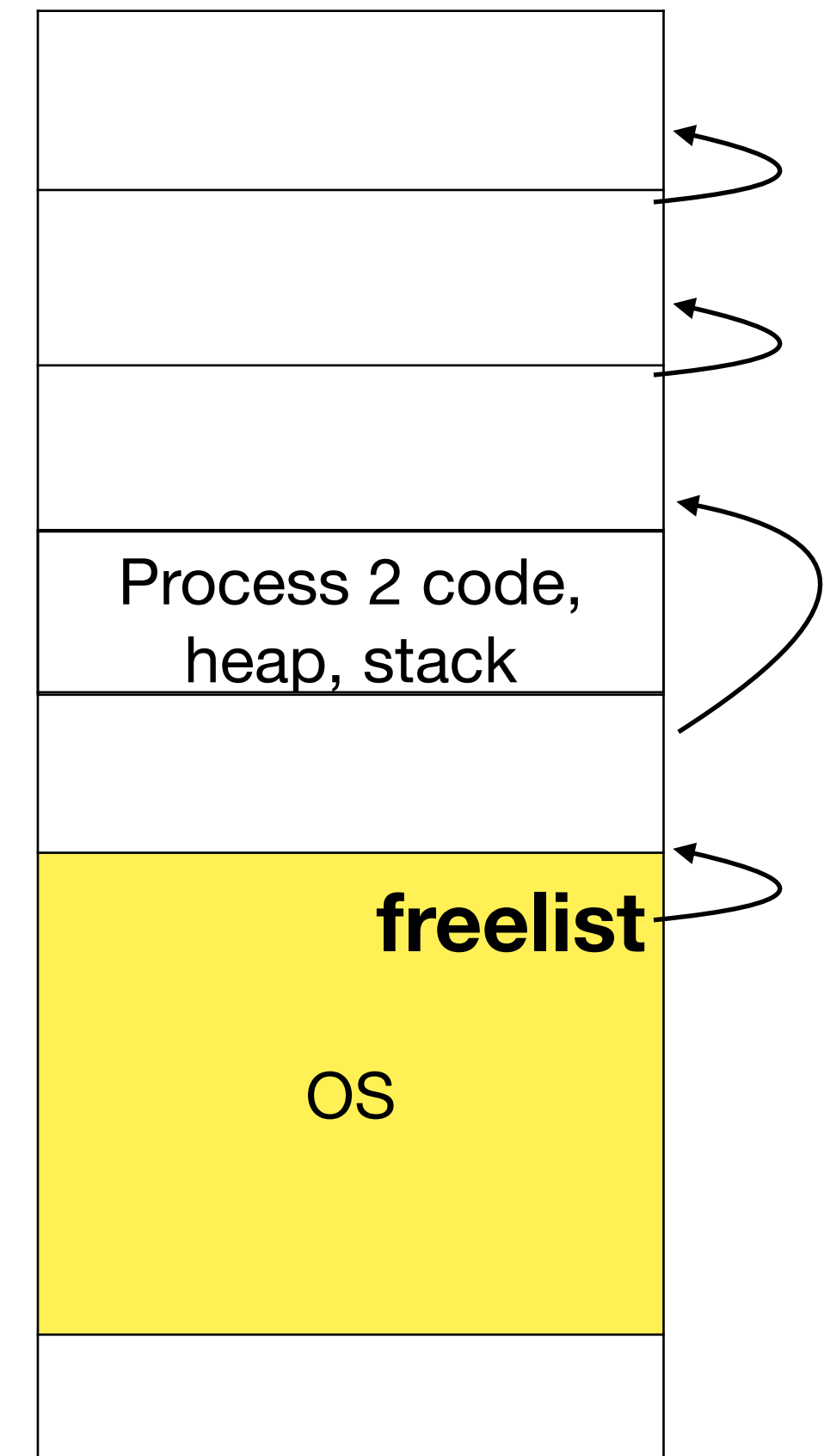
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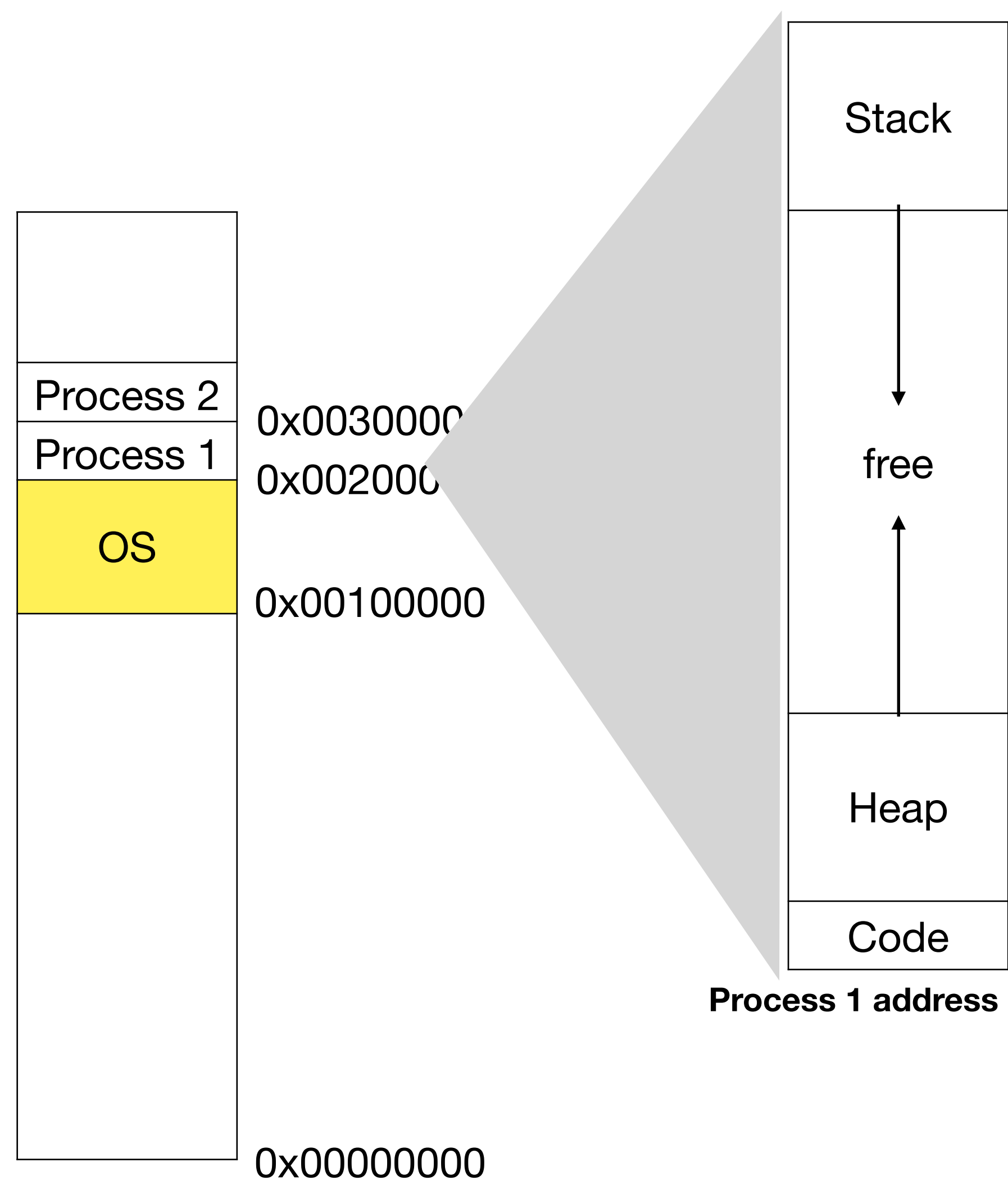
Allocating memory to processes

p17-free-space

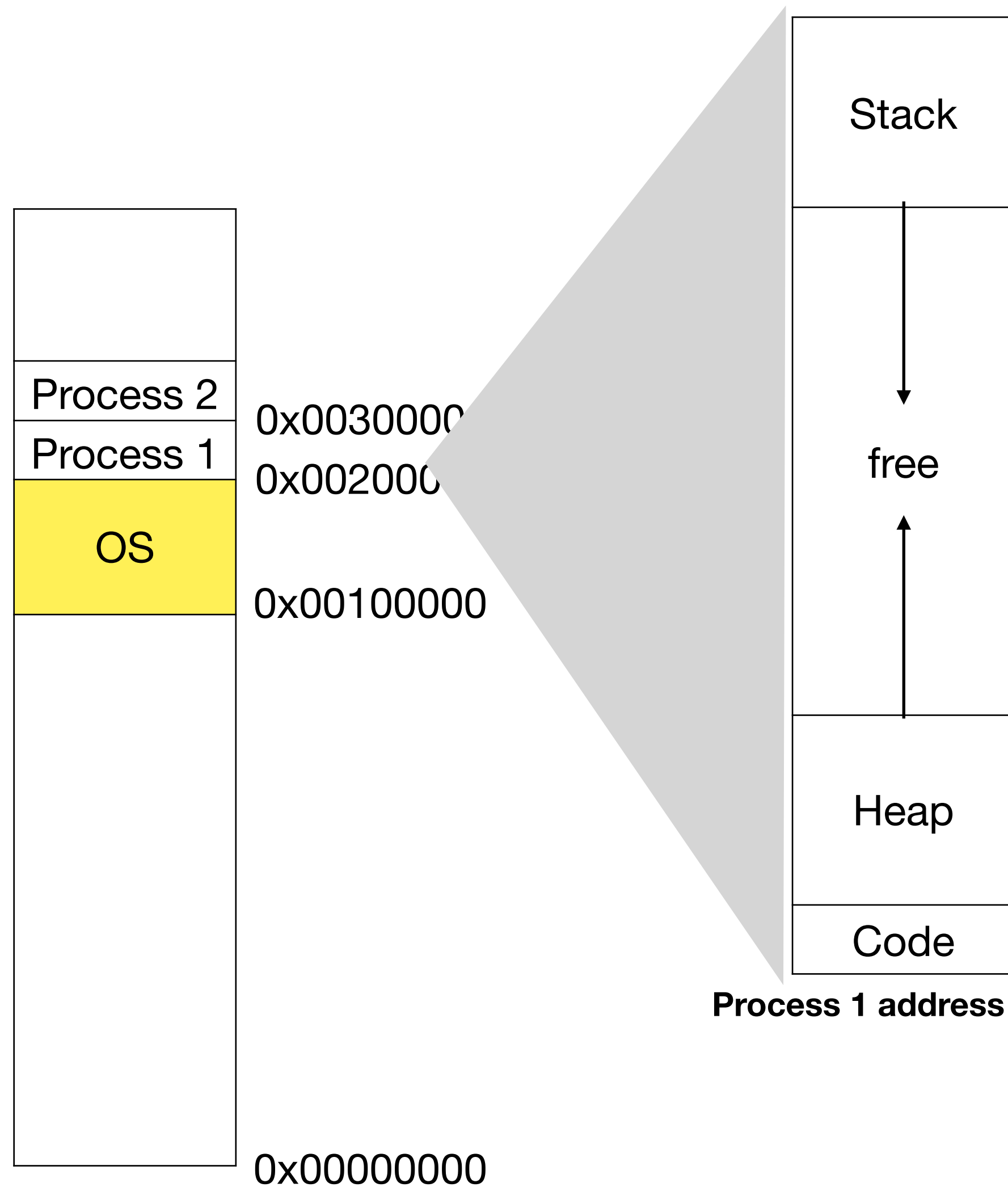
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- Break memory into equal sized parts (say 1MB). OS allocates/deallocates at this granularity by maintaining a free list
- Create (delete) entries in GDT for the new process, finished process
- What if process needed more/less memory?



Segmentation hardware supports multi-segment model

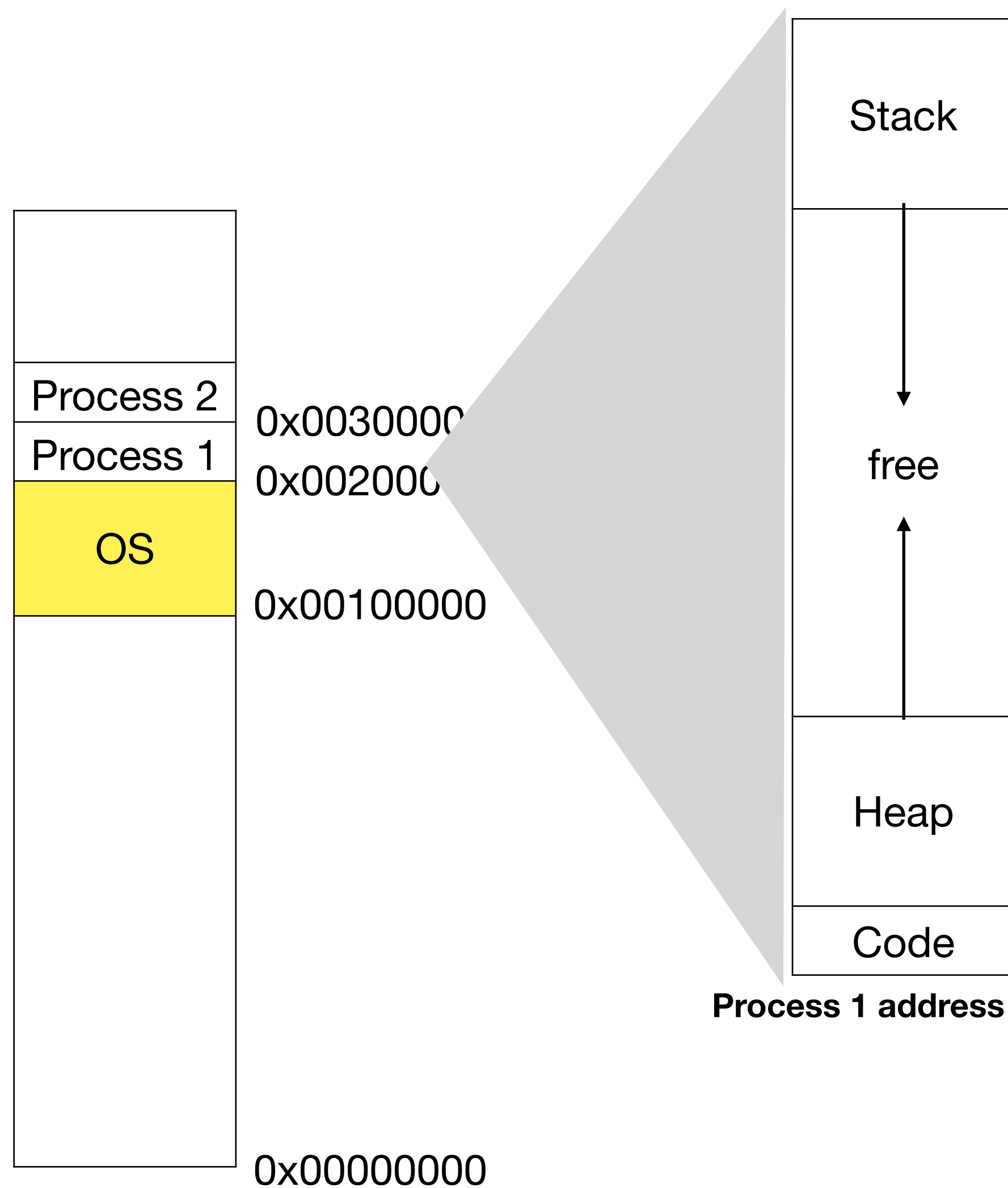


Segmentation hardware supports multi-segment model



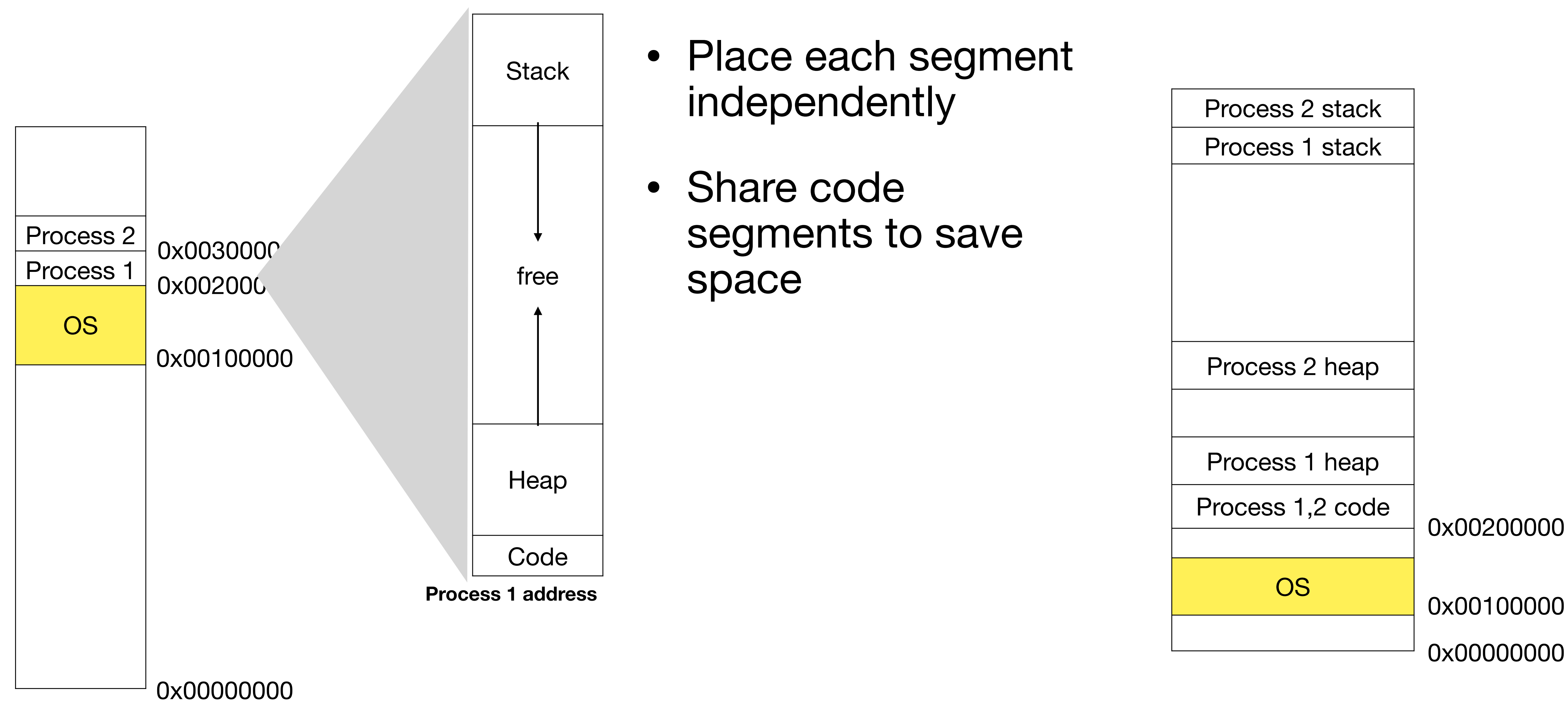
- Place each segment independently

Segmentation hardware supports multi-segment model

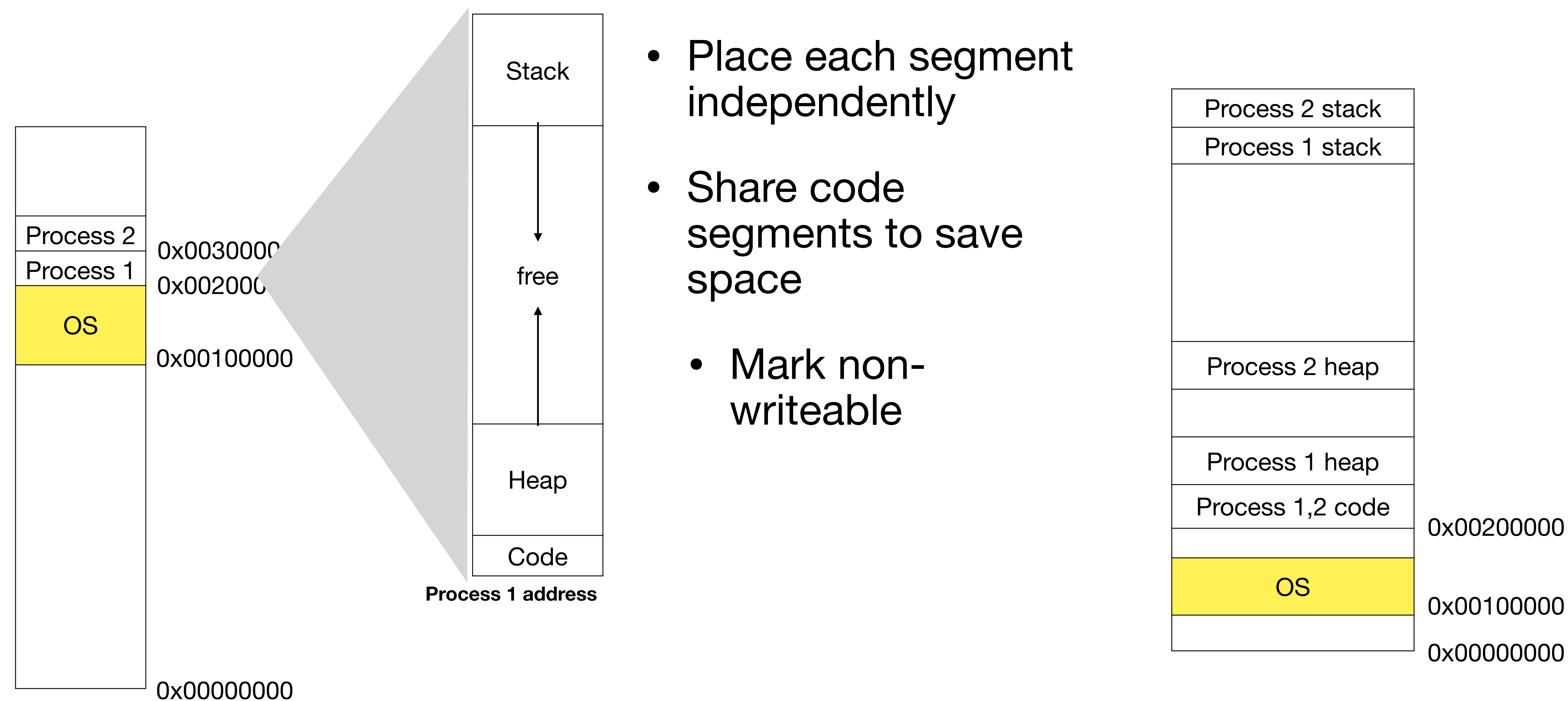


- Place each segment independently
- Share code segments to save space

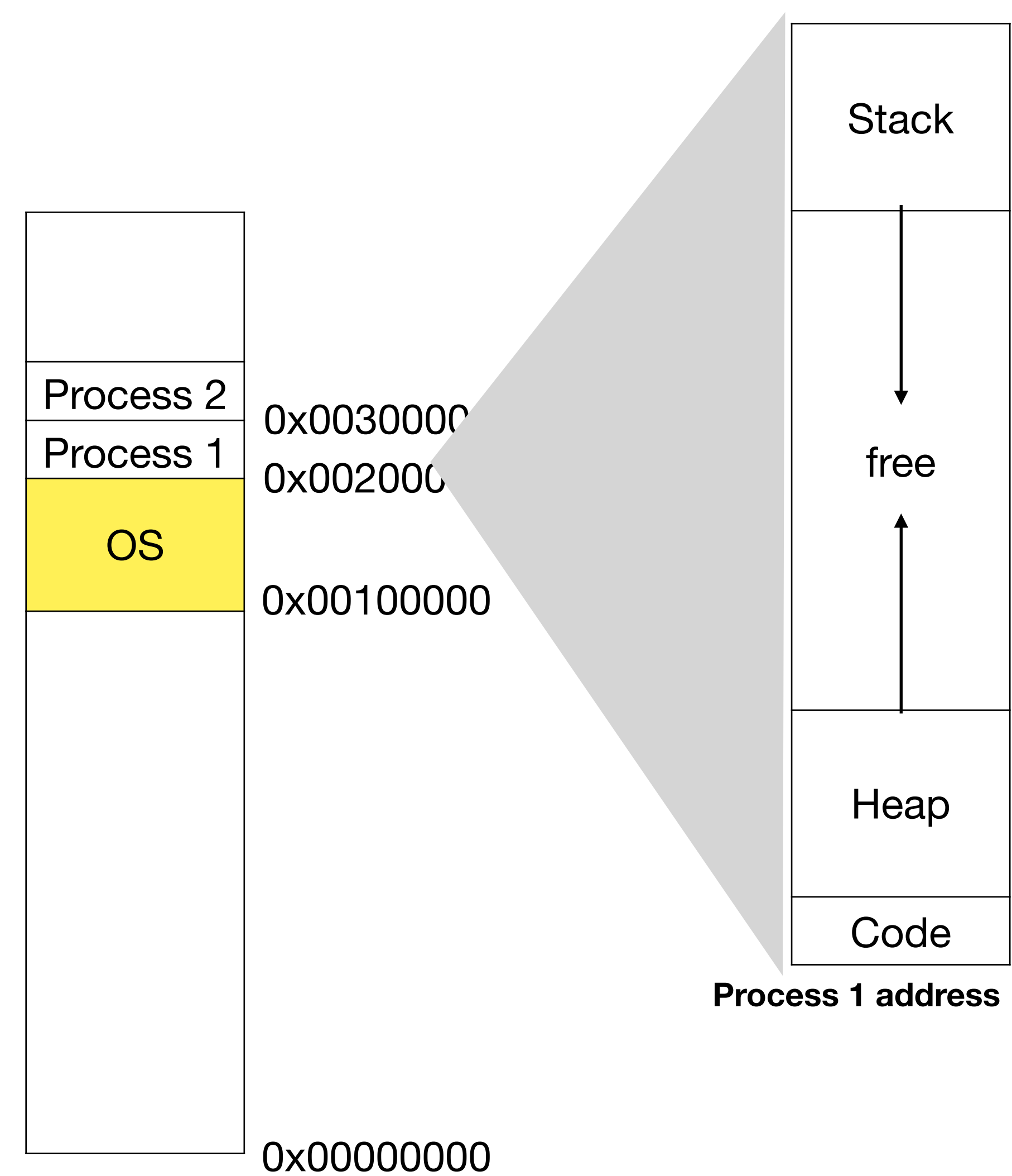
Segmentation hardware supports multi-segment model



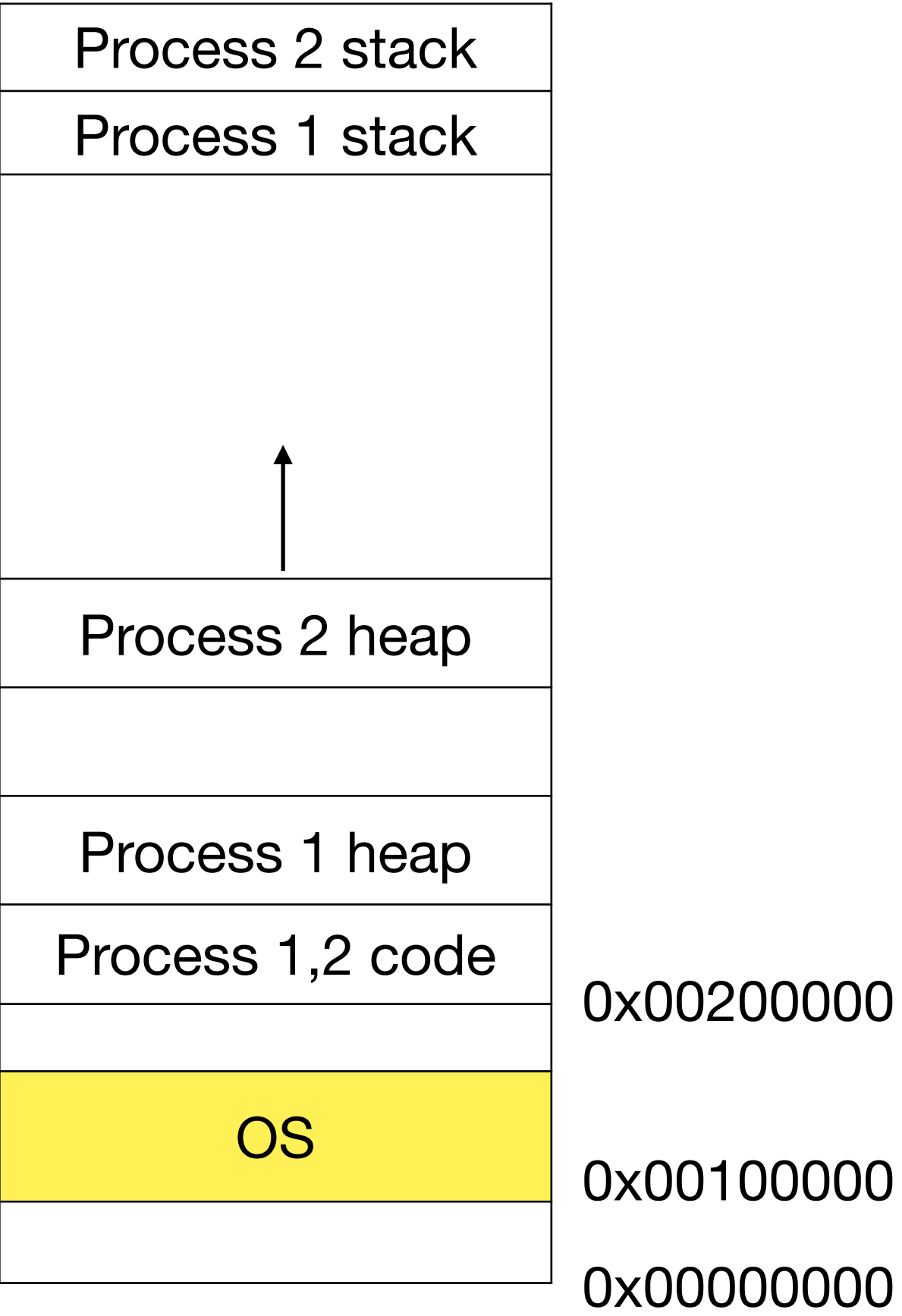
Segmentation hardware supports multi-segment model



Segmentation hardware supports multi-segment model



- Place each segment independently
- Share code segments to save space
- Mark non-writeable
- Process can call `sbrk(N bytes)` system call to grow heap



External fragmentation

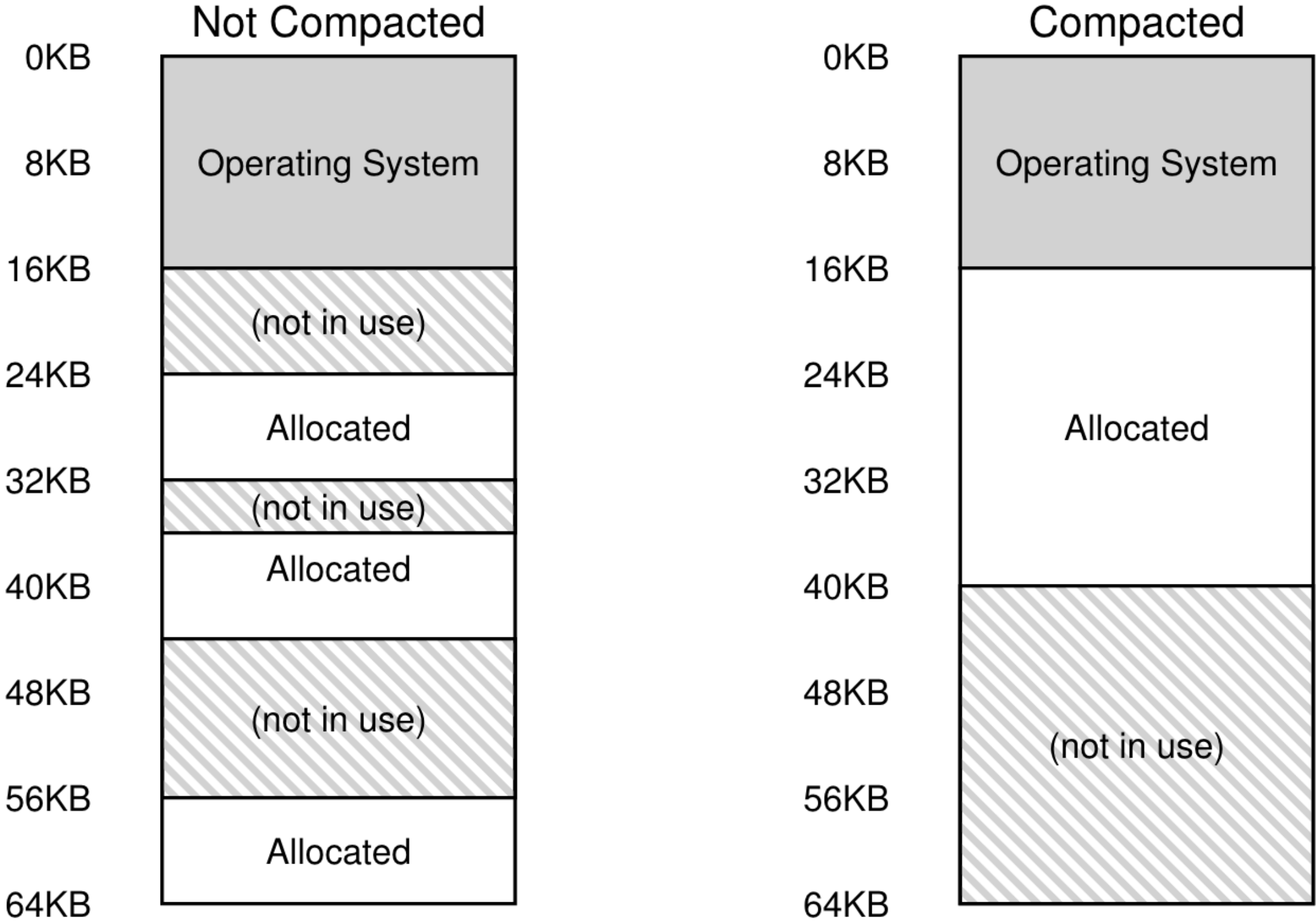


Figure 16.6: Non-compacted and Compacted Memory

External fragmentation

- After many processes start and exit, memory might become “fragmented” (similar to disk)
- Example: cannot allocate 20 KB segment

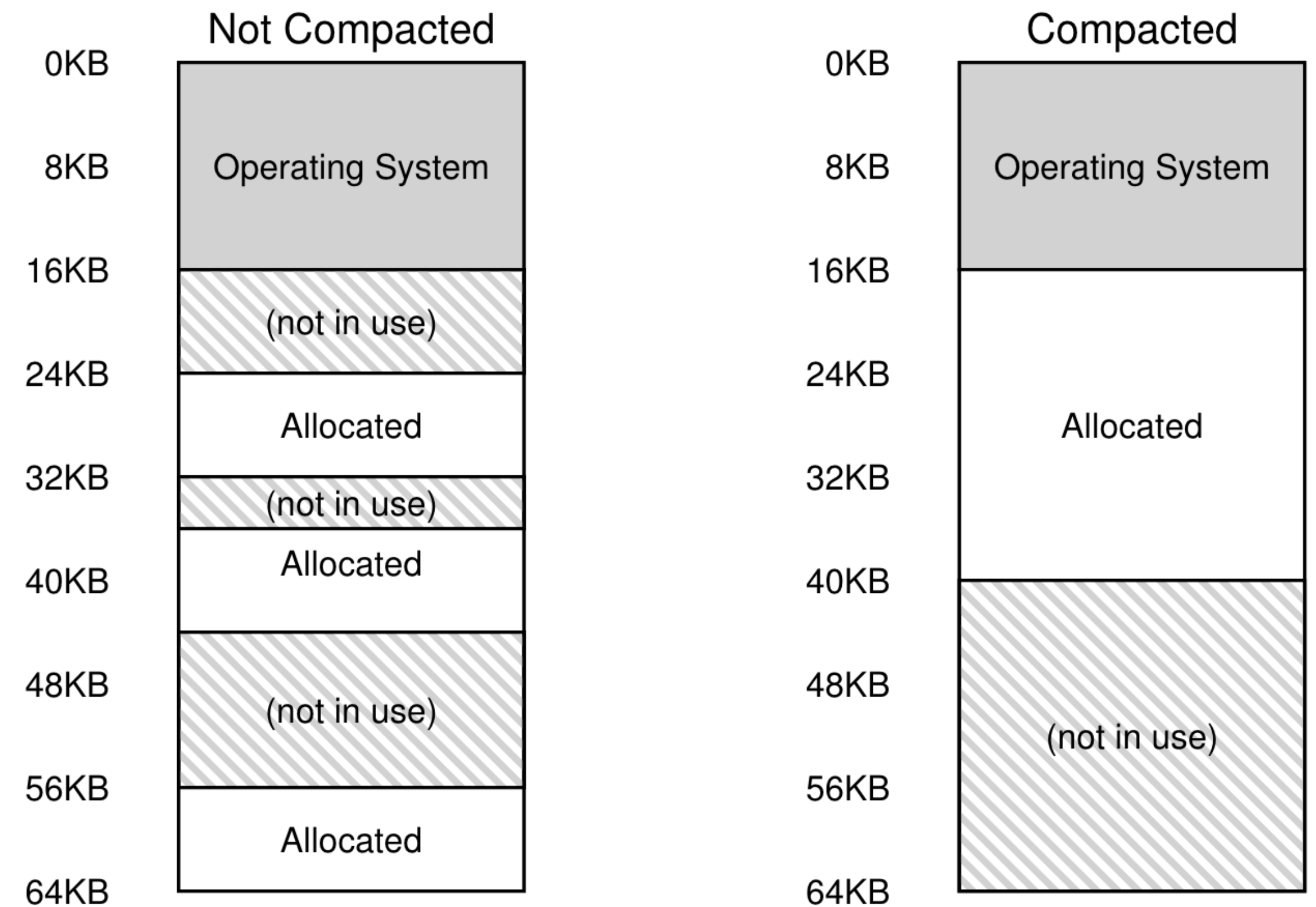


Figure 16.6: Non-compacted and Compacted Memory

External fragmentation

- After many processes start and exit, memory might become “fragmented” (similar to disk)
 - Example: cannot allocate 20 KB segment
- Unlike heap memory allocator, OS can do compaction: copy all allocated regions contiguously, update segment base and bound registers
 - Copying is expensive
 - Growing heap becomes not possible

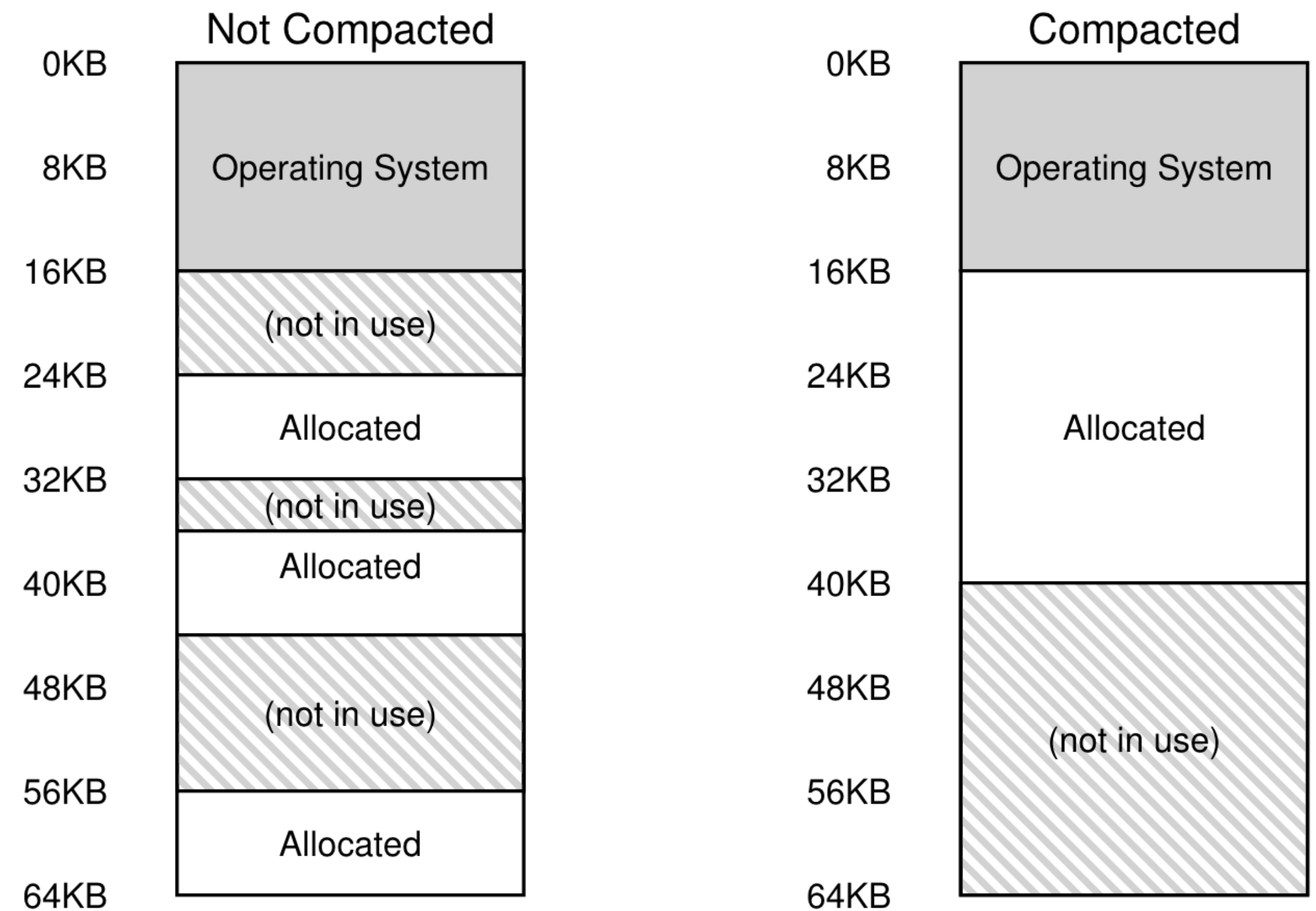
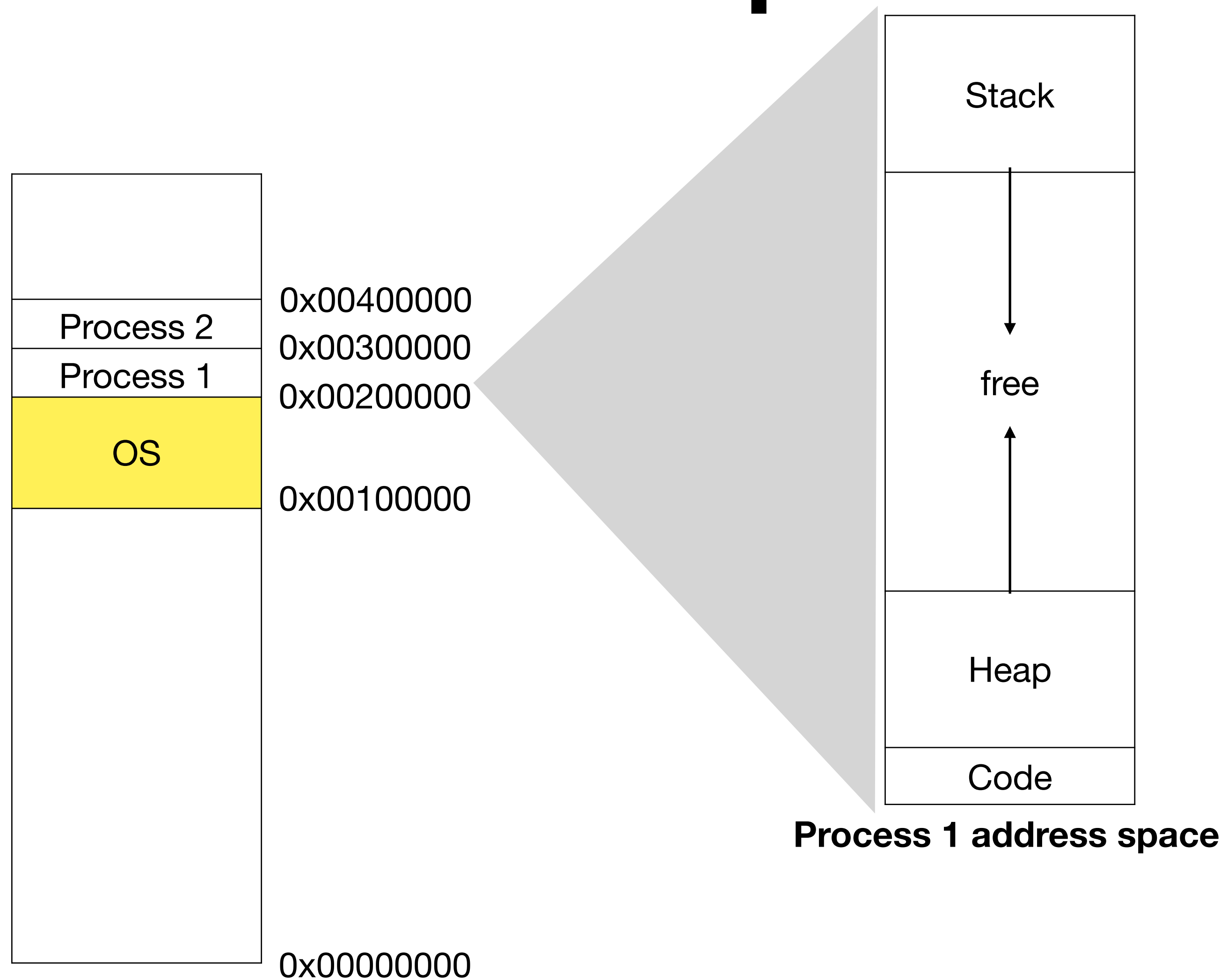


Figure 16.6: Non-compacted and Compacted Memory

Processes in action

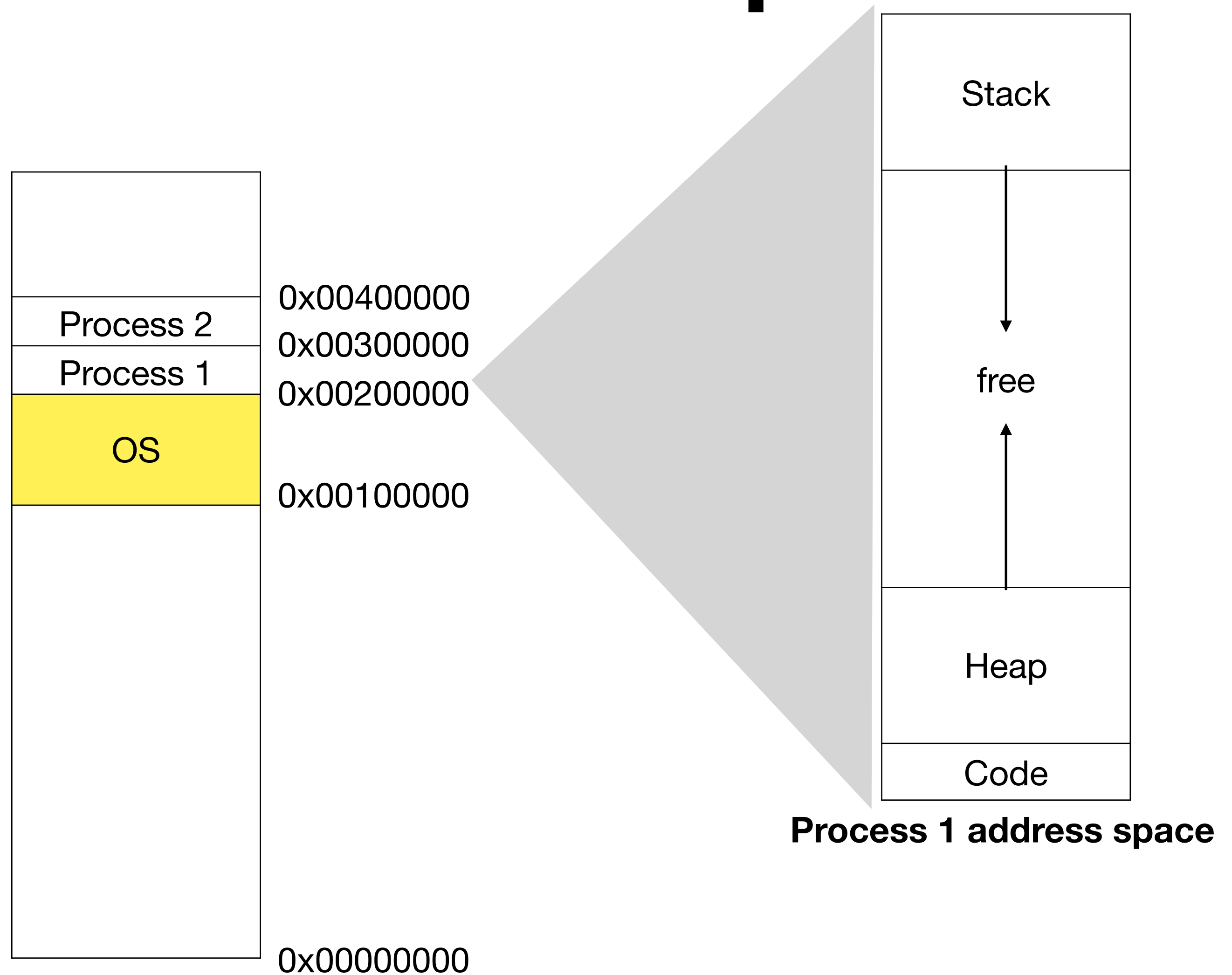
xv6 Ch. 3: system calls, x86 protection, trap handlers

Memory isolation and address space



Memory isolation and address space

- Flat memory model



Protection

Protection

- Process cannot modify OS code since it is not in process' *address space*

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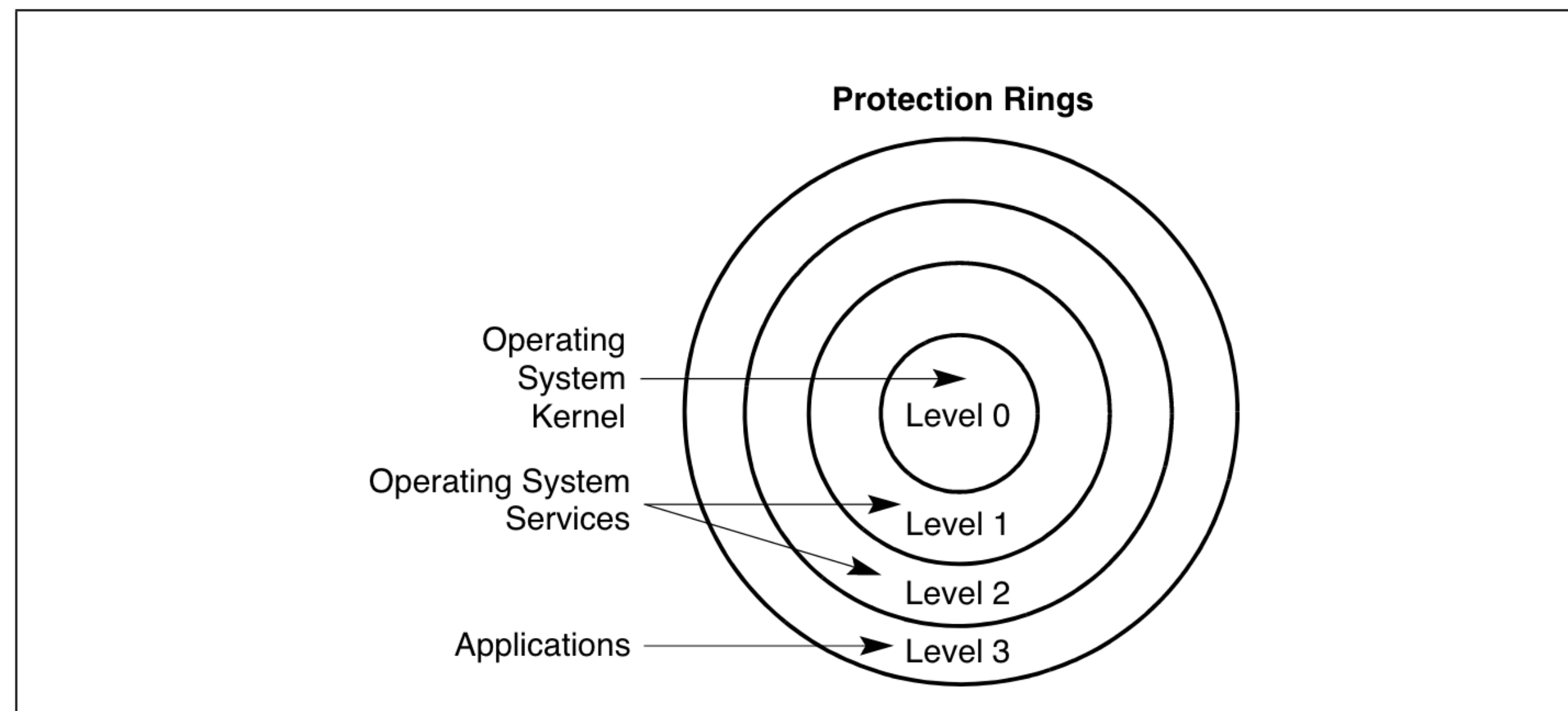


Figure 5-3. Protection Rings

Protection

- Process cannot modify OS code since it is not in process' *address space*
- Process cannot modify GDT and IDT entries since GDT, IDT are not in its address space
- What stops process from calling lgdt and lidt instructions?
 - Ring 0: kernel mode. Ring 3: user mode

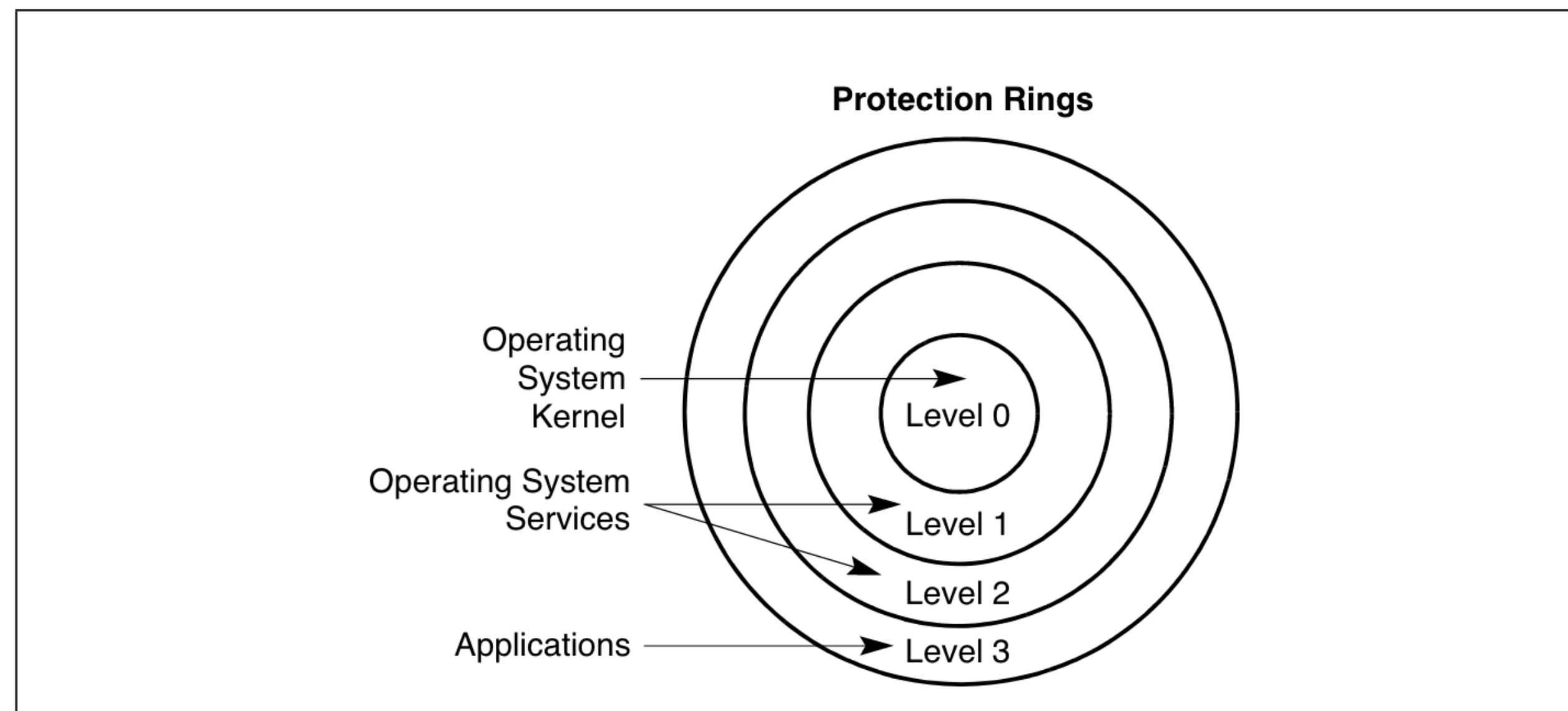


Figure 5-3. Protection Rings

Protection

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- Process cannot modify GDT and IDT entries since GDT, IDT are not in its address space
- What stops process from calling lgdt and lidt instructions?
 - Ring 0: kernel mode. Ring 3: user mode
 - lgdt and lidt are *privileged instructions* only callable from “ring 0”

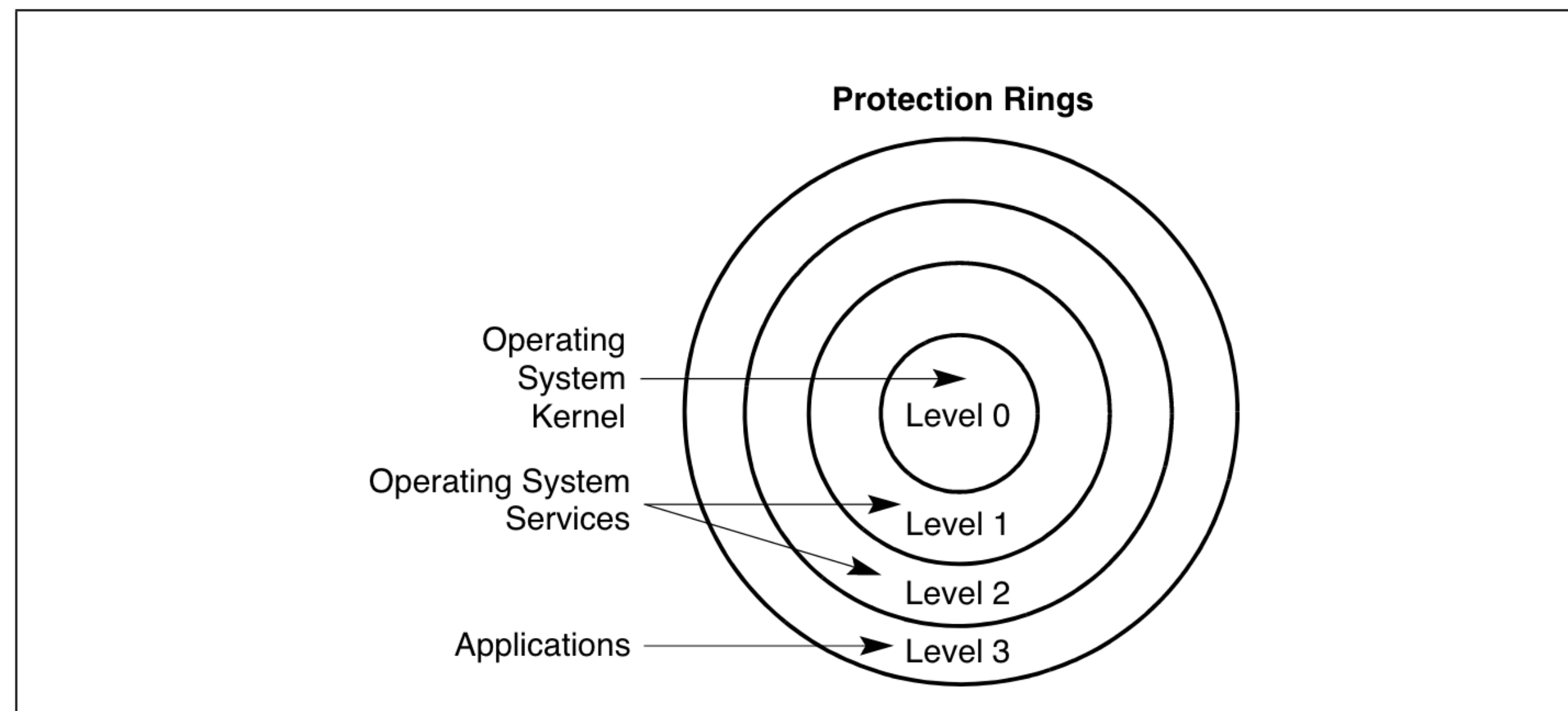


Figure 5-3. Protection Rings

**How does hardware know
the current privilege level?**

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- Two LSBs of %cs determine “current privilege level” (CPL)

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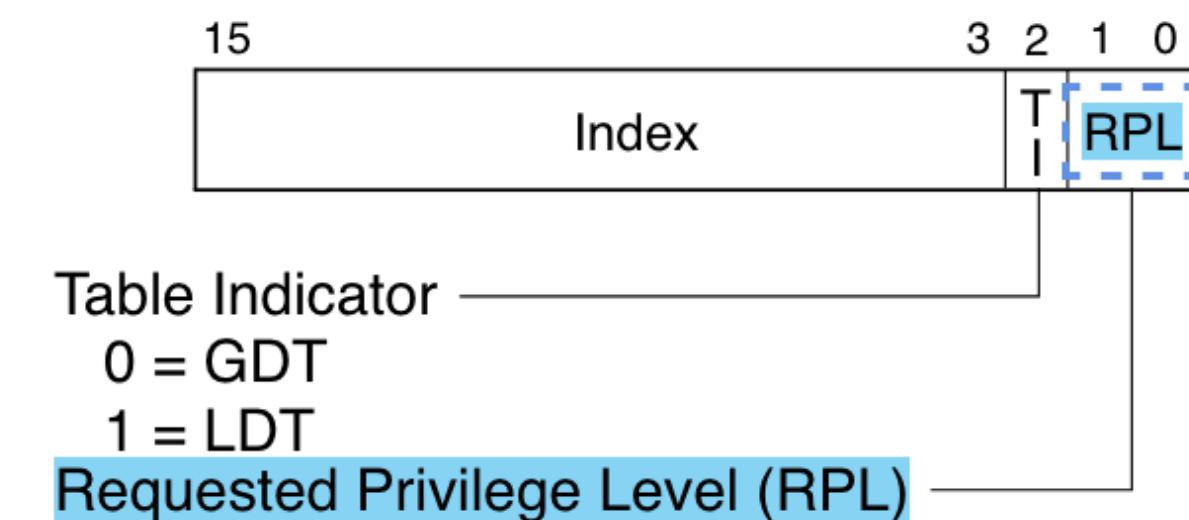


Figure 3-6. Segment Selector

How does hardware know the current privilege level?

- Two LSBs of %cs determine “current privilege level” (CPL)
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- Segment descriptor in GDT specifies “descriptor privilege level” (DPL)

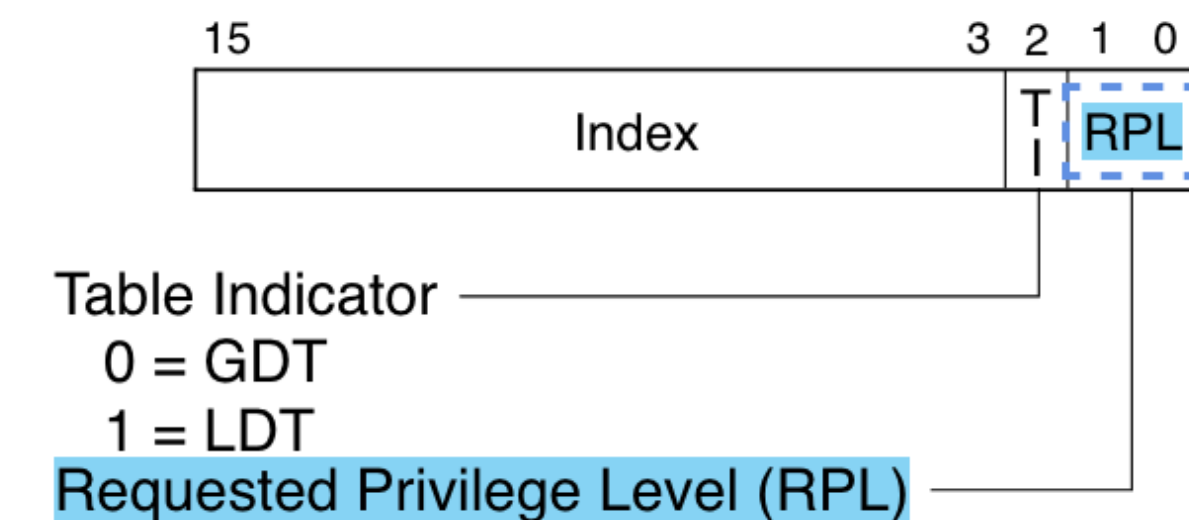
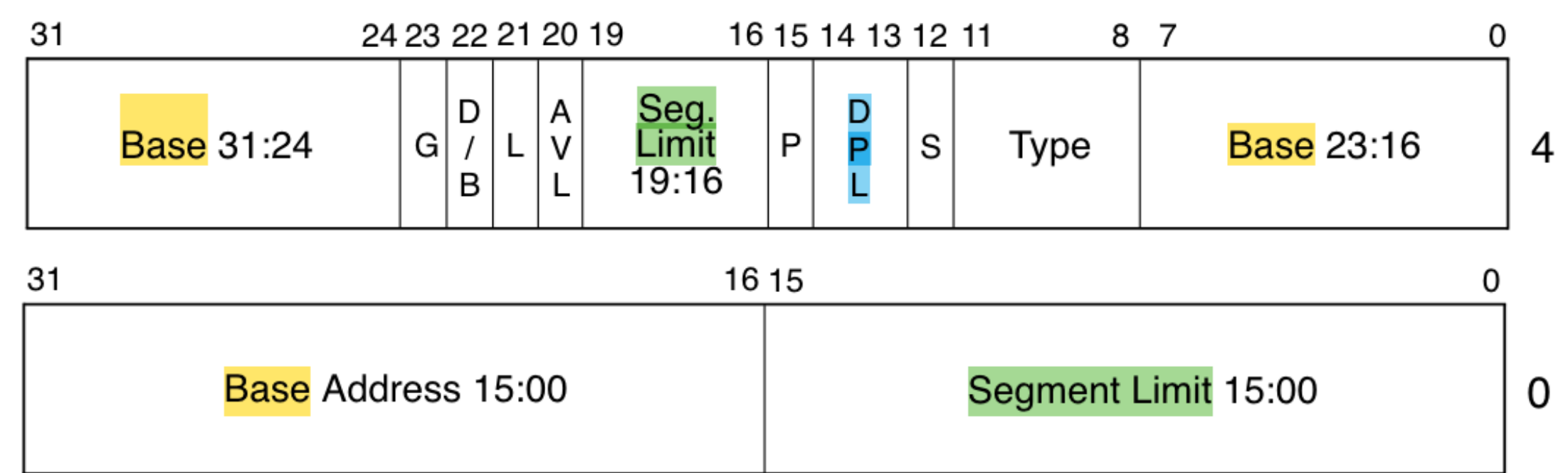


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- Segment descriptor in GDT specifies “descriptor privilege level” (DPL)



- L — 64-bit code segment (IA-32e mode only)
- AVL — Available for use by system software
- BASE — Segment base address
- D/B — Default operation size (0 = 16-bit segment; 1 = 32-bit segment)
- DPL — Descriptor privilege level
- G — Granularity
- LIMIT — Segment Limit
- P — Segment present
- S — Descriptor type (0 = system; 1 = code or data)
- TYPE — Segment type

Figure 3-8. Segment Descriptor

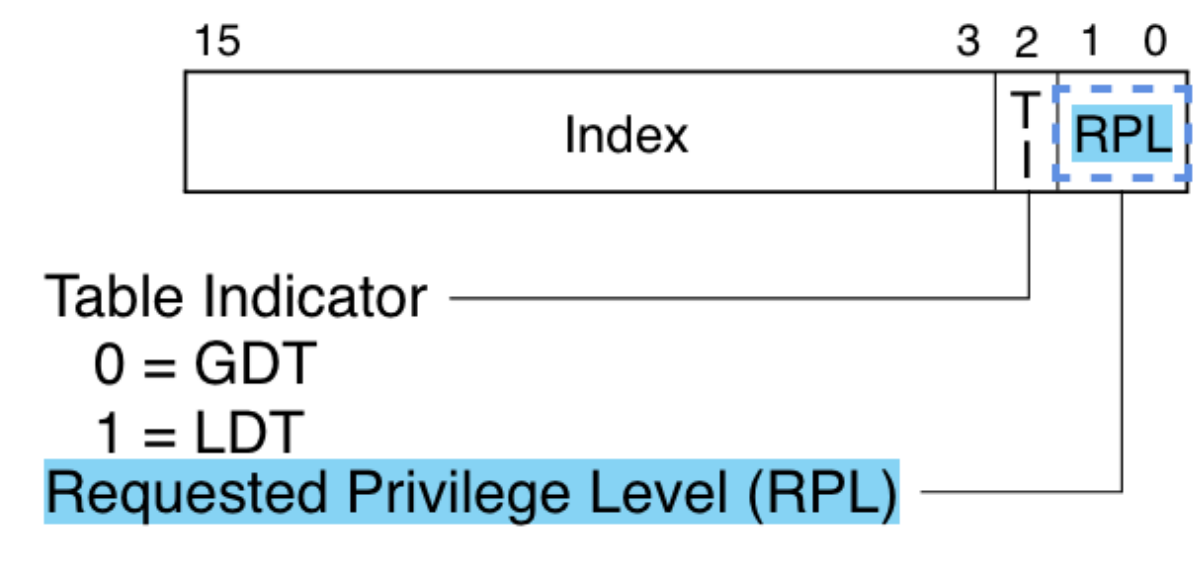


Figure 3-6. Segment Selector

**How does hardware
enforce privilege levels?**

How does hardware enforce privilege levels?

- Privileged instructions can only be called when CPL=0

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- Privileged instructions can only be called when CPL=0
- Programs cannot lower their CPL directly

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- For data access:
 $CPL \leq DPL$ and $RPL \leq DPL$

How does hardware enforce privilege levels?

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- For data access:
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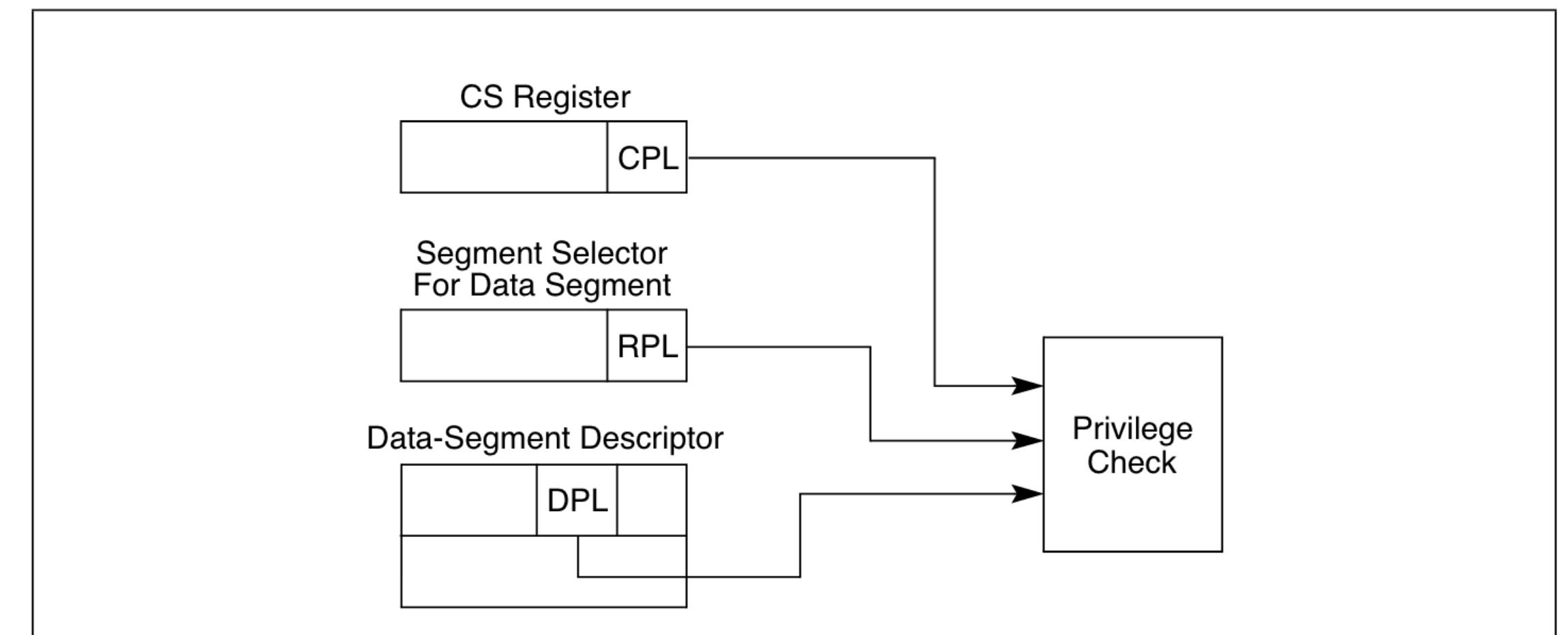


Figure 5-4. Privilege Check for Data Access

How does hardware enforce privilege levels?

- Privileged instructions can only be called when CPL=0
- Programs cannot lower their CPL directly
- INT instruction causes a software interrupt to set ring = 0
- For data access:
 $CPL \leq DPL$ and $RPL \leq DPL$

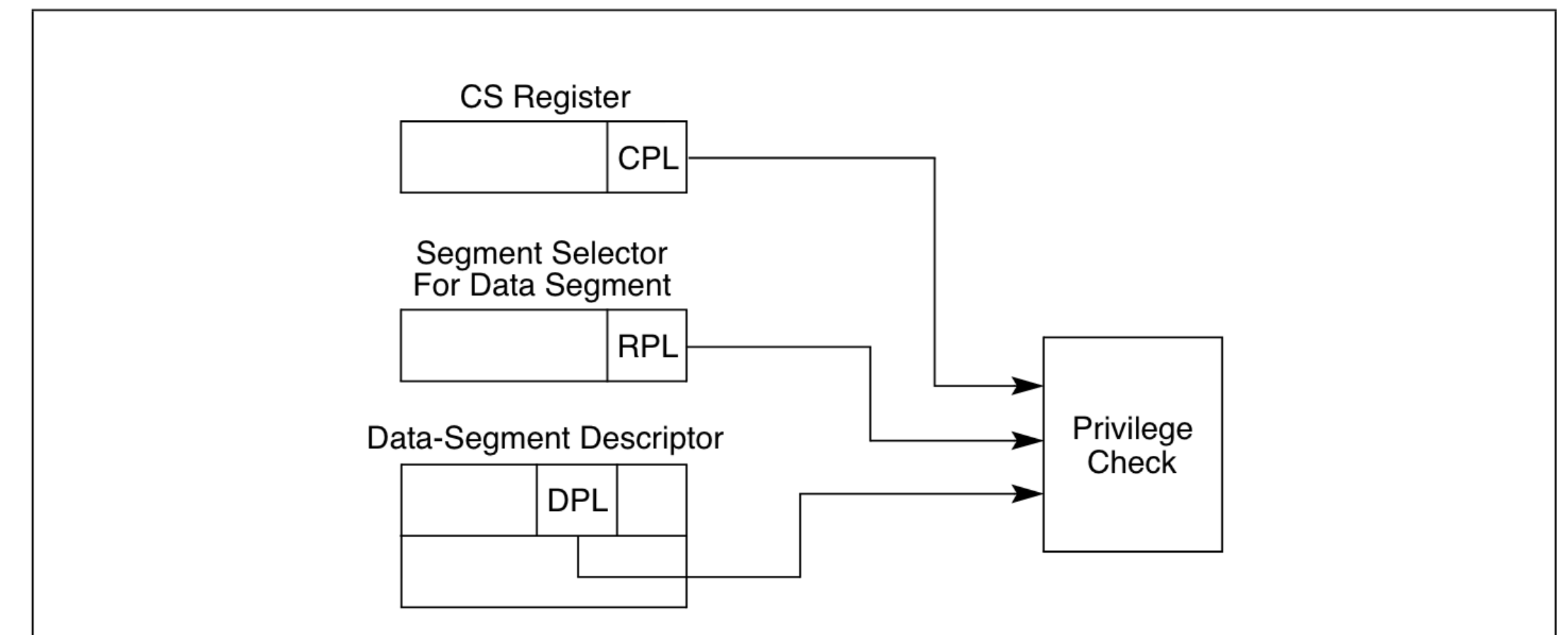


Figure 5-4. Privilege Check for Data Access

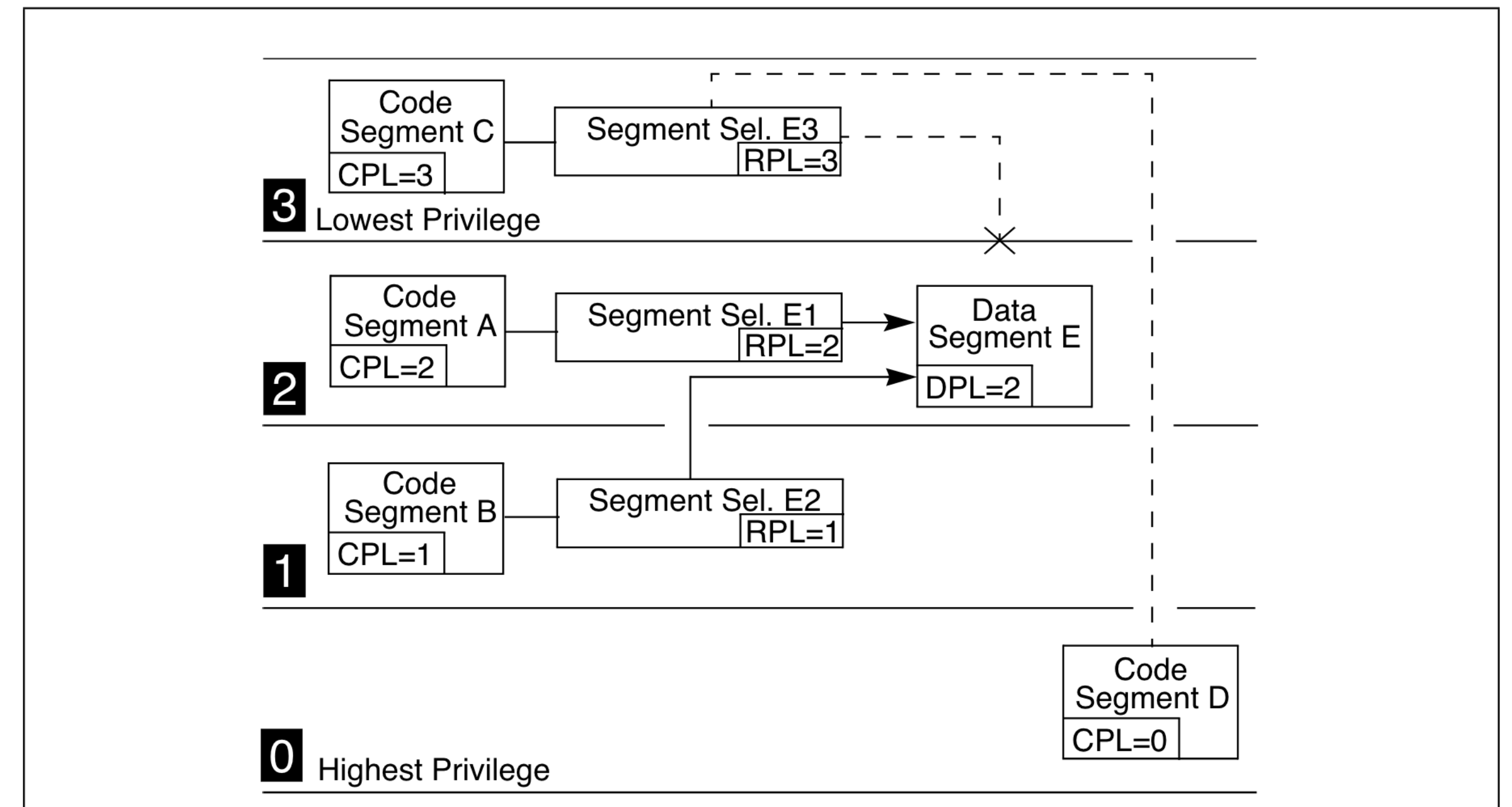
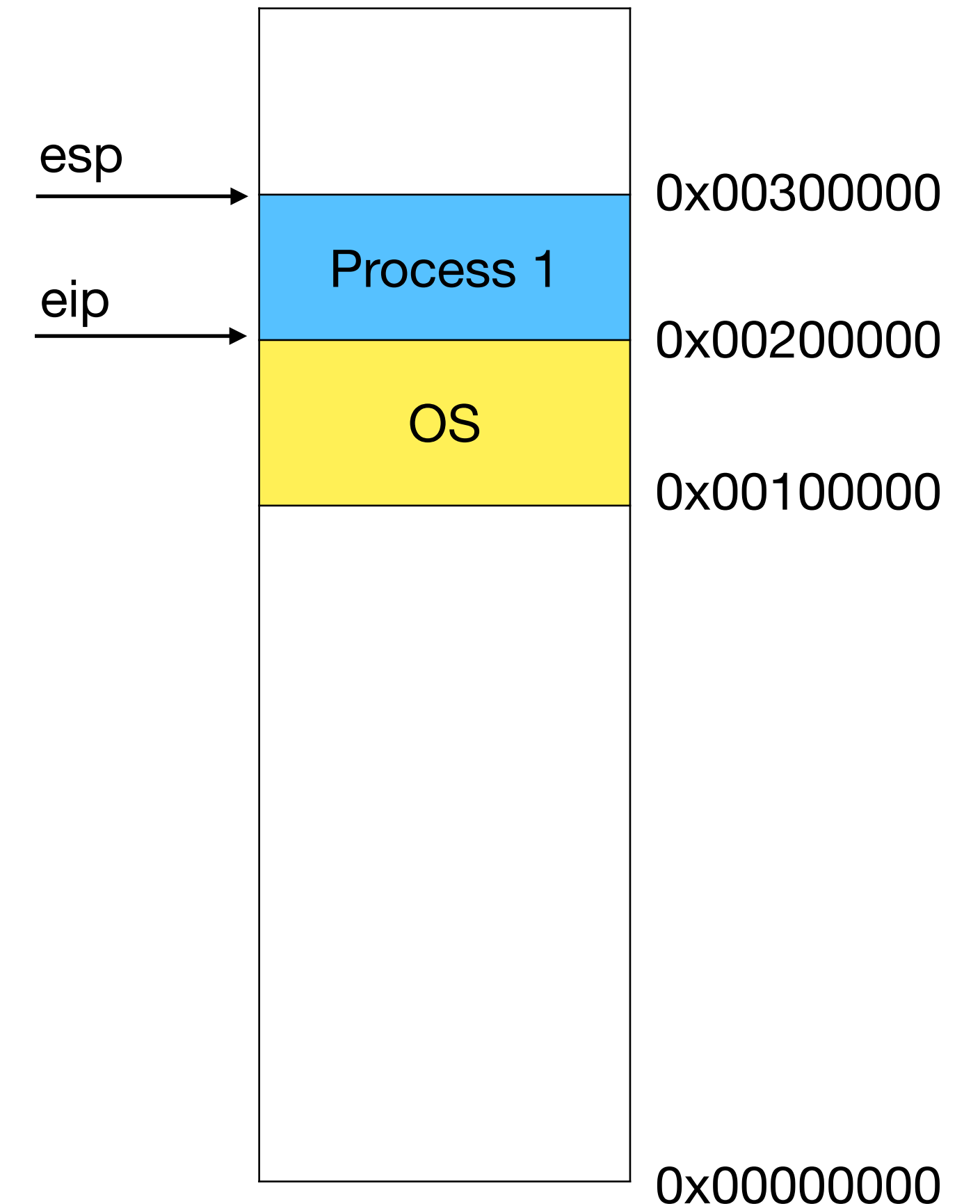


Figure 5-5. Examples of Accessing Data Segments From Various Privilege Levels

Setting up our first process in xv6!

p15-initproc

- Link initcode binary inside the kernel image
- main.c calls seginit, then pinit, then scheduler.
- seginit in vm.c creates code (UCODE) and data segments (UDATA) from STARTPROC (2MB) to 3MB. Flat memory model!
- pinit in proc.c copies program binary to STARTPROC and sets cs,ds,ss,es to the program's segments. Last two bits are set to DPL_USER. Sets eip=0, esp=1MB. Enables interrupt flag in eflags. proc.c maintains list of processes in ptable. We are just starting one process for now.
- scheduler in proc.c selects RUNNABLE process and switches to it.
- Process cannot change GDT entries, IDT entries since they are not in address space of the process! Process cannot call lgdt, lidt since it will run in ring 3.



Link initcode binary inside the OS image

```
.globl start initcode.S
start:
    add $1, %eax
    movl %eax, 0x10001
    jmp start
```

```
initcode: initcode.S Makefile
$(CC) $(CFLAGS) -nostdinc -I. -c initcode.S
$(LD) $(LDFLAGS) -N -e start -Ttext 0 -o initcode.out initcode.o
$(OBJCOPY) -S -O binary initcode.out initcode

kernel: $(OBJJS) entry.o kernel.ld initcode
$(LD) $(LDFLAGS) -T kernel.ld -o kernel entry.o $(OBJJS) -b binary initcode
```

```
kernel.sym
00105418 _binary_initcode_start
0000000a _binary_initcode_size
00105422 _binary_initcode_end
```

```
void pinit(void) { proc.c
    extern char _binary_initcode_start[], _binary_initcode_size[];
    ..
}
```

Link initcode binary inside the OS image

- Make the **entry point as start**, and link the text region **at address 0**

```
.globl start initcode.S  
start:  
    add $1, %eax  
    movl %eax, 0x10001  
    jmp start
```

```
initcode: initcode.S Makefile  
$(CC) $(CFLAGS) -nostdinc -I. -c initcode.S  
$(LD) $(LDFLAGS) -N -e start -Ttext 0 -o initcode.out initcode.o  
$(OBJCOPY) -S -O binary initcode.out initcode  
  
kernel: $(OBSJ) entry.o kernel.ld initcode  
$(LD) $(LDFLAGS) -T kernel.ld -o kernel entry.o $(OBSJ) -b binary initcode
```

```
kernel.sym  
00105418 _binary_initcode_start  
0000000a _binary_initcode_size  
00105422 _binary_initcode_end
```

```
void pinit(void) { proc.c  
    extern char _binary_initcode_start[], _binary_initcode_size[];  
    ..  
}
```

Link initcode binary inside the OS image

- Make the **entry point as start**, and link the text region **at address 0**
- Copy out the binary (drop ELF header etc.), **into the kernel**, i.e., not as ELF

```
.globl start initcode.S
start:
    add $1, %eax
    movl %eax, 0x10001
    jmp start
```

```
initcode: initcode.S Makefile
$(CC) $(CFLAGS) -nostdinc -I. -c initcode.S
$(LD) $(LDFLAGS) -N -e start -Ttext 0 -o initcode.out initcode.o
$(OBJCOPY) -S -O binary initcode.out initcode

kernel: $(OBJS) entry.o kernel.ld initcode
$(LD) $(LDFLAGS) -T kernel.ld -o kernel entry.o $(OBJS) -b binary initcode
```

```
kernel.sym
00105418 _binary_initcode_start
0000000a _binary_initcode_size
00105422 _binary_initcode_end
```

```
void pinit(void) { proc.c
    extern char _binary_initcode_start[], _binary_initcode_size[];
    ..
}
```


Link initcode binary inside the OS image

- Make the **entry point as start**, and link the text region **at address 0**
- Copy out the binary (drop ELF header etc.), **into the kernel**, i.e., not as ELF
- Linker generates **_binary_initcode_start**, etc. for OS code to locate the binary

```
.globl start initcode.S  
start:  
    add $1, %eax  
    movl %eax, 0x10001  
    jmp start
```

```
initcode: initcode.S Makefile  
$(CC) $(CFLAGS) -nostdinc -I. -c initcode.S  
$(LD) $(LDFLAGS) -N -e start -Ttext 0 -o initcode.out initcode.o  
$(OBJCOPY) -S -O binary initcode.out initcode  
  
kernel: $(OBSJ) entry.o kernel.ld initcode  
$(LD) $(LDFLAGS) -T kernel.ld -o kernel entry.o $(OBSJ) -b binary initcode
```

```
kernel.sym  
00105418 _binary_initcode_start  
0000000a _binary_initcode_size  
00105422 _binary_initcode_end
```

```
void pinit(void) { proc.c  
    extern char _binary_initcode_start[], _binary_initcode_size[];  
    ..  
}
```

Setup segments for our process

- Initialize GDT entries for user **code** and **data** segments from **2MB-3MB** with **privilege level 3**

```
int main(void) { // main.c
    ...
    seginit();
    pinit();
    scheduler();
}
```

```
#define DPL_USER    0x3           // User DPL // vm.c
#define STARTPROC  0x200000      // Start allocating process from here
#define PROCSIZE    0x100        // 1MB is the size of each process

void seginit(void) {
    struct cpu *c;
    c->gdt[SEG_KCODE] = SEG(STA_X|STA_R, 0, 0xffffffff, 0);
    c->gdt[SEG_KDATA] = SEG(STA_W, 0, 0xffffffff, 0);
    c->gdt[SEG_UCODE] = SEG(STA_X|STA_R, STARTPROC, (PROCSIZE << 12), DPL_USER);
    c->gdt[SEG_UDATA] = SEG(STA_W, STARTPROC, (PROCSIZE << 12), DPL_USER);
    lgdt(c->gdt, sizeof(c->gdt));
}
```


Understanding swtch

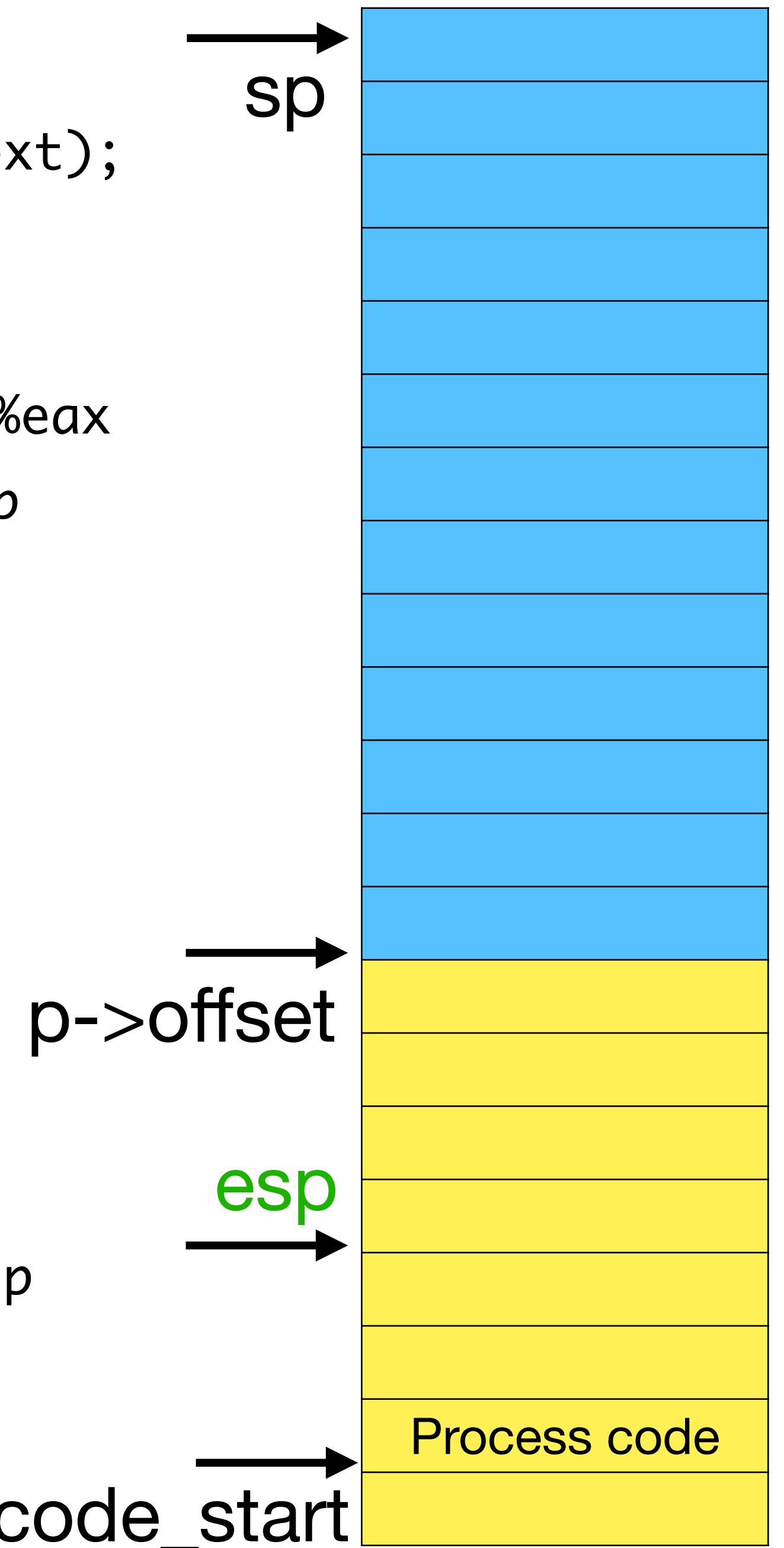
```
pinit(){
    p = allocproc();
    memmove(p->offset, _binary_initcode_start,);
    p->tf->ds,es,ss = (SEG_UDATA<<3) | DPL_USR;
    p->tf->cs = (SEG_UCODE<<3) | DPL_USR;
    p->tf->eflags = FL_IF;
    p->tf->eip = 0;
}
```

```
eip
→ allocproc() {
    sp = (char*)(STARTPROC + (PROCSIZE<<12));
    sp -= sizeof *p->tf;
    p->tf = (struct trapframe*)sp;
    sp -= sizeof *p->context;
    p->context = (struct context*)sp;
    p->context->eip = (uint)trapret;
    return p;
}
```

```
scheduler() {
    ...
    swtch(p->context);
}
```

```
swtch:
    movl 4(%esp), %eax
    movl %eax, %esp
    movl $0, %eax
    ret
```

```
trapret:
    popal
    popl %gs
    popl %fs
    popl %es
    popl %ds
    addl $0x8, %esp
    iret
```



Understanding swtch

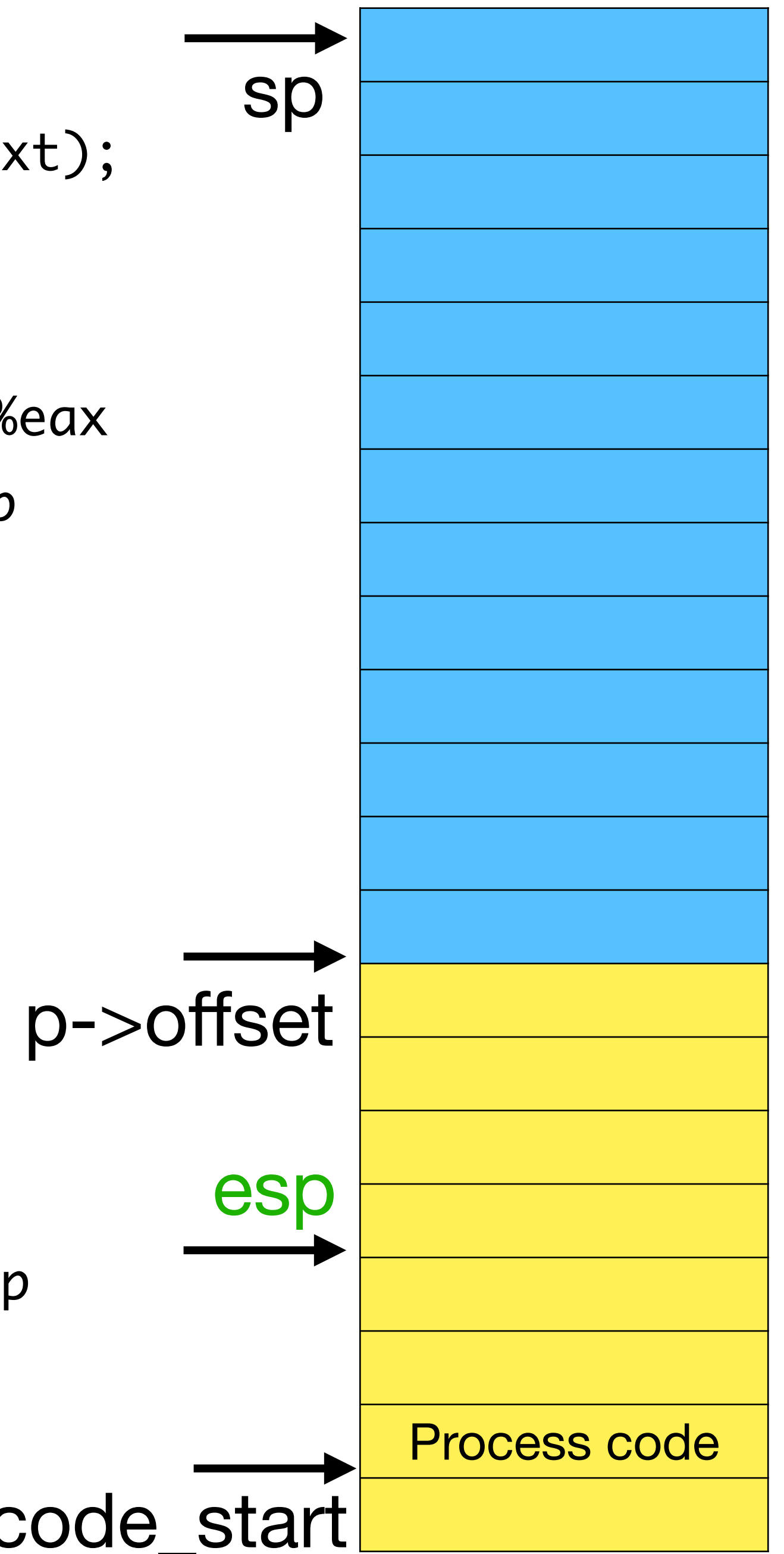
```
pinit(){
    p = allocproc();
    memmove(p->offset, _binary_initcode_start,);
    p->tf->ds,es,ss = (SEG_UDATA<<3) | DPL_USR;
    p->tf->cs = (SEG_UCODE<<3) | DPL_USR;
    p->tf->eflags = FL_IF;
    p->tf->eip = 0;
}
```

```
allocproc() {
    sp = (char*)(STARTPROC + (PROCSIZE<<12));
    eip → sp -= sizeof *p->tf;
    p->tf = (struct trapframe*)sp;
    sp -= sizeof *p->context;
    p->context = (struct context*)sp;
    p->context->eip = (uint)trapret;
    return p;
}
```

```
scheduler() {
    ...
    swtch(p->context);
}
```

```
swtch:
    movl 4(%esp), %eax
    movl %eax, %esp
    movl $0, %eax
    ret
```

```
trapret:
    popal
    popl %gs
    popl %fs
    popl %es
    popl %ds
    addl $0x8, %esp
    iret
```



Understanding swtch

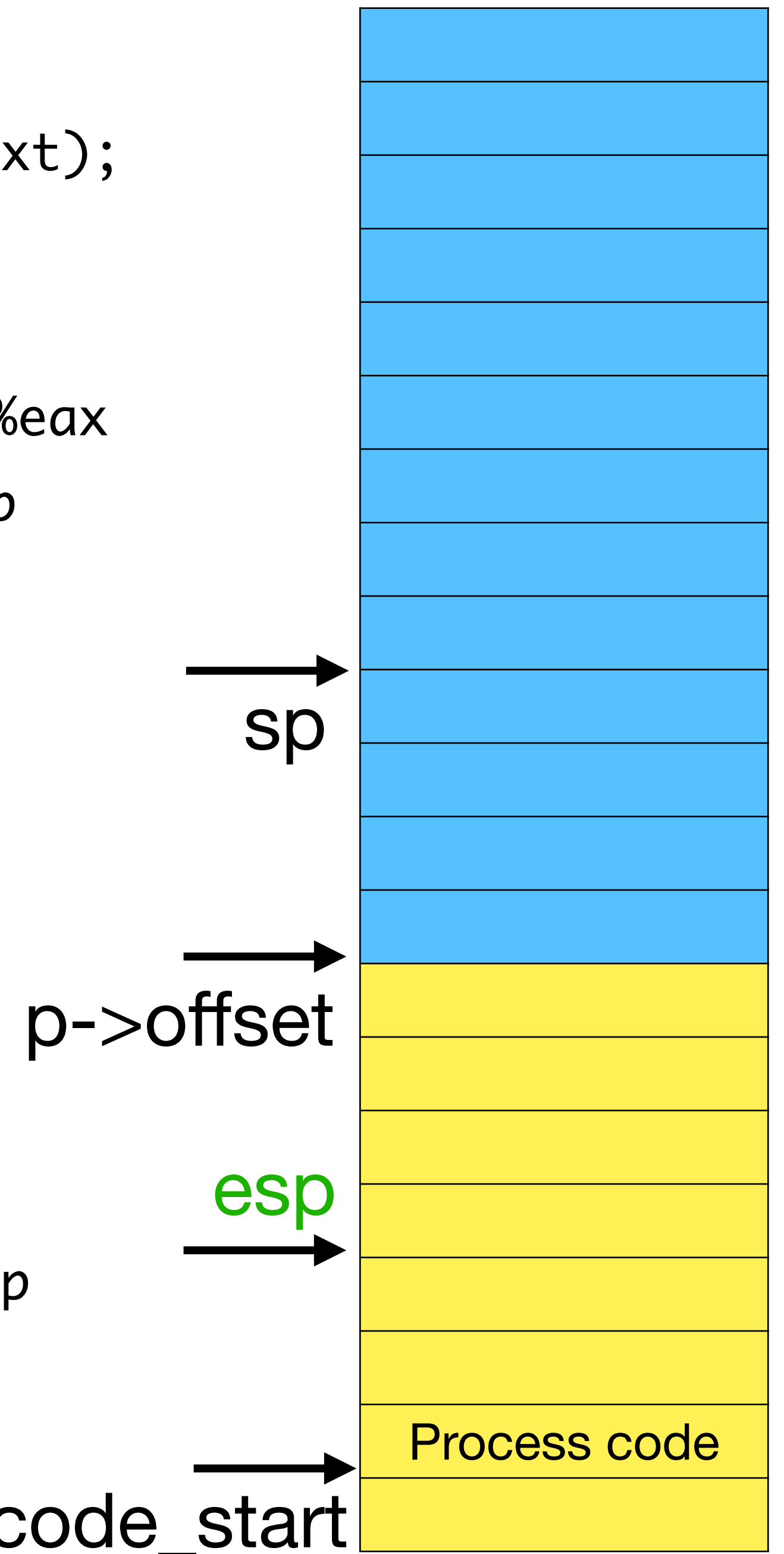
```
pinit(){
    p = allocproc();
    memmove(p->offset, _binary_initcode_start,);
    p->tf->ds,es,ss = (SEG_UDATA<<3) | DPL_USR;
    p->tf->cs = (SEG_UCODE<<3) | DPL_USR;
    p->tf->eflags = FL_IF;
    p->tf->eip = 0;
}
```

```
allocproc() {
    sp = (char*)(STARTPROC + (PROCSIZE<<12));
    sp -= sizeof *p->tf;
    eip → p->tf = (struct trapframe*)sp;
    sp -= sizeof *p->context;
    p->context = (struct context*)sp;
    p->context->eip = (uint)trapret;
    return p;
}
```

```
scheduler() {
    ...
    swtch(p->context);
}
```

```
swtch:
    movl 4(%esp), %eax
    movl %eax, %esp
    movl $0, %eax
    ret
```

```
trapret:
    popal
    popl %gs
    popl %fs
    popl %es
    popl %ds
    addl $0x8, %esp
    iret
```



Understanding swtch

```
pinit(){
    p = allocproc();
    memmove(p->offset, _binary_initcode_start,);
    p->tf->ds,es,ss = (SEG_UDATA<<3) | DPL_USR;
    p->tf->cs = (SEG_UCODE<<3) | DPL_USR;
    p->tf->eflags = FL_IF;
    p->tf->eip = 0;
}
```

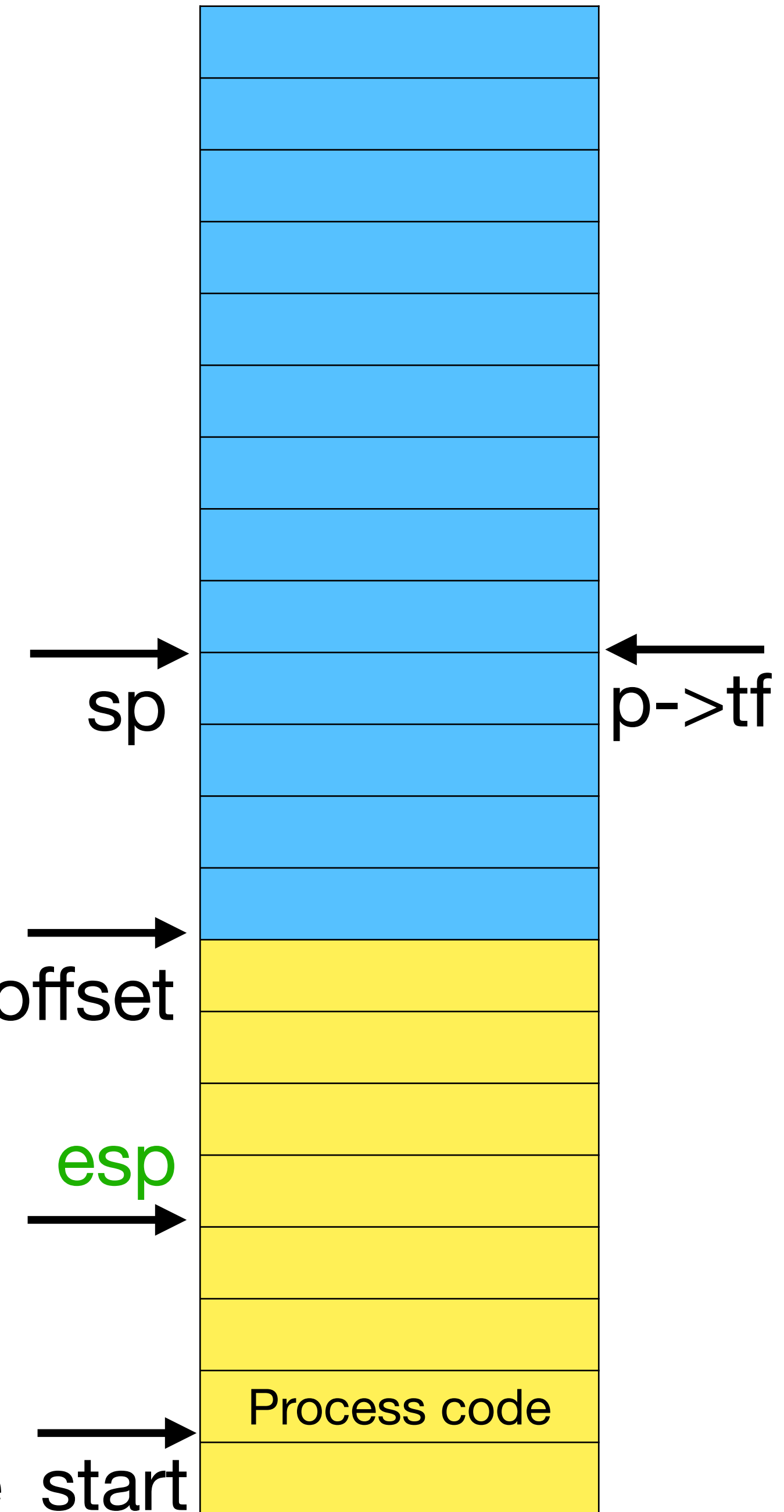
```
allocproc() {
    sp = (char*)(STARTPROC + (PROCSIZE<<12));
    sp -= sizeof *p->tf;
    p->tf = (struct trapframe*)sp;
    sp -= sizeof *p->context;
    p->context = (struct context*)sp;
    p->context->eip = (uint)trapret;
    return p;
}
```

```
scheduler() {  
    ...  
    swtch(p->context);  
}
```

```
switch:
    movl 4(%esp), %eax
    movl %eax, %esp
    movl $0, %eax
    ret
```

```
trapret:
    popal
    popl %gs
    popl %fs
    popl %es
    popl %ds
    addl $0x8, %esp
    iret
```

_binary_initcode_start



Understanding swtch

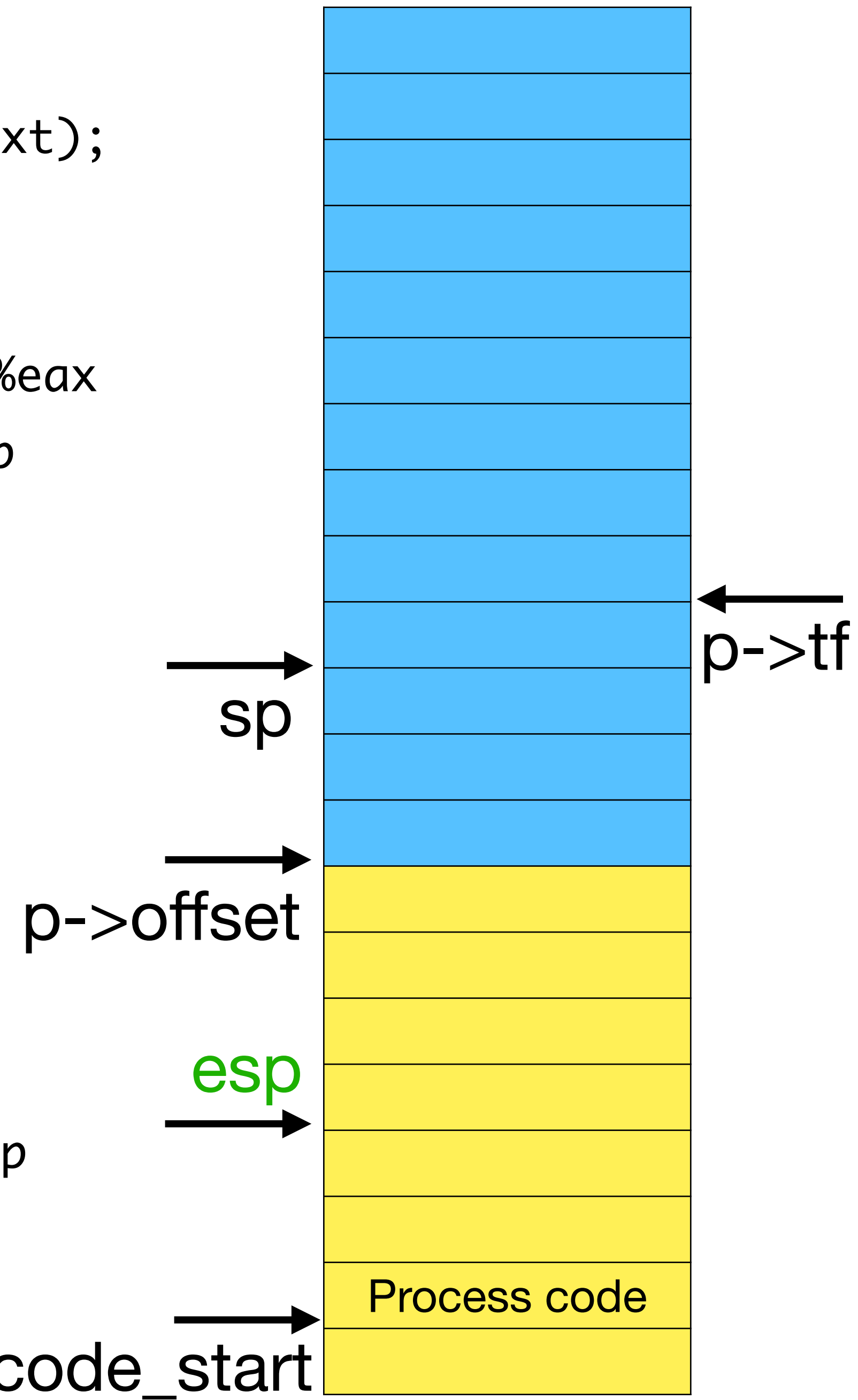
```
pinit(){
    p = allocproc();
    memmove(p->offset, _binary_initcode_start,);
    p->tf->ds,es,ss = (SEG_UDATA<<3) | DPL_USR;
    p->tf->cs = (SEG_UCODE<<3) | DPL_USR;
    p->tf->eflags = FL_IF;
    p->tf->eip = 0;
}
```

```
allocproc() {
    sp = (char*)(STARTPROC + (PROCSIZE<<12));
    sp -= sizeof *p->tf;
    p->tf = (struct trapframe*)sp;
    sp -= sizeof *p->context;
    p->context = (struct context*)sp;
    p->context->eip = (uint)trapret;
    return p;
}
```

```
scheduler() {
    ...
    swtch(p->context);
}
```

```
swtch:
    movl 4(%esp), %eax
    movl %eax, %esp
    movl $0, %eax
    ret
```

```
trapret:
    popal
    popl %gs
    popl %fs
    popl %es
    popl %ds
    addl $0x8, %esp
    iret
```



Understanding swtch

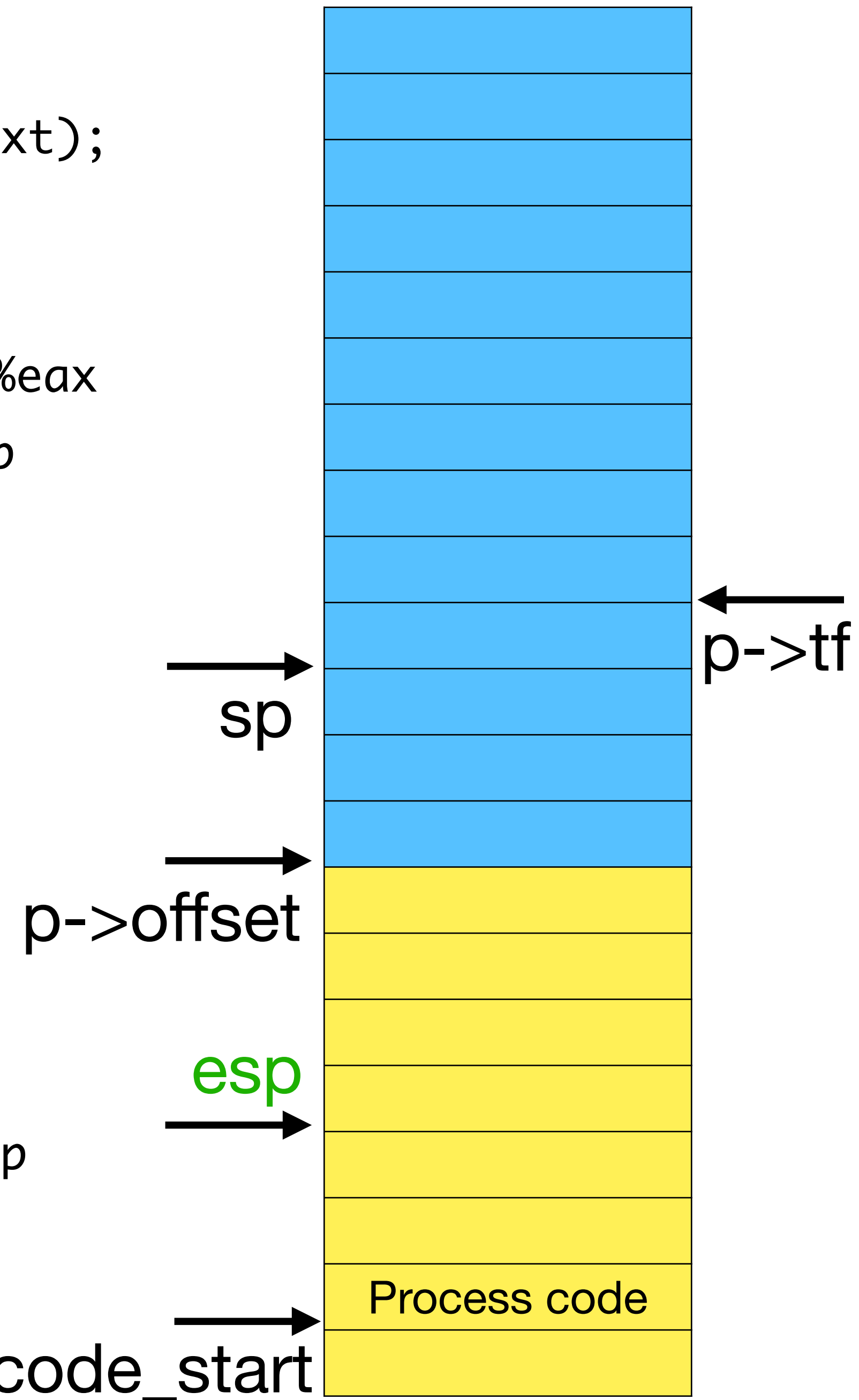
```
pinit(){
    p = allocproc();
    memmove(p->offset, _binary_initcode_start,);
    p->tf->ds,es,ss = (SEG_UDATA<<3) | DPL_USR;
    p->tf->cs = (SEG_UCODE<<3) | DPL_USR;
    p->tf->eflags = FL_IF;
    p->tf->eip = 0;
}
```

```
allocproc() {
    sp = (char*)(STARTPROC + (PROCSIZE<<12));
    sp -= sizeof *p->tf;
    p->tf = (struct trapframe*)sp;
    sp -= sizeof *p->context;
    p->context = (struct context*)sp;
    p->context->eip = (uint)trapret;
    return p;
}
```

```
scheduler() {
    ...
    swtch(p->context);
}
```

```
swtch:
    movl 4(%esp), %eax
    movl %eax, %esp
    movl $0, %eax
    ret
```

```
trapret:
    popal
    popl %gs
    popl %fs
    popl %es
    popl %ds
    addl $0x8, %esp
    iret
```



Understanding swtch

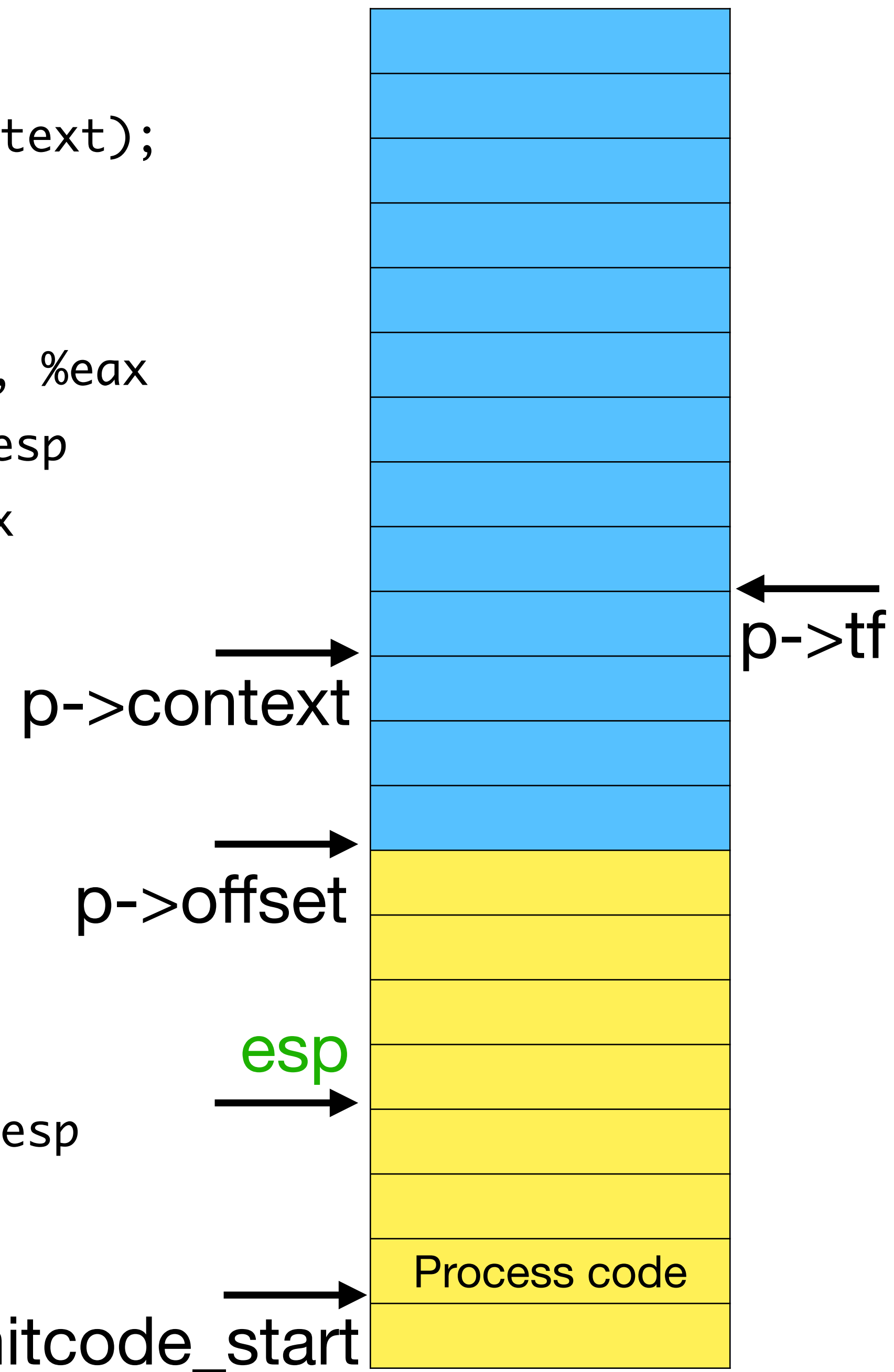
```
pinit(){
    p = allocproc();
    memmove(p->offset, _binary_initcode_start,);
    p->tf->ds,es,ss = (SEG_UDATA<<3) | DPL_USR;
    p->tf->cs = (SEG_UCODE<<3) | DPL_USR;
    p->tf->eflags = FL_IF;
    p->tf->eip = 0;
}
```

```
allocproc() {
    sp = (char*)(STARTPROC + (PROCSIZE<<12));
    sp -= sizeof *p->tf;
    p->tf = (struct trapframe*)sp;
    sp -= sizeof *p->context;
    p->context = (struct context*)sp;
    p->context->eip = (uint)trapret;
    return p;
}
```

```
scheduler() {
    ...
    swtch(p->context);
}
```

```
swtch:
    movl 4(%esp), %eax
    movl %eax, %esp
    movl $0, %eax
    ret
```

```
trapret:
    popal
    popl %gs
    popl %fs
    popl %es
    popl %ds
    addl $0x8, %esp
    iret
```



Understanding swtch

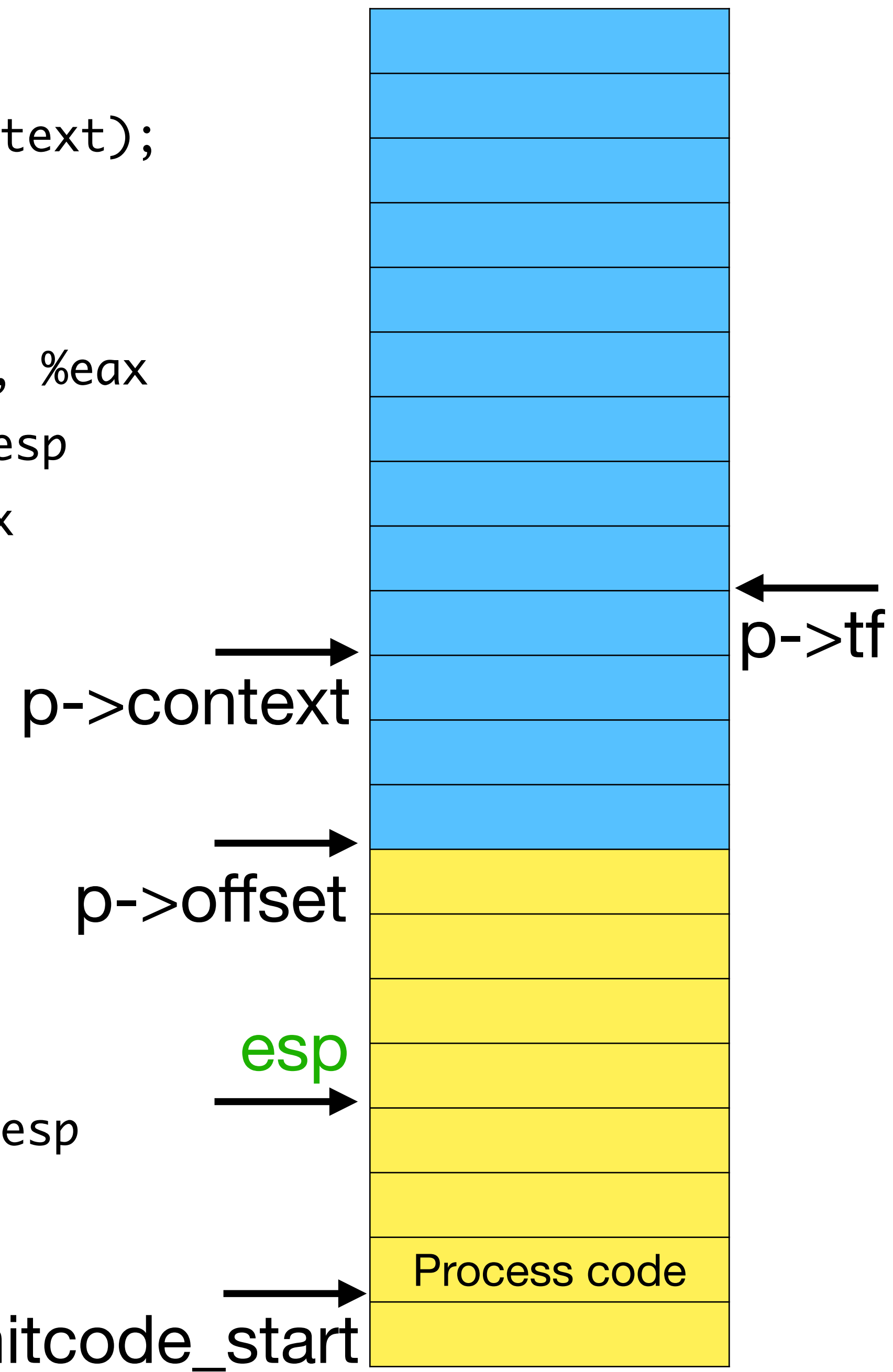
```
pinit(){
    p = allocproc();
    memmove(p->offset, _binary_initcode_start,);
    p->tf->ds,es,ss = (SEG_UDATA<<3) | DPL_USR;
    p->tf->cs = (SEG_UCODE<<3) | DPL_USR;
    p->tf->eflags = FL_IF;
    p->tf->eip = 0;
}
```

```
allocproc() {
    sp = (char*)(STARTPROC + (PROCSIZE<<12));
    sp -= sizeof *p->tf;
    p->tf = (struct trapframe*)sp;
    sp -= sizeof *p->context;
    p->context = (struct context*)sp;
    eip → p->context->eip = (uint)trapret;
    return p;
}
```

```
scheduler() {
    ...
    swtch(p->context);
}
```

```
swtch:
    movl 4(%esp), %eax
    movl %eax, %esp
    movl $0, %eax
    ret
```

```
trapret:
    popal
    popl %gs
    popl %fs
    popl %es
    popl %ds
    addl $0x8, %esp
    iret
```



Understanding swtch

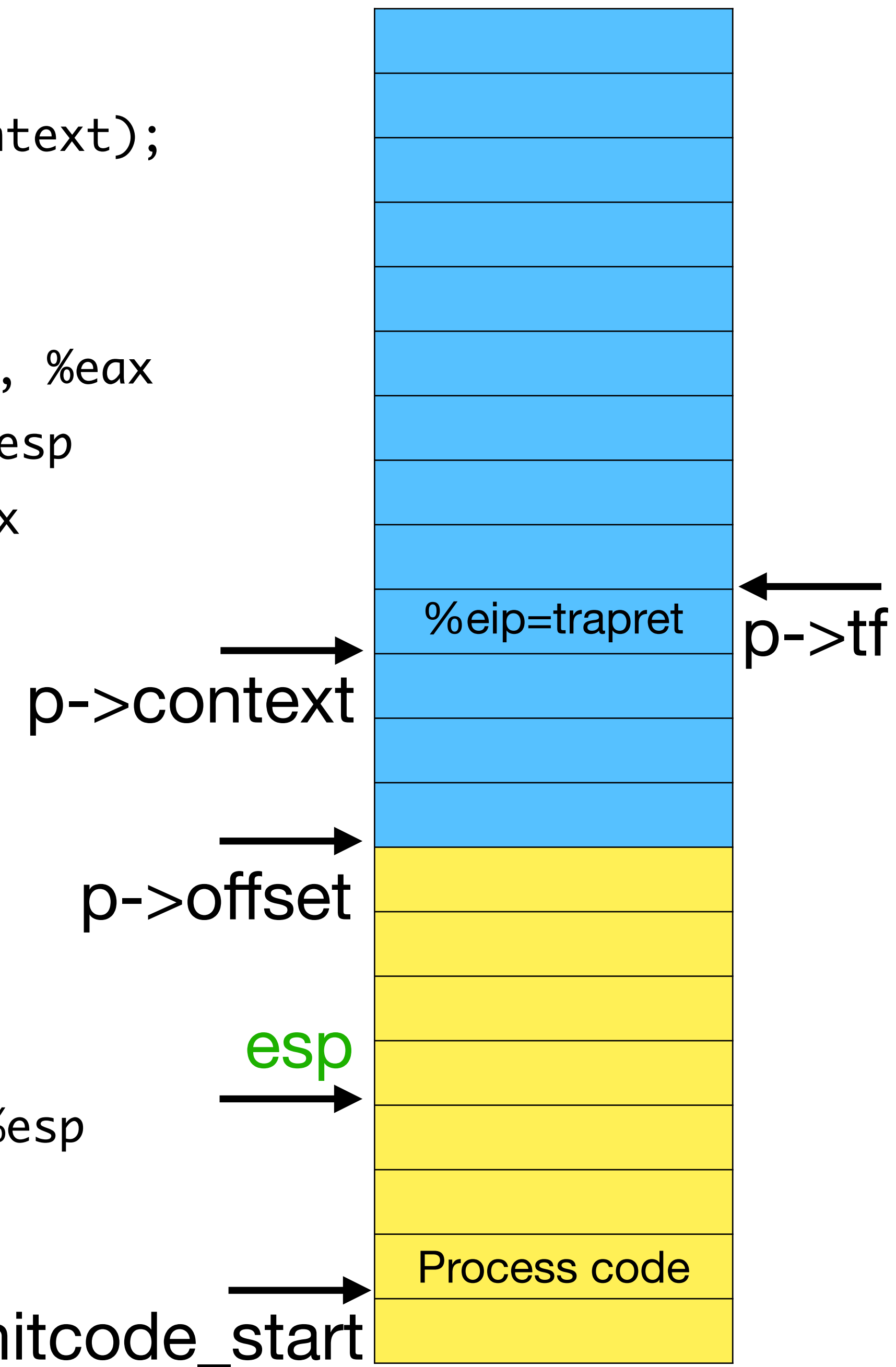
```
pinit(){
    p = allocproc();
    memmove(p->offset, _binary_initcode_start,);
    p->tf->ds,es,ss = (SEG_UDATA<<3) | DPL_USR;
    p->tf->cs = (SEG_UCODE<<3) | DPL_USR;
    p->tf->eflags = FL_IF;
    p->tf->eip = 0;
}
```

```
allocproc() {
    sp = (char*)(STARTPROC + (PROCSIZE<<12));
    sp -= sizeof *p->tf;
    p->tf = (struct trapframe*)sp;
    sp -= sizeof *p->context;
    p->context = (struct context*)sp;
    eip → p->context->eip = (uint)trapret;
    return p;
}
```

```
scheduler() {
    ...
    swtch(p->context);
}
```

```
swtch:
    movl 4(%esp), %eax
    movl %eax, %esp
    movl $0, %eax
    ret
```

```
trapret:
    popal
    popl %gs
    popl %fs
    popl %es
    popl %ds
    addl $0x8, %esp
    iret
```



Understanding swtch

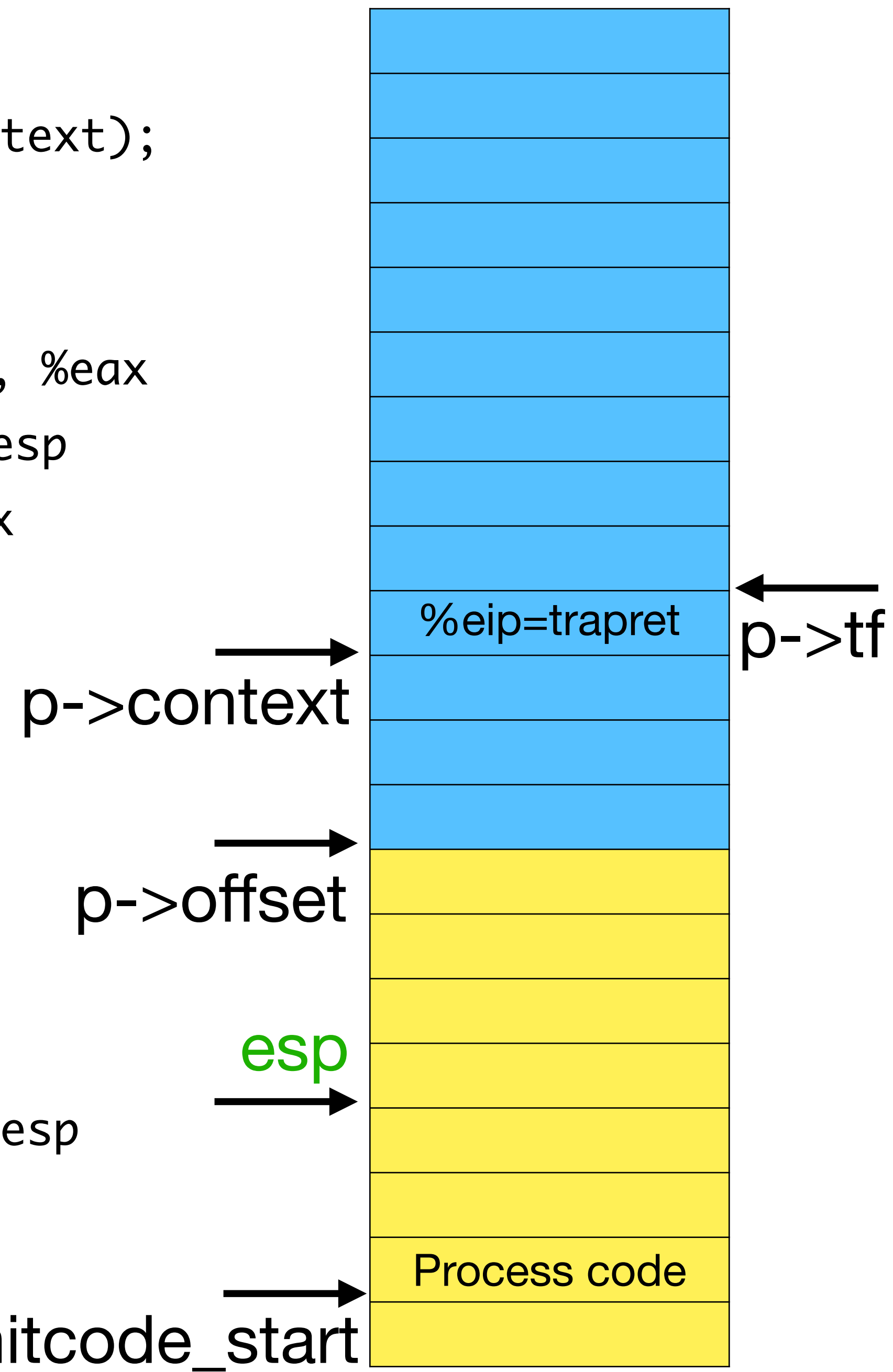
```
pinit(){
    p = allocproc();
    memmove(p->offset, _binary_initcode_start,);
    p->tf->ds,es,ss = (SEG_UDATA<<3) | DPL_USR;
    p->tf->cs = (SEG_UCODE<<3) | DPL_USR;
    p->tf->eflags = FL_IF;
    p->tf->eip = 0;
}
```

```
allocproc() {
    sp = (char*)(STARTPROC + (PROCSIZE<<12));
    sp -= sizeof *p->tf;
    p->tf = (struct trapframe*)sp;
    sp -= sizeof *p->context;
    p->context = (struct context*)sp;
    p->context->eip = (uint)trapret;
    return p;
}
```

```
scheduler() {
    ...
    swtch(p->context);
}
```

```
swtch:
    movl 4(%esp), %eax
    movl %eax, %esp
    movl $0, %eax
    ret
```

```
trapret:
    popal
    popl %gs
    popl %fs
    popl %es
    popl %ds
    addl $0x8, %esp
    iret
```



Understanding swtch

```
pinit(){
```

```
  eip → p = allocproc();
  memmove(p->offset, _binary_initcode_start,);
  p->tf->ds,es,ss = (SEG_UDATA<<3) | DPL_USR;
  p->tf->cs = (SEG_UCODE<<3) | DPL_USR;
  p->tf->eflags = FL_IF;
  p->tf->eip = 0;
}
```

```
allocproc() {
```

```
  sp = (char*)(STARTPROC + (PROCSIZE<<12));
  sp -= sizeof *p->tf;
  p->tf = (struct trapframe*)sp;
  sp -= sizeof *p->context;
  p->context = (struct context*)sp;
  p->context->eip = (uint)trapret;
  return p;
}
```

```
scheduler() {
```

```
  ...
```

```
  swtch(p->context);
```

```
}
```

```
swtch:
```

```
  movl 4(%esp), %eax
```

```
  movl %eax, %esp
```

```
  movl $0, %eax
```

```
  ret
```

```
trapret:
```

```
  popal
```

```
  popl %gs
```

```
  popl %fs
```

```
  popl %es
```

```
  popl %ds
```

```
  addl $0x8, %esp
```

```
  iret
```

```
_binary_initcode_start
```

→ p->context

→ p->offset

→ esp

← %eip=trapret

← p->tf

Process code

Understanding swtch

```
pinit(){
```

```
  eip → p = allocproc();
  memmove(p->offset, _binary_initcode_start,);
  p->tf->ds,es,ss = (SEG_UDATA<<3) | DPL_USR;
  p->tf->cs = (SEG_UCODE<<3) | DPL_USR;
  p->tf->eflags = FL_IF;
  p->tf->eip = 0;
}
```

```
allocproc() {
```

```
  sp = (char*)(STARTPROC + (PROCSIZE<<12));
  sp -= sizeof *p->tf;
  p->tf = (struct trapframe*)sp;
  sp -= sizeof *p->context;
  p->context = (struct context*)sp;
  p->context->eip = (uint)trapret;
  return p;
}
```

```
scheduler() {
```

```
  ...
```

```
  swtch(p->context);
```

```
}
```

```
swtch:
```

```
  movl 4(%esp), %eax
```

```
  movl %eax, %esp
```

```
  movl $0, %eax
```

```
  ret
```

```
trapret:
```

```
  popal
```

```
  popl %gs
```

```
  popl %fs
```

```
  popl %es
```

```
  popl %ds
```

```
  addl $0x8, %esp
```

```
  iret
```

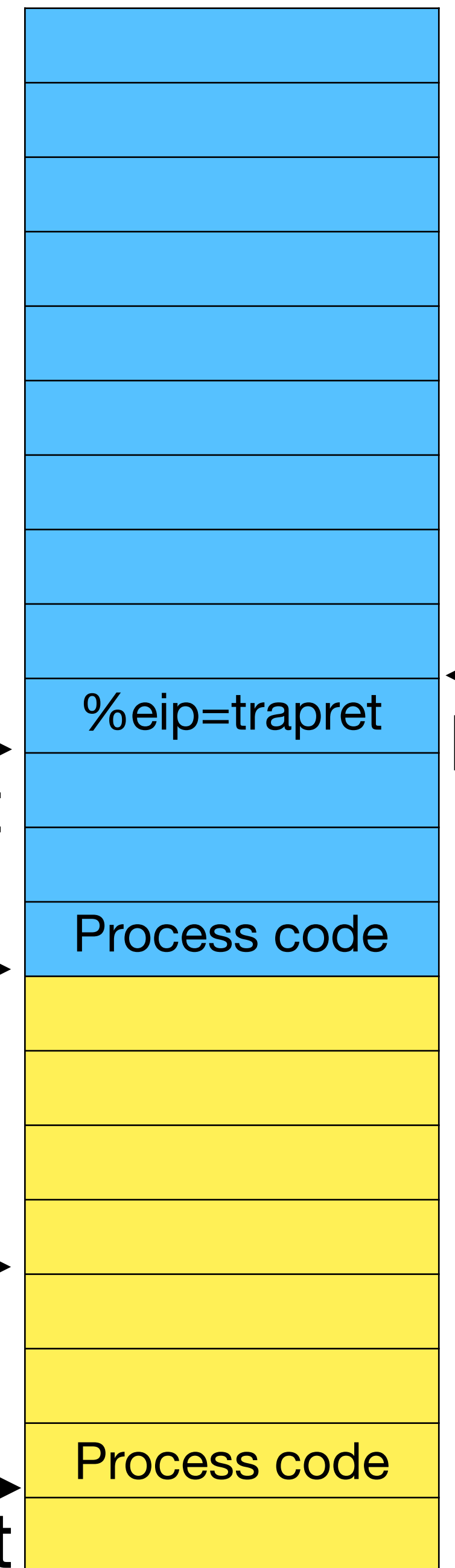
```
_binary_initcode_start
```

→ p->context

→ p->offset

→ esp

← p->tf



Understanding swtch

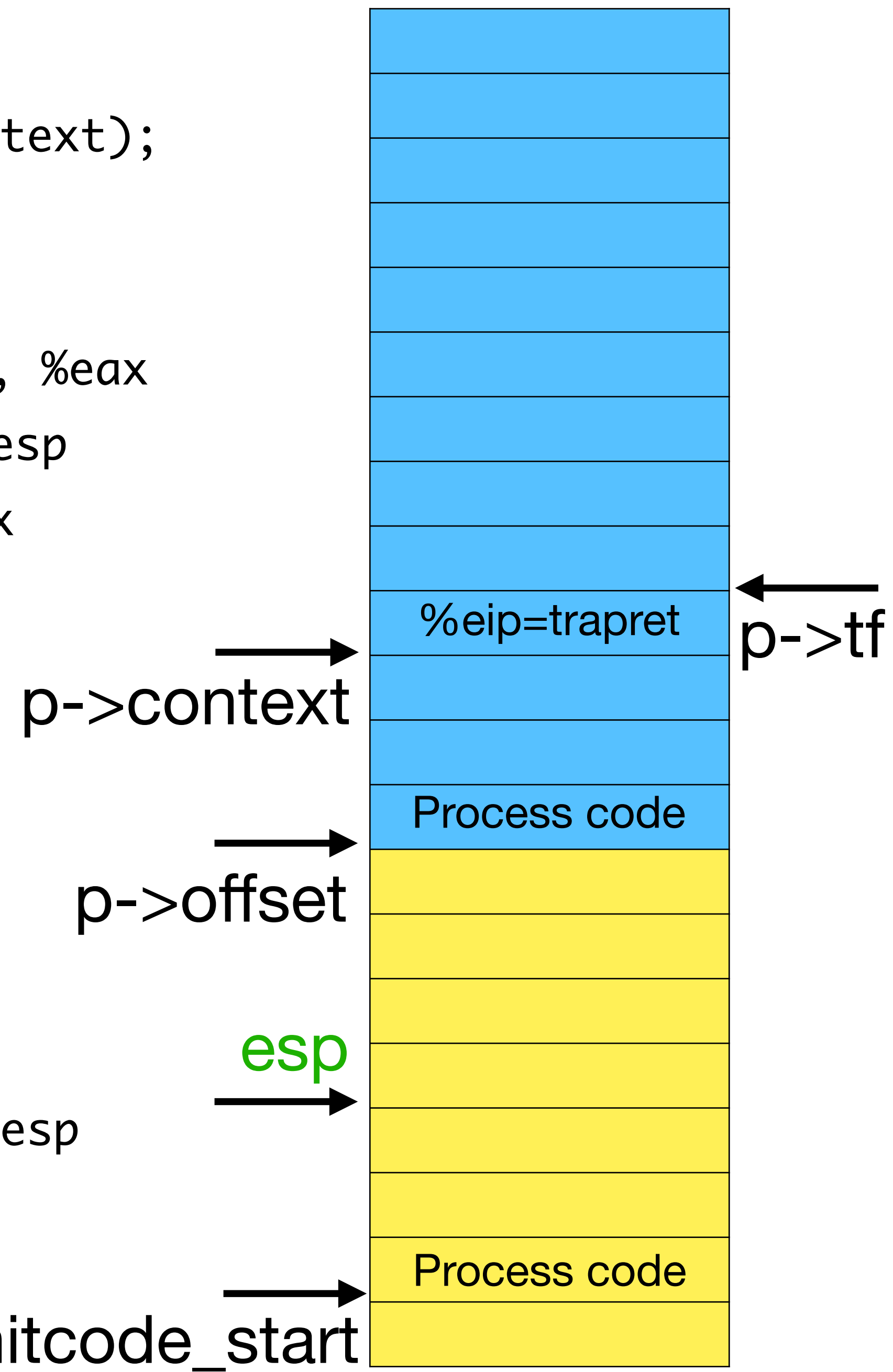
```
pinit(){
    p = allocproc();
    memmove(p->offset, _binary_initcode_start,);
    p->tf->ds,es,ss = (SEG_UDATA<<3) | DPL_USR;
    p->tf->cs = (SEG_UCODE<<3) | DPL_USR;
    p->tf->eflags = FL_IF;
    p->tf->eip = 0;
}
```

```
allocproc() {
    sp = (char*)(STARTPROC + (PROCSIZE<<12));
    sp -= sizeof *p->tf;
    p->tf = (struct trapframe*)sp;
    sp -= sizeof *p->context;
    p->context = (struct context*)sp;
    p->context->eip = (uint)trapret;
    return p;
}
```

```
scheduler() {
    ...
    swtch(p->context);
}
```

```
swtch:
    movl 4(%esp), %eax
    movl %eax, %esp
    movl $0, %eax
    ret
```

```
trapret:
    popal
    popl %gs
    popl %fs
    popl %es
    popl %ds
    addl $0x8, %esp
    iret
```



Understanding swtch

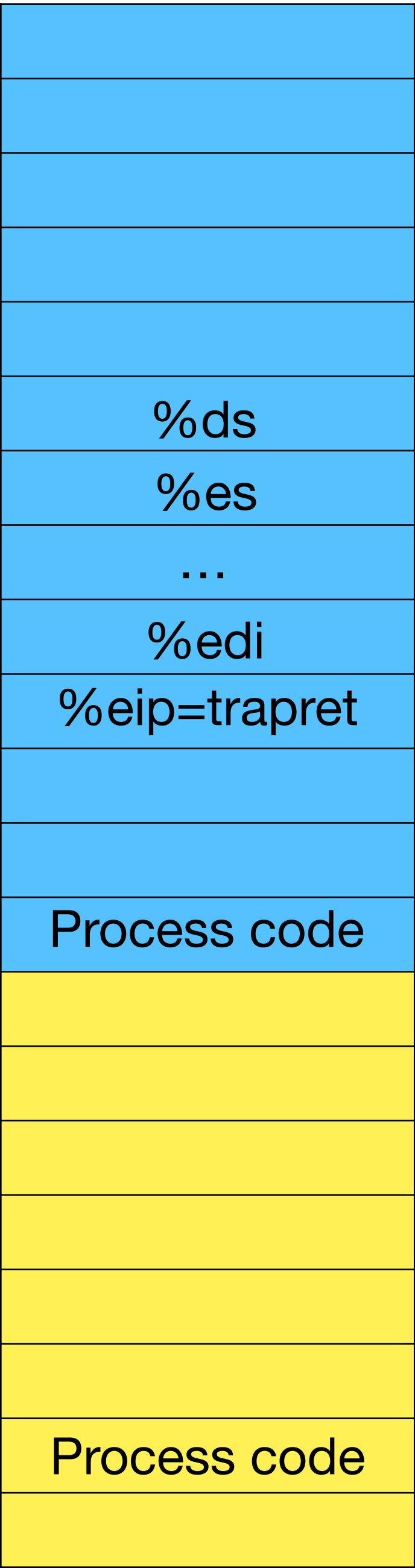
```
pinit(){
    p = allocproc();
    memmove(p->offset, _binary_initcode_start,);
    p->tf->ds,es,ss = (SEG_UDATA<<3) | DPL_USR;
    p->tf->cs = (SEG_UCODE<<3) | DPL_USR;
    p->tf->eflags = FL_IF;
    p->tf->eip = 0;
}

allocproc() {
    sp = (char*)(STARTPROC + (PROCSIZE<<12));
    sp -= sizeof *p->tf;
    p->tf = (struct trapframe*)sp;
    sp -= sizeof *p->context;
    p->context = (struct context*)sp;
    p->context->eip = (uint)trapret;
    return p;
}
```

```
scheduler() {
    ...
    swtch(p->context);
}
```

```
swtch:
    movl 4(%esp), %eax
    movl %eax, %esp
    movl $0, %eax
    ret

trapret:
    popal
    popl %gs
    popl %fs
    popl %es
    popl %ds
    addl $0x8, %esp
    iret
```



Understanding swtch

```
pinit(){
    p = allocproc();
    memmove(p->offset, _binary_initcode_start,);
    p->tf->ds,es,ss = (SEG_UDATA<<3) | DPL_USR;
    p->tf->cs = (SEG_UCODE<<3) | DPL_USR;
    p->tf->eflags = FL_IF;
    p->tf->eip = 0;
}
```

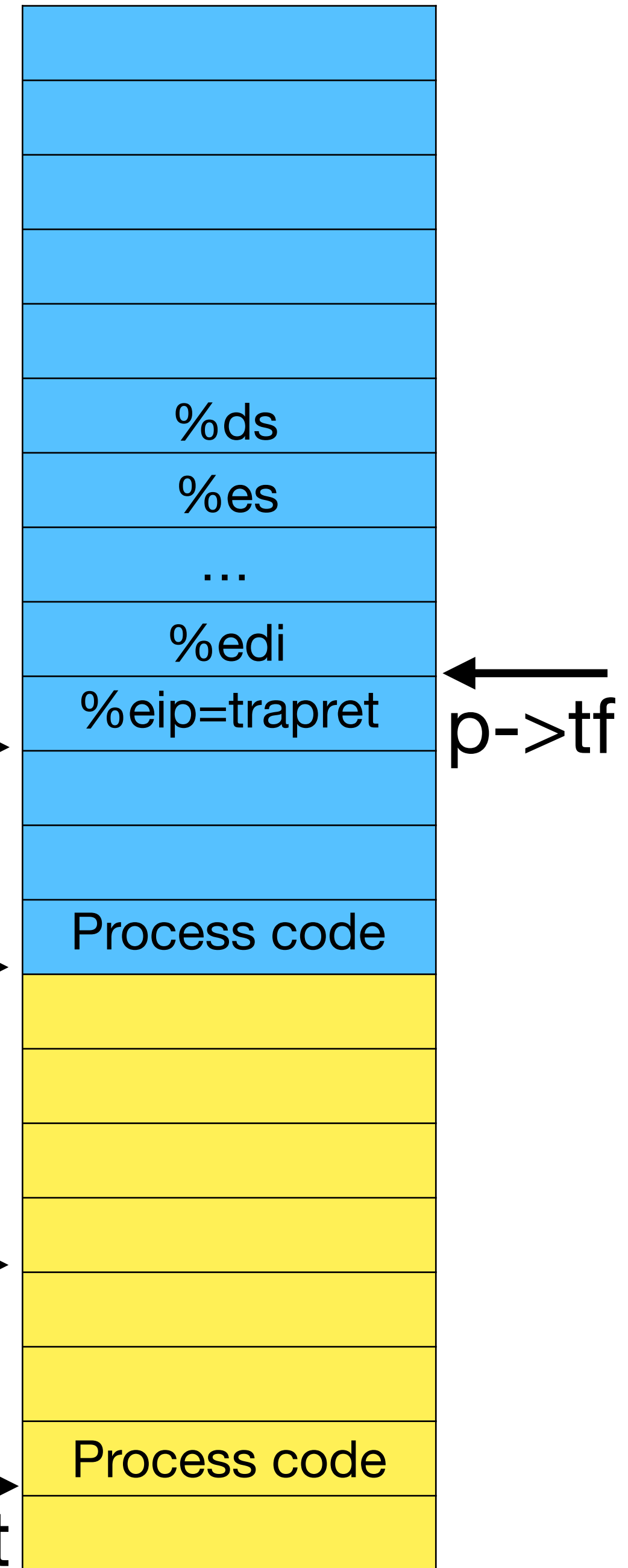
```
allocproc() {
    sp = (char*)(STARTPROC + (PROCSIZE<<12));
    sp -= sizeof *p->tf;
    p->tf = (struct trapframe*)sp;
    sp -= sizeof *p->context;
    p->context = (struct context*)sp;
    p->context->eip = (uint)trapret;
    return p;
}
```

```
scheduler() {  
    ...  
    swtch(p->context);  
}
```

```
switch:
    movl 4(%esp), %eax
    movl %eax, %esp
    movl $0, %eax
    ret
```

```
trapret:                                p-
    popal
    popl %gs
    popl %fs
    popl %es
    popl %ds
    addl $0x8, %esp
    iret
```

_binary_initcode_start



Understanding swtch

```
pinit(){
    p = allocproc();
    memmove(p->offset, _binary_initcode_start,);
    p->tf->ds,es,ss = (SEG_UDATA<<3) | DPL_USR;
    p->tf->cs = (SEG_UCODE<<3) | DPL_USR;
    p->tf->eflags = FL_IF;
    eip → p->tf->eip = 0;
}

allocproc() {
    sp = (char*)(STARTPROC + (PROCSIZE<<12));
    sp -= sizeof *p->tf;
    p->tf = (struct trapframe*)sp;
    sp -= sizeof *p->context;
    p->context = (struct context*)sp;
    p->context->eip = (uint)trapret;
    return p;
}
```

```
scheduler() {
    ...
    swtch(p->context);
}
```

```
swtch:
    movl 4(%esp), %eax
    movl %eax, %esp
    movl $0, %eax
    ret

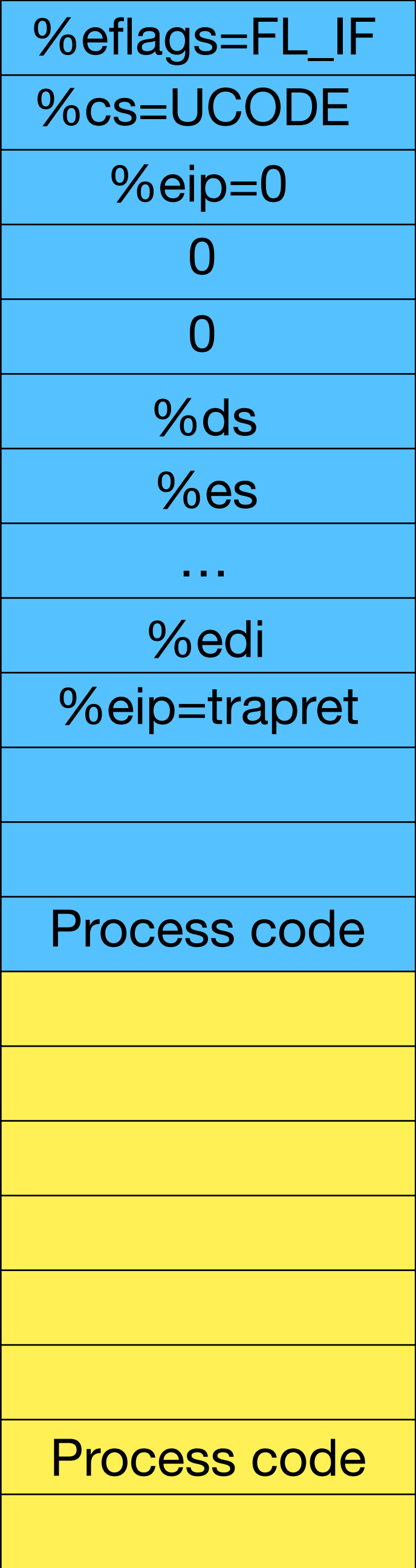
trapret:
    popal
    popl %gs
    popl %fs
    popl %es
    popl %ds
    addl $0x8, %esp
    iret
```

→ p->context

→ p->offset

→ esp

→ _binary_initcode_start



← p->tf

Understanding switch

```
pinit(){
```

```
    p = allocproc();
```

```
    memmove(p->offset, _binary_initcode_start,);
```

```
    p->tf->ds,es,ss = (SEG_UDATA<<3) | DPL_USR;
```

```
    p->tf->cs = (SEG_UCODE<<3) | DPL_USR;
```

```
    p->tf->eflags = FL_IF;
```

```
    p->tf->eip = 0;
```

```
}
```

```
allocproc() {
```

```
    sp = (char*)(STARTPROC + (PROCSIZE<<12));
```

```
    sp -= sizeof *p->tf;
```

```
    p->tf = (struct trapframe*)sp;
```

```
    sp -= sizeof *p->context;
```

```
    p->context = (struct context*)sp;
```

```
    p->context->eip = (uint)trapret;
```

```
    return p;
```

```
}
```

```
    scheduler() {
```

eip



```
        switch(p->context);
```

```
    }
```

```
switch:
```

```
    movl 4(%esp), %eax
```

```
    movl %eax, %esp
```

```
    movl $0, %eax
```

```
    ret
```

```
trapret:
```

```
    popal
```

```
    popl %gs
```

```
    popl %fs
```

```
    popl %es
```

```
    popl %ds
```

```
    addl $0x8, %esp
```

```
    iret
```

_binary_initcode_start

p->context

p->offset

esp

%eflags=FL_IF
%cs=UCODE
%eip=0
0
0
%ds
%es
...
%edi
%eip=trapret
Process code
Process code

p->tf

Understanding swtch

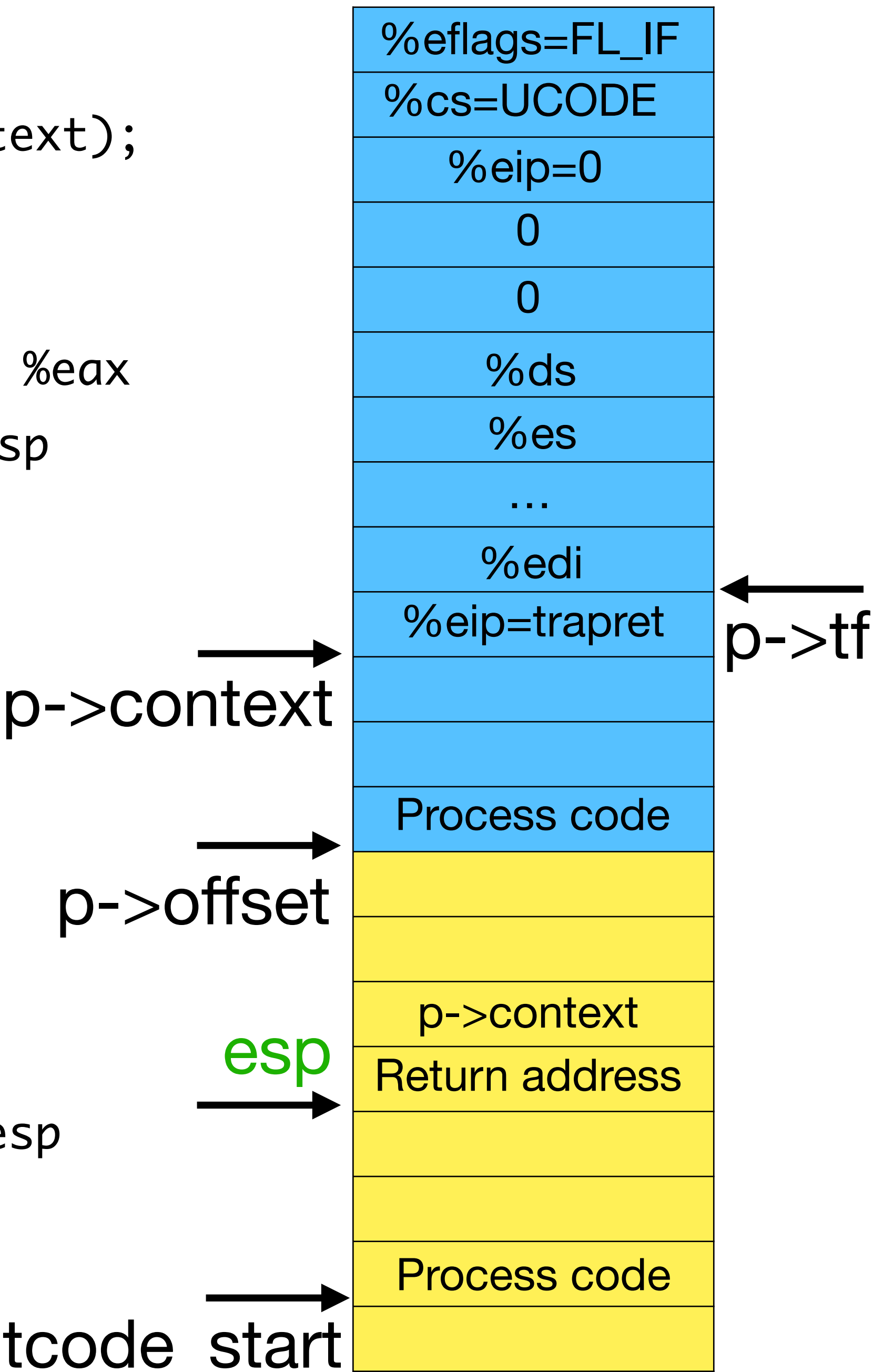
```
pinit(){
    p = allocproc();
    memmove(p->offset, _binary_initcode_start,);
    p->tf->ds,es,ss = (SEG_UDATA<<3) | DPL_USR;
    p->tf->cs = (SEG_UCODE<<3) | DPL_USR;
    p->tf->eflags = FL_IF;
    p->tf->eip = 0;
}

allocproc() {
    sp = (char*)(STARTPROC + (PROCSIZE<<12));
    sp -= sizeof *p->tf;
    p->tf = (struct trapframe*)sp;
    sp -= sizeof *p->context;
    p->context = (struct context*)sp;
    p->context->eip = (uint)trapret;
    return p;
}
```

```
scheduler() {
    ...
    swtch(p->context);
}
```

```
swtch:
    movl 4(%esp), %eax
    movl %eax, %esp
    movl $0, %eax
    ret

trapret:
    popal
    popl %gs
    popl %fs
    popl %es
    popl %ds
    addl $0x8, %esp
    iret
```



Understanding swtch

```
pinit(){
    p = allocproc();
    memmove(p->offset, _binary_initcode_start,);
    p->tf->ds,es,ss = (SEG_UDATA<<3) | DPL_USR;
    p->tf->cs = (SEG_UCODE<<3) | DPL_USR;
    p->tf->eflags = FL_IF;
    p->tf->eip = 0;
}

allocproc() {
    sp = (char*)(STARTPROC + (PROCSIZE<<12));
    sp -= sizeof *p->tf;
    p->tf = (struct trapframe*)sp;
    sp -= sizeof *p->context;
    p->context = (struct context*)sp;
    p->context->eip = (uint)trapret;
    return p;
}
```

```
scheduler() {
    ...
    swtch(p->context);
}
```

```
swtch:
    movl 4(%esp), %eax
    movl %eax, %esp
    movl $0, %eax
    ret
```

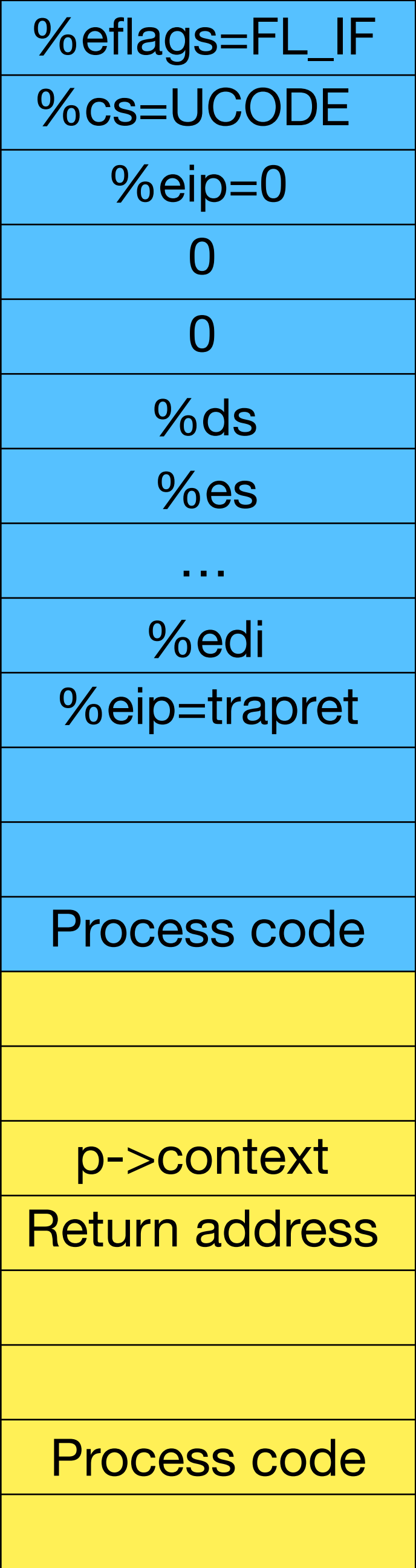
```
trapret:
    popal
    popl %gs
    popl %fs
    popl %es
    popl %ds
    addl $0x8, %esp
    iret
```

p->context

p->offset

esp

_binary_initcode_start



Understanding swtch

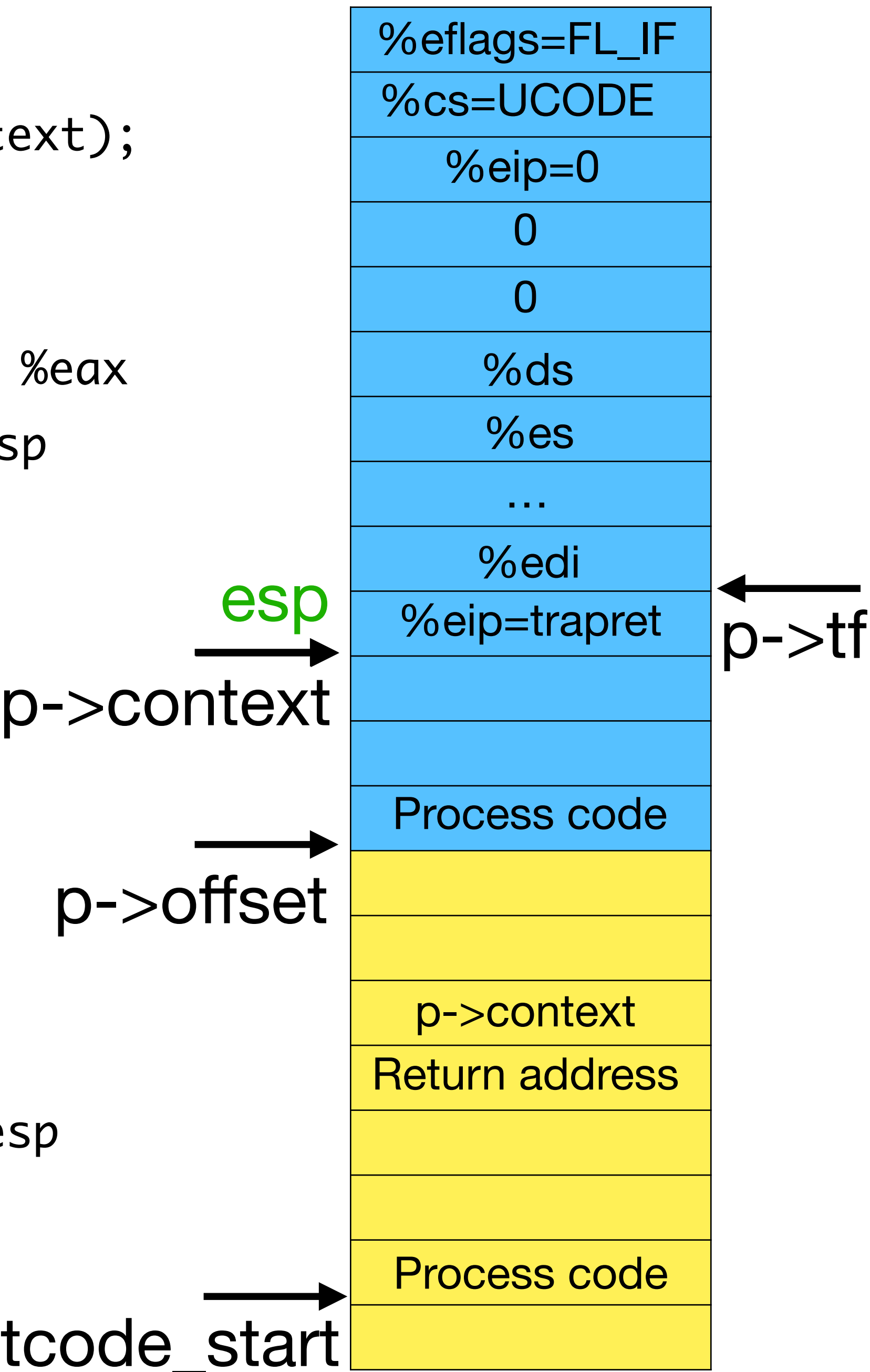
```
pinit(){
    p = allocproc();
    memmove(p->offset, _binary_initcode_start,);
    p->tf->ds,es,ss = (SEG_UDATA<<3) | DPL_USR;
    p->tf->cs = (SEG_UCODE<<3) | DPL_USR;
    p->tf->eflags = FL_IF;
    p->tf->eip = 0;
}

allocproc() {
    sp = (char*)(STARTPROC + (PROCSIZE<<12));
    sp -= sizeof *p->tf;
    p->tf = (struct trapframe*)sp;
    sp -= sizeof *p->context;
    p->context = (struct context*)sp;
    p->context->eip = (uint)trapret;
    return p;
}
```

```
scheduler() {
    ...
    swtch(p->context);
}
```

```
swtch:
    movl 4(%esp), %eax
    movl %eax, %esp
    movl $0, %eax
    ret

trapret:
    popal
    popl %gs
    popl %fs
    popl %es
    popl %ds
    addl $0x8, %esp
    iret
```



Understanding swtch

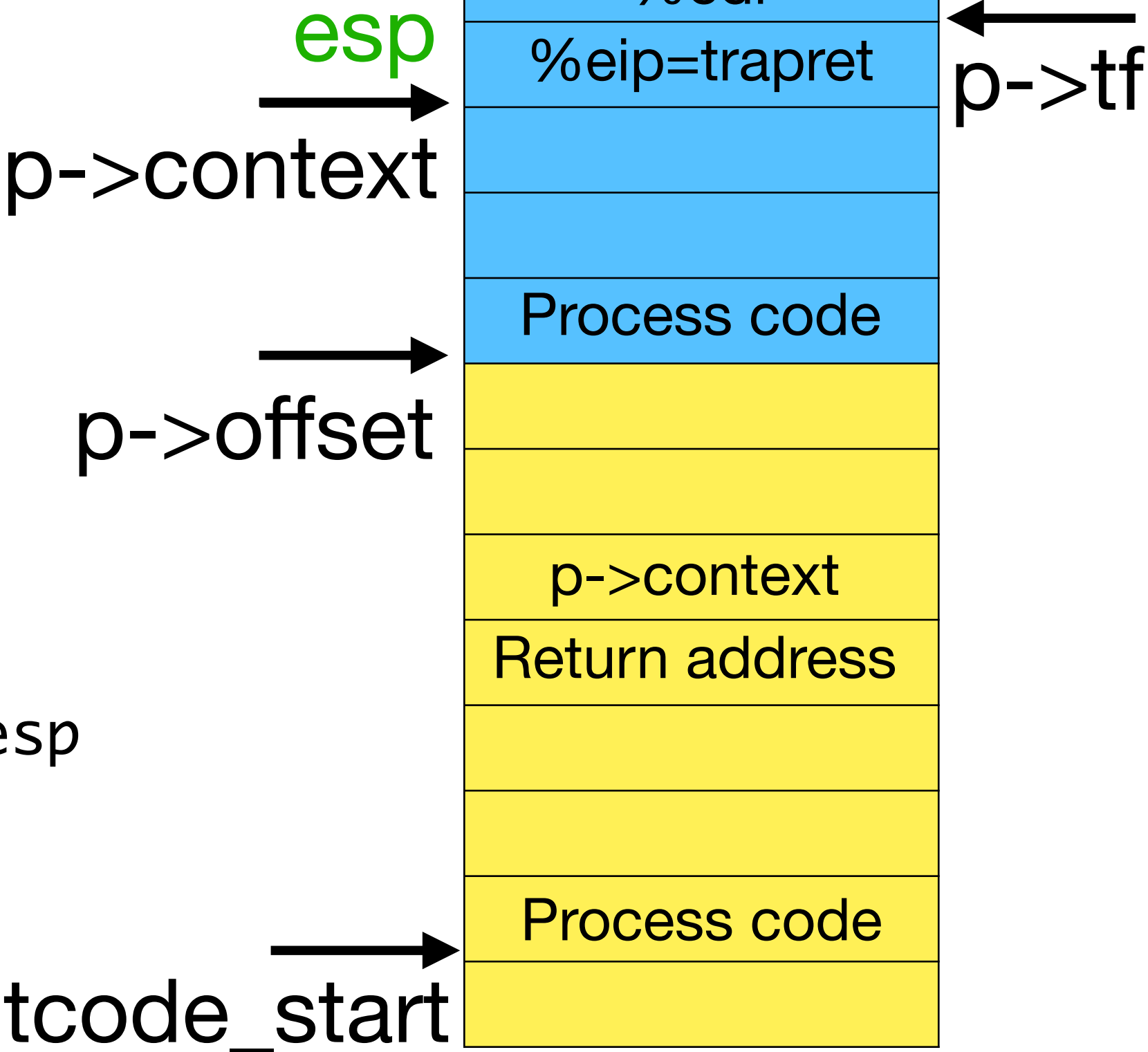
```
pinit(){
    p = allocproc();
    memmove(p->offset, _binary_initcode_start,);
    p->tf->ds,es,ss = (SEG_UDATA<<3) | DPL_USR;
    p->tf->cs = (SEG_UCODE<<3) | DPL_USR;
    p->tf->eflags = FL_IF;
    p->tf->eip = 0;
}

allocproc() {
    sp = (char*)(STARTPROC + (PROCSIZE<<12));
    sp -= sizeof *p->tf;
    p->tf = (struct trapframe*)sp;
    sp -= sizeof *p->context;
    p->context = (struct context*)sp;
    p->context->eip = (uint)trapret;
    return p;
}
```

```
scheduler() {
    ...
    swtch(p->context);
}
```

```
swtch:
    movl 4(%esp), %eax
    movl %eax, %esp
    eip → movl $0, %eax
    ret
```

```
trapret:
    popal
    popl %gs
    popl %fs
    popl %es
    popl %ds
    addl $0x8, %esp
    iret
```

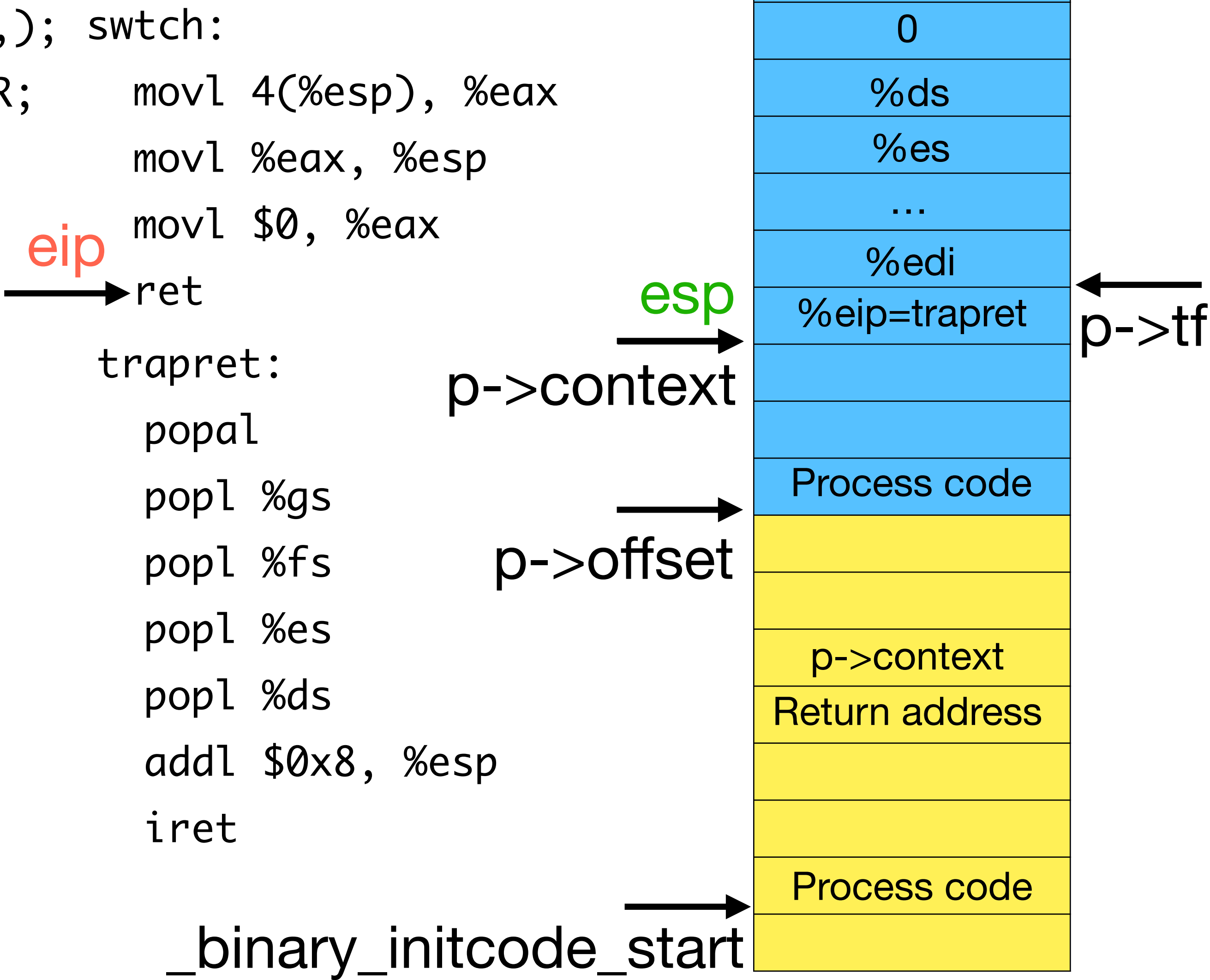


Understanding swtch

```
pinit(){
    p = allocproc();
    memmove(p->offset, _binary_initcode_start,);
    p->tf->ds,es,ss = (SEG_UDATA<<3) | DPL_USR;
    p->tf->cs = (SEG_UCODE<<3) | DPL_USR;
    p->tf->eflags = FL_IF;
    p->tf->eip = 0;
}

allocproc() {
    sp = (char*)(STARTPROC + (PROCSIZE<<12));
    sp -= sizeof *p->tf;
    p->tf = (struct trapframe*)sp;
    sp -= sizeof *p->context;
    p->context = (struct context*)sp;
    p->context->eip = (uint)trapret;
    return p;
}
```

```
scheduler() {
    ...
    swtch(p->context);
}
```

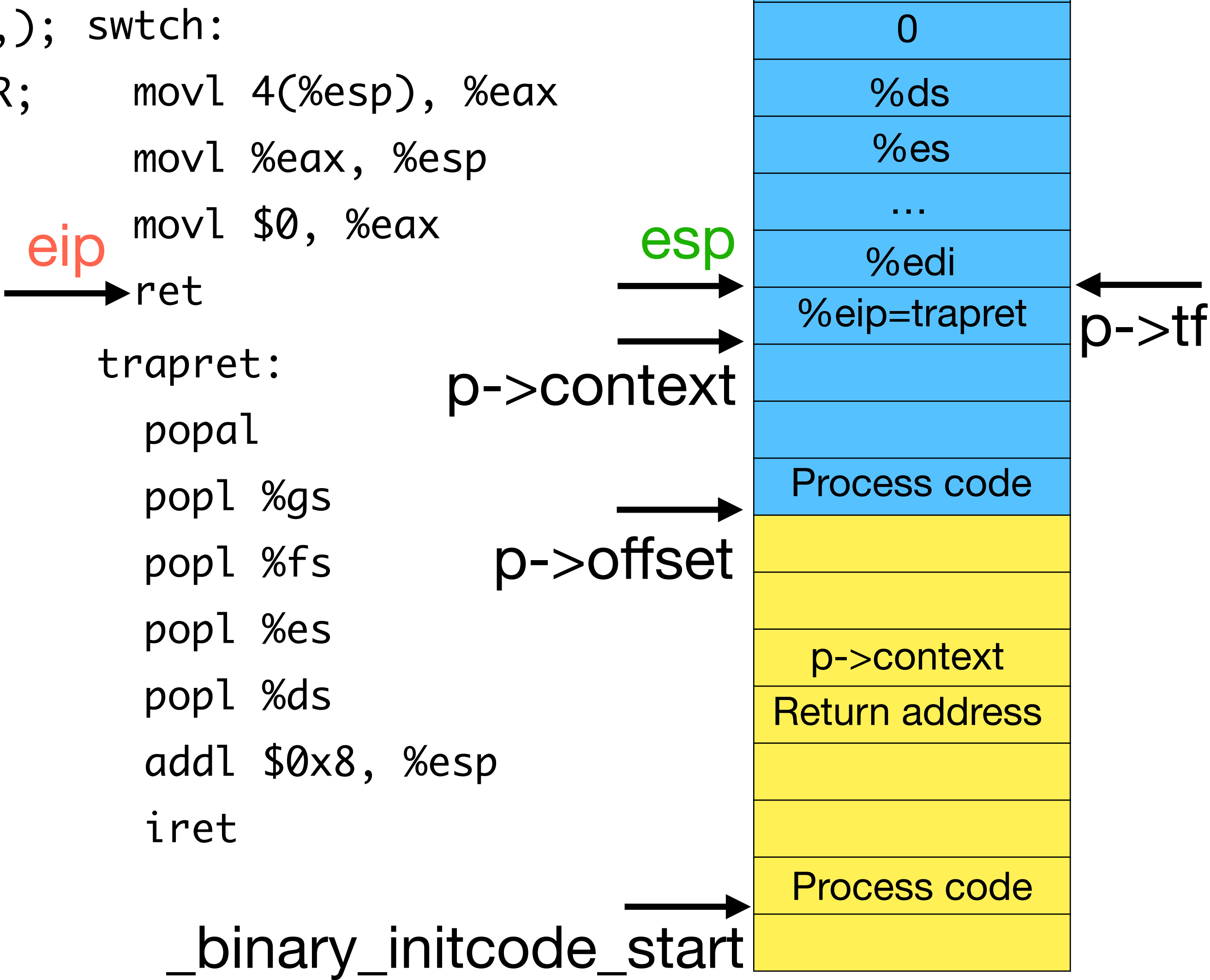


Understanding swtch

```
pinit(){
    p = allocproc();
    memmove(p->offset, _binary_initcode_start,);
    p->tf->ds,es,ss = (SEG_UDATA<<3) | DPL_USR;
    p->tf->cs = (SEG_UCODE<<3) | DPL_USR;
    p->tf->eflags = FL_IF;
    p->tf->eip = 0;
}

allocproc() {
    sp = (char*)(STARTPROC + (PROCSIZE<<12));
    sp -= sizeof *p->tf;
    p->tf = (struct trapframe*)sp;
    sp -= sizeof *p->context;
    p->context = (struct context*)sp;
    p->context->eip = (uint)trapret;
    return p;
}
```

```
scheduler() {
    ...
    swtch(p->context);
}
```



Understanding swtch

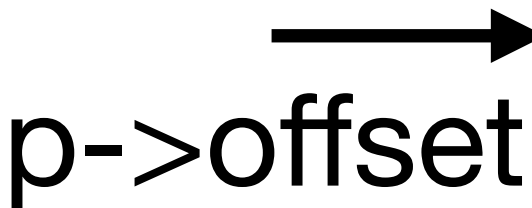
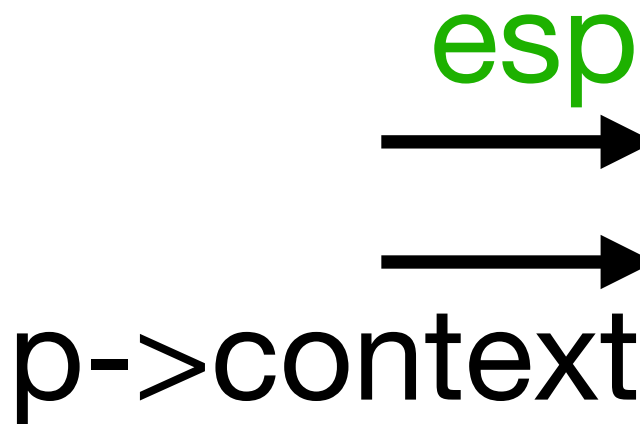
```
pinit(){
    p = allocproc();
    memmove(p->offset, _binary_initcode_start,);
    p->tf->ds,es,ss = (SEG_UDATA<<3) | DPL_USR;
    p->tf->cs = (SEG_UCODE<<3) | DPL_USR;
    p->tf->eflags = FL_IF;
    p->tf->eip = 0;
}

allocproc() {
    sp = (char*)(STARTPROC + (PROCSIZE<<12));
    sp -= sizeof *p->tf;
    p->tf = (struct trapframe*)sp;
    sp -= sizeof *p->context;
    p->context = (struct context*)sp;
    p->context->eip = (uint)trapret;
    return p;
}
```

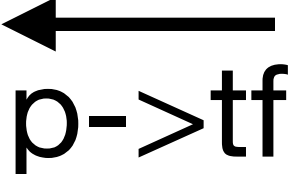
```
scheduler() {
    ...
    swtch(p->context);
}
```

```
swtch:
    movl 4(%esp), %eax
    movl %eax, %esp
    movl $0, %eax
    ret
```

```
trapret:
    popal
    popl %gs
    popl %fs
    popl %es
    popl %ds
    addl $0x8, %esp
    iret
```



%eflags=FL_IF
%cs=UCODE
%eip=0
0
0
%ds
%es
...
%edi
%eip=trapret
Process code
p->context
Return address
Process code



Understanding swtch

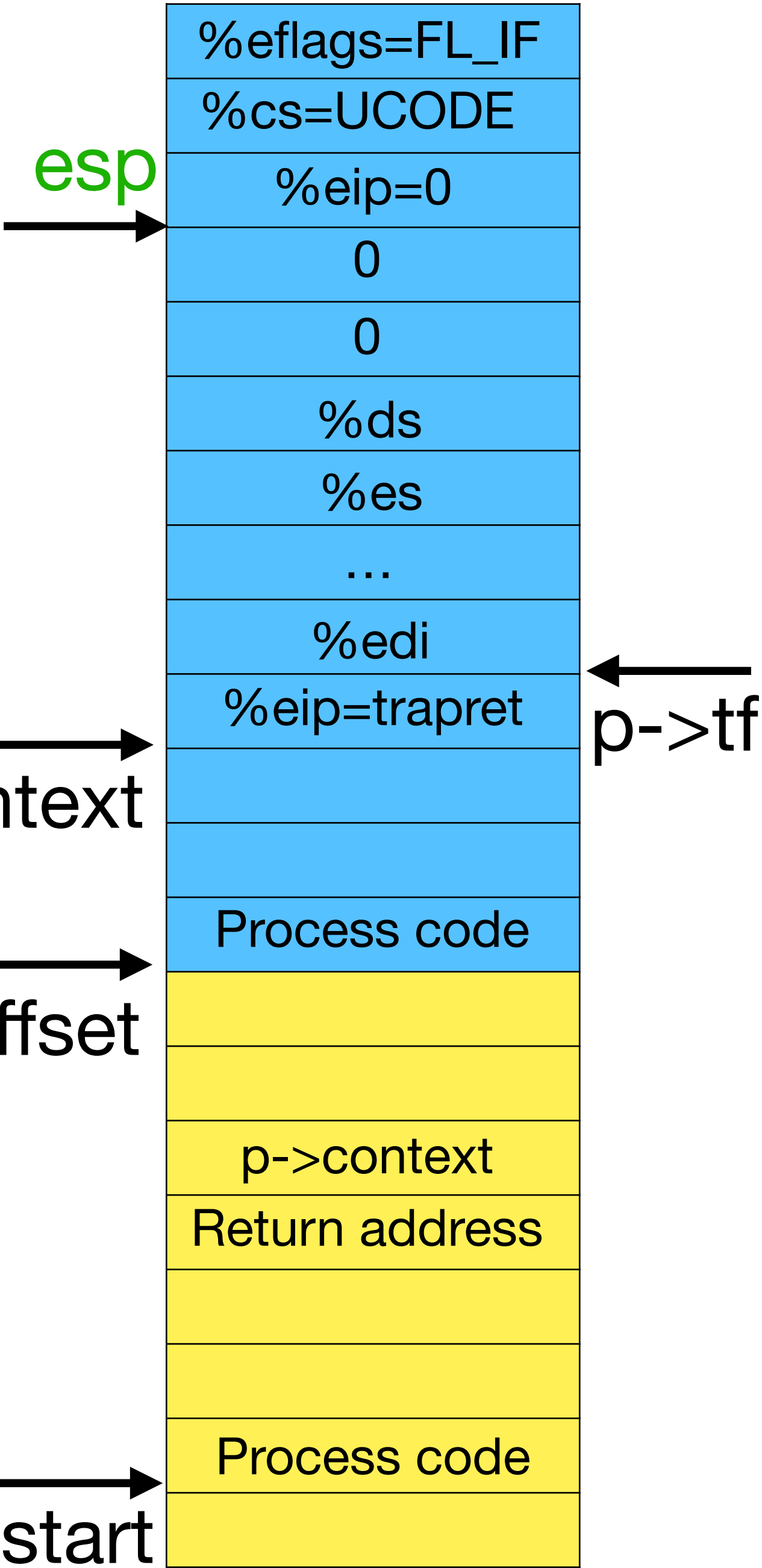
```
pinit(){
    p = allocproc();
    memmove(p->offset, _binary_initcode_start,);
    p->tf->ds,es,ss = (SEG_UDATA<<3) | DPL_USR;
    p->tf->cs = (SEG_UCODE<<3) | DPL_USR;
    p->tf->eflags = FL_IF;
    p->tf->eip = 0;
}

allocproc() {
    sp = (char*)(STARTPROC + (PROCSIZE<<12));
    sp -= sizeof *p->tf;
    p->tf = (struct trapframe*)sp;
    sp -= sizeof *p->context;
    p->context = (struct context*)sp;
    p->context->eip = (uint)trapret;
    return p;
}
```

```
scheduler() {
    ...
    swtch(p->context);
}
```

```
swtch:
    movl 4(%esp), %eax
    movl %eax, %esp
    movl $0, %eax
    ret
```

```
trapret:
    popal
    popl %gs
    popl %fs
    popl %es
    popl %ds
    addl $0x8, %esp
    iret
```



Understanding swtch

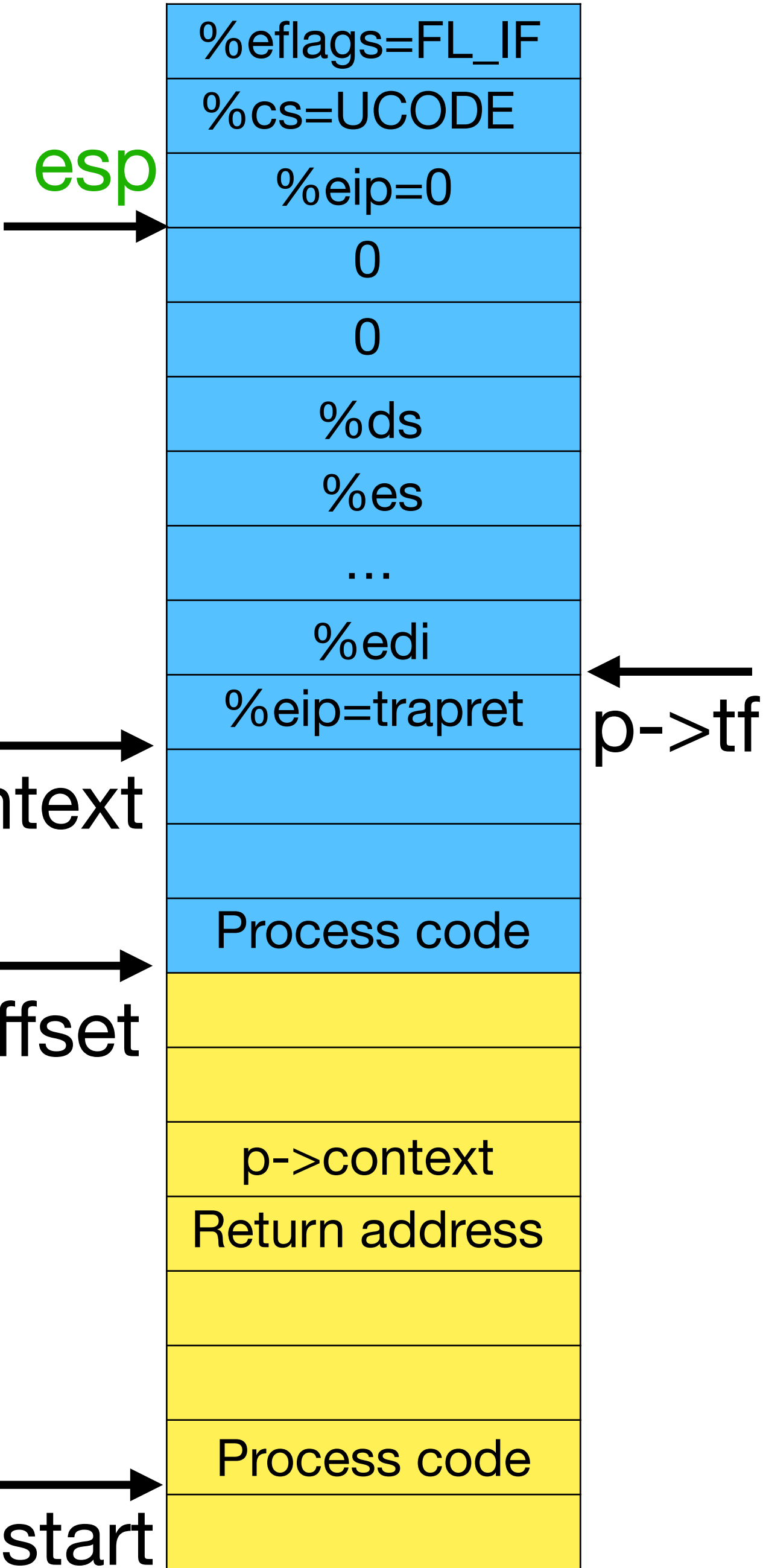
```
pinit(){
    p = allocproc();
    memmove(p->offset, _binary_initcode_start,);
    p->tf->ds,es,ss = (SEG_UDATA<<3) | DPL_USR;
    p->tf->cs = (SEG_UCODE<<3) | DPL_USR;
    p->tf->eflags = FL_IF;
    p->tf->eip = 0;
}

allocproc() {
    sp = (char*)(STARTPROC + (PROCSIZE<<12));
    sp -= sizeof *p->tf;
    p->tf = (struct trapframe*)sp;
    sp -= sizeof *p->context;
    p->context = (struct context*)sp;
    p->context->eip = (uint)trapret;
    return p;
}
```

```
scheduler() {
    ...
    swtch(p->context);
}
```

```
swtch:
    movl 4(%esp), %eax
    movl %eax, %esp
    movl $0, %eax
    ret

trapret:
    popal
    popl %gs
    popl %fs
    popl %es
    popl %ds
    addl $0x8, %esp
    iret
```



Understanding swtch

```
pinit(){
    p = allocproc();
    memmove(p->offset, _binary_initcode_start,);
    p->tf->ds,es,ss = (SEG_UDATA<<3) | DPL_USR;
    p->tf->cs = (SEG_UCODE<<3) | DPL_USR;
    p->tf->eflags = FL_IF;
    p->tf->eip = 0;
}

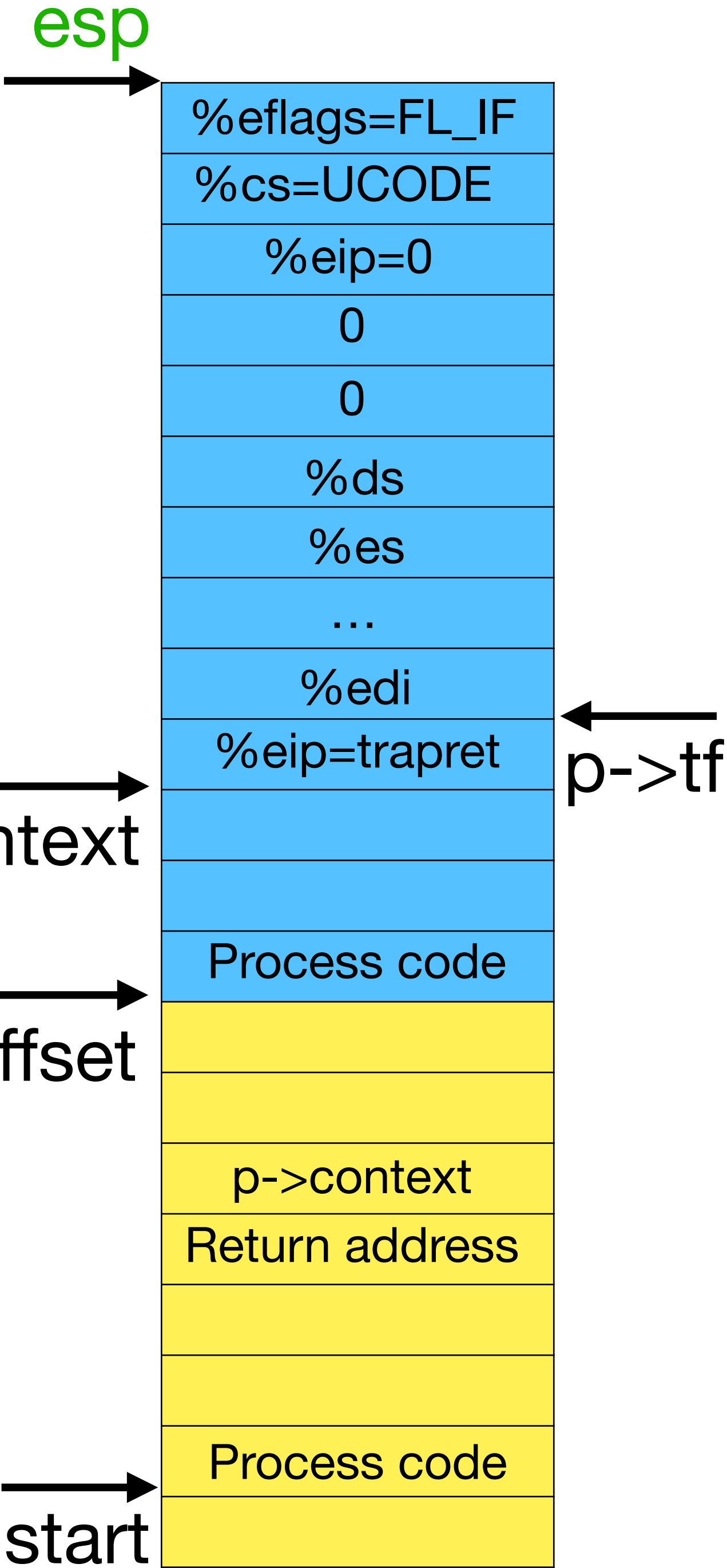
allocproc() {
    sp = (char*)(STARTPROC + (PROCSIZE<<12));
    sp -= sizeof *p->tf;
    p->tf = (struct trapframe*)sp;
    sp -= sizeof *p->context;
    p->context = (struct context*)sp;
    p->context->eip = (uint)trapret;
    return p;
}
```

```
scheduler() {
    ...
    swtch(p->context);
}

swtch:
    movl 4(%esp), %eax
    movl %eax, %esp
    movl $0, %eax
    ret

trapret:
    popal
    popl %gs
    popl %fs
    popl %es
    popl %ds
    addl $0x8, %esp
    iret

_binary_initcode_start
```



Understanding swtch

```
pinit(){
    p = allocproc();
    memmove(p->offset, _binary_initcode_start,);
    p->tf->ds,es,ss = (SEG_UDATA<<3) | DPL_USR;
    p->tf->cs = (SEG_UCODE<<3) | DPL_USR;
    p->tf->eflags = FL_IF;
    p->tf->eip = 0;
}

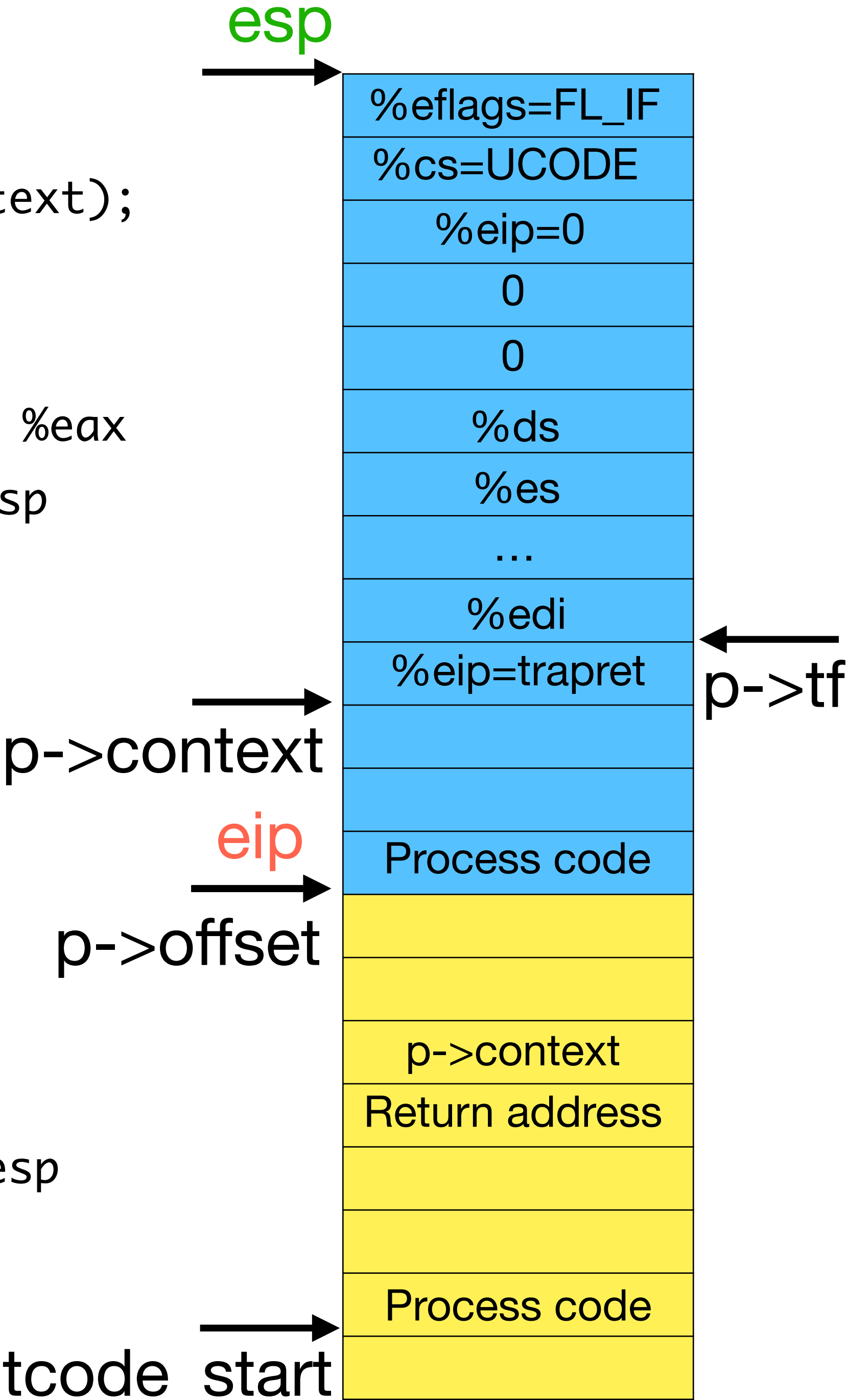
allocproc() {
    sp = (char*)(STARTPROC + (PROCSIZE<<12));
    sp -= sizeof *p->tf;
    p->tf = (struct trapframe*)sp;
    sp -= sizeof *p->context;
    p->context = (struct context*)sp;
    p->context->eip = (uint)trapret;
    return p;
}
```

```
scheduler() {
    ...
    swtch(p->context);
}

swtch:
    movl 4(%esp), %eax
    movl %eax, %esp
    movl $0, %eax
    ret

trapret:
    popal
    popl %gs
    popl %fs
    popl %es
    popl %ds
    addl $0x8, %esp
    iret

_binary_initcode_start
```



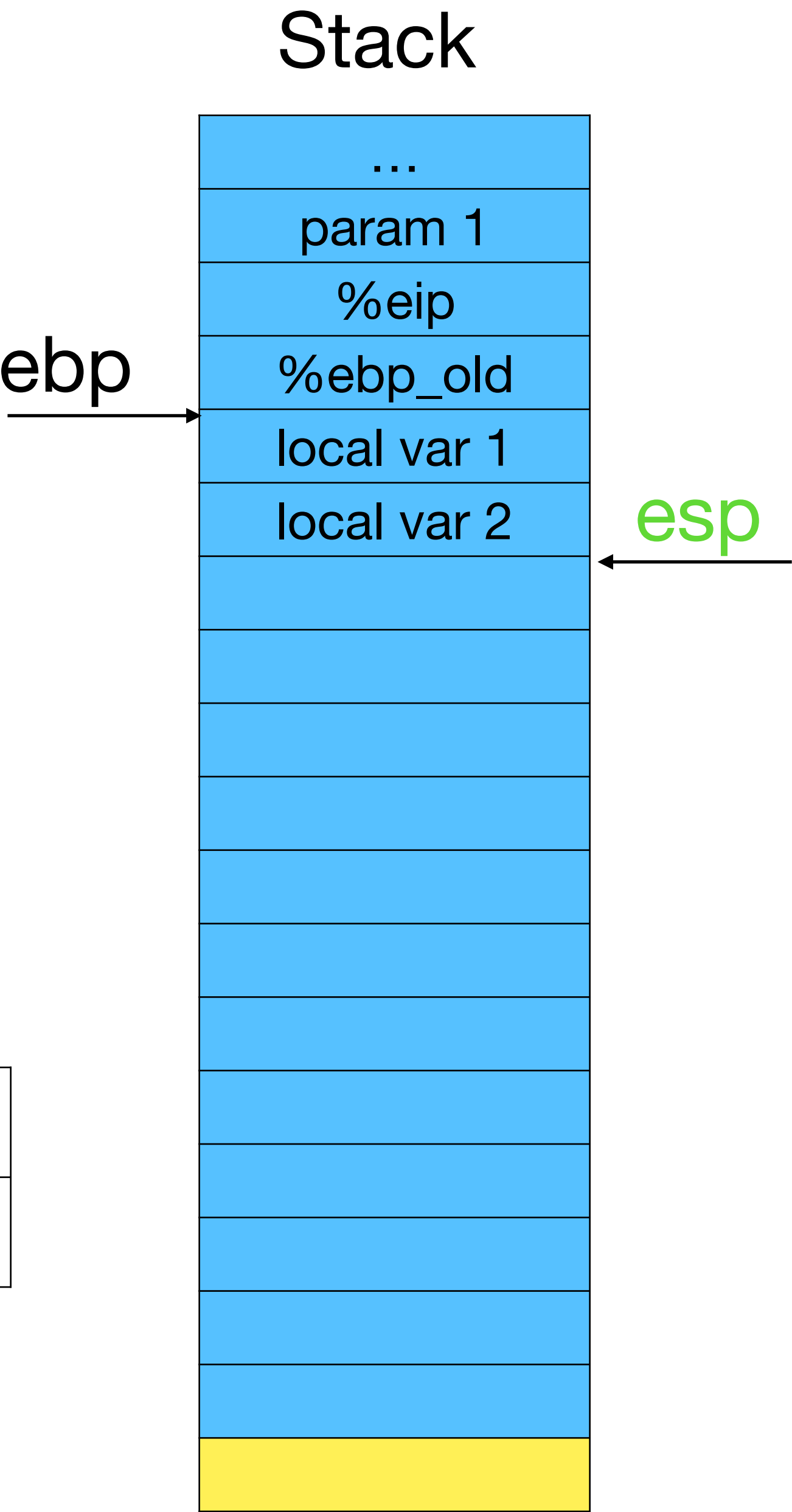
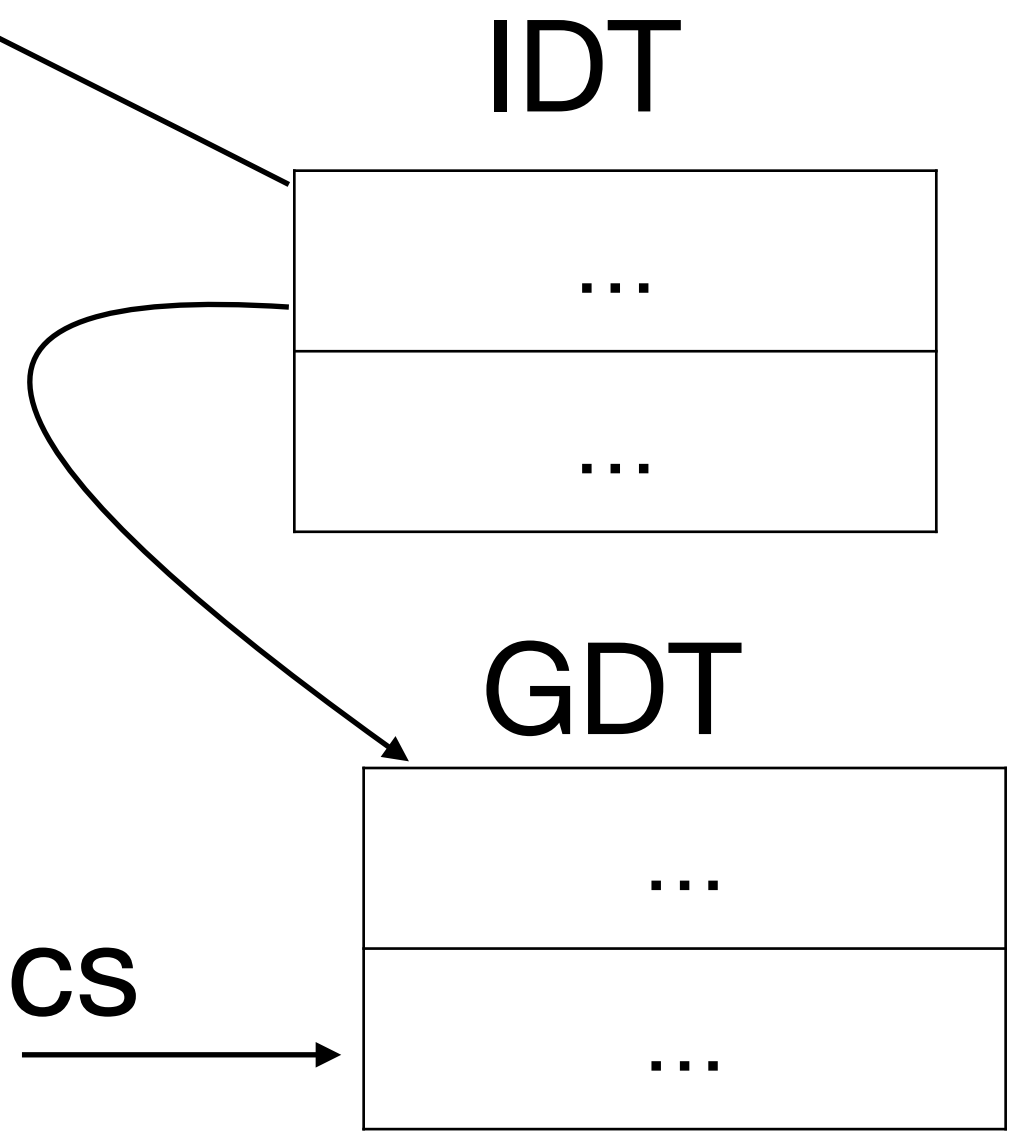
Interrupt handling revisited

eip → for(;;)
;

trap.c
void
trap(struct trapframe *tf)
{
 switch(tf->trapno){
 case T_IRQ0 + IRQ_TIMER:
 ticks++;
 cprintf("Tick! %d\n\0", ticks);
 lapiceoi();
 ..
 return

vectors.S
.globl vector0
vector0:
 pushl \$0
 pushl \$0
 jmp alltraps

trapasm.S
alltraps:
 pushal
 pushl %esp
 call trap
 addl \$4, %esp
 popal
 addl \$0x8, %esp
 iret



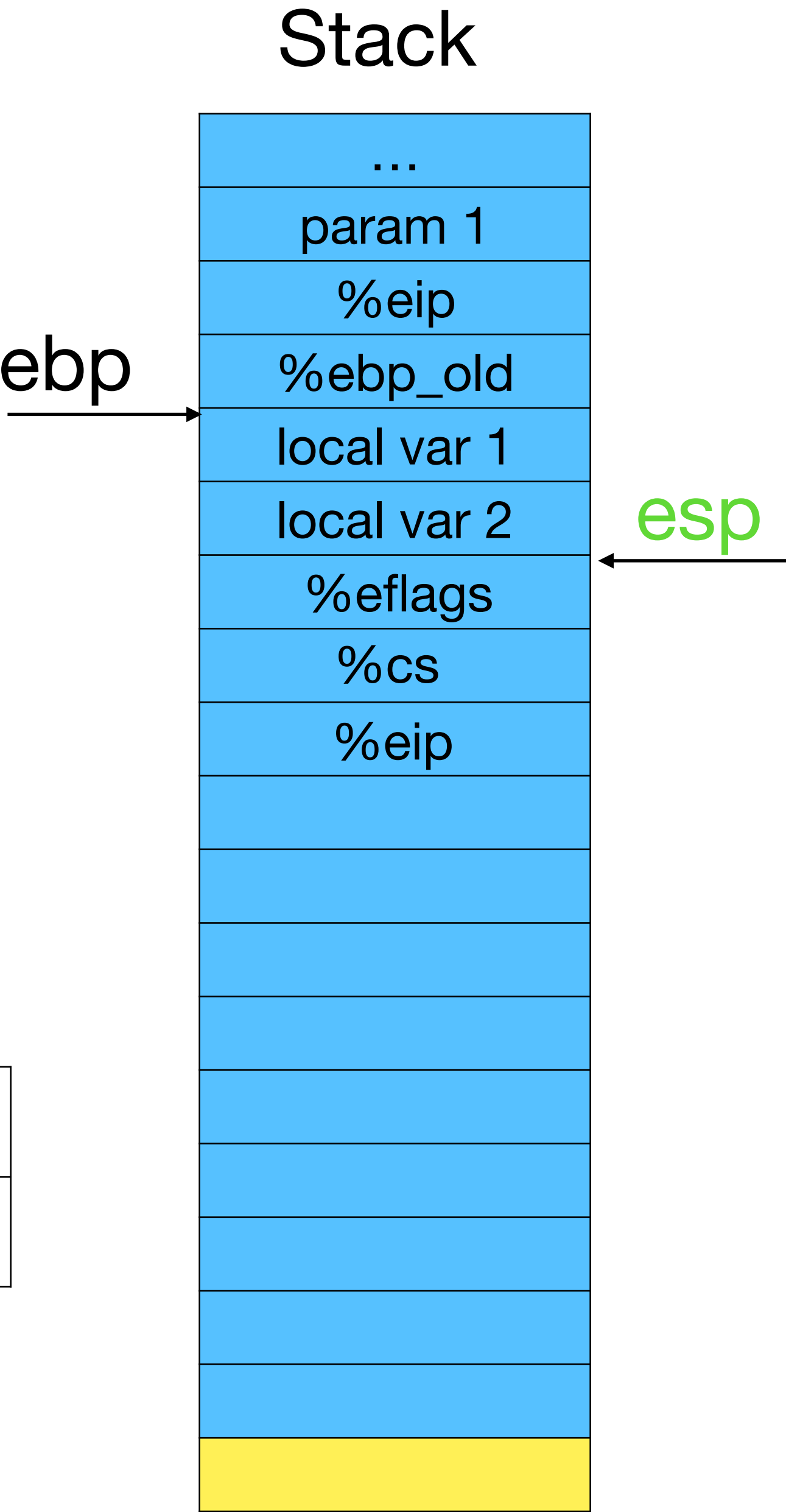
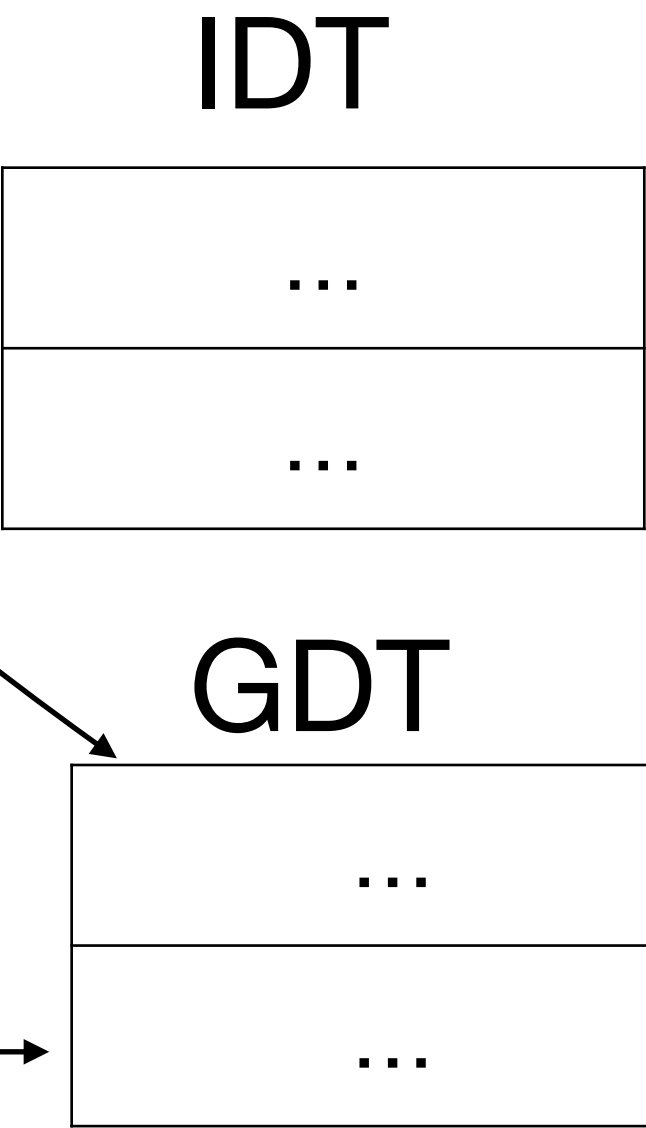
Interrupt handling revisited

eip → for(;;)
;

trap.c
void
trap(struct trapframe *tf)
{
 switch(tf->trapno){
 case T_IRQ0 + IRQ_TIMER:
 ticks++;
 cprintf("Tick! %d\n\0", ticks);
 lapiceoi();
 ..
 return

vectors.S
.globl vector0
vector0:
 pushl \$0
 pushl \$0
 jmp alltraps

trapasm.S
alltraps:
 pushal
 pushl %esp
 call trap
 addl \$4, %esp
 popal
 addl \$0x8, %esp
 iret



Interrupt handling revisited

eip

→

for(;;)

;

trap.c

void

trap(struct trapframe *tf)

{

switch(tf->trapno){

case T_IRQ0 + IRQ_TIMER:

ticks++;

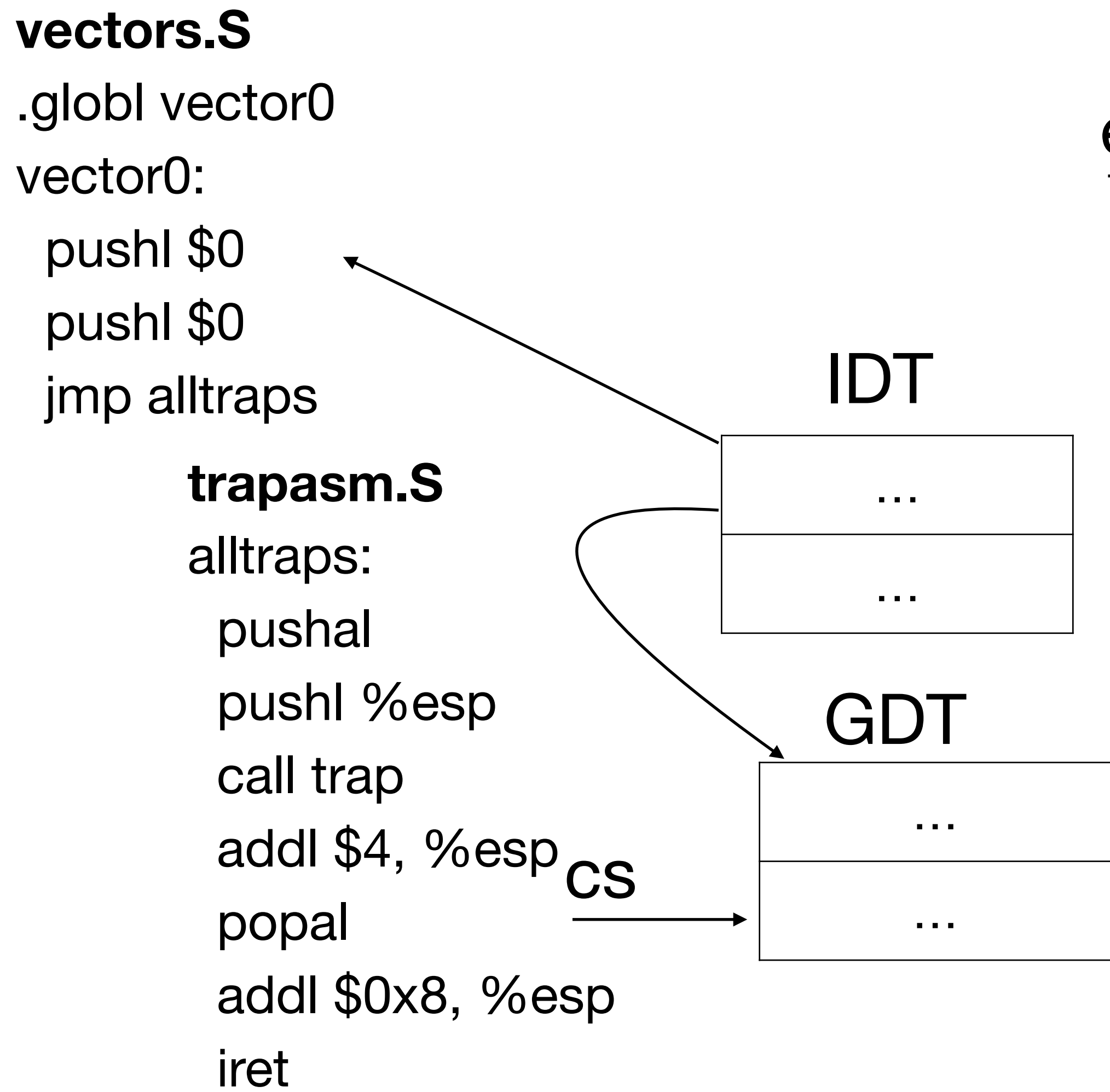
cprintf("Tick! %d\n\0", ticks);

lapiceoi();

..

return

}



Interrupt handling revisited

eip

→

for(;;)

;

trap.c

void

trap(struct trapframe *tf)

{

switch(tf->trapno){

case T_IRQ0 + IRQ_TIMER:

ticks++;

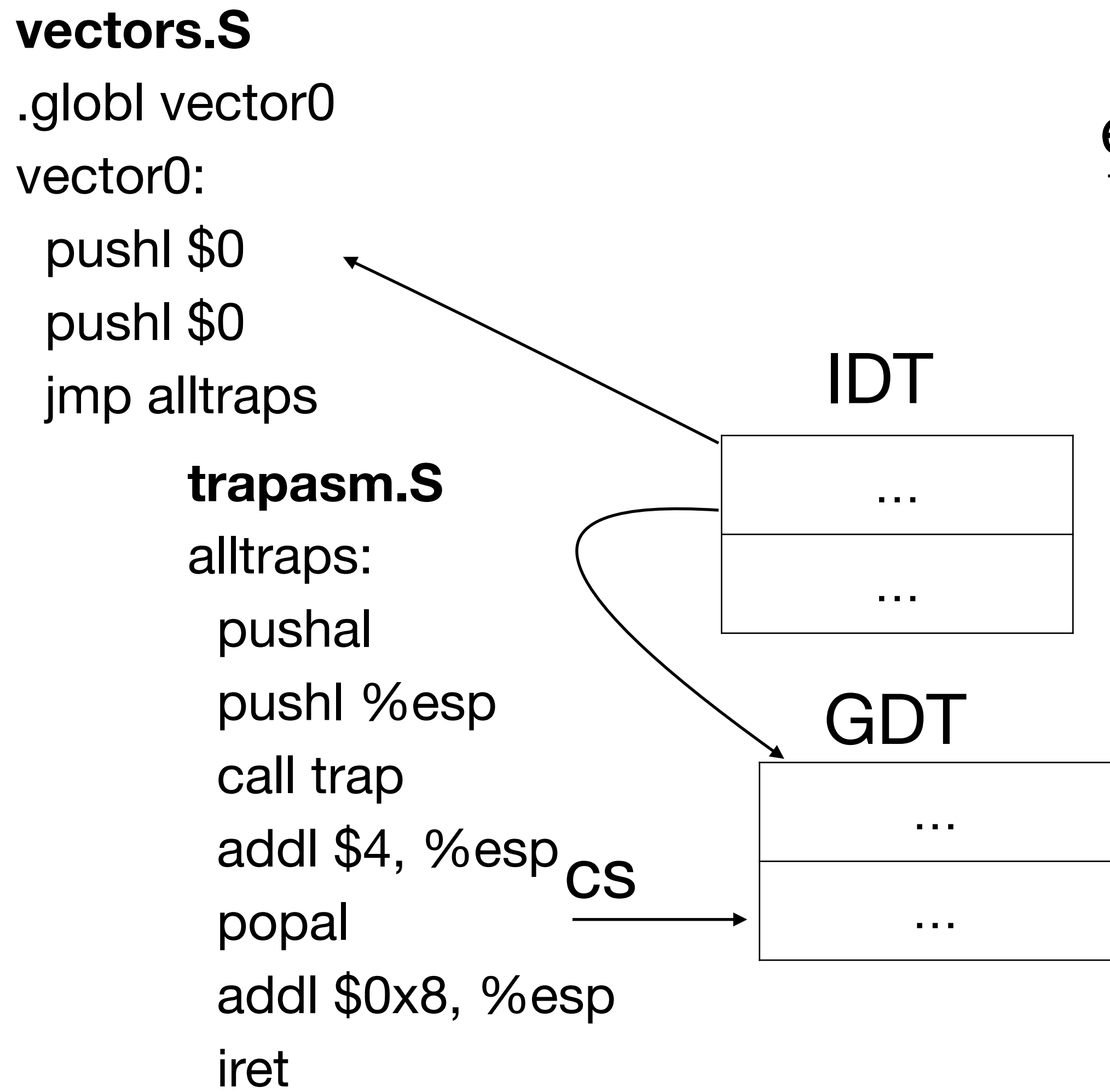
cprintf("Tick! %d\n\0", ticks);

lapiceoi();

..

return

}



Interrupt handling revisited

eip → for(;;)
;

trap.c

```
void
trap(struct trapframe *tf)
{
    switch(tf->trapno){
    case T_IRQ0 + IRQ_TIMER:
        ticks++;
        cprintf("Tick! %d\n", ticks);
        lapiceoi();
        ..
    }
    return
}
```

vectors.S

```
.globl vector0
vector0:
    pushl $0
    pushl $0
    jmp alltraps
```

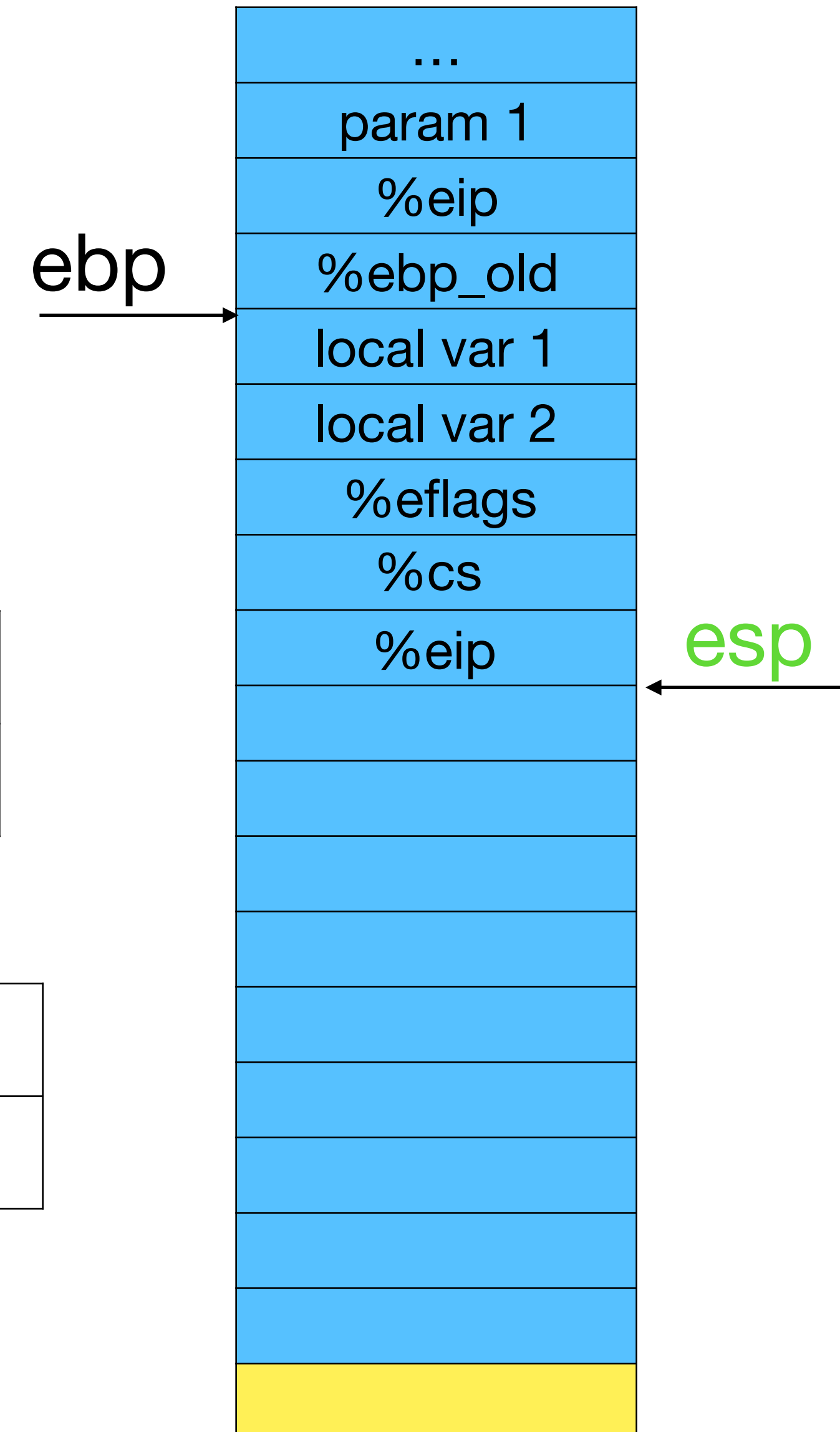
trapasm.S

```
alltraps:
    pushal
    pushl %esp
    call trap
    addl $4, %esp
    popal
    addl $0x8, %esp
    iret
```

IDT

GDT

Stack



Interrupt handling revisited

```
for(;;)
;
```

trap.c

```
void
trap(struct trapframe *tf)
{
    switch(tf->trapno){
    case T_IRQ0 + IRQ_TIMER:
        ticks++;
        cprintf("Tick! %d\n\0", ticks);
        lapiceoi();
    ..
    }
    return
}
```

vectors.S

```
.globl vector0
```

vector0:

```

pushl $0
pushl $0
jmp alltraps

```

trapasm.S

```
alltraps:
    pushal
    pushl %esp
    call trap      CS
    addl $4, %esp
    popal
    addl $0x8, %esp
    iret
```

IDT

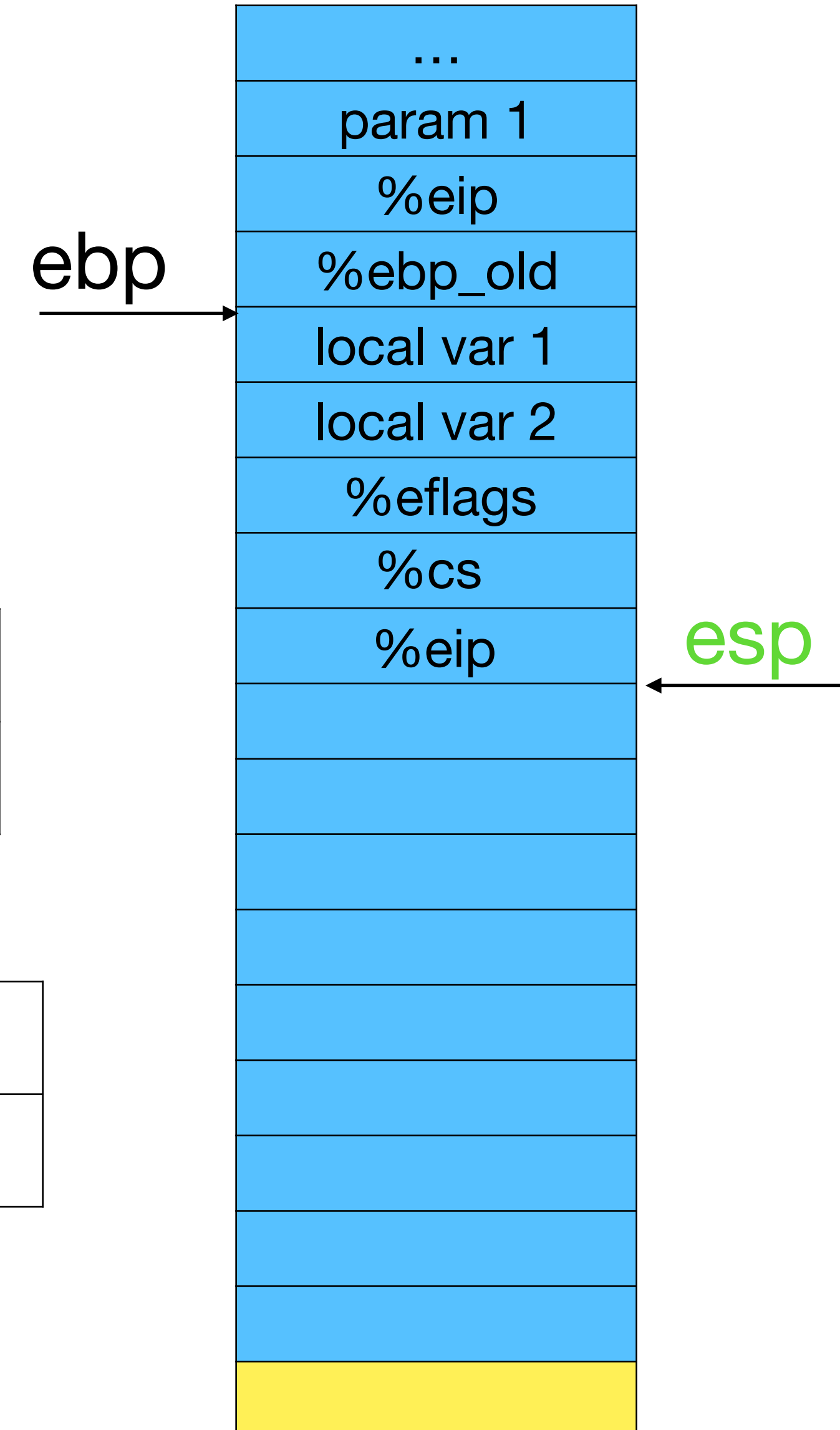
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GDT

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Interrupt handling revisited

```
for(;;)
;
```

trap.c

```
void
trap(struct trapframe *tf)
{
    switch(tf->trapno){
    case T_IRQ0 + IRQ_TIMER:
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vectors.S

```
.globl vector0
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vector0:

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trapasm.S

```
alltraps:
    pushal
    pushl %esp
    call trap      CS
    addl $4, %esp
    popal
    addl $0x8, %esp
    iret
```

IDT

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param 1

```
%eip
```

```
%ebp_old
```

local var 1

local var 2

```
%eflags
```

%CS

```
%eip
```

0

esp



ebp

CS

Interrupt handling revisited

```
for(;;)  
;
```

trap.c

```
void  
trap(struct trapframe *tf)  
{  
    switch(tf->trapno){  
    case T_IRQ0 + IRQ_TIMER:  
        ticks++;  
        cprintf("Tick! %d\n\0", ticks);  
        lapiceoi();  
    ..  
    return
```

vectors.S

```
.globl vector0  
vector0:  
    pushl $0  
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    jmp alltraps
```

eip

trapasm.S

```
alltraps:  
    pushal  
    pushl %esp  
    call trap  
    addl $4, %esp  
    popal  
    addl $0x8, %esp  
    iret
```

IDT

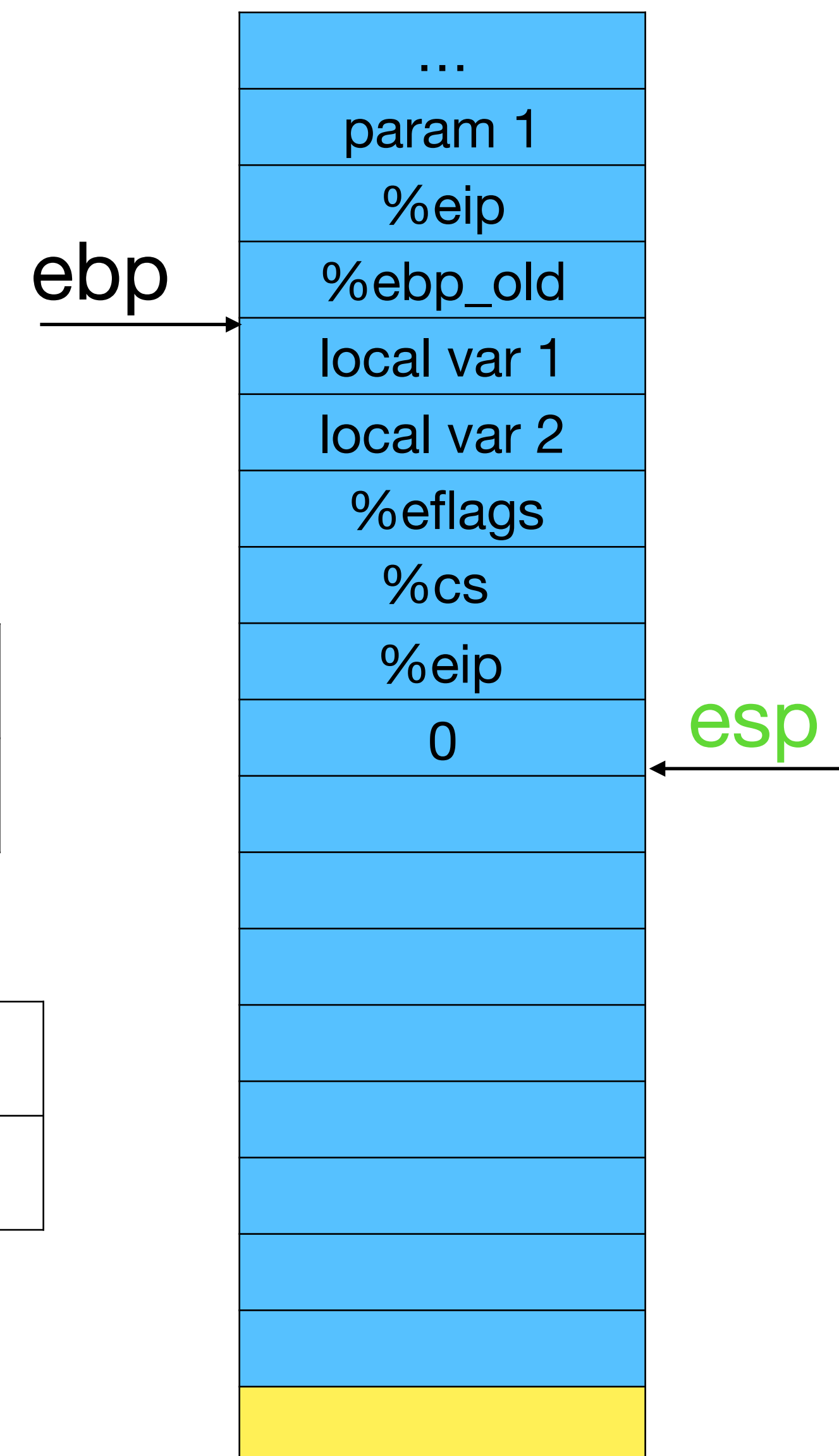


GDT



CS

Stack



ebp

esp

Interrupt handling revisited

```
for(;;)  
;
```

trap.c

```
void  
trap(struct trapframe *tf)  
{  
    switch(tf->trapno){  
    case T_IRQ0 + IRQ_TIMER:  
        ticks++;  
        cprintf("Tick! %d\n\0", ticks);  
        lapiceoi();  
    ..  
    return
```

vectors.S

```
.globl vector0  
vector0:  
    pushl $0  
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    jmp alltraps
```

eip

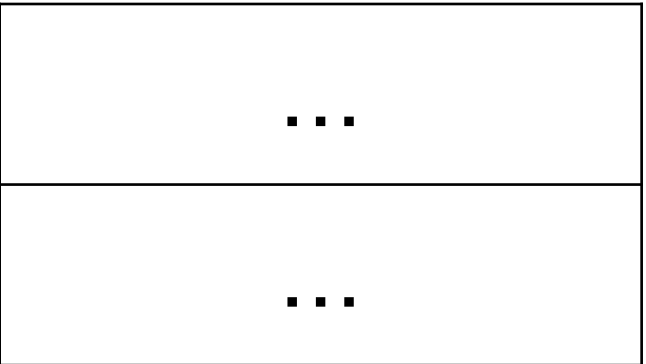
trapasm.S

```
alltraps:  
    pushal  
    pushl %esp  
    call trap  
    addl $4, %esp  
    popal  
    addl $0x8, %esp  
    iret
```

IDT

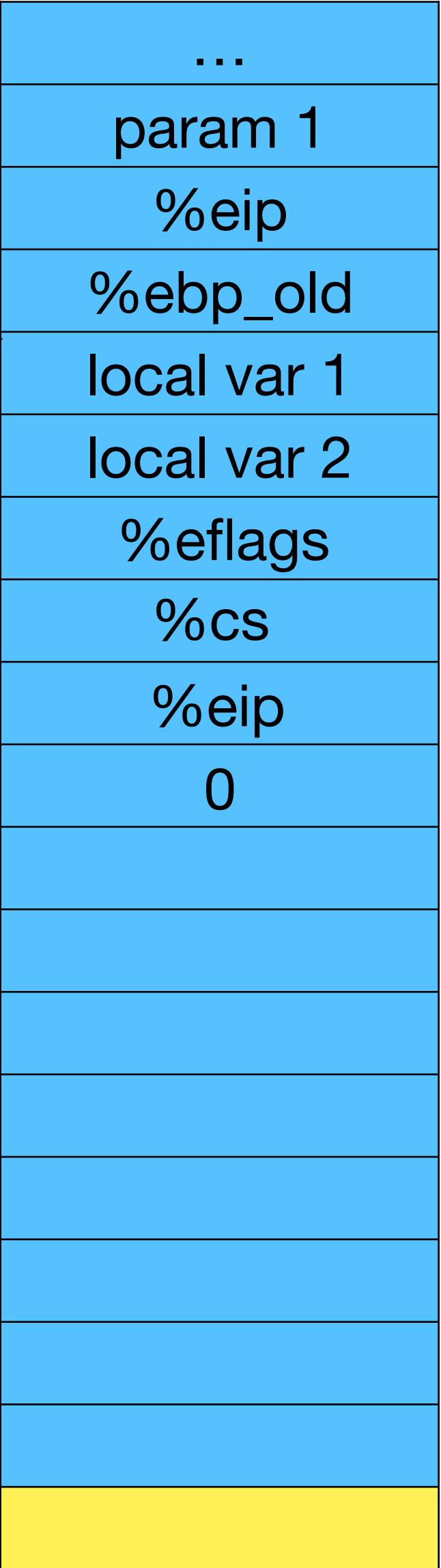


GDT



CS

Stack



ebp

esp

Interrupt handling revisited

```
for(;;)
;
```

trap.c

```
void
trap(struct trapframe *tf)
{
    switch(tf->trapno){
    case T_IRQ0 + IRQ_TIMER:
        ticks++;
        cprintf("Tick! %d\n\0", ticks);
        lapiceoi();
    ..
    }
    return
}
```

vectors.S

```
.globl vector0
```

vector0:

eip

```
pushl $0
pushl $0
jmp alltraps
```

trapasm.S

```
alltraps:
    pushal
    pushl %esp
    call trap      CS
    addl $4, %esp
    popal
    addl $0x8, %esp
    iret
```

IDT

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GDT

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■ ■ ■

Stack

■ ■ ■

param 1

```
%eip
```

```
%ebp_old
```

local var 1

local var 2

```
%eflags
```

%CS

```
%eip
```

0

0

esp



Interrupt handling revisited

```
for(;;)
;
```

trap.c

```
void
trap(struct trapframe *tf)
{
    switch(tf->trapno){
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    }
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}
```

vectors.S

```
.globl vector0
```

vector0:

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pushl $0
pushl $0
jmp alltraps
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trapasm.S

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alltraps:
    pushal
    pushl %esp
    call trap      CS
    addl $4, %esp
    popal
    addl $0x8, %esp
    iret
```

IDT

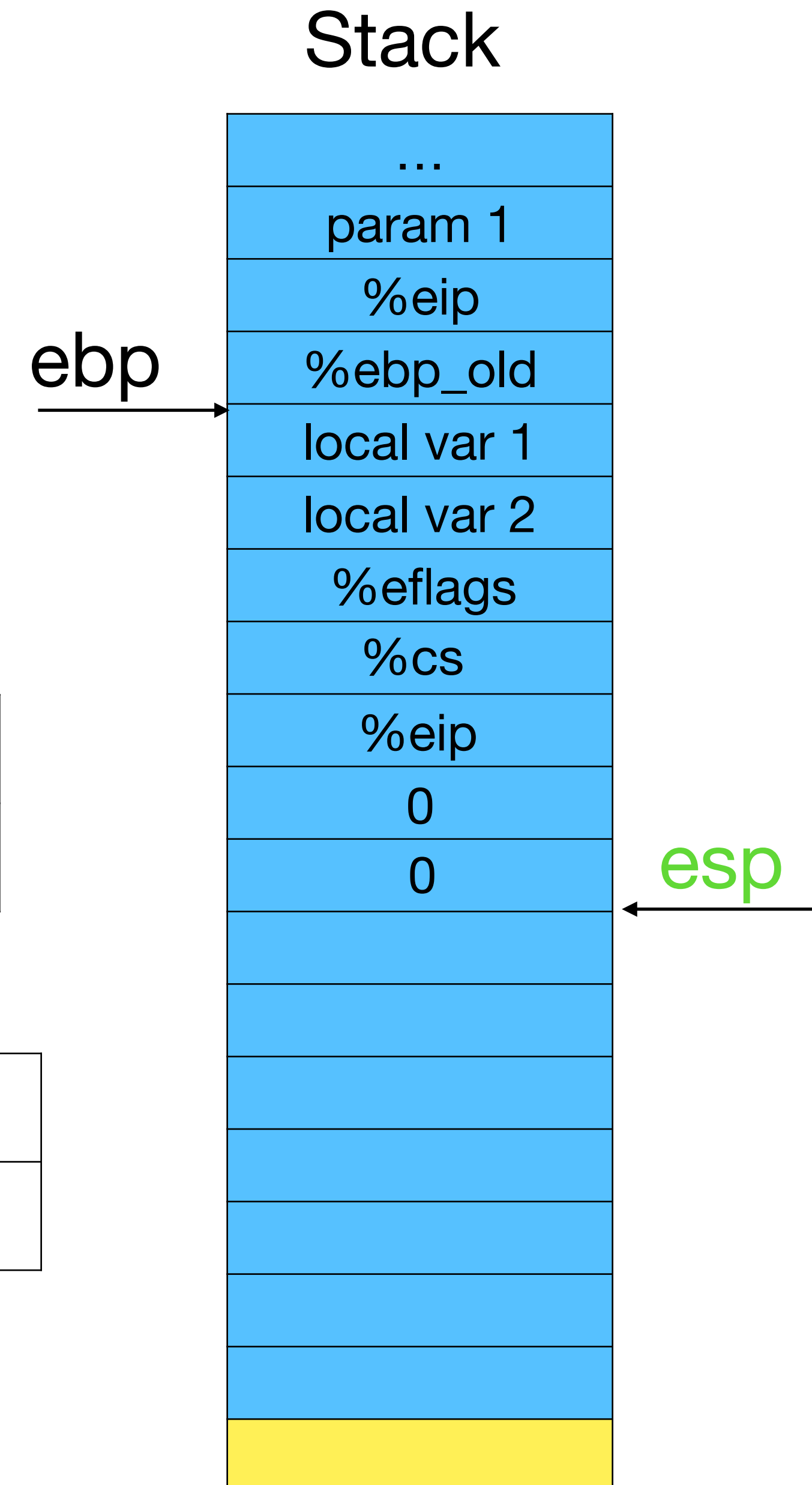
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GDT

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Interrupt handling revisited

```
for(;;)  
;
```

trap.c

```
void  
trap(struct trapframe *tf)  
{  
    switch(tf->trapno){  
    case T_IRQ0 + IRQ_TIMER:  
        ticks++;  
        cprintf("Tick! %d\n\0", ticks);  
        lapiceoi();  
    ..  
    return
```

vectors.S

```
.globl vector0  
vector0:  
    pushl $0  
    pushl $0  
    jmp alltraps
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eip

trapasm.S

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alltraps:  
    pushal  
    pushl %esp  
    call trap  
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    popal  
    addl $0x8, %esp  
    iret
```

IDT

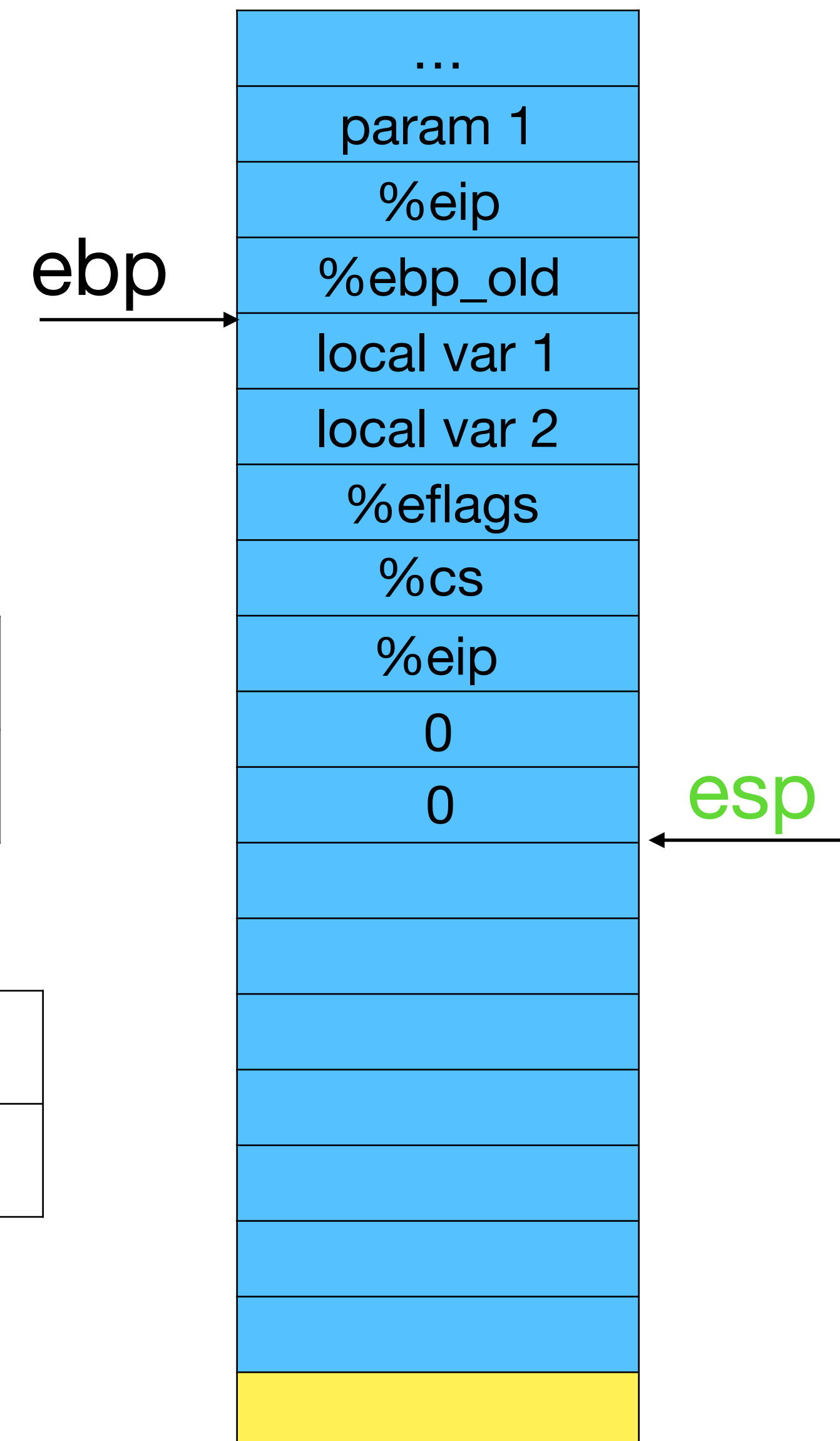


GDT



CS

Stack



ebp

esp

Interrupt handling revisited

```
for(;;)  
;
```

trap.c

```
void  
trap(struct trapframe *tf)  
{  
    switch(tf->trapno){  
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        cprintf("Tick! %d\n\0", ticks);  
        lapiceoi();  
    ..  
    return
```

vectors.S

```
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vector0:  
    pushl $0  
    pushl $0  
    jmp alltraps
```

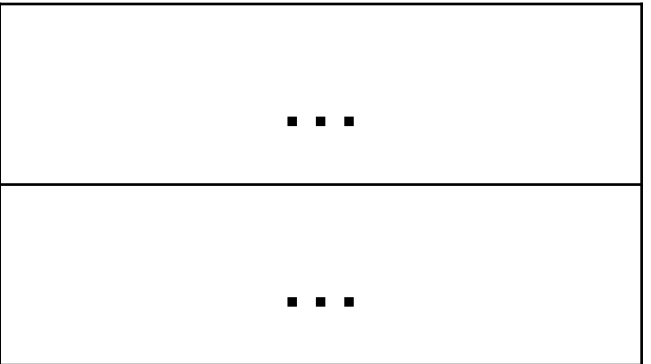
trapasm.S

```
alltraps:  
    eip → pushal  
        pushl %esp  
        call trap  
        addl $4, %esp  
        popal  
        addl $0x8, %esp  
        iret
```

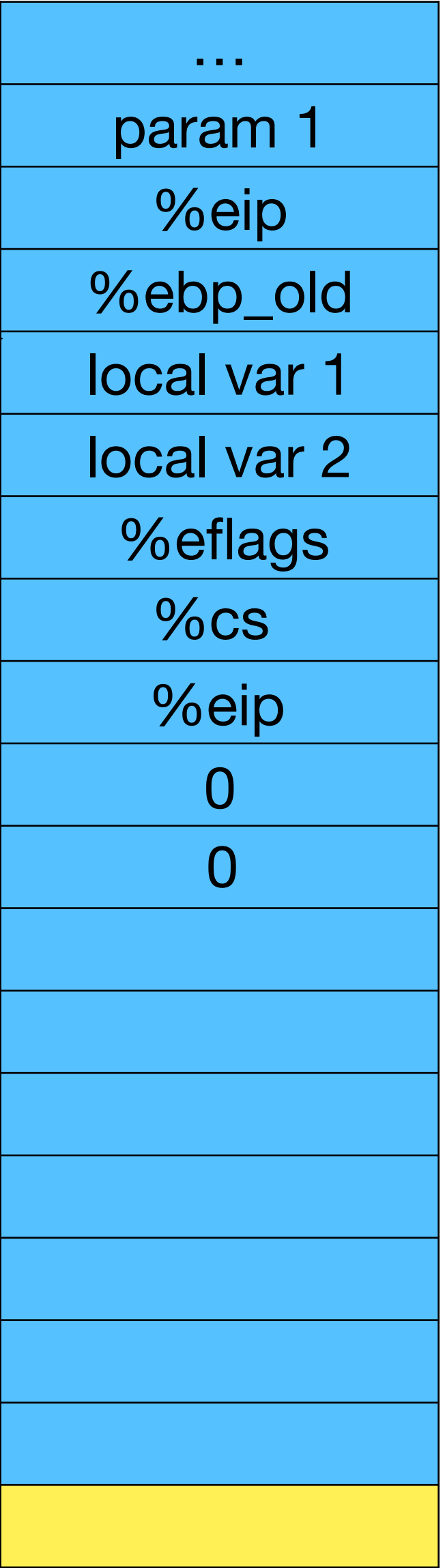
IDT



GDT



Stack



Interrupt handling revisited

```
for(;;)  
;
```

trap.c

```
void  
trap(struct trapframe *tf)  
{  
    switch(tf->trapno){  
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vectors.S

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    pushl $0  
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```

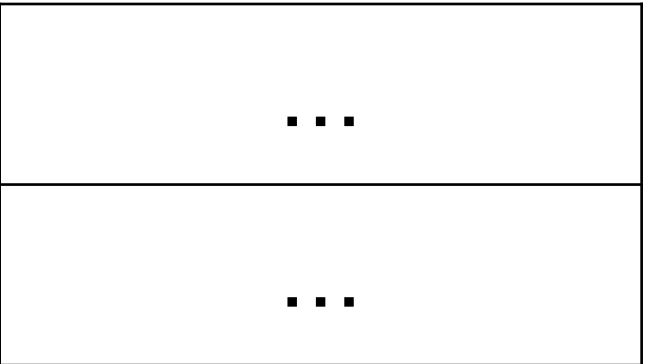
trapasm.S

```
alltraps:  
    eip → pushal  
    pushl %esp  
    call trap  
    addl $4, %esp  
    popal  
    addl $0x8, %esp  
    iret
```

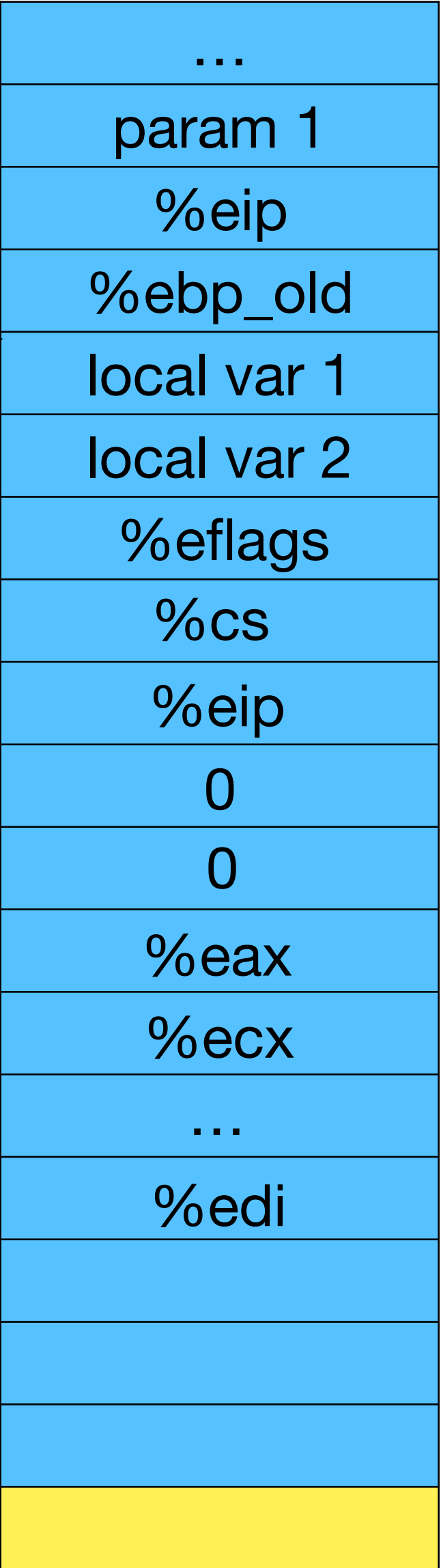
IDT



GDT



Stack



ebp →

esp ←

Interrupt handling revisited

```
for(;;)  
;
```

trap.c

```
void  
trap(struct trapframe *tf)  
{  
    switch(tf->trapno){  
    case T_IRQ0 + IRQ_TIMER:  
        ticks++;  
        cprintf("Tick! %d\n\0", ticks);  
        lapiceoi();  
    ..  
    return
```

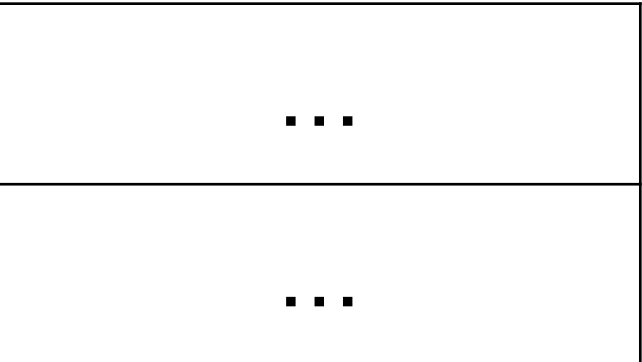
vectors.S

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.globl vector0  
vector0:  
    pushl $0  
    pushl $0  
    jmp alltraps
```

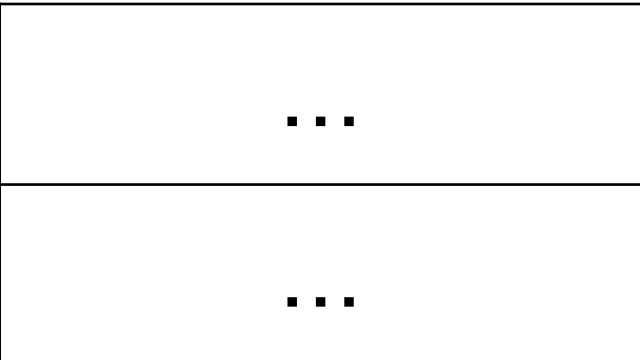
trapasm.S

```
alltraps:  
    eip → pushal  
           pushl %esp  
           call trap  
           addl $4, %esp  
           popal  
           addl $0x8, %esp  
           iret
```

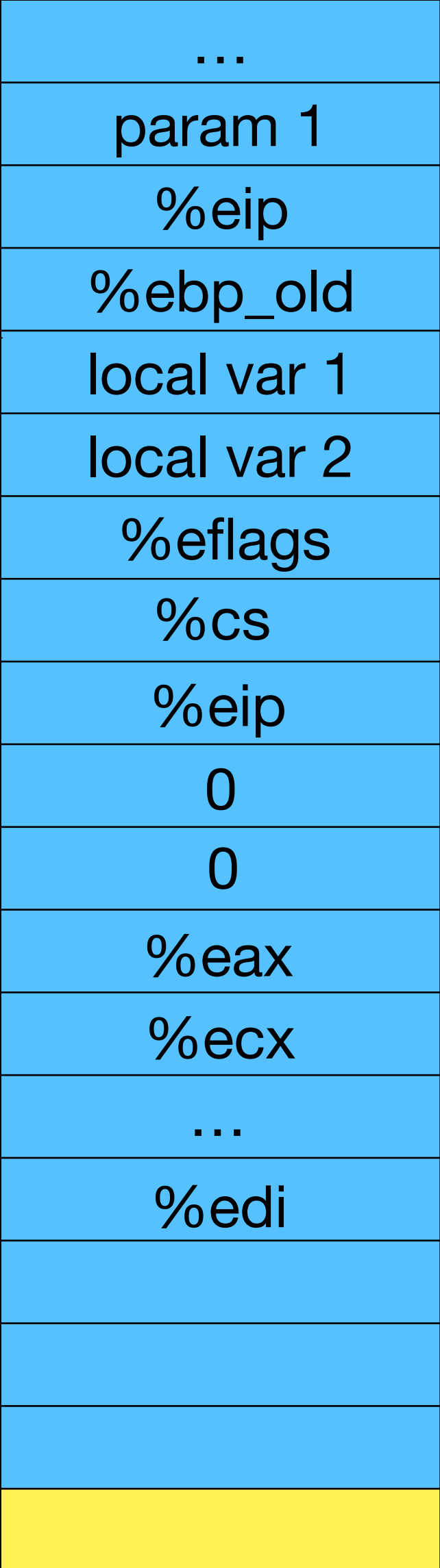
IDT



GDT



Stack



ebp →

esp ←

Interrupt handling revisited

```
for(;;)  
;
```

trap.c

```
void  
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vectors.S

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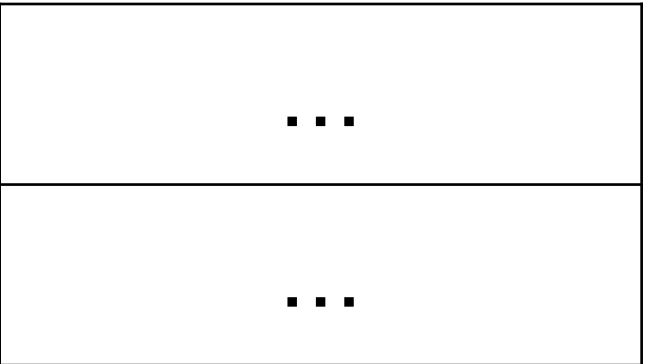
trapasm.S

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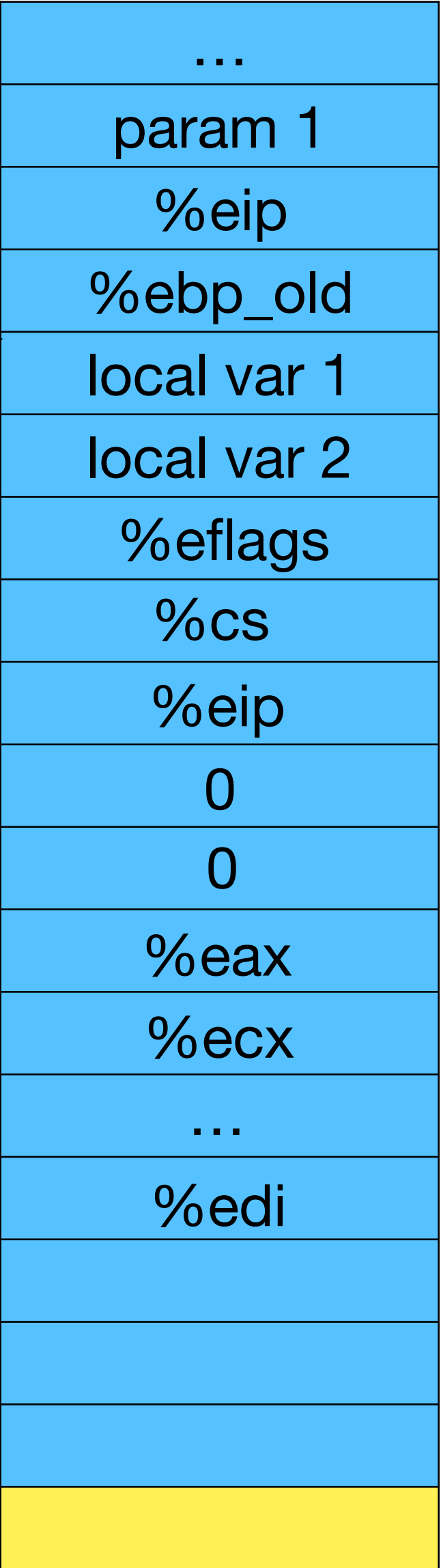
IDT



GDT



Stack



ebp →

esp ←

CS →

Interrupt handling revisited

```
for(;;)
;
```

trap.c

```
void
trap(struct trapframe *tf)
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    switch(tf->trapno){
    case T_IRQ0 + IRQ_TIMER:
        ticks++;
        cprintf("Tick! %d\n\0", ticks);
        lapiceoi();
    ..
    }
    return
}
```

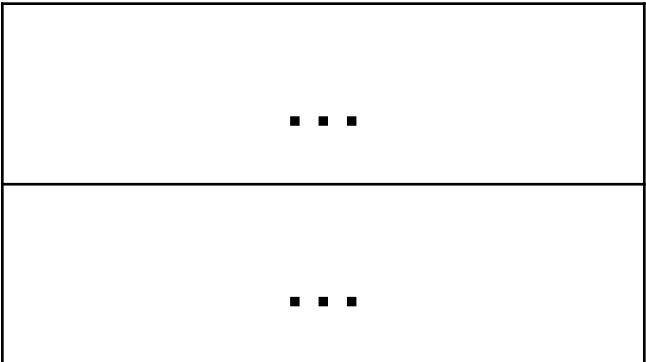
vectors.S

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vector0:
    pushl $0
    pushl $0
    jmp alltraps
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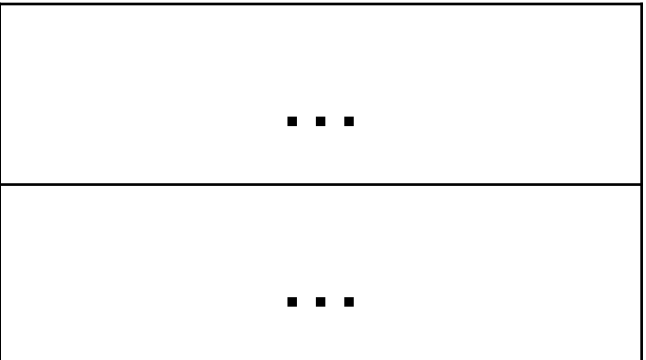
trapasm.S

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alltraps:
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    call trap
    addl $4, %esp
    popal
    addl $0x8, %esp
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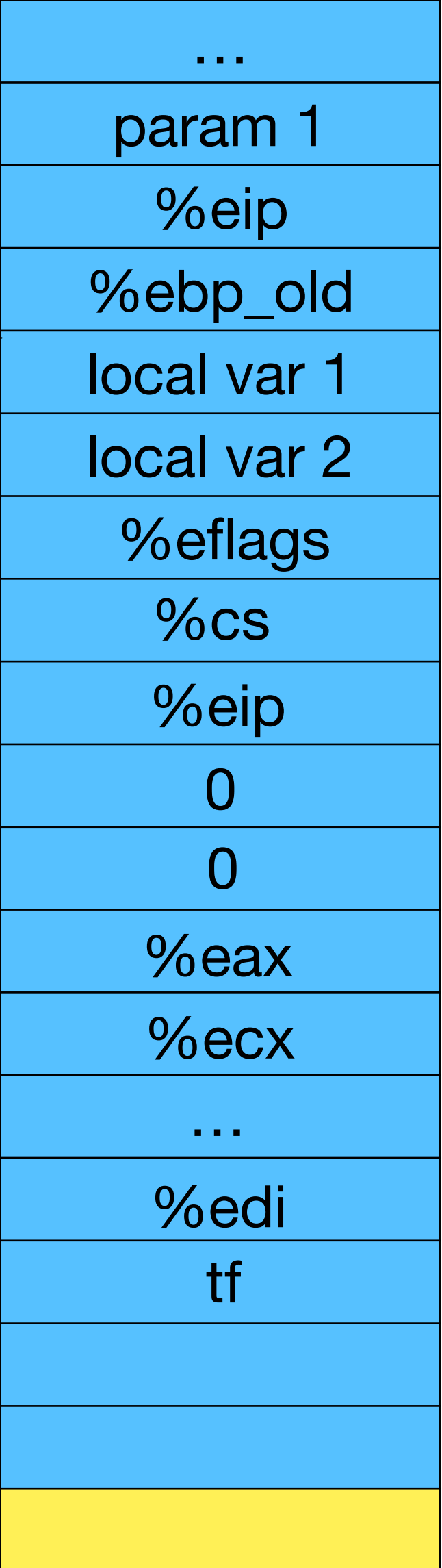
IDT



GDT



Stack



ebp →

esp ←

CS →

Interrupt handling revisited

```
for(;;)
;
```

trap.c

```
void
trap(struct trapframe *tf)
{
    switch(tf->trapno){
    case T_IRQ0 + IRQ_TIMER:
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        cprintf("Tick! %d\n\0", ticks);
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    }
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vectors.S

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    pushl $0
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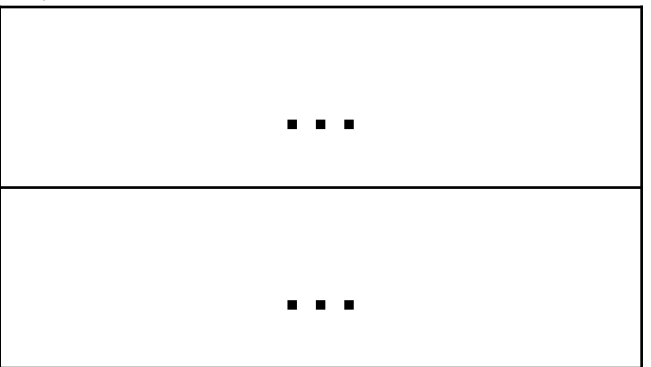
trapasm.S

```
alltraps:
    eip → pushal
    pushl %esp
    call trap
    addl $4, %esp
    CS → popal
    addl $0x8, %esp
    iret
```

IDT



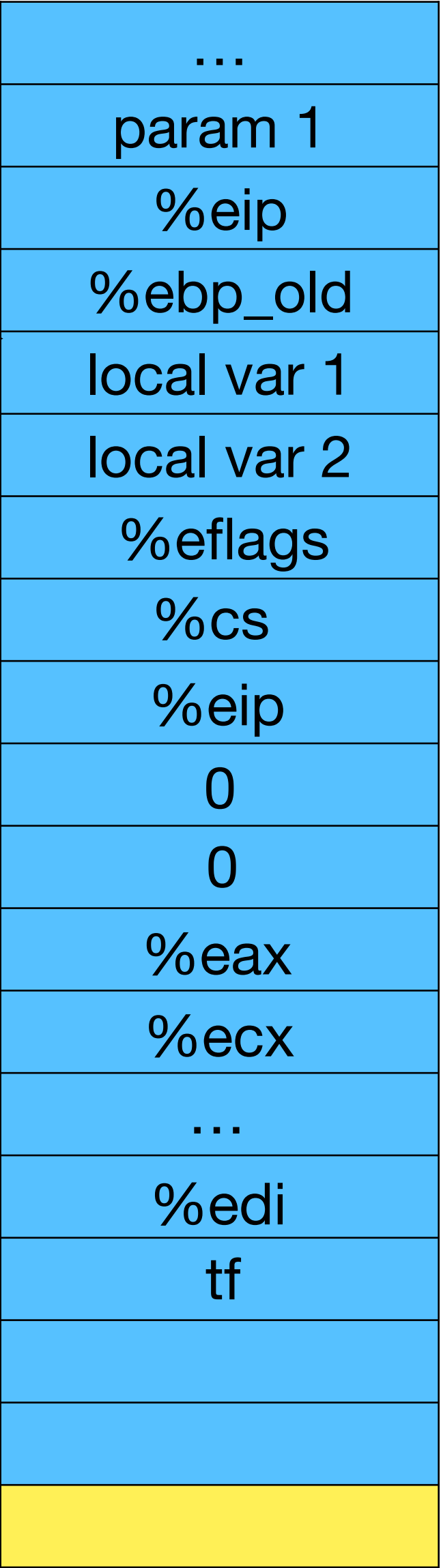
GDT



ebp →

trap frame

Stack



esp ←

Interrupt handling revisited

```
for(;;)  
;
```

trap.c

```
void  
trap(struct trapframe *tf)  
{  
    switch(tf->trapno){  
    case T_IRQ0 + IRQ_TIMER:  
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vectors.S

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```

IDT

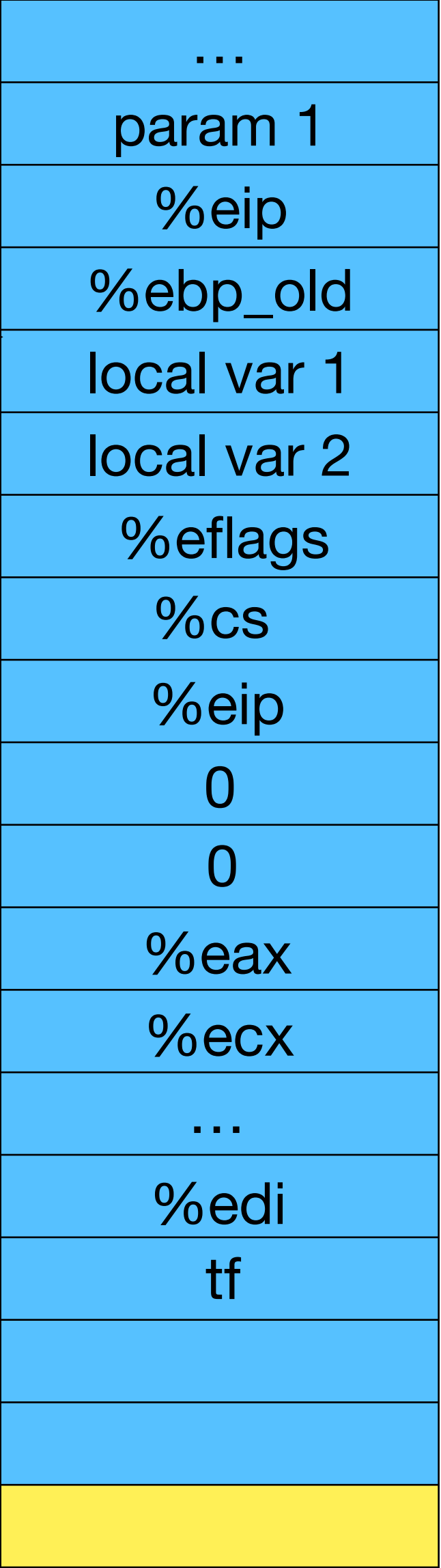
GDT

Stack

ebp

trap frame

esp



Interrupt handling revisited

```
for(;;)
;
```

trap.c

```
void
trap(struct trapframe *tf)
{
    switch(tf->trapno){
    case T_IRQ0 + IRQ_TIMER:
        ticks++;
        cprintf("Tick! %d\n\0", ticks);
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vectors.S

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.globl vector0
vector0:
    pushl $0
    pushl $0
    jmp alltraps
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trapasm.S

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alltraps:
    pushal
    pushl %esp
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    addl $4, %esp
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```

eip

CS

IDT

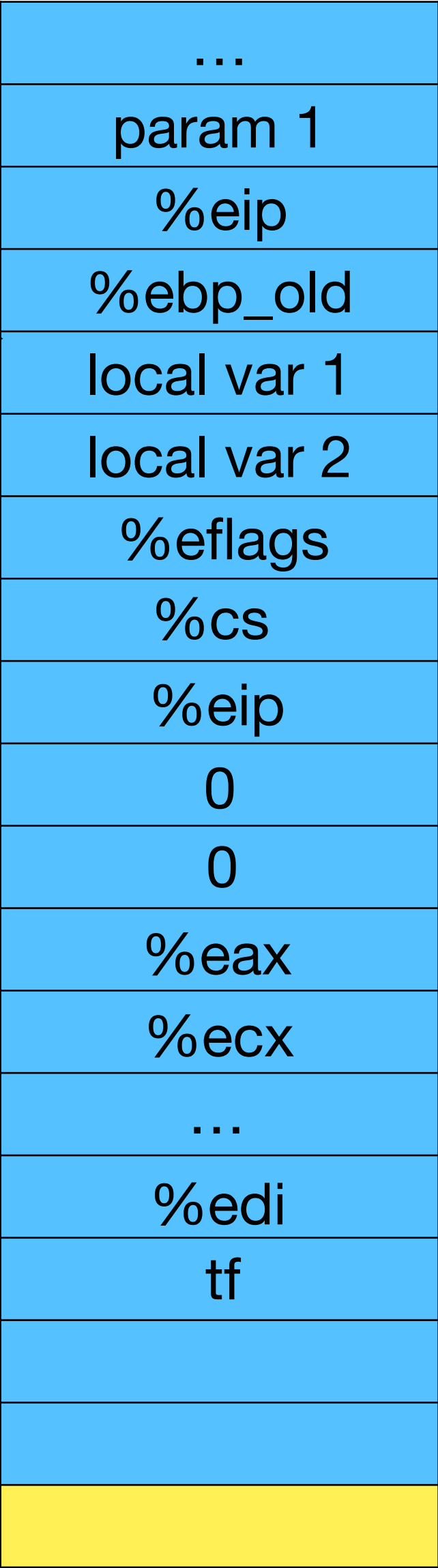
GDT

Stack

ebp

trap frame

esp



Interrupt handling revisited

```
for(;;)
;
```

trap.c

```
void
trap(struct trapframe *tf)
{
    switch(tf->trapno){
    case T_IRQ0 + IRQ_TIMER:
        ticks++;
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vectors.S

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    iret
```

eip

CS

IDT

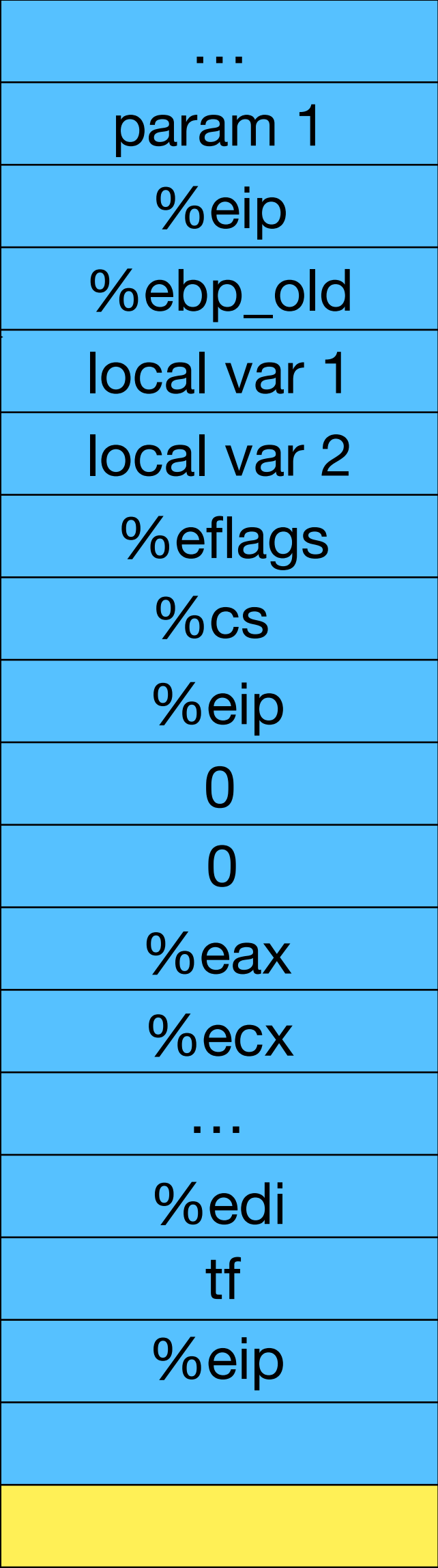
GDT

Stack

ebp

trap frame

esp



Interrupt handling revisited

```
for(;;)
;
```

trap.c

```
void
trap(struct trapframe *tf)
{
    switch(tf->trapno){
    case T_IRQ0 + IRQ_TIMER:
        ticks++;
        cprintf("Tick! %d\n\0", ticks);
        lapiceoi();
    ..
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```

vectors.S

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vector0:
    pushl $0
    pushl $0
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```

trapasm.S

```
alltraps:
    pushal
    pushl %esp
    call trap
    addl $4, %esp
    popal
    addl $0x8, %esp
    iret
```

eip

CS

IDT

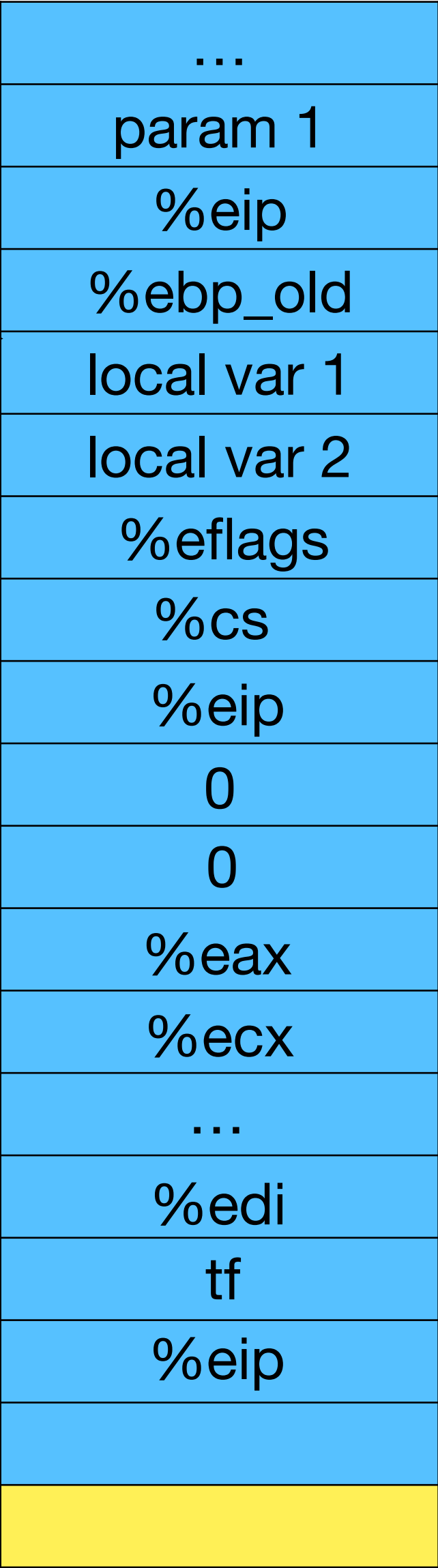
GDT

Stack

ebp

trap frame

esp



Interrupt handling revisited

```
for(;;)  
;
```

trap.c

```
void  
trap(struct trapframe *tf)  
{  
  switch(tf->trapno){  
    case T_IRQ0 + IRQ_TIMER:  
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vectors.S

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vector0:  
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  pushl $0  
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```

trapasm.S

```
alltraps:  
  pushal  
  pushl %esp  
  call trap  
  addl $4, %esp  
  popal  
  addl $0x8, %esp  
  iret
```

IDT

GDT

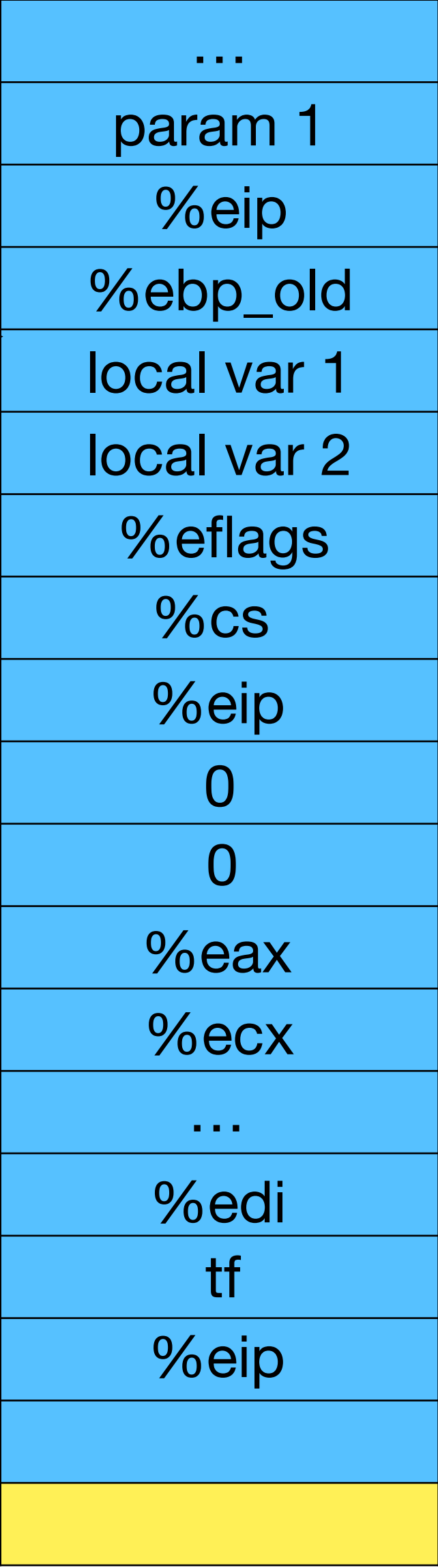
CS

Stack

ebp

trap frame

esp



Interrupt handling revisited

```
for(;;)
;
```

trap.c

```
void
trap(struct trapframe *tf)
{
  switch(tf->trapno){
  case T_IRQ0 + IRQ_TIMER:
    ticks++;
    cprintf("Tick! %d\n\0", ticks);
    lapiceoi();
  ..
  }
  return
}
```

vectors.S

```
.globl vector0
vector0:
  pushl $0
  pushl $0
  jmp alltraps
```

trapasm.S

```
alltraps:
  pushal
  pushl %esp
  call trap
  addl $4, %esp
  popal
  addl $0x8, %esp
  iret
```

IDT

GDT

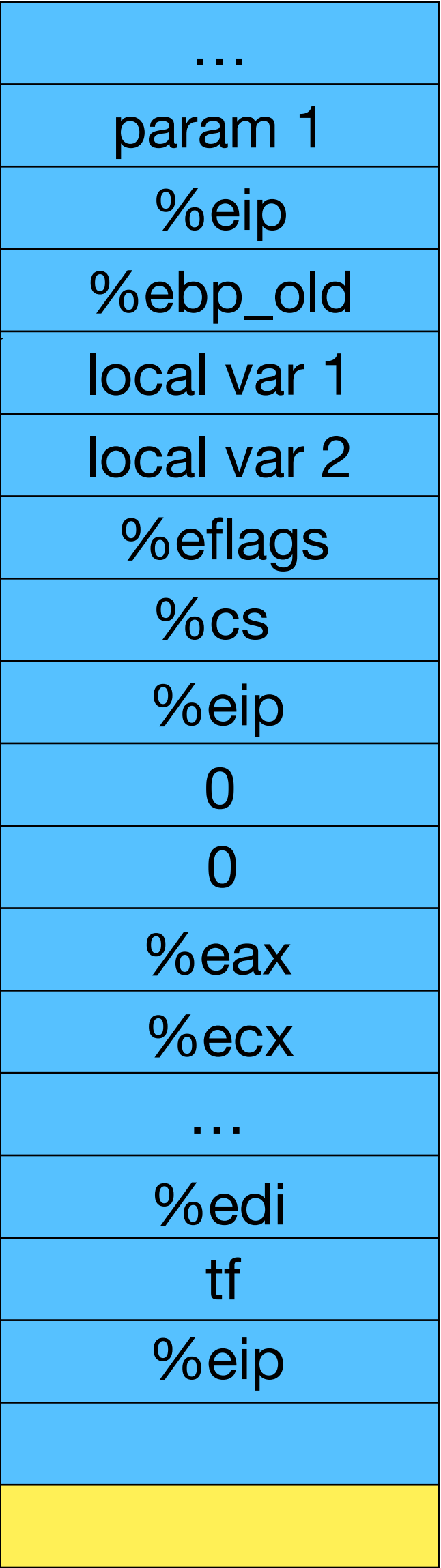
CS

Stack

ebp

trap frame

esp



Interrupt handling revisited

```
for(;;)  
;
```

trap.c

```
void  
trap(struct trapframe *tf)  
{  
    switch(tf->trapno){  
    case T_IRQ0 + IRQ_TIMER:  
        ticks++;  
        cprintf("Tick! %d\n\0", ticks);  
        lapiceoi();  
        ..  
    }  
}
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vectors.S

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trapasm.S

```
alltraps:  
    pushal  
    pushl %esp  
    call trap  
    addl $4, %esp  
    popal  
    addl $0x8, %esp  
    iret
```

IDT

GDT

CS

Stack

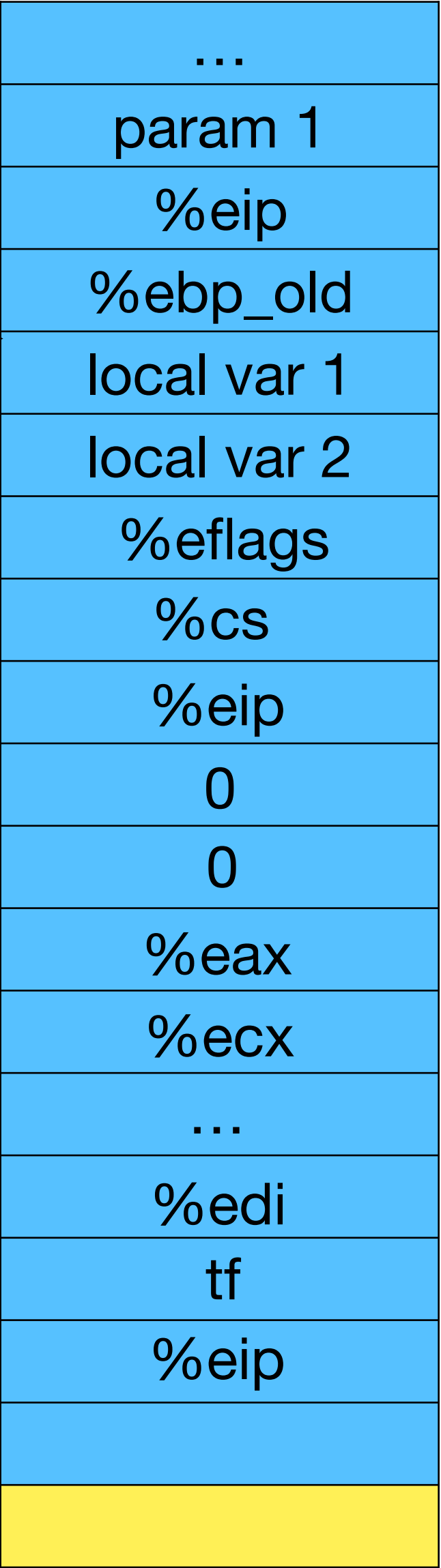
ebp

trap frame

esp

eip

return



Interrupt handling revisited

```
for(;;)  
;
```

trap.c

```
void  
trap(struct trapframe *tf)  
{  
    switch(tf->trapno){  
    case T_IRQ0 + IRQ_TIMER:  
        ticks++;  
        cprintf("Tick! %d\n\0", ticks);  
        lapiceoi();  
        ..  
    }  
}
```

eip → return

vectors.S

```
.globl vector0  
vector0:  
    pushl $0  
    pushl $0  
    jmp alltraps
```

trapasm.S

```
alltraps:  
    pushal  
    pushl %esp  
    call trap  
    addl $4, %esp  
    popal  
    addl $0x8, %esp  
    iret
```

IDT

GDT

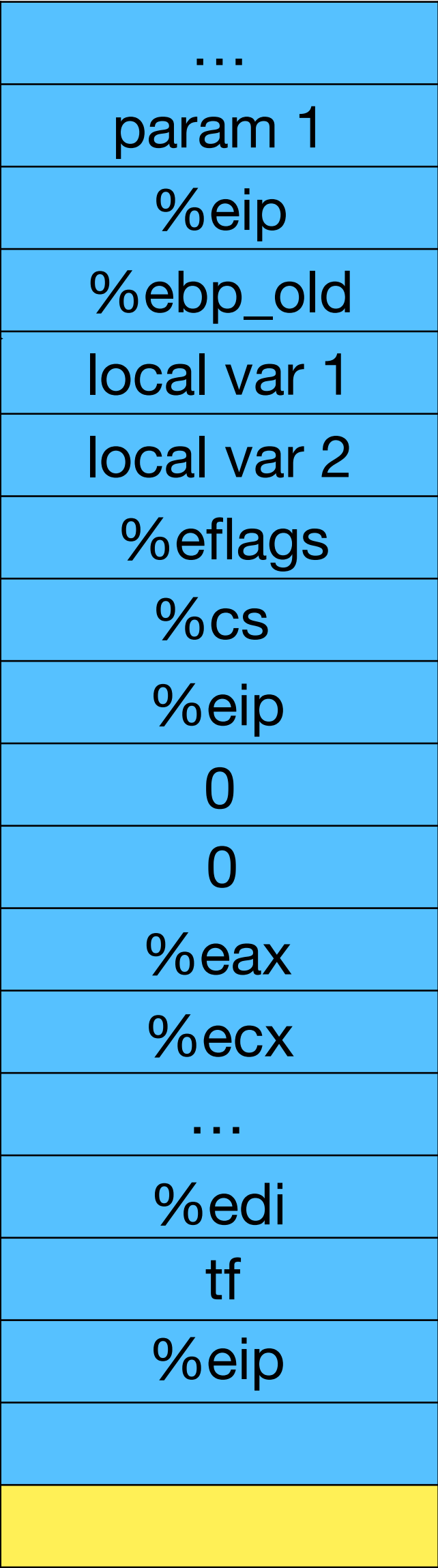
CS

Stack

ebp

trap frame

esp



Interrupt handling revisited

```
for(;;)
;
```

trap.c

```
void
trap(struct trapframe *tf)
{
    switch(tf->trapno){
    case T_IRQ0 + IRQ_TIMER:
        ticks++;
        cprintf("Tick! %d\n\0", ticks);
        lapiceoi();
    ..
    }
    return
}
```

vectors.S

```
.globl vector0
vector0:
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trapasm.S

```
alltraps:
    pushal
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    call trap
    addl $4, %esp
    popal
    addl $0x8, %esp
    iret
```

eip

CS

IDT

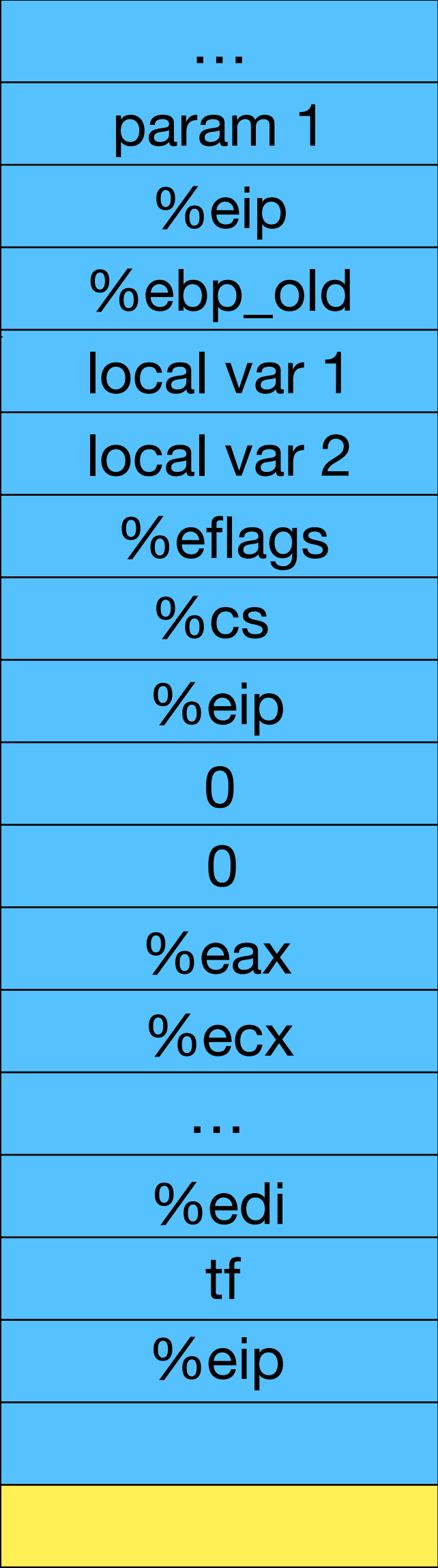
GDT

Stack

ebp

trap frame

esp



Interrupt handling revisited

```
for(;;)  
;
```

trap.c

```
void  
trap(struct trapframe *tf)  
{  
    switch(tf->trapno){  
    case T_IRQ0 + IRQ_TIMER:  
        ticks++;  
        cprintf("Tick! %d\n\0", ticks);  
        lapiceoi();  
    ..  
    return
```

vectors.S

```
.globl vector0  
vector0:  
    pushl $0  
    pushl $0  
    jmp alltraps
```

trapasm.S

```
alltraps:  
    pushal  
    pushl %esp  
    call trap  
    addl $4, %esp  
    popal  
    addl $0x8, %esp  
    iret
```

eip

CS

IDT

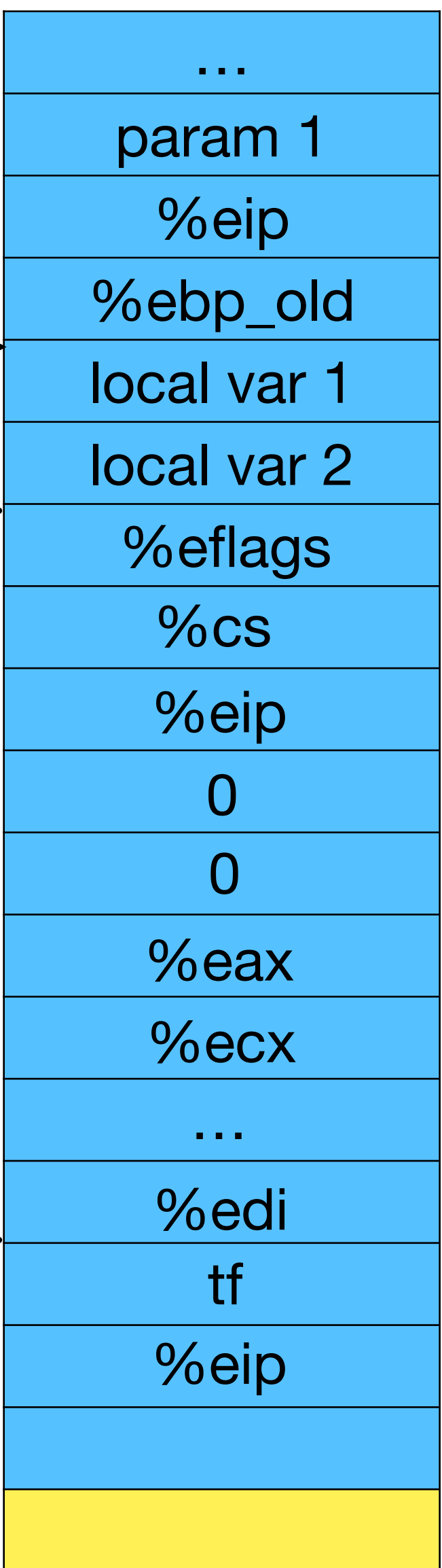
GDT

Stack

ebp

trap frame

esp



Interrupt handling revisited

```
for(;;)
;
```

trap.c

```
void
trap(struct trapframe *tf)
{
    switch(tf->trapno){
    case T_IRQ0 + IRQ_TIMER:
        ticks++;
        cprintf("Tick! %d\n\0", ticks);
        lapiceoi();
    ..
    return
```

vectors.S

```
.globl vector0
vector0:
    pushl $0
    pushl $0
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```

trapasm.S

```
alltraps:
    pushal
    pushl %esp
    call trap
    addl $4, %esp
    popal
    addl $0x8, %esp
    iret
```

eip

CS

IDT

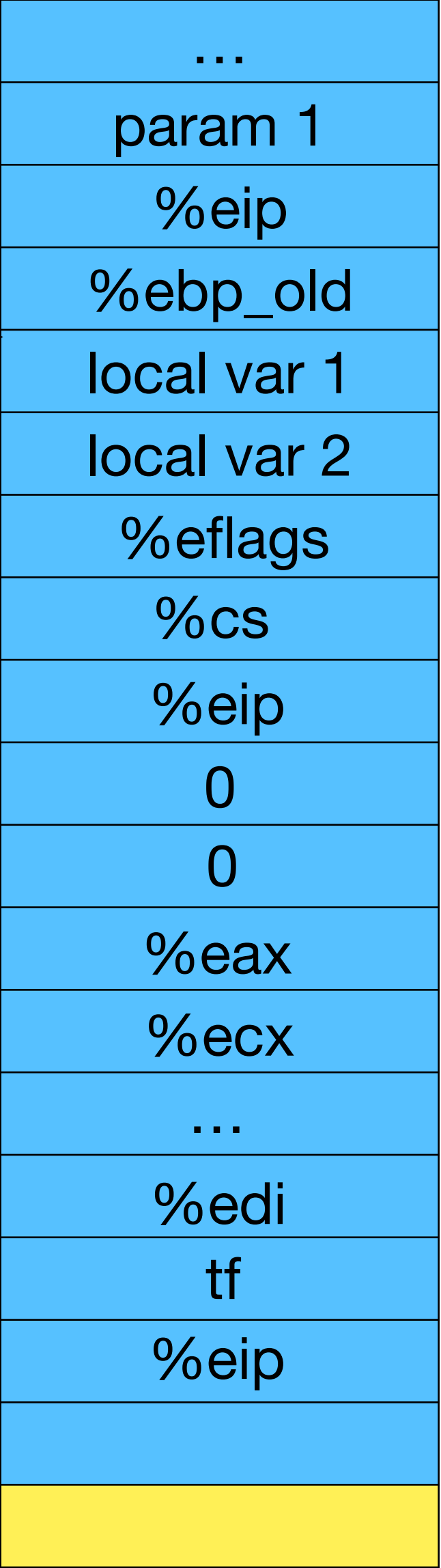
GDT

Stack

ebp

trap frame

esp



Interrupt handling revisited

```
for(;;)  
;
```

trap.c

```
void  
trap(struct trapframe *tf)  
{  
    switch(tf->trapno){  
    case T_IRQ0 + IRQ_TIMER:  
        ticks++;  
        cprintf("Tick! %d\n\0", ticks);  
        lapiceoi();  
    ..  
    return
```

vectors.S

```
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trapasm.S

```
alltraps:  
    pushal  
    pushl %esp  
    call trap  
    addl $4, %esp  
    popal  
    addl $0x8, %esp  
    iret
```

eip

CS

IDT

GDT

ebp

trap frame

Stack

...
param 1
%eip
%ebp_old
local var 1
local var 2
%eflags
%cs
%eip
0
0
%eax
%ecx
...
%edi
tf
%eip

esp

Interrupt handling revisited

```
for(;;)
;
```

trap.c

```
void
trap(struct trapframe *tf)
{
    switch(tf->trapno){
    case T_IRQ0 + IRQ_TIMER:
        ticks++;
        cprintf("Tick! %d\n\0", ticks);
        lapiceoi();
    ..
    return
```

vectors.S

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trapasm.S

```
alltraps:
    pushal
    pushl %esp
    call trap
    addl $4, %esp
    popal
    addl $0x8, %esp
    iret
```

eip →

CS →

IDT

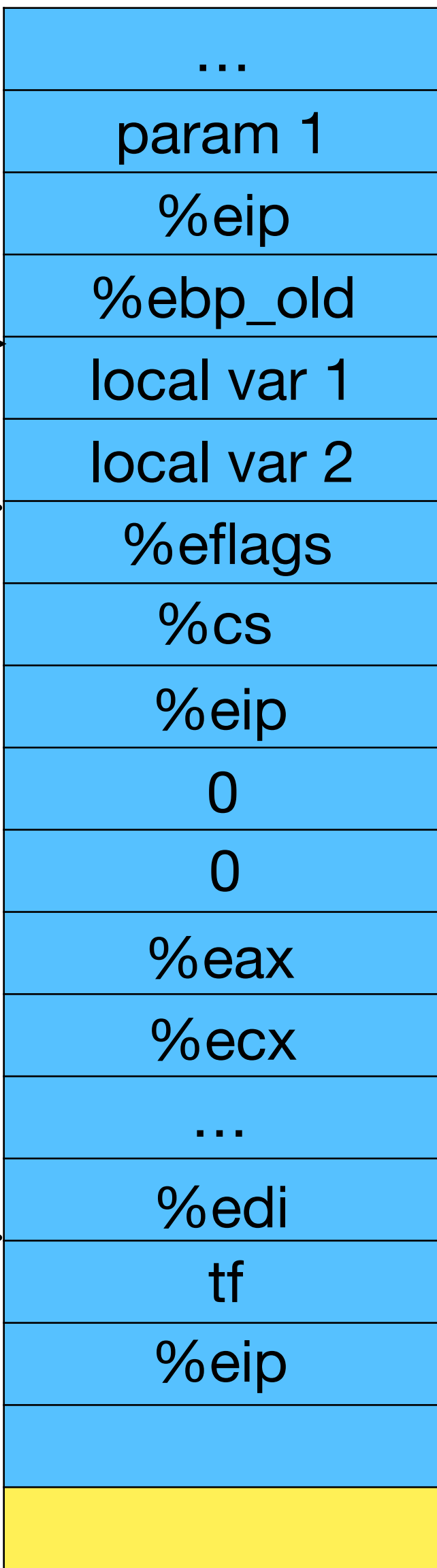
GDT

Stack

ebp →

trap frame

esp ←



Interrupt handling revisited

```
for(;;)  
;
```

trap.c

```
void  
trap(struct trapframe *tf)  
{  
    switch(tf->trapno){  
    case T_IRQ0 + IRQ_TIMER:  
        ticks++;  
        cprintf("Tick! %d\n\0", ticks);  
        lapiceoi();  
    ..  
    return
```

vectors.S

```
.globl vector0  
vector0:  
    pushl $0  
    pushl $0  
    jmp alltraps
```

trapasm.S

```
alltraps:  
    pushal  
    pushl %esp  
    call trap  
    addl $4, %esp  
    popal  
    addl $0x8, %esp  
    iret
```

IDT

GDT

CS

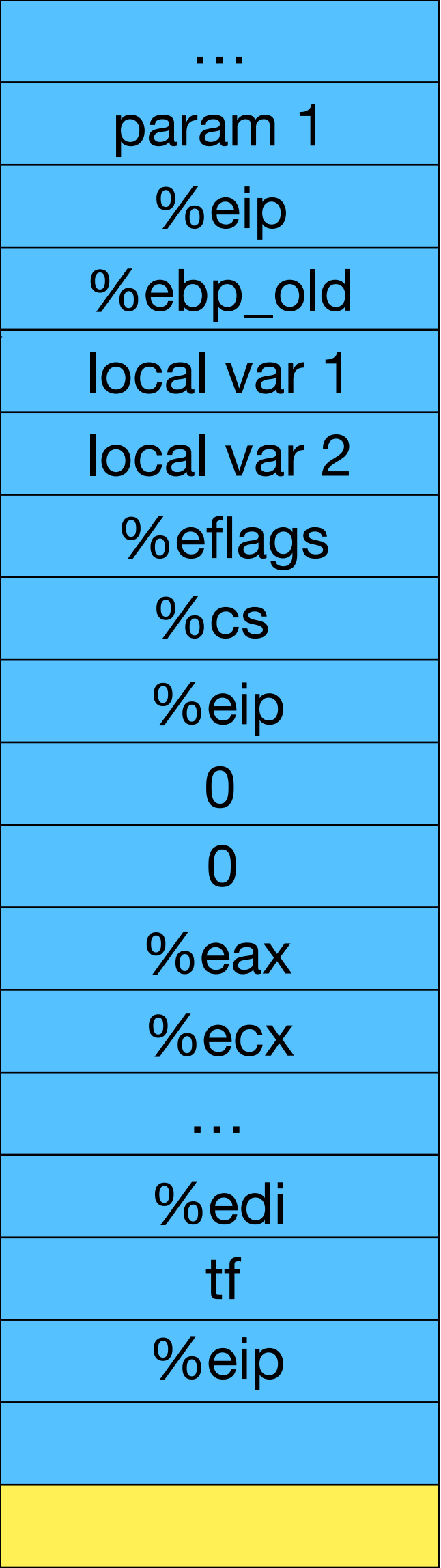
eip

Stack

ebp

esp

trap frame



Interrupt handling revisited

```
for(;;)  
;
```

trap.c

```
void  
trap(struct trapframe *tf)  
{  
    switch(tf->trapno){  
    case T_IRQ0 + IRQ_TIMER:  
        ticks++;  
        cprintf("Tick! %d\n\0", ticks);  
        lapiceoi();  
    ..  
    return
```

vectors.S

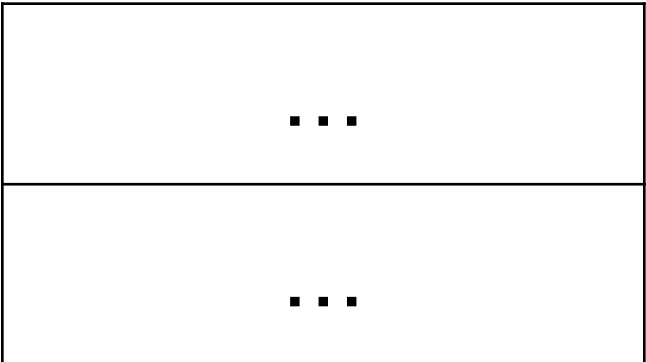
```
.globl vector0  
vector0:  
    pushl $0  
    pushl $0  
    jmp alltraps
```

trapasm.S

```
alltraps:  
    pushal  
    pushl %esp  
    call trap  
    addl $4, %esp  
    popal  
    addl $0x8, %esp  
    iret
```

eip →

IDT



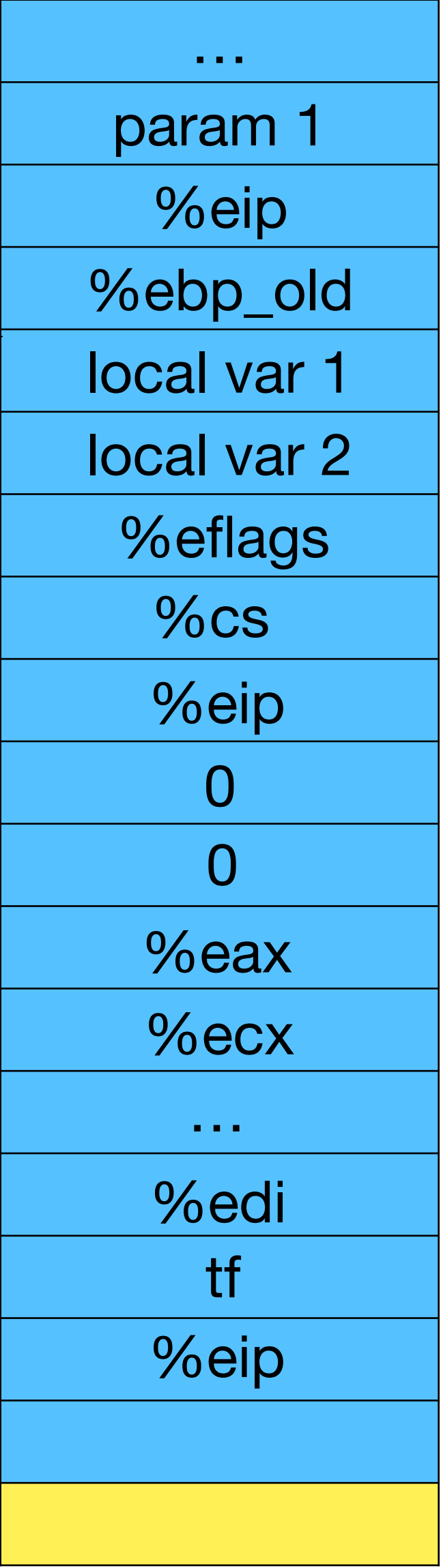
GDT



CS →

ebp →

Stack



esp ←

Interrupt handling revisited

```
for(;;)  
;
```

trap.c

```
void  
trap(struct trapframe *tf)  
{  
    switch(tf->trapno){  
    case T_IRQ0 + IRQ_TIMER:  
        ticks++;  
        cprintf("Tick! %d\n\0", ticks);  
        lapiceoi();  
    ..  
    return
```

vectors.S

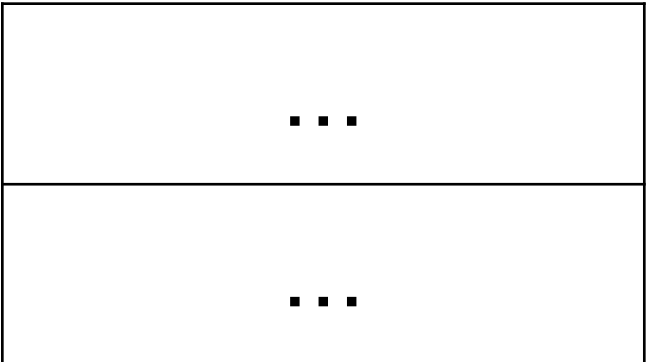
```
.globl vector0  
vector0:  
    pushl $0  
    pushl $0  
    jmp alltraps
```

trapasm.S

```
alltraps:  
    pushal  
    pushl %esp  
    call trap  
    addl $4, %esp  
    popal  
    addl $0x8, %esp  
    iret
```

eip →

IDT



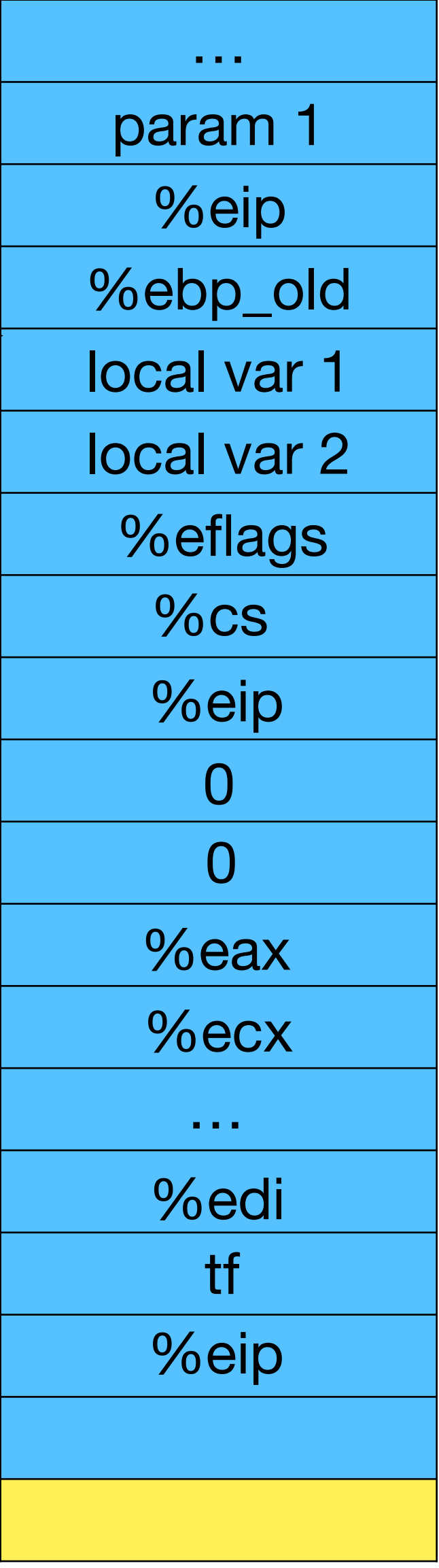
GDT



CS →

ebp →

Stack



esp ←

trap frame

Interrupt handling revisited

```
for(;;)  
;
```

trap.c

```
void  
trap(struct trapframe *tf)  
{  
    switch(tf->trapno){  
    case T_IRQ0 + IRQ_TIMER:  
        ticks++;  
        cprintf("Tick! %d\n\0", ticks);  
        lapiceoi();  
    ..  
    return
```

vectors.S

```
.globl vector0  
vector0:  
    pushl $0  
    pushl $0  
    jmp alltraps
```

trapasm.S

```
alltraps:  
    pushal  
    pushl %esp  
    call trap  
    addl $4, %esp  
    popal  
    addl $0x8, %esp  
    iret
```

eip →

IDT

GDT

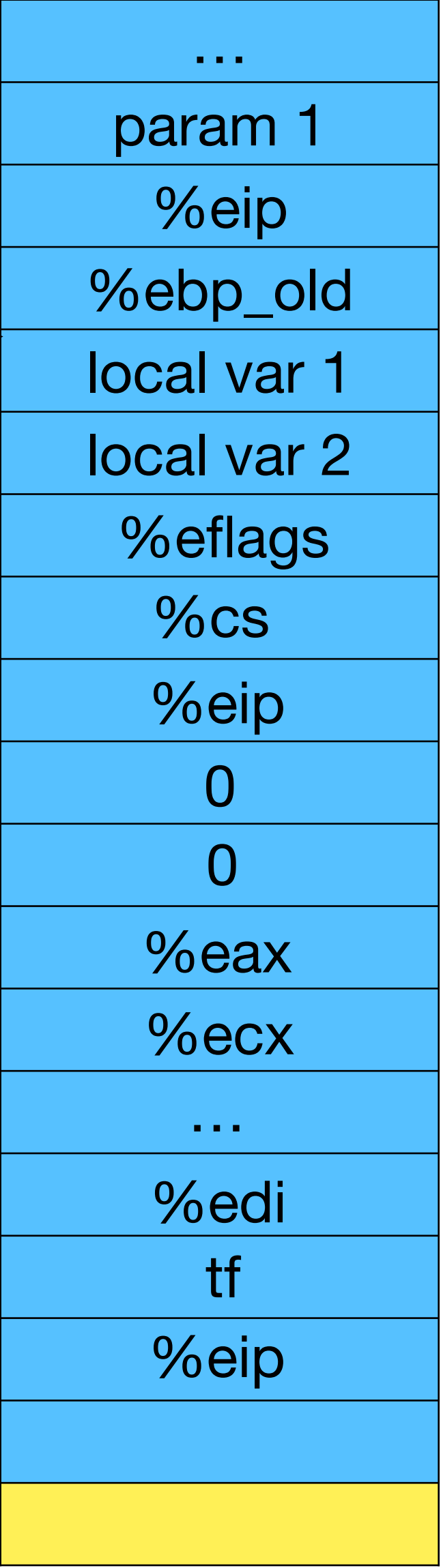
CS →

Stack

ebp →

esp ←

trap frame



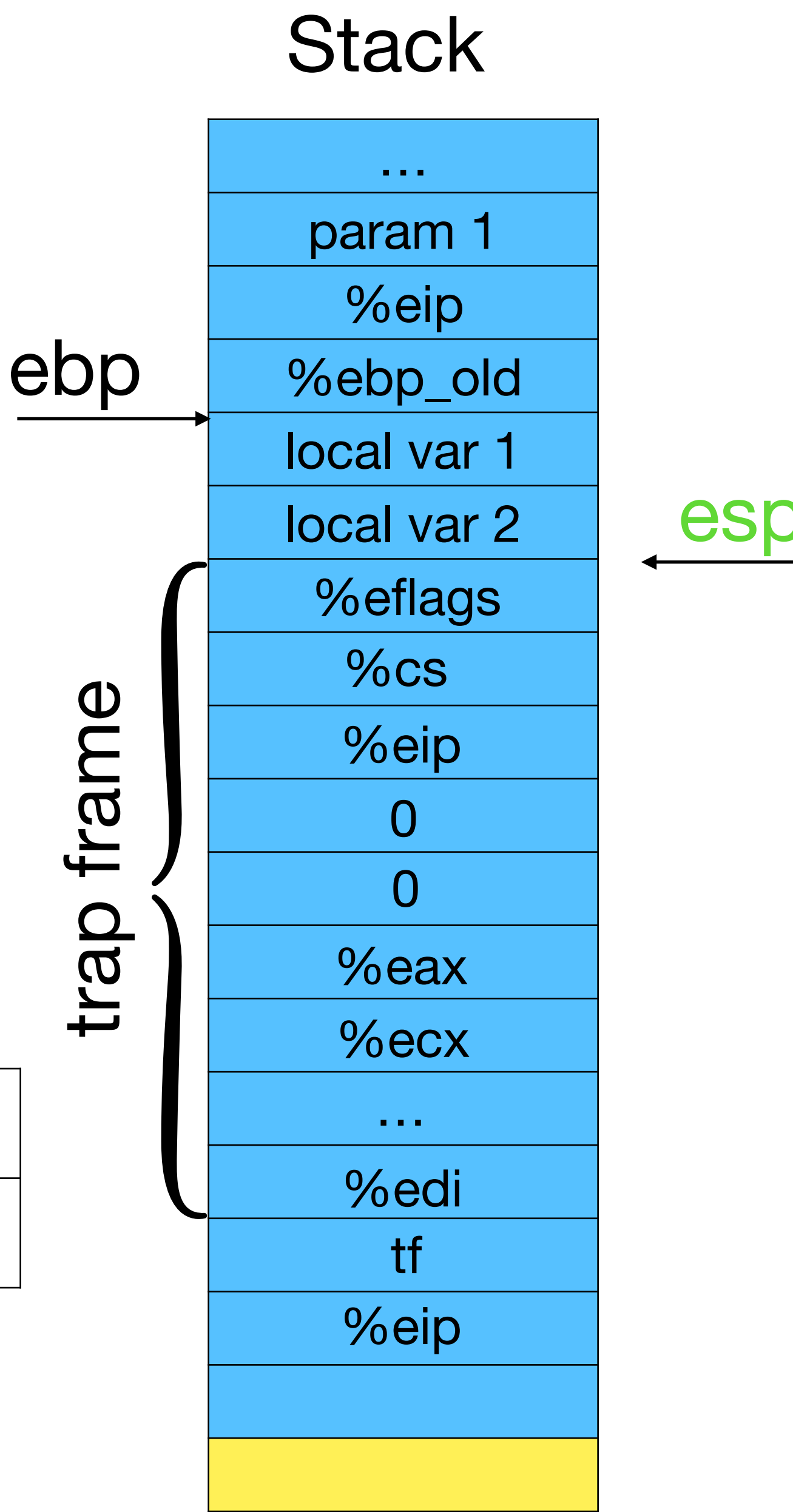
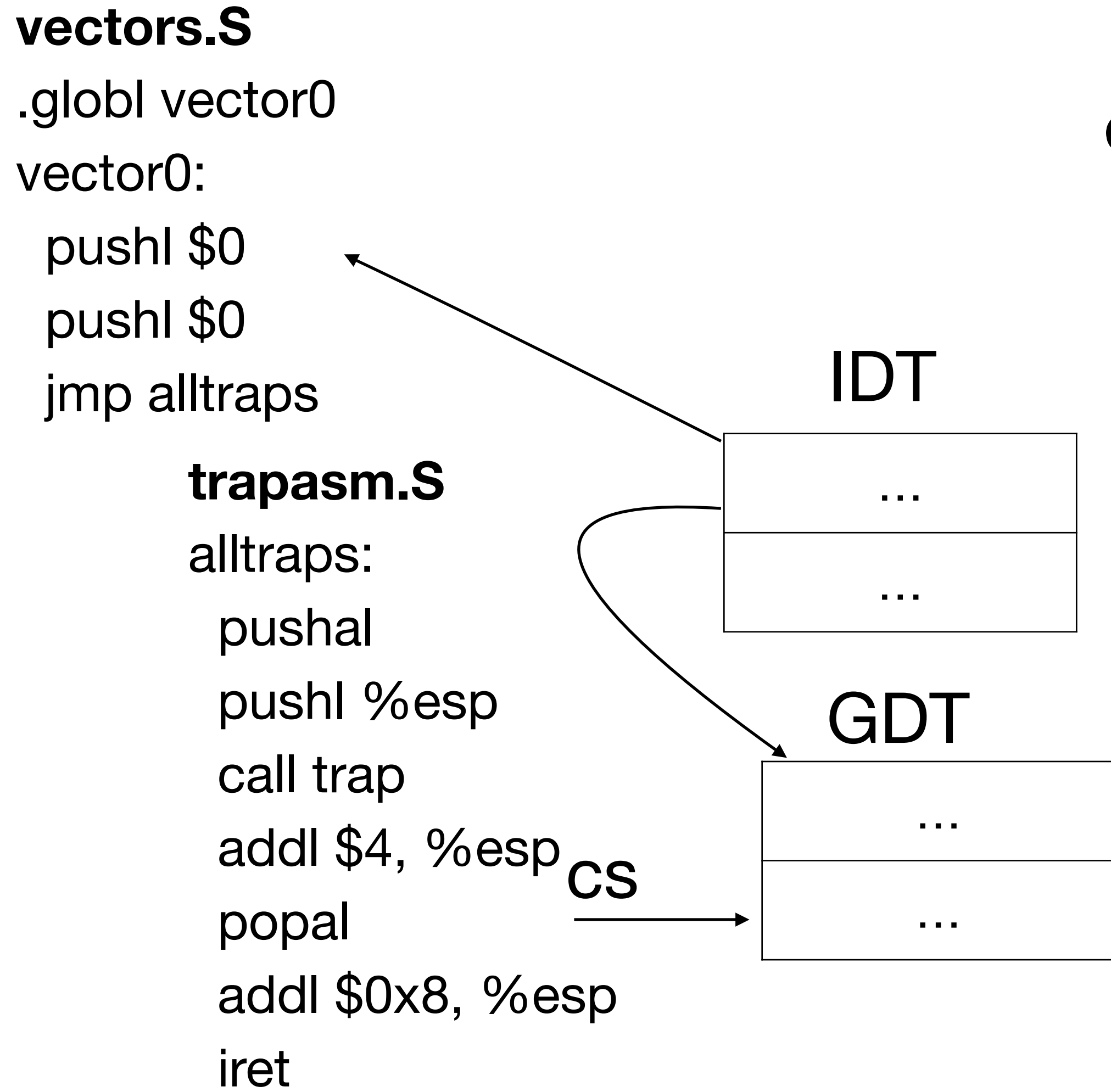
Interrupt handling revisited

eip

→

for(;;)
;

trap.c
void
trap(struct trapframe *tf)
{
 switch(tf->trapno){
 case T_IRQ0 + IRQ_TIMER:
 ticks++;
 cprintf("Tick! %d\n\0", ticks);
 lapiceoi();
 ..
 return



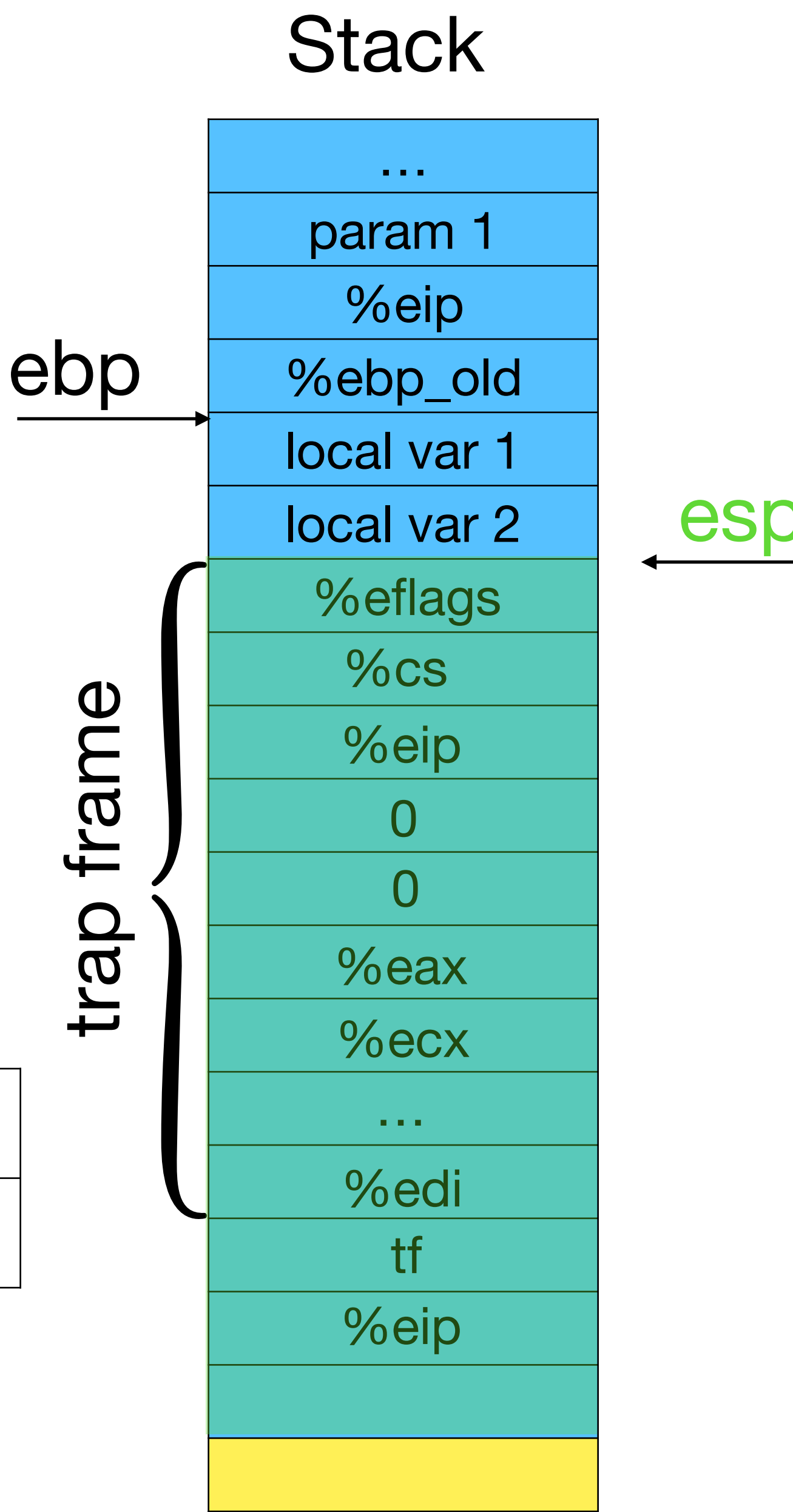
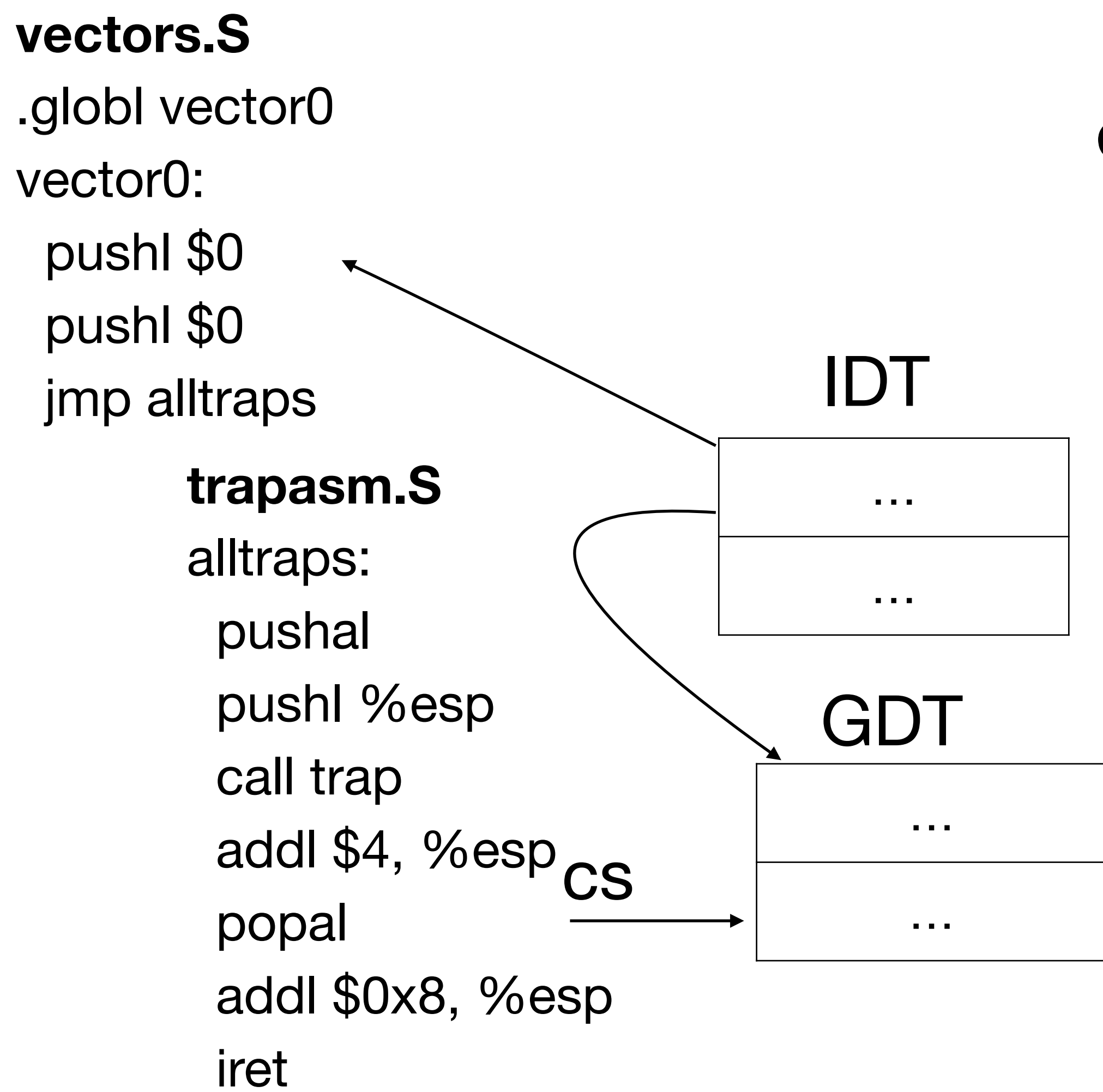
Interrupt handling revisited

eip

→

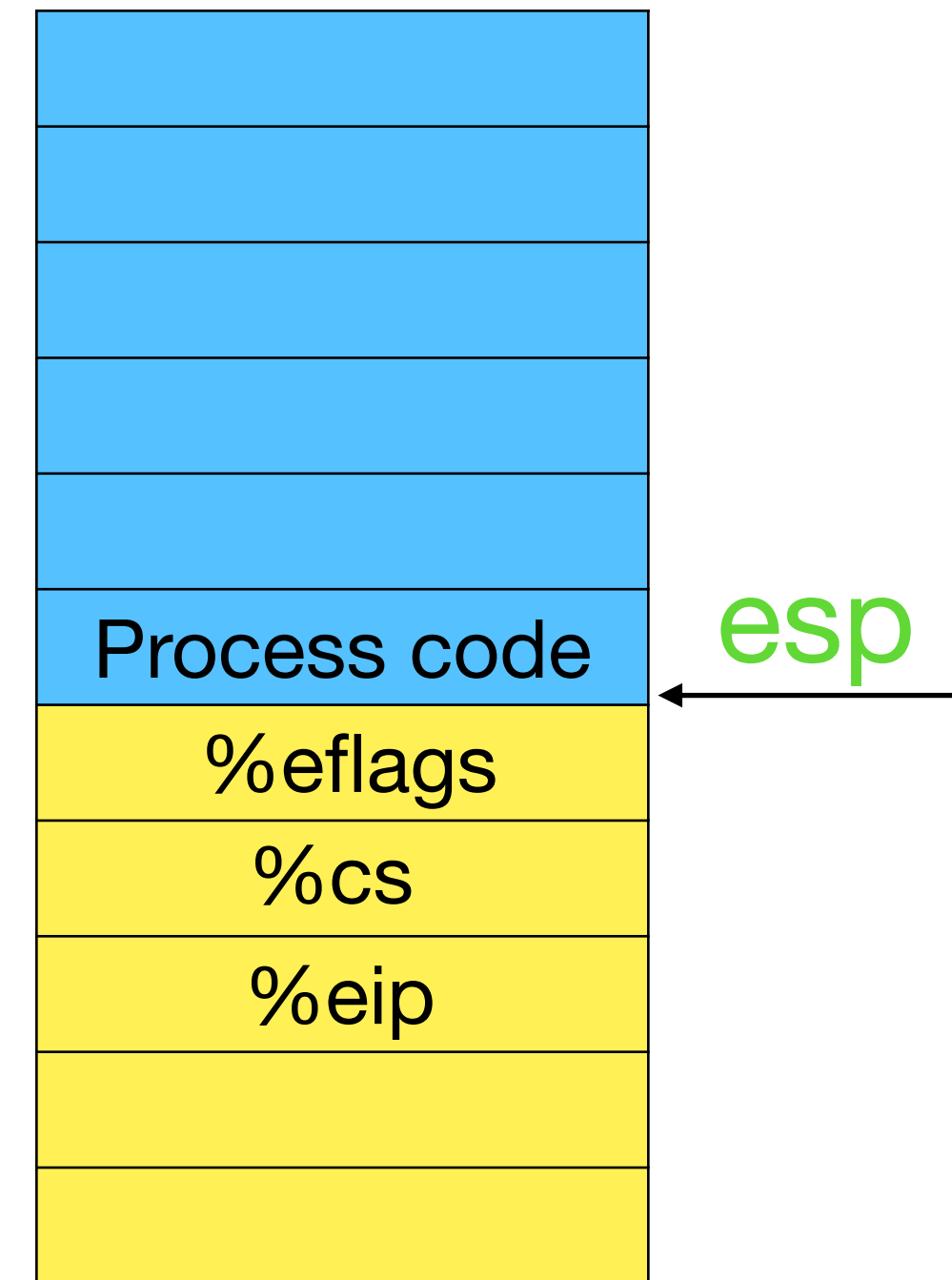
for(;;)
;

trap.c
void
trap(struct trapframe *tf)
{
 switch(tf->trapno){
 case T_IRQ0 + IRQ_TIMER:
 ticks++;
 cprintf("Tick! %d\n\0", ticks);
 lapiceoi();
 ..
 return



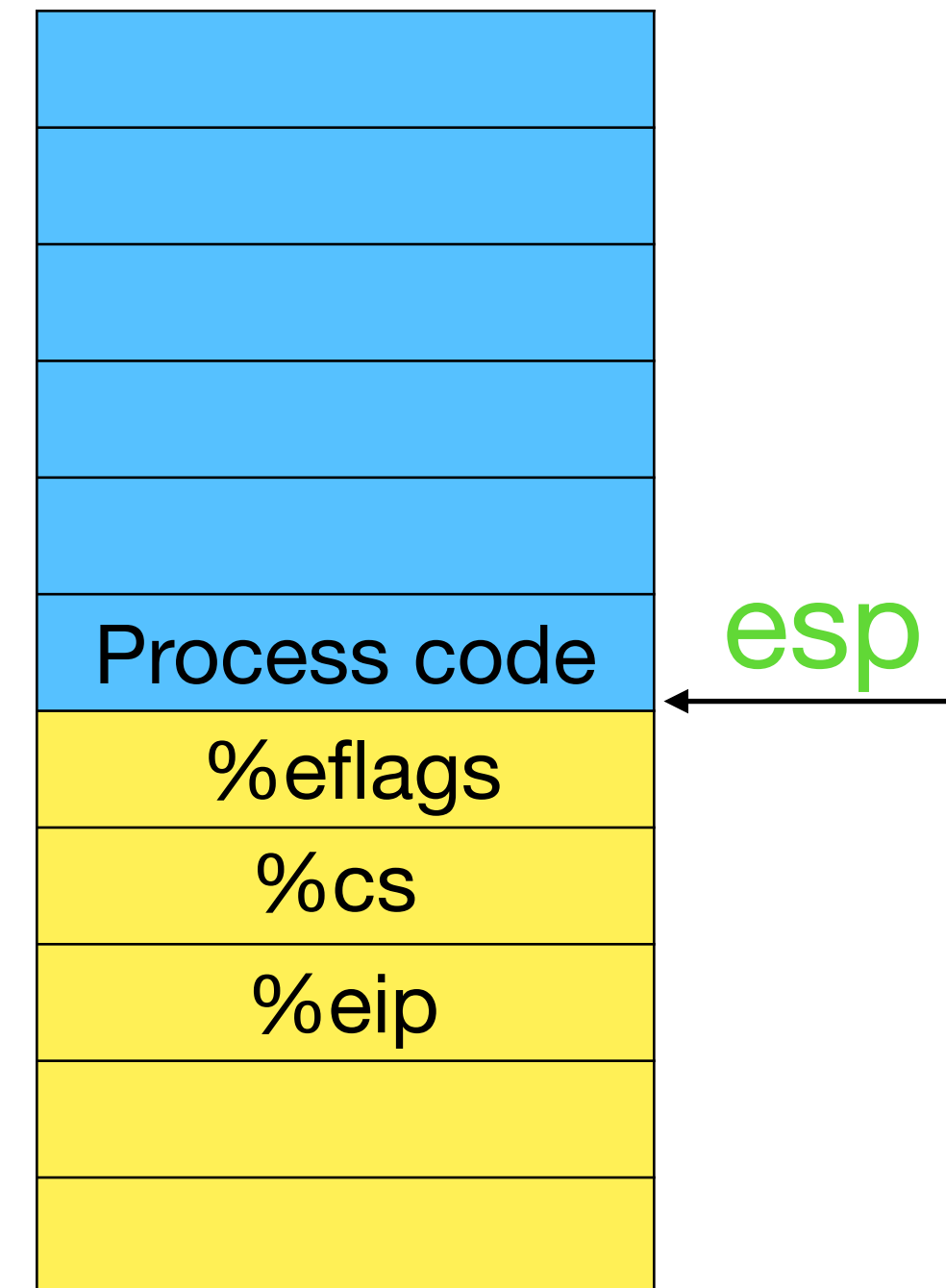
Interrupt handling problem

- We cannot trust %esp of the process. Hardware might write (%eflags, %cs, %eip) into another process' address space or into OS memory.



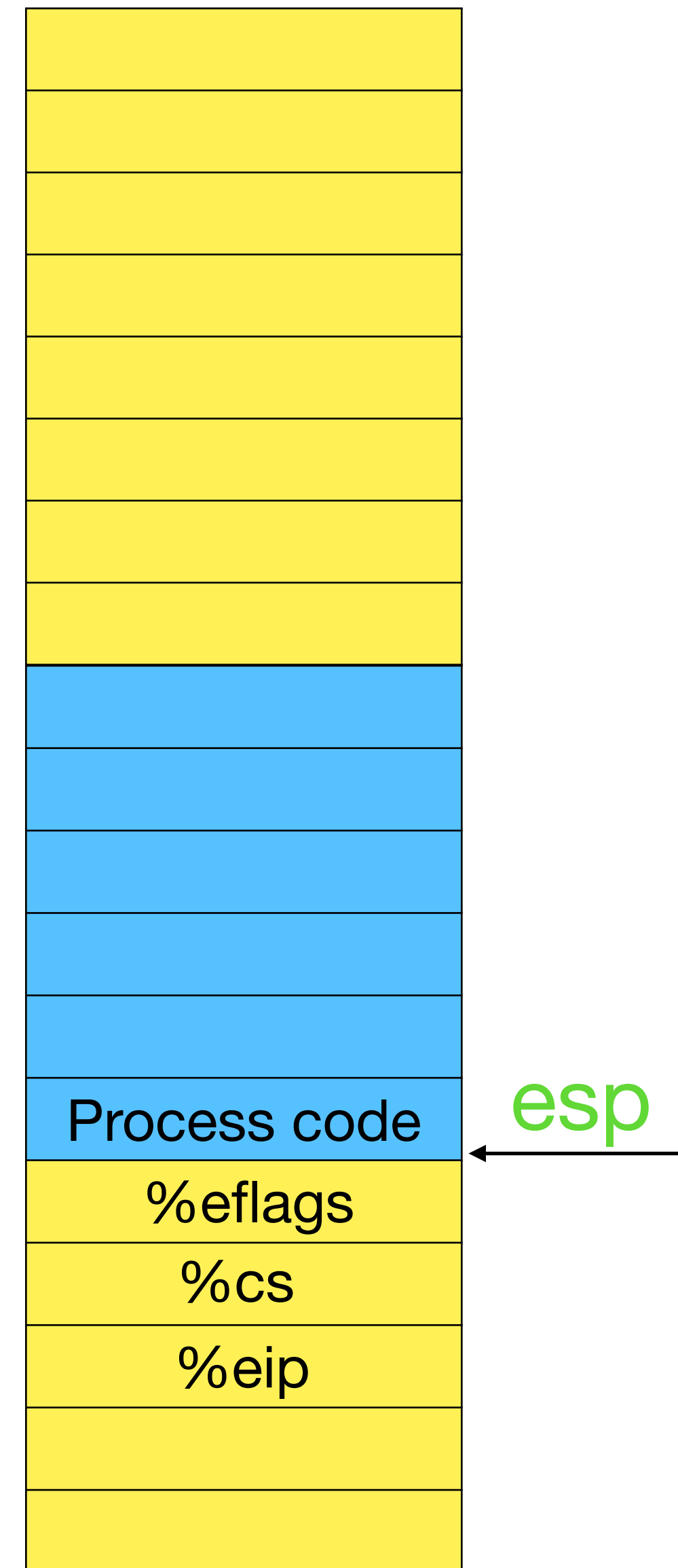
Interrupt handling problem

- We cannot trust %esp of the process. Hardware might write (%eflags, %cs, %eip) into another process' address space or into OS memory.
- Use a separate stack set up by the OS!



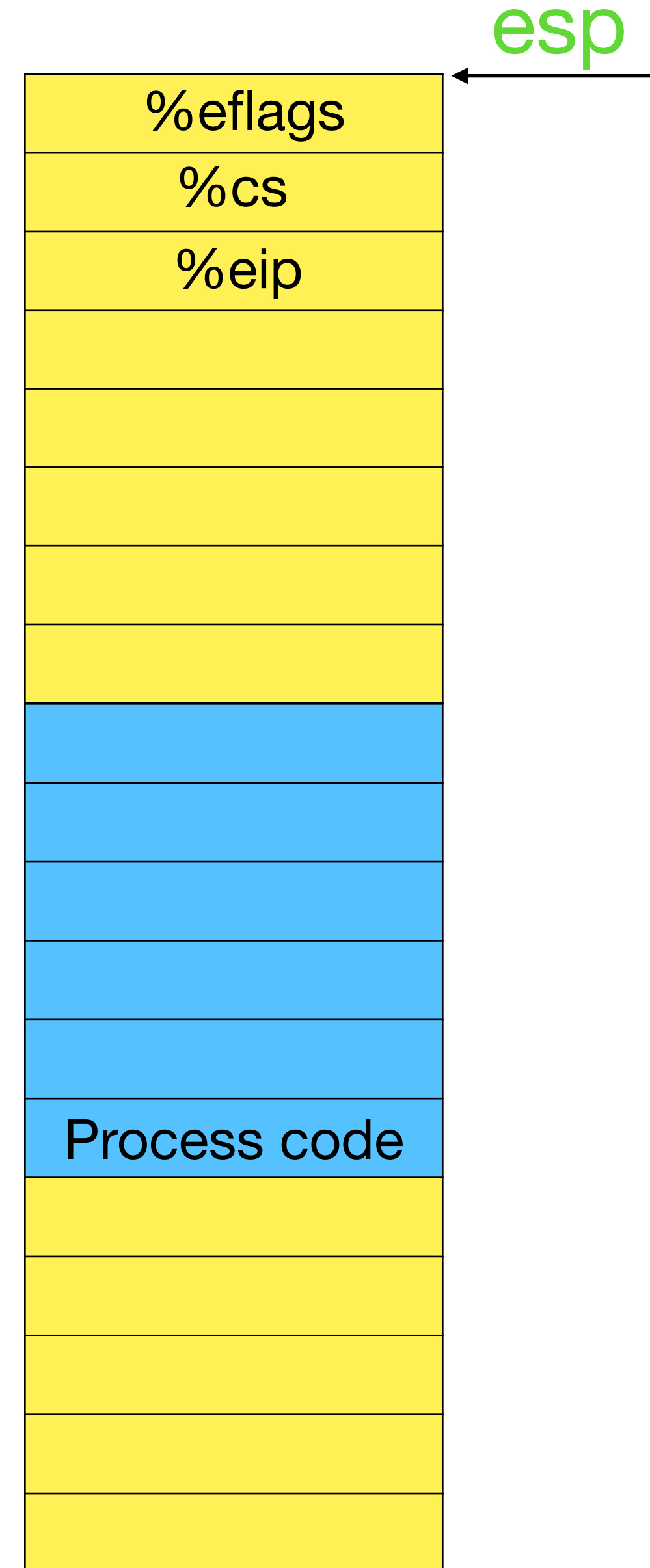
Interrupt handling problem

- We cannot trust `%esp` of the process. Hardware might write (`%eflags`, `%cs`, `%eip`) into another process' address space or into OS memory.
- Use a separate stack set up by the OS!



Interrupt handling problem

- We cannot trust %esp of the process. Hardware might write (%eflags, %cs, %eip) into another process' address space or into OS memory.
- Use a separate stack set up by the OS!



OS tells hardware where the kernel stack is

- In task-state segment (TSS), OS sets up
 - Stack segment for transition to ring 0 in SS0 and
 - Stack pointer for transition to ring 0 in ESP0

31	15	0	
I/O Map Base Address		Reserved	T 100
Reserved		LDT Segment Selector 96	
Reserved		GS 92	
Reserved		FS 88	
Reserved		DS 84	
Reserved		SS 80	
Reserved		CS 76	
Reserved		ES 72	
EDI			68
ESI			64
EBP			60
ESP			56
EBX			52
EDX			48
ECX			44
EAX			40
EFLAGS			36
EIP			32
CR3 (PDBR)			28
Reserved		SS2	24
ESP2			20
Reserved		SS1	16
ESP1			12
Reserved		SS0	8
ESP0			4
Reserved		Previous Task Link 0	

Reserved bits. Set to 0.

Figure 7-2. 32-Bit Task-State Segment (TSS)

OS tells hardware where the kernel stack is (2)

- OS sets up an entry in GDT that contains location of TSS
- LTR instruction changes task register

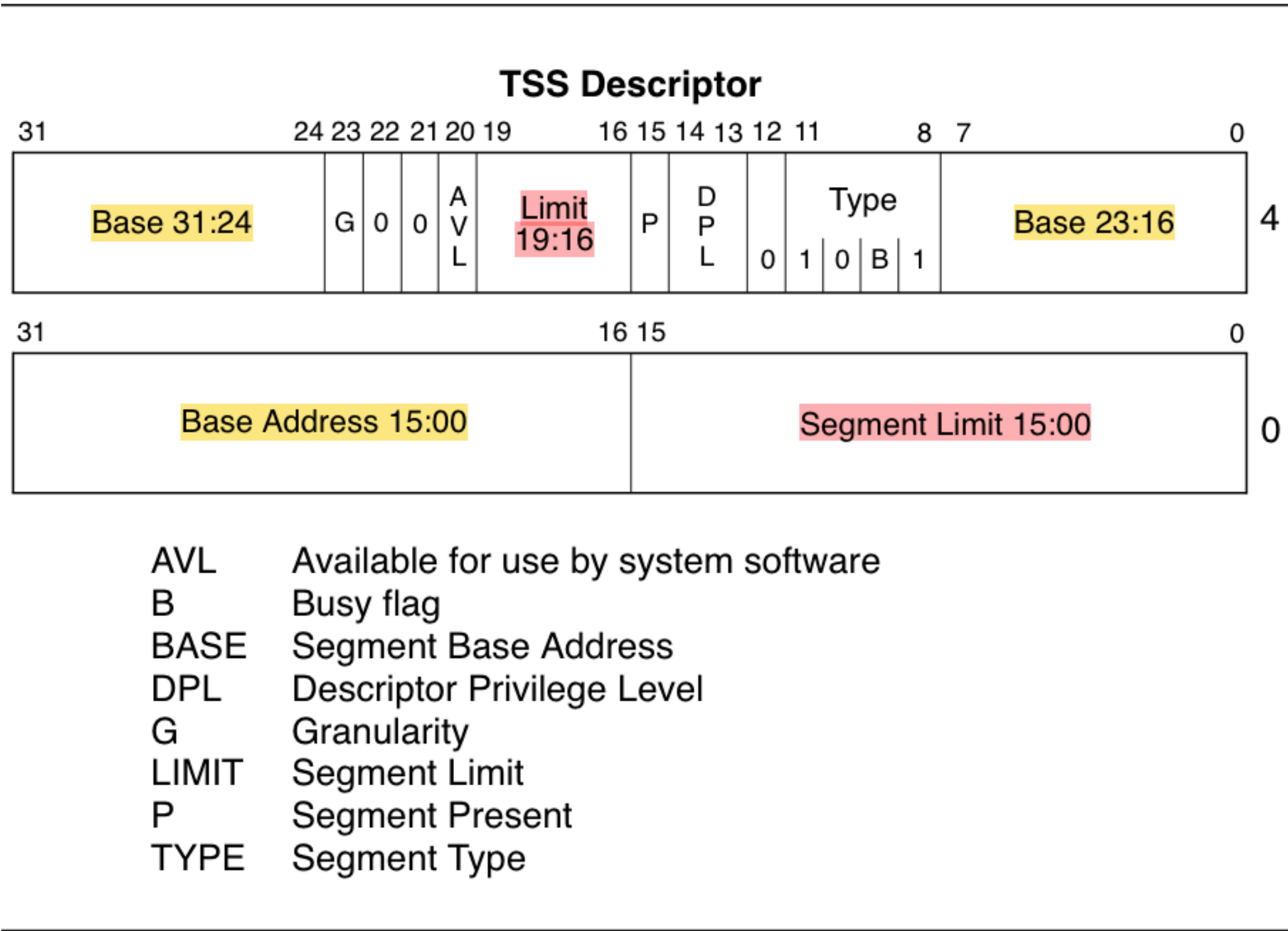


Figure 7-3. TSS Descriptor

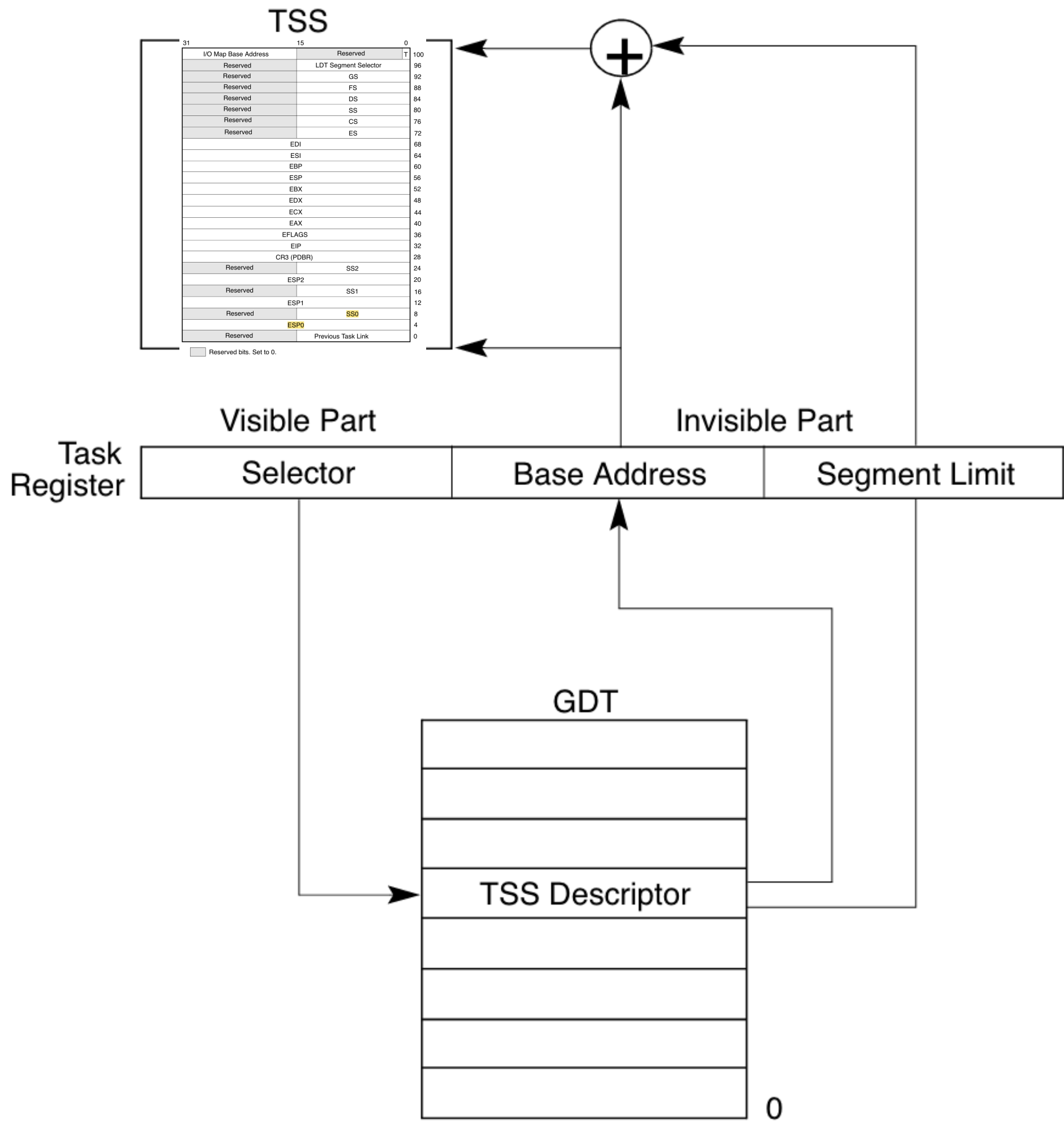
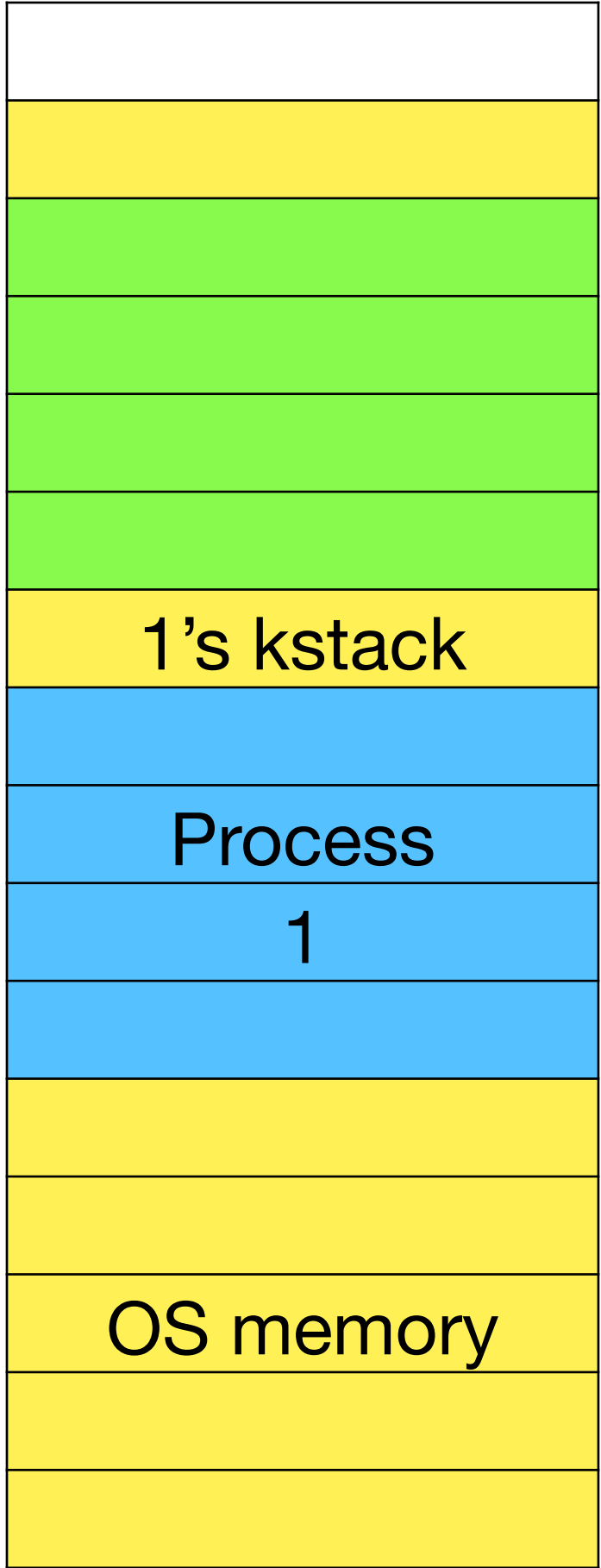


Figure 7-5. Task Register

Code walkthrough: setting up kernel stack

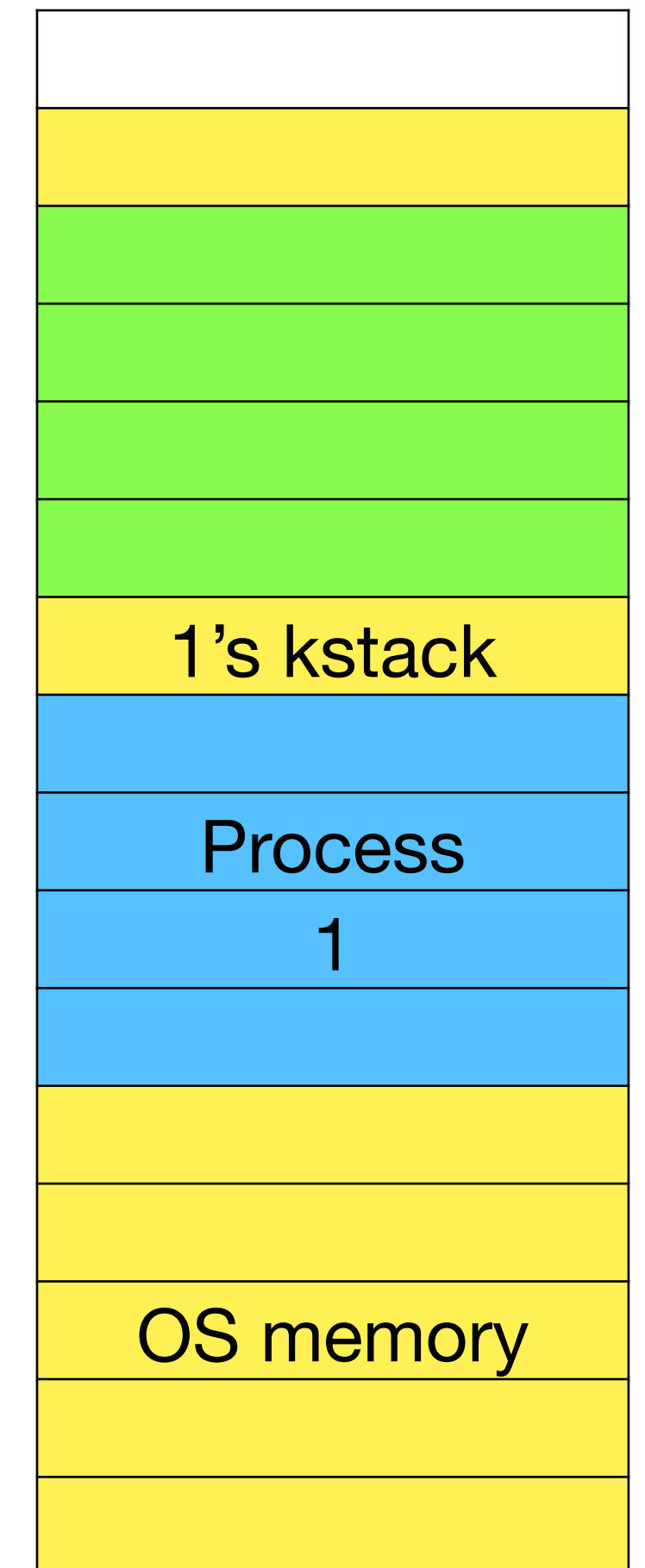
p16-tss

```
void seginit(void) {
    c->gdt[SEG_UCODE] = SEG(.., STARTPROC, (PROCSIZE-1) << 12, DPL_USER);
    c->gdt[SEG_UDATA] = SEG(.., STARTPROC, (PROCSIZE-1) << 12, DPL_USER);
}
```



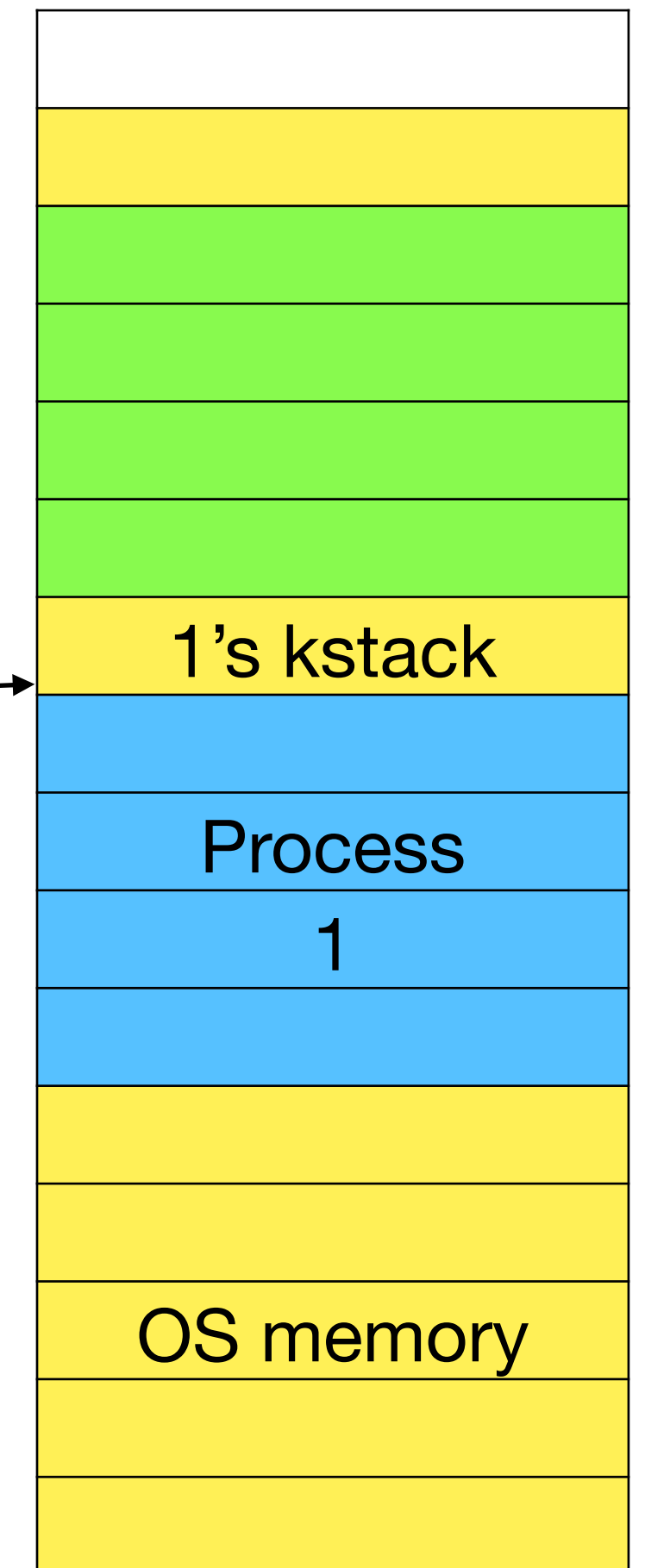
Code walkthrough: setting up kernel stack p16-tss

```
void seginit(void) {  
    c->gdt[SEG_UCODE] = SEG(.., STARTPROC, (PROCSIZE-1) << 12, DPL_USER);  
    c->gdt[SEG_UDATA] = SEG(.., STARTPROC, (PROCSIZE-1) << 12, DPL_USER);  
}  
  
static struct proc* allocproc(void) {  
    sp = (char*)(STARTPROC + (PROCSIZE<<12));  
    p->kstack = sp - KSTACKSIZE;  
}
```



Code walkthrough: setting up kernel stack p16-tss

```
void seginit(void) {  
    c->gdt[SEG_UCODE] = SEG(.., STARTPROC, (PROCSIZE-1) << 12, DPL_USER);  
    c->gdt[SEG_UDATA] = SEG(.., STARTPROC, (PROCSIZE-1) << 12, DPL_USER);  
}  
  
static struct proc* allocproc(void) {  
    sp = (char*)(STARTPROC + (PROCSIZE<<12));  
    p->kstack = sp - KSTACKSIZE;  
}
```



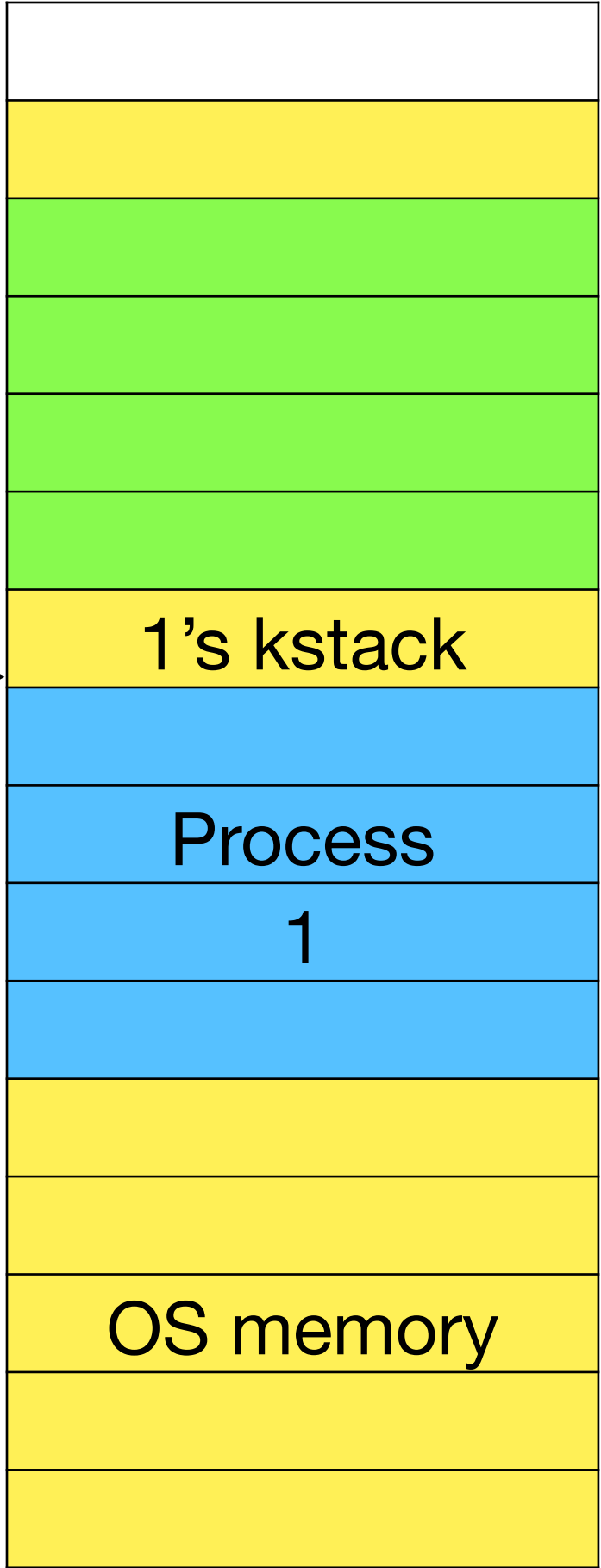
Code walkthrough: setting up kernel stack

p16-tss

```
void seginit(void) {
    c->gdt[SEG_UCODE] = SEG(.., STARTPROC, (PROCSIZE-1) << 12, DPL_USER);
    c->gdt[SEG_UDATA] = SEG(.., STARTPROC, (PROCSIZE-1) << 12, DPL_USER);
}

static struct proc* allocproc(void) {
    sp = (char*)(STARTPROC + (PROCSIZE<<12));
    p->kstack = sp - KSTACKSIZE;
}

void scheduler(void) {
    // pick RUNNABLE process p
    switchvm(p);
    swtch(p->context);
}
```



Code walkthrough: setting up kernel stack p16-tss

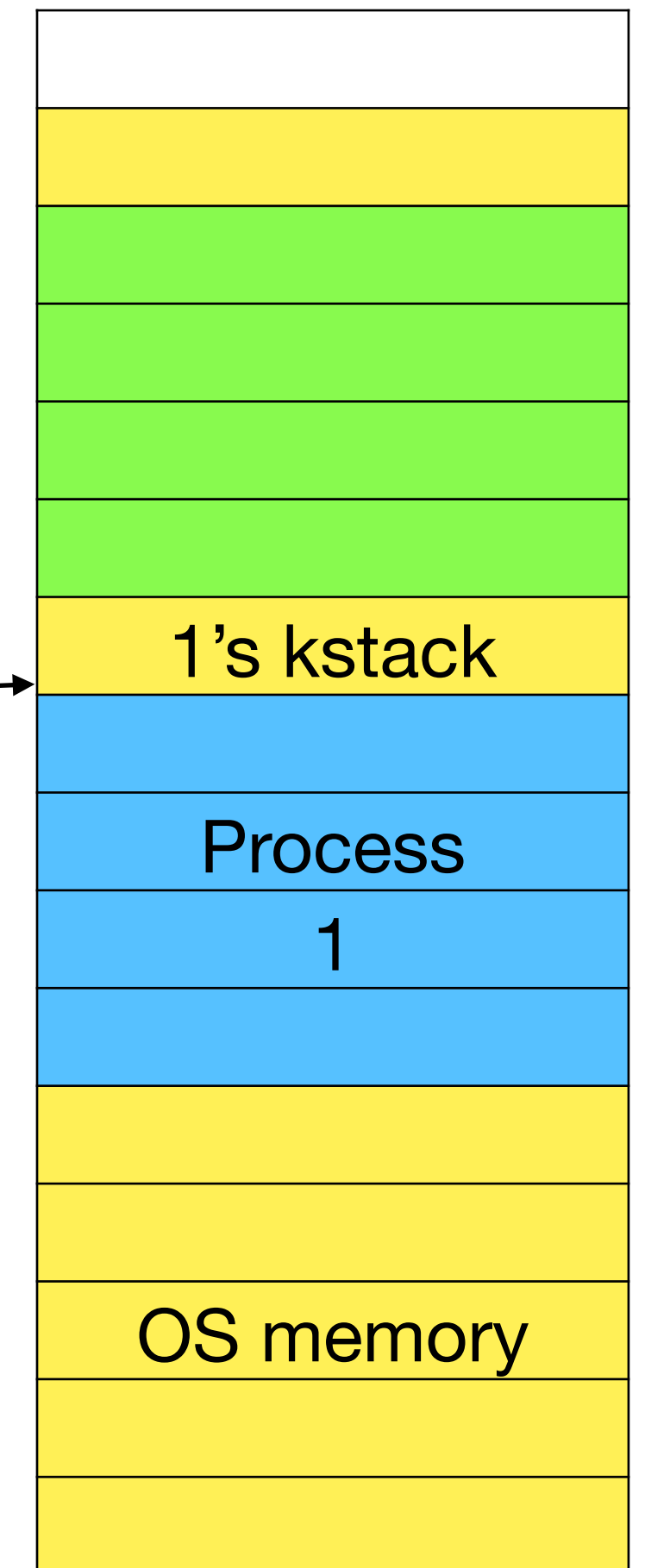
```
void seginit(void) {
    c->gdt[SEG_UCODE] = SEG(.., STARTPROC, (PROCSIZE-1) << 12, DPL_USER);
    c->gdt[SEG_UDATA] = SEG(.., STARTPROC, (PROCSIZE-1) << 12, DPL_USER);
}

static struct proc* allocproc(void) {
    sp = (char*)(STARTPROC + (PROCSIZE<<12));
    p->kstack = sp - KSTACKSIZE;
}

void scheduler(void) {
    // pick RUNNABLE process p
    switchvm(p);
    swtch(p->context);
}

void switchvm(struct proc *p) {
    mycpu()->gdt[SEG_TSS] = SEG16(STS_T32A, &mycpu()->ts,
                                sizeof(mycpu()->ts)-1, 0);

    mycpu()->ts.ss0 = SEG_KDATA << 3;
    mycpu()->ts.esp0 = (uint)p->kstack + KSTACKSIZE;
    ltr(SEG_TSS << 3);
}
```



Code walkthrough: setting up kernel stack

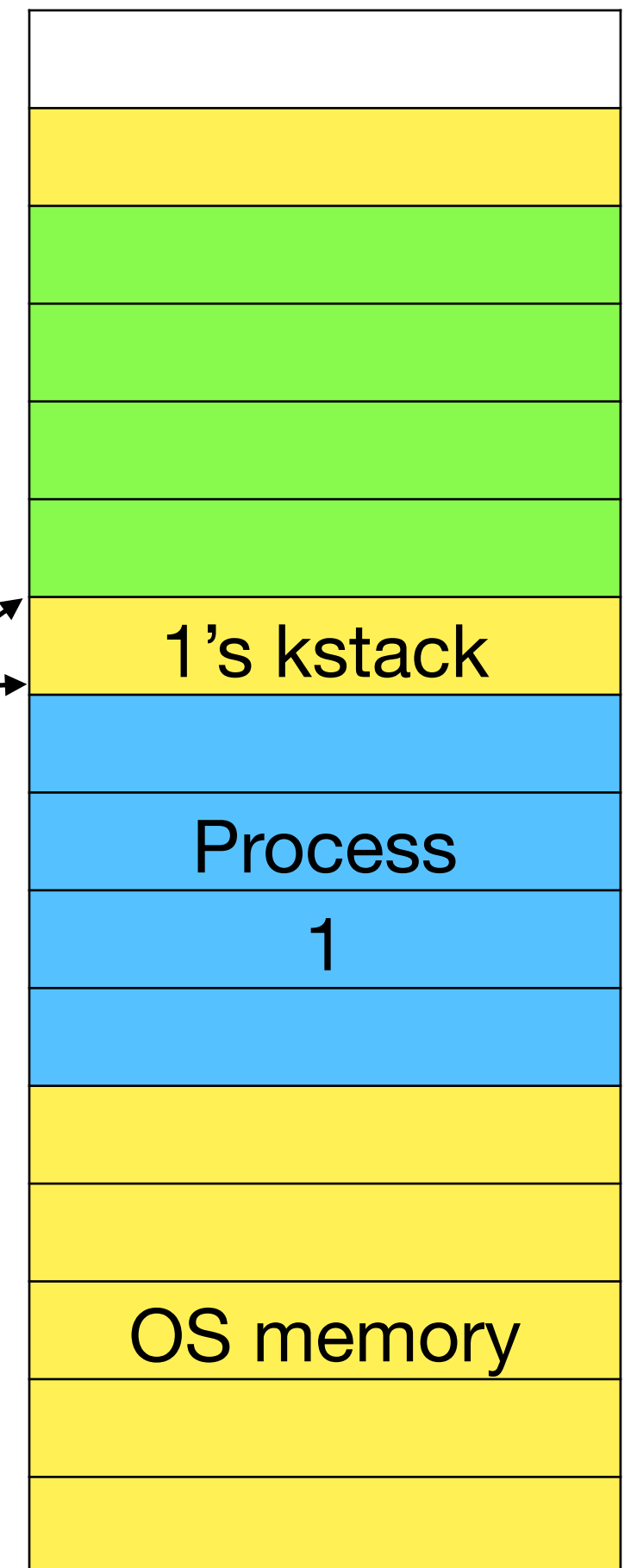
p16-tss

```
void seginit(void) {
    c->gdt[SEG_UCODE] = SEG(.., STARTPROC, (PROCSIZE-1) << 12, DPL_USER);
    c->gdt[SEG_UDATA] = SEG(.., STARTPROC, (PROCSIZE-1) << 12, DPL_USER);
}

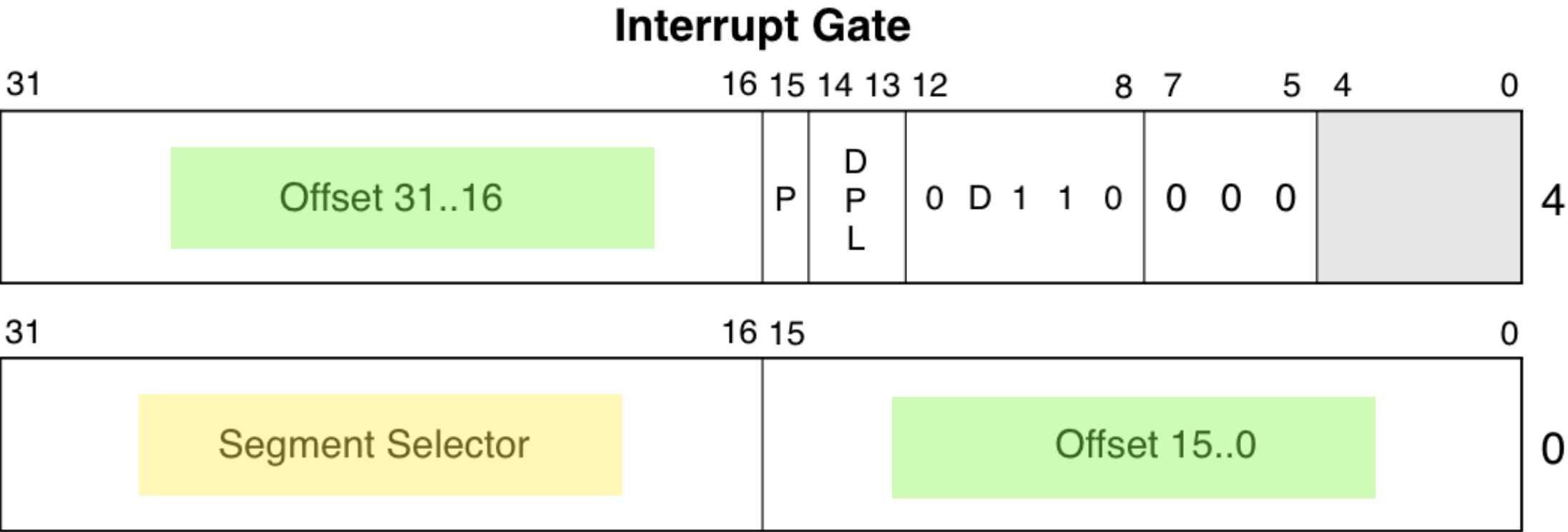
static struct proc* allocproc(void) {
    sp = (char*)(STARTPROC + (PROCSIZE<<12));
    p->kstack = sp - KSTACKSIZE;
}

void scheduler(void) {
    // pick RUNNABLE process p
    switchvm(p);
    swtch(p->context);
}

void switchvm(struct proc *p) {
    mycpu()->gdt[SEG_TSS] = SEG16(STS_T32A, &mycpu()->ts,
                                sizeof(mycpu()->ts)-1, 0);
    mycpu()->ts.ss0 = SEG_KDATA << 3;
    mycpu()->ts.esp0 = (uint)p->kstack + KSTACKSIZE;
    ltr(SEG_TSS << 3);
}
```

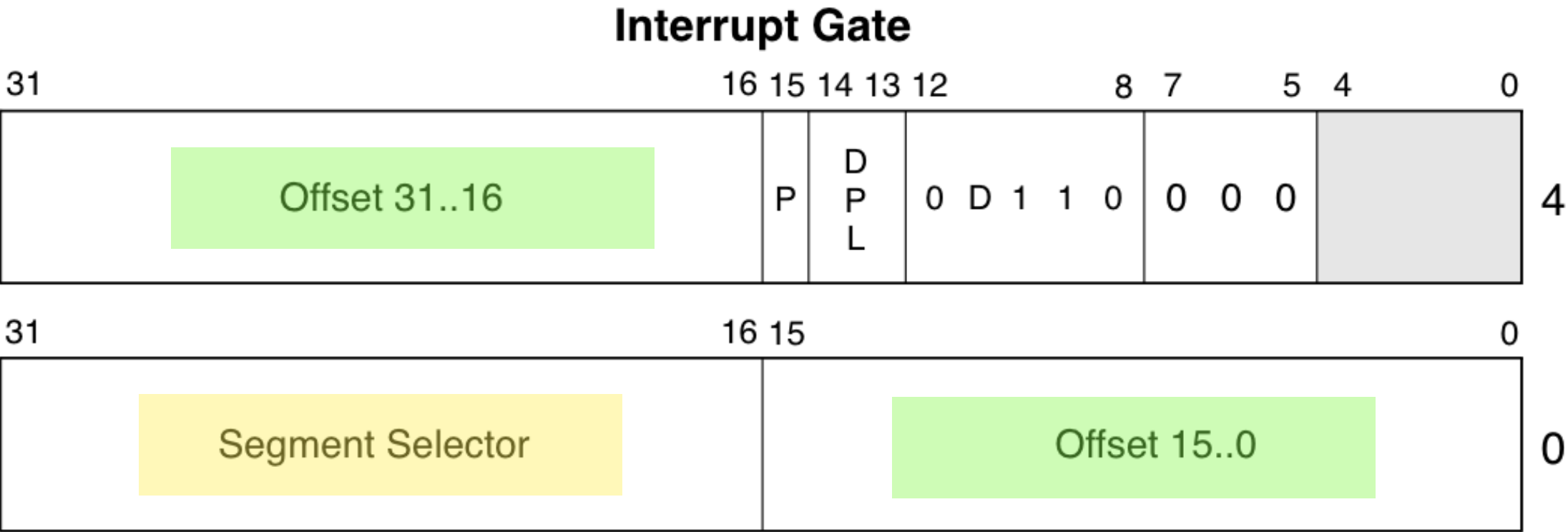


Interrupt handling



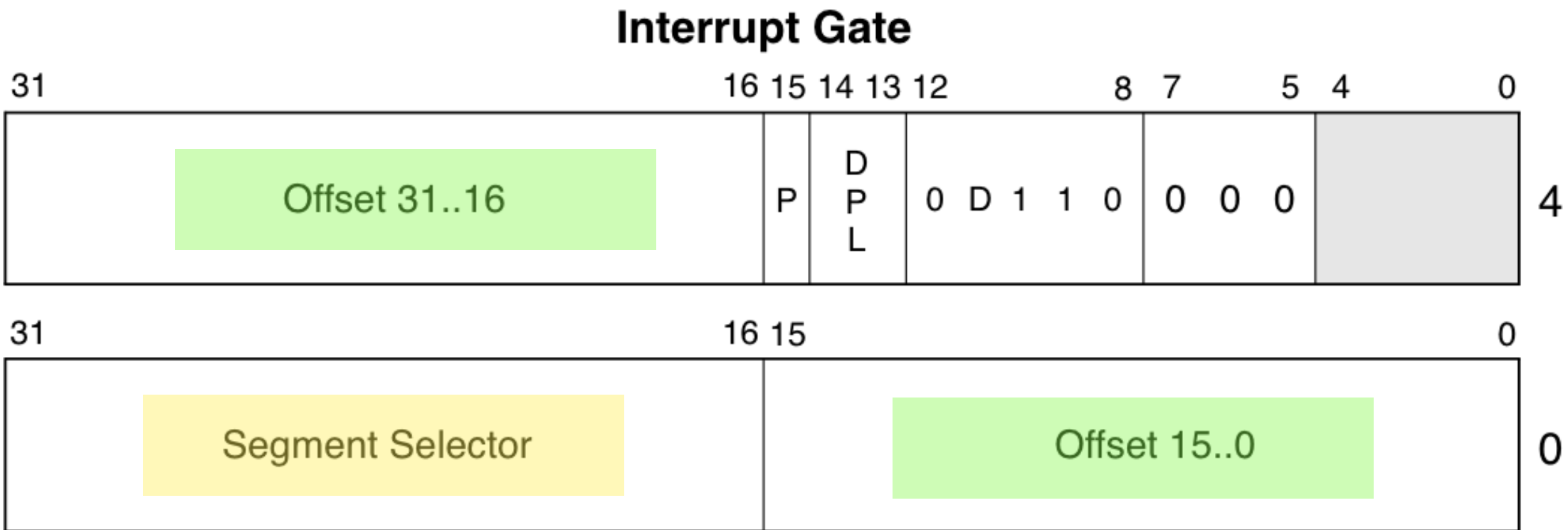
Interrupt handling

- Each IDT entry is 64-bits. Contains code segment and eip



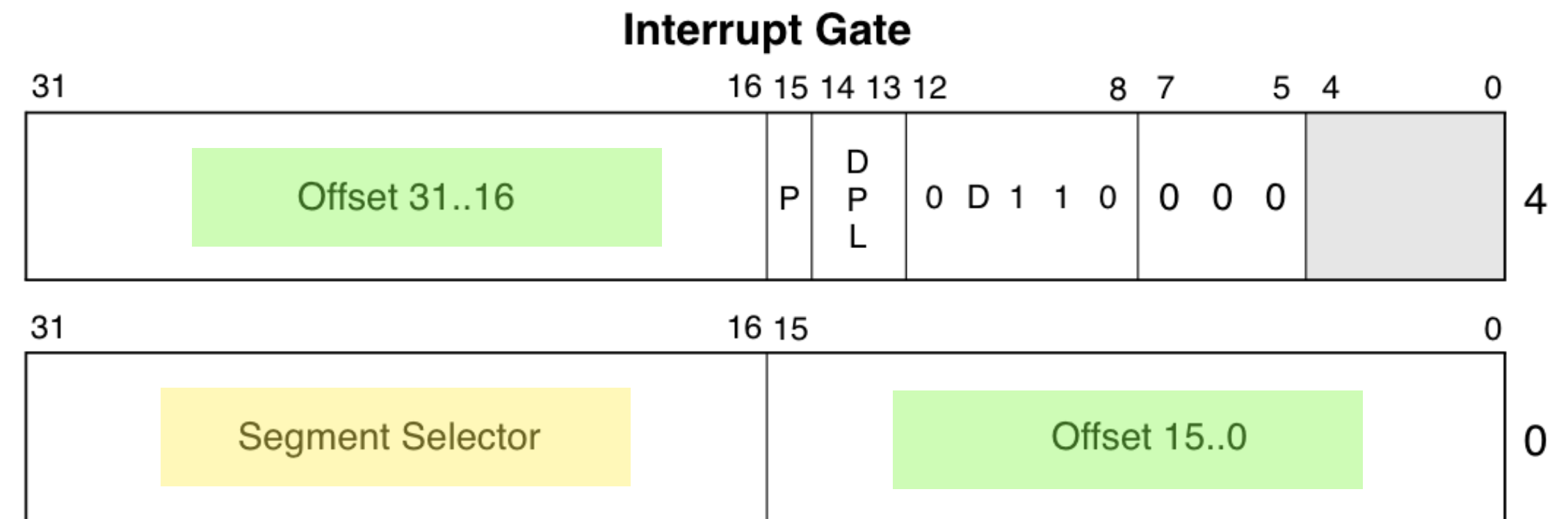
Interrupt handling

- Each IDT entry is 64-bits. Contains **code segment** and **eip**
- When interrupt appears,



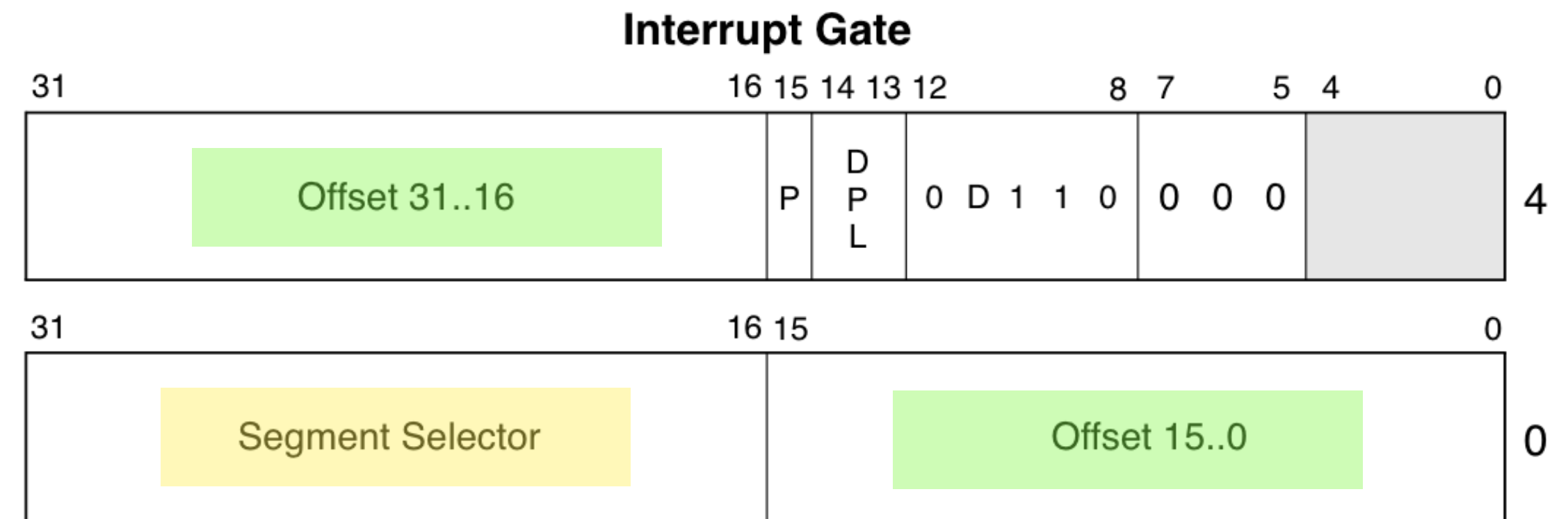
Interrupt handling

- Each IDT entry is 64-bits. Contains **code segment** and **eip**
- When interrupt appears,
 - Hardware changes **SS and ESP** according to the TSS



Interrupt handling

- Each IDT entry is 64-bits. Contains **code segment** and **eip**
- When interrupt appears,
 - Hardware changes **SS and ESP** according to the TSS
 - Hardware changes CS and EIP to the one pointed by IDT entry.



Interrupt handling

- Each IDT entry is 64-bits. Contains **code segment** and **eip**
- When interrupt appears,
 - Hardware changes **SS and ESP** according to the TSS
 - Hardware changes CS and EIP to the one pointed by IDT entry.

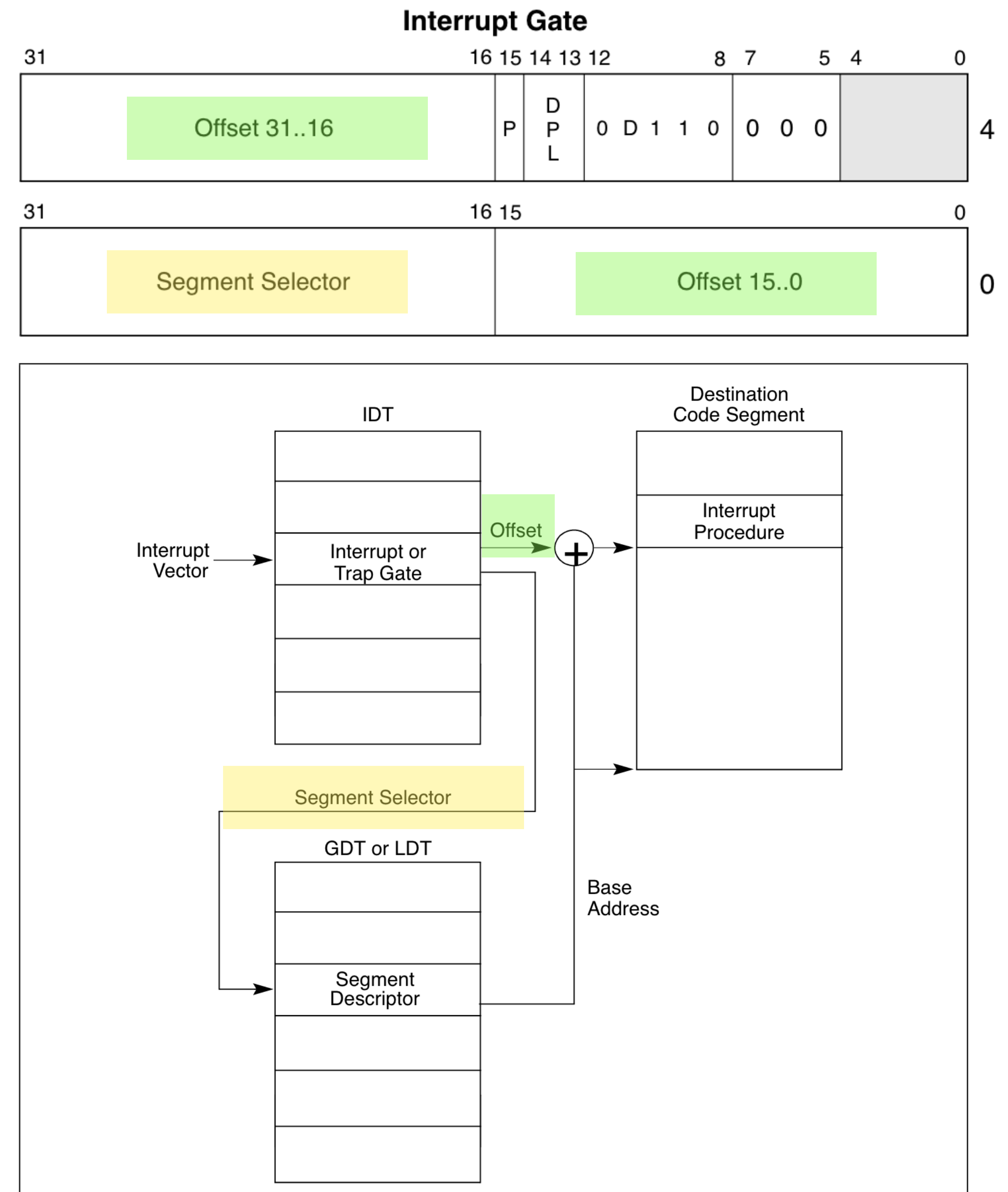


Figure 6-3. Interrupt Procedure Call

Interrupt handling with user process running

eip

```
for(;;)
;
```

trapasm.S

alltraps:

pushl %ds..

pushal

```
movw $(SEG_KDATA<<3), %ax
```

```
movw %ax, %ds..
```

pushl %esp

call trap

```
addl $4, %esp
```

popal

popl %ds..

```
addl $0x8, %esp
```

iret

vectors.S

```
.globl vector0
```

vector0:

pushl \$0

pushl \$0

```
jmp alltraps
```

IDT

%cs, %eip

■ ■ ■

GDT

■ ■ ■

■ ■ ■

...

CS

TR

SS

TSS

%ss0, %esp0

Process code

esp

Interrupt handling with user process running

eip

```
for(;;)
;
```

trapasm.S

alltraps:

pushl %ds..

pushal

```
movw $(SEG_KDATA<<3), %ax
```

```
movw %ax, %ds..
```

pushl %esp

call trap

```
addl $4, %esp
```

popal

popl %ds..

```
addl $0x8, %esp
```

iret

vectors.S

```
.globl vector0
```

vector0:

pushl \$0

pushl \$0

```
jmp alltraps
```

IDT

%cs, %eip

■ ■ ■

GDT

■ ■ ■

■ ■ ■

...

CS

TR

SS

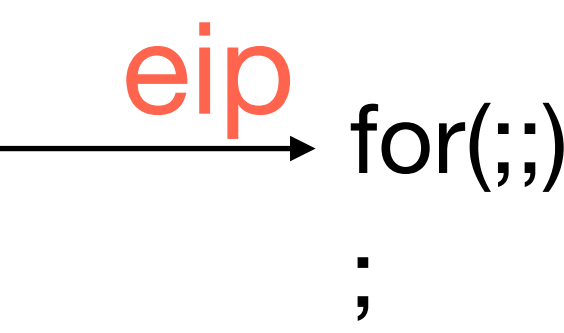
TSS

%ss0, %esp0

esp

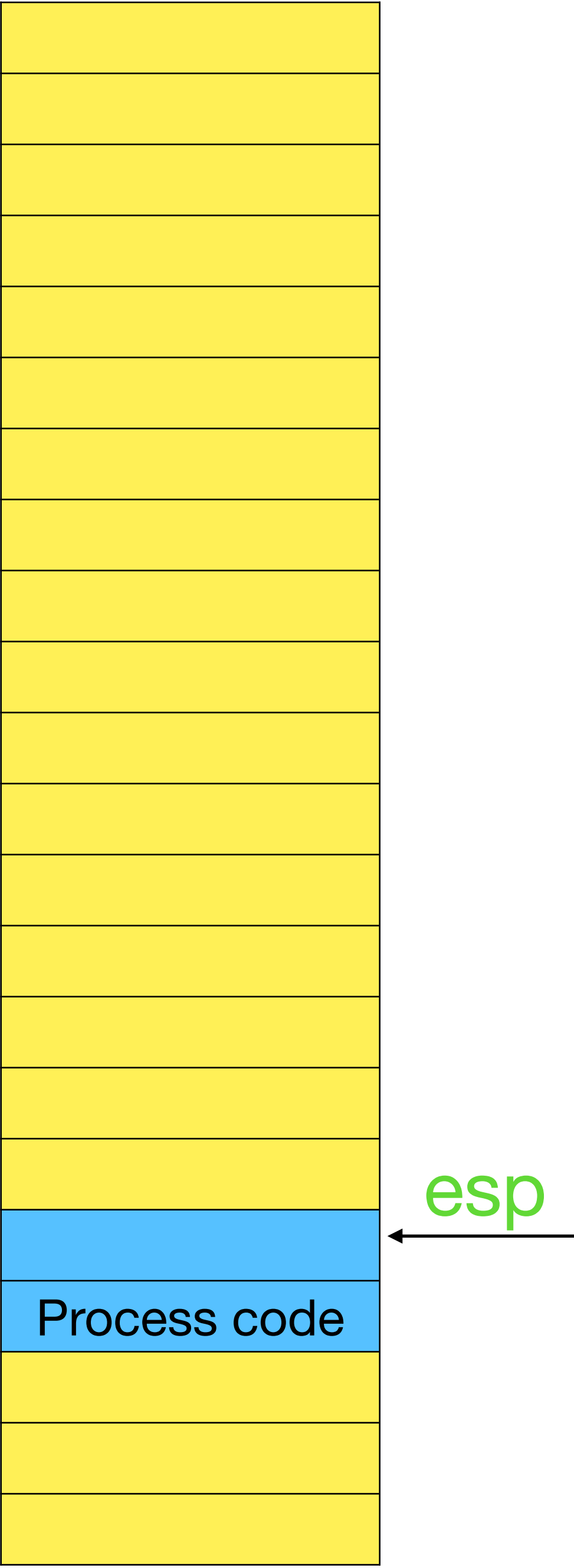
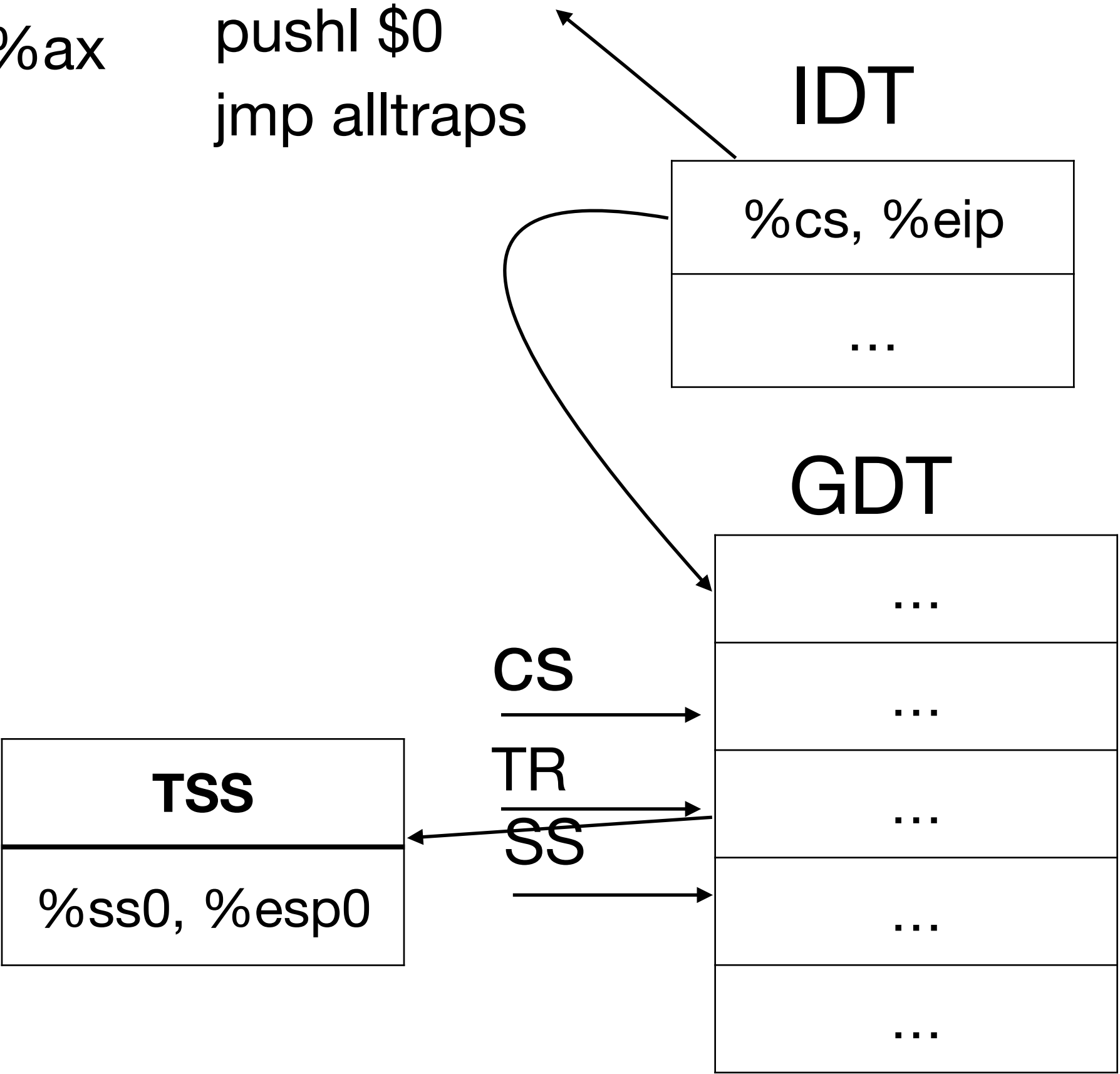
Process code

Interrupt handling with user process running

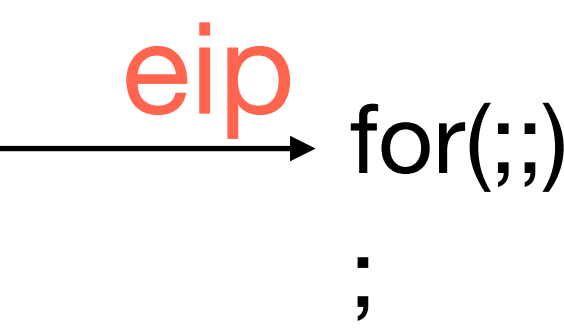


```
trapasm.S
alltraps:
    pushl %ds..
    pushal
    movw $(SEG_KDATA<<3), %ax
    movw %ax, %ds..
    pushl %esp
    call trap
    addl $4, %esp
    popal
    popl %ds..
    addl $0x8, %esp
    iret
```

```
vectors.S
.globl vector0
vector0:
    pushl $0
    pushl $0
    jmp alltraps
```

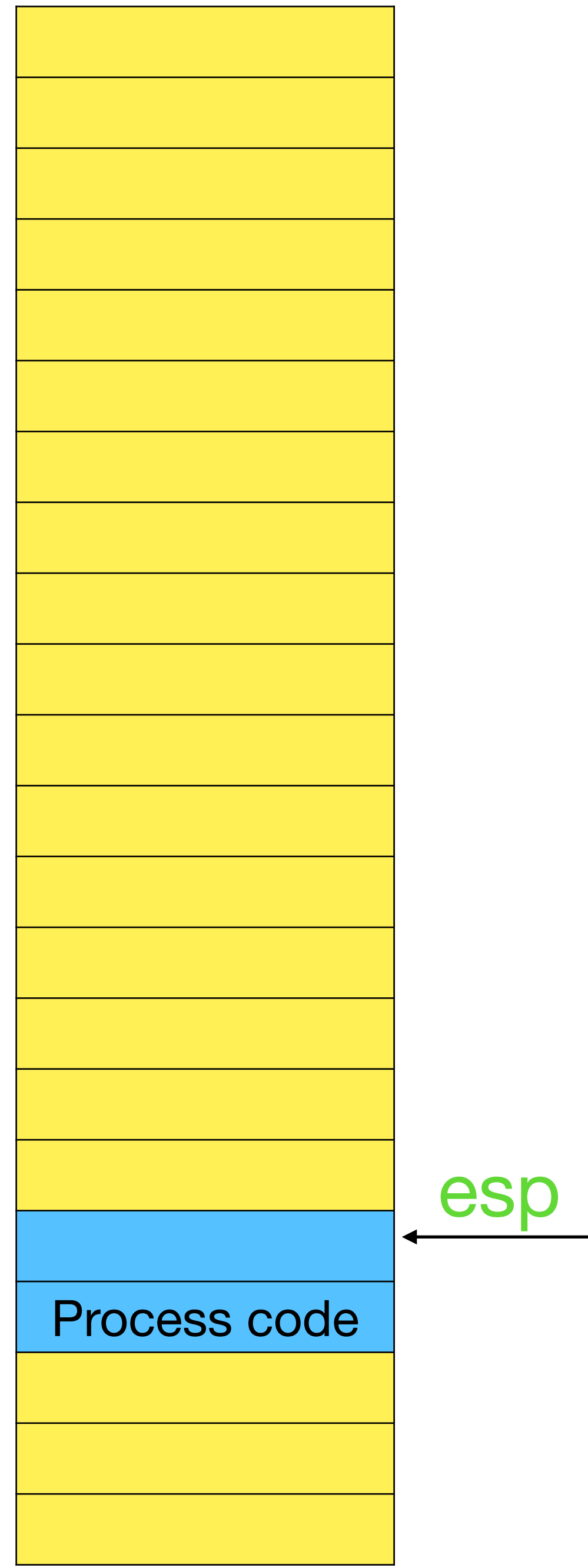
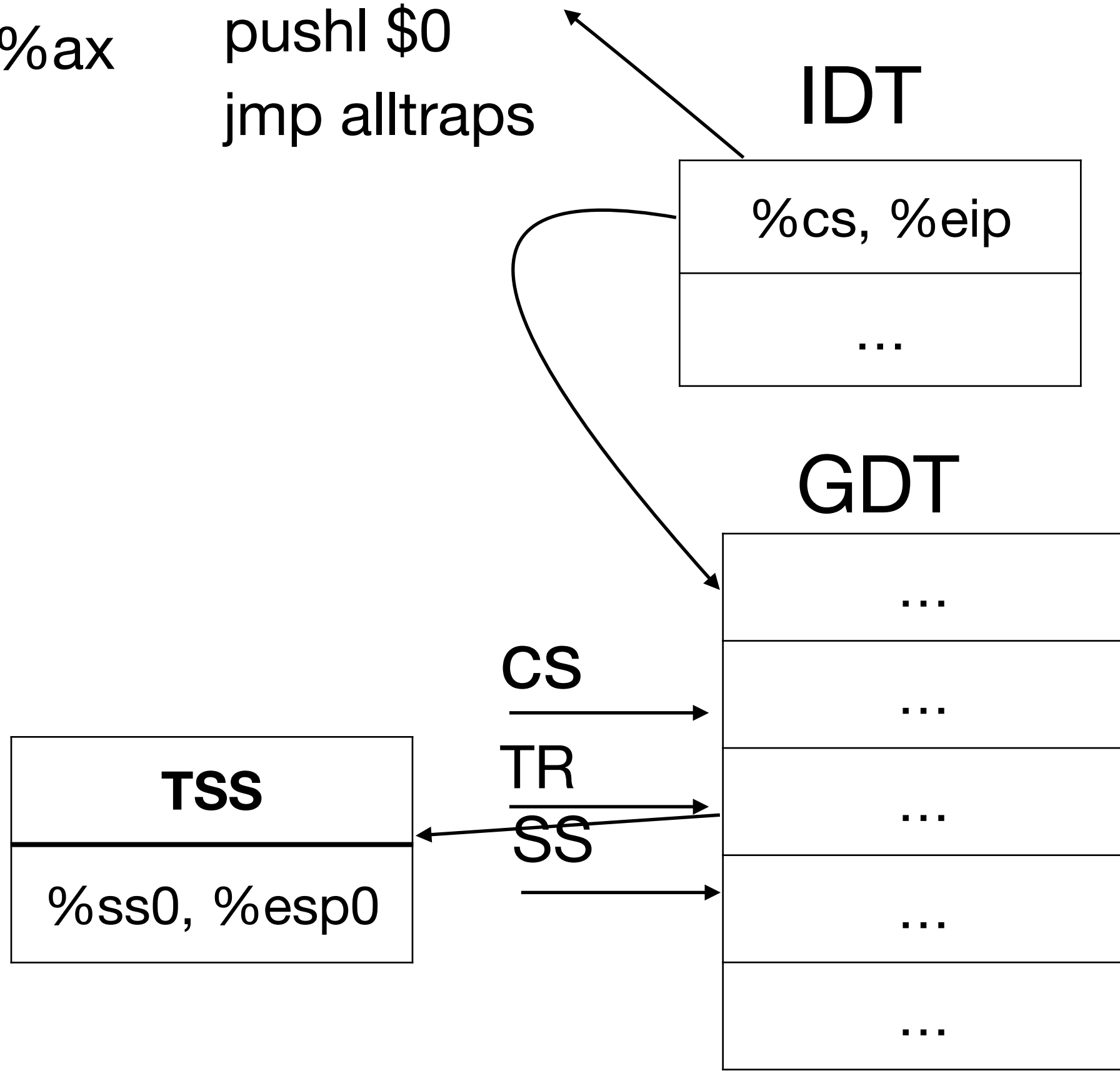


Interrupt handling with user process running



```
trapasm.S
alltraps:
    pushl %ds..
    pushal
    movw $(SEG_KDATA<<3), %ax
    movw %ax, %ds..
    pushl %esp
    call trap
    addl $4, %esp
    popal
    popl %ds..
    addl $0x8, %esp
    iret
```

```
vectors.S
.globl vector0
vector0:
    pushl $0
    pushl $0
    jmp alltraps
```



Interrupt handling with user process running

eip

```
for(;;)
;
```

trapasm.S

alltraps:

pushl %ds..

pushal

```
movw $(SEG_KDATA<<3), %ax
```

```
movw %ax, %ds..
```

pushl %esp

call trap

addl \$4, %esp

popal

popl %ds..

```
addl $0x8, %esp
```

iret

vectors.S

```
.globl vector0
```

vector0:

pushl \$0

pushl \$0

```
jmp alltraps
```

IDT

%cs, %eip

■ ■ ■

GDT

■ ■ ■

...

CS

TR

~~SS~~

TSS

%ss0, %esp0

esp

Process code

Interrupt handling with user process running

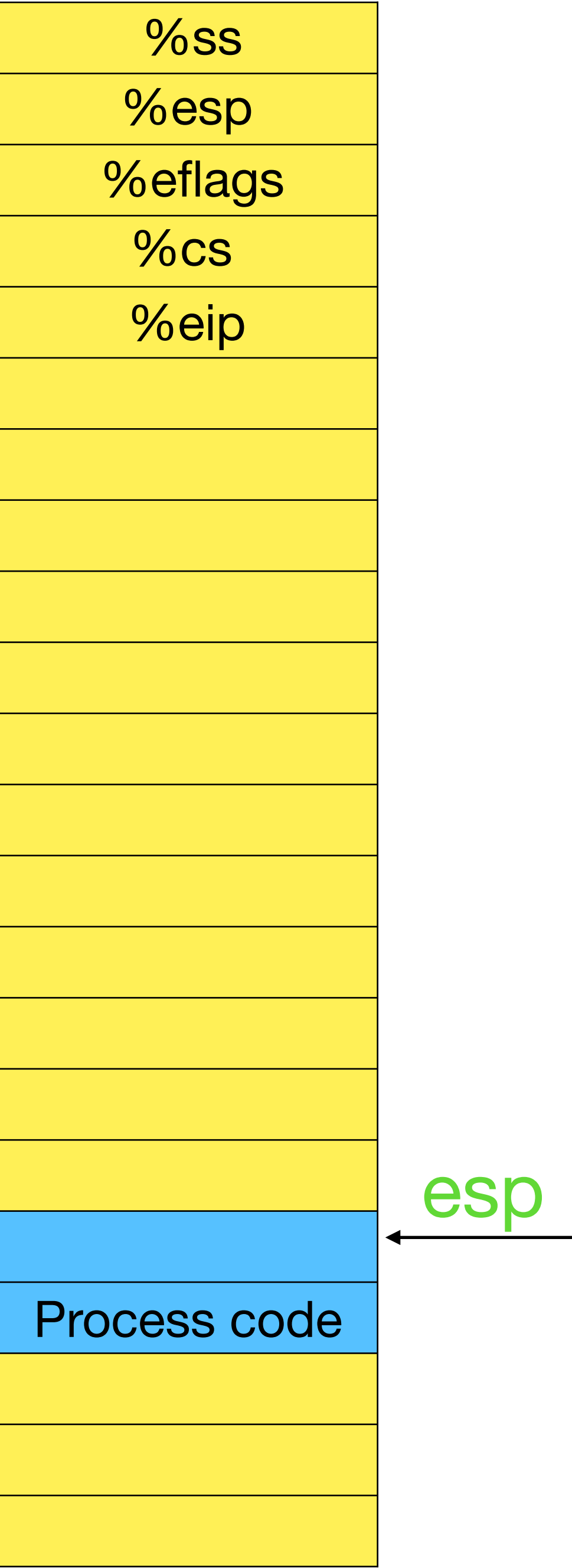
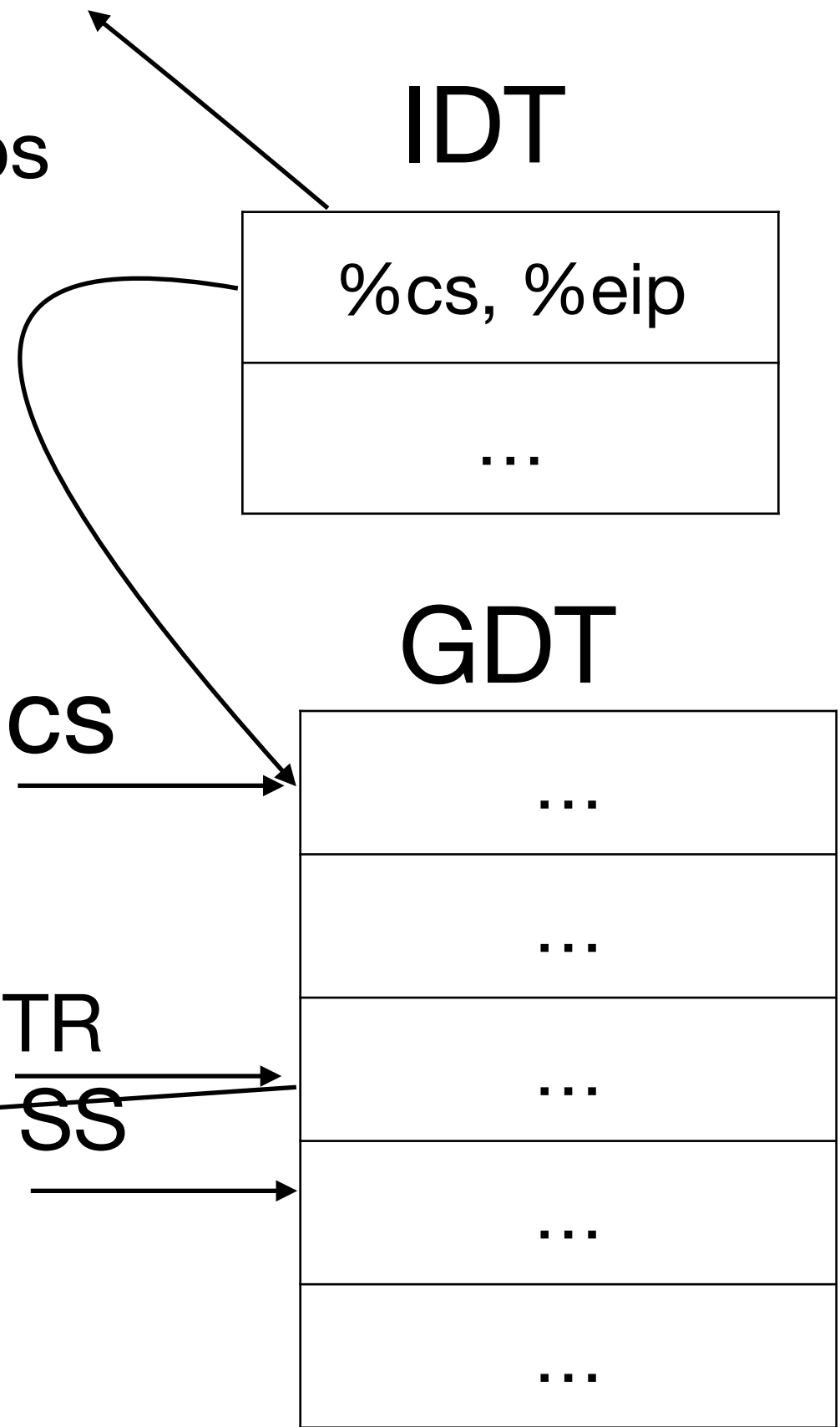
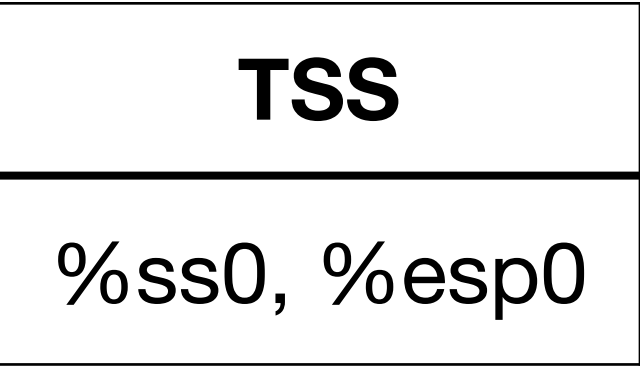
eip → for(;;)
;

trapasm.S

```
alltraps:
    pushl %ds..
    pushal
    movw $(SEG_KDATA<<3), %ax
    movw %ax, %ds..
    pushl %esp
    call trap
    addl $4, %esp
    popal
    popl %ds..
    addl $0x8, %esp
    iret
```

vectors.S

```
.globl vector0
vector0:
    pushl $0
    pushl $0
    jmp alltraps
```

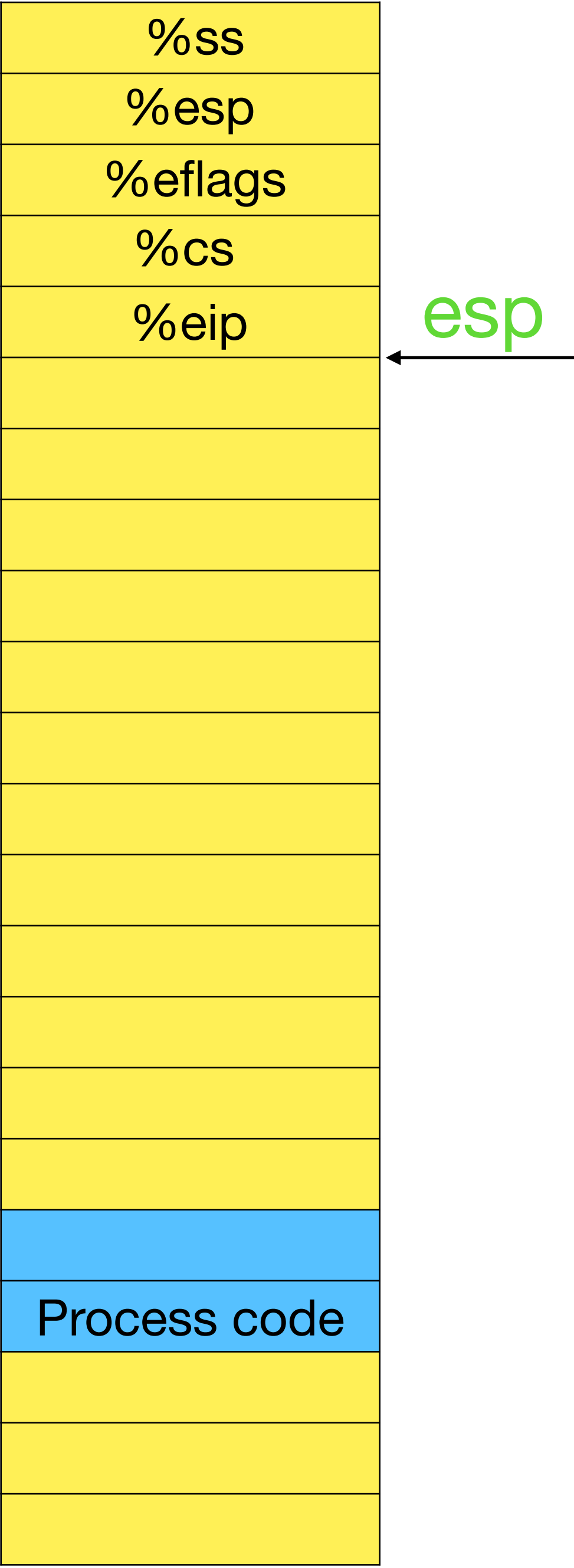
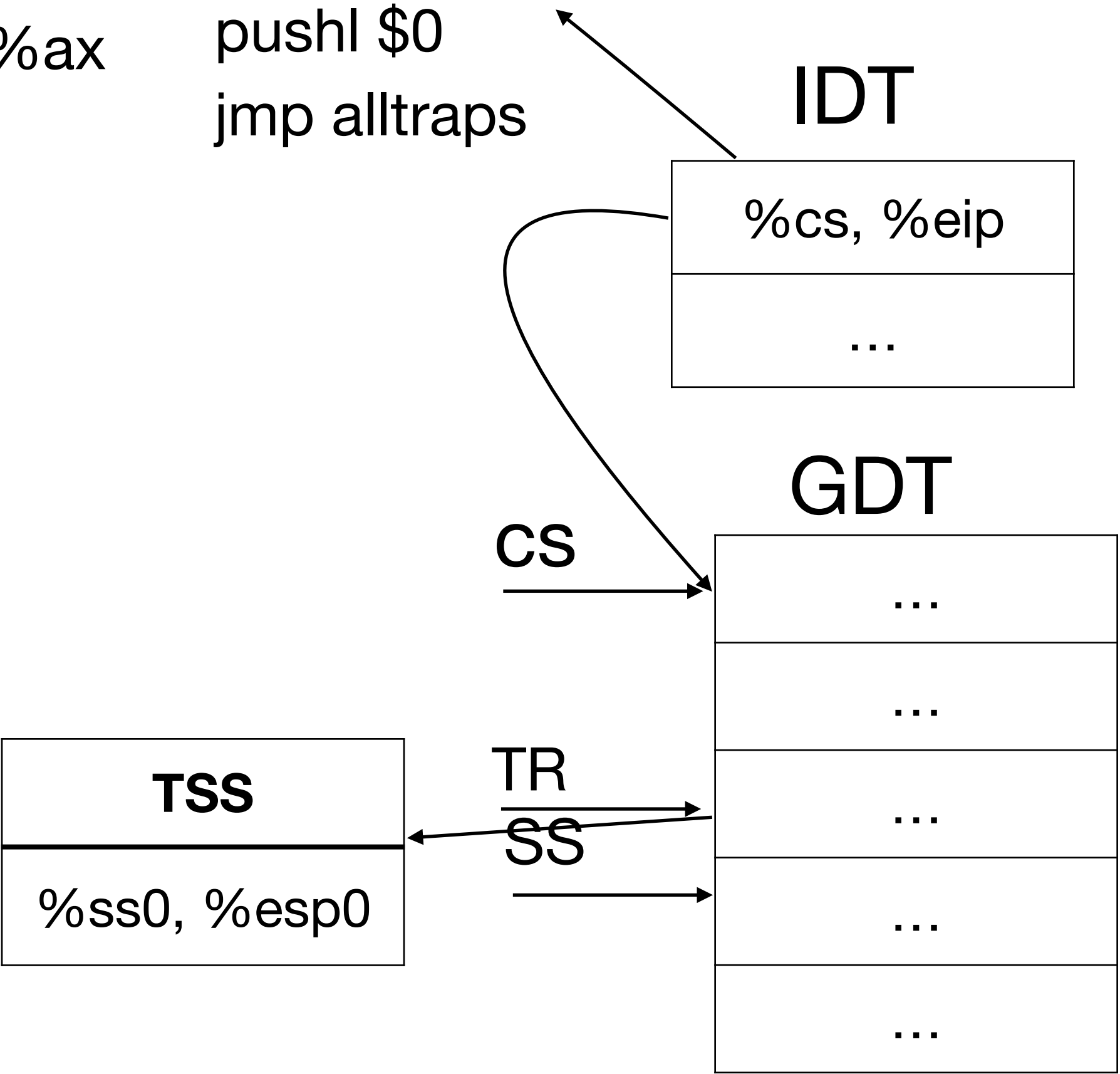


Interrupt handling with user process running

eip → for(;;)
;

```
trapasm.S
alltraps:
    pushl %ds..
    pushal
    movw $(SEG_KDATA<<3), %ax
    movw %ax, %ds..
    pushl %esp
    call trap
    addl $4, %esp
    popal
    popl %ds..
    addl $0x8, %esp
    iret
```

```
vectors.S
.globl vector0
vector0:
    pushl $0
    pushl $0
    jmp alltraps
```



Interrupt handling with user process running

```
for(;;)
;
```

trapasm.S

alltraps:

pushl %ds..

pushal

```
movw $(SEG_KDATA<<3), %ax
```

```
movw %ax, %ds..
```

pushl %esp

call trap

```
addl $4, %esp
```

popal

popl %ds..

```
addl $0x8, %esp
```

iret

vectors.S

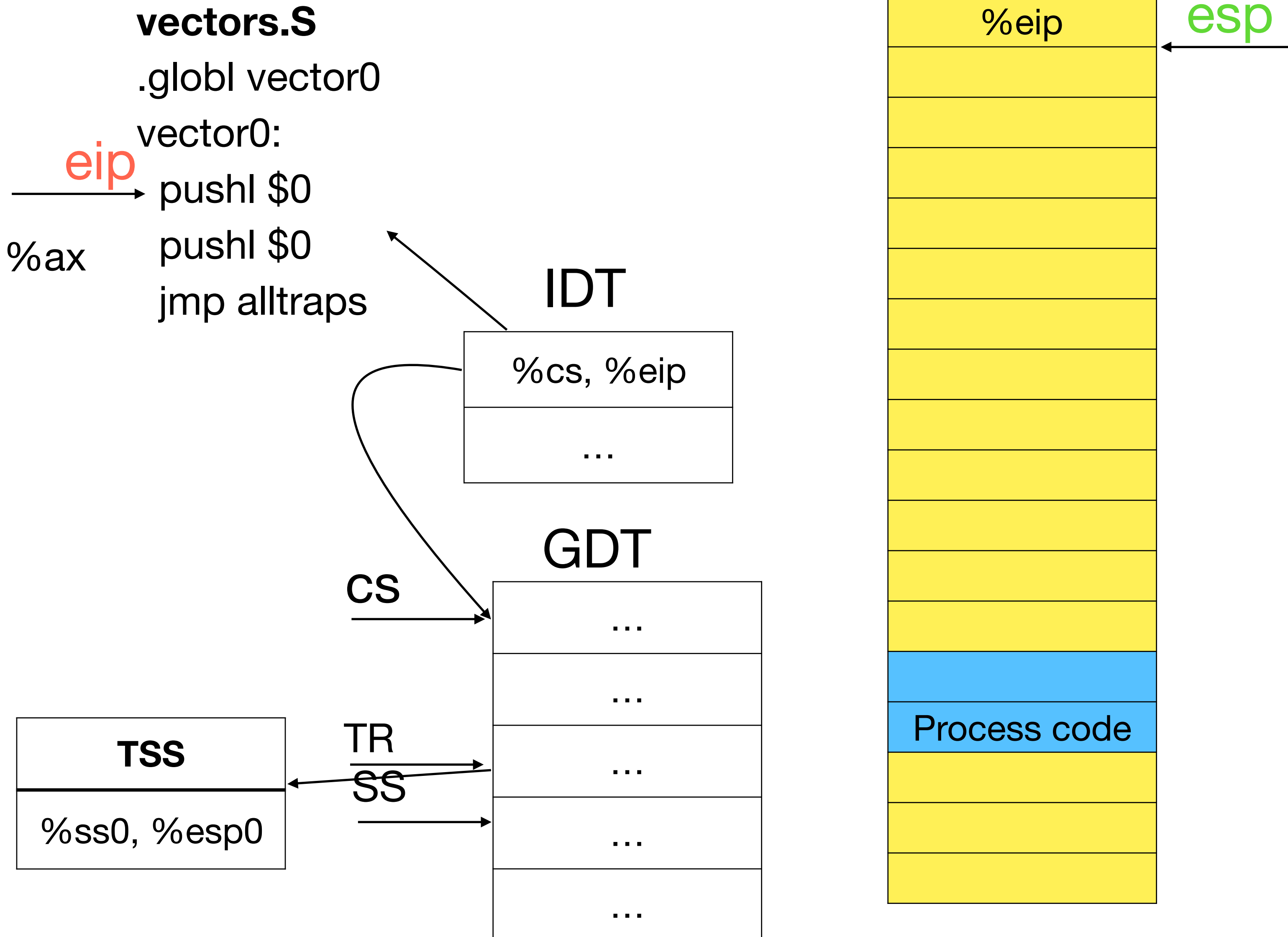
```
.globl vector0
```

vector0:

```
pushl $0
```

pushl \$0

```
jmp alltraps
```



Interrupt handling with user process running

```
for(;;)
;
```

trapasm.S

alltraps:

pushl %ds..

pushal

```
movw $(SEG_KDATA<<3), %ax
```

```
movw %ax, %ds..
```

pushl %esp

call trap

```
addl $4, %esp
```

popal

popl %ds..

```
addl $0x8, %esp
```

iret

vectors.S

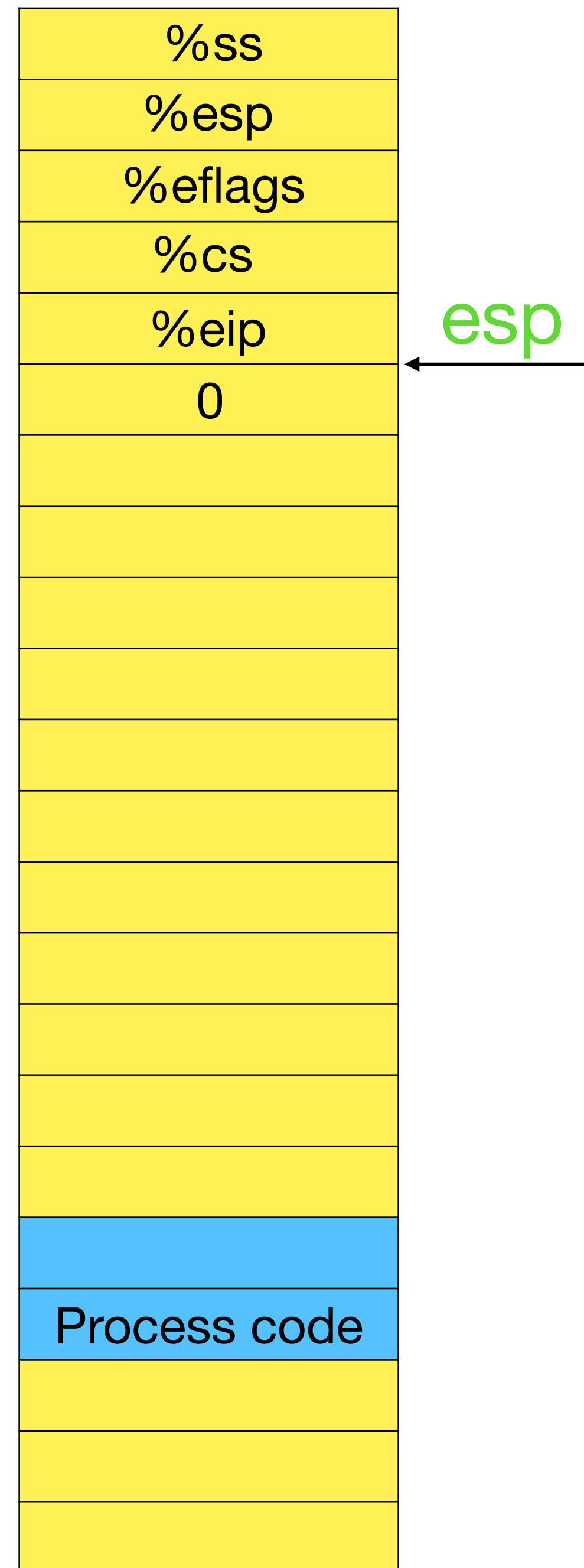
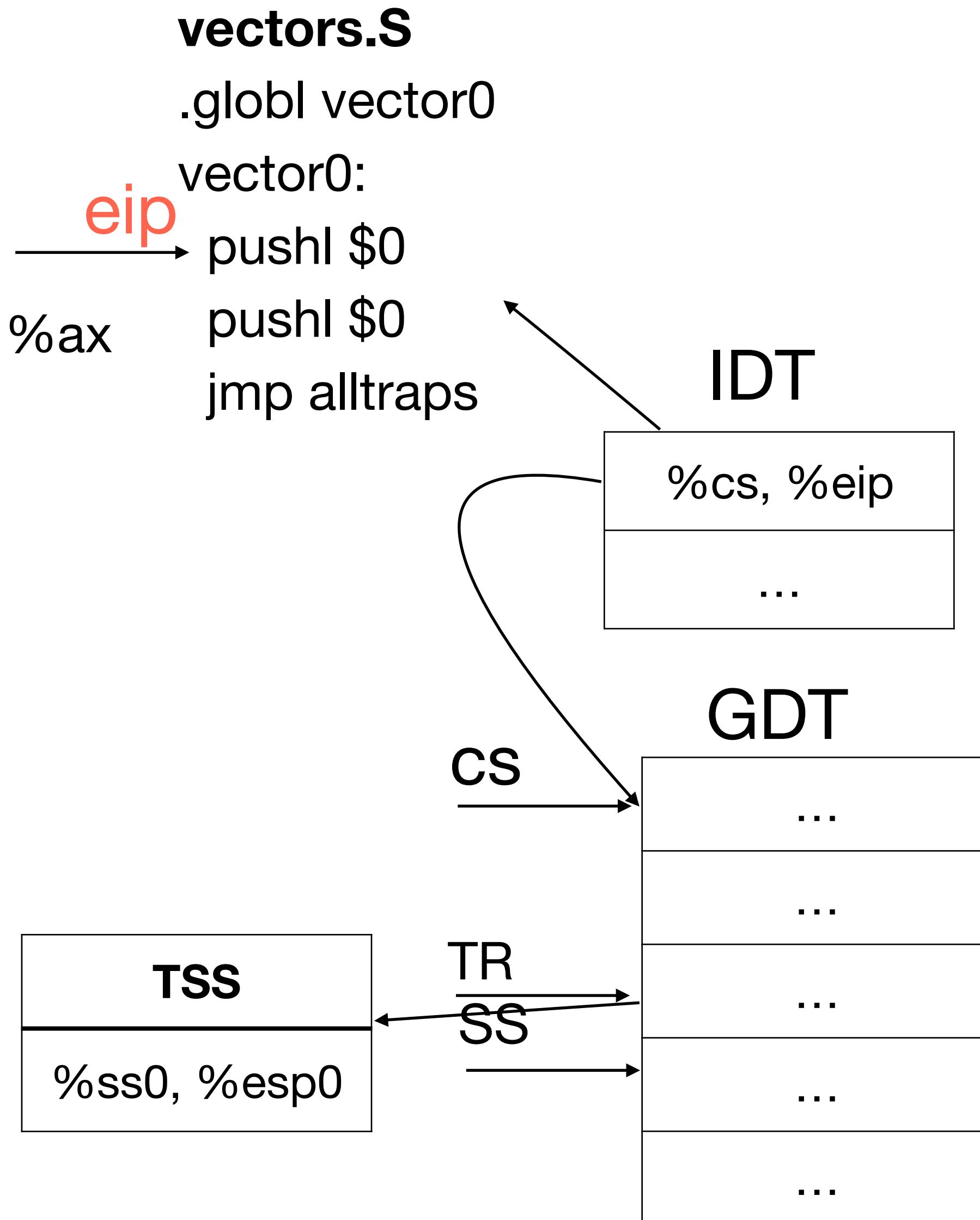
```
.globl vector0
```

vector0:

```
pushl $0
```

pushl \$0

```
jmp alltraps
```



Interrupt handling with user process running

```

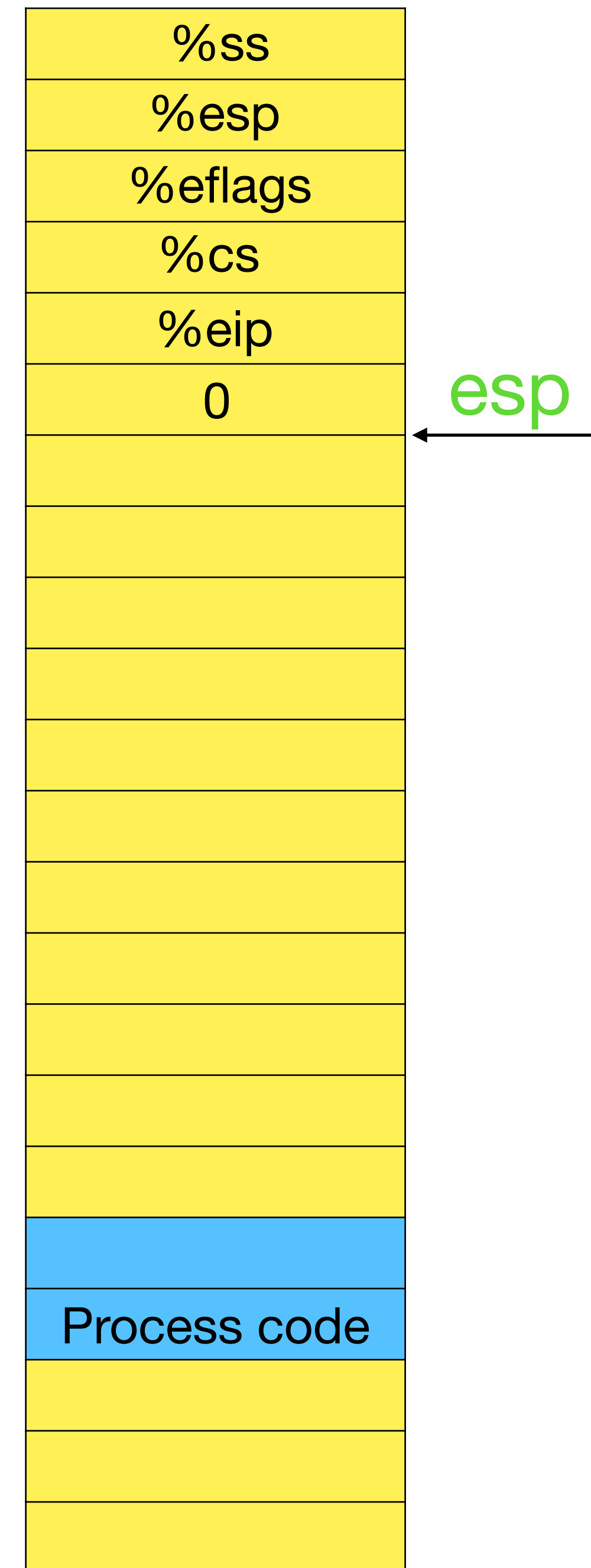
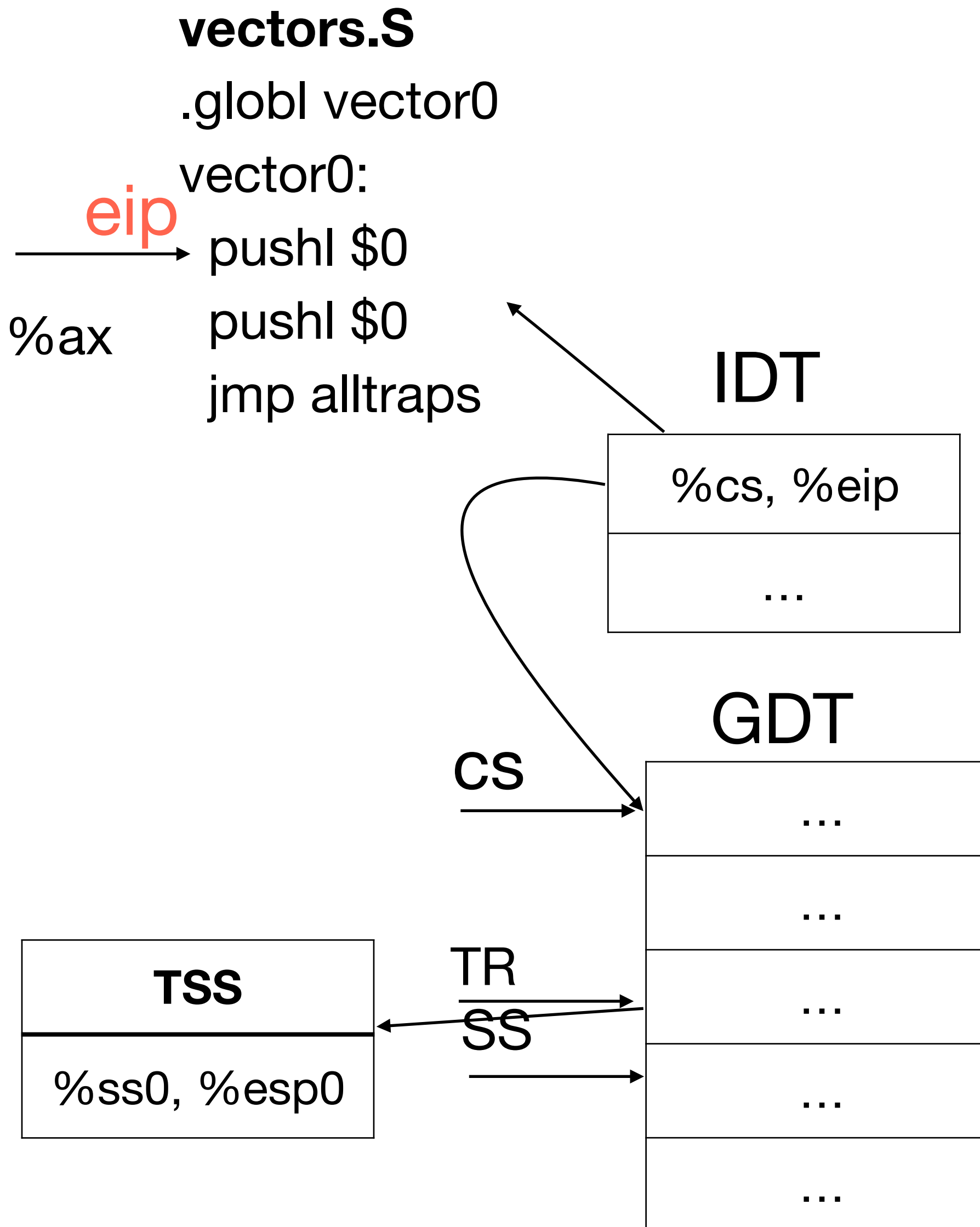
trapasm.S
for(;;)
;    pushl %ds..
    pushal
    movw $(SEG_KD
    movw %ax, %ds
    pushl %esp
    call trap
    addl $4, %esp
    popal
    popl %ds..
    addl $0x8, %esp
    iret

```

```

vectors.S
.globl vector0
vector0:
→ pushl $0
  pushl $0
  jmp alltraps

```



Interrupt handling with user process running

```
for(;;)
;
```

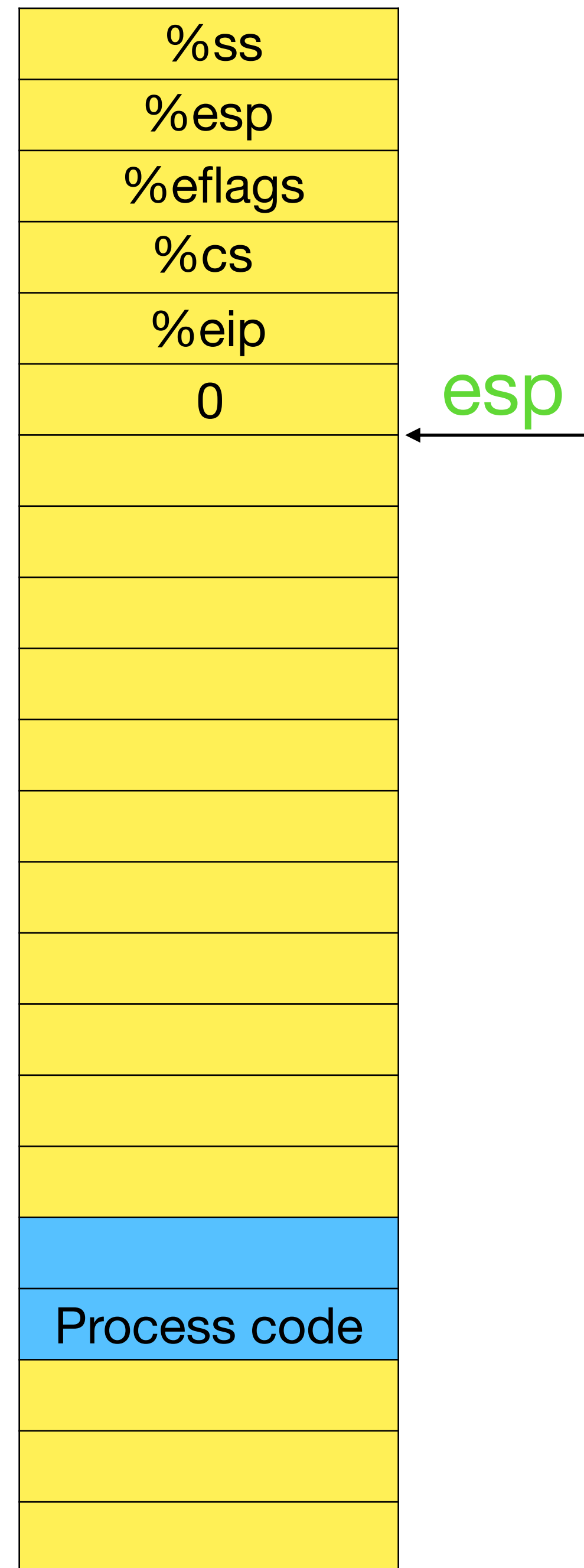
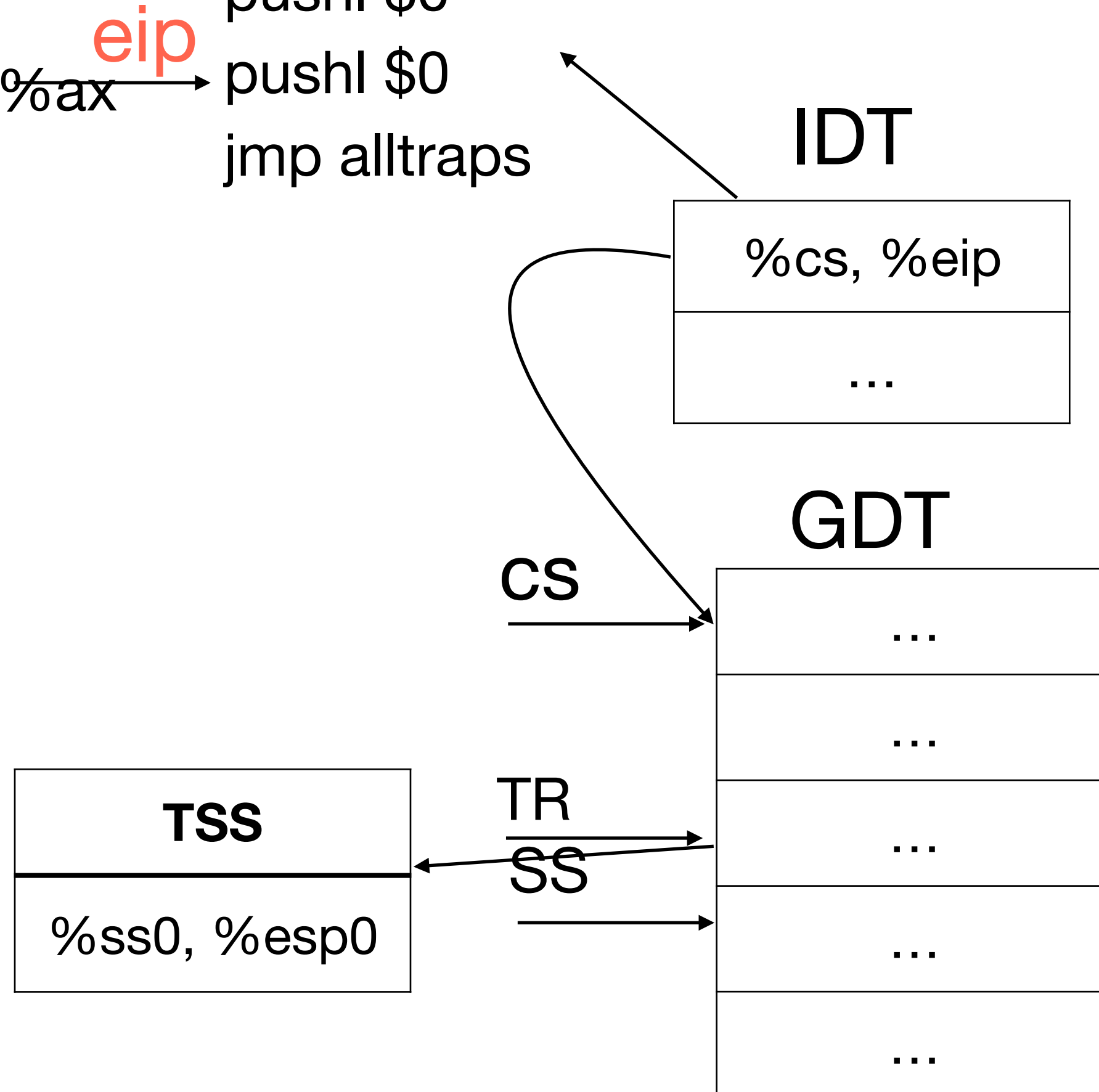
trapasm.S

```
alltraps:
    pushl %ds..
    pushal
    movw $(SEG_KD
    movw %ax, %ds
    pushl %esp
    call trap
    addl $4, %esp
    popal
    popl %ds..
    addl $0x8, %esp
    iret
```

vectors.S

```
.globl vector0
vector0:
    pushl $0
    → pushl $0
    jmp alltraps
```

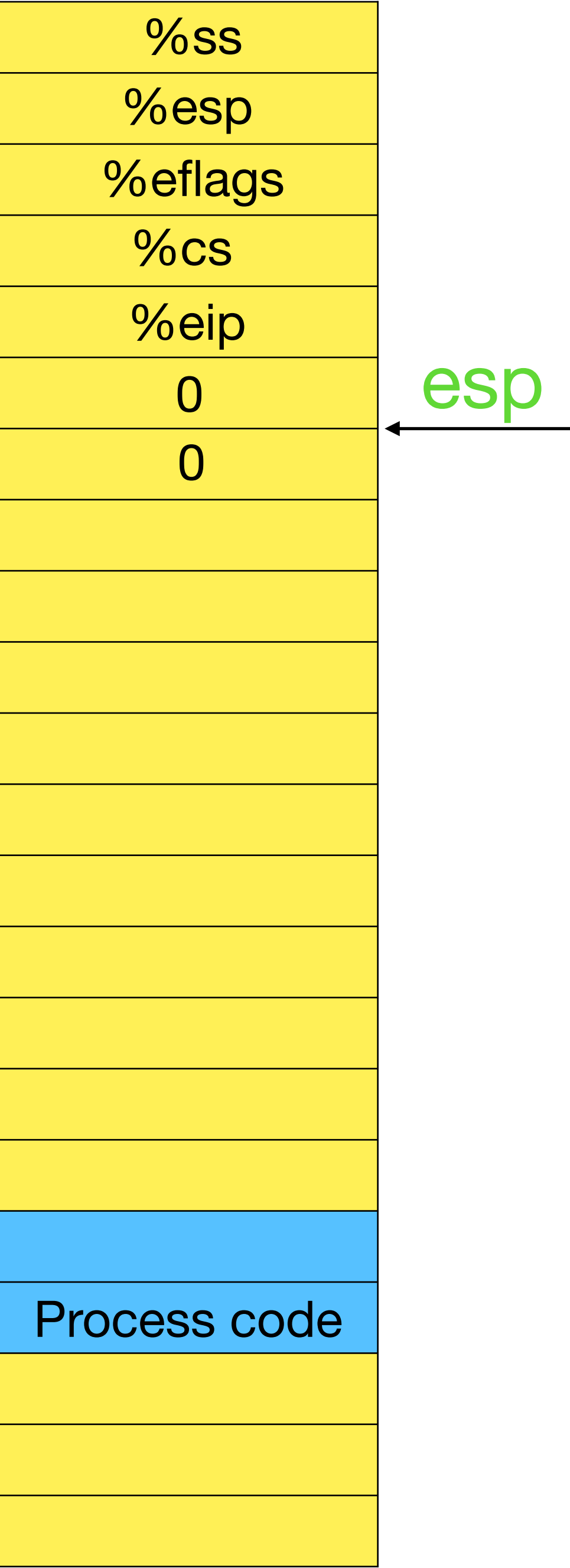
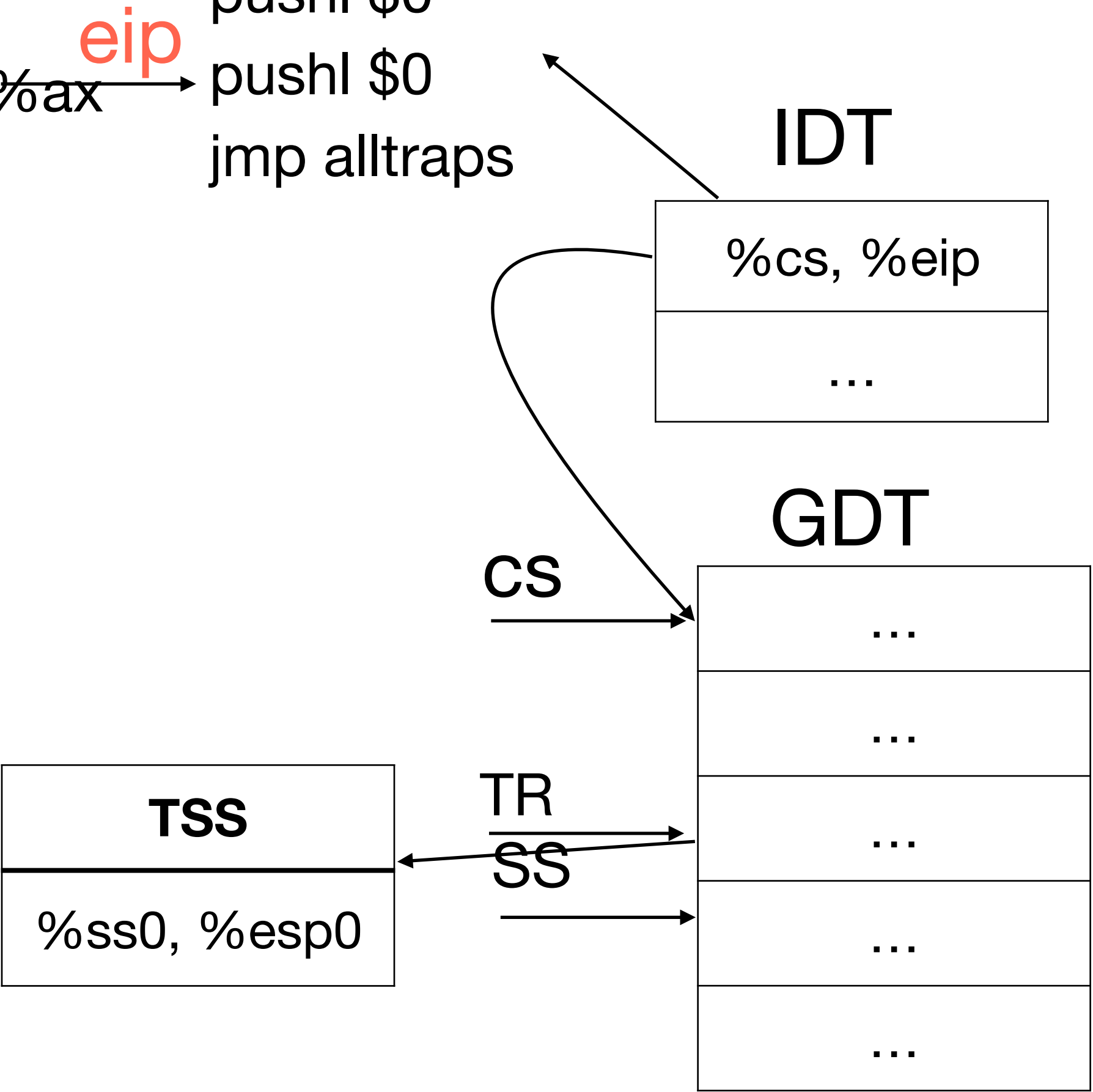
eip



Interrupt handling with user process running

```
trapasm.S
for(;;)
;
alltraps:
    pushl %ds..
    pushal
    movw $(SEG_KDATA<<3), %ax
    movw %ax, %ds..
    pushl %esp
    call trap
    addl $4, %esp
    popal
    popl %ds..
    addl $0x8, %esp
    iret
```

```
vectors.S
.globl vector0
vector0:
    pushl $0
    pushl $0
    jmp alltraps
```



Interrupt handling with user process running

```

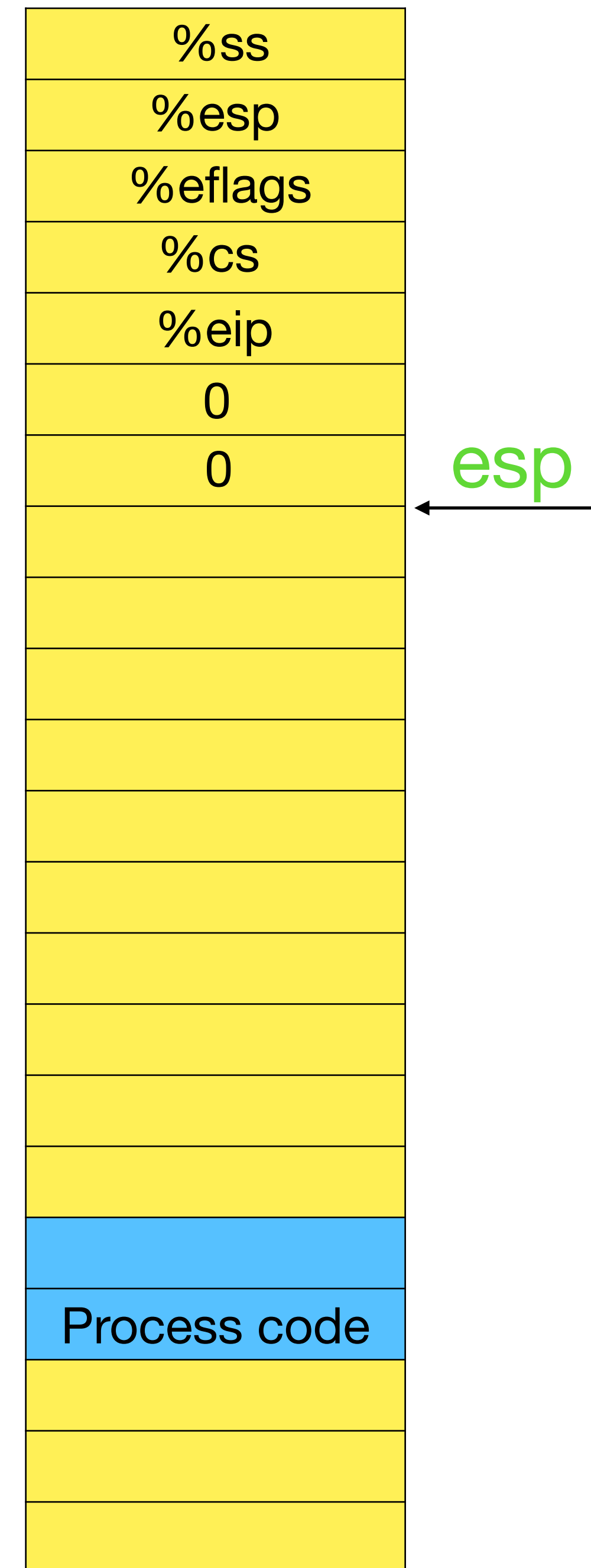
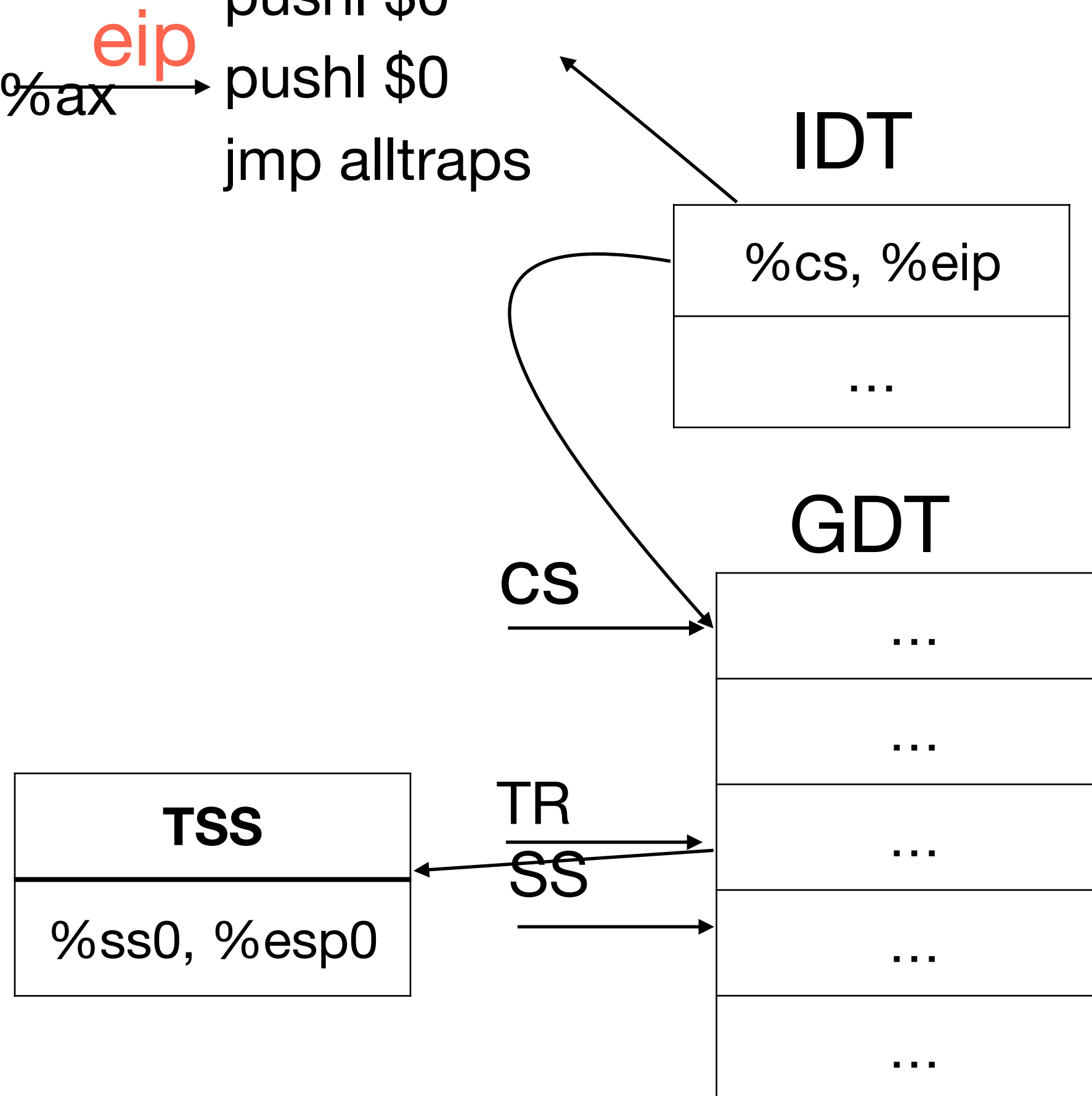
trapasm.S
for(;;)
;
    pushl %ds..
    pushal
    movw $(SEG_KD
    movw %ax, %ds
    pushl %esp
    call trap
    addl $4, %esp
    popal
    popl %ds..
    addl $0x8, %esp
    iret

```

```

vectors.S
.globl vector0
vector0:
    pushl $0
    → pushl $0
    jmp alltraps

```



Interrupt handling with user process running

```

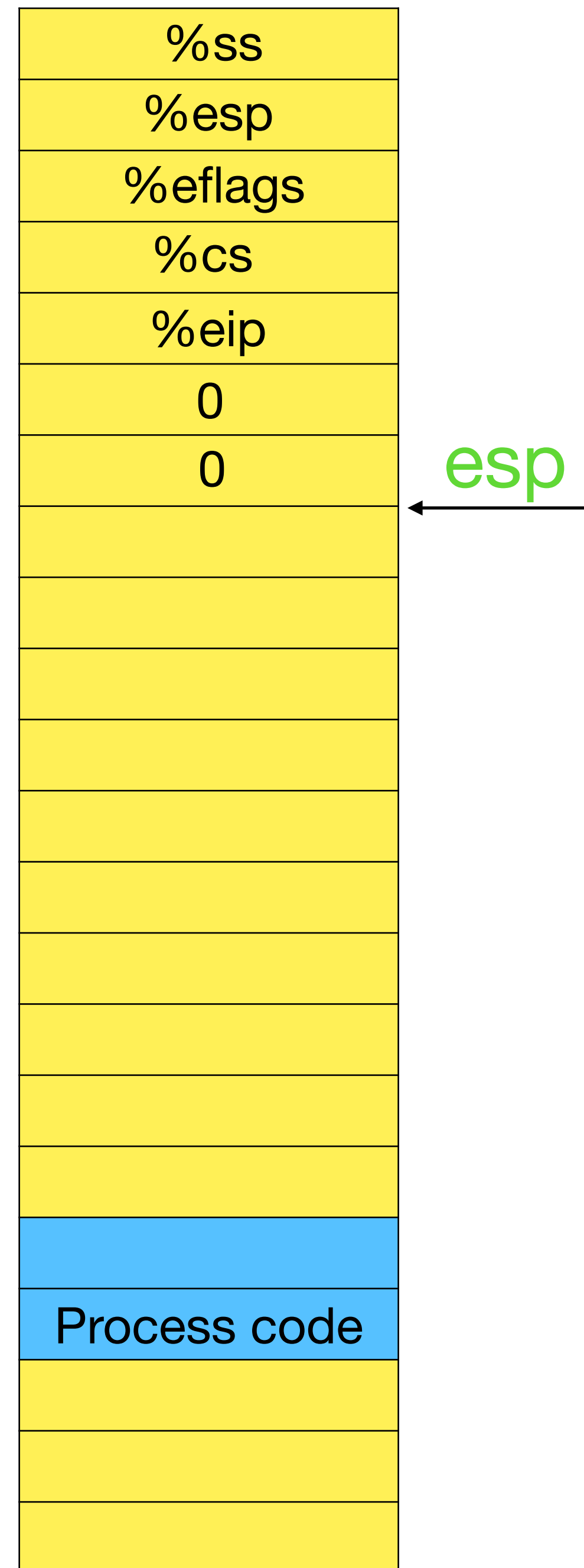
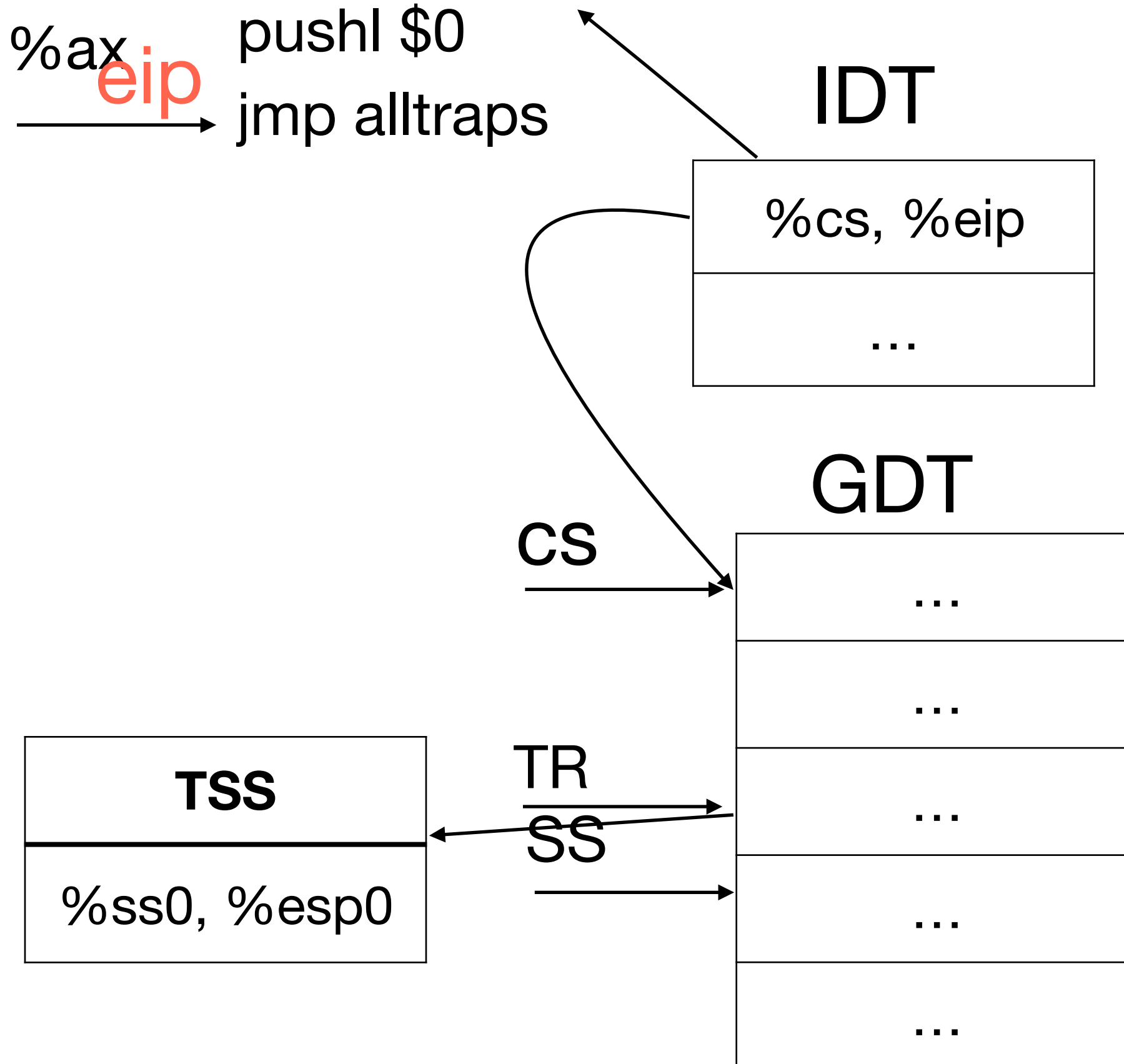
trapasm.S
for(;;)
;    pushl %ds..
    pushal
    movw $(SEG_KD
    movw %ax, %ds
    pushl %esp
    call trap
    addl $4, %esp
    popal
    popl %ds..
    addl $0x8, %esp
    iret

```

```

vectors.S
.globl vector0
vector0:
    pushl $0
    pushl $0
    → jmp alltraps

```



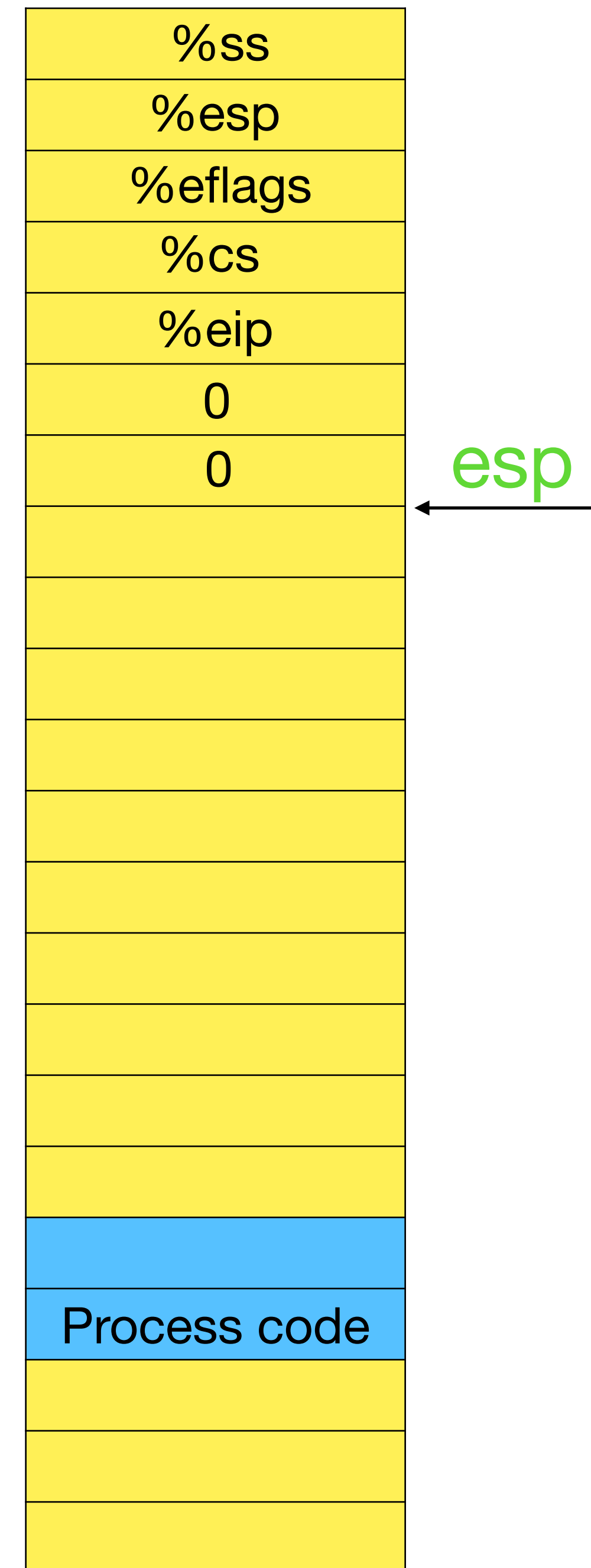
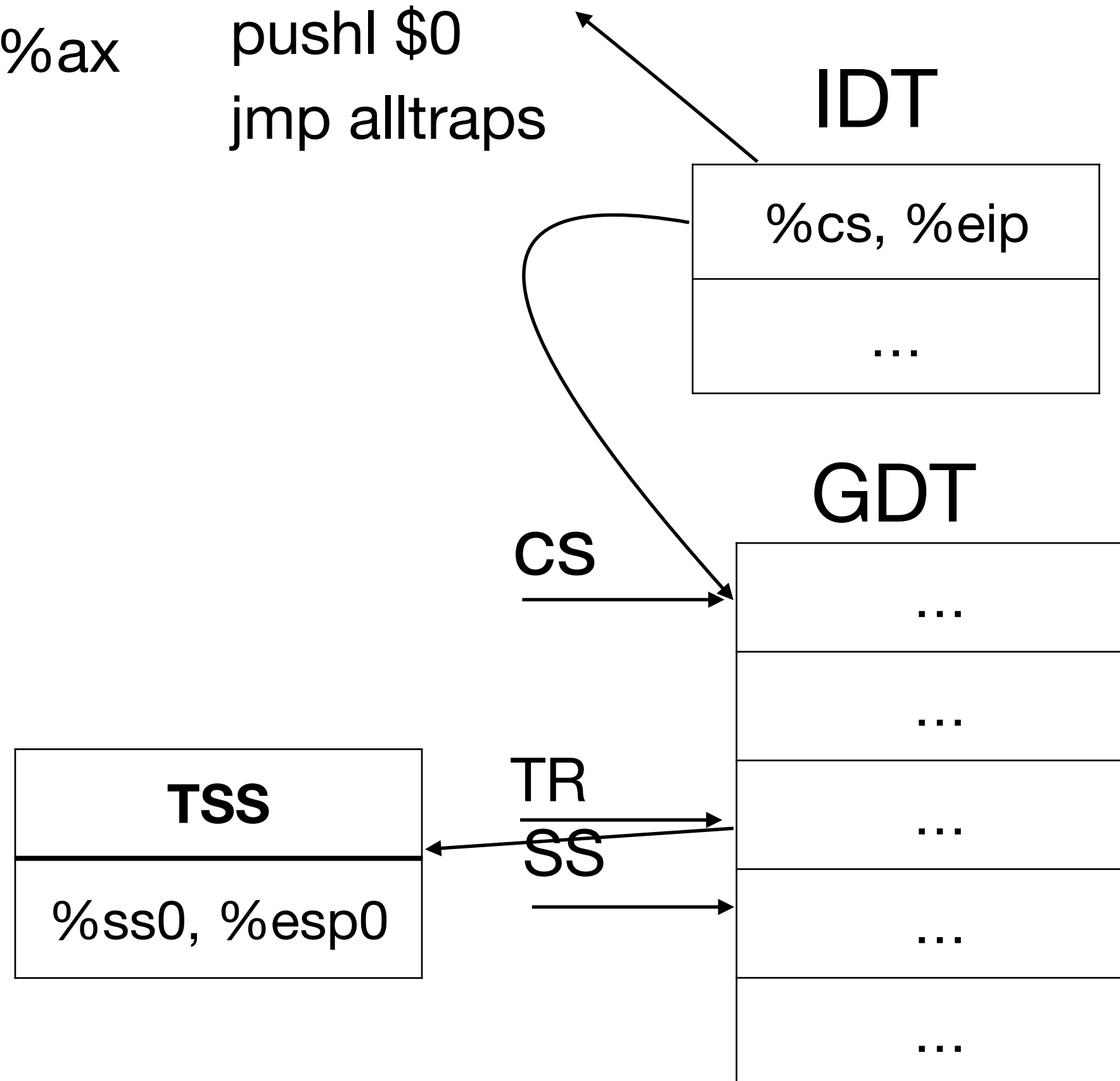
Interrupt handling with user process running

trapasm.S

```
for(;;) eip → alltraps:
;      → pushl %ds..
      pushal
      movw $(SEG_KDATA<<3), %ax
      movw %ax, %ds..
      pushl %esp
      call trap
      addl $4, %esp
      popal
      popl %ds..
      addl $0x8, %esp
      iret
```

vectors.S

```
.globl vector0
vector0:
    pushl $0
    pushl $0
    jmp alltraps
```



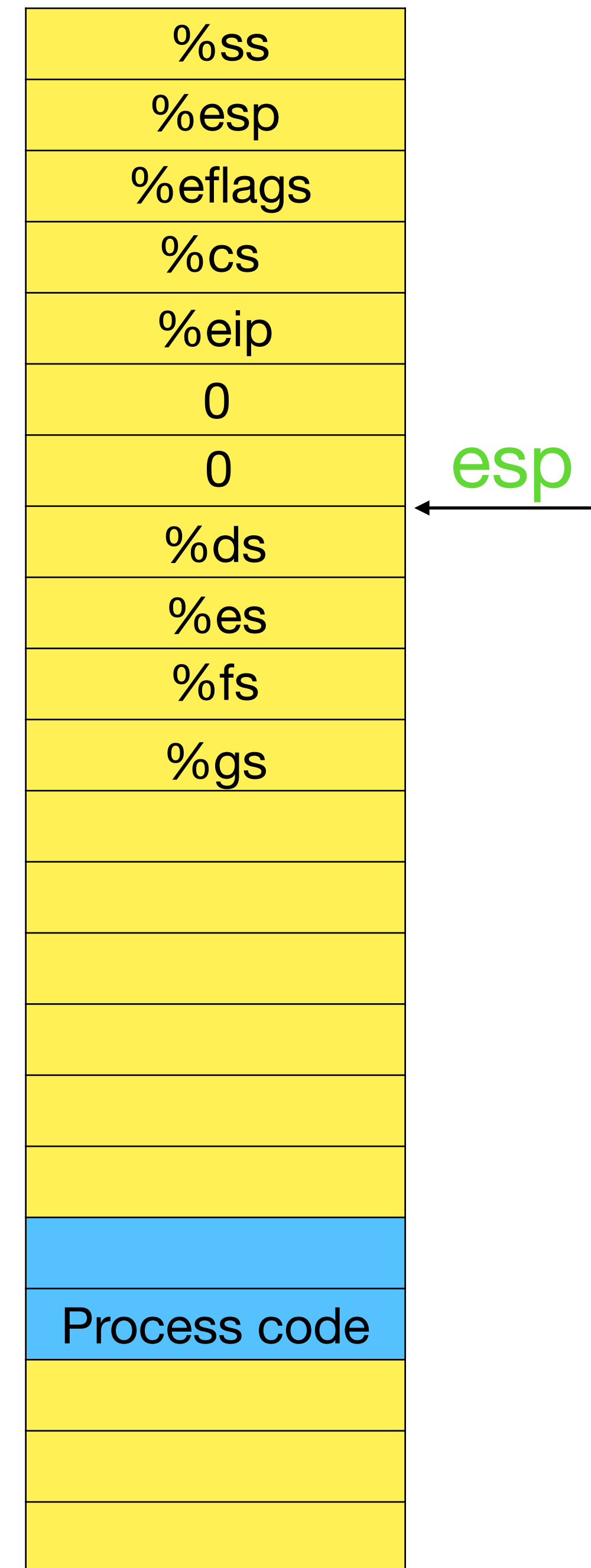
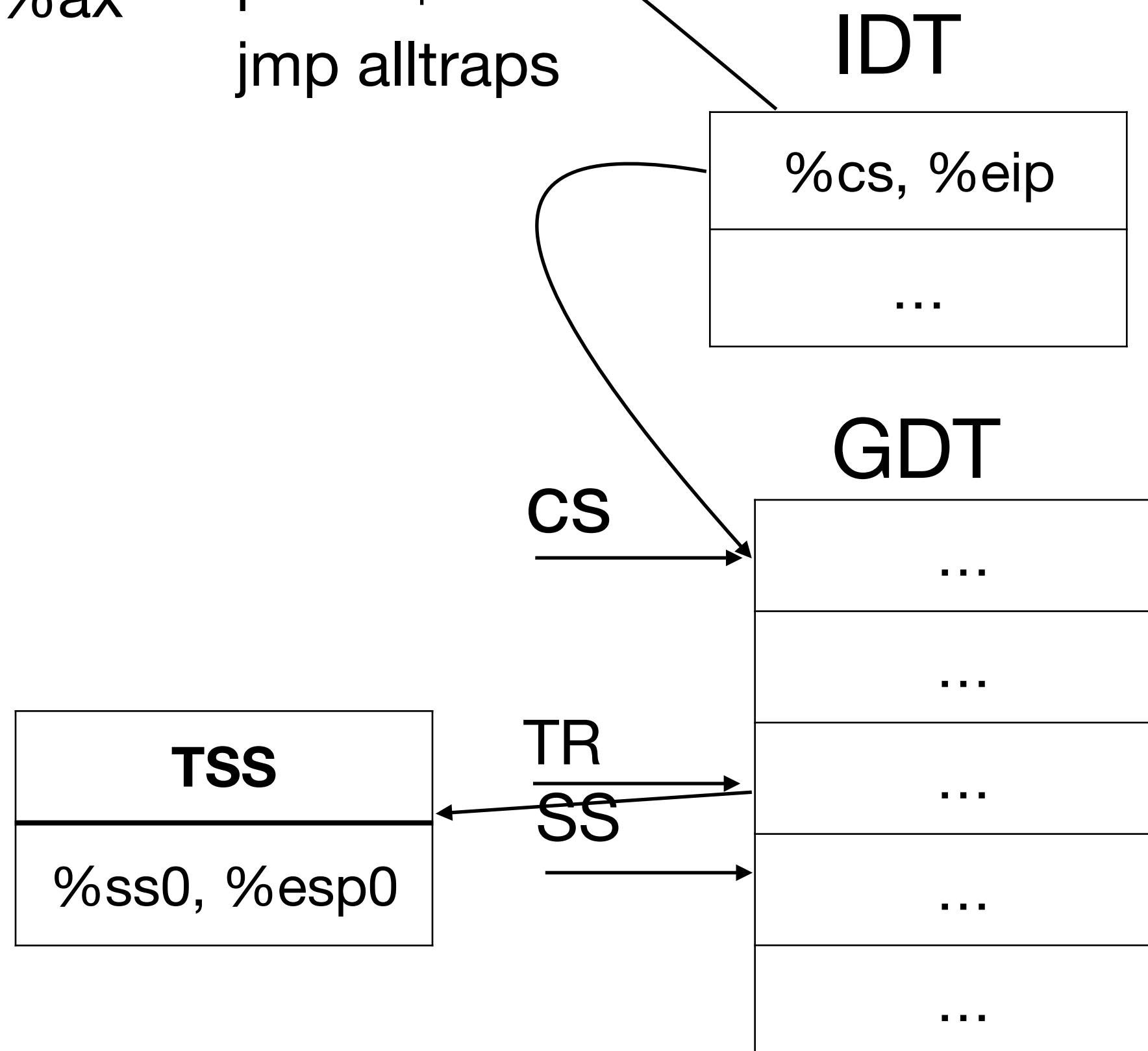
Interrupt handling with user process running

```
trapasm.S
for(;;) alltraps:
;  
    pushl %ds..
    pushal
    movw $(SEG_KDATA<<3), %ax
    movw %ax, %ds..
    pushl %esp
    call trap
    addl $4, %esp
    popal
    popl %ds..
    addl $0x8, %esp
    iret
```

```

vectors.S
.globl vector0
vector0:
    pushl $0
    pushl $0
    jmp alltraps

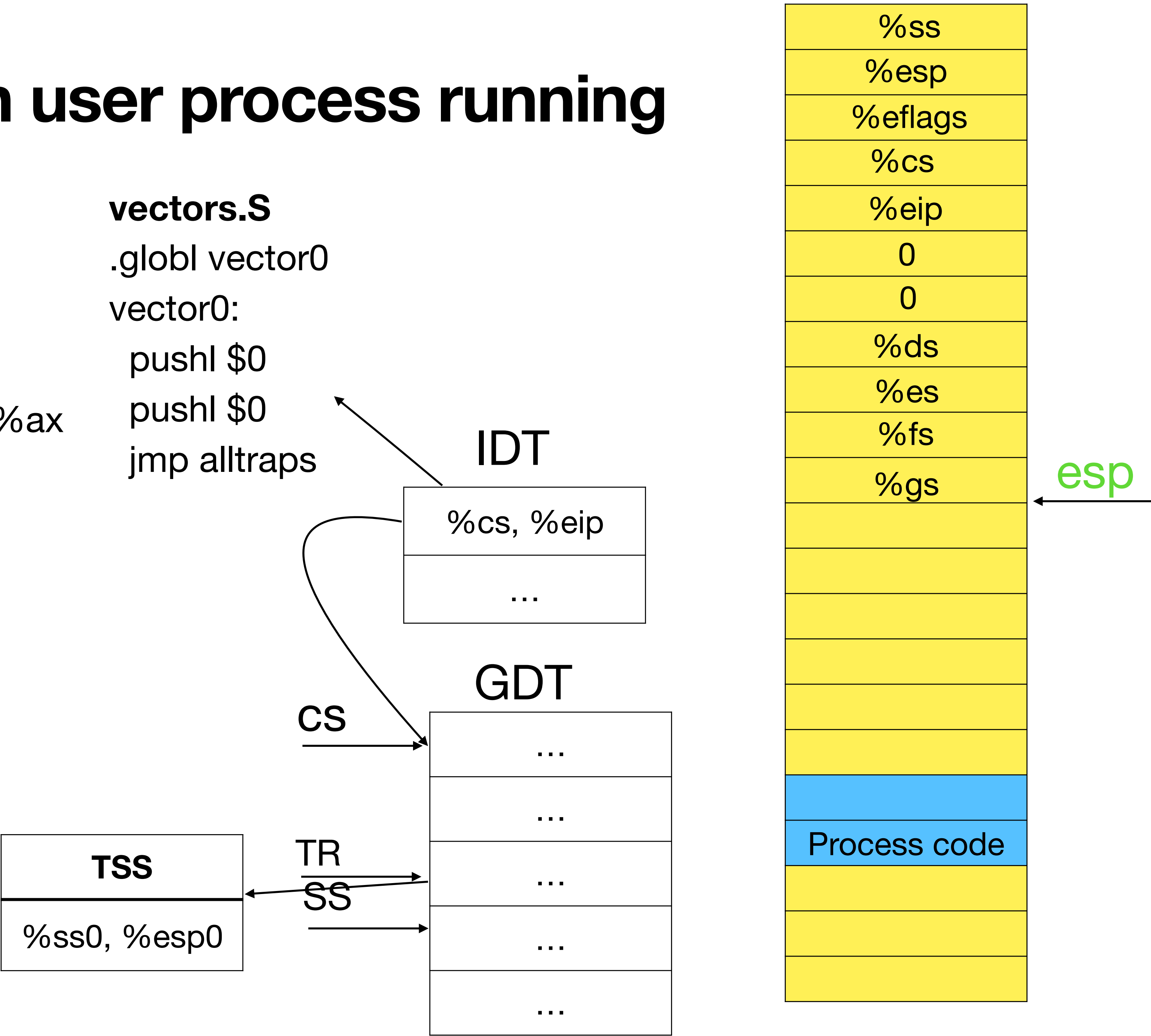
```



Interrupt handling with user process running

```
trapasm.S
for(;;) alltraps:
;   eip → pushl %ds..
    pushal
    movw $(SEG_KDATA<<3), %ax
    movw %ax, %ds..
    pushl %esp
    call trap
    addl $4, %esp
    popal
    popl %ds..
    addl $0x8, %esp
    iret
```

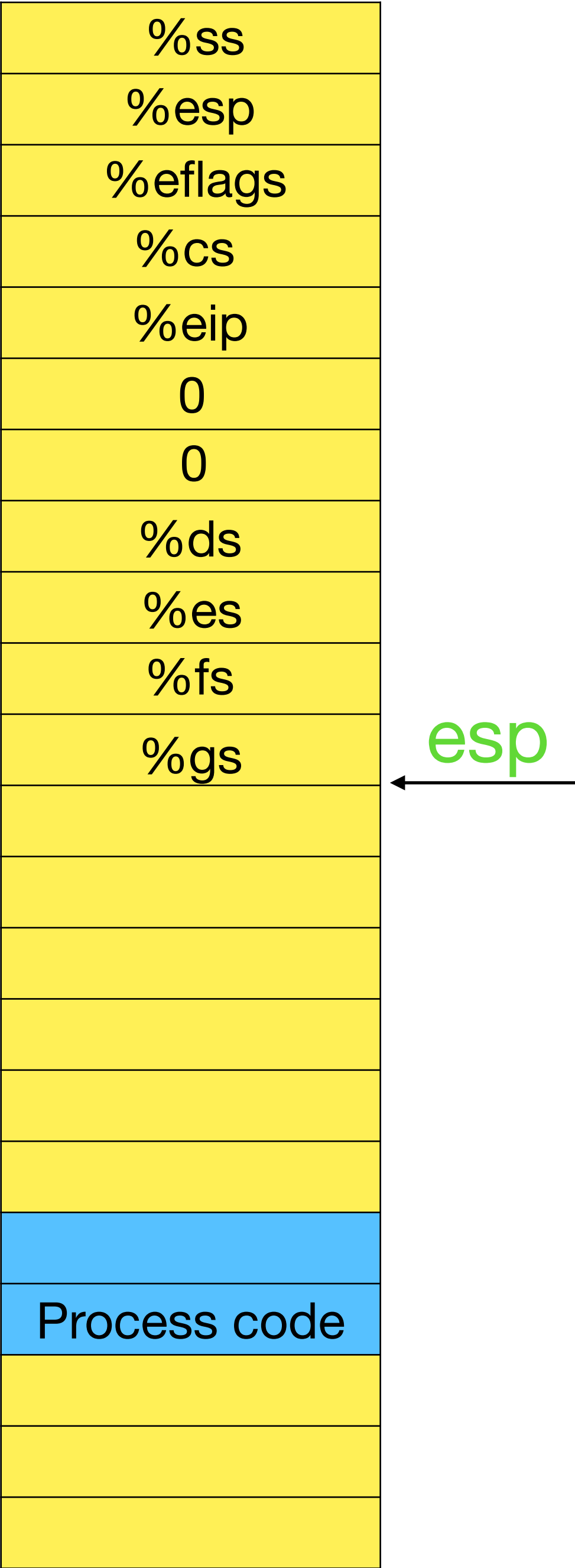
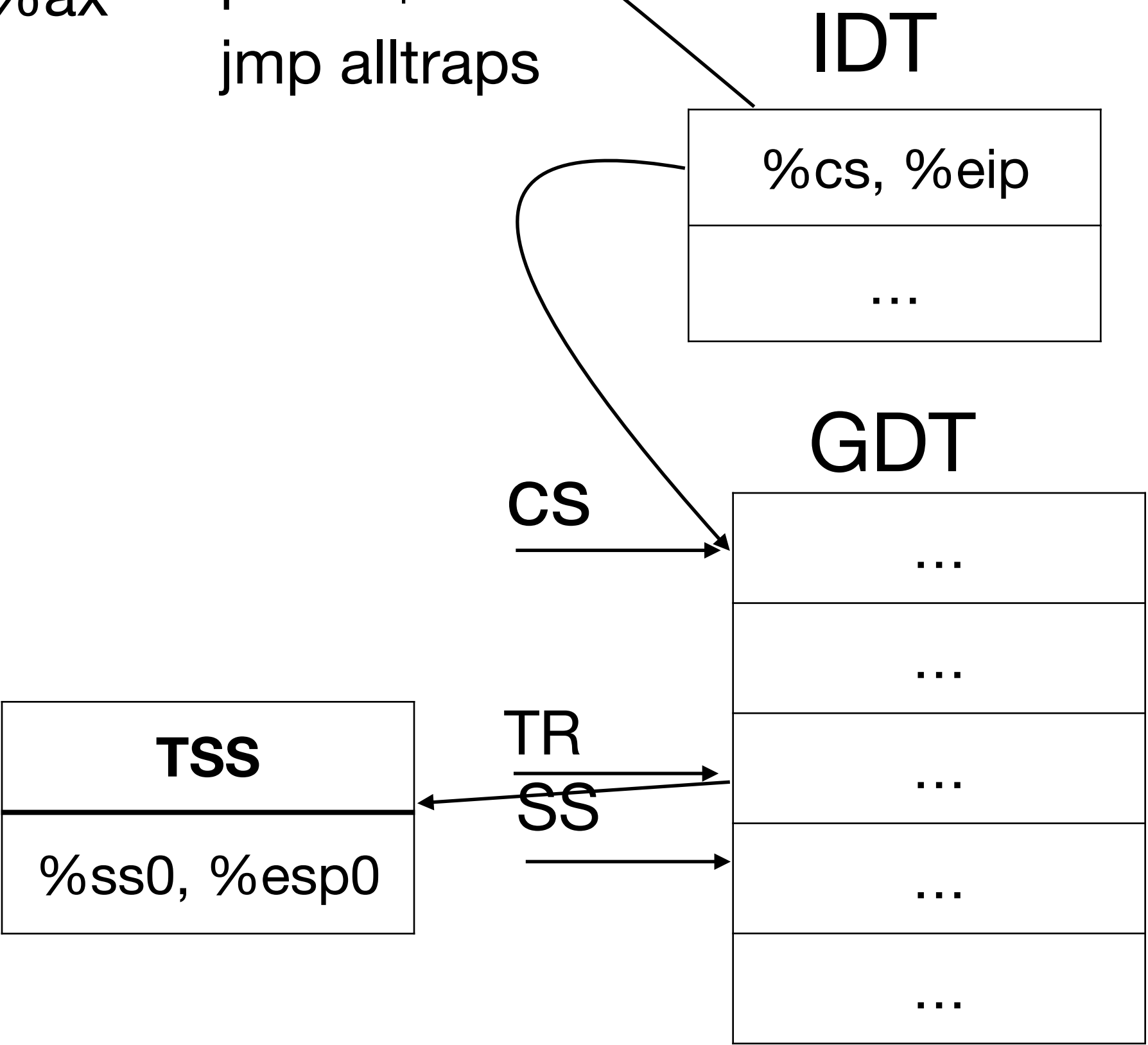
```
vectors.S
.globl vector0
vector0:
    pushl $0
    pushl $0
    jmp alltraps
```



Interrupt handling with user process running

```
trapasm.S
for(;;) alltraps:
;      pushl %ds..
;      eip → pushal
;      movw $(SEG_KDATA<<3), %ax
;      movw %ax, %ds..
;      pushl %esp
;      call trap
;      addl $4, %esp
;      popal
;      popl %ds..
;      addl $0x8, %esp
;      iret
```

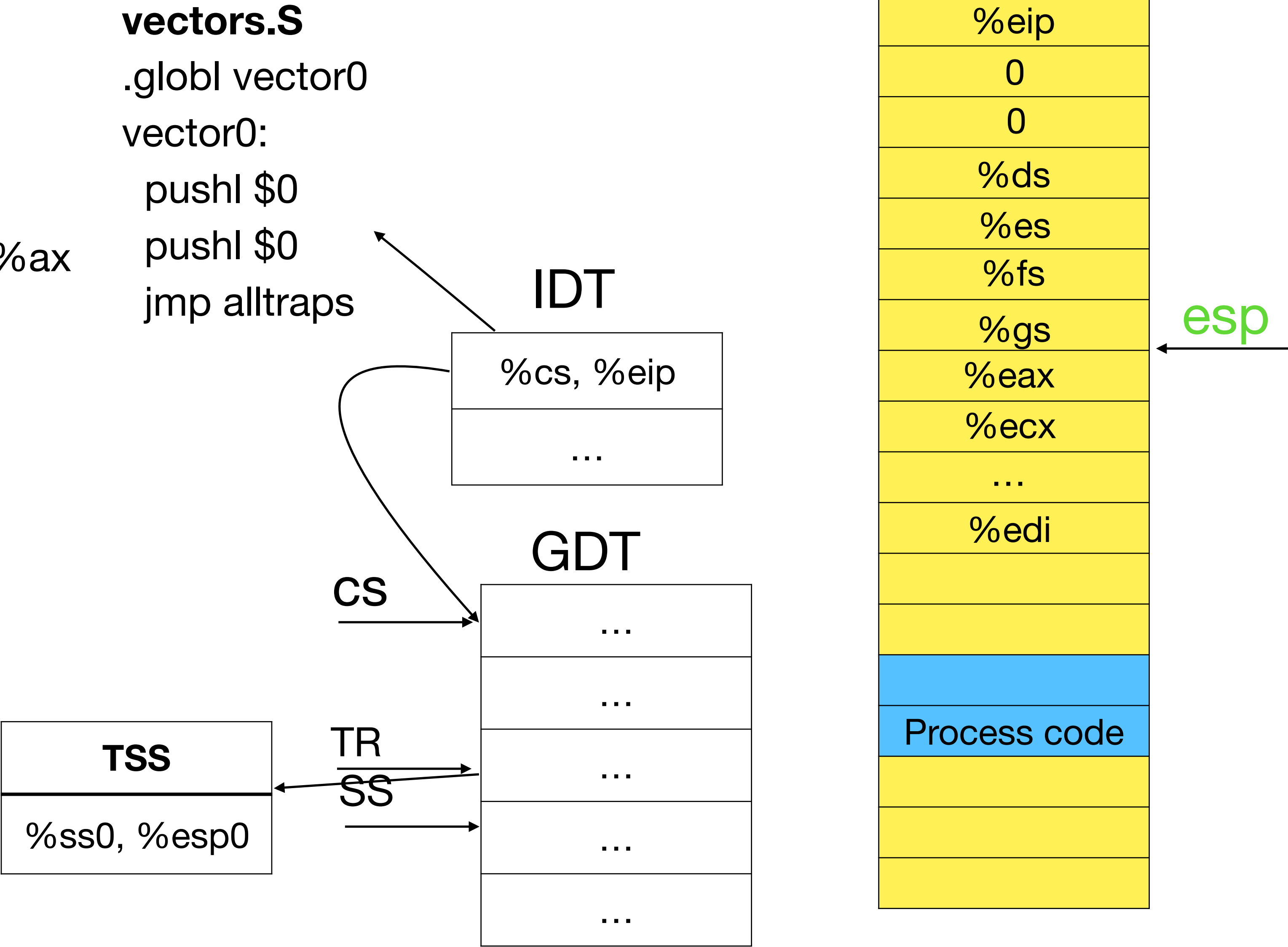
```
vectors.S
.globl vector0
vector0:
    pushl $0
    pushl $0
    jmp alltraps
```



Interrupt handling with user process running

```
trapasm.S
for(;;) alltraps:
;      pushl %ds..
;      eip → pushal
;      movw $(SEG_KDATA<<3), %ax
;      movw %ax, %ds..
;      pushl %esp
;      call trap
;      addl $4, %esp
;      popal
;      popl %ds..
;      addl $0x8, %esp
;      iret
```

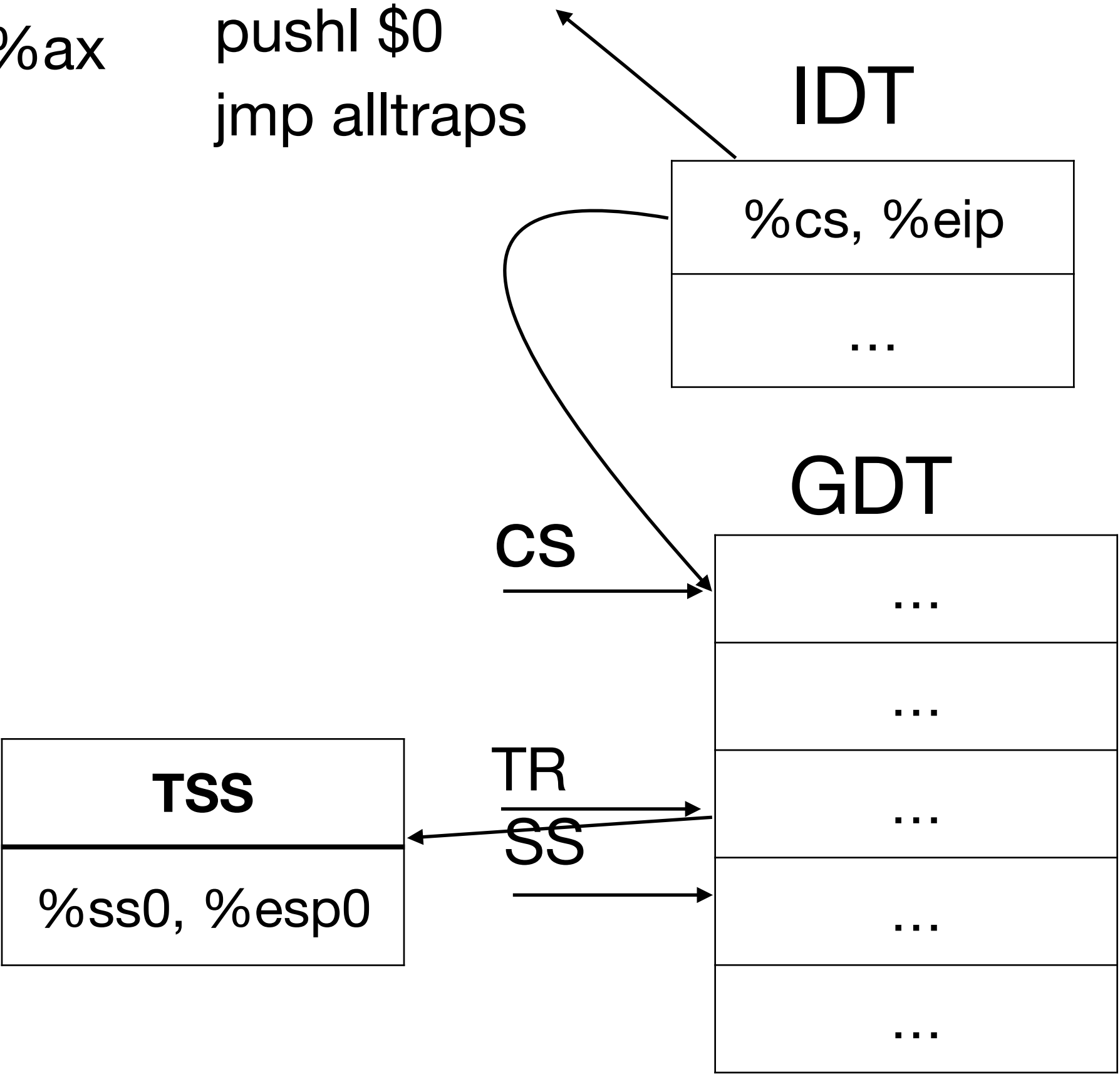
```
vectors.S
.globl vector0
vector0:
    pushl $0
    pushl $0
    jmp alltraps
```



Interrupt handling with user process running

```
trapasm.S
for(;;) alltraps:
;      pushl %ds..
;      eip → pushal
;      movw $(SEG_KDATA<<3), %ax
;      movw %ax, %ds..
;      pushl %esp
;      call trap
;      addl $4, %esp
;      popal
;      popl %ds..
;      addl $0x8, %esp
;      iret
```

```
vectors.S
.globl vector0
vector0:
    pushl $0
    pushl $0
    jmp alltraps
```



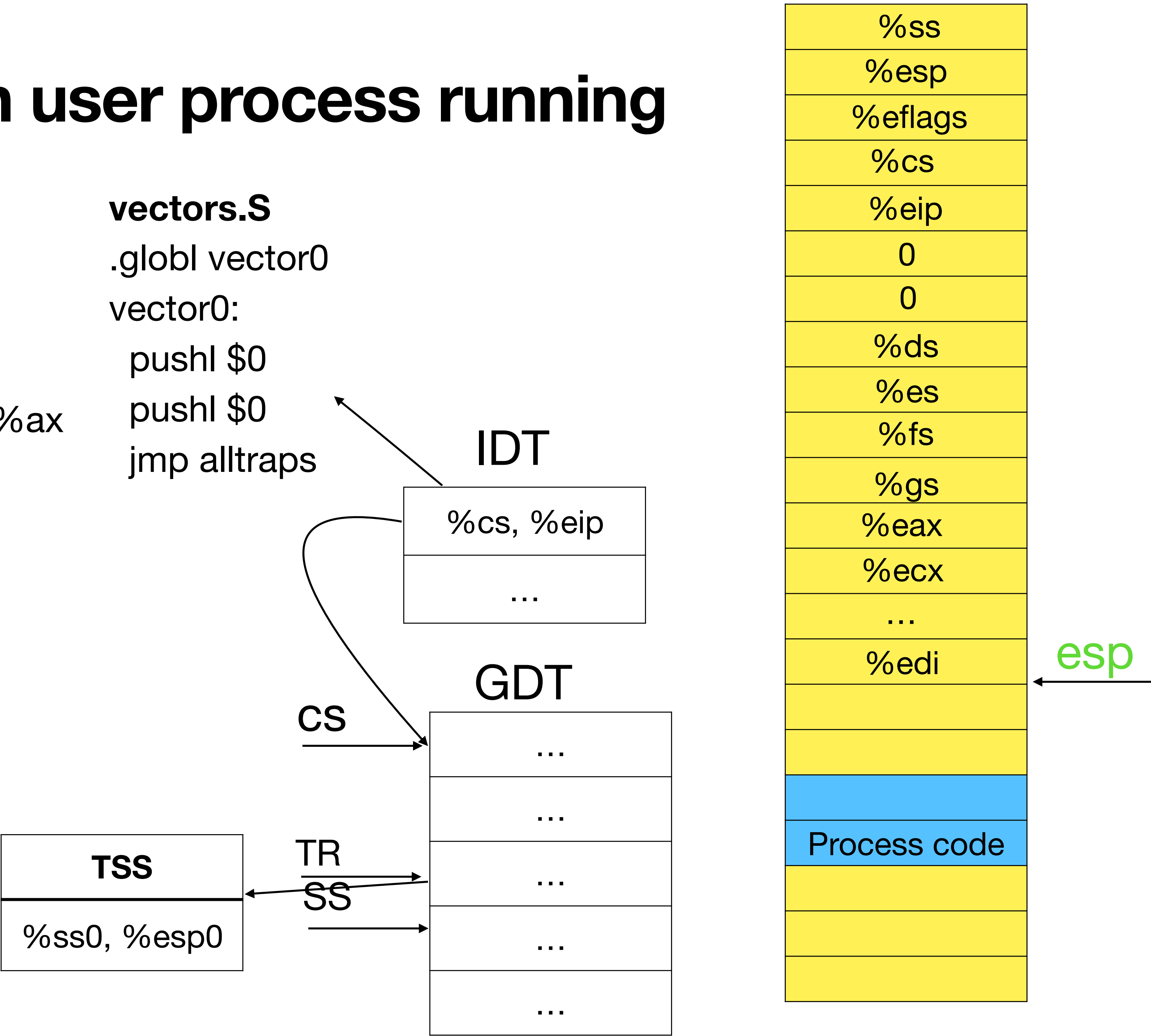
%ss
%esp
%eflags
%cs
%eip
0
0
%ds
%es
%fs
%gs
%eax
%ecx
...
%edi
Process code

esp →

Interrupt handling with user process running

```
trapasm.S
for(;;)
;   alltraps:
;       pushl %ds..
;       pushal
;       movw $(SEG_KDATA<<3), %ax
;       movw %ax, %ds..
;       eip → pushl %esp
;       call trap
;       addl $4, %esp
;       popal
;       popl %ds..
;       addl $0x8, %esp
;       iret
```

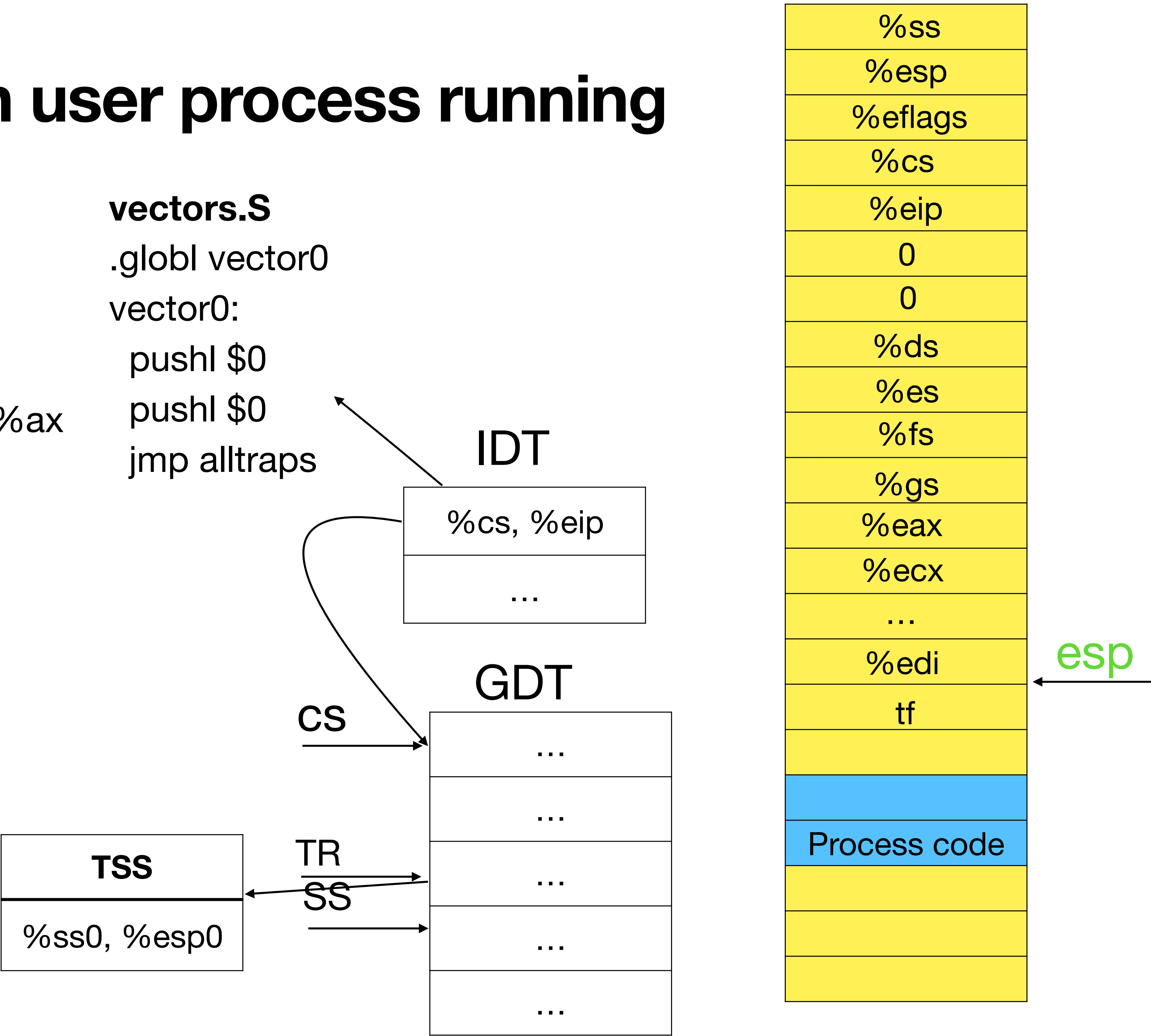
```
vectors.S
.globl vector0
vector0:
    pushl $0
    pushl $0
    jmp alltraps
```



Interrupt handling with user process running

```
trapasm.S
for(;;)
;
;   alltraps:
;       pushl %ds..
;       pushal
;       movw $(SEG_KDATA<<3), %ax
;       movw %ax, %ds..
;       eip → pushl %esp
;       call trap
;       addl $4, %esp
;       popal
;       popl %ds..
;       addl $0x8, %esp
;       iret
```

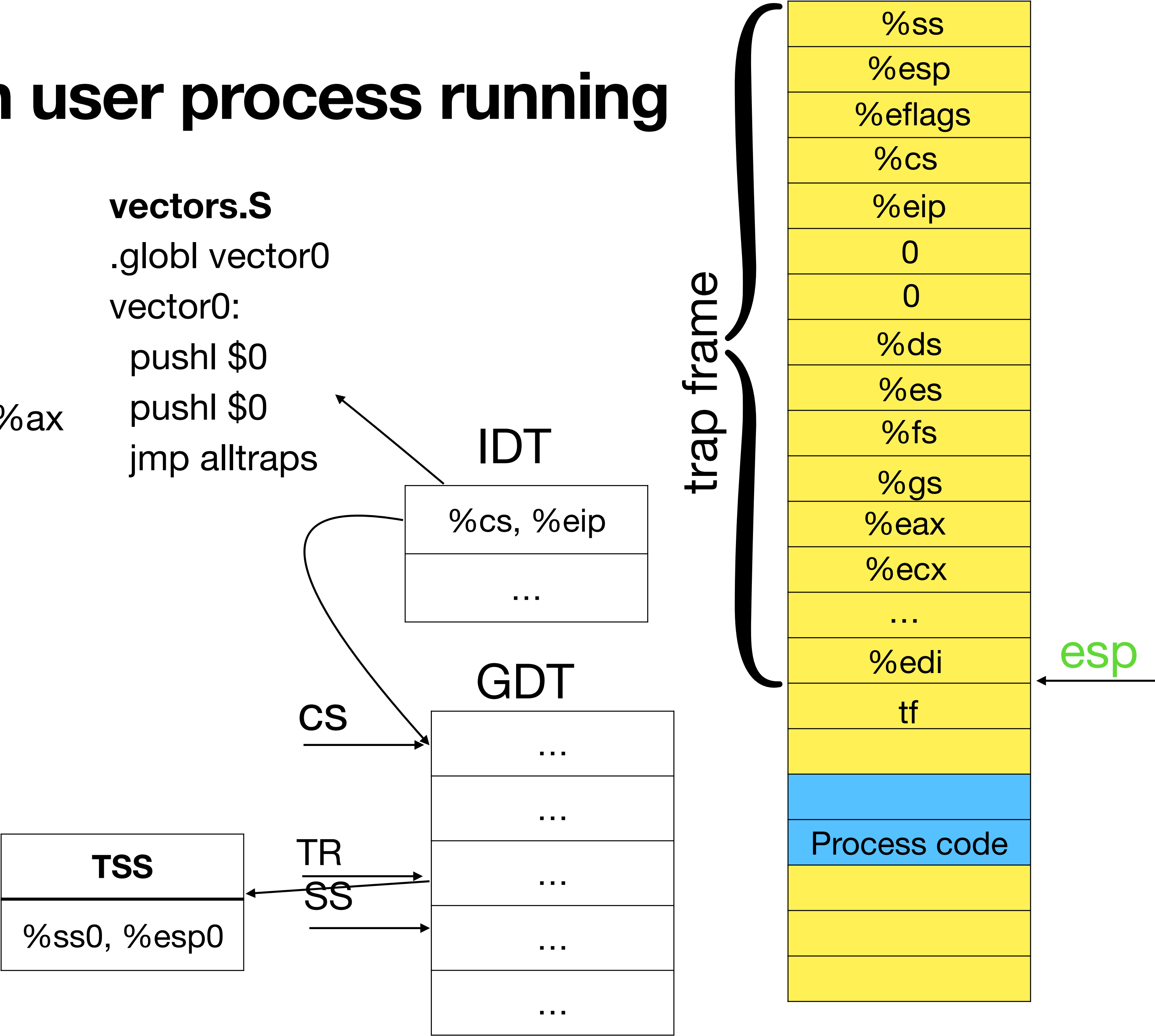
```
vectors.S
.globl vector0
vector0:
    pushl $0
    pushl $0
    jmp alltraps
```



Interrupt handling with user process running

```
trapasm.S
for(;;)
;
;   alltraps:
;       pushl %ds..
;       pushal
;       movw $(SEG_KDATA<<3), %ax
;       movw %ax, %ds..
;       eip→ pushl %esp
;       call trap
;       addl $4, %esp
;       popal
;       popl %ds..
;       addl $0x8, %esp
;       iret
```

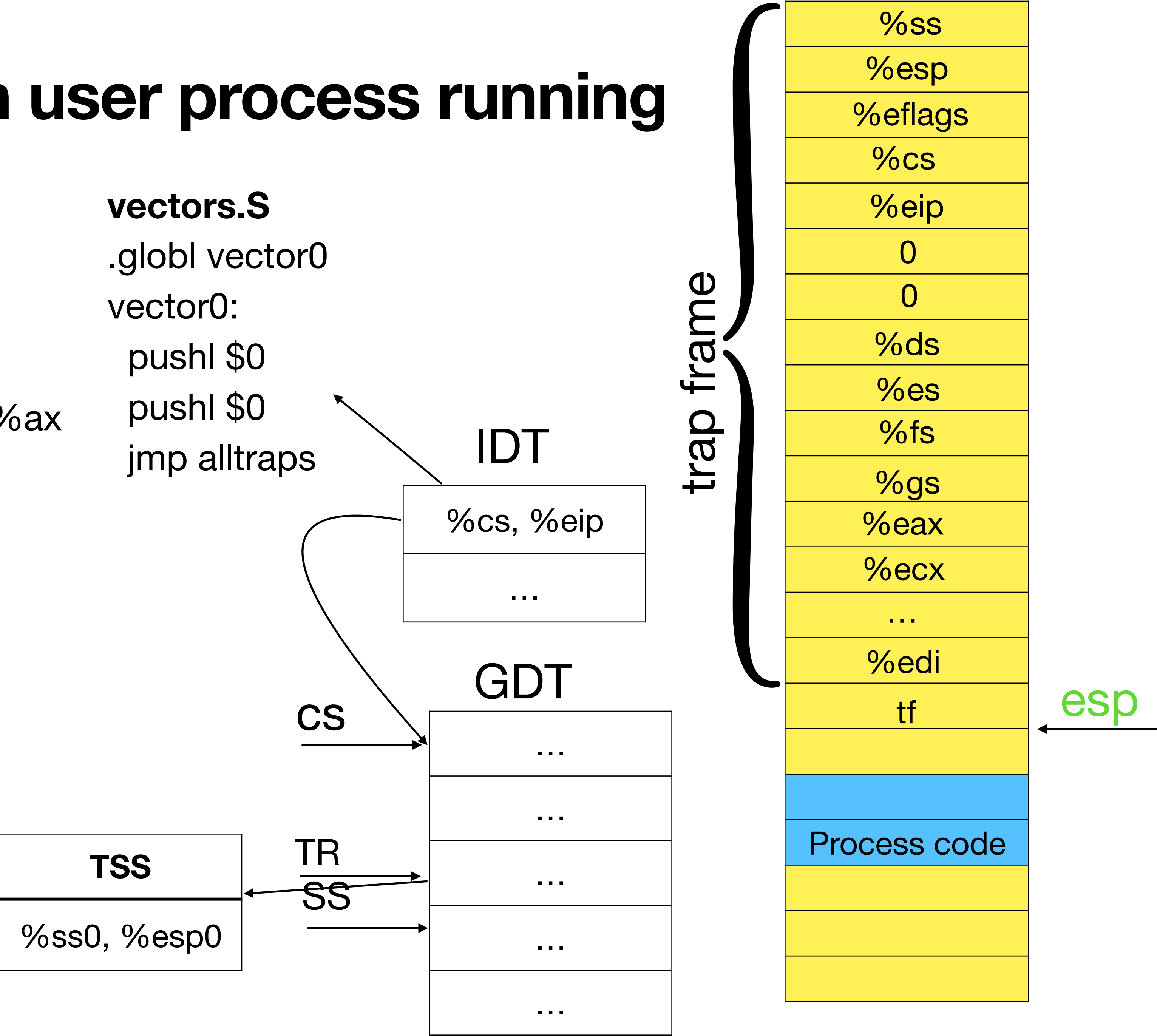
```
vectors.S
.globl vector0
vector0:
    pushl $0
    pushl $0
    jmp alltraps
```



Interrupt handling with user process running

```
trapasm.S
for(;;)
;
;   alltraps:
;       pushl %ds..
;       pushal
;       movw $(SEG_KDATA<<3), %ax
;       movw %ax, %ds..
;       eip → pushl %esp
;       call trap
;       addl $4, %esp
;       popal
;       popl %ds..
;       addl $0x8, %esp
;       iret
```

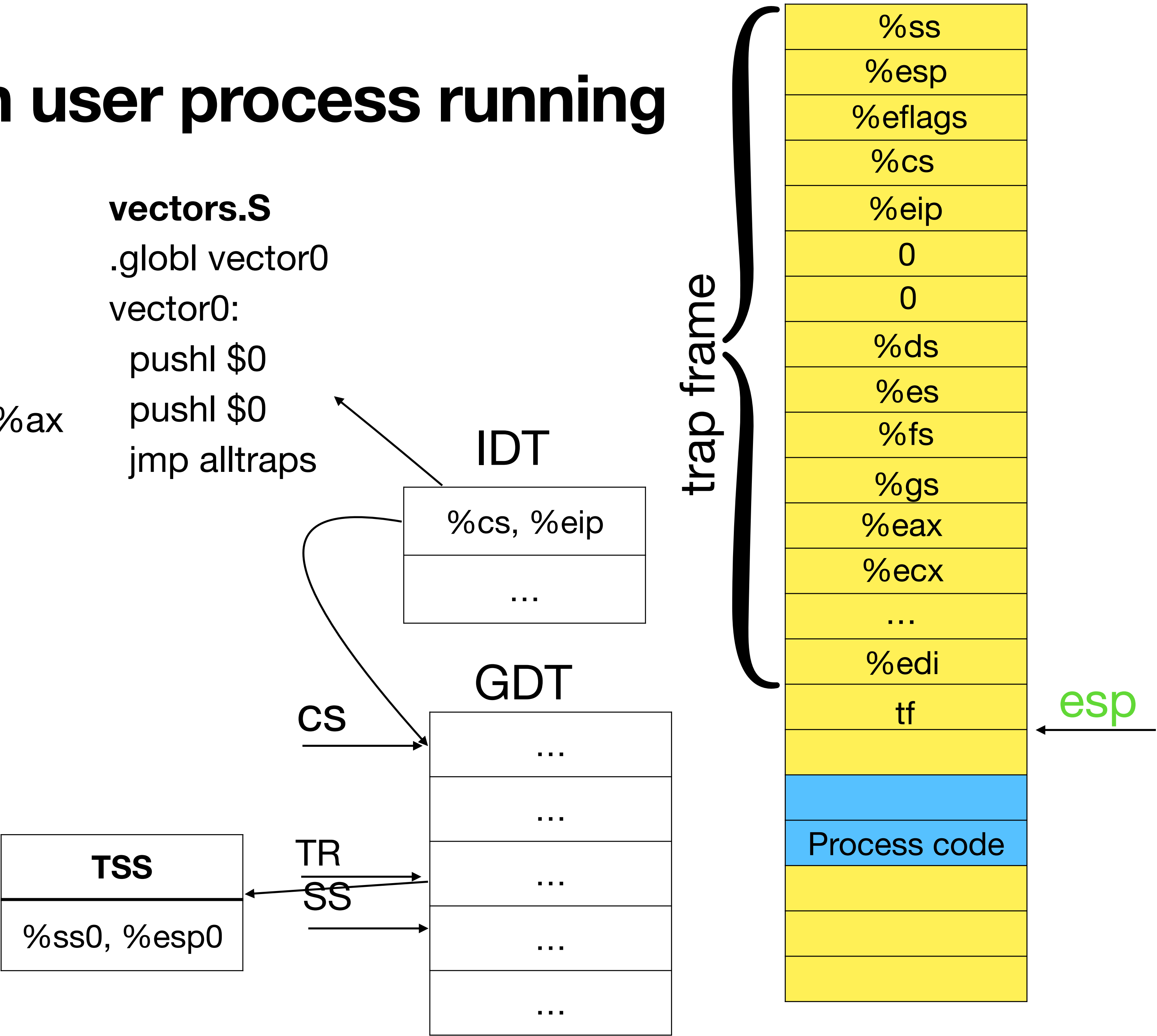
```
vectors.S
.globl vector0
vector0:
    pushl $0
    pushl $0
    jmp alltraps
```



Interrupt handling with user process running

```
trapasm.S
for(;;)
;
;   alltraps:
;       pushl %ds..
;       pushal
;       movw $(SEG_KDATA<<3), %ax
;       movw %ax, %ds..
;       pushl %esp
;       call trap
;       addl $4, %esp
;       popal
;       popl %ds..
;       addl $0x8, %esp
;       iret
```

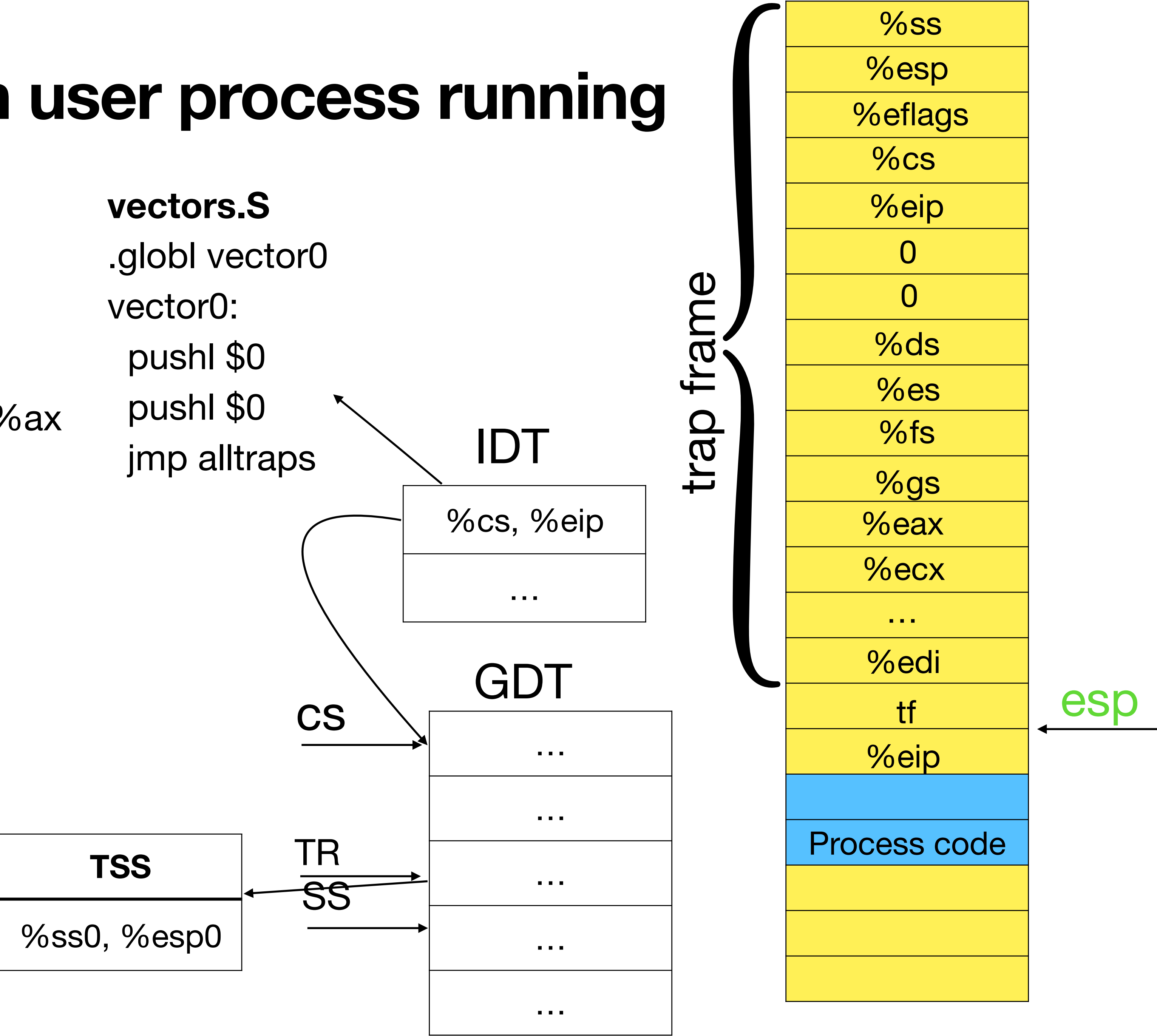
```
vectors.S
.globl vector0
vector0:
    pushl $0
    pushl $0
    jmp alltraps
```



Interrupt handling with user process running

```
trapasm.S
for(;;)
;
alltraps:
    pushl %ds..
    pushal
    movw $(SEG_KDATA<<3), %ax
    movw %ax, %ds..
    pushl %esp
    call trap
    addl $4, %esp
    popal
    popl %ds..
    addl $0x8, %esp
    iret
```

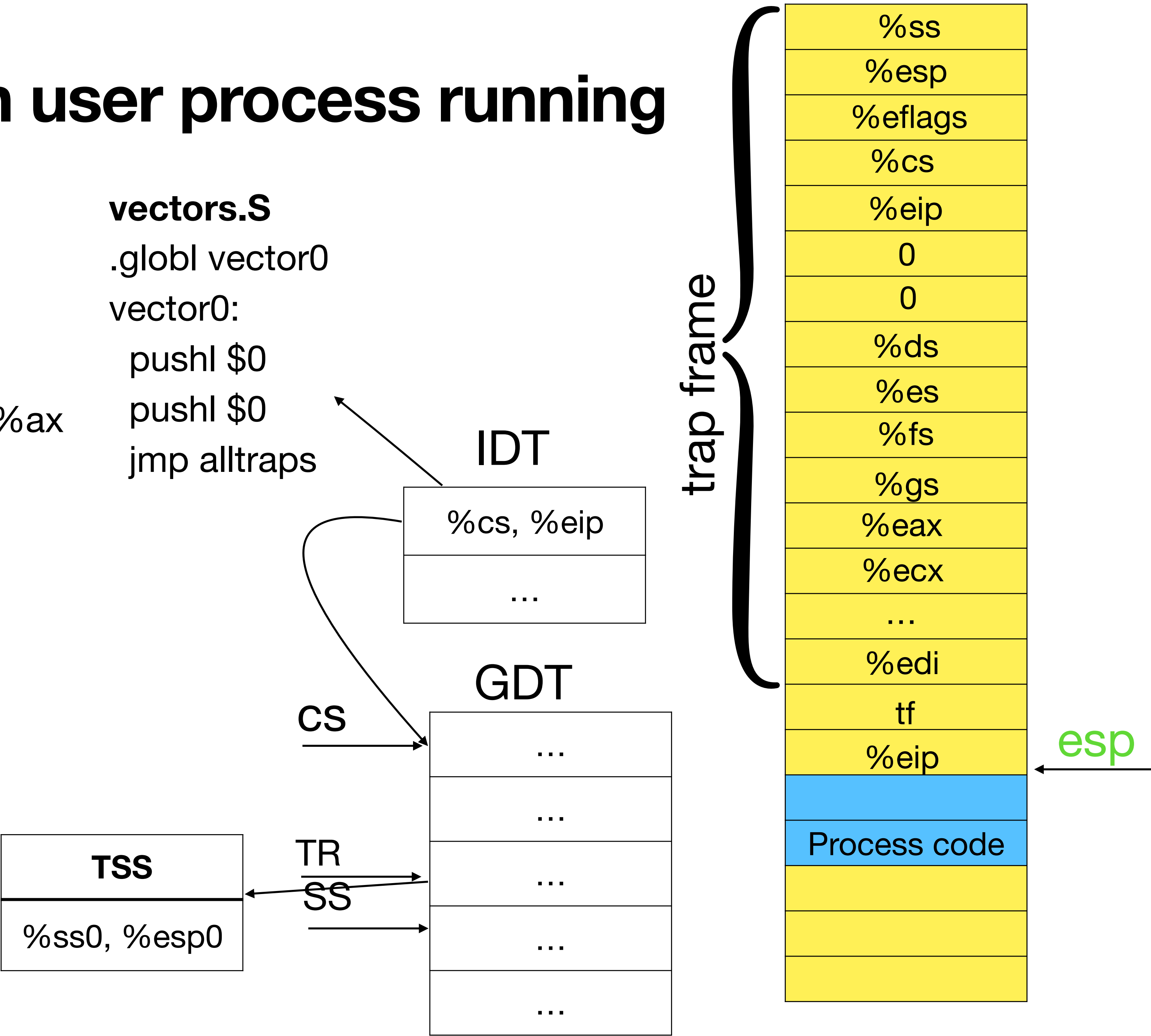
```
vectors.S
.globl vector0
vector0:
    pushl $0
    pushl $0
    jmp alltraps
```



Interrupt handling with user process running

```
trapasm.S
for(;;)
;
;   alltraps:
;       pushl %ds..
;       pushal
;       movw $(SEG_KDATA<<3), %ax
;       movw %ax, %ds..
;       pushl %esp
;       call trap
;       addl $4, %esp
;       popal
;       popl %ds..
;       addl $0x8, %esp
;       iret
```

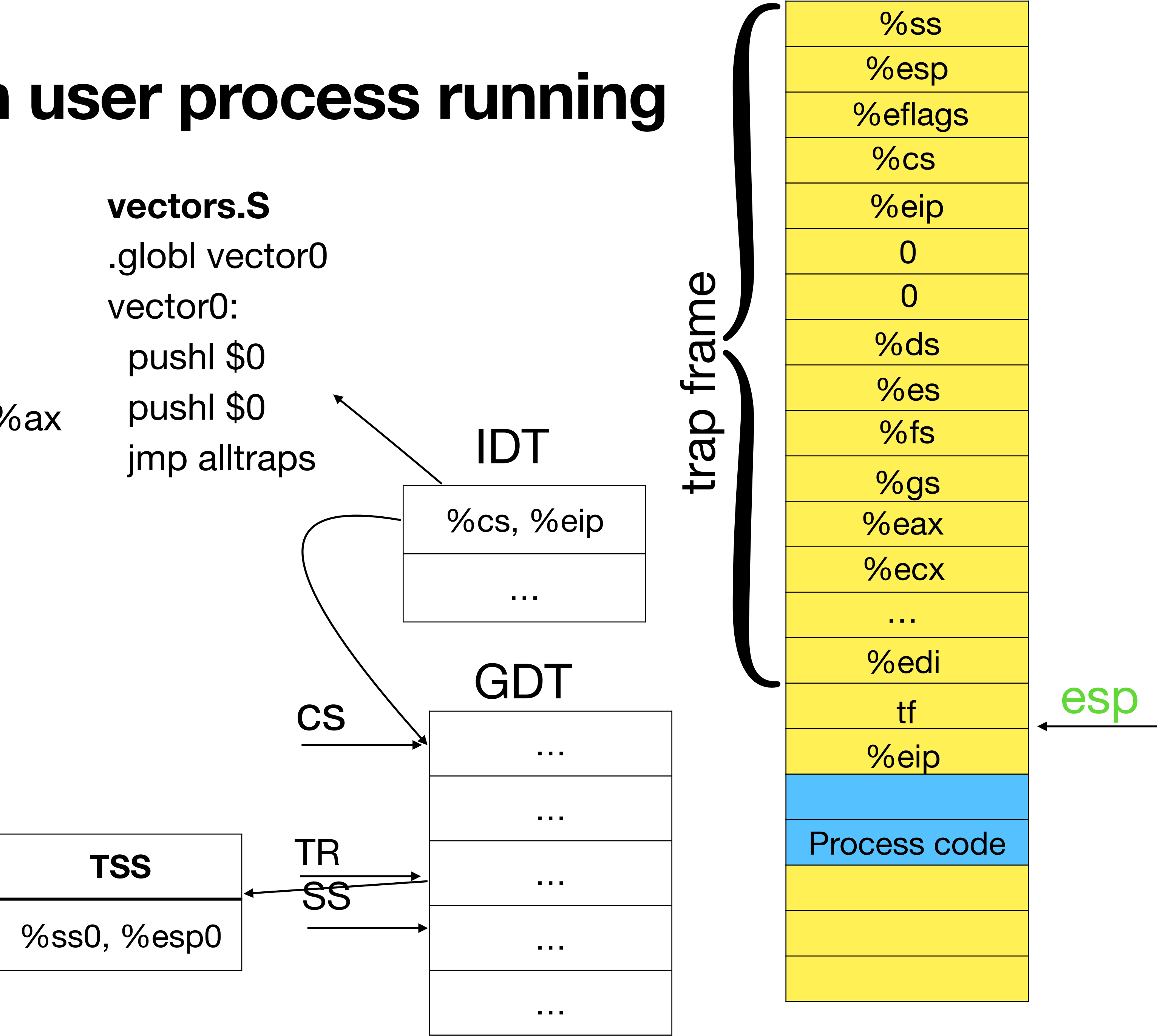
```
vectors.S
.globl vector0
vector0:
    pushl $0
    pushl $0
    jmp alltraps
```



Interrupt handling with user process running

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trapasm.S
for(;;)
;
;   alltraps:
;       pushl %ds..
;       pushal
;       movw $(SEG_KDATA<<3), %ax
;       movw %ax, %ds..
;       pushl %esp
;       call trap
;       addl $4, %esp
;       popal
;       popl %ds..
;       addl $0x8, %esp
;       iret
```

```
vectors.S
.globl vector0
vector0:
    pushl $0
    pushl $0
    jmp alltraps
```



Interrupt handling with user process running

```

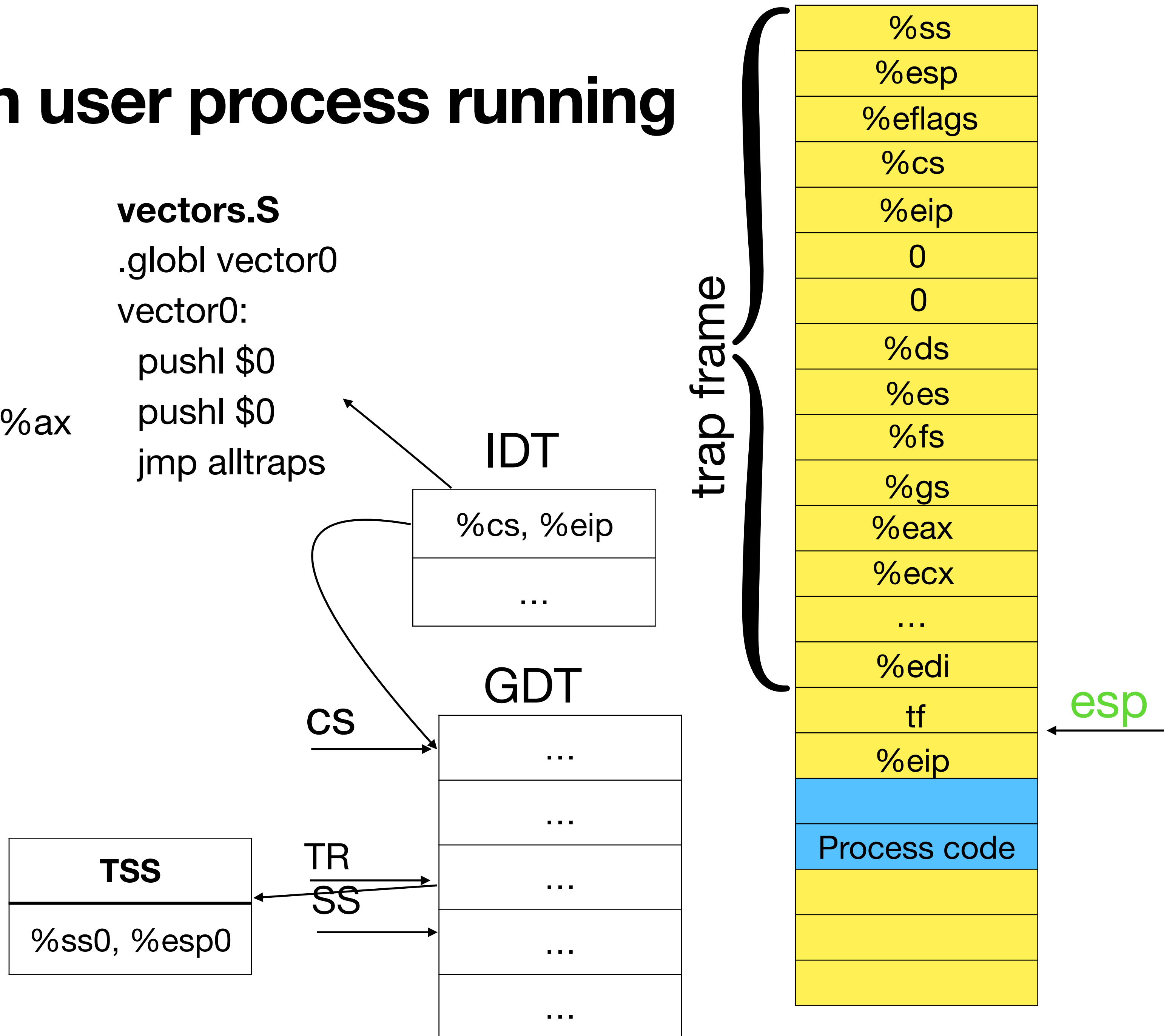
trapasm.S
for(;;)
;    pushl %ds..
    pushal
    movw $(SEG_KDATA<<3), %ax
    movw %ax, %ds..
    pushl %esp
    call trap
    
    addl $4, %esp
    popal
    popl %ds..
    addl $0x8, %esp
    iret

```

```

vectors.S
.globl vector0
vector0:
    pushl $0
    pushl $0
    jmp alltraps

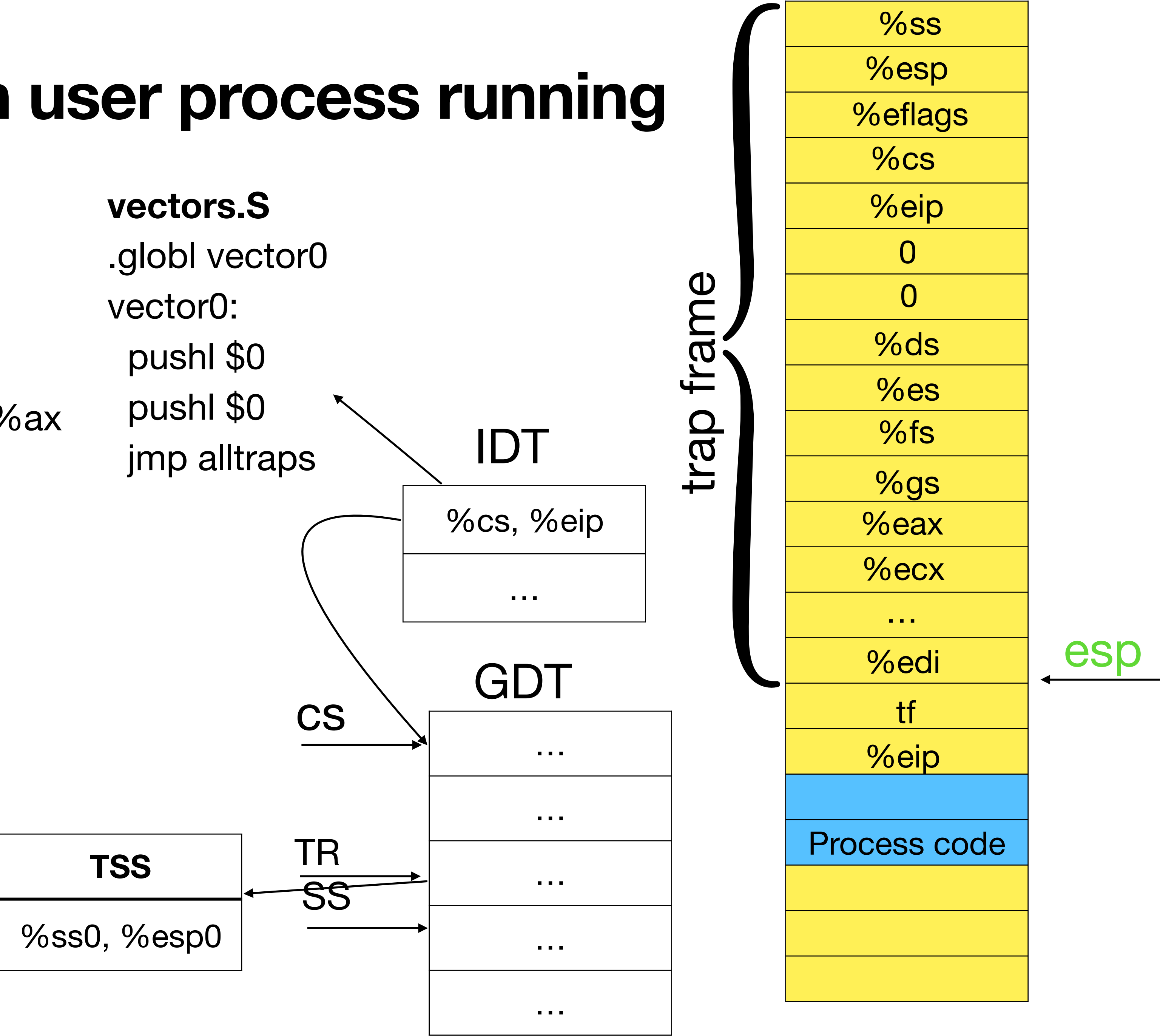
```



Interrupt handling with user process running

```
trapasm.S
for(;;)
;
alltraps:
    pushl %ds..
    pushal
    movw $(SEG_KDATA<<3), %ax
    movw %ax, %ds..
    pushl %esp
    call trap
    eip → addl $4, %esp
    popal
    popl %ds..
    addl $0x8, %esp
    iret
```

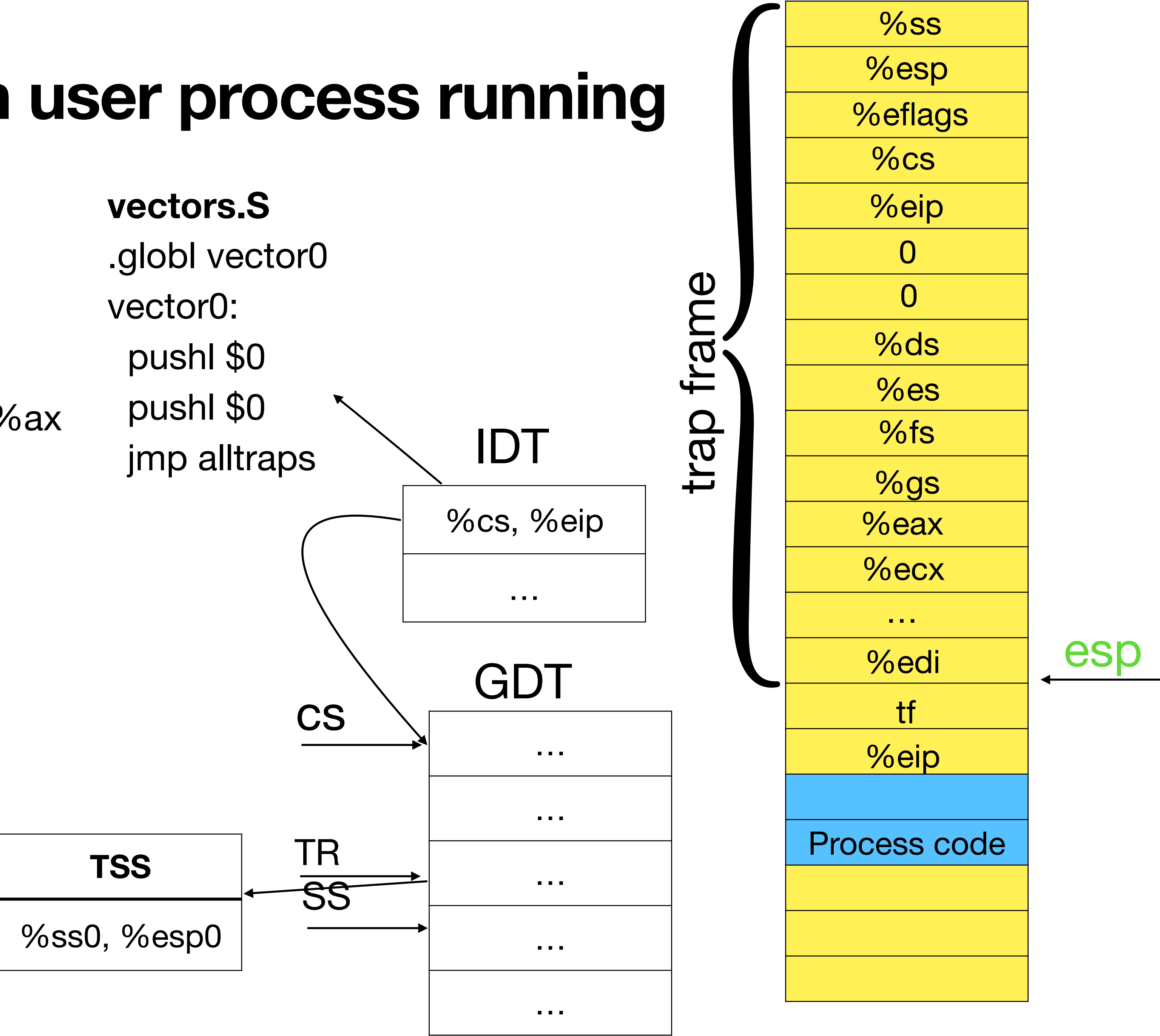
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Interrupt handling with user process running

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trapasm.S
for(;;)
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alltraps:
    pushl %ds..
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    movw $(SEG_KDATA<<3), %ax
    movw %ax, %ds..
    pushl %esp
    call trap
    addl $4, %esp
    popal
    popl %ds..
    addl $0x8, %esp
    iret
```

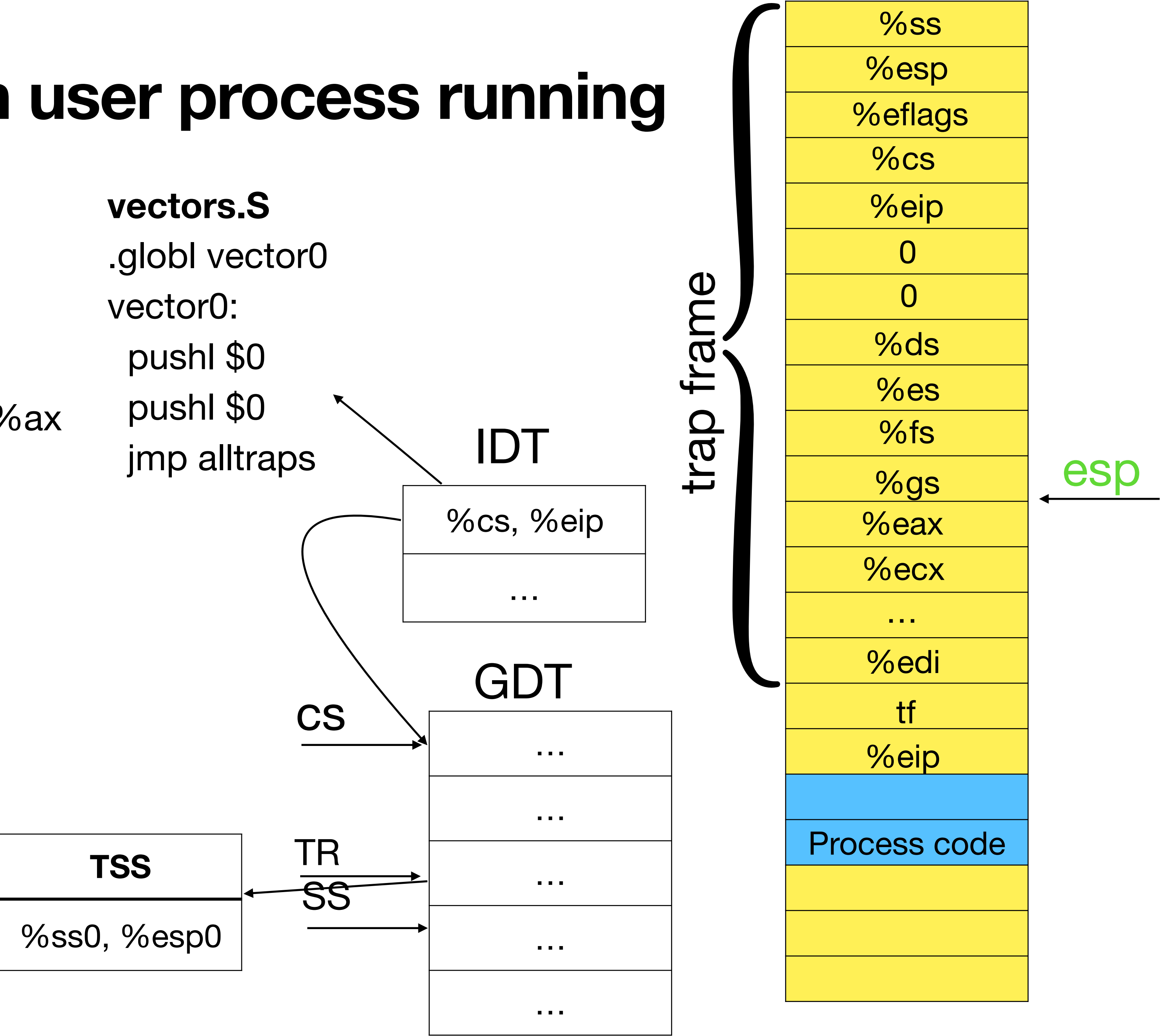
```
vectors.S
.globl vector0
vector0:
    pushl $0
    pushl $0
    jmp alltraps
```



Interrupt handling with user process running

```
trapasm.S
for(;;)
;
alltraps:
    pushl %ds..
    pushal
    movw $(SEG_KDATA<<3), %ax
    movw %ax, %ds..
    pushl %esp
    call trap
    addl $4, %esp
    popal
    popl %ds..
    addl $0x8, %esp
    iret
```

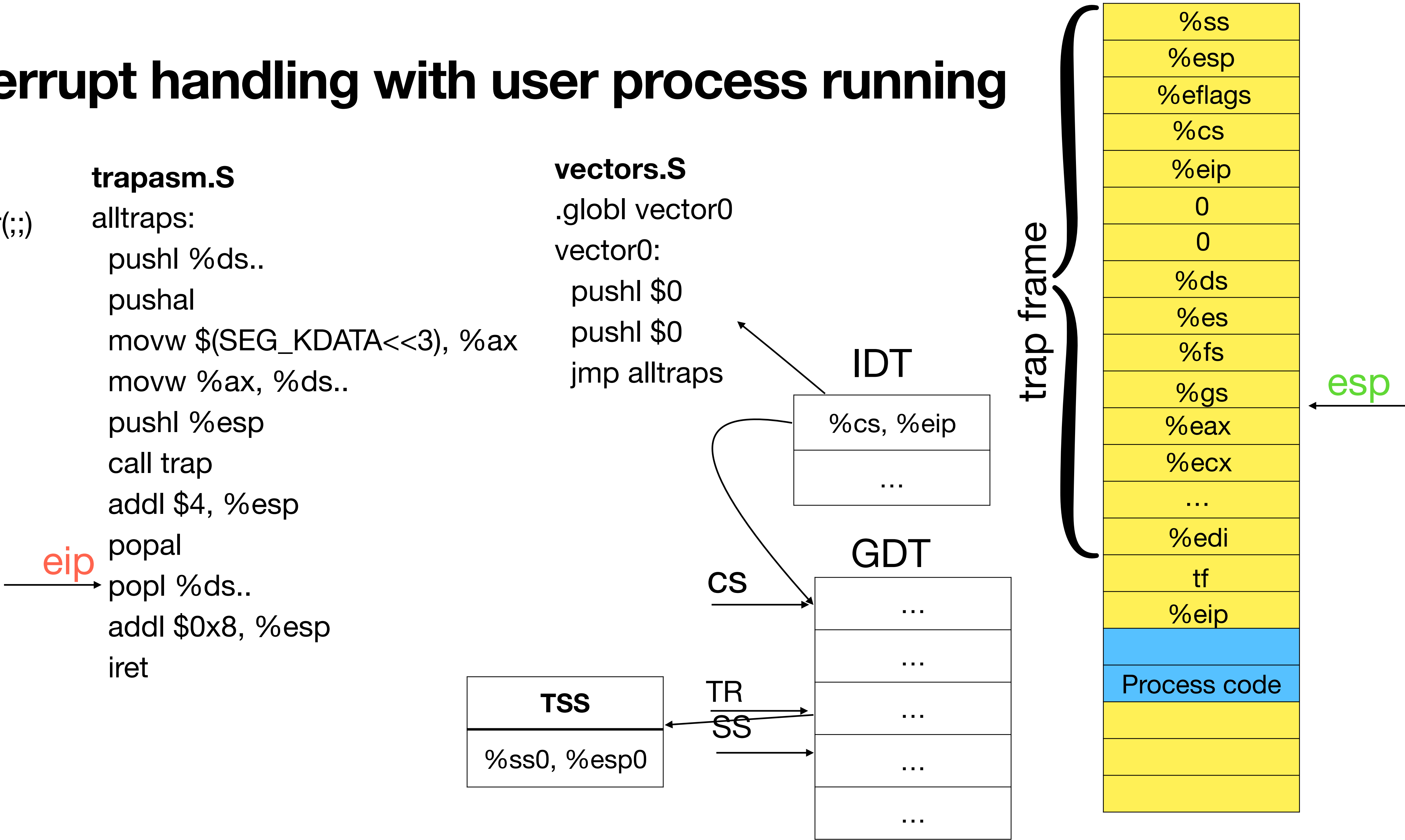
```
vectors.S
.globl vector0
vector0:
    pushl $0
    pushl $0
    jmp alltraps
```



Interrupt handling with user process running

```
trapasm.S
for(;;)
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alltraps:
    pushl %ds..
    pushal
    movw $(SEG_KDATA<<3), %ax
    movw %ax, %ds..
    pushl %esp
    call trap
    addl $4, %esp
    popal
    popl %ds..
    addl $0x8, %esp
    iret
```

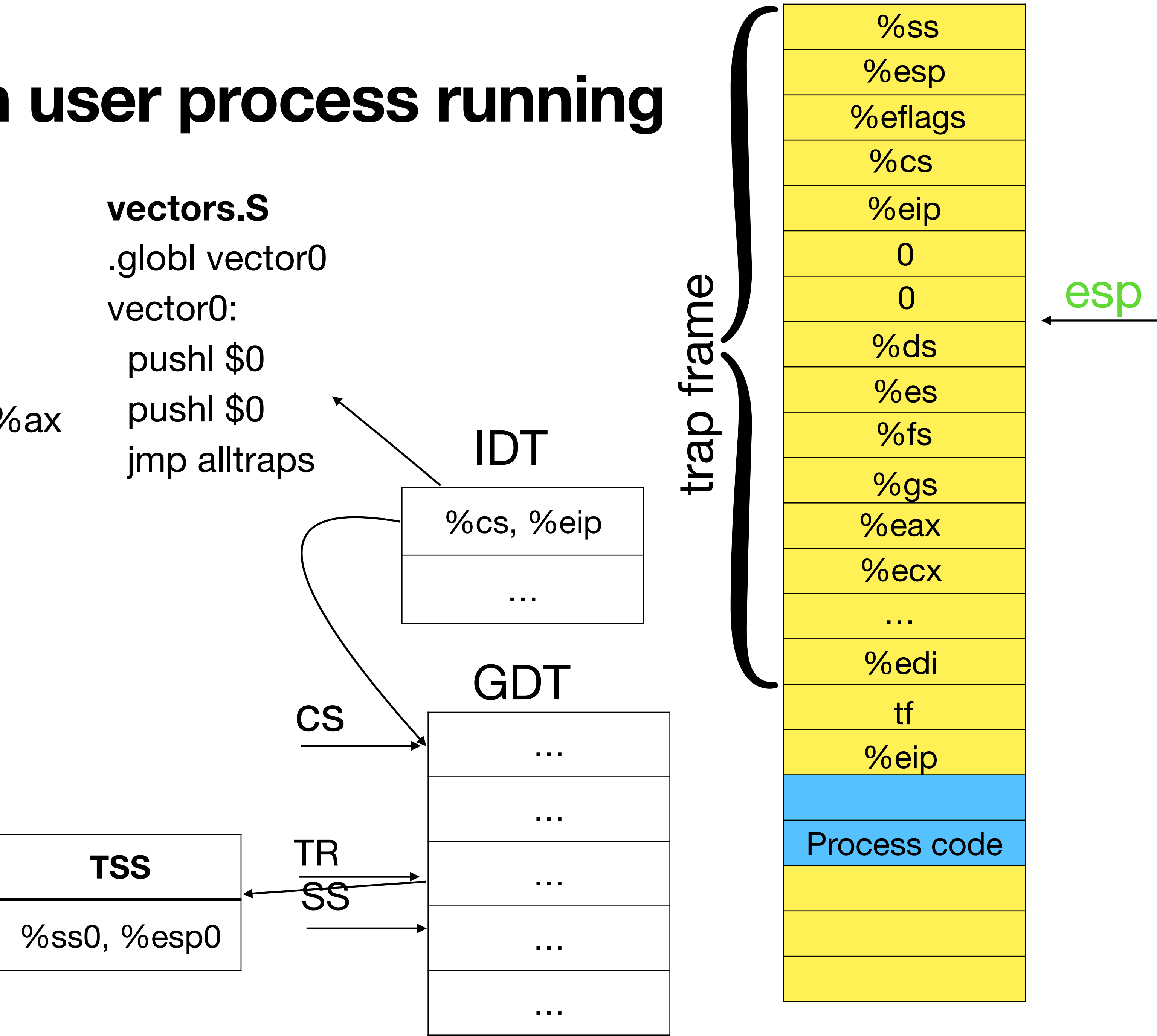
```
vectors.S
.globl vector0
vector0:
    pushl $0
    pushl $0
    jmp alltraps
```



Interrupt handling with user process running

```
trapasm.S
for(;;)
;
alltraps:
    pushl %ds..
    pushal
    movw $(SEG_KDATA<<3), %ax
    movw %ax, %ds..
    pushl %esp
    call trap
    addl $4, %esp
    popal
    popl %ds..
    addl $0x8, %esp
    iret
```

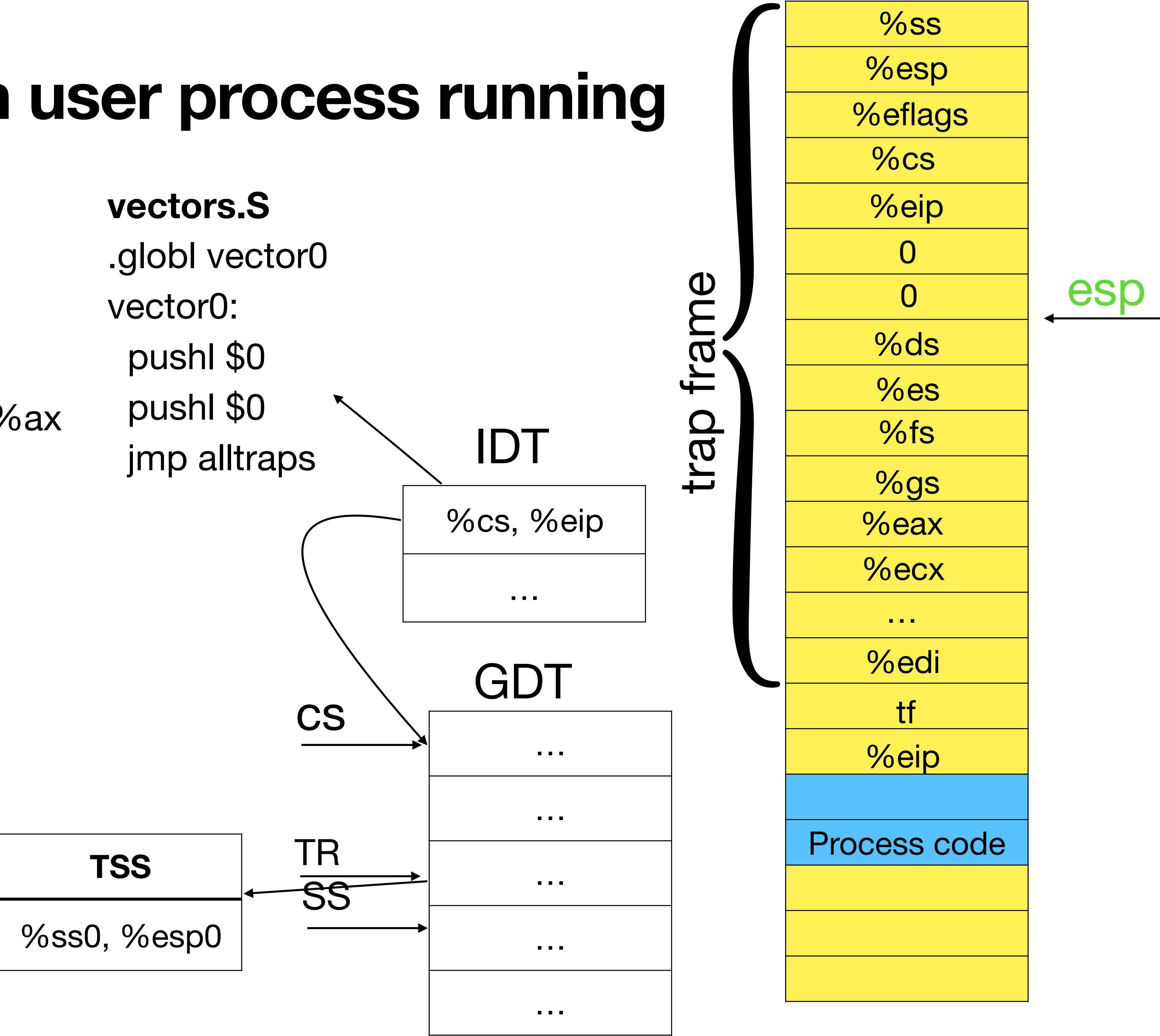
```
vectors.S
.globl vector0
vector0:
    pushl $0
    pushl $0
    jmp alltraps
```



Interrupt handling with user process running

```
trapasm.S
for(;;)
;
alltraps:
    pushl %ds..
    pushal
    movw $(SEG_KDATA<<3), %ax
    movw %ax, %ds..
    pushl %esp
    call trap
    addl $4, %esp
    popal
    popl %ds..
    addl $0x8, %esp
    iret
```

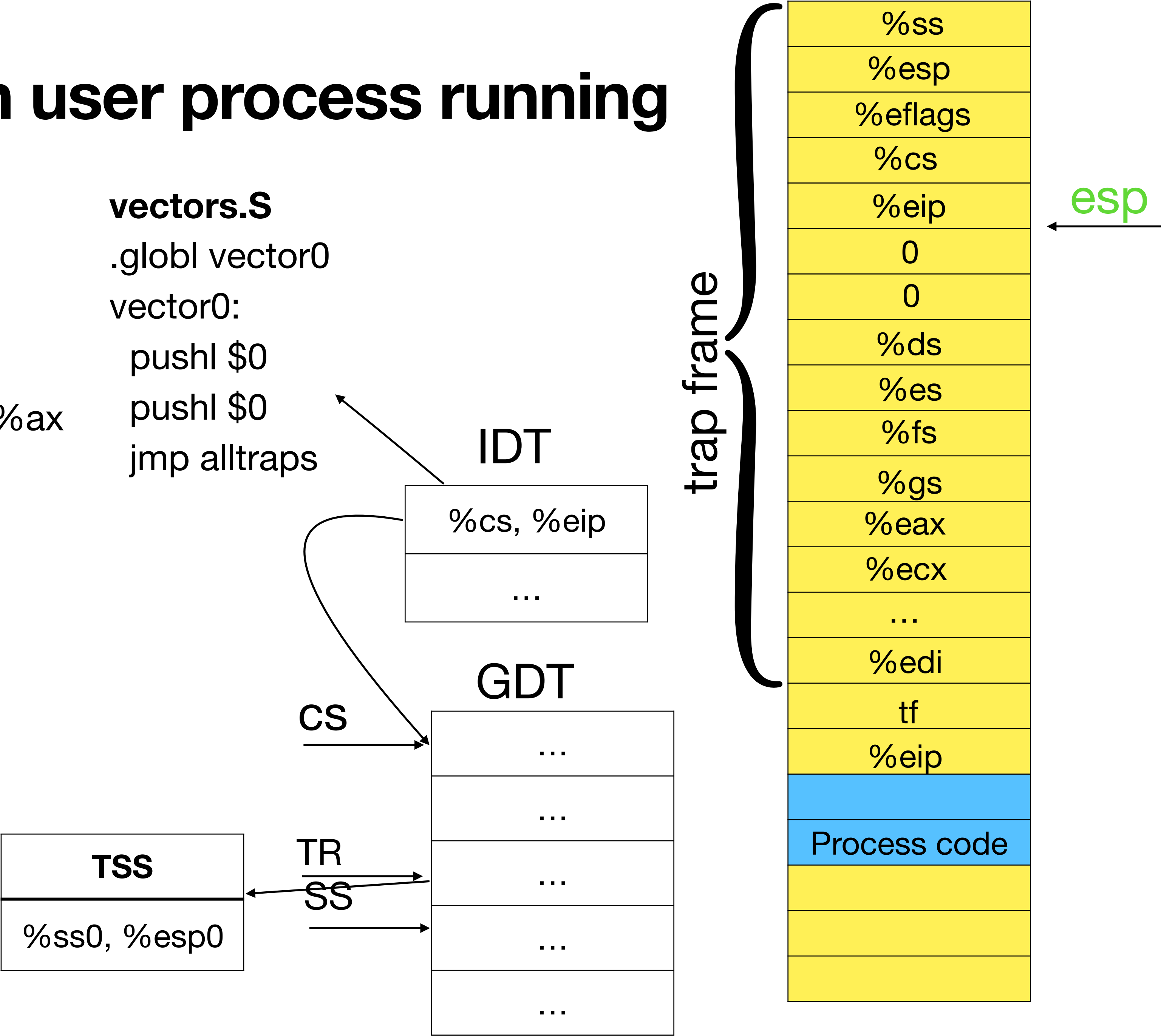
```
vectors.S
.globl vector0
vector0:
    pushl $0
    pushl $0
    jmp alltraps
```



Interrupt handling with user process running

```
trapasm.S
for(;;)
;
alltraps:
    pushl %ds..
    pushal
    movw $(SEG_KDATA<<3), %ax
    movw %ax, %ds..
    pushl %esp
    call trap
    addl $4, %esp
    popal
    popl %ds..
    addl $0x8, %esp
    iret
```

```
vectors.S
.globl vector0
vector0:
    pushl $0
    pushl $0
    jmp alltraps
```

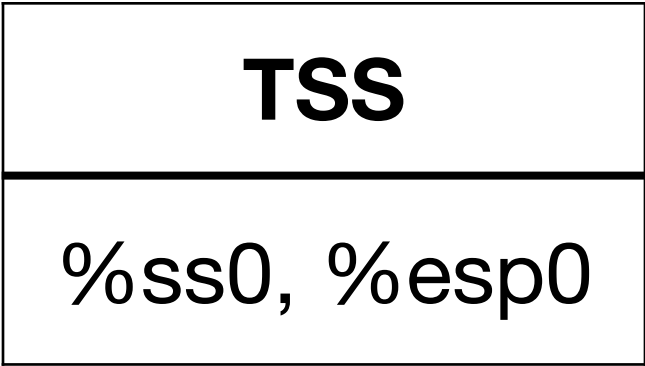


Interrupt handling with user process running

```
trapasm.S
for(;;)
;
alltraps:
    pushl %ds..
    pushal
    movw $(SEG_KDATA<<3), %ax
    movw %ax, %ds..
    pushl %esp
    call trap
    addl $4, %esp
    popal
    popl %ds..
    addl $0x8, %esp
    iret
```

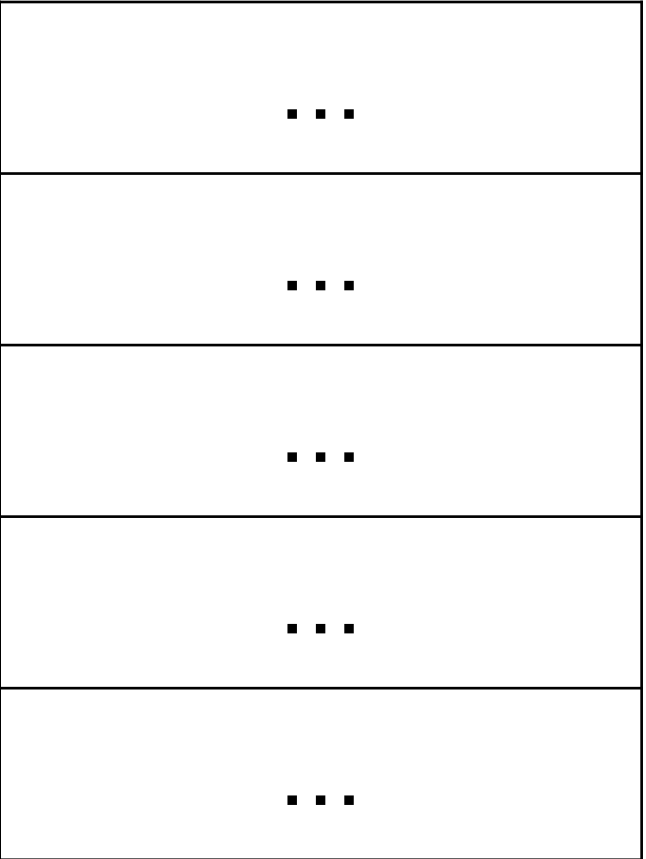
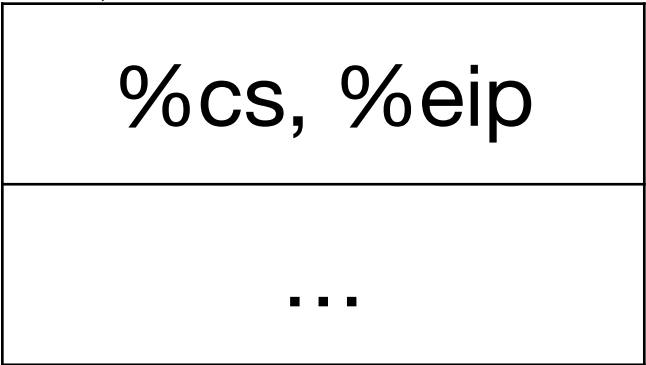
eip →

```
vectors.S
.globl vector0
vector0:
    pushl $0
    pushl $0
    jmp alltraps
```

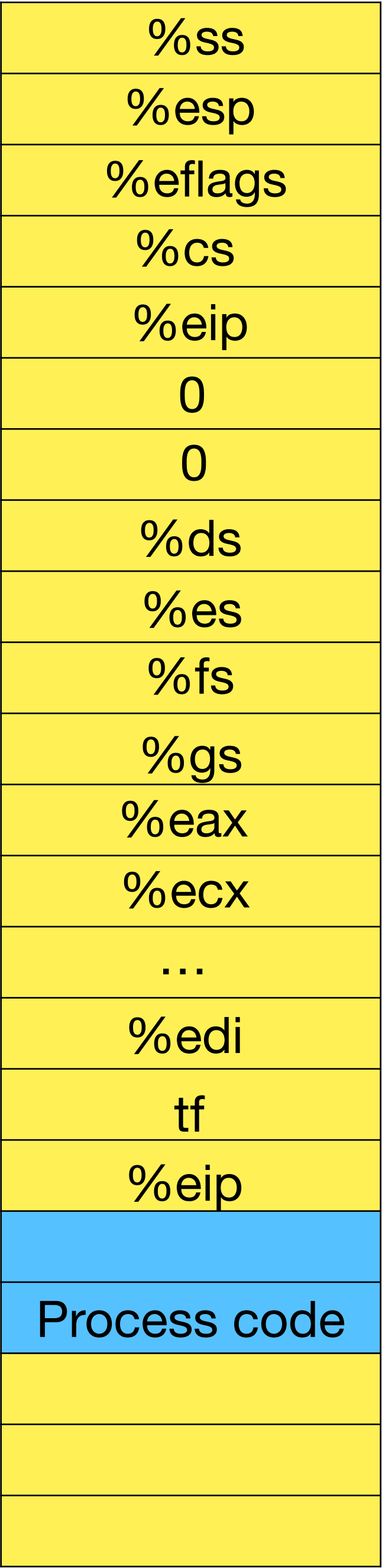


CS →

TR
SS →



trap frame

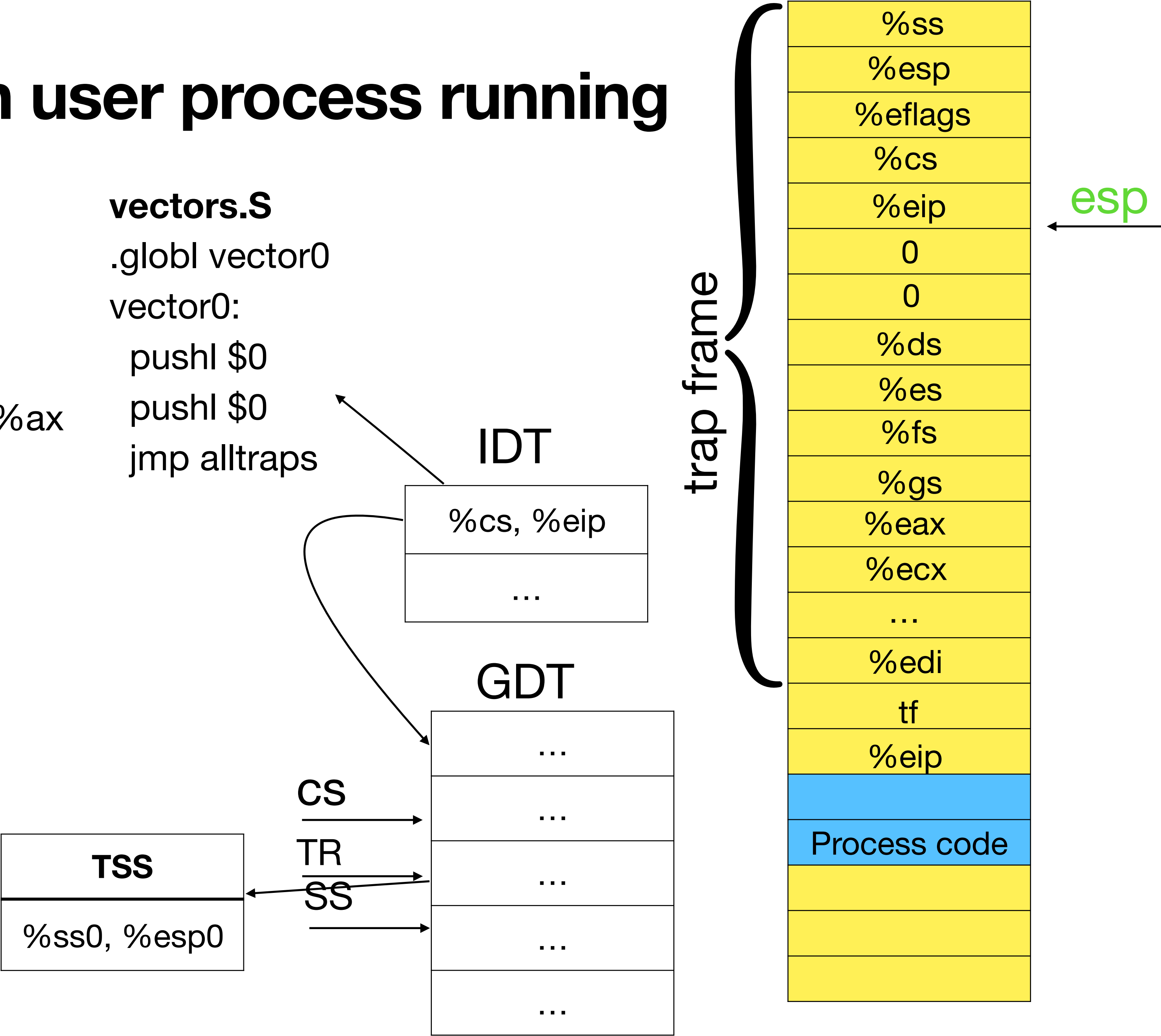


esp ←

Interrupt handling with user process running

```
trapasm.S
for(;;)
;
alltraps:
    pushl %ds..
    pushal
    movw $(SEG_KDATA<<3), %ax
    movw %ax, %ds..
    pushl %esp
    call trap
    addl $4, %esp
    popal
    popl %ds..
    addl $0x8, %esp
    eip → iret
```

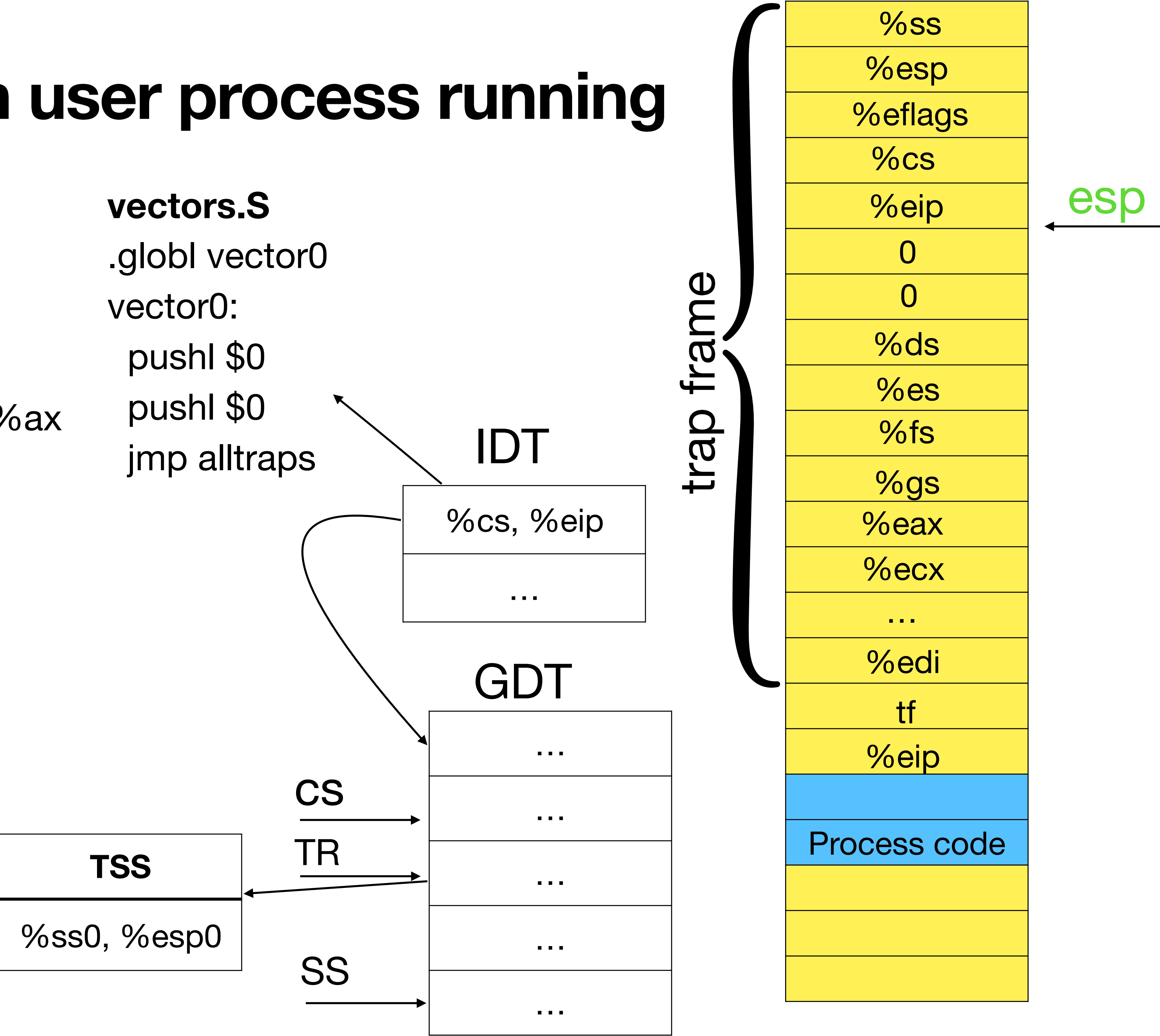
```
vectors.S
.globl vector0
vector0:
    pushl $0
    pushl $0
    jmp alltraps
```



Interrupt handling with user process running

```
trapasm.S
for(;;)
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    pushl %ds..
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    movw $(SEG_KDATA<<3), %ax
    movw %ax, %ds..
    pushl %esp
    call trap
    addl $4, %esp
    popal
    popl %ds..
    addl $0x8, %esp
    eip → iret
```

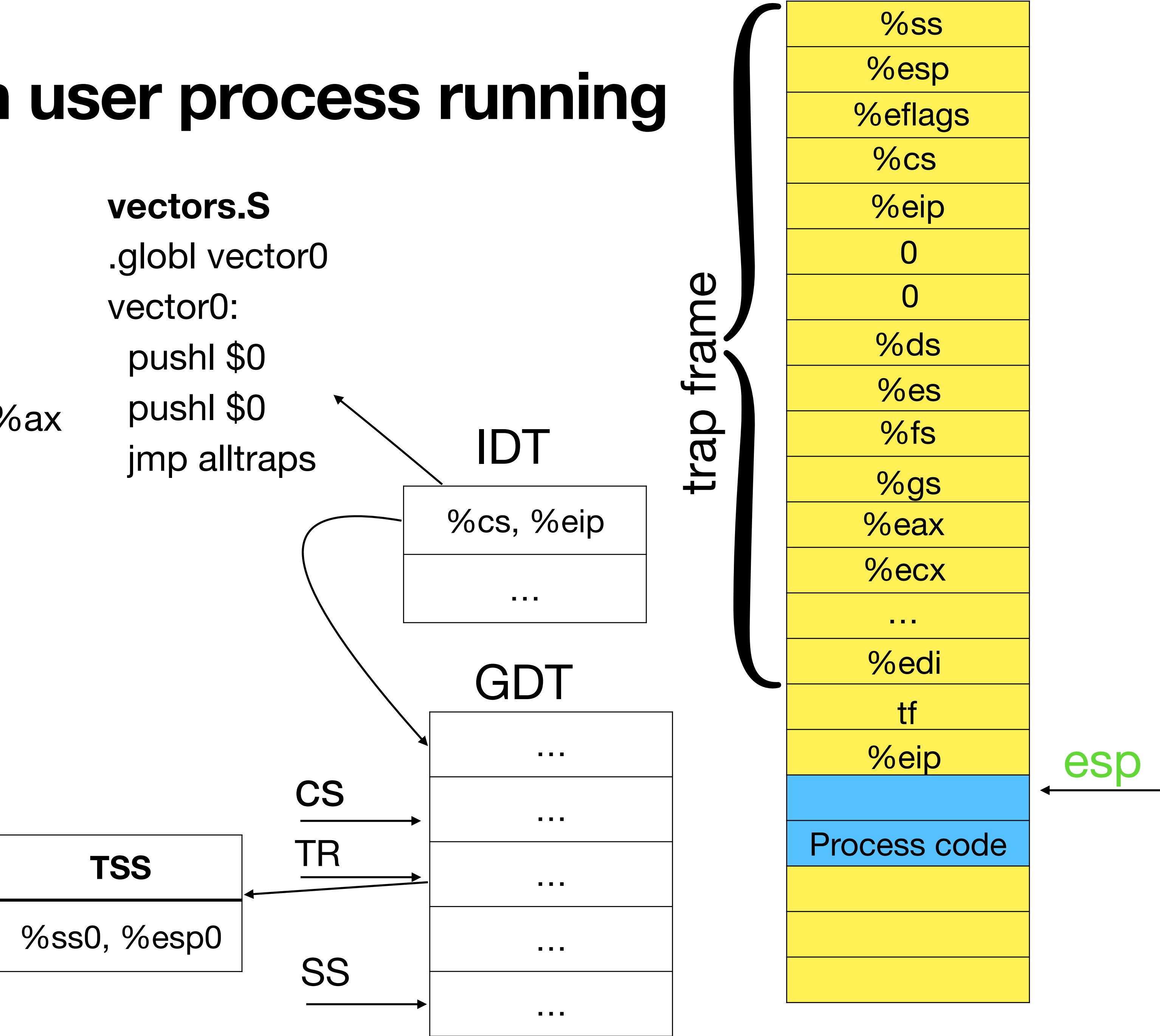
```
vectors.S
.globl vector0
vector0:
    pushl $0
    pushl $0
    jmp alltraps
```



Interrupt handling with user process running

```
trapasm.S
for(;;)
;
alltraps:
    pushl %ds..
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    movw $(SEG_KDATA<<3), %ax
    movw %ax, %ds..
    pushl %esp
    call trap
    addl $4, %esp
    popal
    popl %ds..
    addl $0x8, %esp
    eip → iret
```

```
vectors.S
.globl vector0
vector0:
    pushl $0
    pushl $0
    jmp alltraps
```



Interrupt handling with user process running

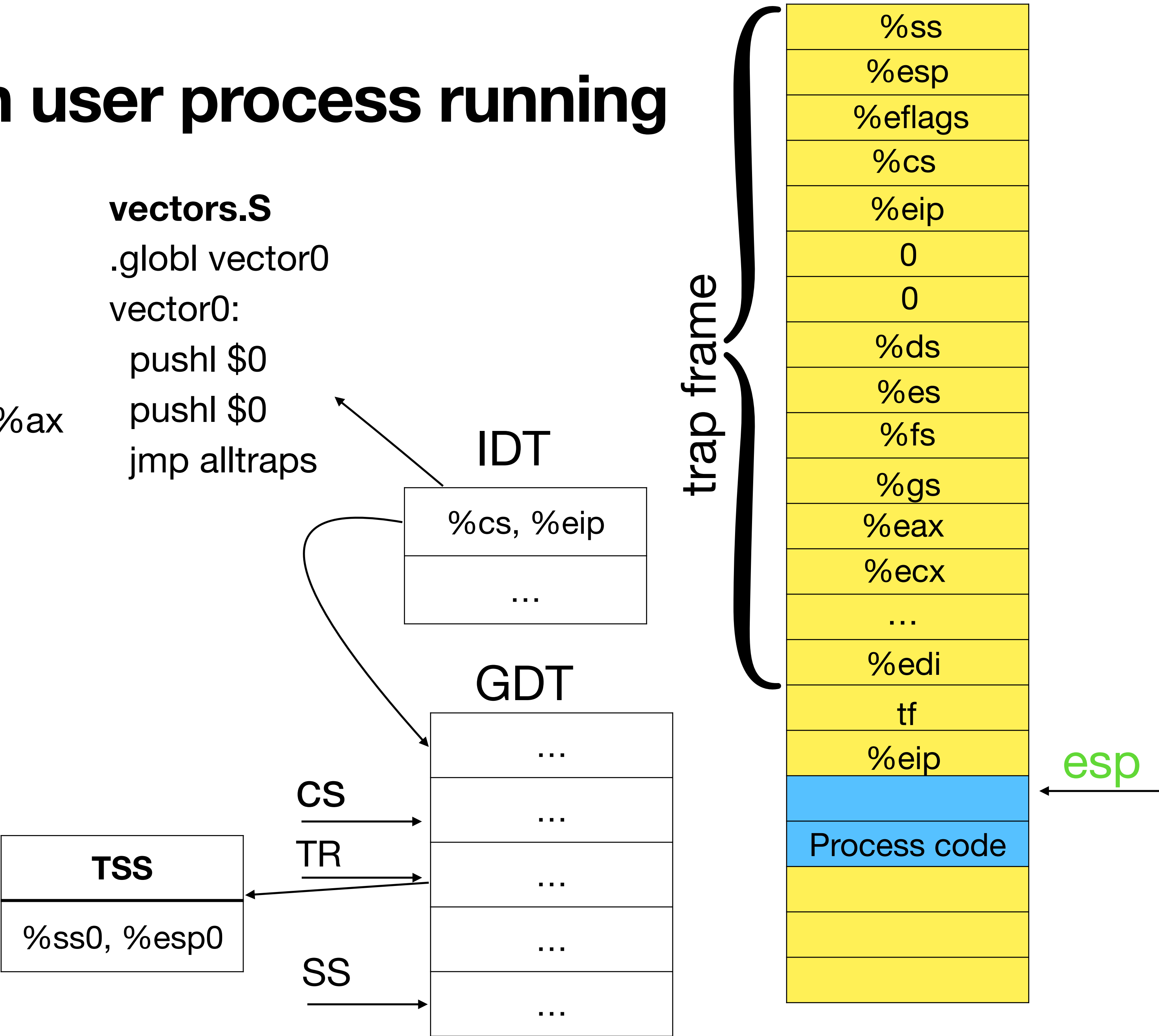
eip → for(;;)
;

trapasm.S

```
alltraps:
    pushl %ds..
    pushal
    movw $(SEG_KDATA<<3), %ax
    movw %ax, %ds..
    pushl %esp
    call trap
    addl $4, %esp
    popal
    popl %ds..
    addl $0x8, %esp
    iret
```

vectors.S

```
.globl vector0
vector0:
    pushl $0
    pushl $0
    jmp alltraps
```



Protection so far

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- Can processes read/write each other's memory?
 - No! The other process' memory is not *addressable* by (not in the *address space* of) the process

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- Can process take over IDT, GDT, TSS?
 - No! IDT, GDT, TSS are not in process' address space.
 - Process cannot run LIDT, LGDT, LTR instructions since they are *privileged instructions* (only runnable from ring 0)

Protection so far

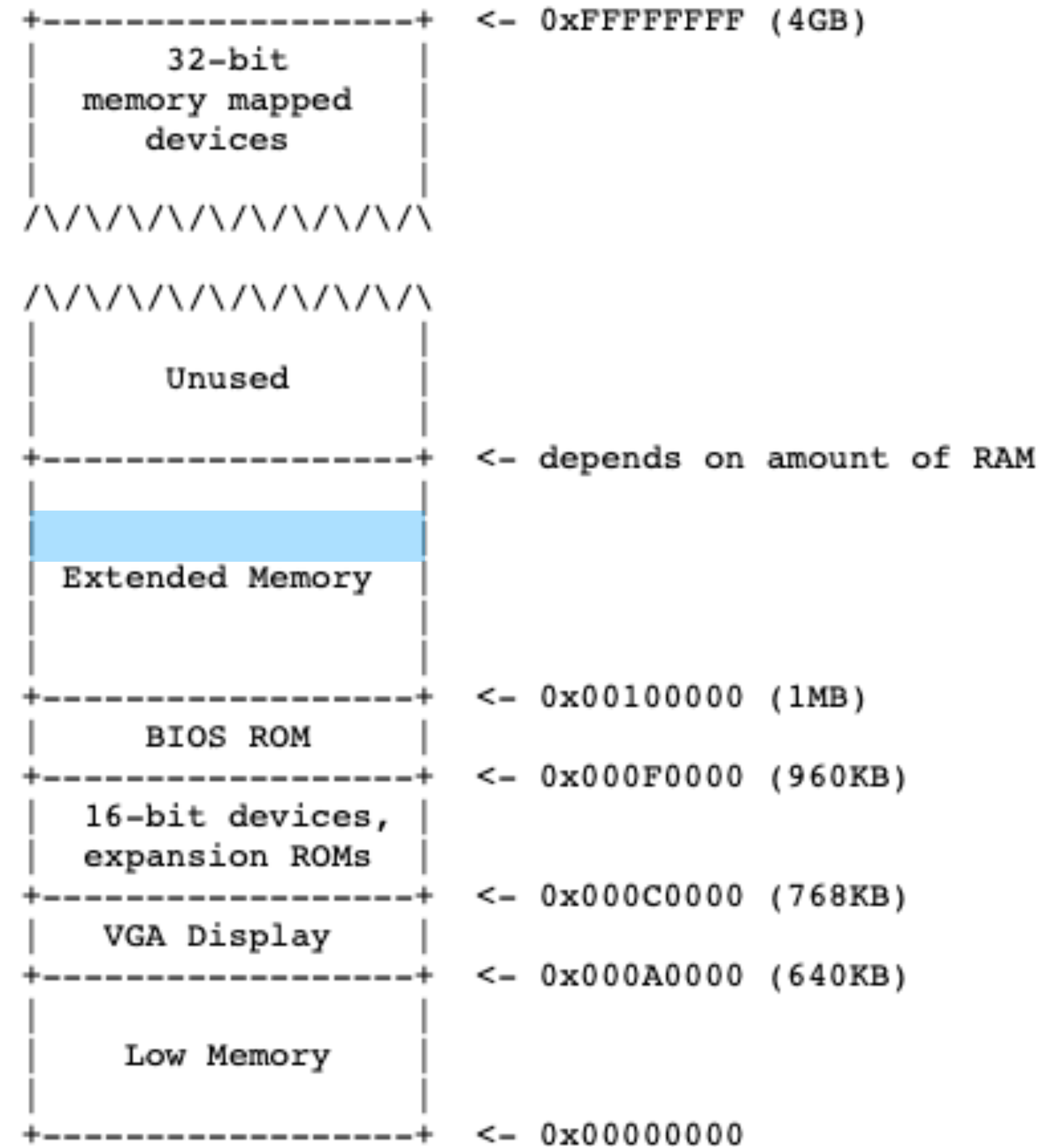
- Can processes read/write each other's memory?
 - No! The other process' memory is not *addressable* by (not in the *address space* of) the process
- Can process take over IDT, GDT, TSS?
 - No! IDT, GDT, TSS are not in process' address space.
 - Process cannot run LIDT, LGDT, LTR instructions since they are *privileged instructions* (only runnable from ring 0)
- Can process change its privilege to ring 0?
 - Not directly. It can change only via trap handling mechanism. OS code runs upon a trap. Trap handling writes to kernel stack.

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 - No! The other process' memory is not *addressable* by (not in the *address space* of) the process
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 - No! IDT, GDT, TSS are not in process' address space.
 - Process cannot run LIDT, LGDT, LTR instructions since they are *privileged instructions* (only runnable from ring 0)
- Can process change its privilege to ring 0?
 - Not directly. It can change only via trap handling mechanism. OS code runs upon a trap. Trap handling writes to kernel stack.
- Can process run away with the CPU?
 - CPU will be snatched after timer interrupt. Process cannot disable interrupts

IO protection

- Memory mapped IO
 - Can directly read from / write to IO devices if OS keeps it in *process address space*



Port-mapped IO protection

`inb(0x1F7), outb(0x1F2, 1), ..`

Port-mapped IO protection

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- Option 1: Make in and out instruction privileged. Processes must do IO via kernel.

Port-mapped IO protection

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Port-mapped IO protection

`inb(0x1F7), outb(0x1F2, 1), ..`

- Option 1: Make in and out instruction privileged. Processes must do IO via kernel.
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- Option 2: Set `EFLAGS.IOPL = 3`

Port-mapped IO protection

`inb(0x1F7), outb(0x1F2, 1), ..`

- Option 1: Make in and out instruction privileged. Processes must do IO via kernel.
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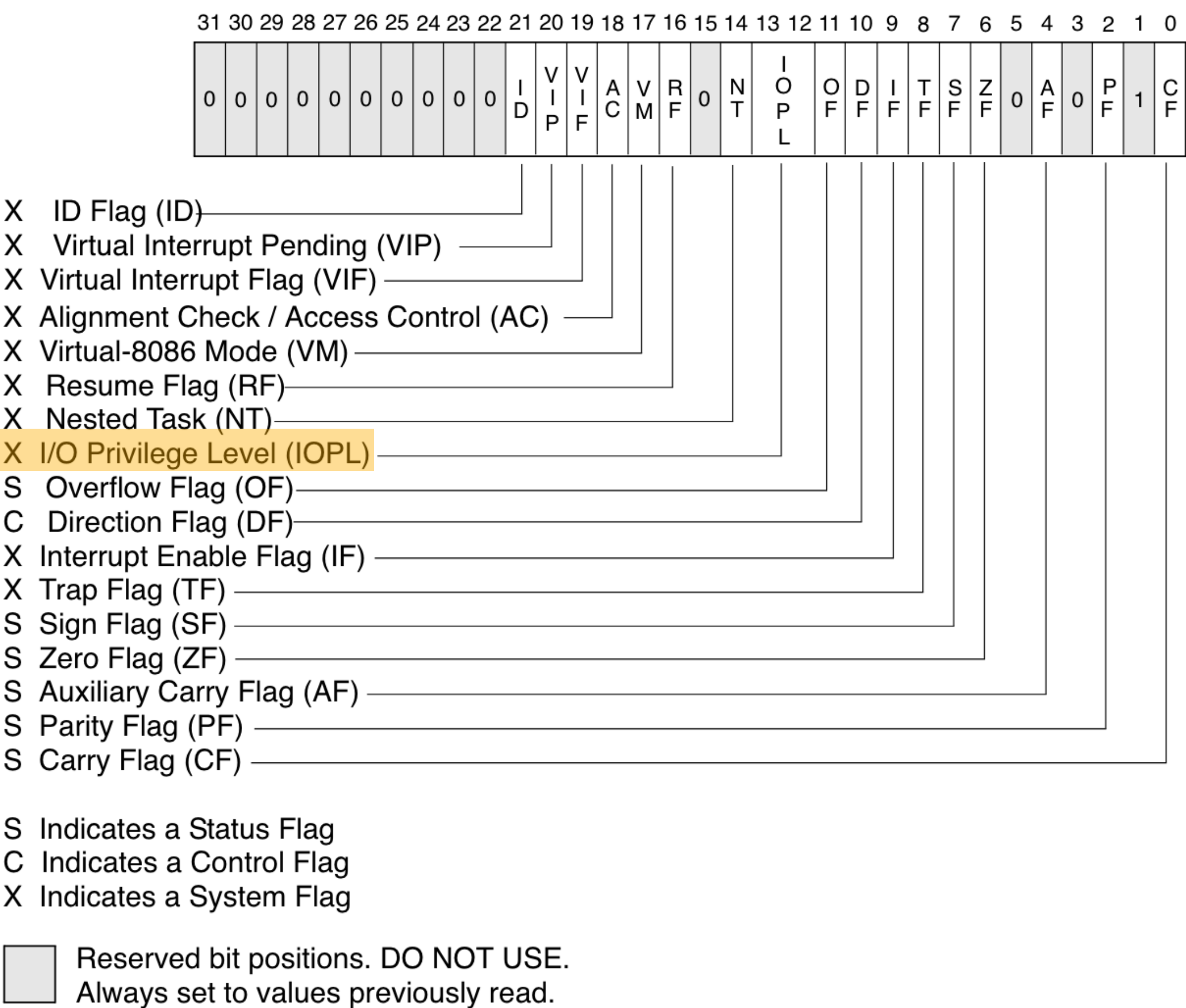


Figure 3-8. EFLAGS Register

Port-mapped IO protection

```
inb(0x1F7), outb(0x1F2, 1), ..
```

- Option 1: Make in and out instruction privileged. Processes must do IO via kernel.
- Expensive user mode-kernel mode transitions for IO intensive processes.
- Option 2: Set EFLAGS.IOPL = 3
 - Processes (CPL=3) cannot modify EFLAGS.IOPL

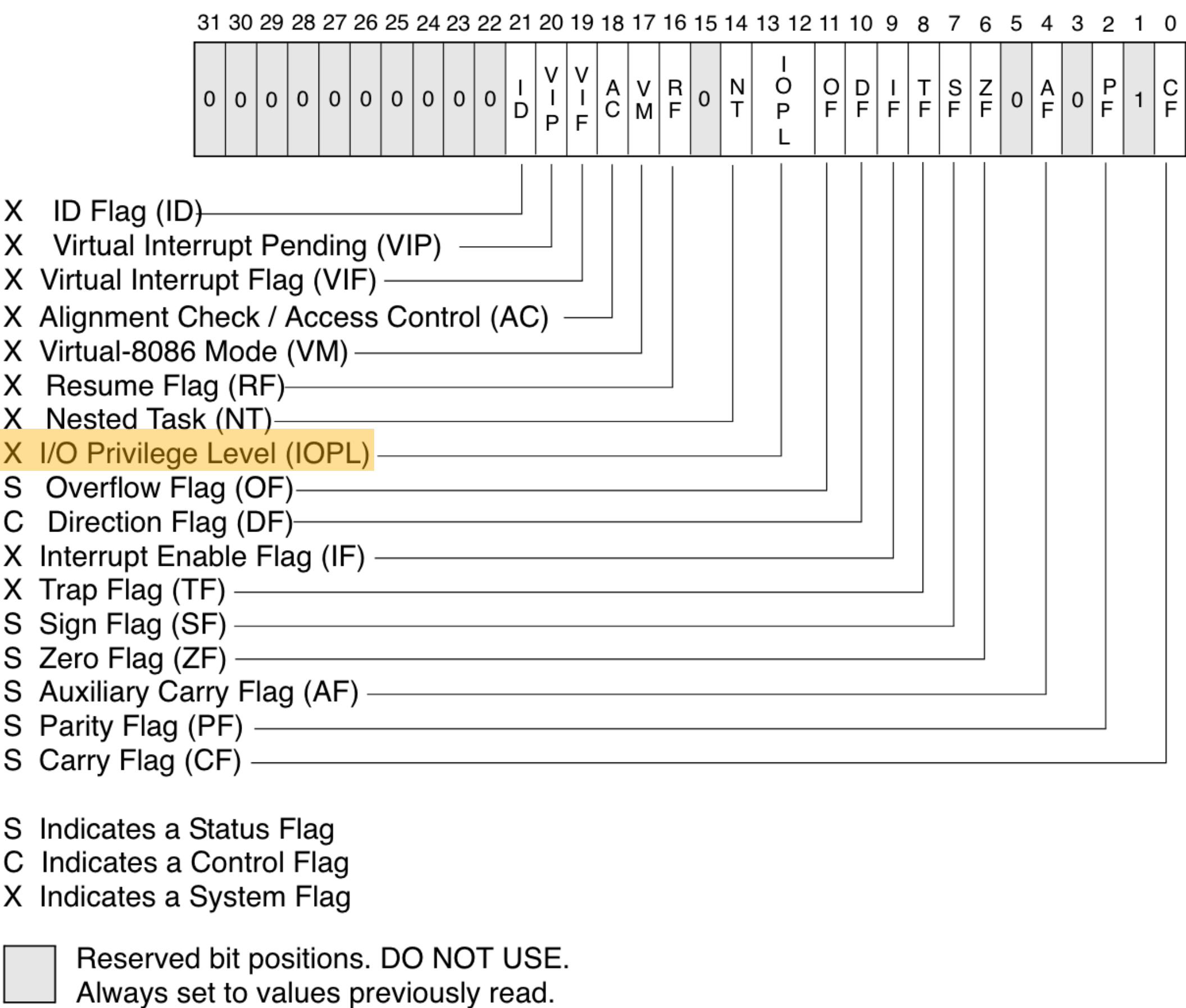


Figure 3-8. EFLAGS Register

Port-mapped IO protection

```
inb(0x1F7), outb(0x1F2, 1), ..
```

- Option 1: Make in and out instruction privileged. Processes must do IO via kernel.
- Expensive user mode-kernel mode transitions for IO intensive processes.
- Option 2: Set EFLAGS.IOPL = 3
 - Processes (CPL=3) cannot modify EFLAGS.IOPL
 - Gives permission to access all IO ports

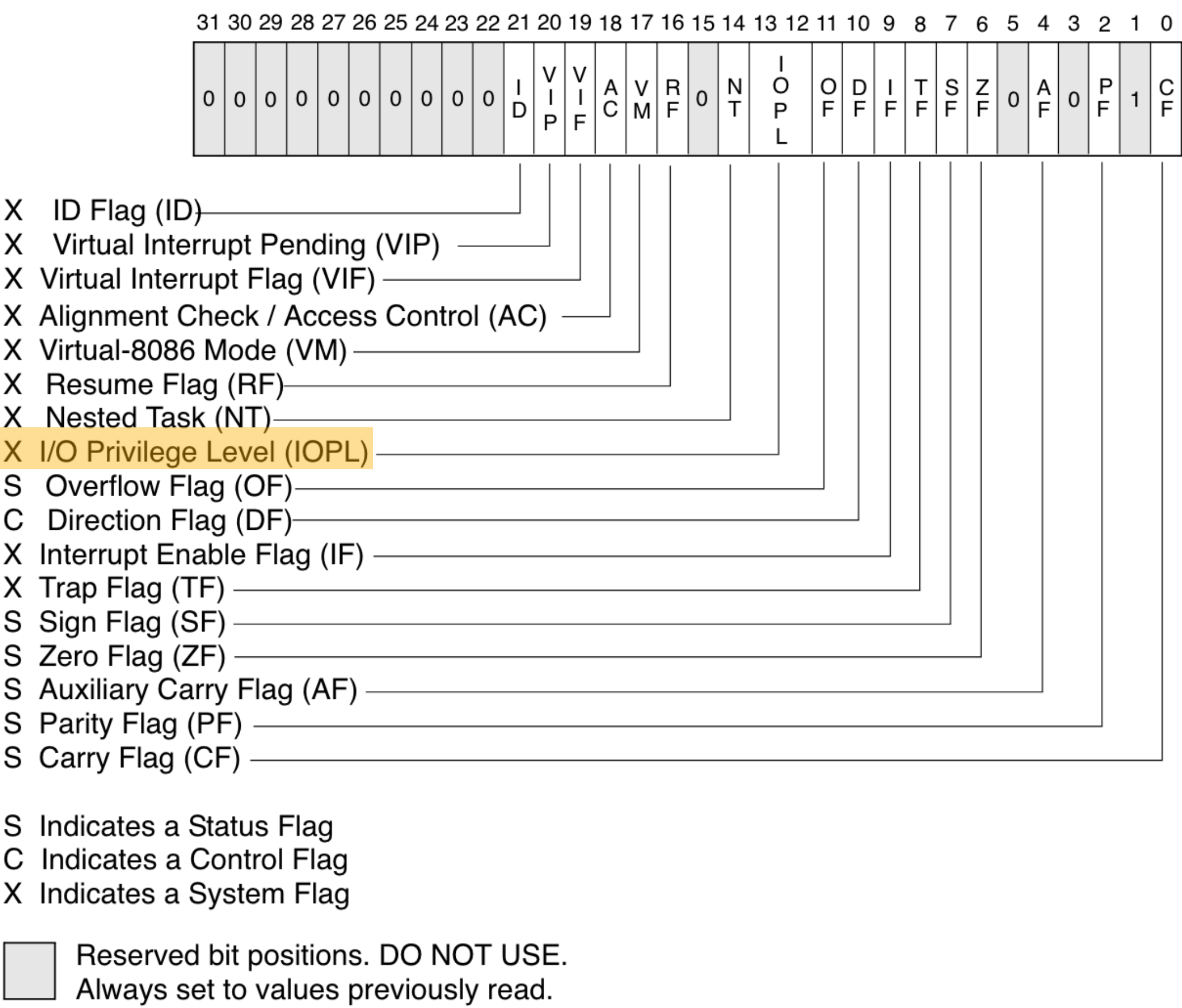


Figure 3-8. EFLAGS Register

Port-mapped IO protection (2)

Port-mapped IO protection (2)

- Option 3: If $CPL > EFLAGS.IOPL$, use a bitmap to decide if in, out instruction should generate a trap

Port-mapped IO protection (2)

- Option 3: If $CPL > EFLAGS.IOPL$, use a bitmap to decide if in, out instruction should generate a trap

31	15	0	
I/O Map Base Address		Reserved	T 100
Reserved		LDT Segment Selector 96	
Reserved		GS 92	
Reserved		FS 88	
Reserved		DS 84	
Reserved		SS 80	
Reserved		CS 76	
Reserved		ES 72	
EDI			68
ESI			64
EBP			60
ESP			56
EBX			52
EDX			48
ECX			44
EAX			40
EFLAGS			36
EIP			32
CR3 (PDBR)			28
Reserved		SS2	24
ESP2			20
Reserved		SS1	16
ESP1			12
Reserved		SS0	8
ESP0			4
Reserved		Previous Task Link 0	

Reserved bits. Set to 0.

Figure 7-2. 32-Bit Task-State Segment (TSS)

Port-mapped IO protection (2)

- Option 3: If $CPL > EFLAGS.IOPL$, use a bitmap to decide if in, out instruction should generate a trap

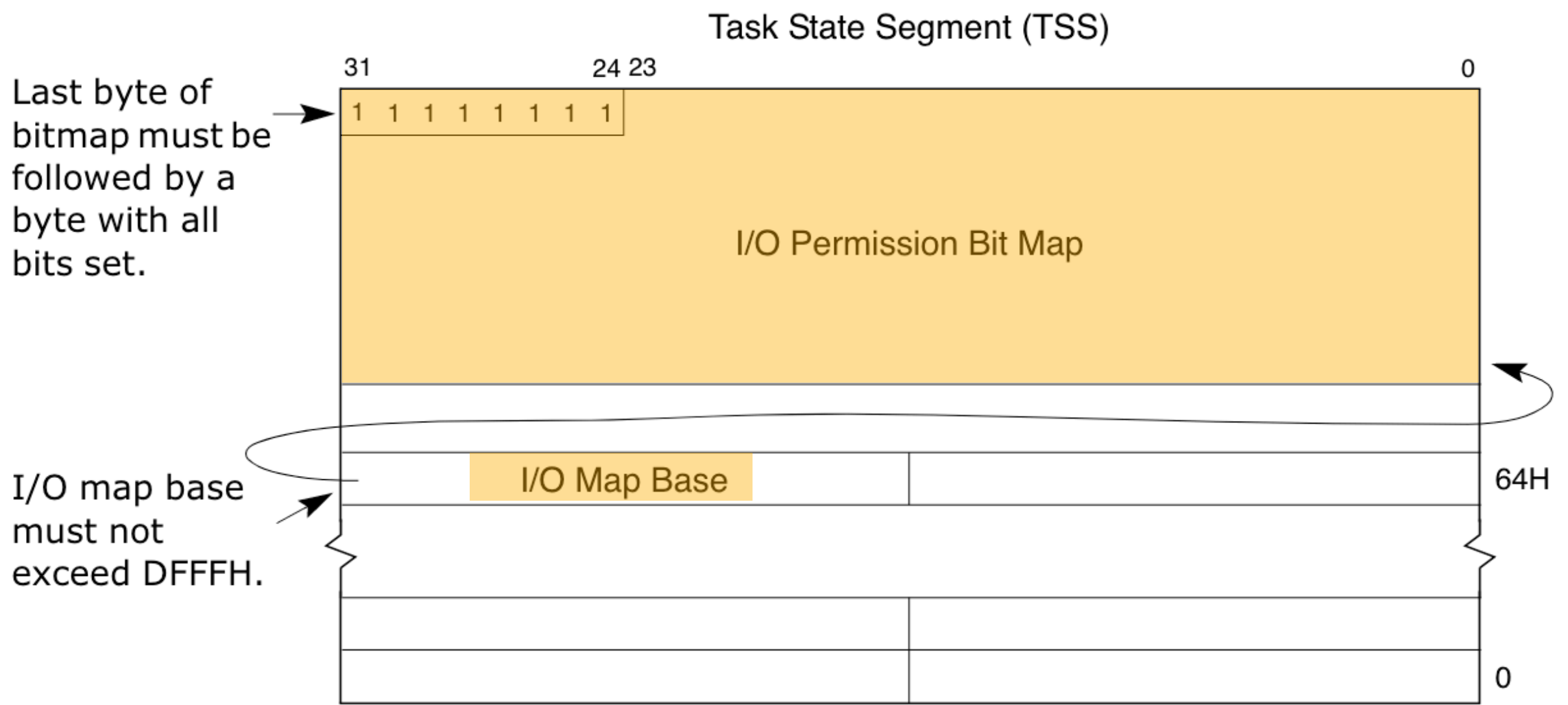


Figure 18-2. I/O Permission Bit Map

31	15	0	
I/O Map Base Address		Reserved	T 100
Reserved		LDT Segment Selector	96
Reserved		GS	92
Reserved		FS	88
Reserved		DS	84
Reserved		SS	80
Reserved		CS	76
Reserved		ES	72
EDI			68
ESI			64
EBP			60
ESP			56
EBX			52
EDX			48
ECX			44
EAX			40
EFLAGS			36
EIP			32
CR3 (PDBR)			28
Reserved		SS2	24
ESP2			20
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ESP1			12
Reserved		SS0	8
ESP0			4
Reserved		Previous Task Link	0

Reserved bits. Set to 0.

Figure 7-2. 32-Bit Task-State Segment (TSS)

Port-mapped IO protection (2)

- Option 3: If $CPL > EFLAGS.IOPL$, use a bitmap to decide if in, out instruction should generate a trap
- Example: `outb(41, 1)` is allowed if 41st bit is set to 0

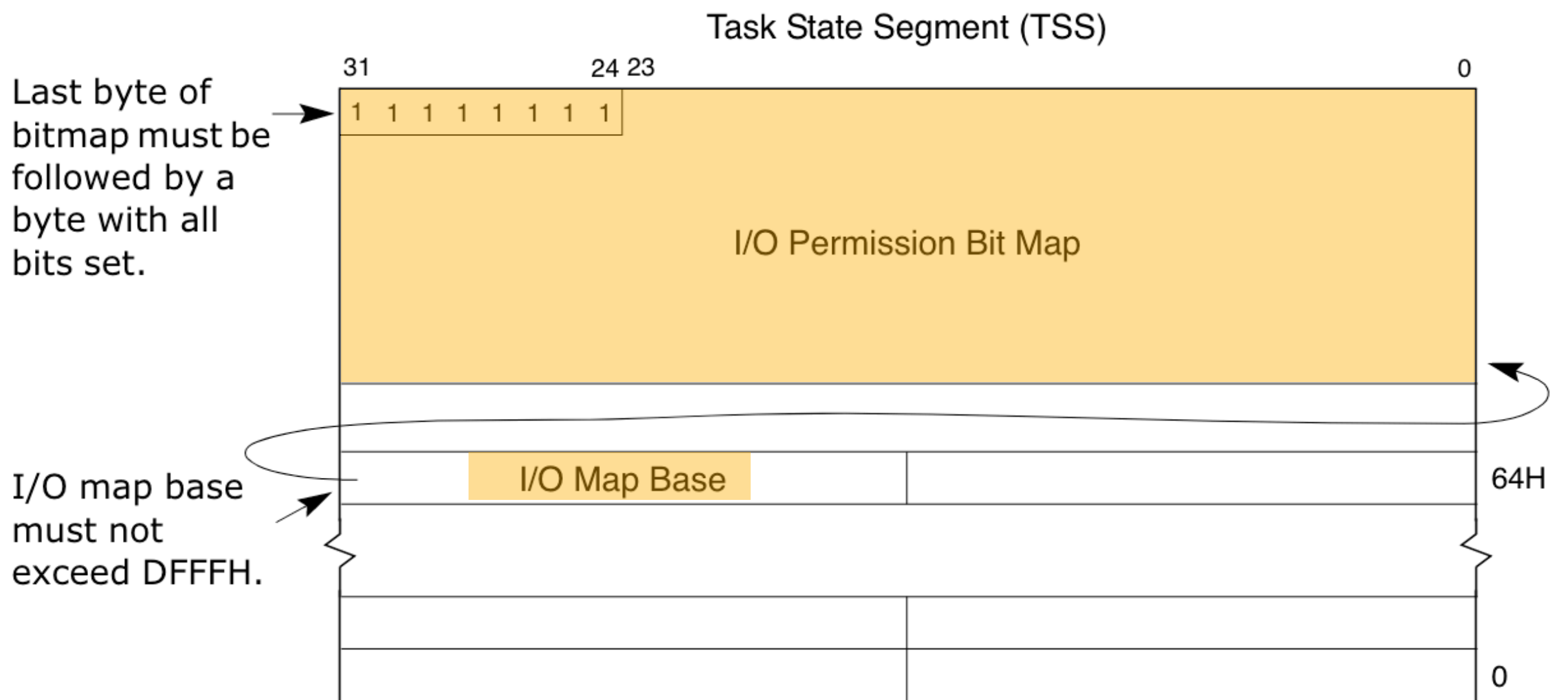


Figure 18-2. I/O Permission Bit Map

31	15	0	
I/O Map Base Address		Reserved	T 100
Reserved		LDT Segment Selector	96
Reserved		GS	92
Reserved		FS	88
Reserved		DS	84
Reserved		SS	80
Reserved		CS	76
Reserved		ES	72
EDI			68
ESI			64
EBP			60
ESP			56
EBX			52
EDX			48
ECX			44
EAX			40
EFLAGS			36
EIP			32
CR3 (PDBR)			28
Reserved		SS2	24
ESP2			20
Reserved		SS1	16
ESP1			12
Reserved		SS0	8
ESP0			4
Reserved		Previous Task Link	0

Reserved bits. Set to 0.

Figure 7-2. 32-Bit Task-State Segment (TSS)

IO protection in xv6

- Processes cannot directly do IO
 - MMIO: Do not map IO addresses in process address space
 - PIO: Running in and out instructions will generate a trap (jump to OS)

IO protection in xv6

```
void pinit(void) {
```

```
    ..
```

```
    p->tf->eflags = FL_IF;
```

```
}
```

```
void switchvm(struct proc *p) {
```

```
    mycpu()->gdt[SEG_TSS] = SEG16(STS_T32A, &mycpu()->ts,  
                                  sizeof(mycpu()->ts)-1, 0);
```

```
    mycpu()->ts.ss0 = SEG_KDATA << 3;
```

```
    mycpu()->ts.esp0 = (uint)p->kstack + KSTACKSIZE;
```

```
    // setting IOPL=0 in eflags *and* iomb beyond the tss segment limit
```

```
    // forbids I/O instructions (e.g., inb and outb) from user space
```

```
    mycpu()->ts.iomb = (ushort) 0xFFFF;
```

```
    ltr(SEG_TSS << 3);
```

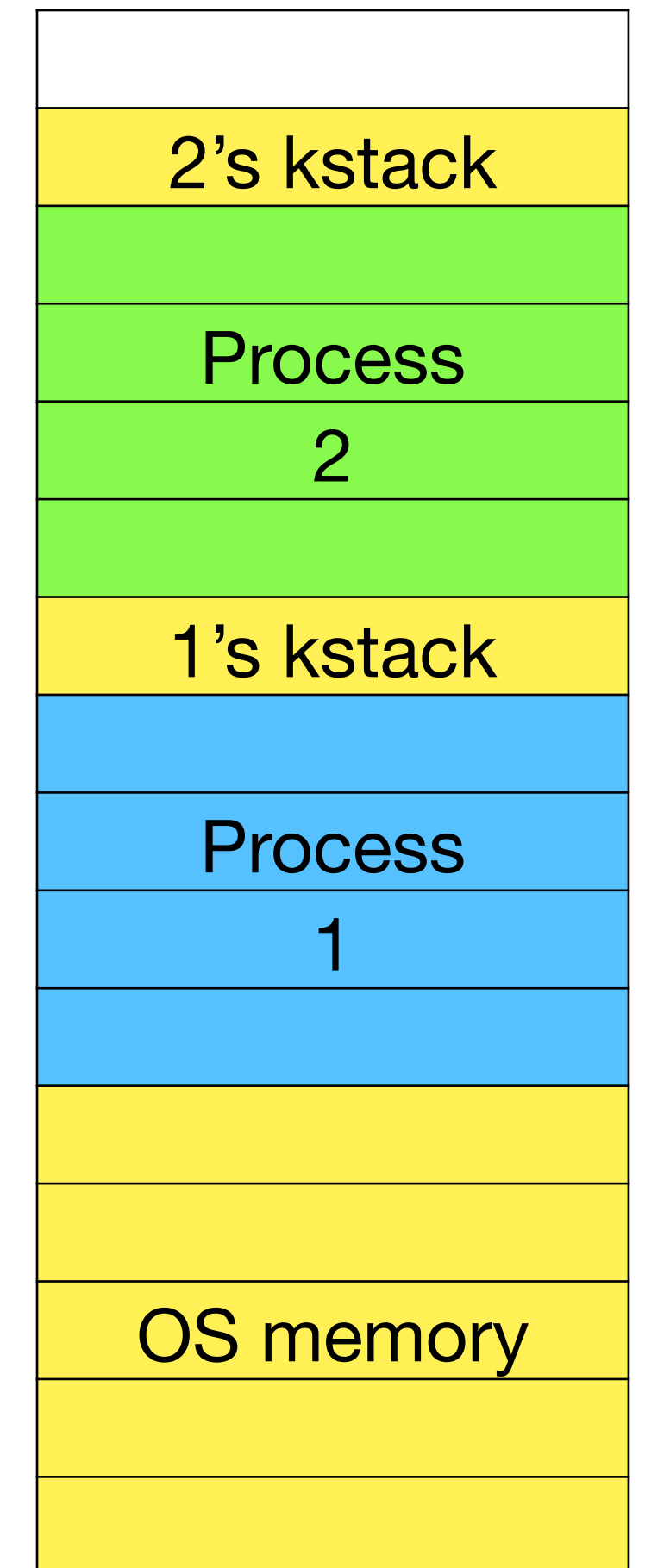
```
}
```

- Processes **cannot** directly do IO
- MMIO: Do not map IO addresses in process address space
- PIO: Running in and out instructions will generate a trap (jump to OS)

Code walkthrough: Setting up multiple processes

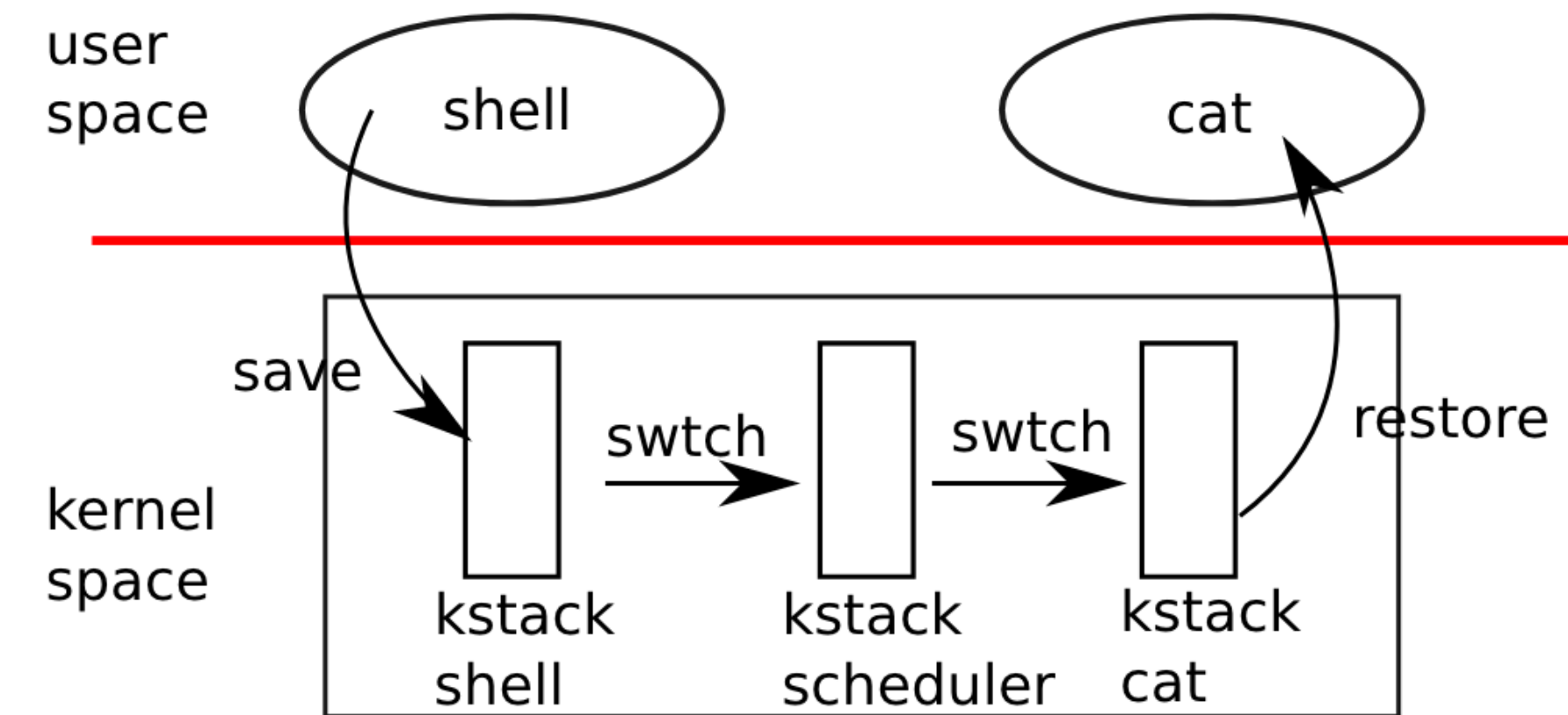
p17-free-space

- main.c
 - Calls pinit twice to start two processes and then calls scheduler
- proc.c
 - pinit calls allocproc which sets up processes like earlier with the added difference that it calls kalloc to find empty 1MB of space
- kalloc.c
 - Maintains free list in chunks of 1MB. kalloc returns the first element from free list. There is no coalescing and splitting since our OS always asks for 1MB.
- No support for context switching. Ctrl+P shows process 1 running forever



Code walkthrough: context switching

p18-sched



- main.c
 - Calls scheduler to run the first RUNNABLE process. When giving control, we *also remember “scheduler context” to come back to scheduler*. Earlier, we were only running one process and *were never coming back to the scheduler*.
- trap.c
 - Calls *yield* when timer interrupt happens
- proc.c
 - yield calls sched which switches control from the process to the scheduler
 - scheduler looks for the next RUNNABLE process and switches to it
- Ctrl+P now shows that processes are switching

Context switching in action: giving control

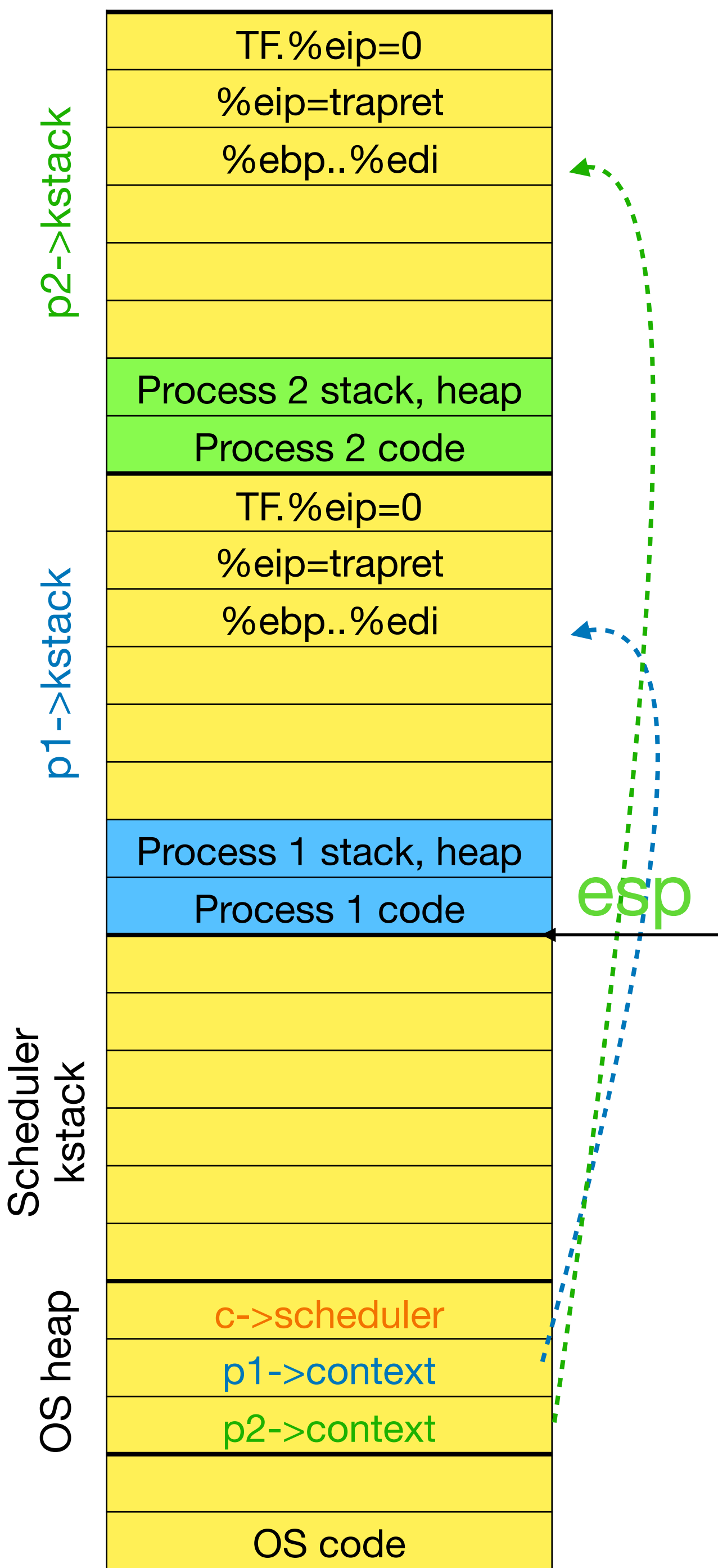
p18-sched

```
.globl swtch
swtch:
    movl 4(%esp), %eax
    movl 8(%esp), %edx
    pushl %ebp
    pushl %ebx
    pushl %esi
    pushl %edi
    movl %esp, (%eax)
    movl %edx, %esp
    popl %edi
    popl %esi
    popl %ebx
    popl %ebp
    ret

void scheduler(void) {
    struct proc *p; struct cpu *c = mycpu();
    for(;;){
        for(p = ptable.proc; p < &ptable.proc[NPROC]; p++){
            if(p->state != RUNNABLE)
                continue;
            p->state = RUNNING;
            switchvm(p);
            swtch(&(c->scheduler), p->context);
        }
    }
}

void yield(void) {
    struct proc *p = myproc();
    p->state = RUNNABLE;
    swtch(&p->context, c->scheduler);
}

void trap(struct trapframe *tf) {
    ..
    if(tf->trapno == T_IRQ0+IRQ_TIMER)
        yield();
}
```



Context switching in action: giving control

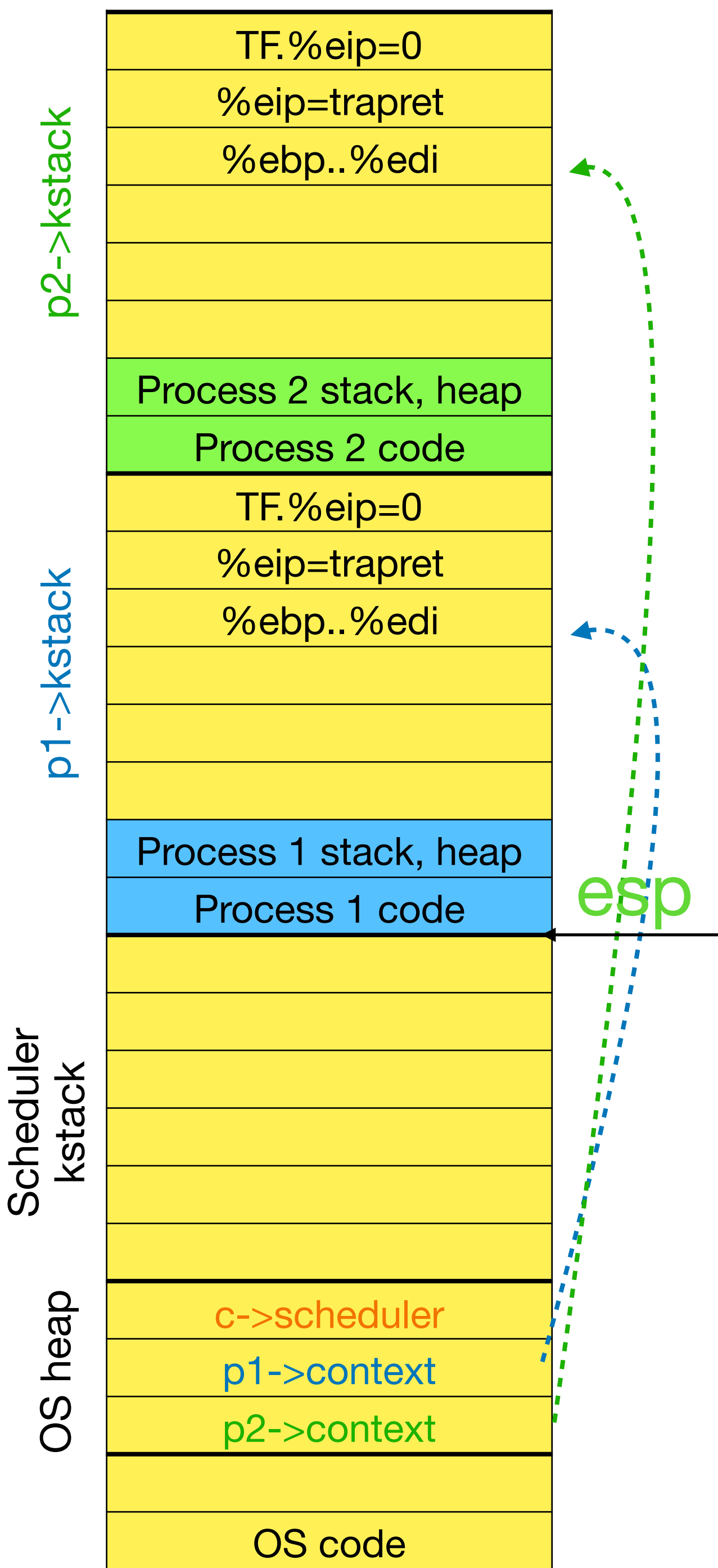
p18-sched

```
.globl swtch
swtch:
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    movl 8(%esp), %edx
    pushl %ebp
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    pushl %edi
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    movl %edx, %esp
    popl %edi
    popl %esi
    popl %ebx
    popl %ebp
    ret

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            p->state = RUNNING;
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        }
    }
}

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    swtch(&p->context, c->scheduler);
}

void trap(struct trapframe *tf) {
    ..
    if(tf->trapno == T_IRQ0+IRQ_TIMER)
        yield();
}
```



Context switching in action: giving control

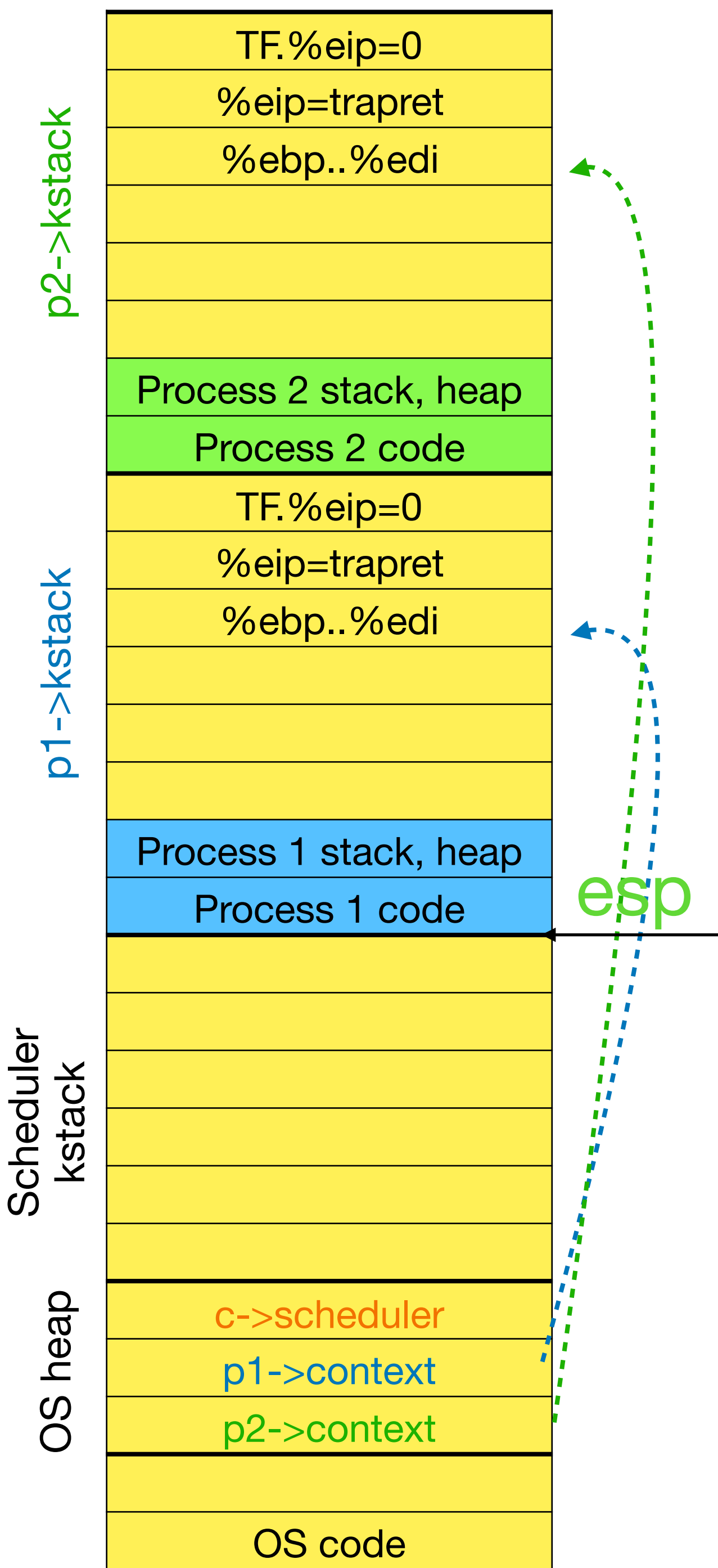
p18-sched

```
.globl swtch
swtch:
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    movl 8(%esp), %edx
    pushl %ebp
    pushl %ebx
    pushl %esi
    pushl %edi
    movl %esp, (%eax)
    movl %edx, %esp
    popl %edi
    popl %esi
    popl %ebx
    popl %ebp
    ret

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                continue;
            p->state = RUNNING;
            switchvm(p);
            swtch(&(c->scheduler), p->context);
        }
    }
}

void yield(void) {
    struct proc *p = myproc();
    p->state = RUNNABLE;
    swtch(&p->context, c->scheduler);
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void trap(struct trapframe *tf) {
    ..
    if(tf->trapno == T_IRQ0+IRQ_TIMER)
        yield();
}
```



Context switching in action: giving control

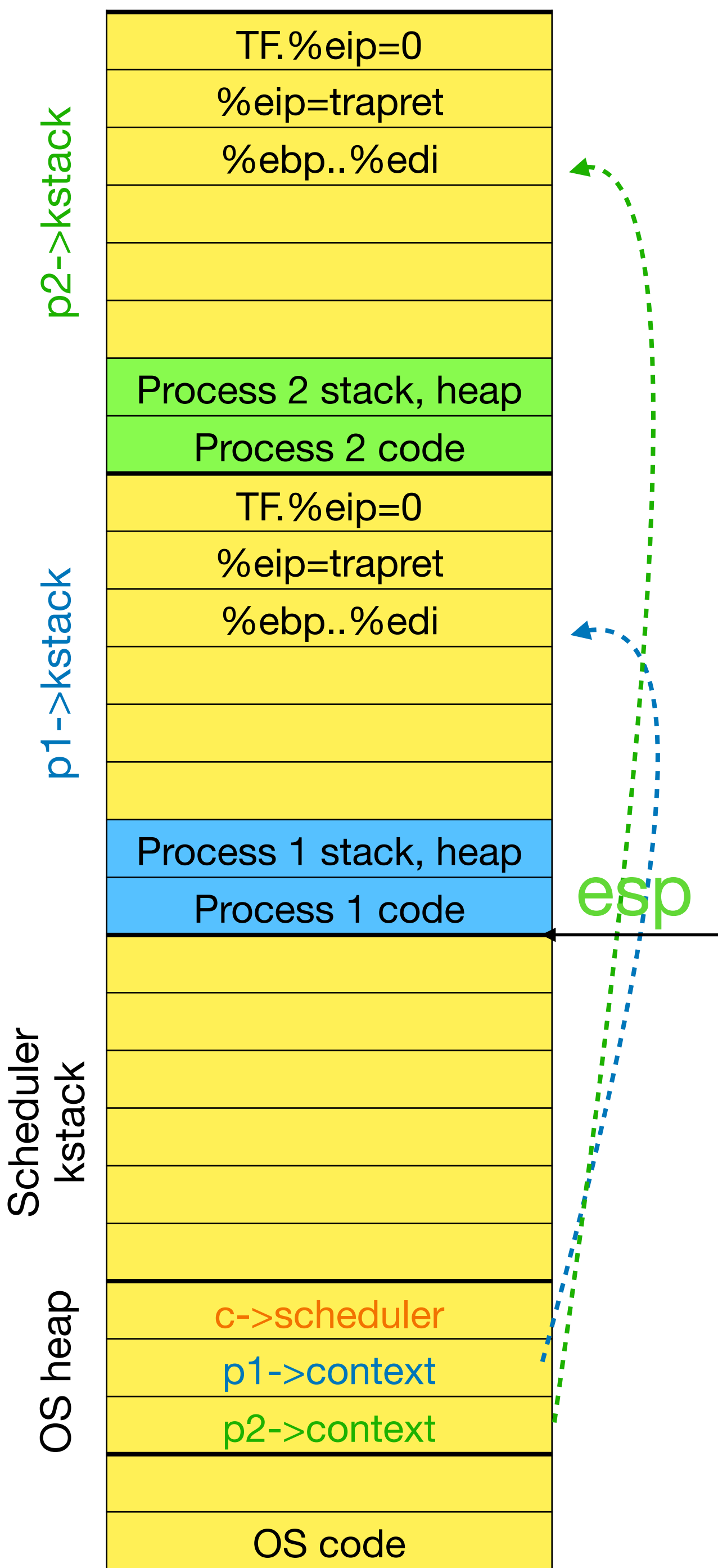
p18-sched

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    popl %esi
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    swtch(&p->context, c->scheduler);
}

void trap(struct trapframe *tf) {
    ..
    if(tf->trapno == T_IRQ0+IRQ_TIMER)
        yield();
}
```



Context switching in action: giving control

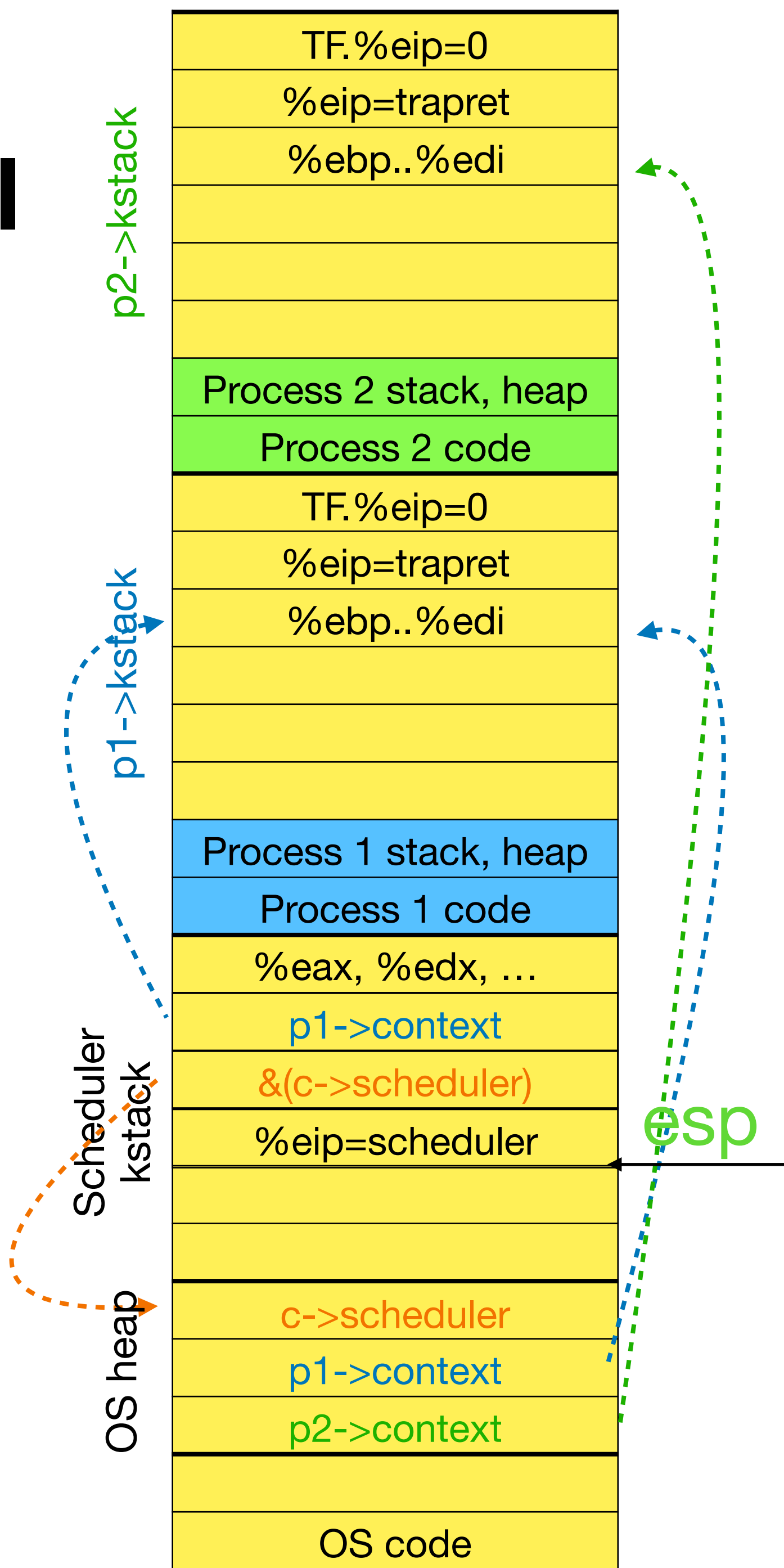
p18-sched

```
.globl swtch
swtch:
    movl 4(%esp), %eax
    movl 8(%esp), %edx
    pushl %ebp
    pushl %ebx
    pushl %esi
    pushl %edi
    movl %esp, (%eax)
    movl %edx, %esp
    popl %edi
    popl %esi
    popl %ebx
    popl %ebp
    ret

void scheduler(void) {
    struct proc *p; struct cpu *c = mycpu();
    for(;;){
        for(p = ptable.proc; p < &ptable.proc[NPROC]; p++){
            if(p->state != RUNNABLE)
                continue;
            p->state = RUNNING;
            switchvm(p);
            swtch(&(c->scheduler), p->context);
        }
    }
}

void yield(void) {
    struct proc *p = myproc();
    p->state = RUNNABLE;
    swtch(&p->context, c->scheduler);
}

void trap(struct trapframe *tf) {
    ..
    if(tf->trapno == T_IRQ0+IRQ_TIMER)
        yield();
}
```



Context switching in action: giving control

p18-sched

.globl swtch

swtch:

eip

movl 4(%esp), %eax
movl 8(%esp), %edx
pushl %ebp
pushl %ebx
pushl %esi
pushl %edi
movl %esp, (%eax)
movl %edx, %esp
popl %edi
popl %esi
popl %ebx
popl %ebp
ret

void scheduler(void) {

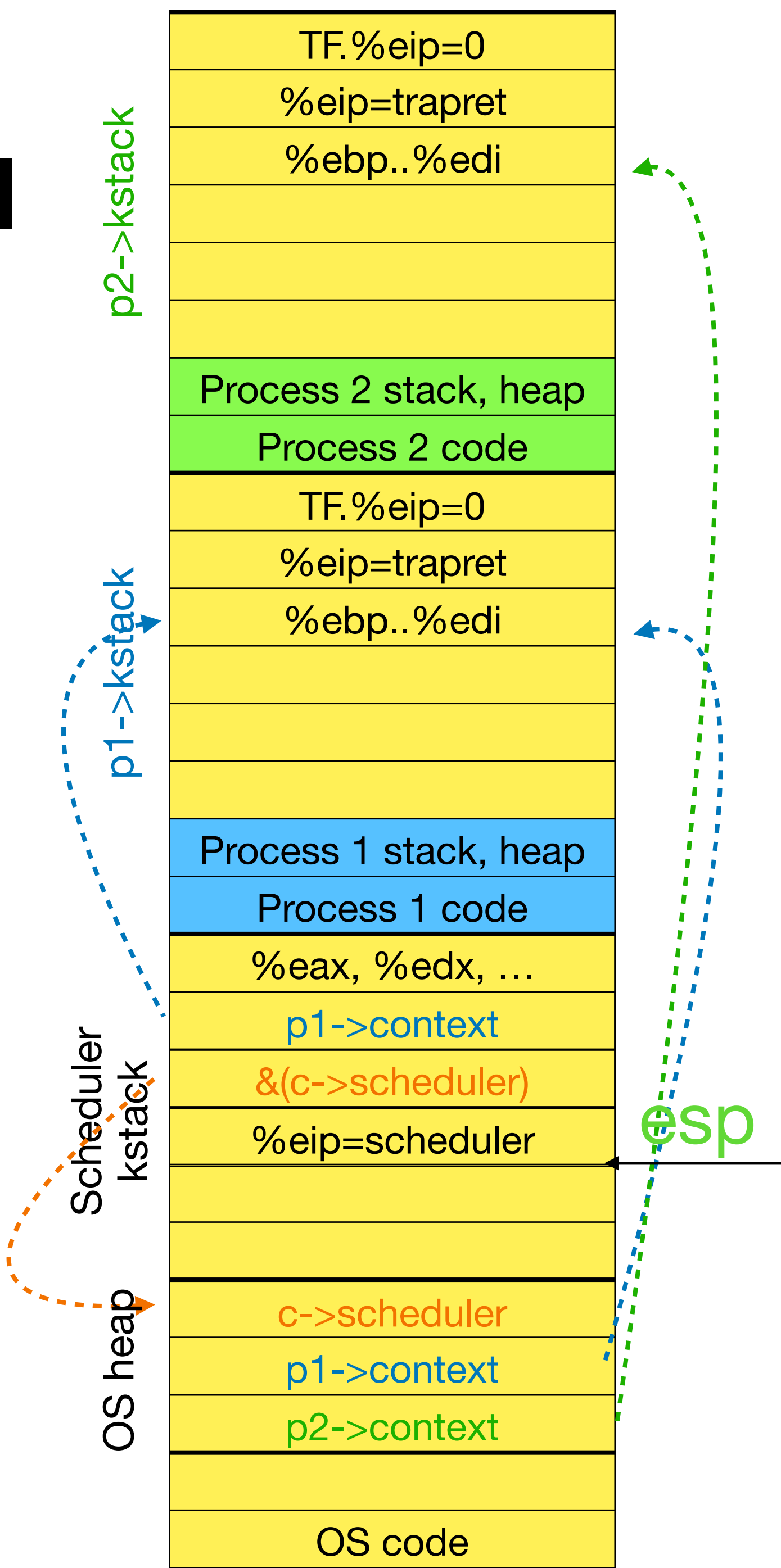
struct proc *p; struct cpu *c = mycpu();
for(;;){
for(p = ptable.proc; p < &ptable.proc[NPROC]; p++){
if(p->state != RUNNABLE)
continue;
p->state = RUNNING;
switchvm(p);
swtch(&(c->scheduler), p->context);
}
}

void yield(void) {

struct proc *p = myproc();
p->state = RUNNABLE;
swtch(&p->context, c->scheduler);
}

void trap(struct trapframe *tf) {

..
if(tf->trapno == T_IRQ0+IRQ_TIMER)
yield();
}



Context switching in action: giving control

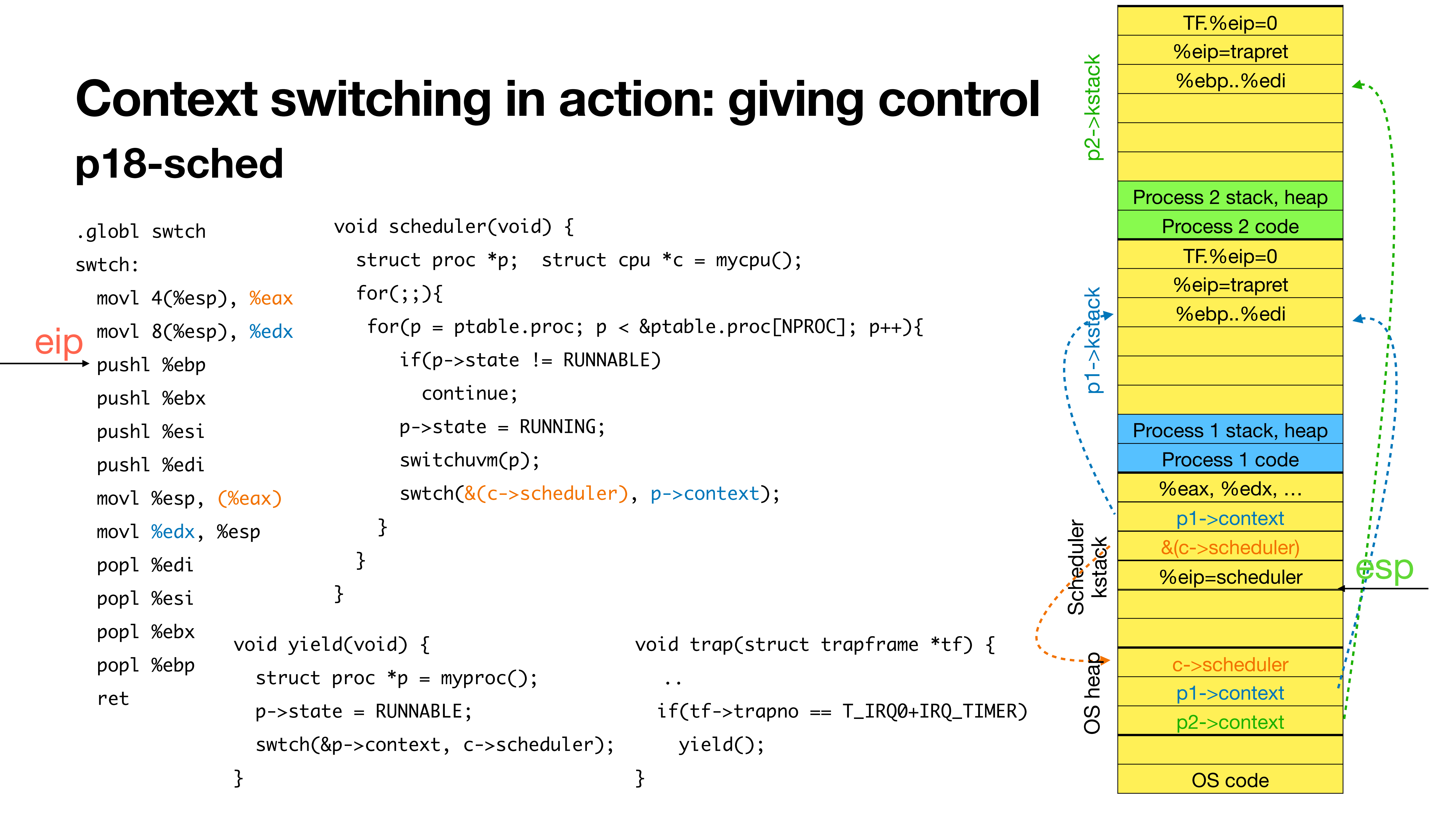
p18-sched

```
.globl swtch
swtch:
    movl 4(%esp), %eax
    movl 8(%esp), %edx
    pushl %ebp
    pushl %ebx
    pushl %esi
    pushl %edi
    movl %esp, (%eax)
    movl %edx, %esp
    popl %edi
    popl %esi
    popl %ebx
    popl %ebp
    ret
```

```
void scheduler(void) {
    struct proc *p; struct cpu *c = mycpu();
    for(;;){
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            swtch(&(c->scheduler), p->context);
        }
    }
}
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```
void yield(void) {
    struct proc *p = myproc();
    p->state = RUNNABLE;
    swtch(&p->context, c->scheduler);
}
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```
void trap(struct trapframe *tf) {
    ..
    if(tf->trapno == T_IRQ0+IRQ_TIMER)
        yield();
}
```



Context switching in action: giving control

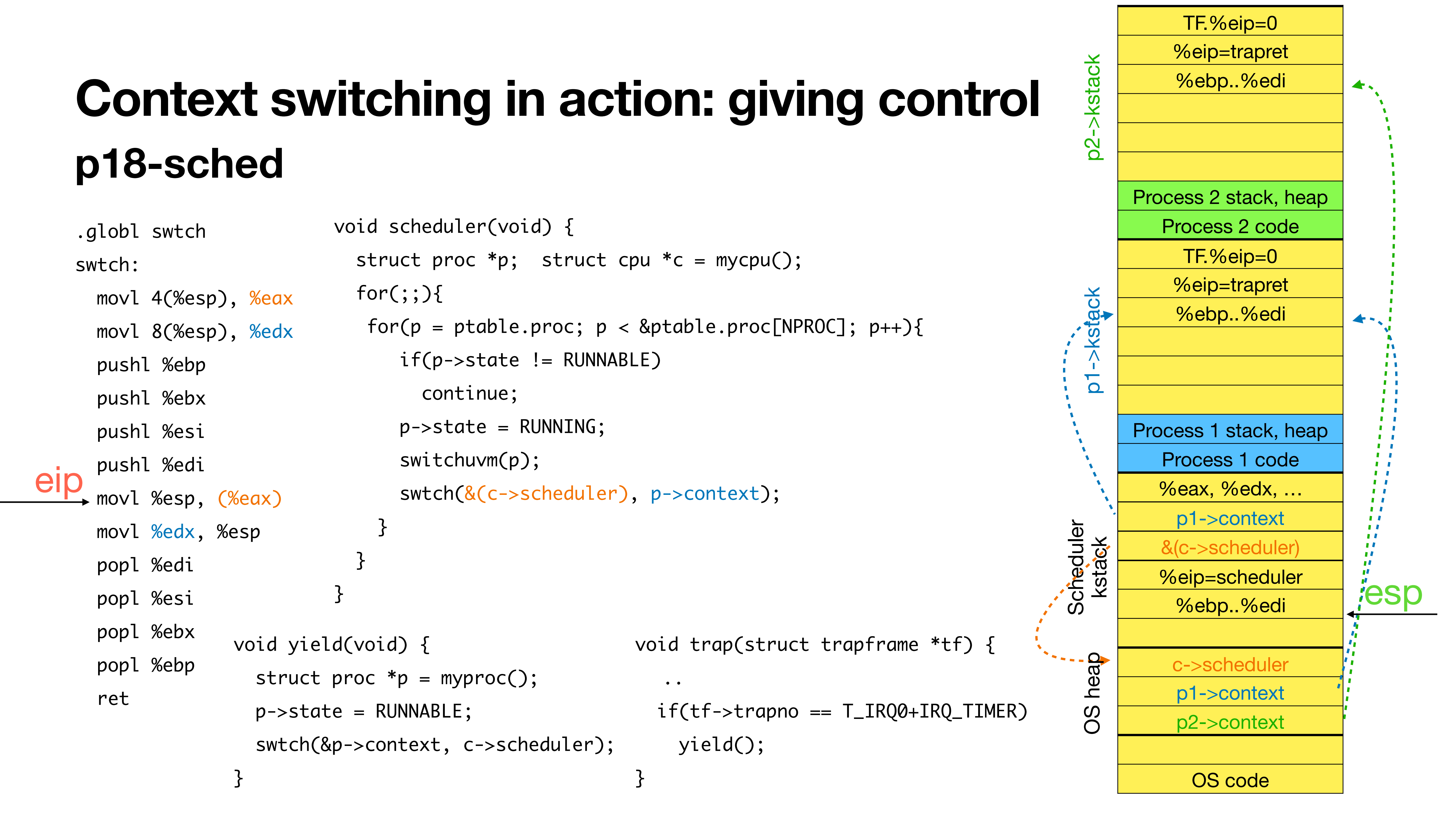
p18-sched

```
.globl swtch
swtch:
    movl 4(%esp), %eax
    movl 8(%esp), %edx
    pushl %ebp
    pushl %ebx
    pushl %esi
    pushl %edi
    movl %esp, (%eax)
    movl %edx, %esp
    popl %edi
    popl %esi
    popl %ebx
    popl %ebp
    ret

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            swtch(&(c->scheduler), p->context);
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void yield(void) {
    struct proc *p = myproc();
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    if(tf->trapno == T_IRQ0+IRQ_TIMER)
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```



Context switching in action: giving control

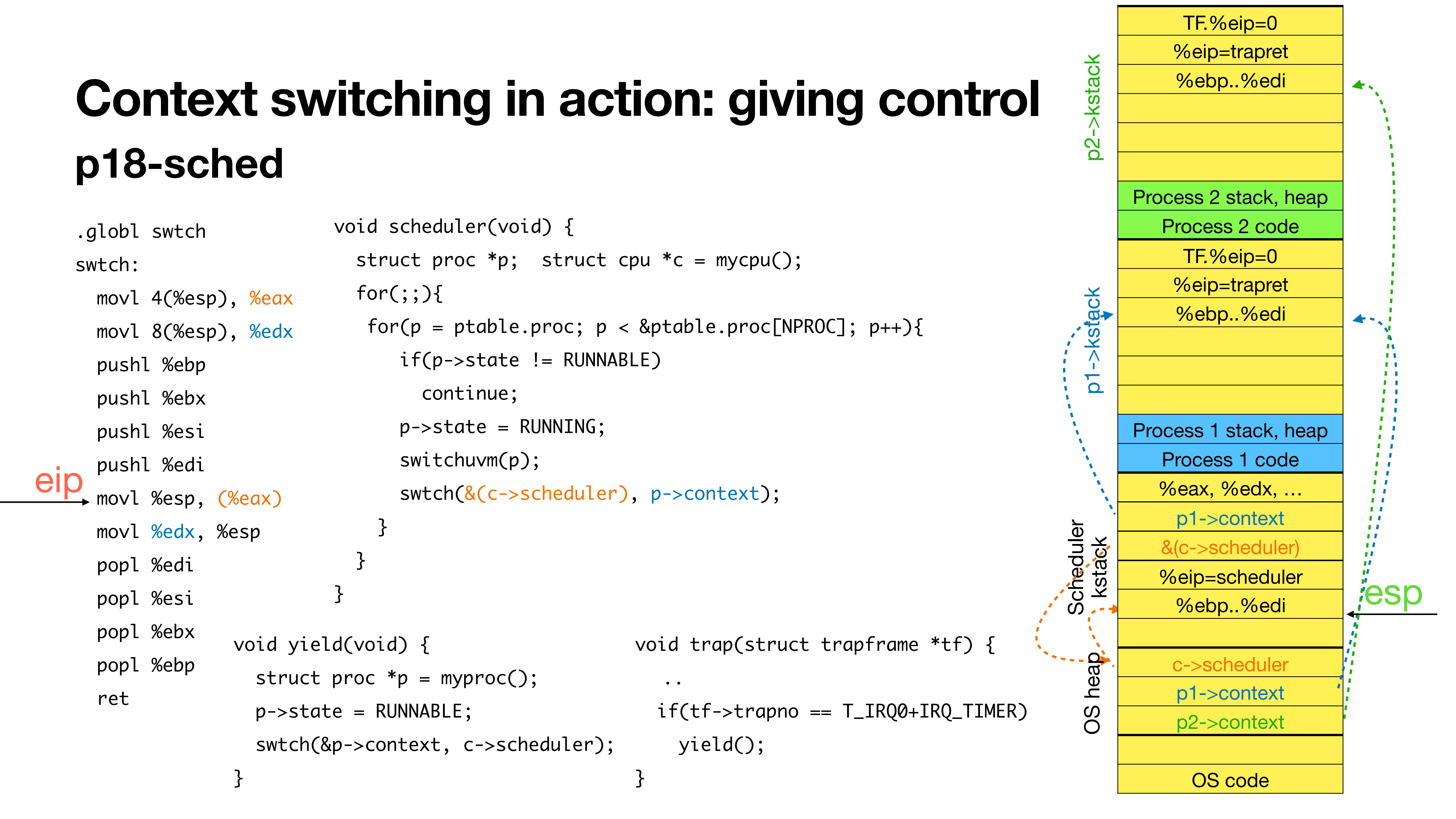
p18-sched

```
.globl swtch
swtch:
    movl 4(%esp), %eax
    movl 8(%esp), %edx
    pushl %ebp
    pushl %ebx
    pushl %esi
    pushl %edi
    movl %esp, (%eax)
    movl %edx, %esp
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    popl %esi
    popl %ebx
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    ..
    if(tf->trapno == T_IRQ0+IRQ_TIMER)
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}
```



Context switching in action: giving control

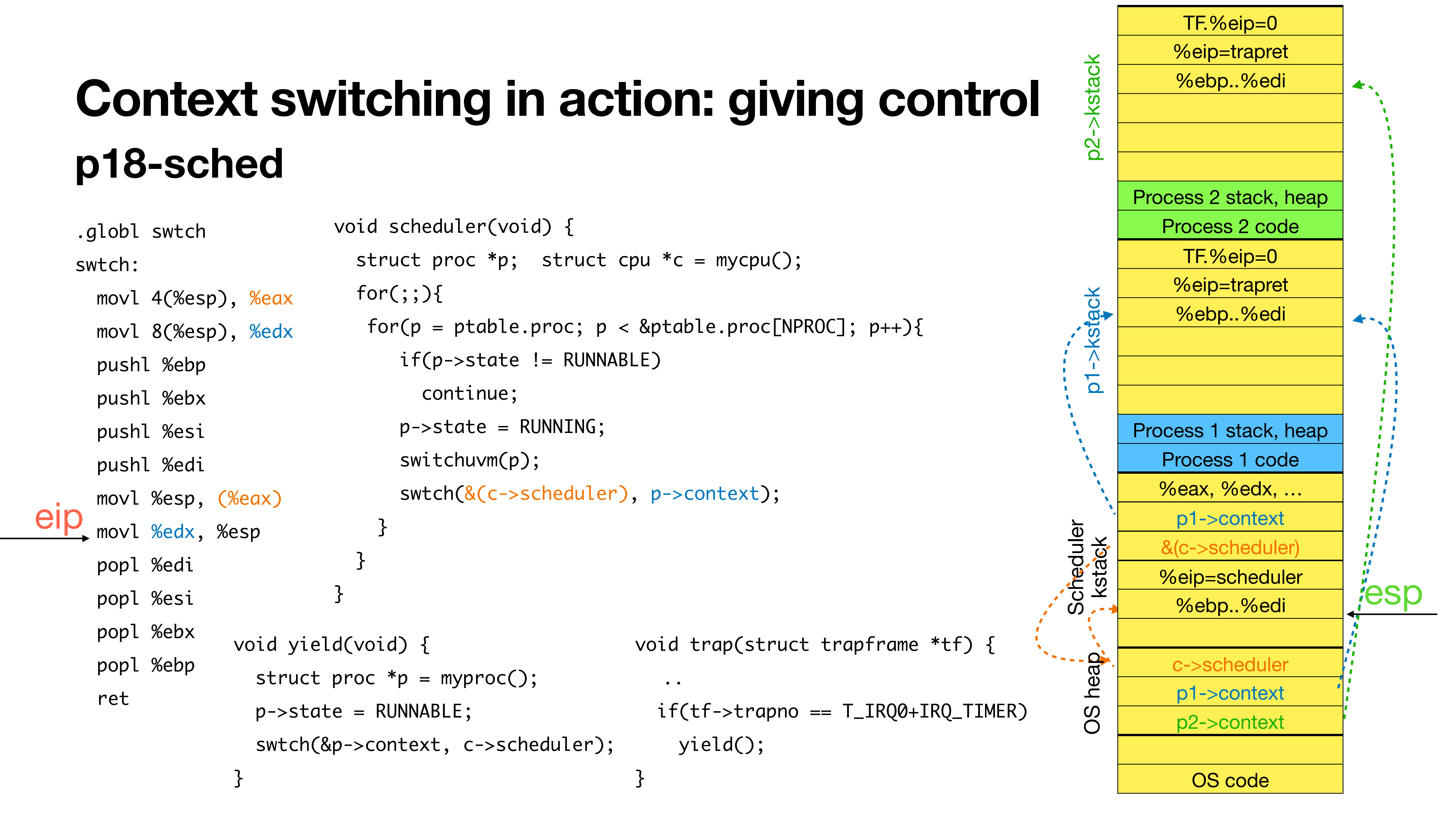
p18-sched

```
.globl swtch
swtch:
    movl 4(%esp), %eax
    movl 8(%esp), %edx
    pushl %ebp
    pushl %ebx
    pushl %esi
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    movl %esp, (%eax)
    movl %edx, %esp
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    popl %esi
    popl %ebx
    popl %ebp
    ret

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        }
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void yield(void) {
    struct proc *p = myproc();
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    swtch(&p->context, c->scheduler);
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void trap(struct trapframe *tf) {
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    if(tf->trapno == T_IRQ0+IRQ_TIMER)
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Context switching in action: giving control

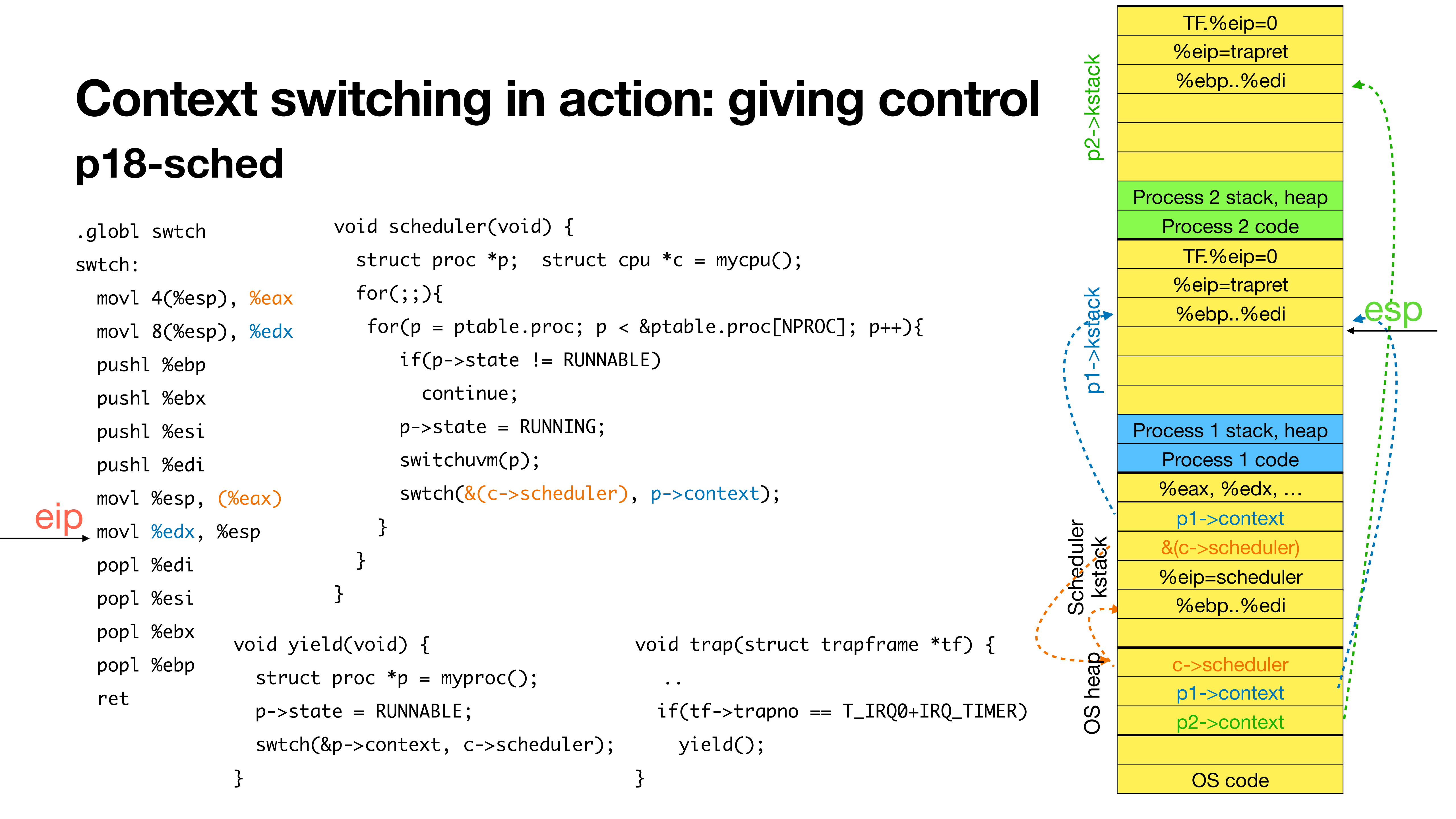
p18-sched

```
.globl swtch
swtch:
    movl 4(%esp), %eax
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    pushl %ebp
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Context switching in action: giving control

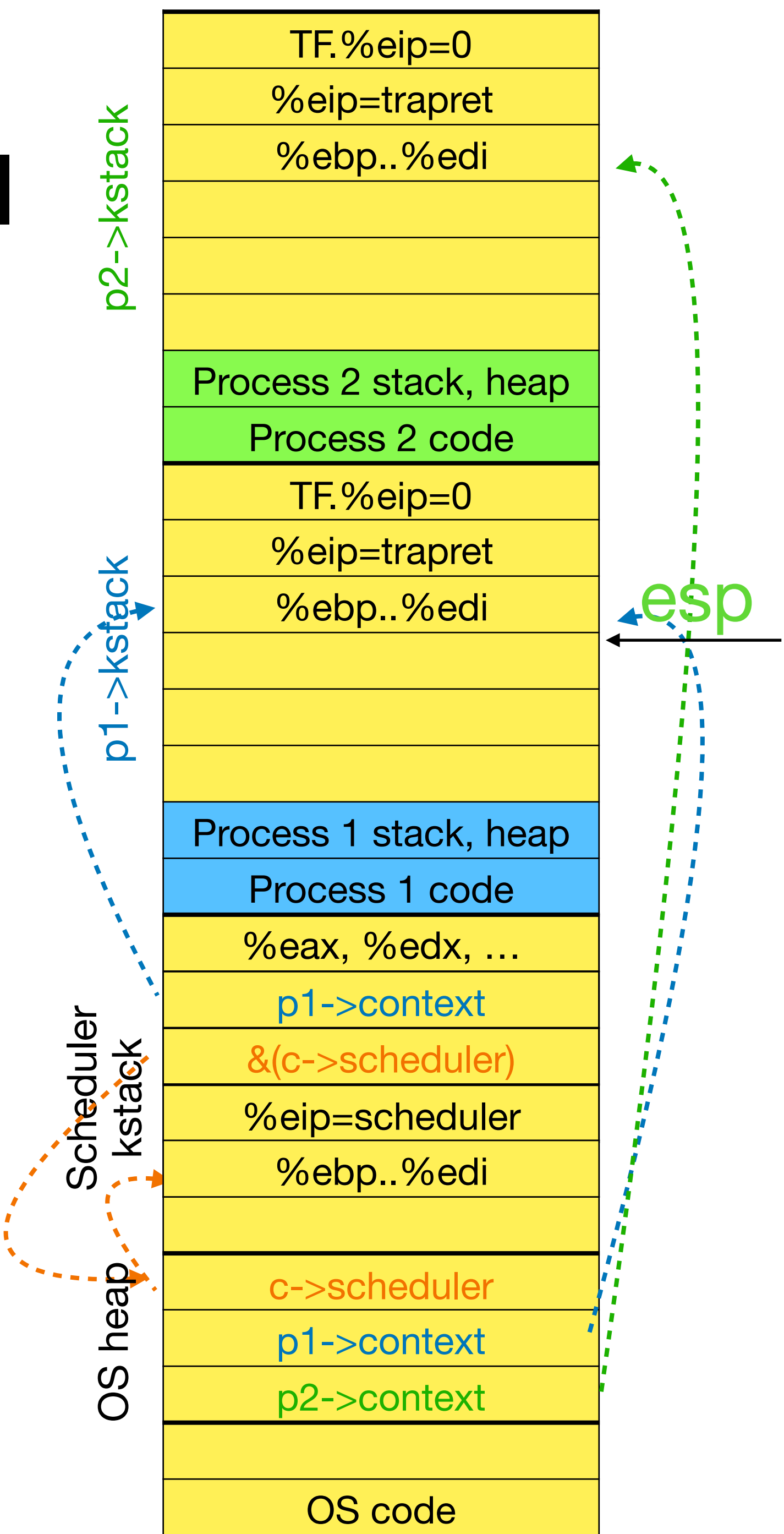
p18-sched

```
.globl swtch
swtch:
    movl 4(%esp), %eax
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    movl %esp, (%eax)
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```

```
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        }
    }
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```
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void trap(struct trapframe *tf) {
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Context switching in action: giving control

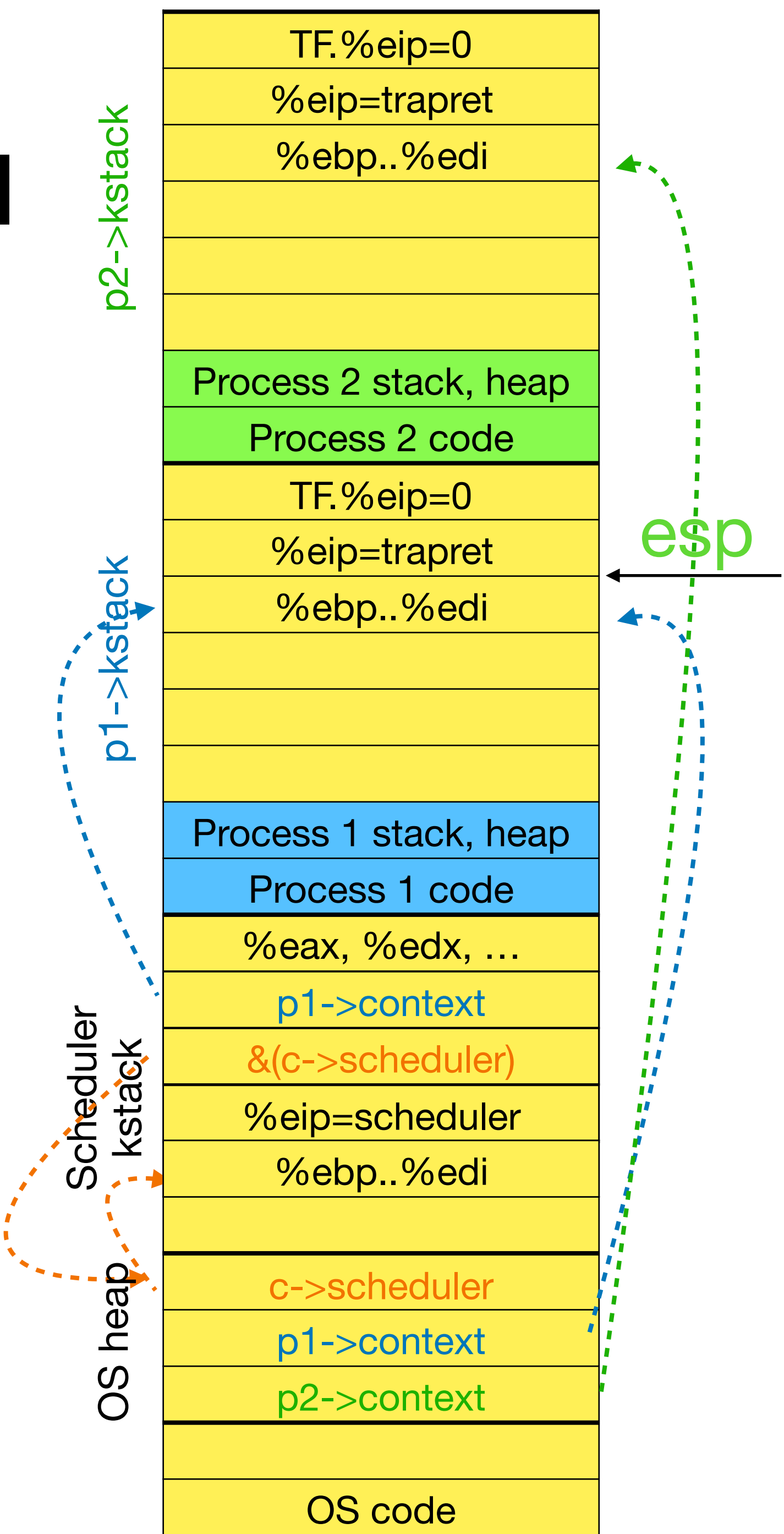
p18-sched

```
.globl swtch
swtch:
    movl 4(%esp), %eax
    movl 8(%esp), %edx
    pushl %ebp
    pushl %ebx
    pushl %esi
    pushl %edi
    movl %esp, (%eax)
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Context switching in action: giving control

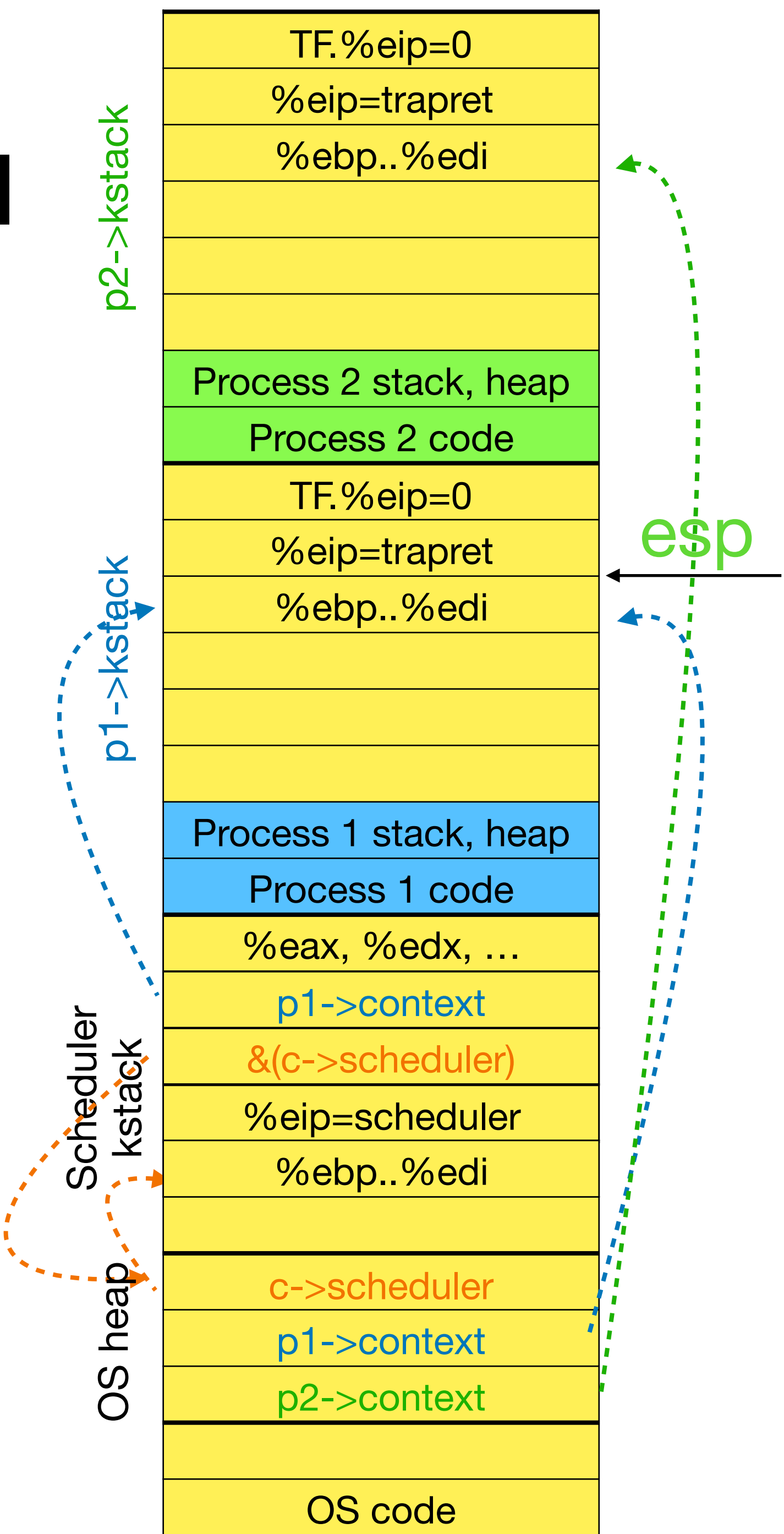
p18-sched

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Context switching in action: giving control

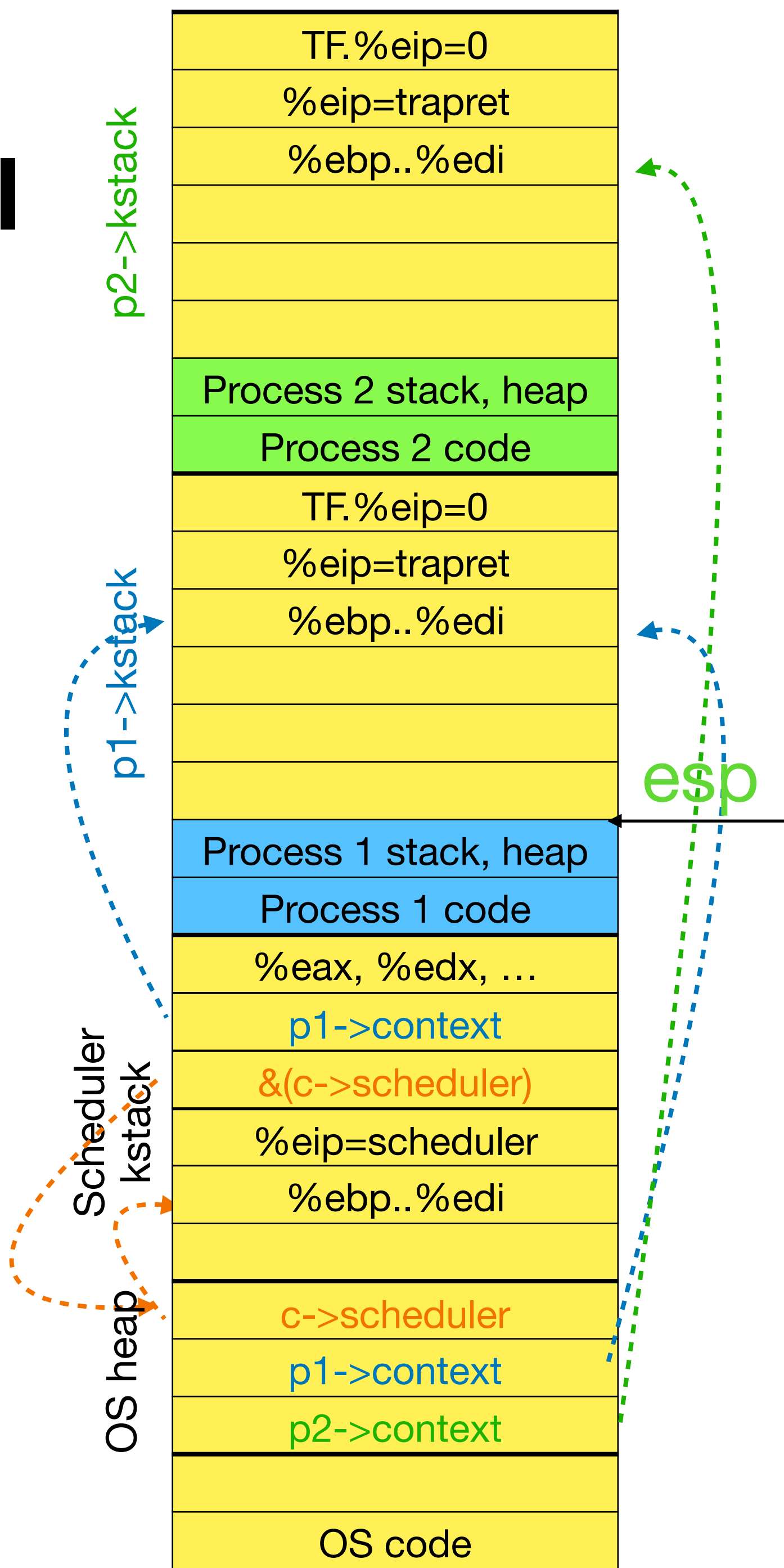
p18-sched

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Context switching in action: giving control

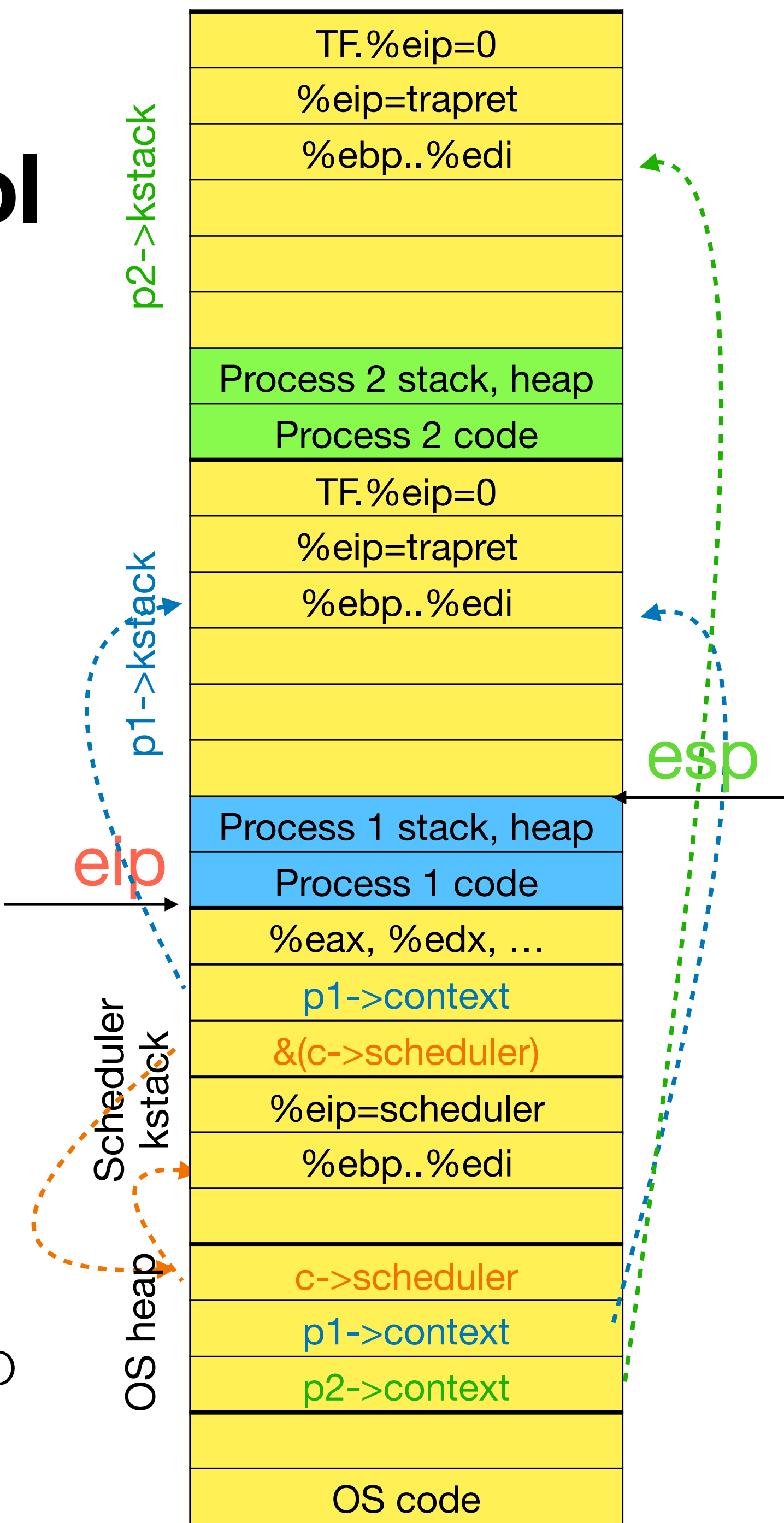
p18-sched

```
.globl swtch
swtch:
    movl 4(%esp), %eax
    movl 8(%esp), %edx
    pushl %ebp
    pushl %ebx
    pushl %esi
    pushl %edi
    movl %esp, (%eax)
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    popl %ebx
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    ..
    if(tf->trapno == T_IRQ0+IRQ_TIMER)
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}
```



Context switching in action: taking control

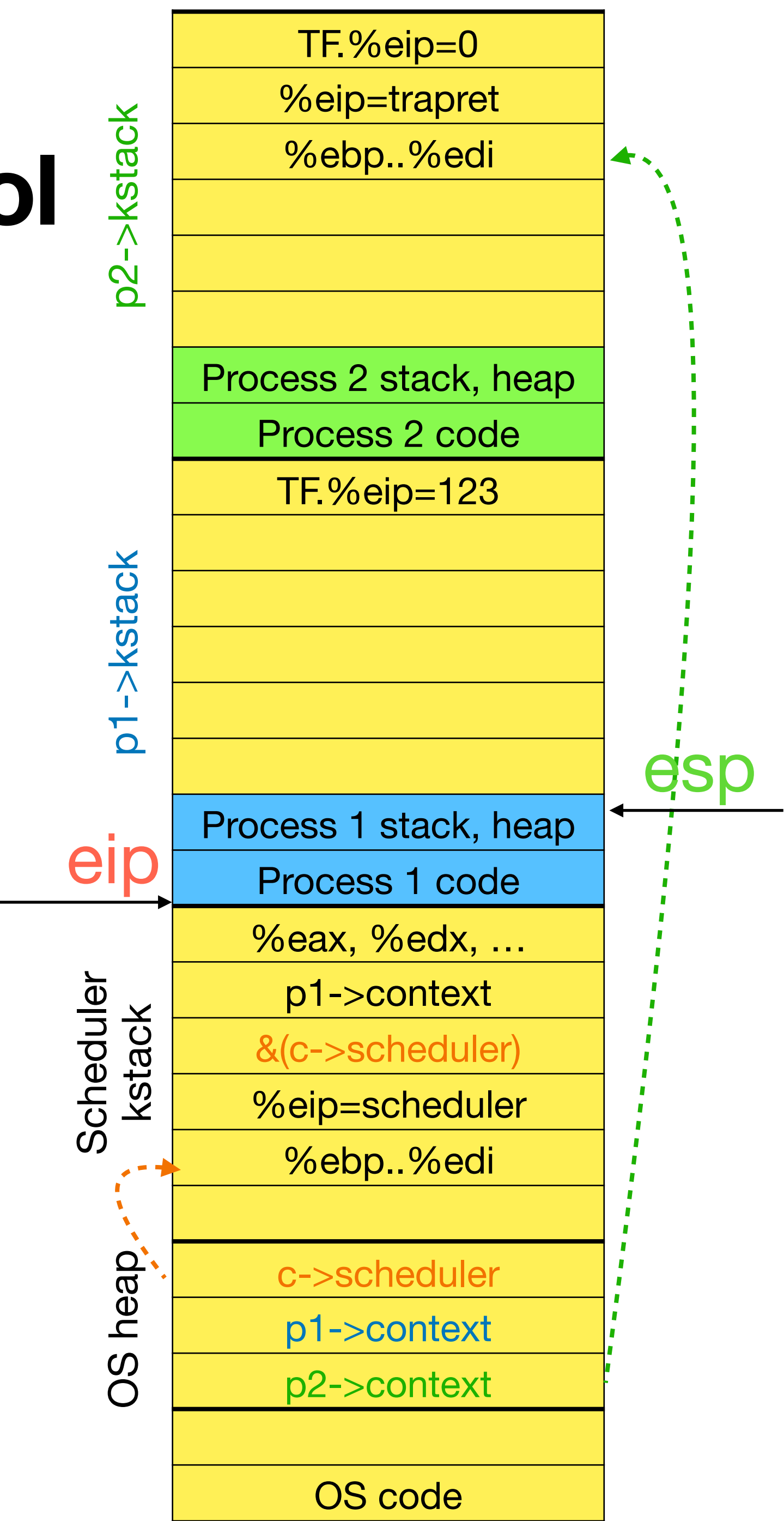
p18-sched

```
.globl swtch
swtch:
    movl 4(%esp), %eax
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    pushl %ebx
    pushl %esi
    pushl %edi
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    swtch(&p->context, c->scheduler);
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void trap(struct trapframe *tf) {
    ..
    if(tf->trapno == T_IRQ0+IRQ_TIMER)
        yield();
}
```



Context switching in action: taking control

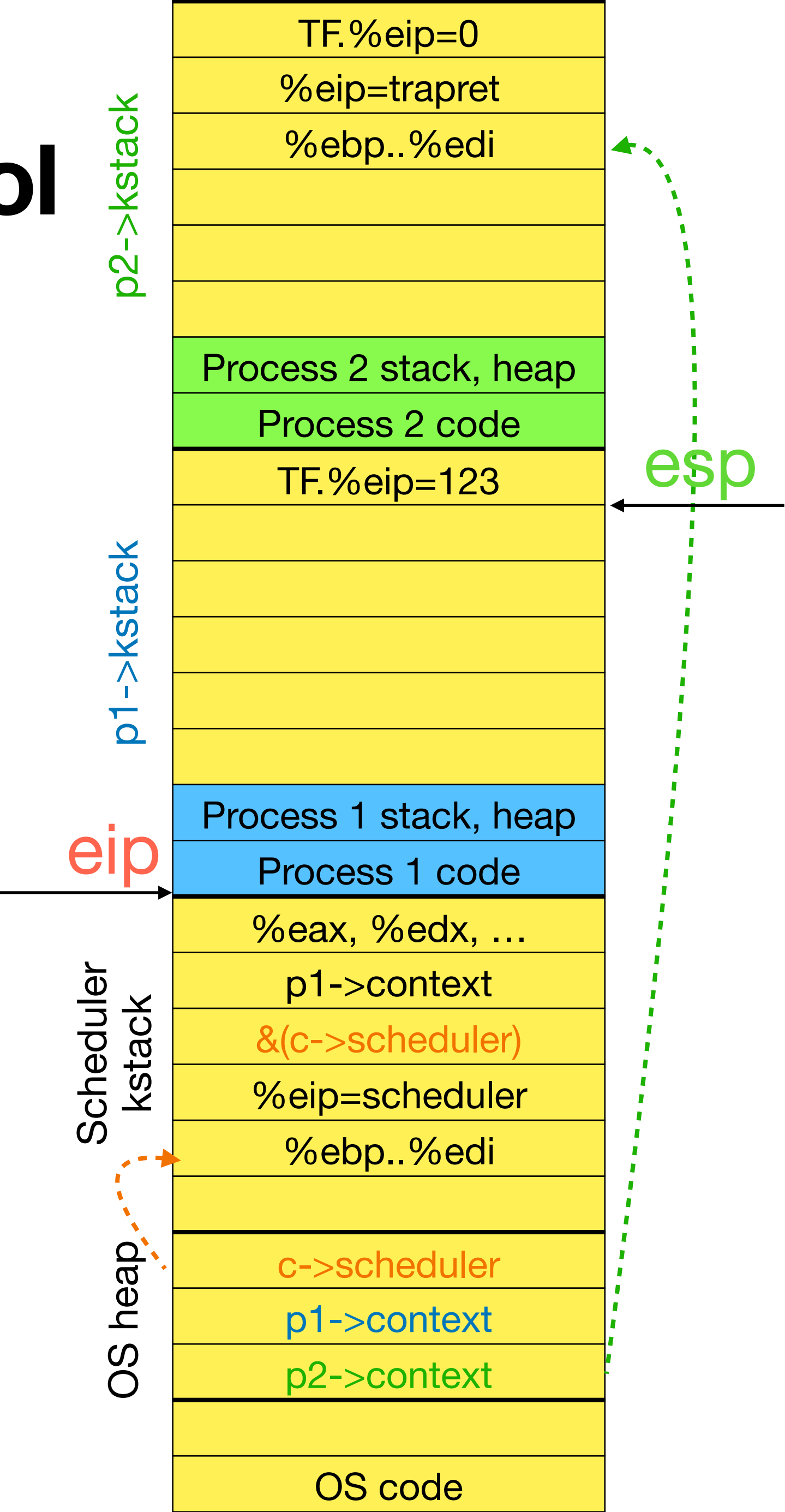
p18-sched

```
.globl swtch
swtch:
    movl 4(%esp), %eax
    movl 8(%esp), %edx
    pushl %ebp
    pushl %ebx
    pushl %esi
    pushl %edi
    movl %esp, (%eax)
    movl %edx, %esp
    popl %edi
    popl %esi
    popl %ebx
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    ret

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    ..
    if(tf->trapno == T_IRQ0+IRQ_TIMER)
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}
```



Context switching in action: taking control

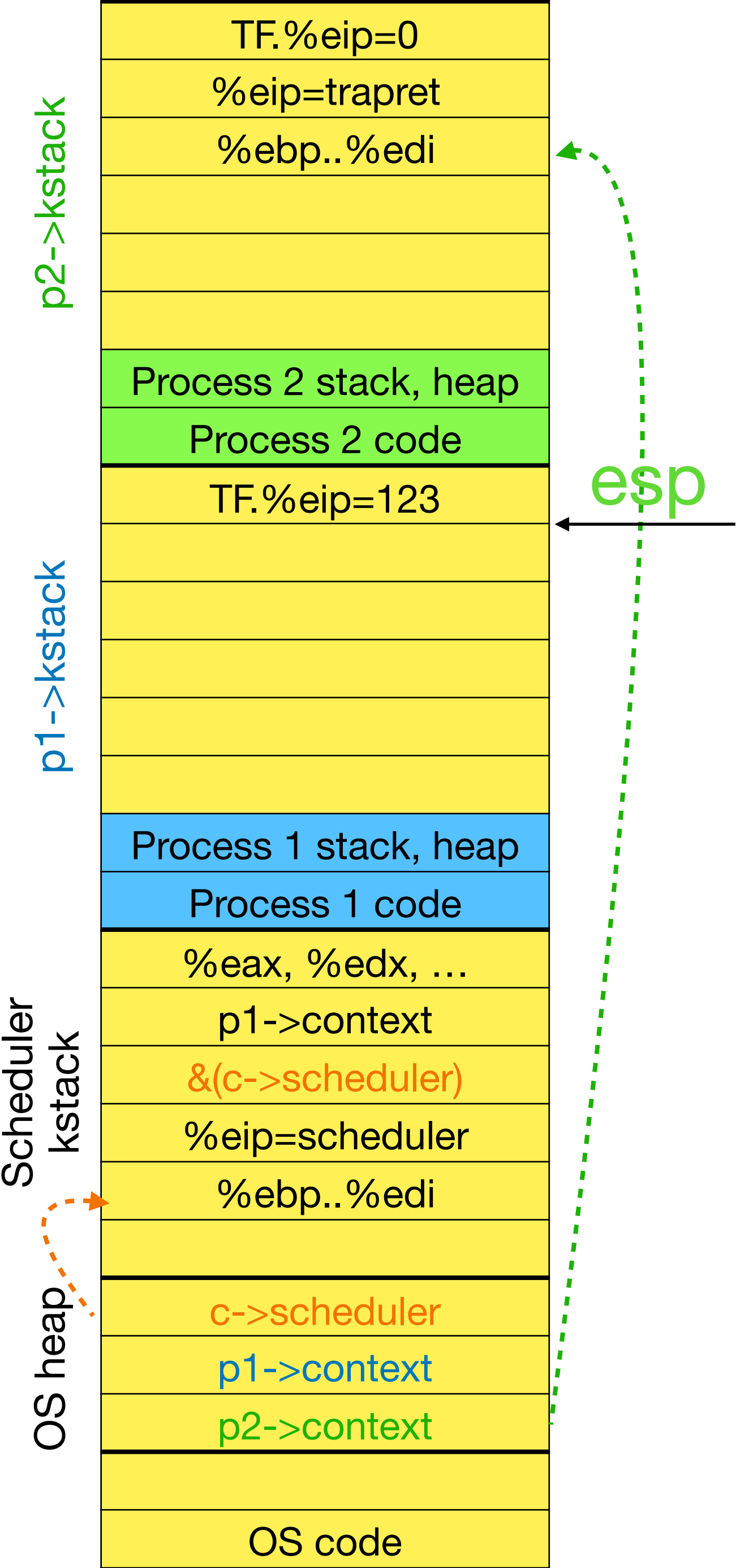
p18-sched

```
.globl swtch
swtch:
    movl 4(%esp), %eax
    movl 8(%esp), %edx
    pushl %ebp
    pushl %ebx
    pushl %esi
    pushl %edi
    movl %esp, (%eax)
    movl %edx, %esp
    popl %edi
    popl %esi
    popl %ebx
    popl %ebp
    ret

void scheduler(void) {
    struct proc *p; struct cpu *c = mycpu();
    for(;;){
        for(p = ptable.proc; p < &ptable.proc[NPROC]; p++){
            if(p->state != RUNNABLE)
                continue;
            p->state = RUNNING;
            switchvm(p);
            swtch(&(c->scheduler), p->context);
        }
    }
}

void yield(void) {
    struct proc *p = myproc();
    p->state = RUNNABLE;
    swtch(&p->context, c->scheduler);
}

void trap(struct trapframe *tf) {
    ..
    if(tf->trapno == T_IRQ0+IRQ_TIMER)
        yield();
}
```



Context switching in action: taking control

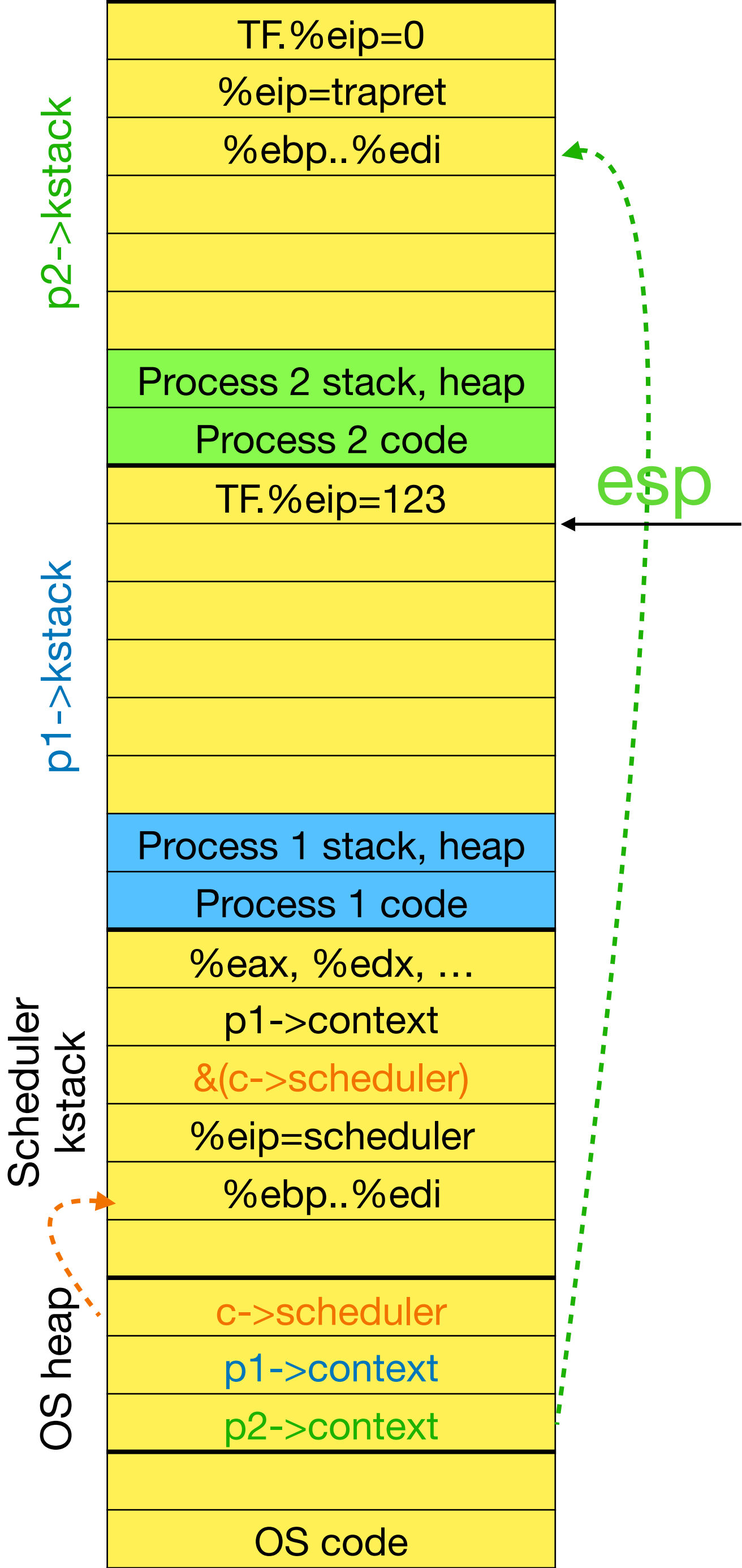
p18-sched

```
.globl swtch
swtch:
    movl 4(%esp), %eax
    movl 8(%esp), %edx
    pushl %ebp
    pushl %ebx
    pushl %esi
    pushl %edi
    movl %esp, (%eax)
    movl %edx, %esp
    popl %edi
    popl %esi
    popl %ebx
    popl %ebp
    ret

void scheduler(void) {
    struct proc *p; struct cpu *c = mycpu();
    for(;;){
        for(p = ptable.proc; p < &ptable.proc[NPROC]; p++){
            if(p->state != RUNNABLE)
                continue;
            p->state = RUNNING;
            switchvm(p);
            swtch(&(c->scheduler), p->context);
        }
    }
}

void yield(void) {
    struct proc *p = myproc();
    p->state = RUNNABLE;
    swtch(&p->context, c->scheduler);
}

void trap(struct trapframe *tf) {
    ..
    if(tf->trapno == T_IRQ0+IRQ_TIMER)
        yield();
}
```



Context switching in action: taking control

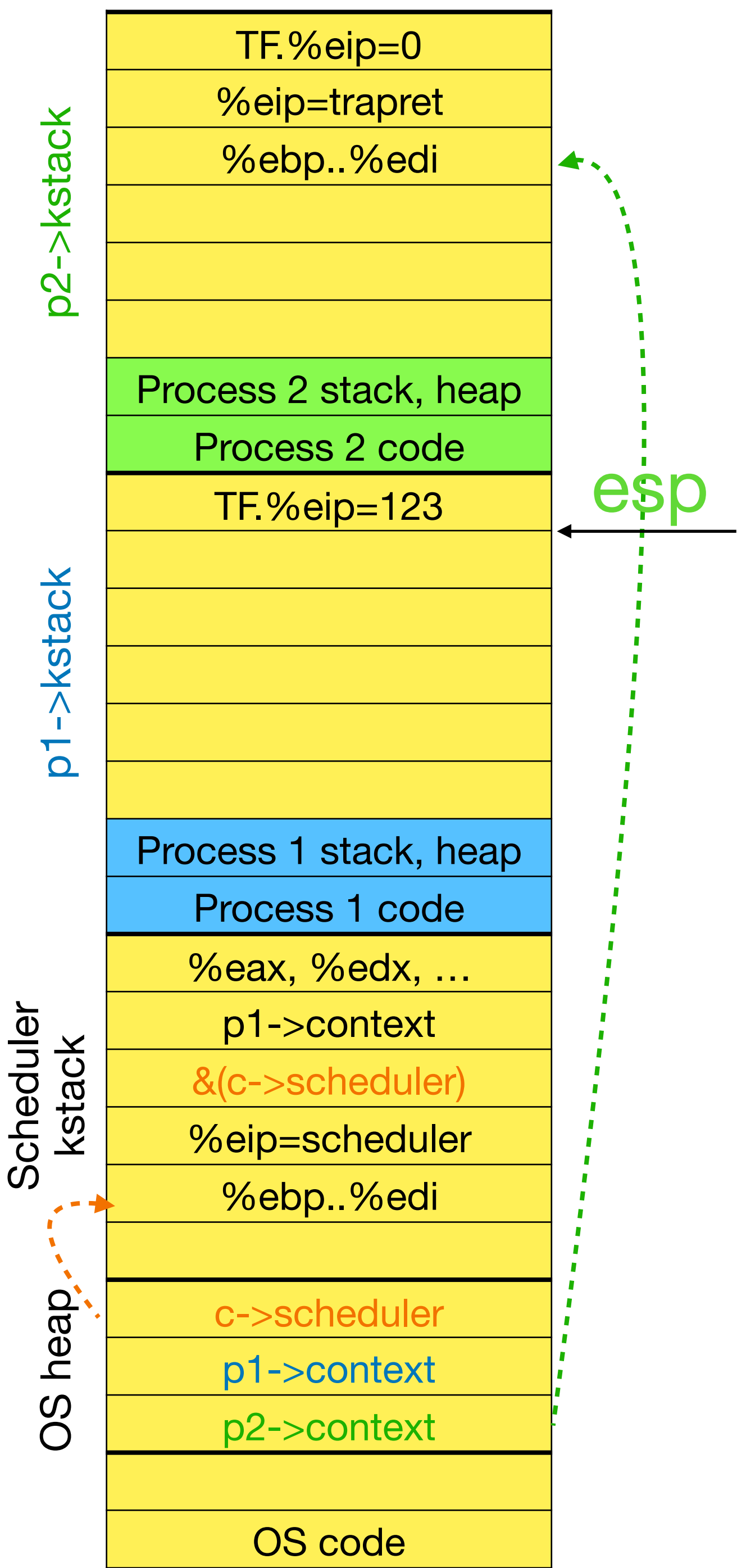
p18-sched

```
.globl swtch
swtch:
    movl 4(%esp), %eax
    movl 8(%esp), %edx
    pushl %ebp
    pushl %ebx
    pushl %esi
    pushl %edi
    movl %esp, (%eax)
    movl %edx, %esp
    popl %edi
    popl %esi
    popl %ebx
    popl %ebp
    ret

void scheduler(void) {
    struct proc *p; struct cpu *c = mycpu();
    for(;;){
        for(p = ptable.proc; p < &ptable.proc[NPROC]; p++){
            if(p->state != RUNNABLE)
                continue;
            p->state = RUNNING;
            switchvm(p);
            swtch(&(c->scheduler), p->context);
        }
    }
}

void yield(void) {
    struct proc *p = myproc();
    p->state = RUNNABLE;
    swtch(&p->context, c->scheduler);
}

void trap(struct trapframe *tf) {
    ..
    if(tf->trapno == T_IRQ0+IRQ_TIMER)
        yield();
}
```



Context switching in action: taking control

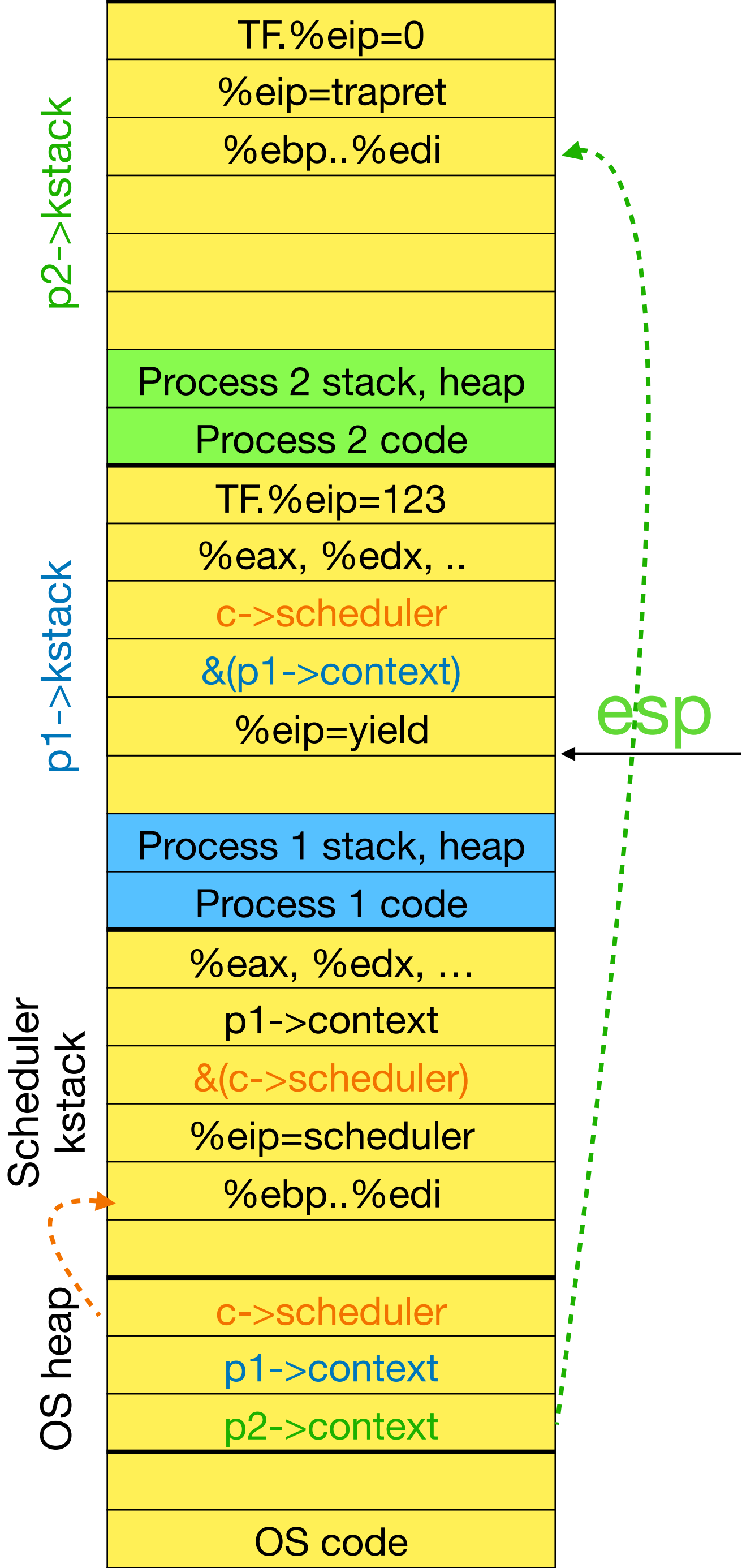
p18-sched

```
.globl swtch
swtch:
    movl 4(%esp), %eax
    movl 8(%esp), %edx
    pushl %ebp
    pushl %ebx
    pushl %esi
    pushl %edi
    movl %esp, (%eax)
    movl %edx, %esp
    popl %edi
    popl %esi
    popl %ebx
    popl %ebp
    ret

void scheduler(void) {
    struct proc *p; struct cpu *c = mycpu();
    for(;;){
        for(p = ptable.proc; p < &ptable.proc[NPROC]; p++){
            if(p->state != RUNNABLE)
                continue;
            p->state = RUNNING;
            switchvm(p);
            swtch(&(c->scheduler), p->context);
        }
    }
}

void yield(void) {
    struct proc *p = myproc();
    p->state = RUNNABLE;
    swtch(&p->context, c->scheduler);
}

void trap(struct trapframe *tf) {
    ..
    if(tf->trapno == T_IRQ0+IRQ_TIMER)
        yield();
}
```



Context switching in action: taking control

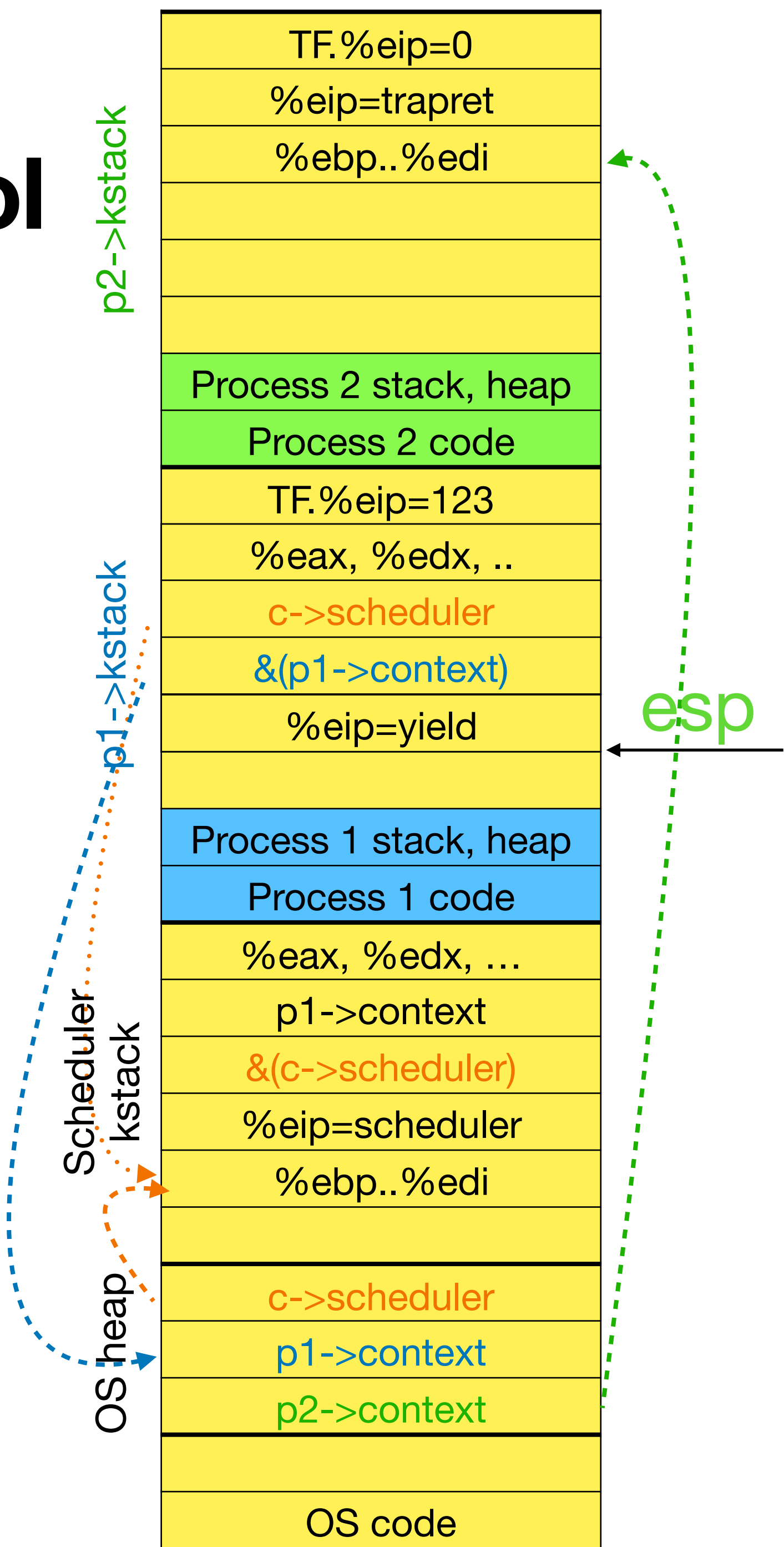
p18-sched

```
.globl switch
switch:
    movl 4(%esp), %eax
    movl 8(%esp), %edx
    pushl %ebp
    pushl %ebx
    pushl %esi
    pushl %edi
    movl %esp, (%eax)
    movl %edx, %esp
    popl %edi
    popl %esi
    popl %ebx
    popl %ebp
    ret

void scheduler(void) {
    struct proc *p; struct cpu *c = mycpu();
    for(;;){
        for(p = ptable.proc; p < &ptable.proc[NPROC]; p++){
            if(p->state != RUNNABLE)
                continue;
            p->state = RUNNING;
            switchvm(p);
            swtch(&(c->scheduler), p->context);
        }
    }
}

void yield(void) {
    struct proc *p = myproc();
    p->state = RUNNABLE;
    swtch(&p->context, c->scheduler);
}

void trap(struct trapframe *tf) {
    ..
    if(tf->trapno == T_IRQ0+IRQ_TIMER)
        yield();
}
```



Context switching in action: taking control

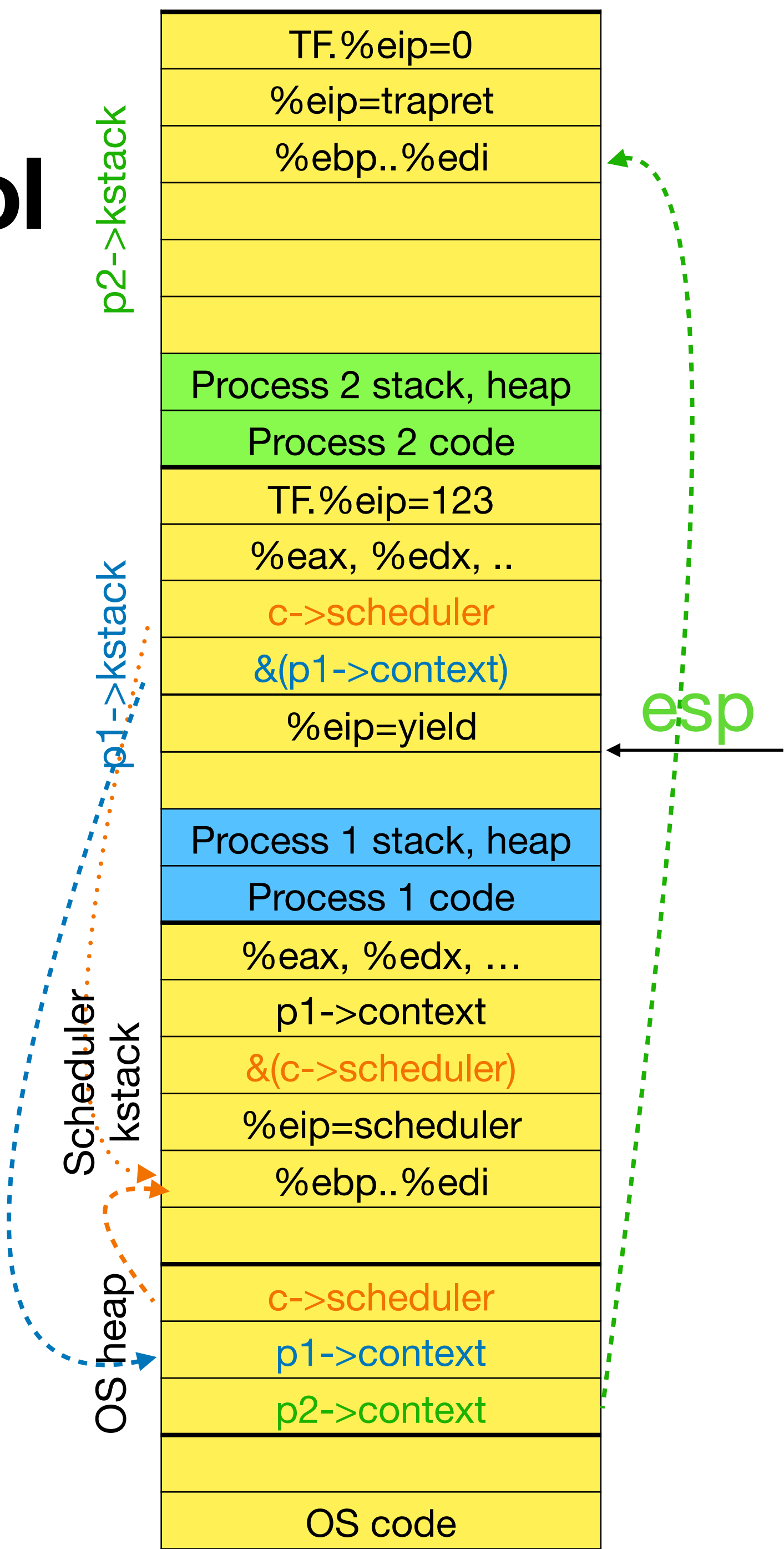
p18-sched

```
.globl swtch
swtch:
    movl 4(%esp), %eax
    movl 8(%esp), %edx
    pushl %ebp
    pushl %ebx
    pushl %esi
    pushl %edi
    movl %esp, (%eax)
    movl %edx, %esp
    popl %edi
    popl %esi
    popl %ebx
    popl %ebp
    ret

void scheduler(void) {
    struct proc *p; struct cpu *c = mycpu();
    for(;;){
        for(p = ptable.proc; p < &ptable.proc[NPROC]; p++){
            if(p->state != RUNNABLE)
                continue;
            p->state = RUNNING;
            switchvm(p);
            swtch(&(c->scheduler), p->context);
        }
    }
}

void yield(void) {
    struct proc *p = myproc();
    p->state = RUNNABLE;
    swtch(&p->context, c->scheduler);
}

void trap(struct trapframe *tf) {
    ..
    if(tf->trapno == T_IRQ0+IRQ_TIMER)
        yield();
}
```



Context switching in action: taking control

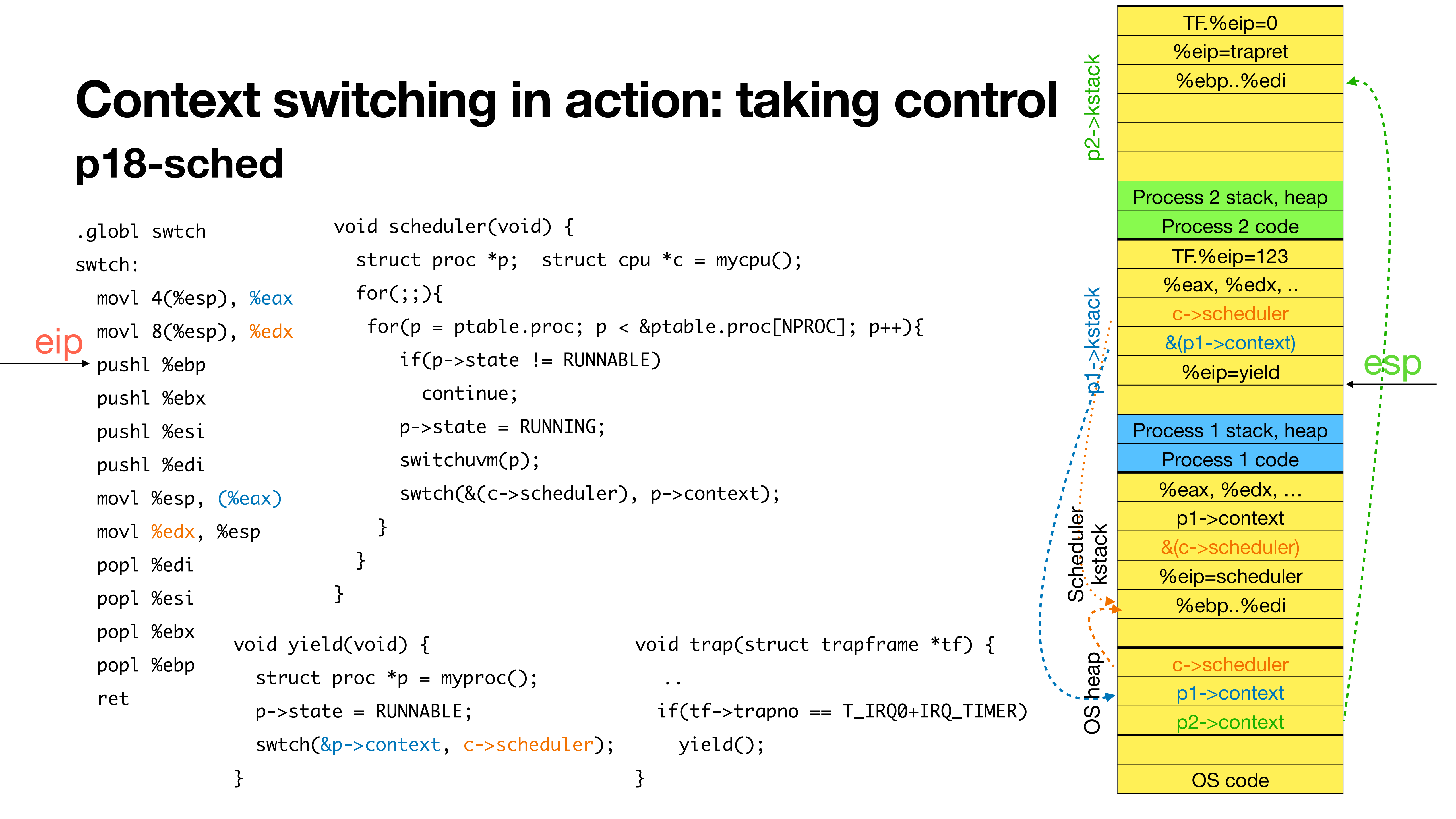
p18-sched

```
.globl swtch
swtch:
    movl 4(%esp), %eax
    movl 8(%esp), %edx
    pushl %ebp
    pushl %ebx
    pushl %esi
    pushl %edi
    movl %esp, (%eax)
    movl %edx, %esp
    popl %edi
    popl %esi
    popl %ebx
    popl %ebp
    ret

void scheduler(void) {
    struct proc *p; struct cpu *c = mycpu();
    for(;;){
        for(p = ptable.proc; p < &ptable.proc[NPROC]; p++){
            if(p->state != RUNNABLE)
                continue;
            p->state = RUNNING;
            switchvm(p);
            swtch(&(c->scheduler), p->context);
        }
    }
}

void yield(void) {
    struct proc *p = myproc();
    p->state = RUNNABLE;
    swtch(&p->context, c->scheduler);
}

void trap(struct trapframe *tf) {
    ..
    if(tf->trapno == T_IRQ0+IRQ_TIMER)
        yield();
}
```



Context switching in action: taking control

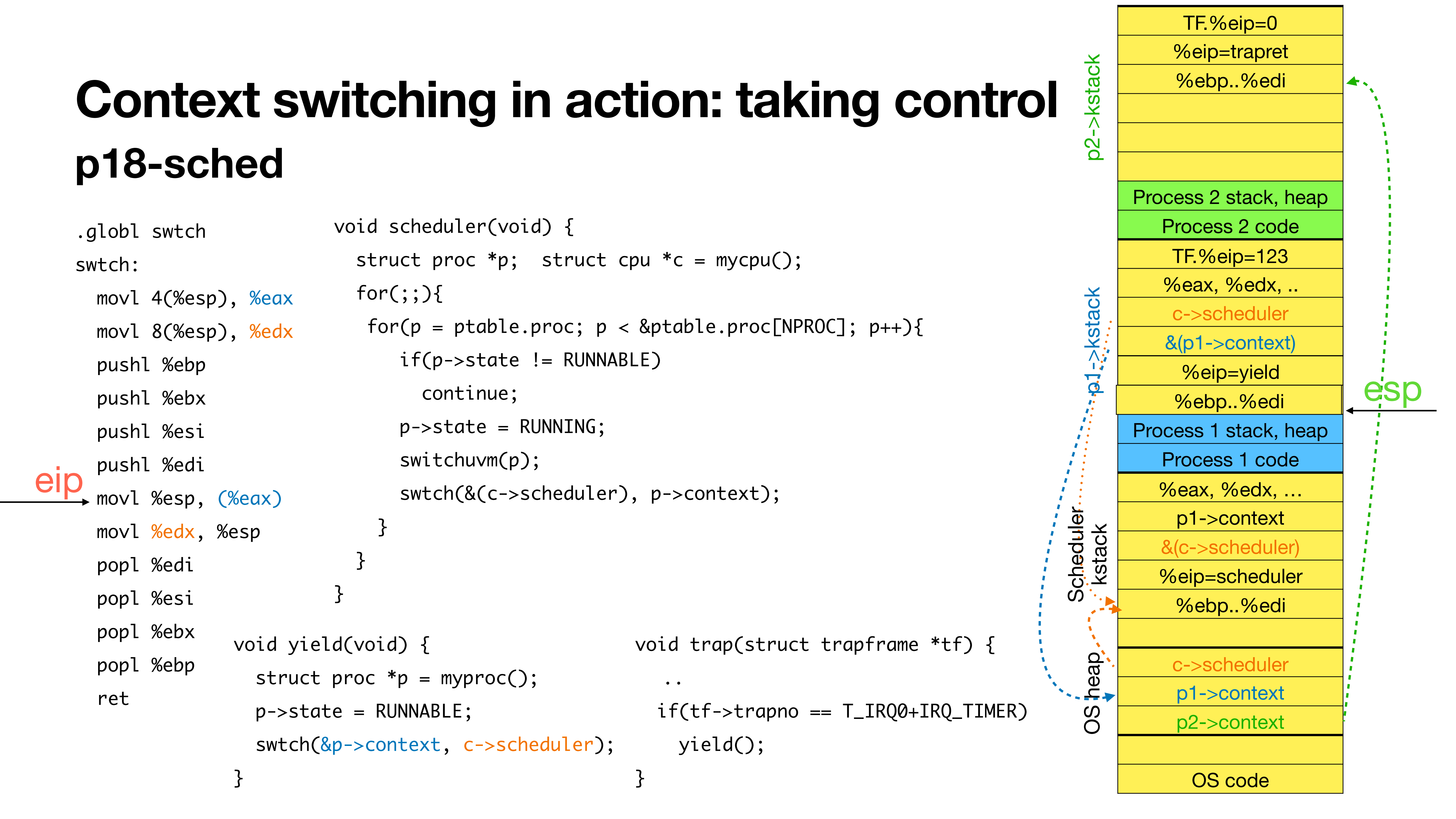
p18-sched

```
.globl swtch
swtch:
    movl 4(%esp), %eax
    movl 8(%esp), %edx
    pushl %ebp
    pushl %ebx
    pushl %esi
    pushl %edi
    movl %esp, (%eax)
    movl %edx, %esp
    popl %edi
    popl %esi
    popl %ebx
    popl %ebp
    ret

void scheduler(void) {
    struct proc *p; struct cpu *c = mycpu();
    for(;;){
        for(p = ptable.proc; p < &ptable.proc[NPROC]; p++){
            if(p->state != RUNNABLE)
                continue;
            p->state = RUNNING;
            switchvm(p);
            swtch(&(c->scheduler), p->context);
        }
    }
}

void yield(void) {
    struct proc *p = myproc();
    p->state = RUNNABLE;
    swtch(&p->context, c->scheduler);
}

void trap(struct trapframe *tf) {
    ..
    if(tf->trapno == T_IRQ0+IRQ_TIMER)
        yield();
}
```



Context switching in action: taking control

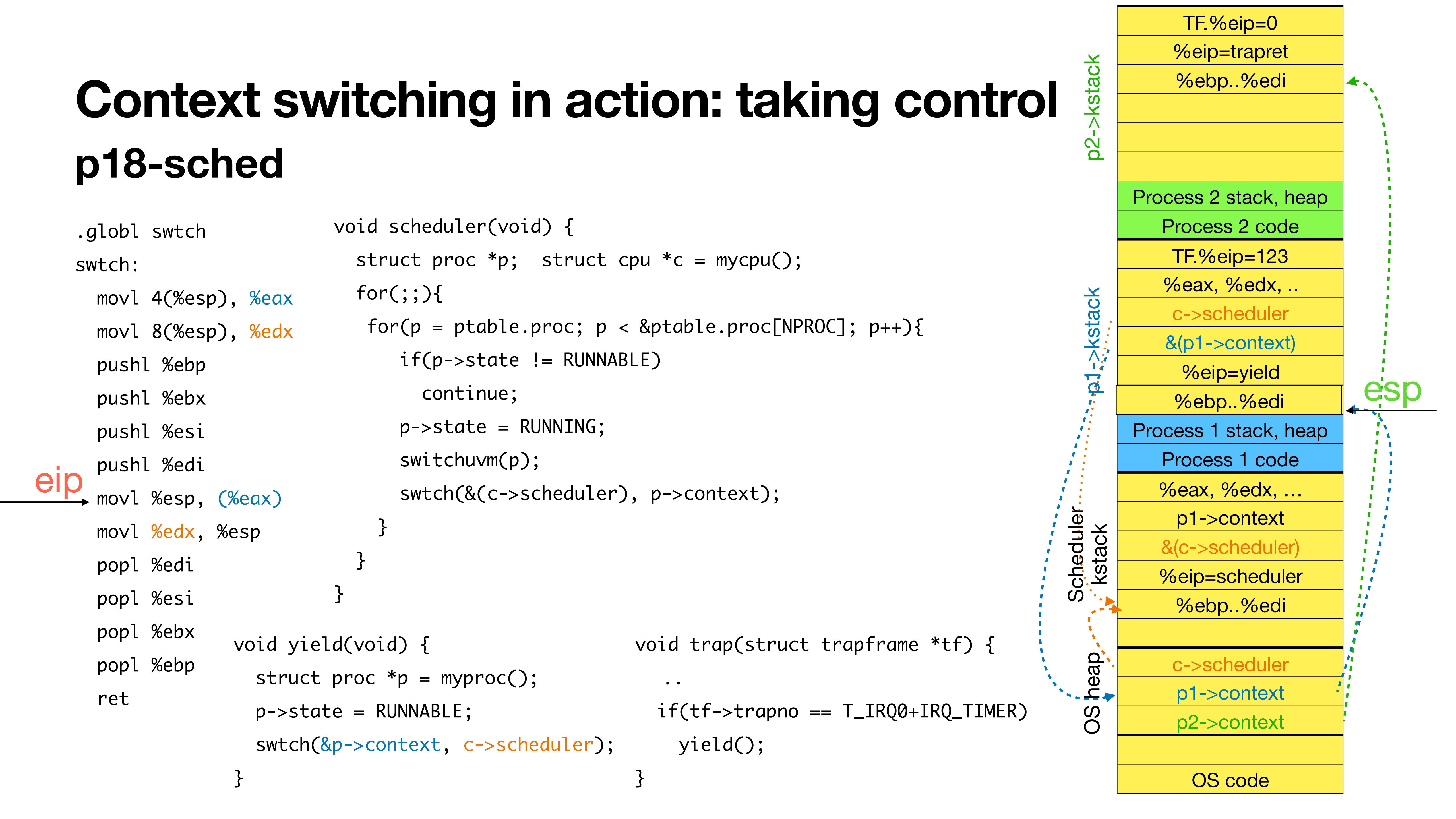
p18-sched

```
.globl swtch
swtch:
    movl 4(%esp), %eax
    movl 8(%esp), %edx
    pushl %ebp
    pushl %ebx
    pushl %esi
    pushl %edi
    movl %esp, (%eax)
    movl %edx, %esp
    popl %edi
    popl %esi
    popl %ebx
    popl %ebp
    ret

void scheduler(void) {
    struct proc *p; struct cpu *c = mycpu();
    for(;;){
        for(p = ptable.proc; p < &ptable.proc[NPROC]; p++){
            if(p->state != RUNNABLE)
                continue;
            p->state = RUNNING;
            switchvm(p);
            swtch(&(c->scheduler), p->context);
        }
    }
}

void yield(void) {
    struct proc *p = myproc();
    p->state = RUNNABLE;
    swtch(&p->context, c->scheduler);
}

void trap(struct trapframe *tf) {
    ..
    if(tf->trapno == T_IRQ0+IRQ_TIMER)
        yield();
}
```



Context switching in action: taking control

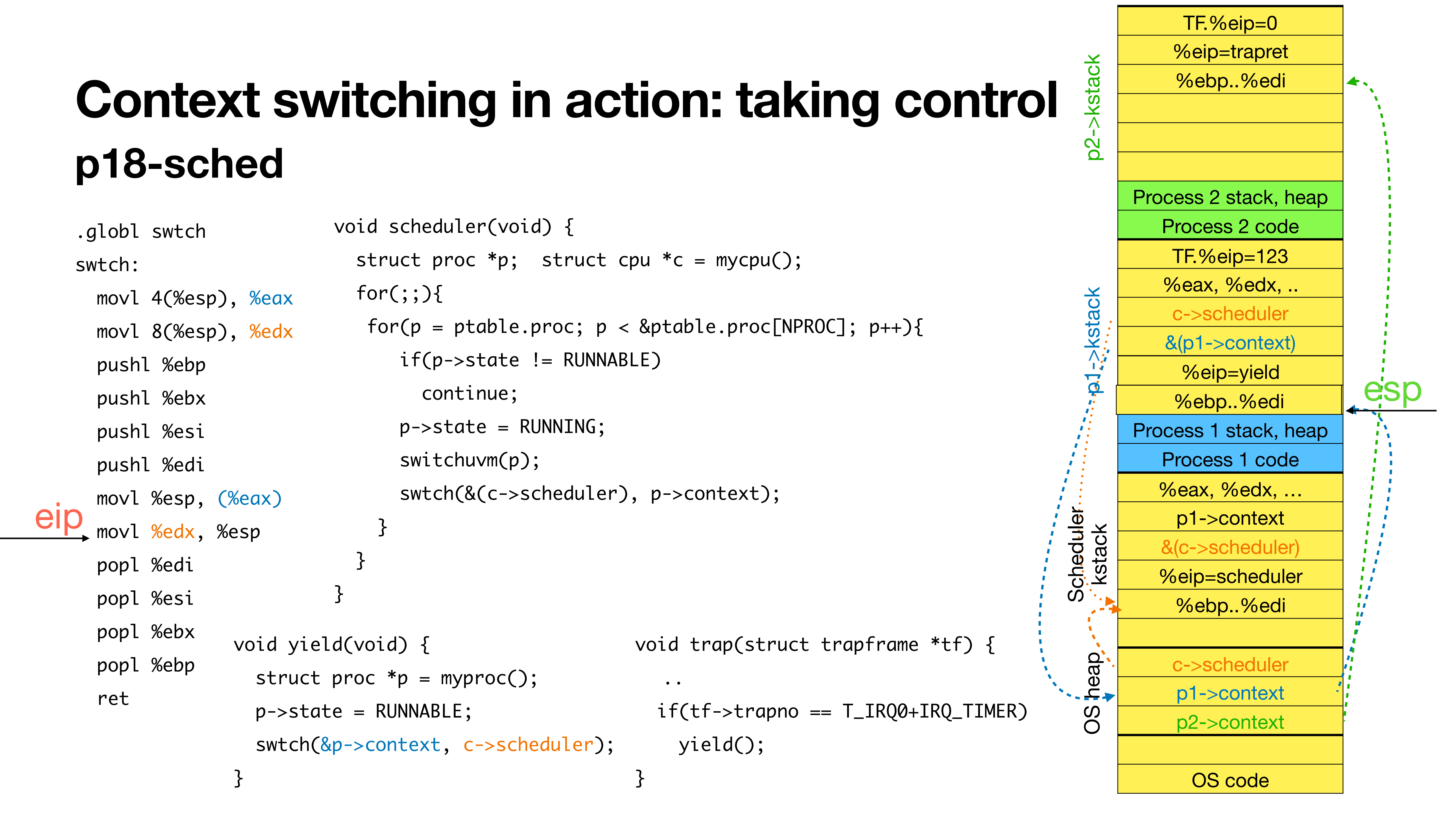
p18-sched

```
.globl swtch
swtch:
    movl 4(%esp), %eax
    movl 8(%esp), %edx
    pushl %ebp
    pushl %ebx
    pushl %esi
    pushl %edi
    movl %esp, (%eax)
    movl %edx, %esp
    popl %edi
    popl %esi
    popl %ebx
    popl %ebp
    ret

void scheduler(void) {
    struct proc *p; struct cpu *c = mycpu();
    for(;;){
        for(p = ptable.proc; p < &ptable.proc[NPROC]; p++){
            if(p->state != RUNNABLE)
                continue;
            p->state = RUNNING;
            switchvm(p);
            swtch(&(c->scheduler), p->context);
        }
    }
}

void yield(void) {
    struct proc *p = myproc();
    p->state = RUNNABLE;
    swtch(&p->context, c->scheduler);
}

void trap(struct trapframe *tf) {
    ..
    if(tf->trapno == T_IRQ0+IRQ_TIMER)
        yield();
}
```



Context switching in action: taking control

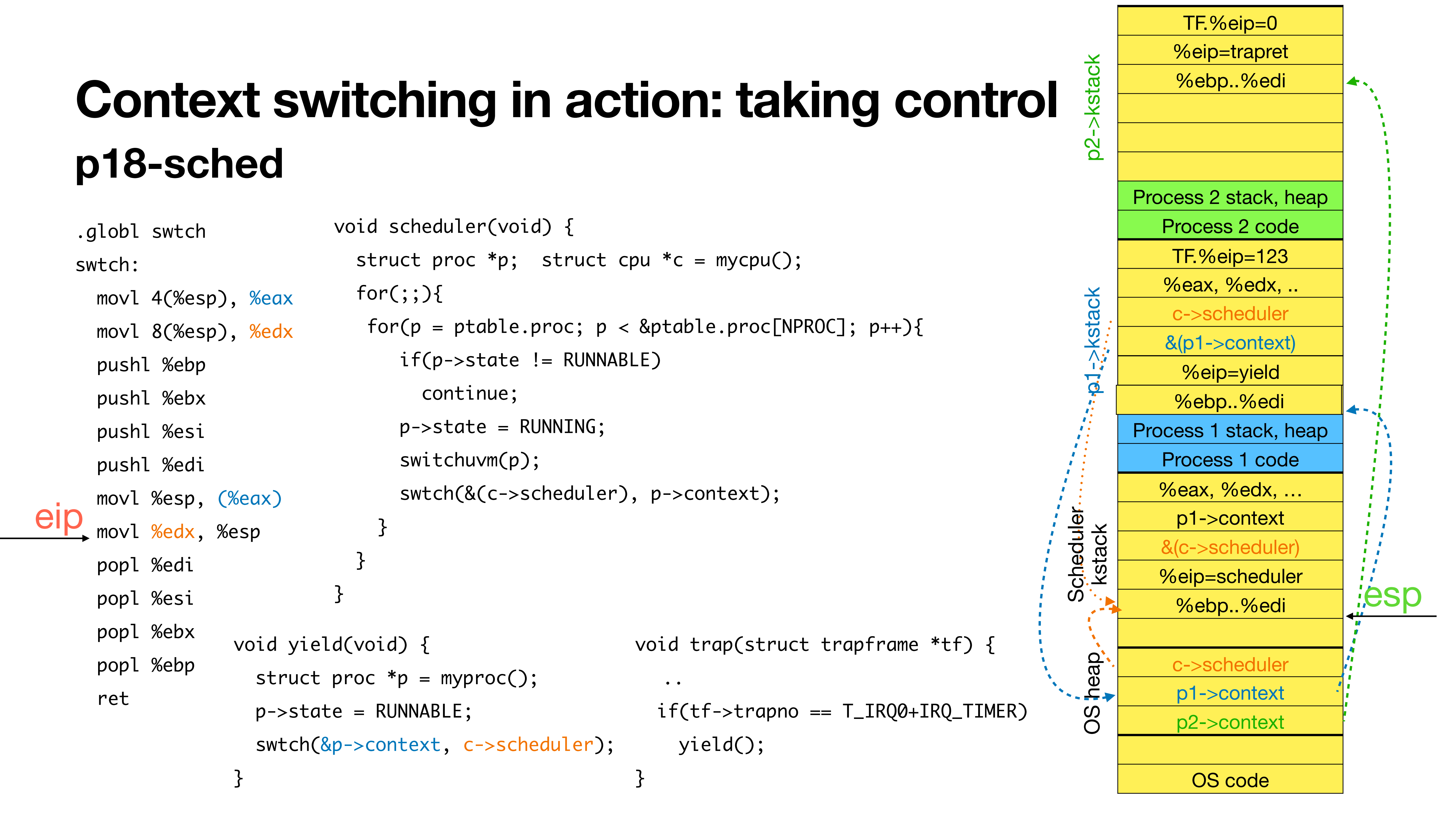
p18-sched

```
.globl swtch
swtch:
    movl 4(%esp), %eax
    movl 8(%esp), %edx
    pushl %ebp
    pushl %ebx
    pushl %esi
    pushl %edi
    movl %esp, (%eax)
    movl %edx, %esp
    popl %edi
    popl %esi
    popl %ebx
    popl %ebp
    ret

void scheduler(void) {
    struct proc *p; struct cpu *c = mycpu();
    for(;;){
        for(p = ptable.proc; p < &ptable.proc[NPROC]; p++){
            if(p->state != RUNNABLE)
                continue;
            p->state = RUNNING;
            switchvm(p);
            swtch(&(c->scheduler), p->context);
        }
    }
}

void yield(void) {
    struct proc *p = myproc();
    p->state = RUNNABLE;
    swtch(&p->context, c->scheduler);
}

void trap(struct trapframe *tf) {
    ..
    if(tf->trapno == T_IRQ0+IRQ_TIMER)
        yield();
}
```



Context switching in action: taking control

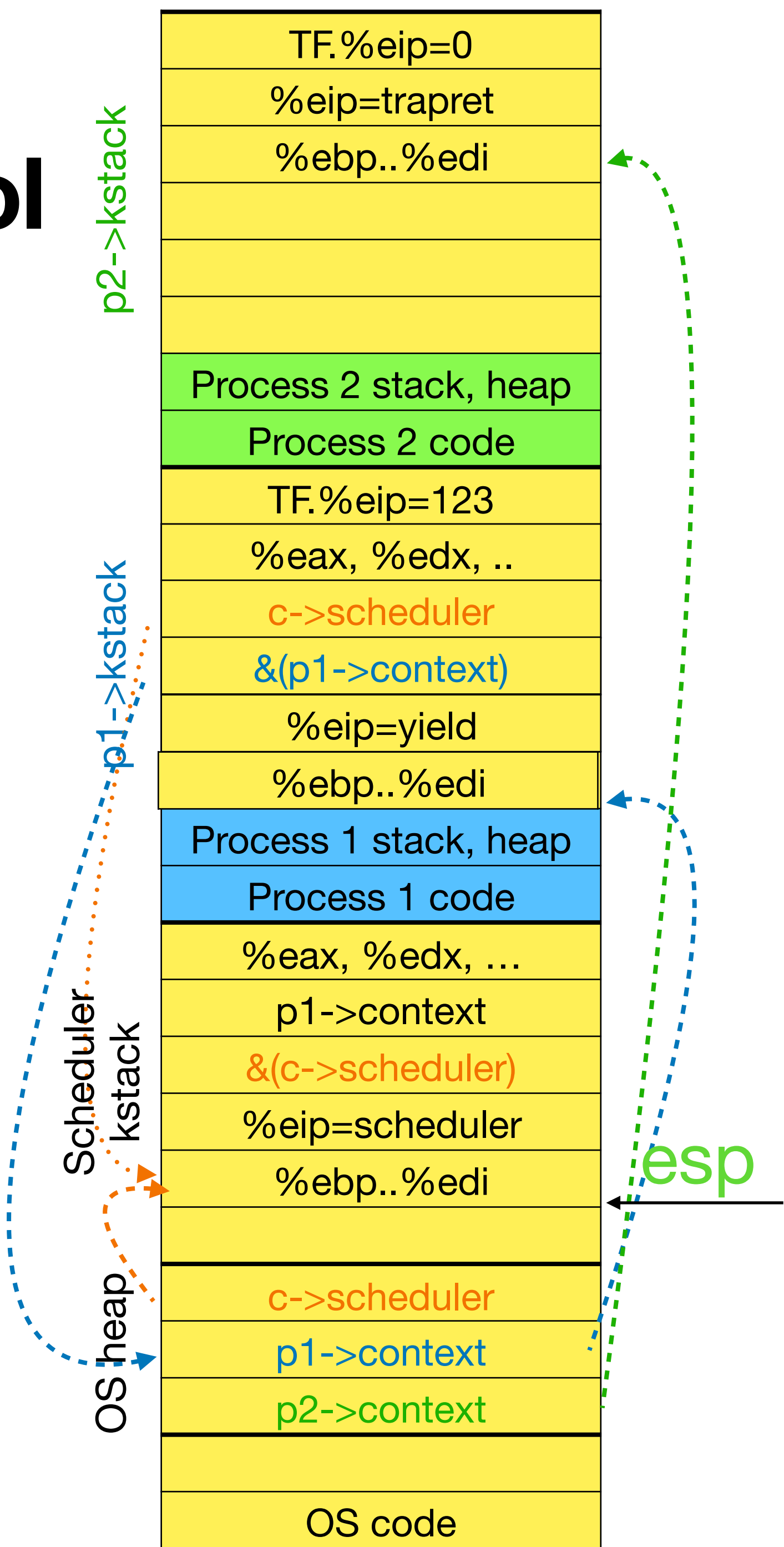
p18-sched

```
.globl swtch
swtch:
    movl 4(%esp), %eax
    movl 8(%esp), %edx
    pushl %ebp
    pushl %ebx
    pushl %esi
    pushl %edi
    movl %esp, (%eax)
    movl %edx, %esp
    popl %edi
    popl %esi
    popl %ebx
    popl %ebp
    ret

void scheduler(void) {
    struct proc *p; struct cpu *c = mycpu();
    for(;;){
        for(p = ptable.proc; p < &ptable.proc[NPROC]; p++){
            if(p->state != RUNNABLE)
                continue;
            p->state = RUNNING;
            switchvm(p);
            swtch(&(c->scheduler), p->context);
        }
    }
}

void yield(void) {
    struct proc *p = myproc();
    p->state = RUNNABLE;
    swtch(&p->context, c->scheduler);
}

void trap(struct trapframe *tf) {
    ..
    if(tf->trapno == T_IRQ0+IRQ_TIMER)
        yield();
}
```



Context switching in action: taking control

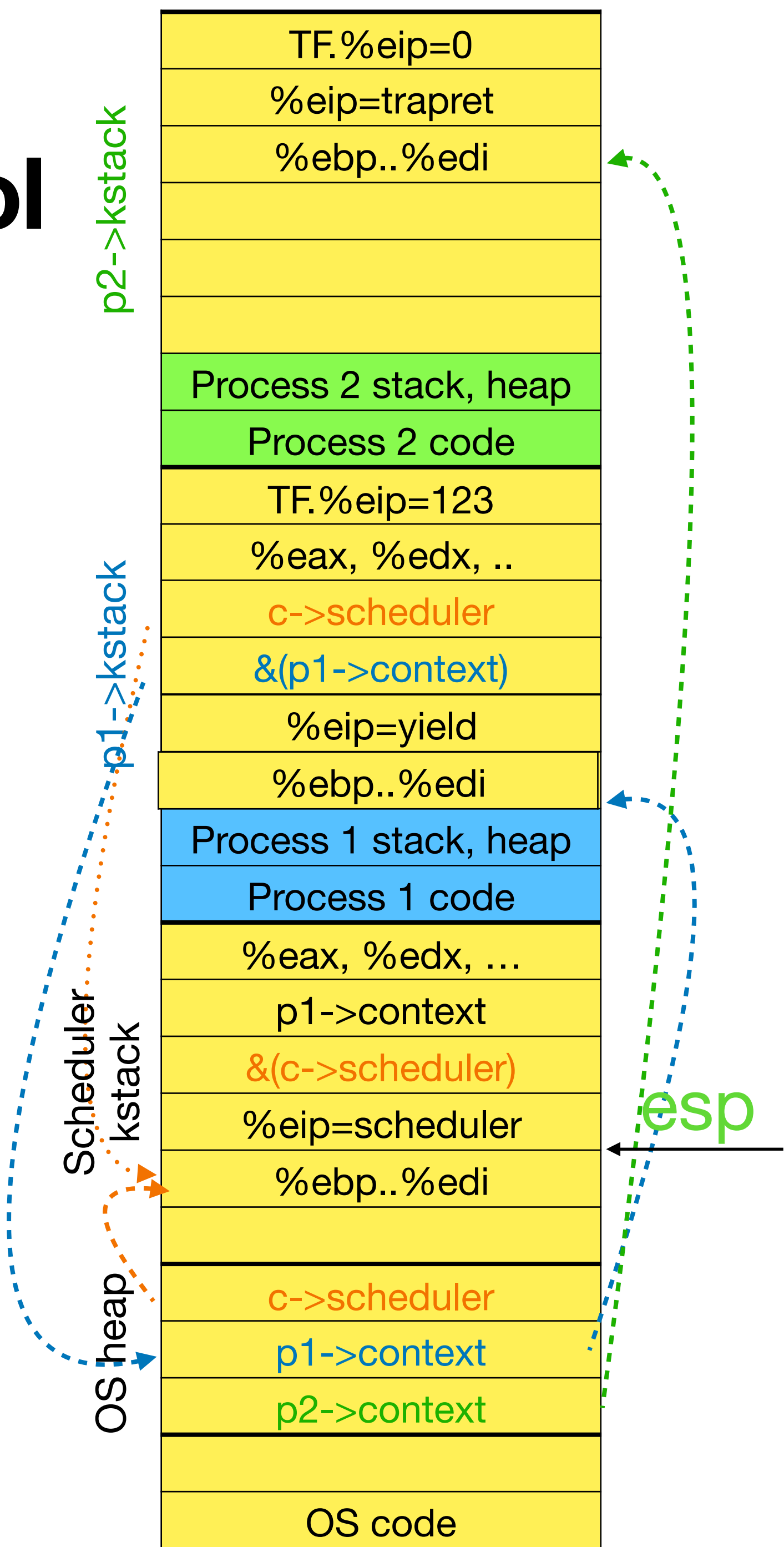
p18-sched

```
.globl swtch
swtch:
    movl 4(%esp), %eax
    movl 8(%esp), %edx
    pushl %ebp
    pushl %ebx
    pushl %esi
    pushl %edi
    movl %esp, (%eax)
    movl %edx, %esp
    popl %edi
    popl %esi
    popl %ebx
    popl %ebp
    ret

void scheduler(void) {
    struct proc *p; struct cpu *c = mycpu();
    for(;;){
        for(p = ptable.proc; p < &ptable.proc[NPROC]; p++){
            if(p->state != RUNNABLE)
                continue;
            p->state = RUNNING;
            switchvm(p);
            swtch(&(c->scheduler), p->context);
        }
    }
}

void yield(void) {
    struct proc *p = myproc();
    p->state = RUNNABLE;
    swtch(&p->context, c->scheduler);
}

void trap(struct trapframe *tf) {
    ..
    if(tf->trapno == T_IRQ0+IRQ_TIMER)
        yield();
}
```



Context switching in action: taking control

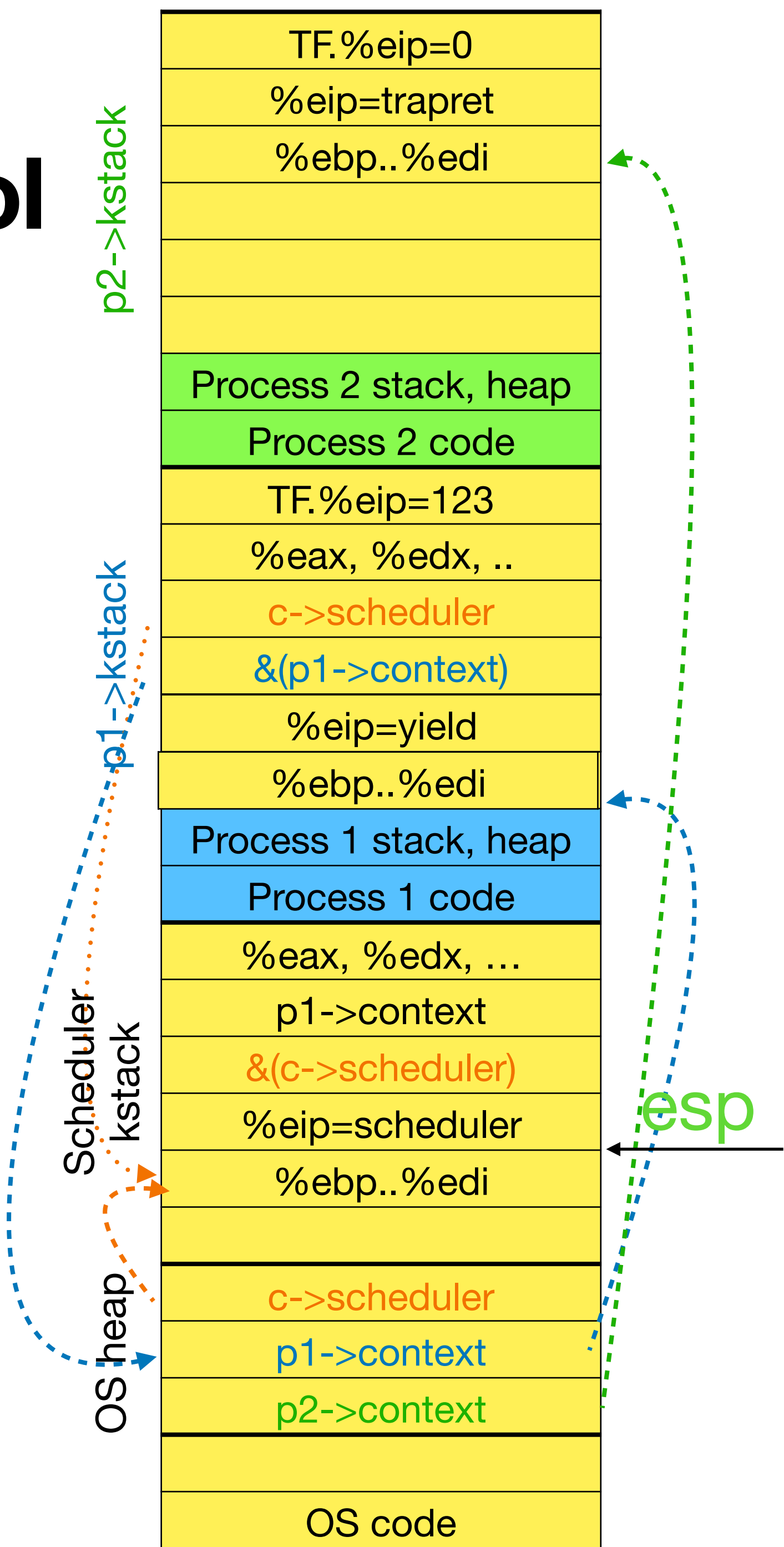
p18-sched

```
.globl swtch
swtch:
    movl 4(%esp), %eax
    movl 8(%esp), %edx
    pushl %ebp
    pushl %ebx
    pushl %esi
    pushl %edi
    movl %esp, (%eax)
    movl %edx, %esp
    popl %edi
    popl %esi
    popl %ebx
    popl %ebp
    ret

void scheduler(void) {
    struct proc *p; struct cpu *c = mycpu();
    for(;;){
        for(p = ptable.proc; p < &ptable.proc[NPROC]; p++){
            if(p->state != RUNNABLE)
                continue;
            p->state = RUNNING;
            switchvm(p);
            swtch(&(c->scheduler), p->context);
        }
    }
}

void yield(void) {
    struct proc *p = myproc();
    p->state = RUNNABLE;
    swtch(&p->context, c->scheduler);
}

void trap(struct trapframe *tf) {
    ..
    if(tf->trapno == T_IRQ0+IRQ_TIMER)
        yield();
}
```



Context switching in action: taking control

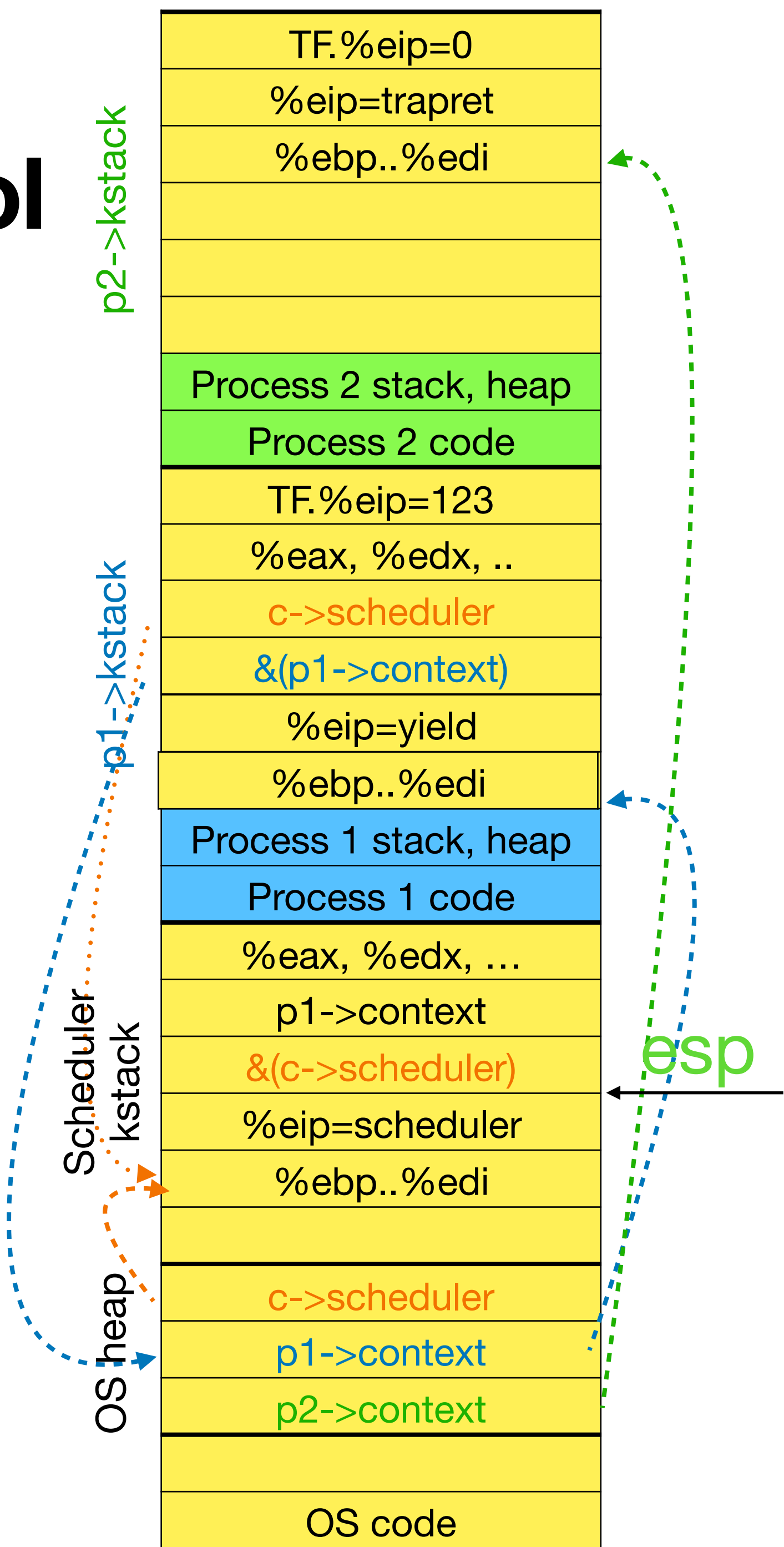
p18-sched

```
.globl swtch
swtch:
    movl 4(%esp), %eax
    movl 8(%esp), %edx
    pushl %ebp
    pushl %ebx
    pushl %esi
    pushl %edi
    movl %esp, (%eax)
    movl %edx, %esp
    popl %edi
    popl %esi
    popl %ebx
    popl %ebp
    ret

void scheduler(void) {
    struct proc *p; struct cpu *c = mycpu();
    for(;;){
        for(p = ptable.proc; p < &ptable.proc[NPROC]; p++){
            if(p->state != RUNNABLE)
                continue;
            p->state = RUNNING;
            switchvm(p);
            swtch(&(c->scheduler), p->context);
        }
    }
}

void yield(void) {
    struct proc *p = myproc();
    p->state = RUNNABLE;
    swtch(&p->context, c->scheduler);
}

void trap(struct trapframe *tf) {
    ..
    if(tf->trapno == T_IRQ0+IRQ_TIMER)
        yield();
}
```



Context switching in action: taking control

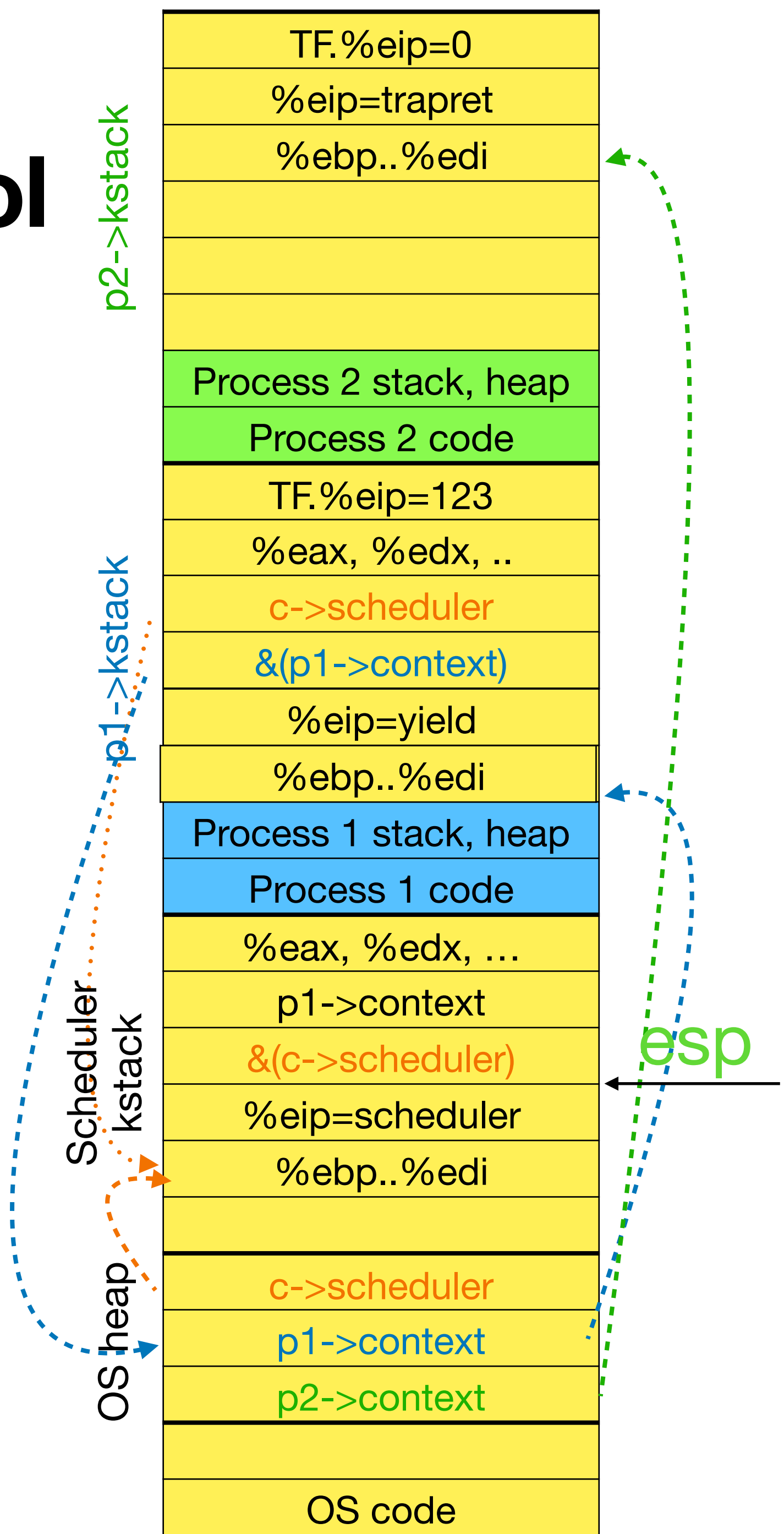
p18-sched

```
.globl swtch
swtch:
    movl 4(%esp), %eax
    movl 8(%esp), %edx
    pushl %ebp
    pushl %ebx
    pushl %esi
    pushl %edi
    movl %esp, (%eax)
    movl %edx, %esp
    popl %edi
    popl %esi
    popl %ebx
    popl %ebp
    ret

void scheduler(void) {
    struct proc *p; struct cpu *c = mycpu();
    for(;;){
        for(p = ptable.proc; p < &ptable.proc[NPROC]; p++){
            if(p->state != RUNNABLE)
                continue;
            p->state = RUNNING;
            switchvm(p);
            swtch(&(c->scheduler), p->context);
        }
    }
}

void yield(void) {
    struct proc *p = myproc();
    p->state = RUNNABLE;
    swtch(&p->context, c->scheduler);
}

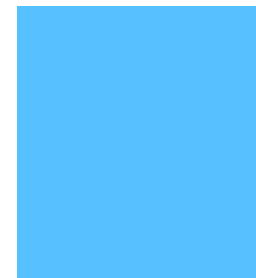
void trap(struct trapframe *tf) {
    ..
    if(tf->trapno == T_IRQ0+IRQ_TIMER)
        yield();
}
```



System calls

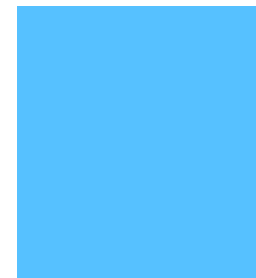
System calls

- Process wants OS to work on its behalf



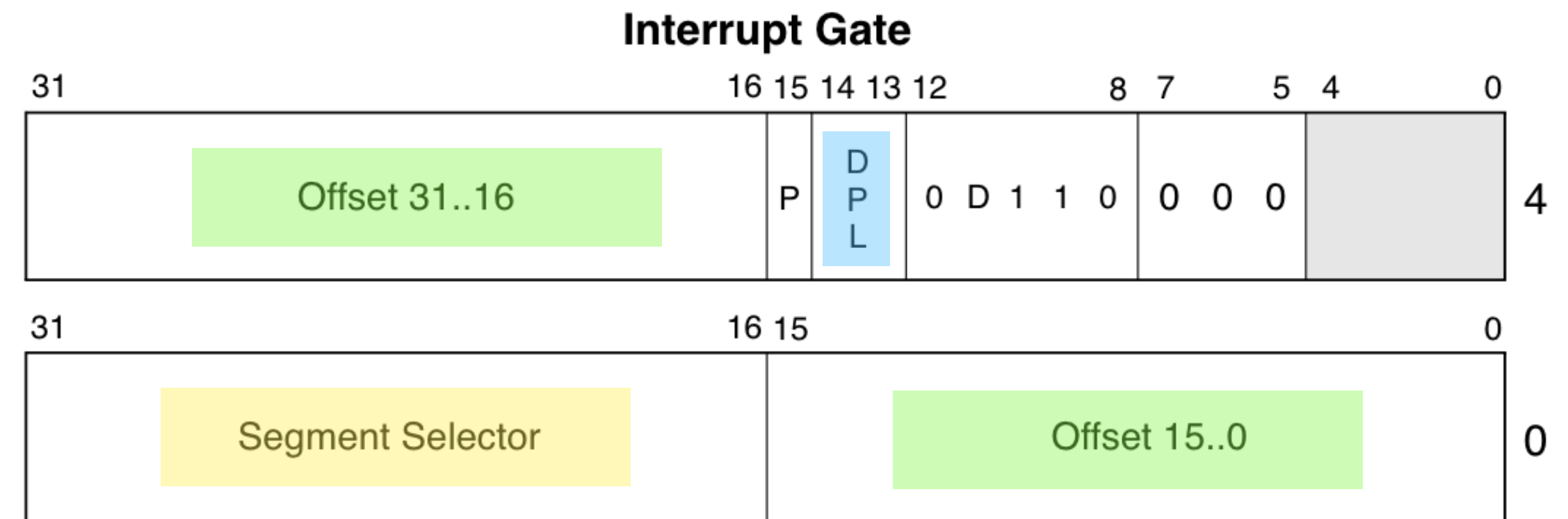
System calls

- Process wants OS to work on its behalf
 - Run INT xx instruction to trigger interrupt number xx



System calls

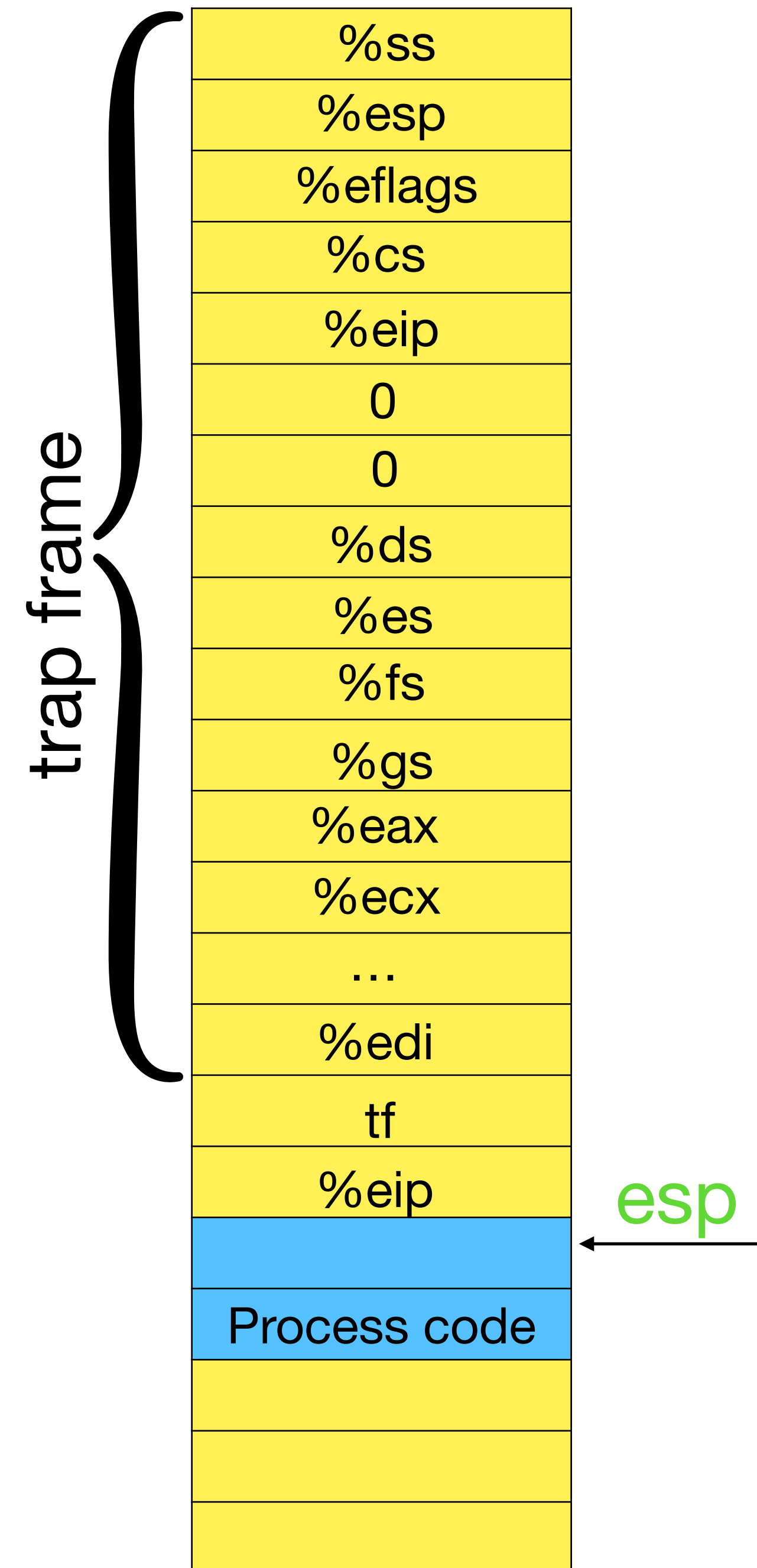
- Process wants OS to work on its behalf
 - Run INT xx instruction to trigger interrupt number xx
 - INT xx is allowed if $CPL \leq DPL$ for the xx interrupt vector



Code walkthrough

p19-syscall

- trap.c
 - tvinit sets DPL of IDT entry for T_SYSCALL to 3
 - trap calls syscall if interrupt vector is T_SYSCALL
- syscall.c
 - syscall reads eax from trap frame to find which syscall is made and calls that particular call
 - Return value of sys call is set in trap frame's eax
- initproc.S
 - Calls three system calls: open “console” file, write “hello world”, close



Syscall handling

```
void tvinit(void) {  
    int i;  
    for(i = 0; i < 256; i++)  
        SETGATE(idt[i], 0, SEG_KCODE<<3, vectors[i], 0);  
    SETGATE(idt[T_SYSCALL], 0, SEG_KCODE<<3, vectors[T_SYSCALL], DPL_USER);  
}
```

trap.c

```
void trap(struct trapframe *tf) {  
    if(tf->trapno == T_SYSCALL){  
        myproc()->tf = tf;  
        syscall();  
        return;  
    }  
    ...  
}
```

```
void syscall(void) {  
    int num;  
    struct proc *curproc = myproc();  
  
    num = curproc->tf->eax;  
    if(num > 0 && num < NELEM(syscalls) && syscalls[num]) {  
        curproc->tf->eax = syscalls[num]();  
    } ...  
}
```

syscall.c

Visualising syscall handling p19-syscall

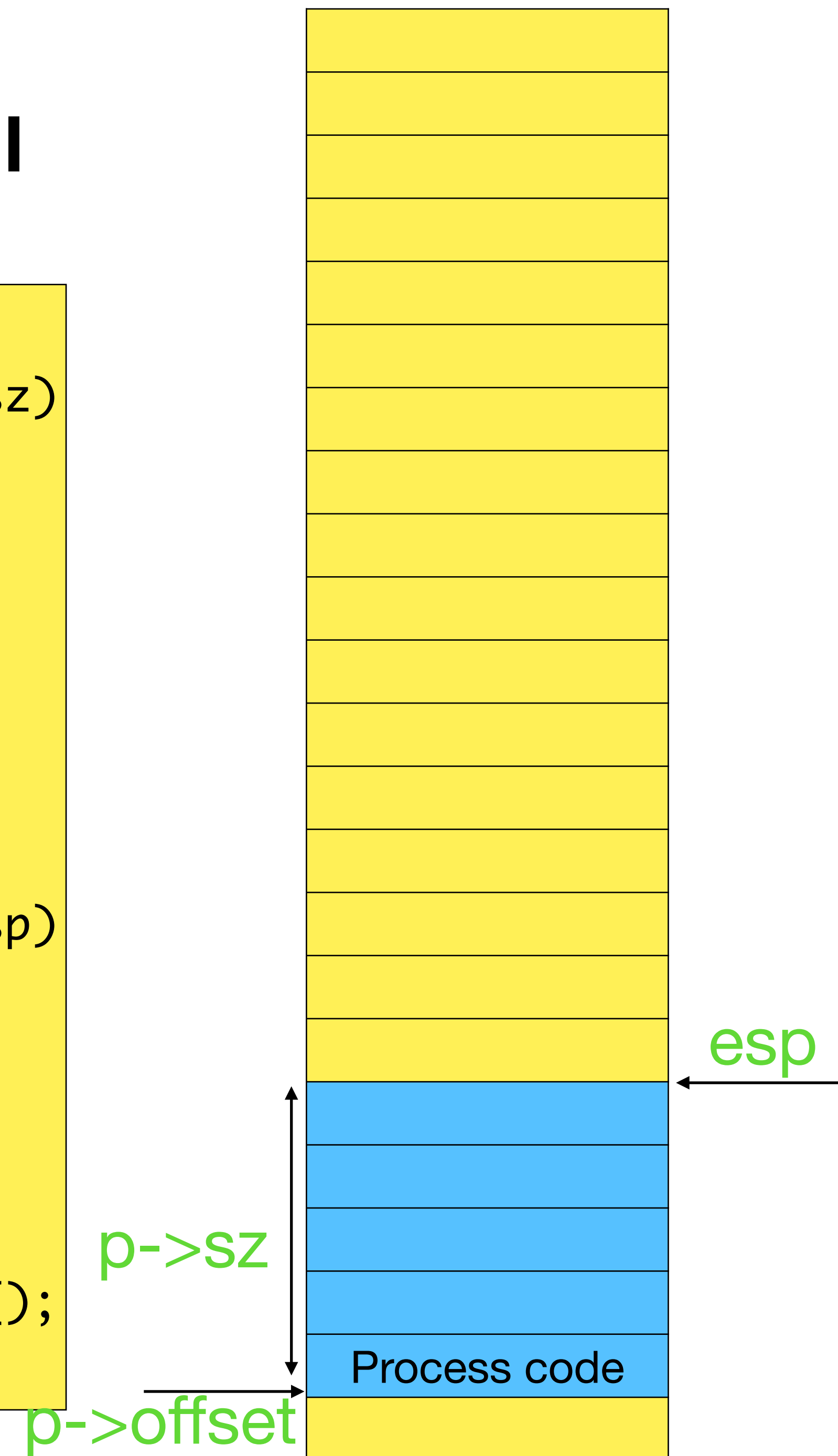
```
# fd=open("console", O_WRONLY)
  eip→ pushl $1
    pushl $console
    pushl $0
    movl $SYS_open, %eax
    int $T_SYSCALL
    # fd = %eax
```

```
int sys_open(void) {
    int fd, omode;
    if(argint(1, &omode) < 0) {
        return -1;
    }
    ..
    return fd;
}
```

```
int fetchint(uint addr, int *ip) {
    if(addr >= p->sz || addr+4 > p->sz)
        return -1;
    *ip = *(int*)(addr + p->offset);
    return 0;
}
```

```
int argint(int n, int *ip) {
    return fetchint((myproc()->tf->esp)
                    + 4 + 4*n, ip);
}

void syscall(void) {
    int num = curproc->tf->eax;
    curproc->tf->eax = syscalls[num]();
}
```



Visualising syscall handling p19-syscall

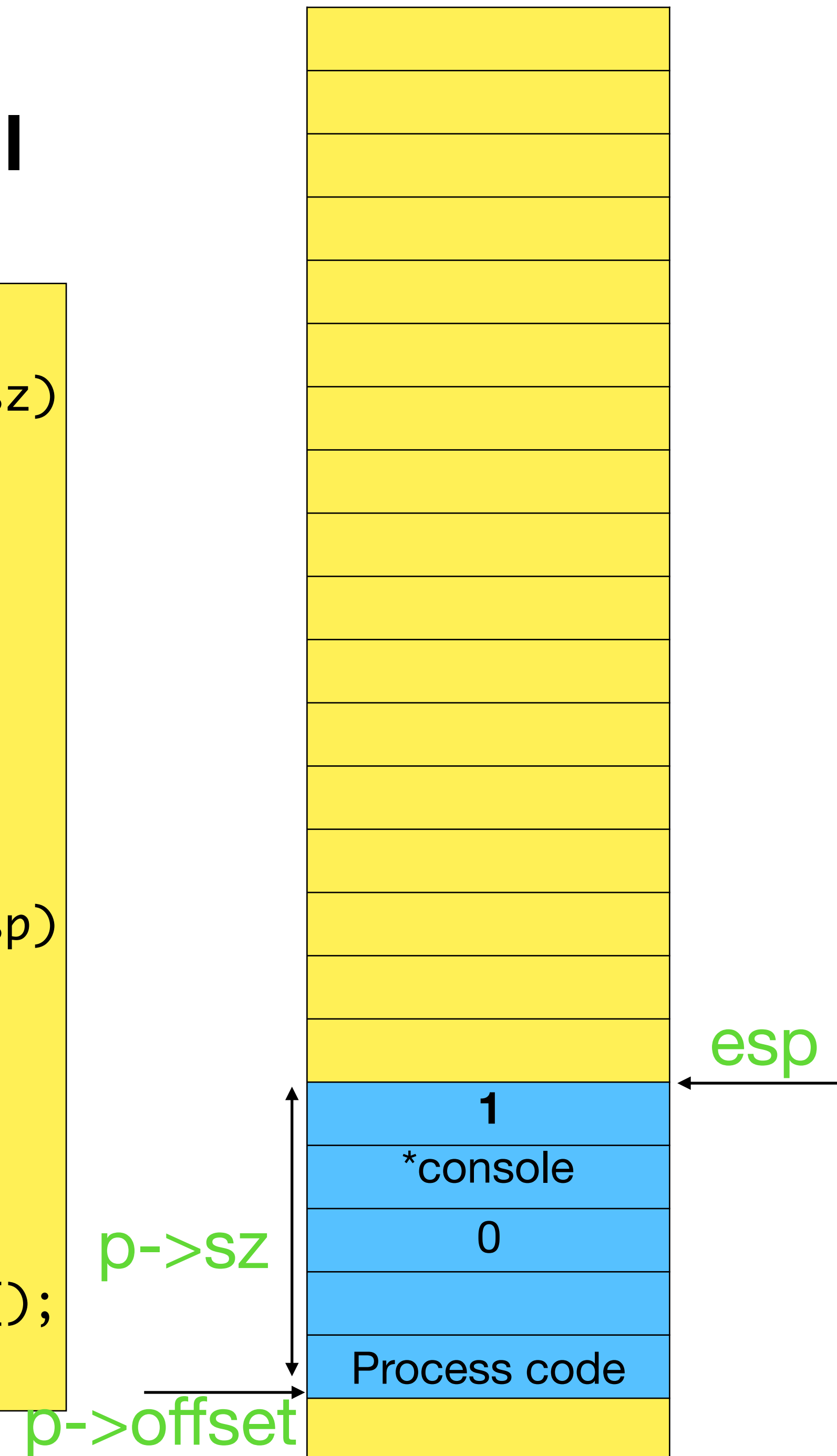
```
# fd=open("console", O_WRONLY)
  eip→ pushl $1
    pushl $console
    pushl $0
    movl $SYS_open, %eax
    int $T_SYSCALL
    # fd = %eax
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int sys_open(void) {
    int fd, omode;
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        return -1;
    }
    ..
    return fd;
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int fetchint(uint addr, int *ip) {
    if(addr >= p->sz || addr+4 > p->sz)
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int argint(int n, int *ip) {
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    int num = curproc->tf->eax;
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Visualising syscall handling p19-syscall

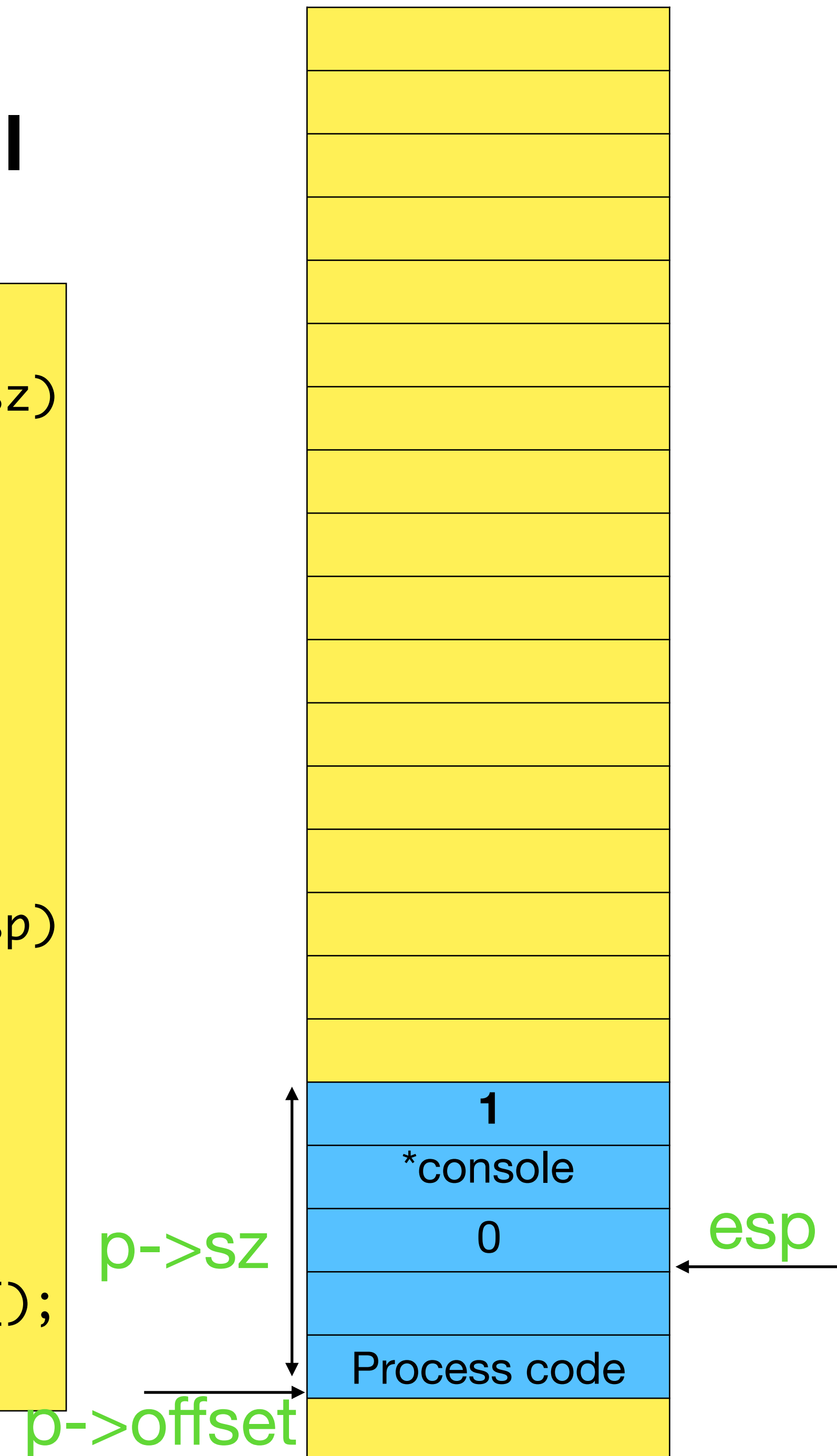
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Visualising syscall handling p19-syscall

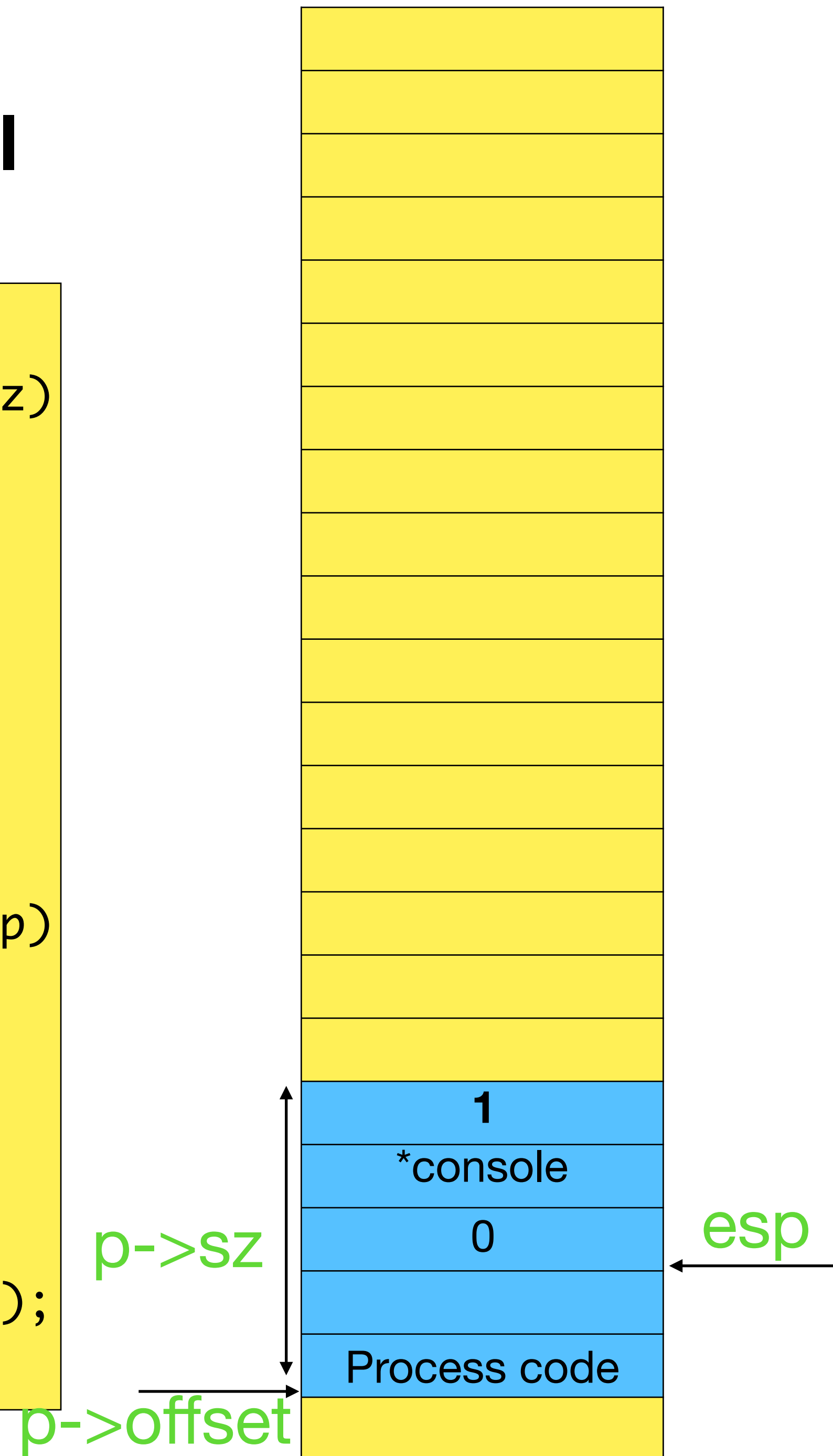
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Visualising syscall handling p19-syscall

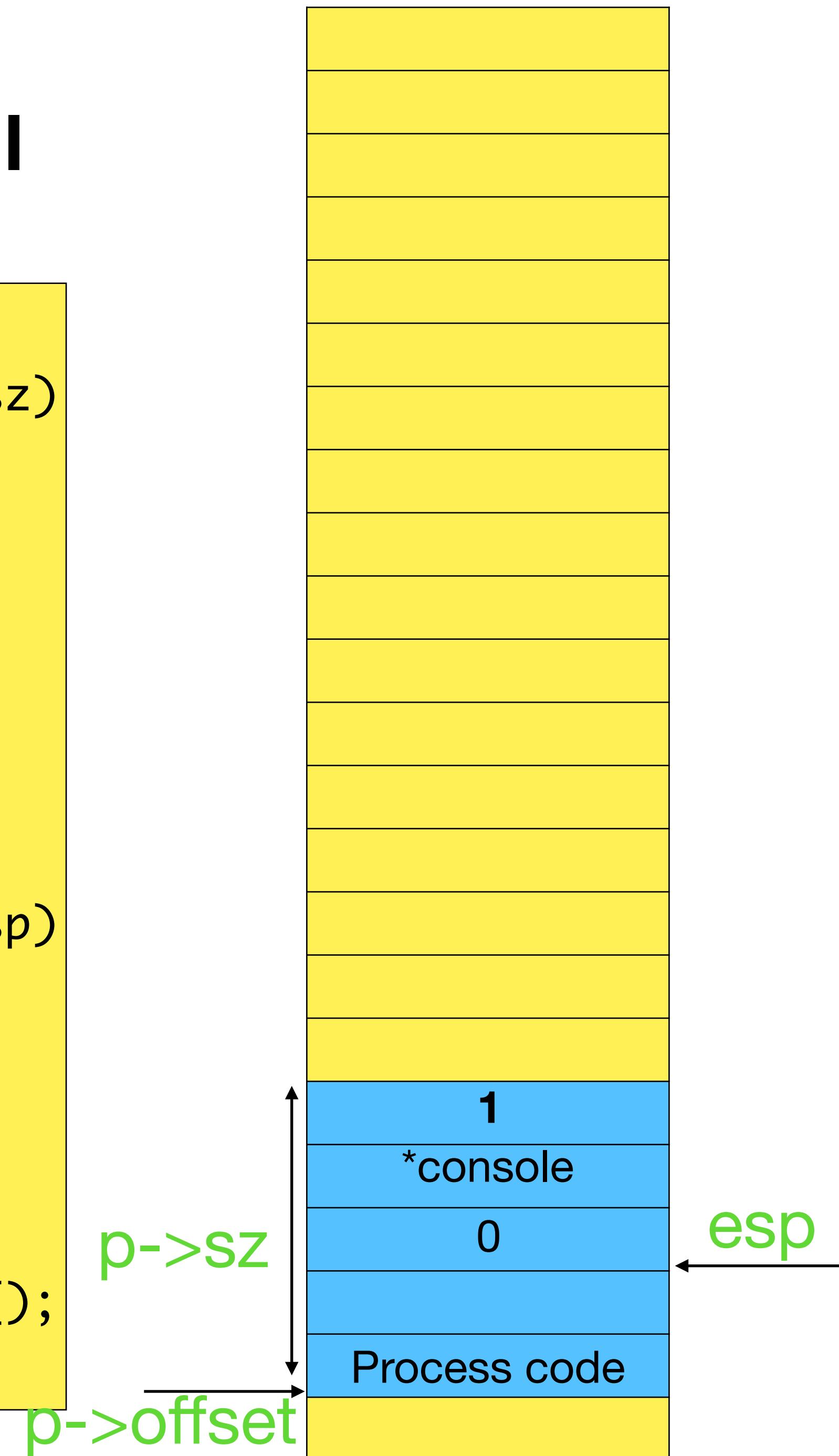
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    pushl $1
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    pushl $0
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Visualising syscall handling p19-syscall

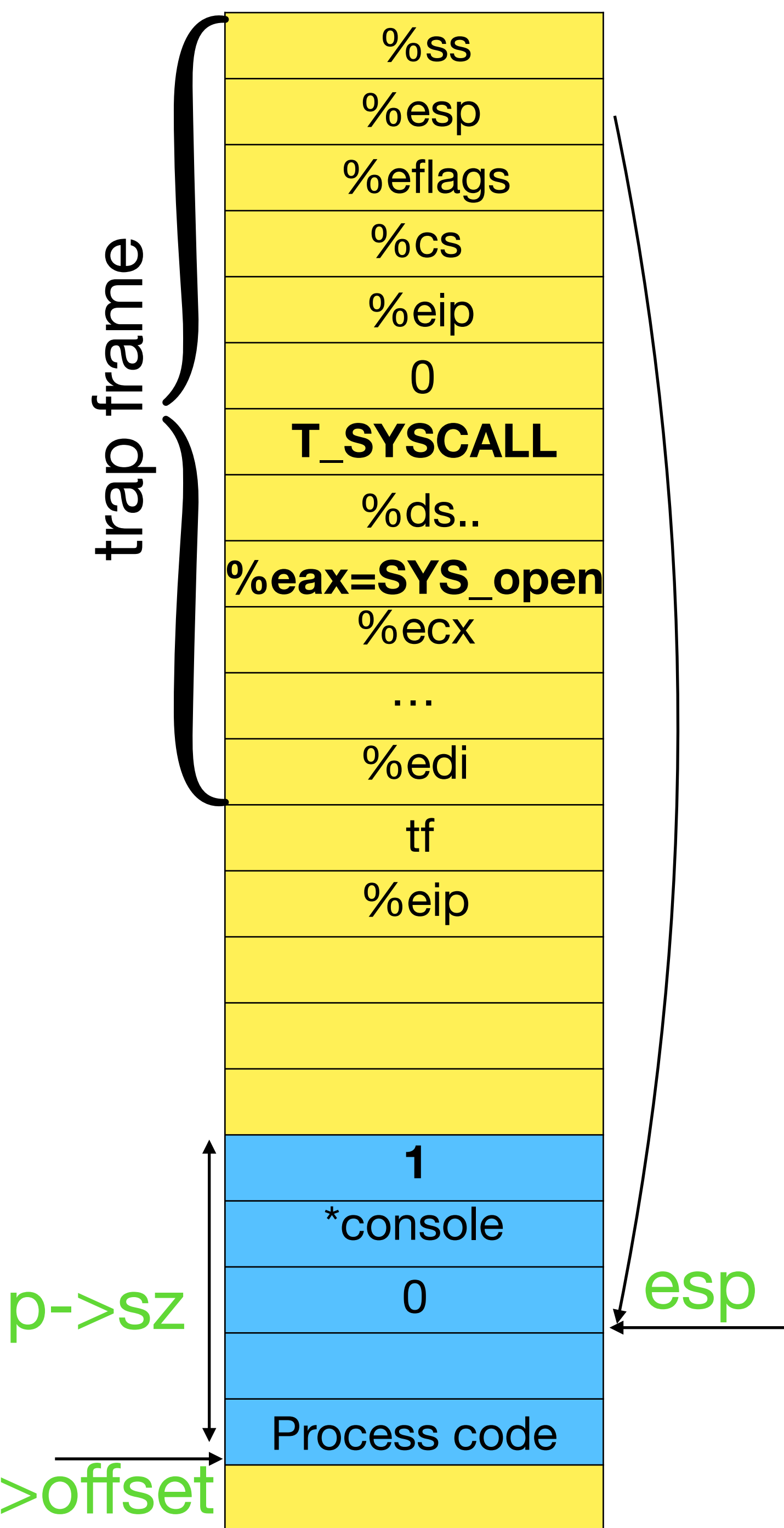
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    pushl $console
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    movl $SYS_open, %eax
    → eip int $T_SYSCALL
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Visualising syscall handling p19-syscall

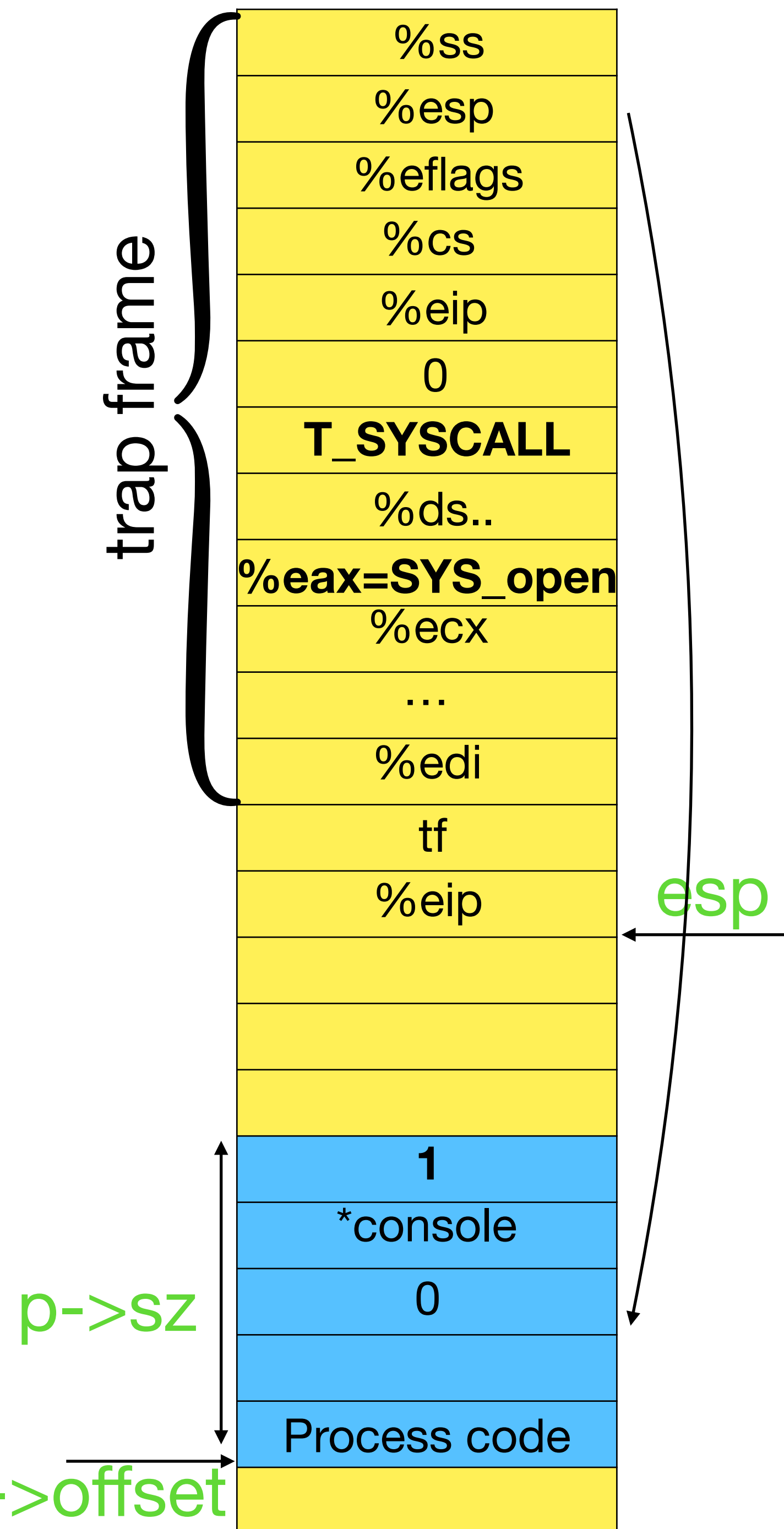
```
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    pushl $1
    pushl $console
    pushl $0
    movl $SYS_open, %eax
    eip → int $T_SYSCALL
    # fd = %eax
```

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    int fd, omode;
    if(argint(1, &omode) < 0) {
        return -1;
    }
    ..
    return fd;
}
```

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Visualising syscall handling p19-syscall

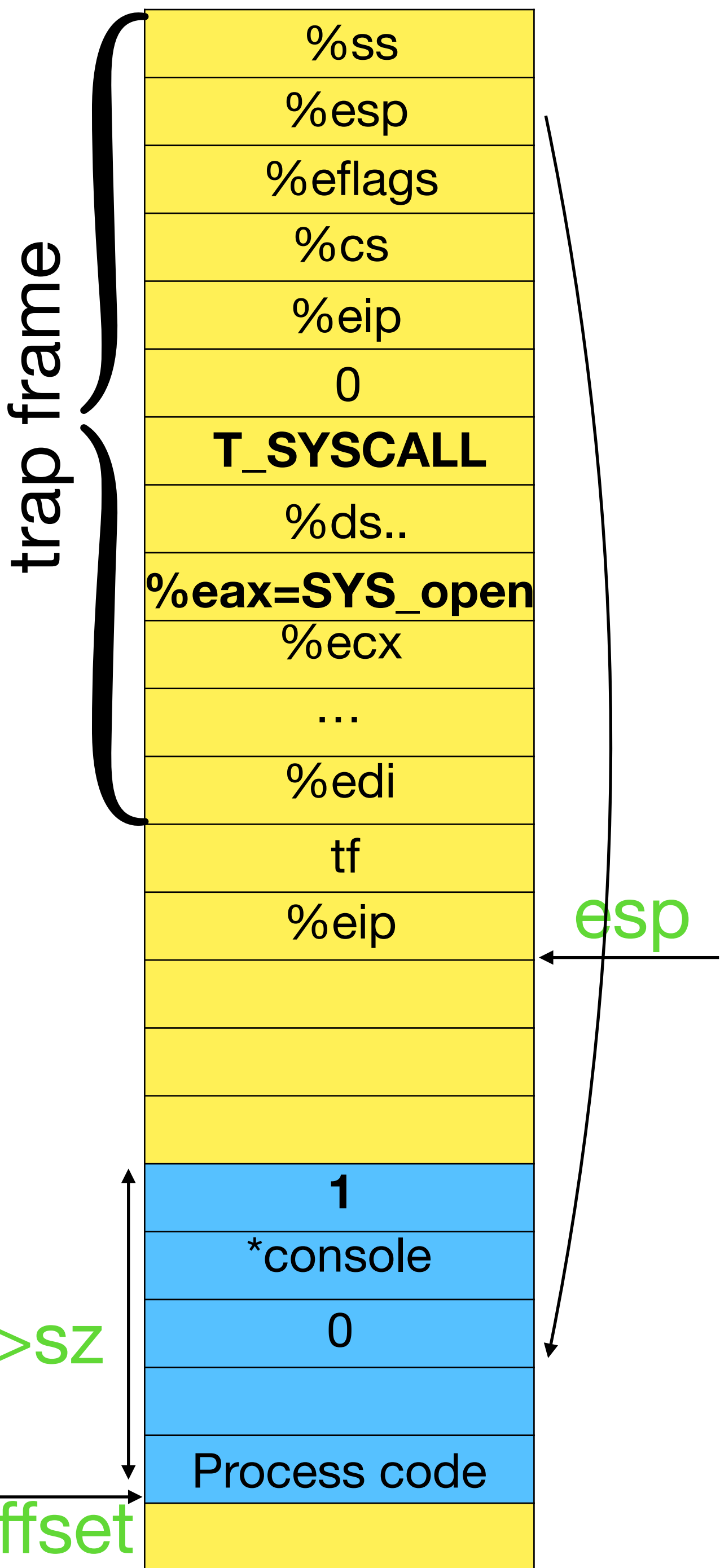
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    pushl $0
    movl $SYS_open, %eax
    int $T_SYSCALL
    # fd = %eax
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int sys_open(void) {
    int fd, omode;
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    }
    ..
    return fd;
}
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int fetchint(uint addr, int *ip) {
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Visualising syscall handling p19-syscall

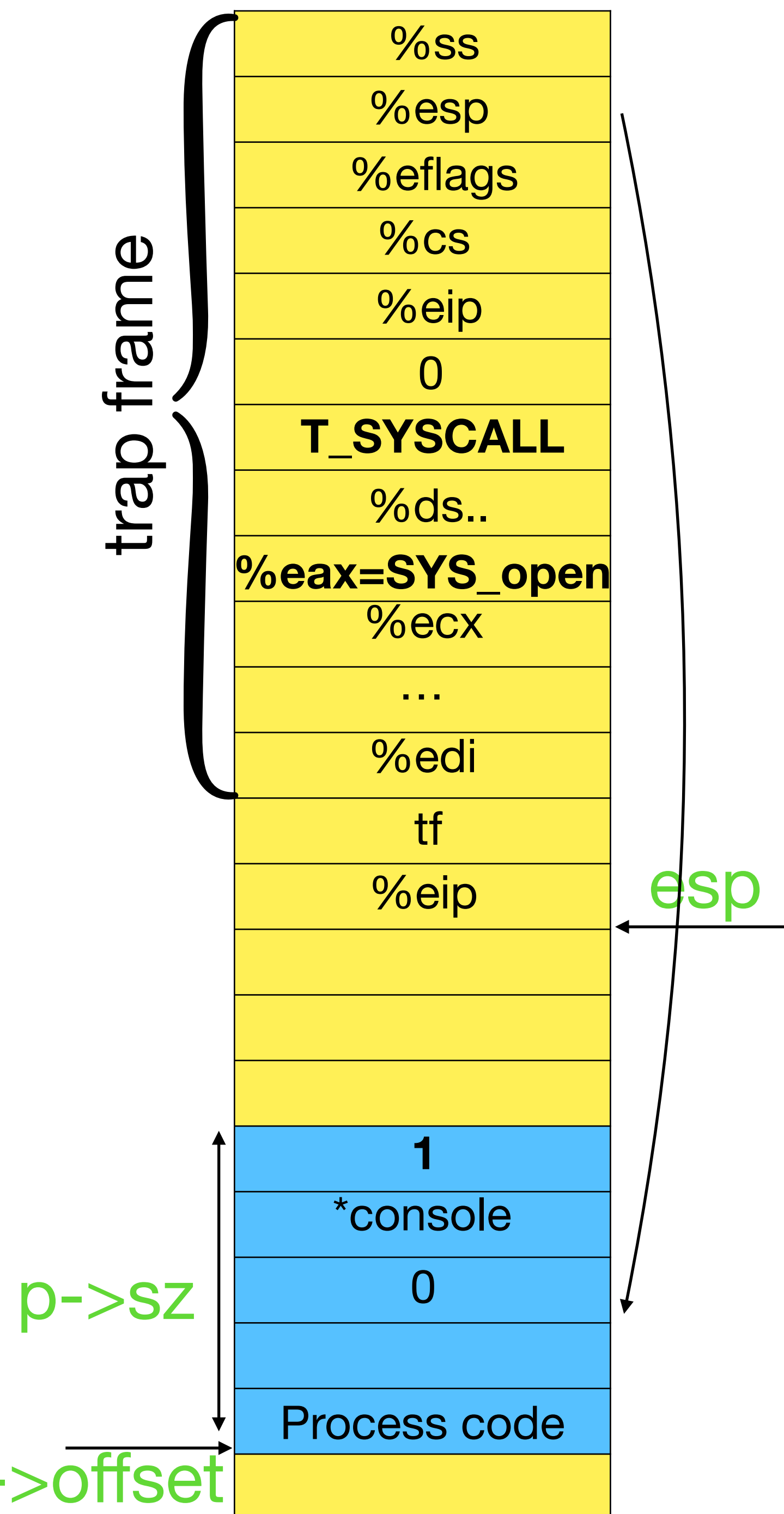
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    pushl $0
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Visualising syscall handling p19-syscall

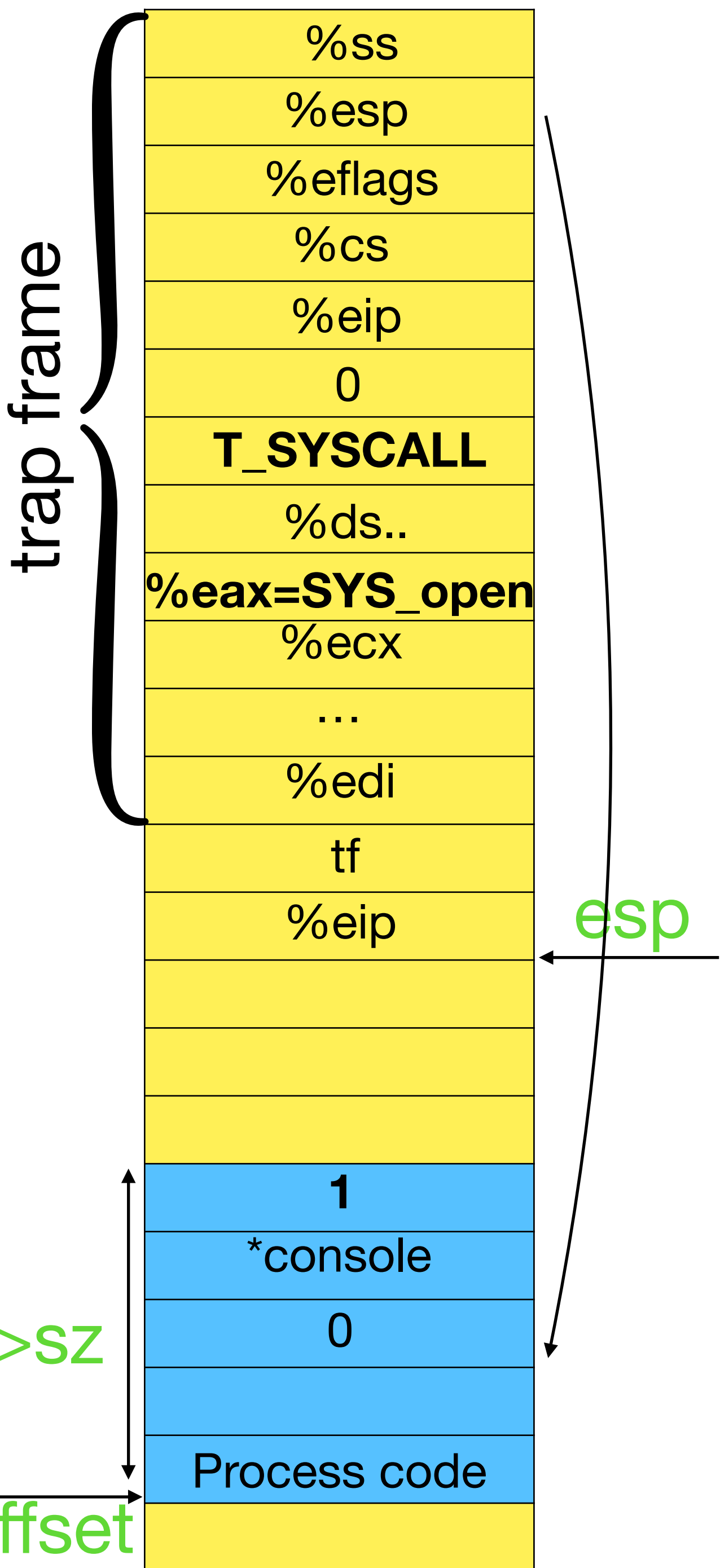
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Visualising syscall handling p19-syscall

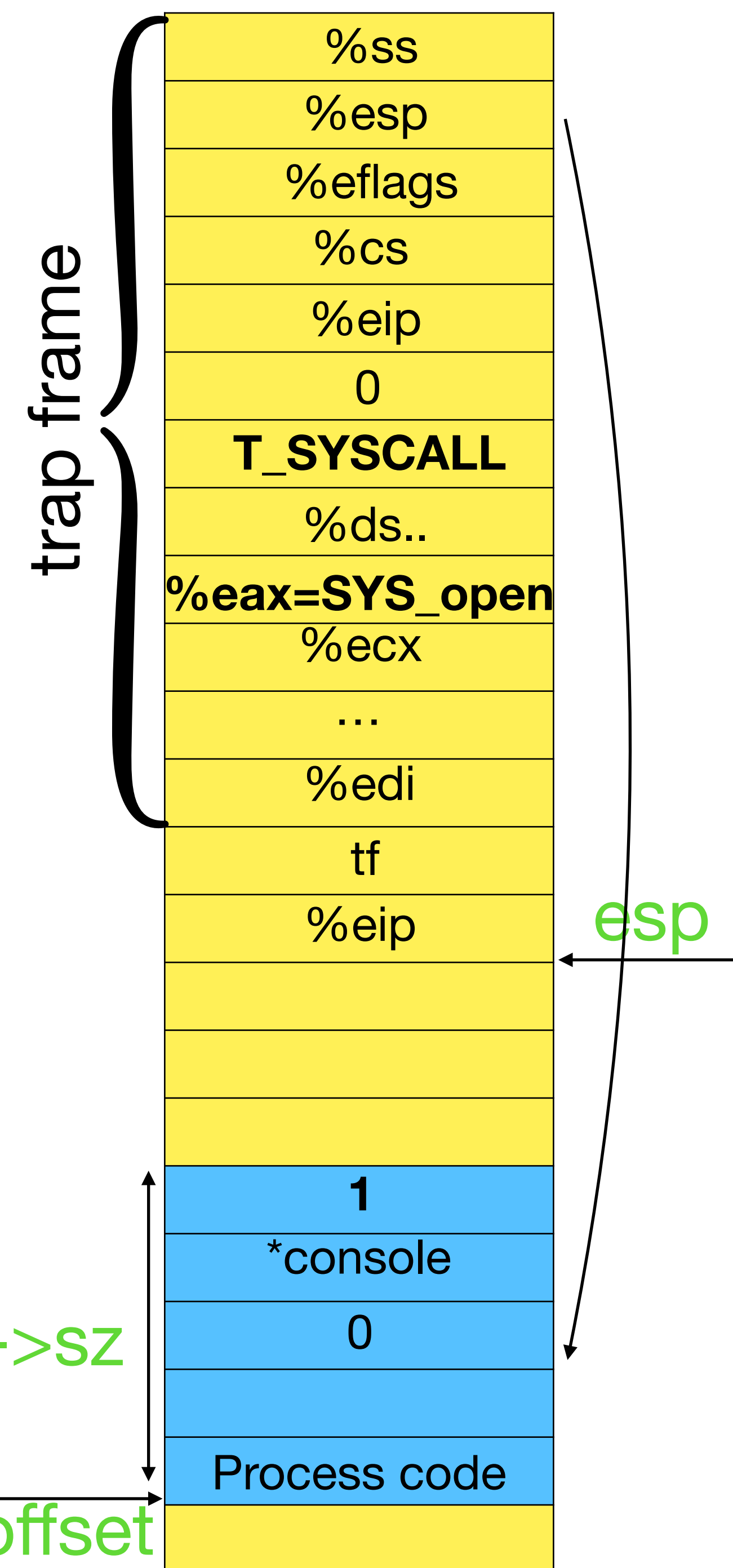
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Visualising syscall handling p19-syscall

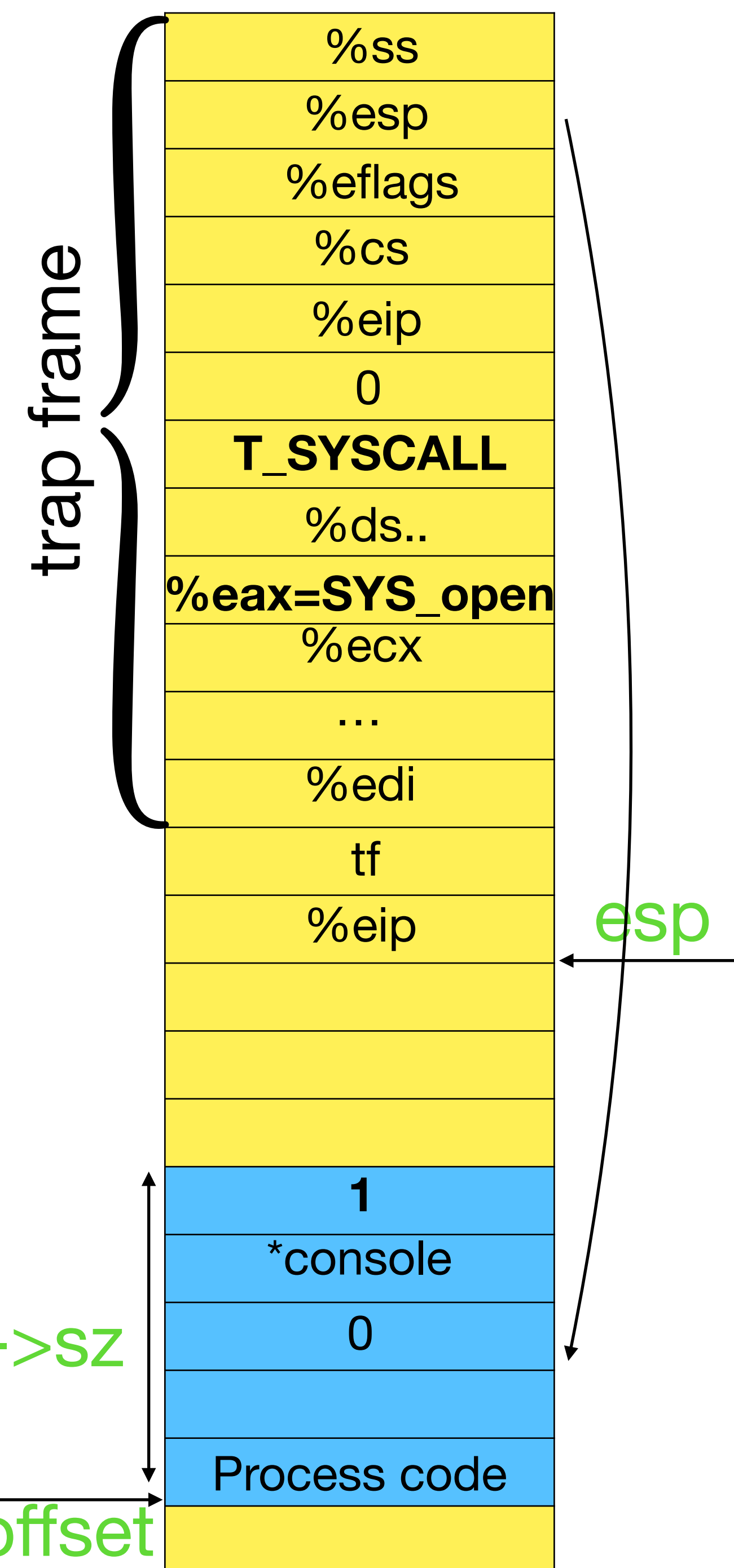
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    pushl $0
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Visualising syscall handling p19-syscall

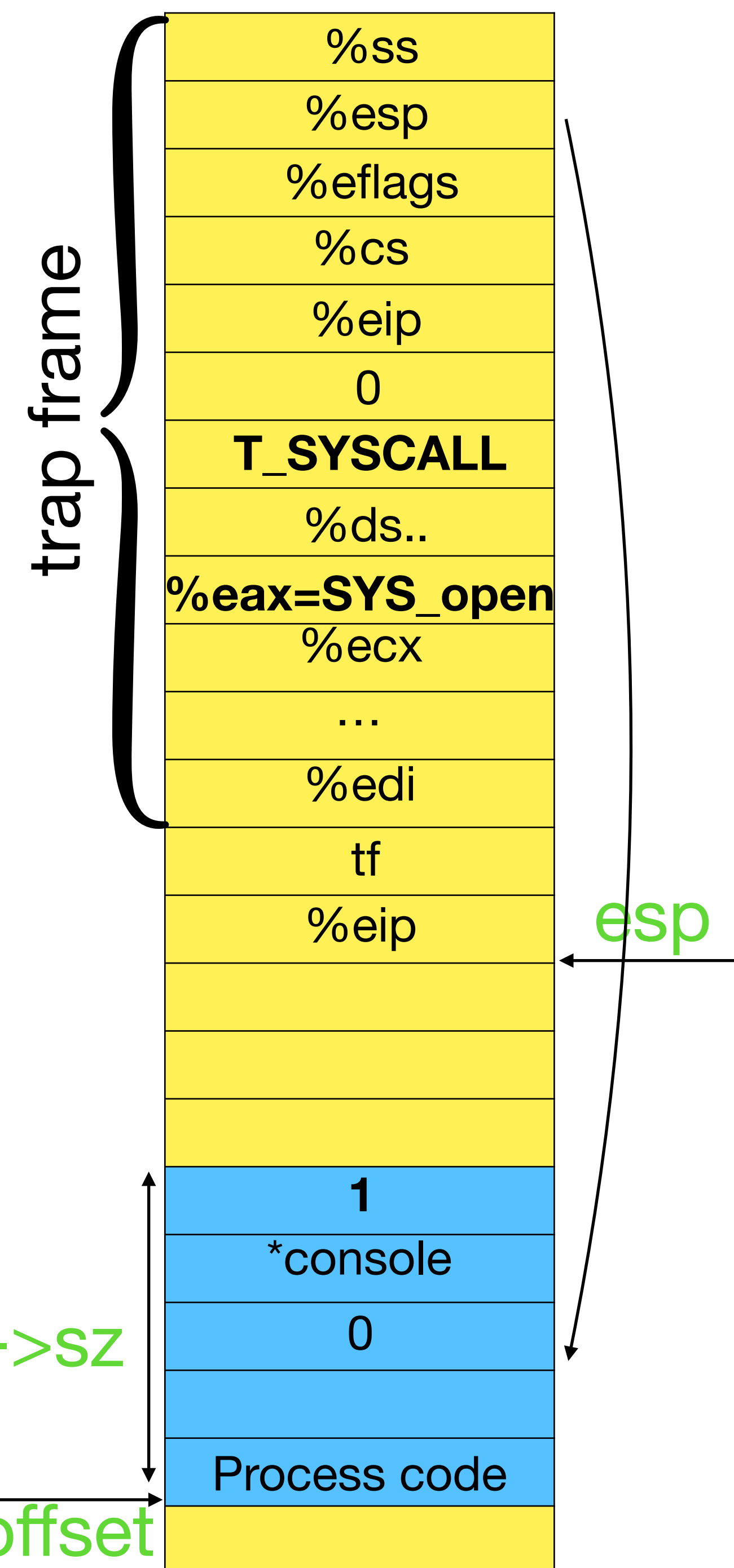
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Visualising syscall handling p19-syscall

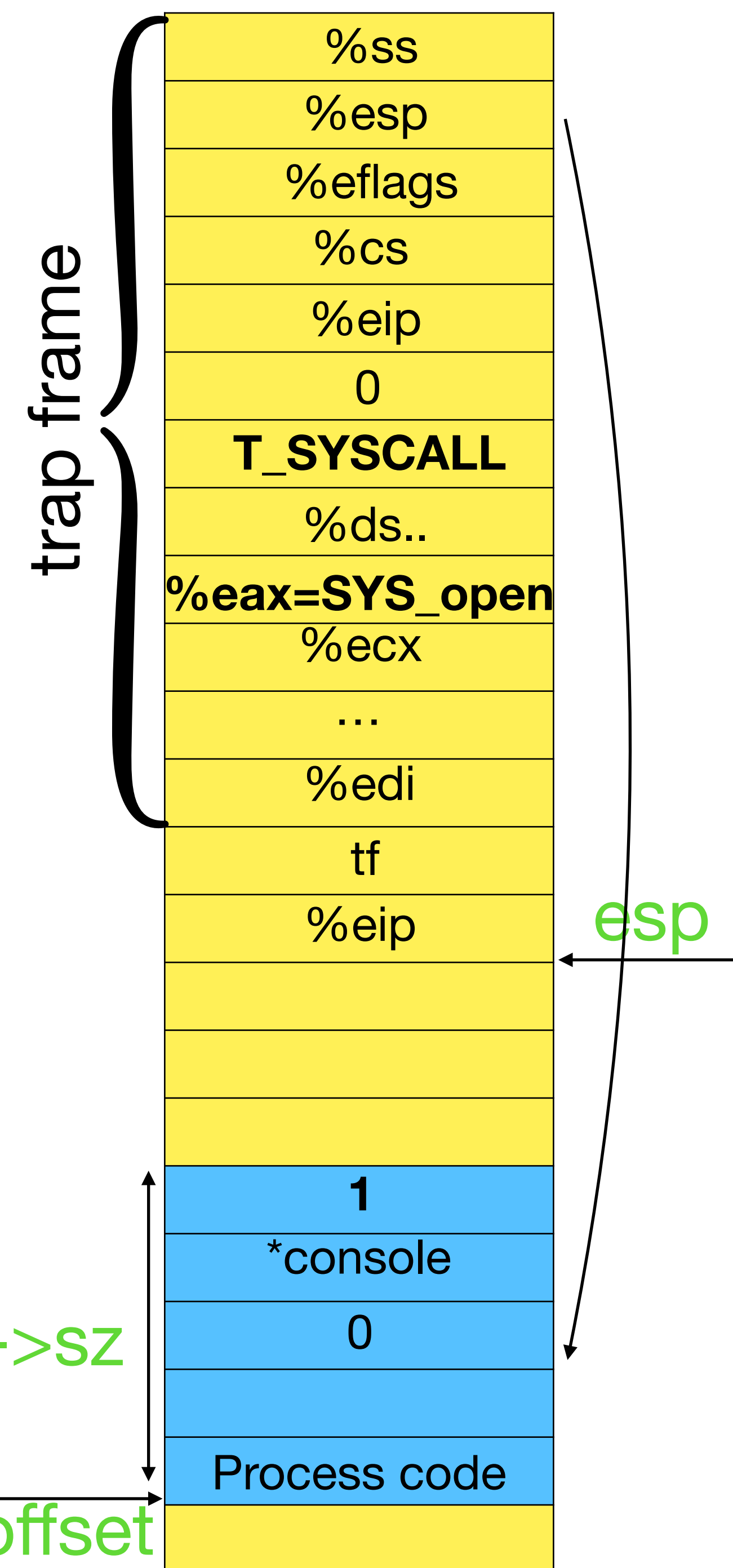
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Visualising syscall handling p19-syscall

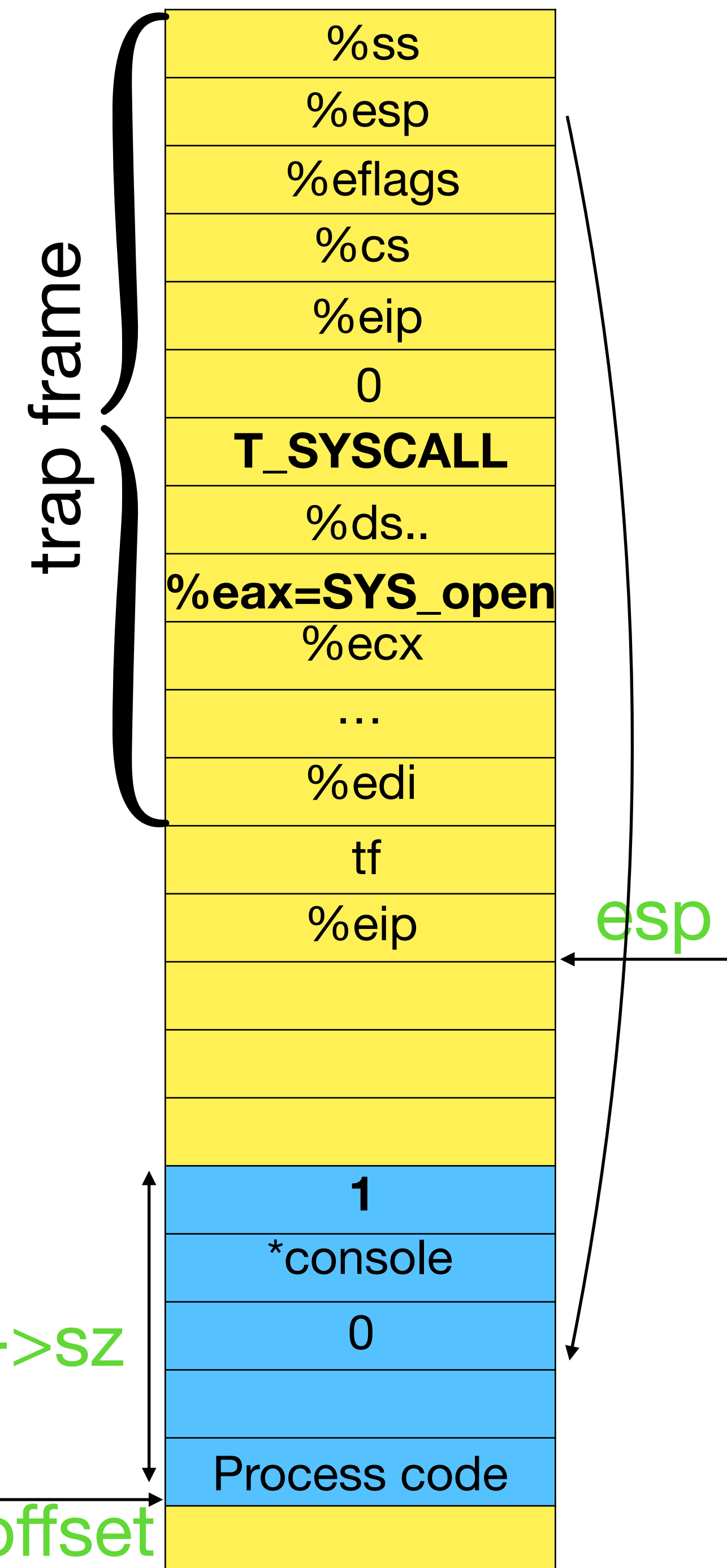
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Visualising syscall handling p19-syscall

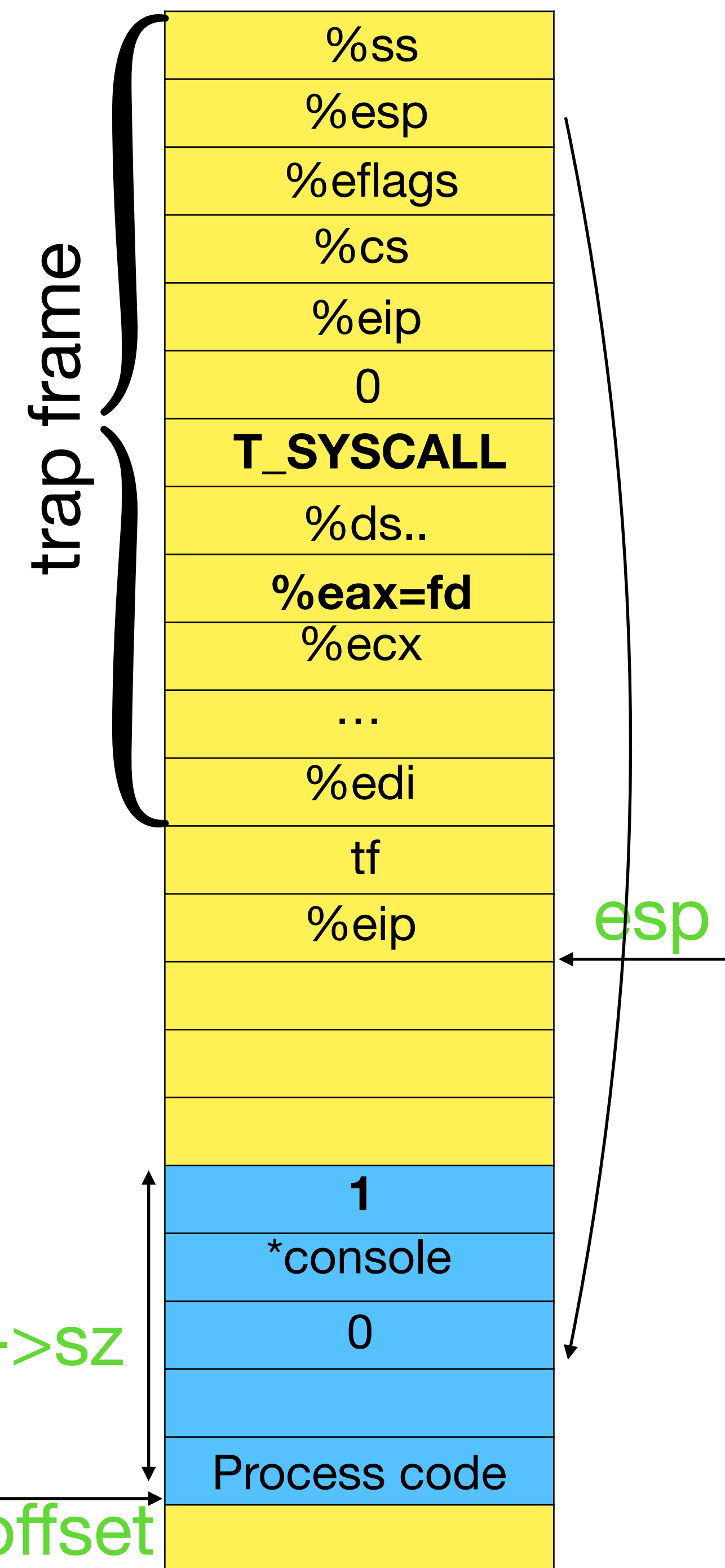
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Visualising syscall handling p19-syscall

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    curproc->tf->eax = syscalls[num]();
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```

trap frame

%ss
%esp
%eflags
%cs
%eip
0
T_SYSCALL
%ds..
%eax=fd
%ecx
...
%edi
tf
%eip
1
*console
0
Process code

eip

p->sz

esp

p->offset

Visualising syscall handling p19-syscall

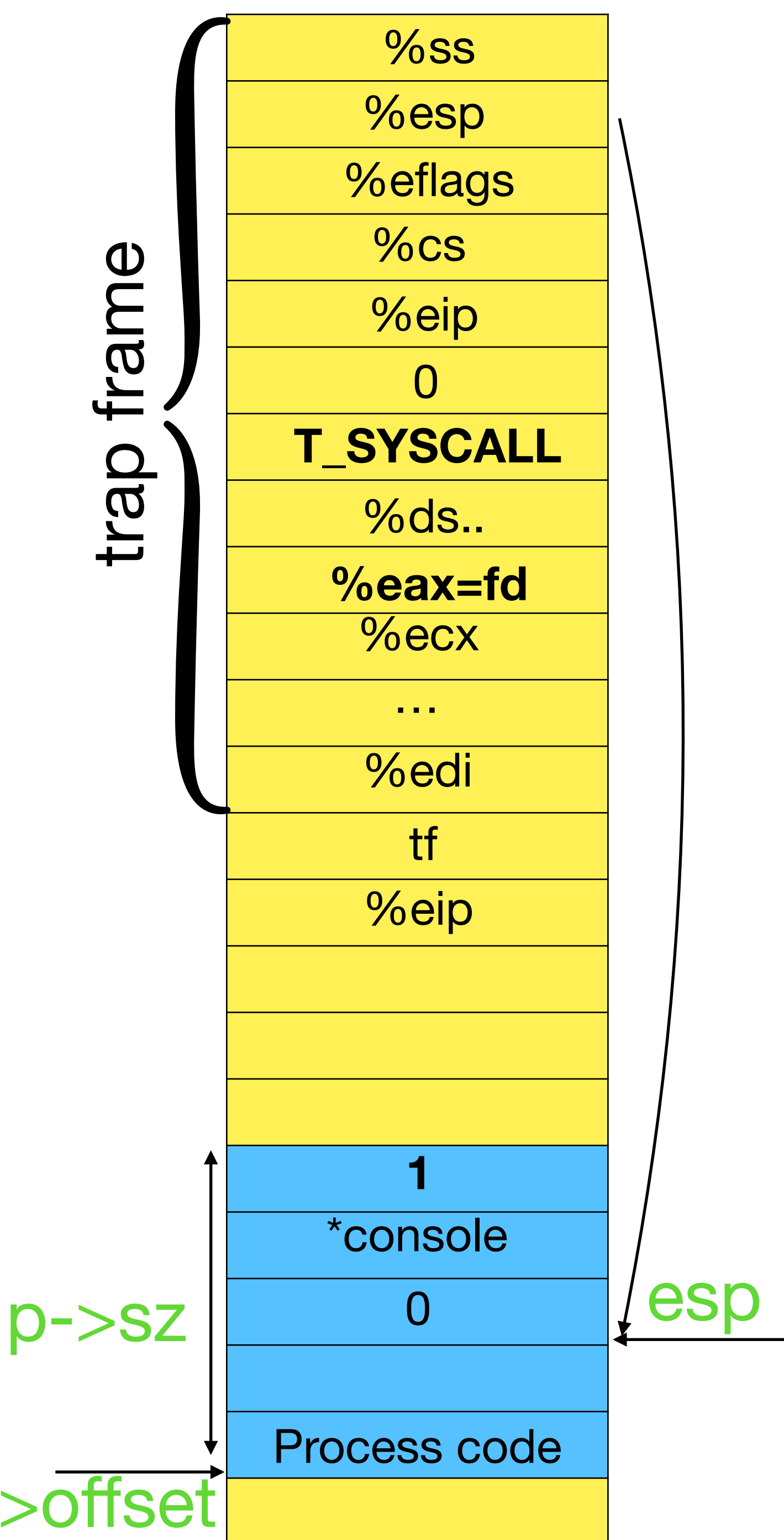
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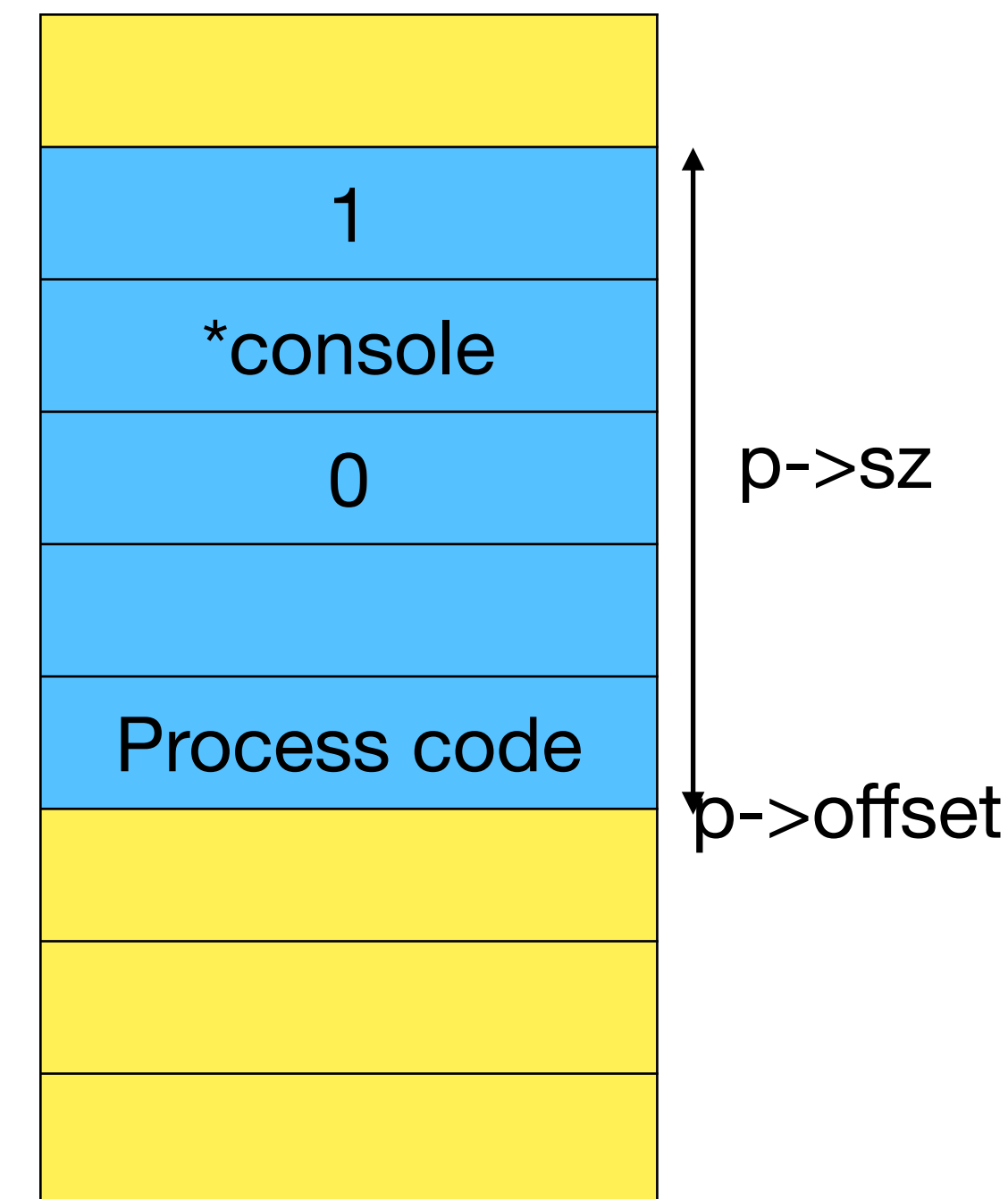
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int argint(int n, int *ip) {
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Syscall parameter checking and translation

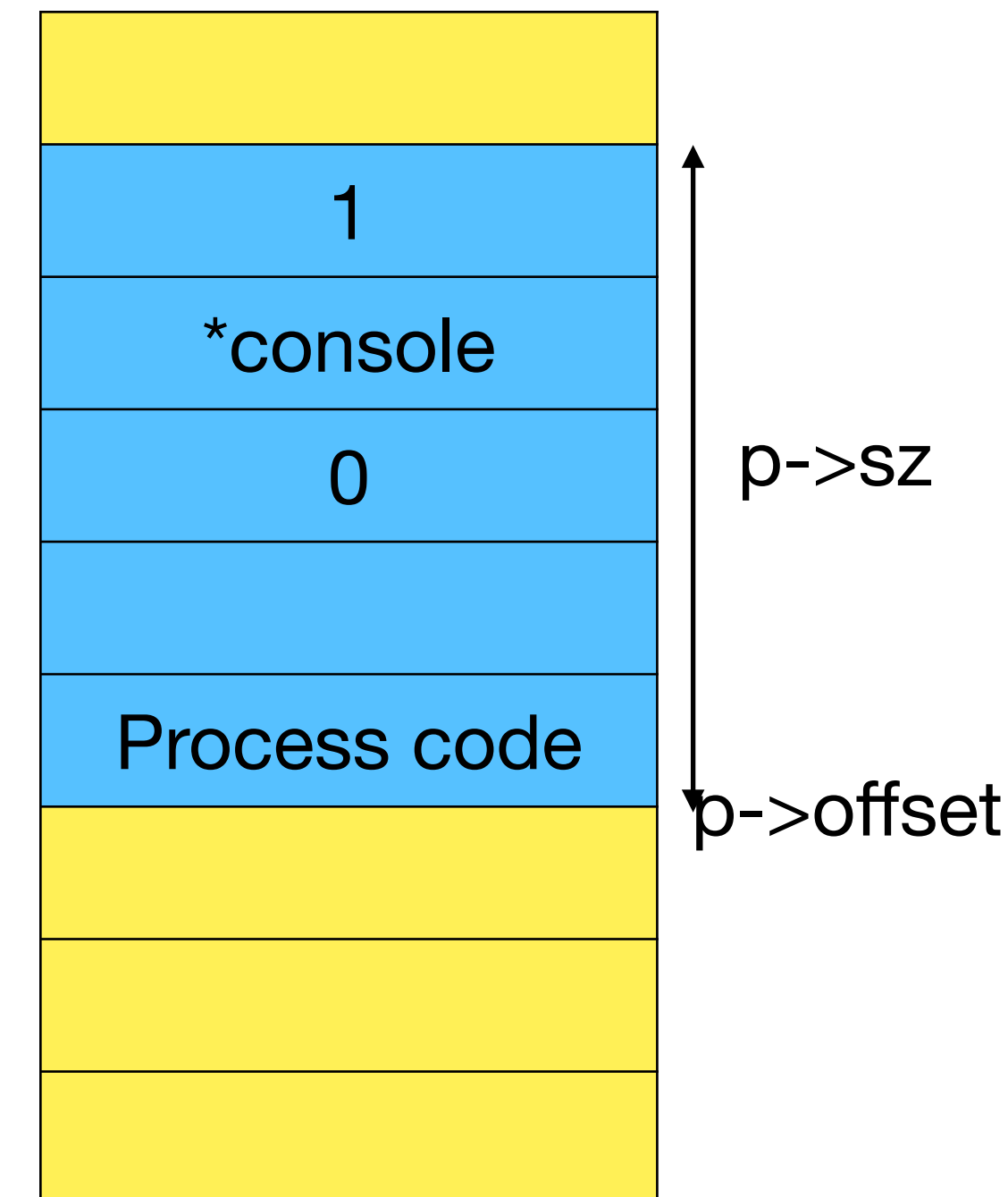
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Syscall parameter checking and translation

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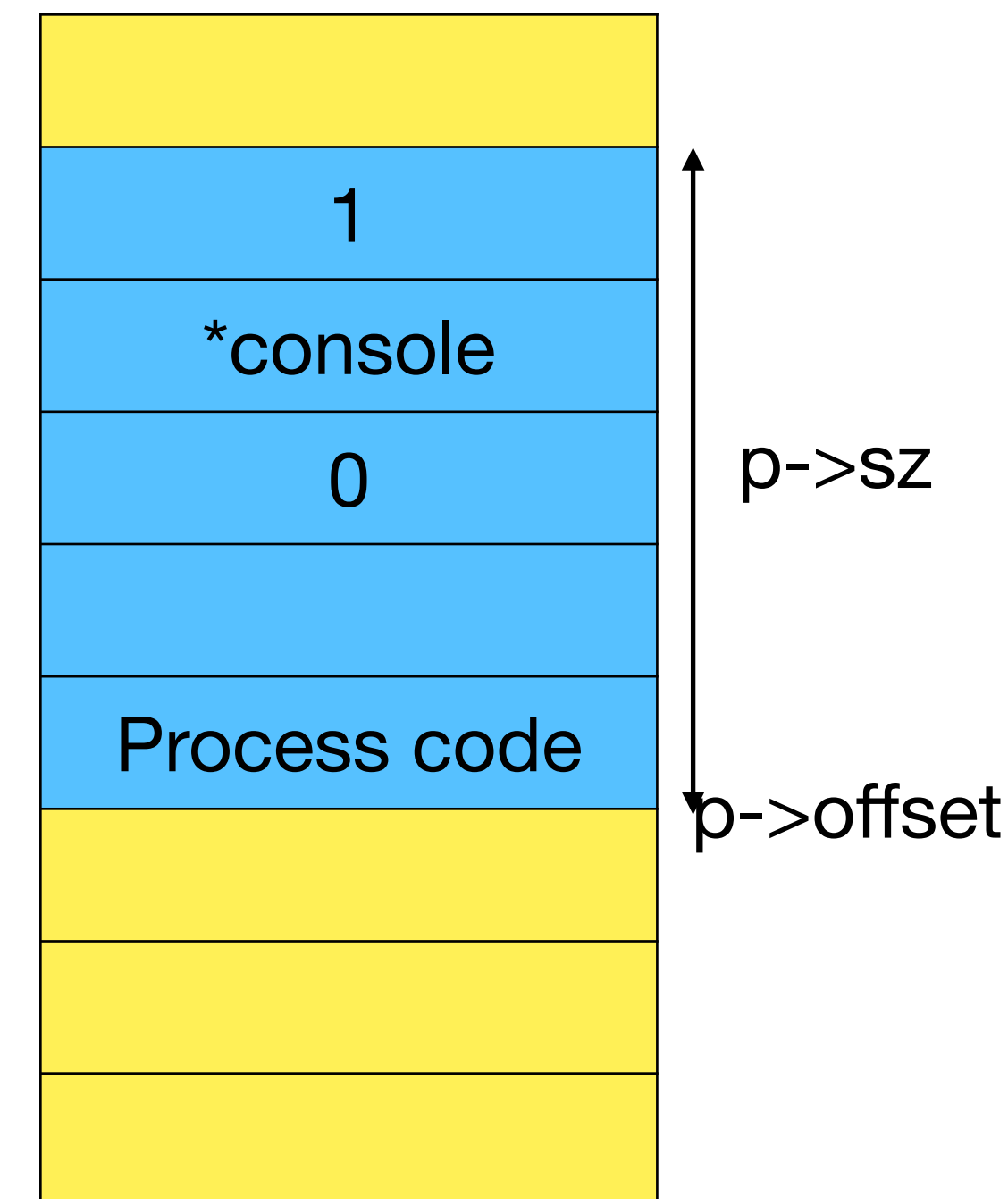
- Syscall parameters must be in process' address space:



Syscall parameter checking and translation

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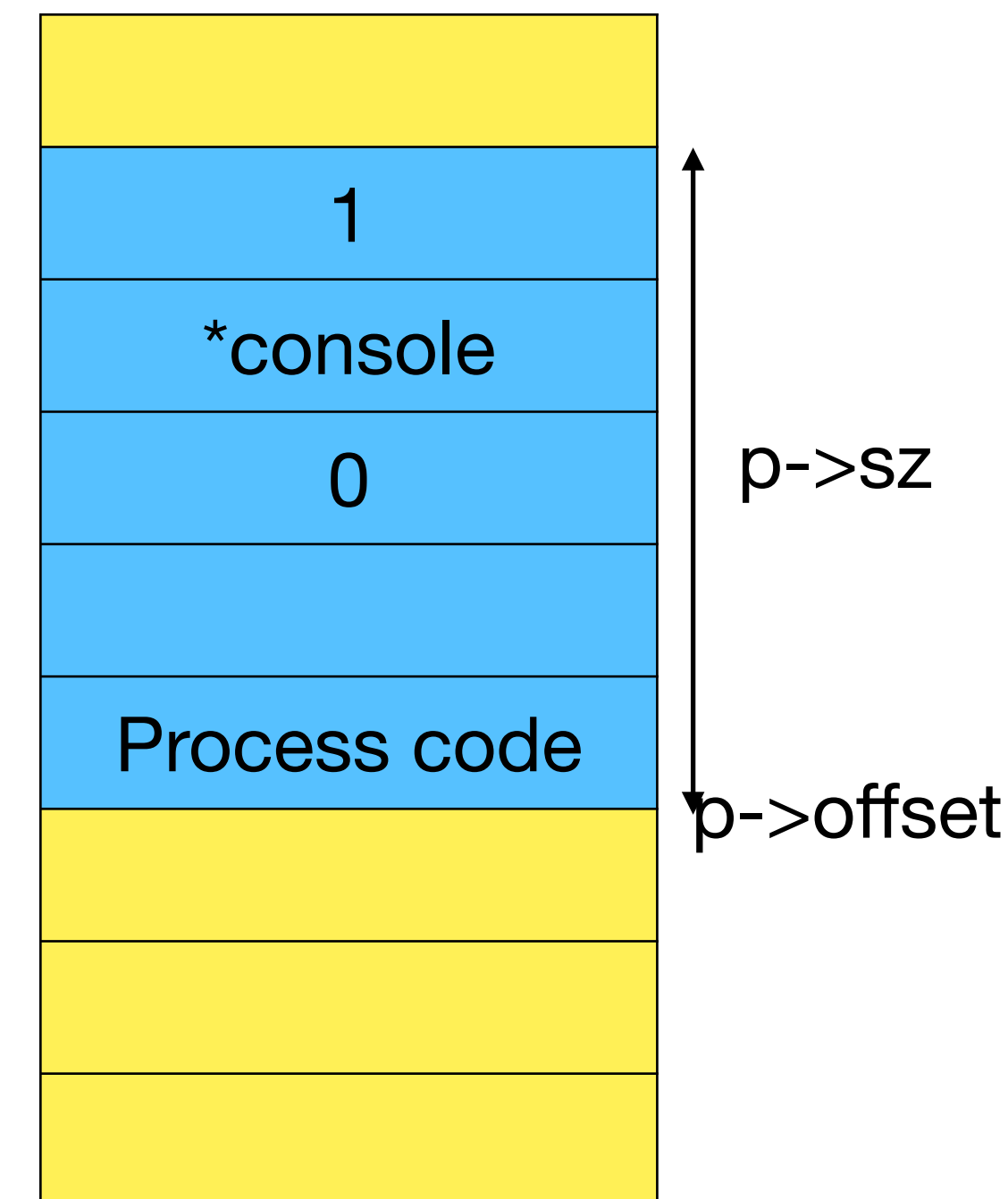
- Syscall parameters must be in process' address space:
 - Check virtual address (VA) is within limit



Syscall parameter checking and translation

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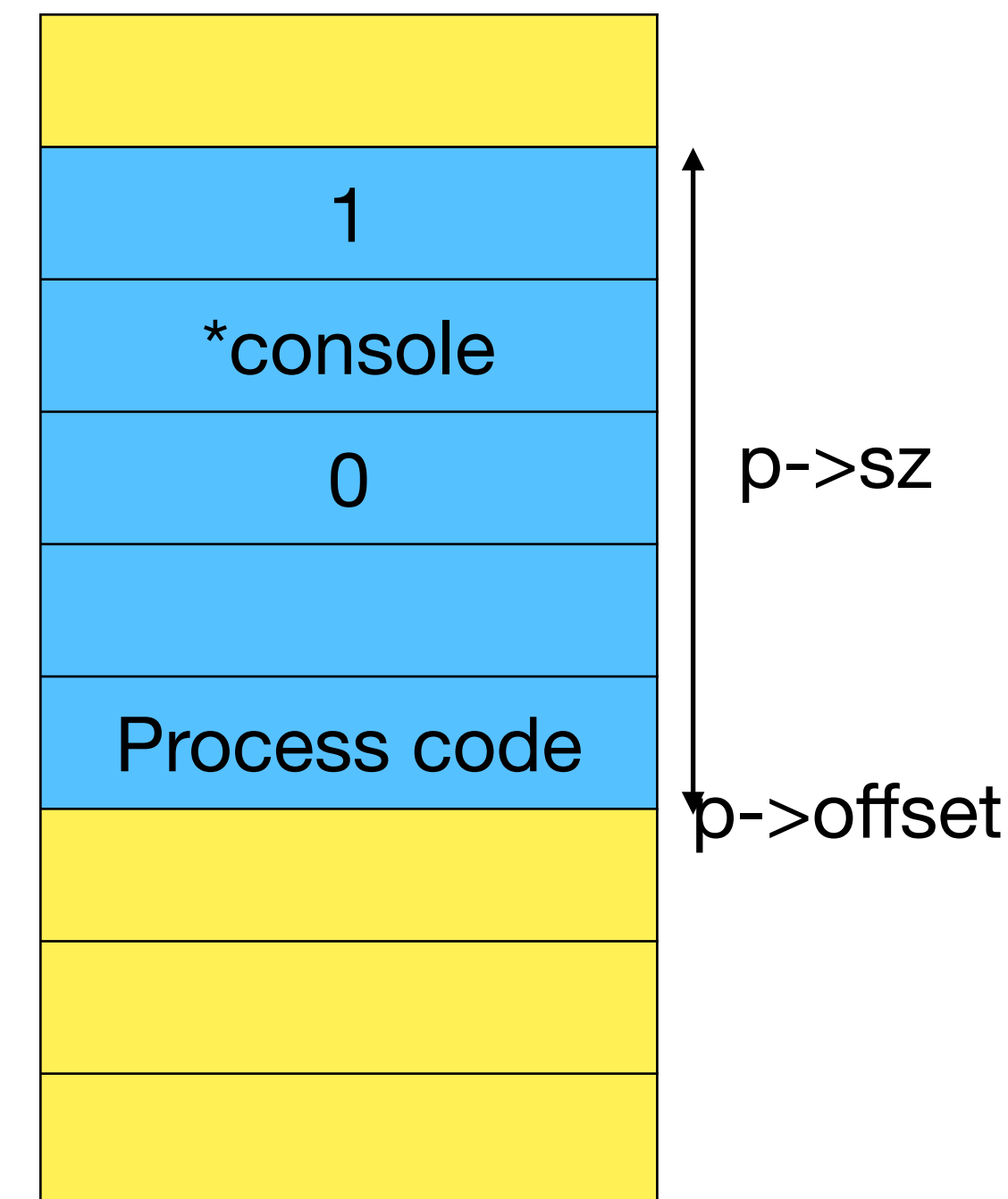
- Syscall parameters must be in process' address space:
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Syscall parameter checking and translation

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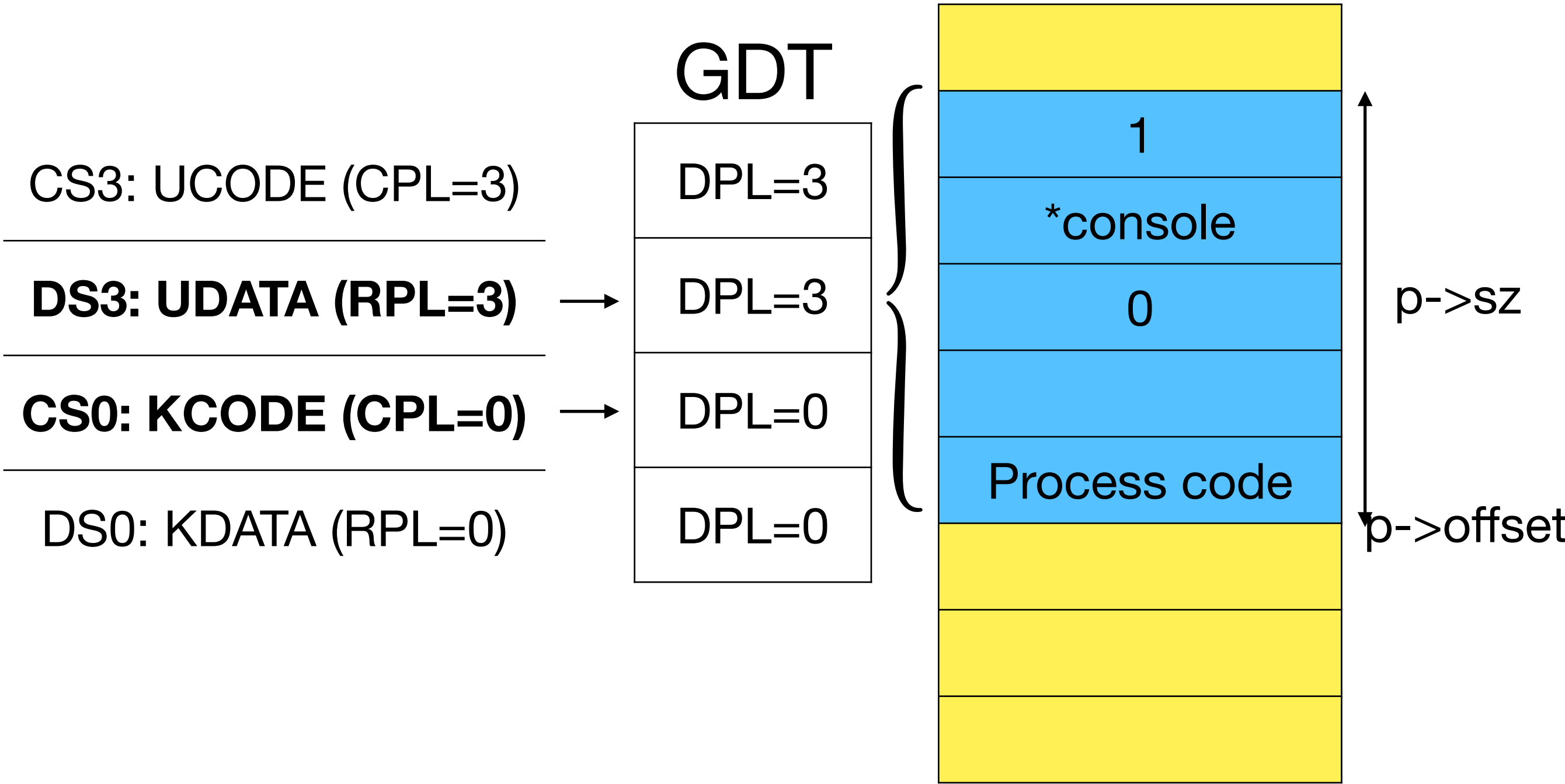
- Syscall parameters must be in process' address space:
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- Alternative design: use process' DS to read syscall parameters



Syscall parameter checking and translation

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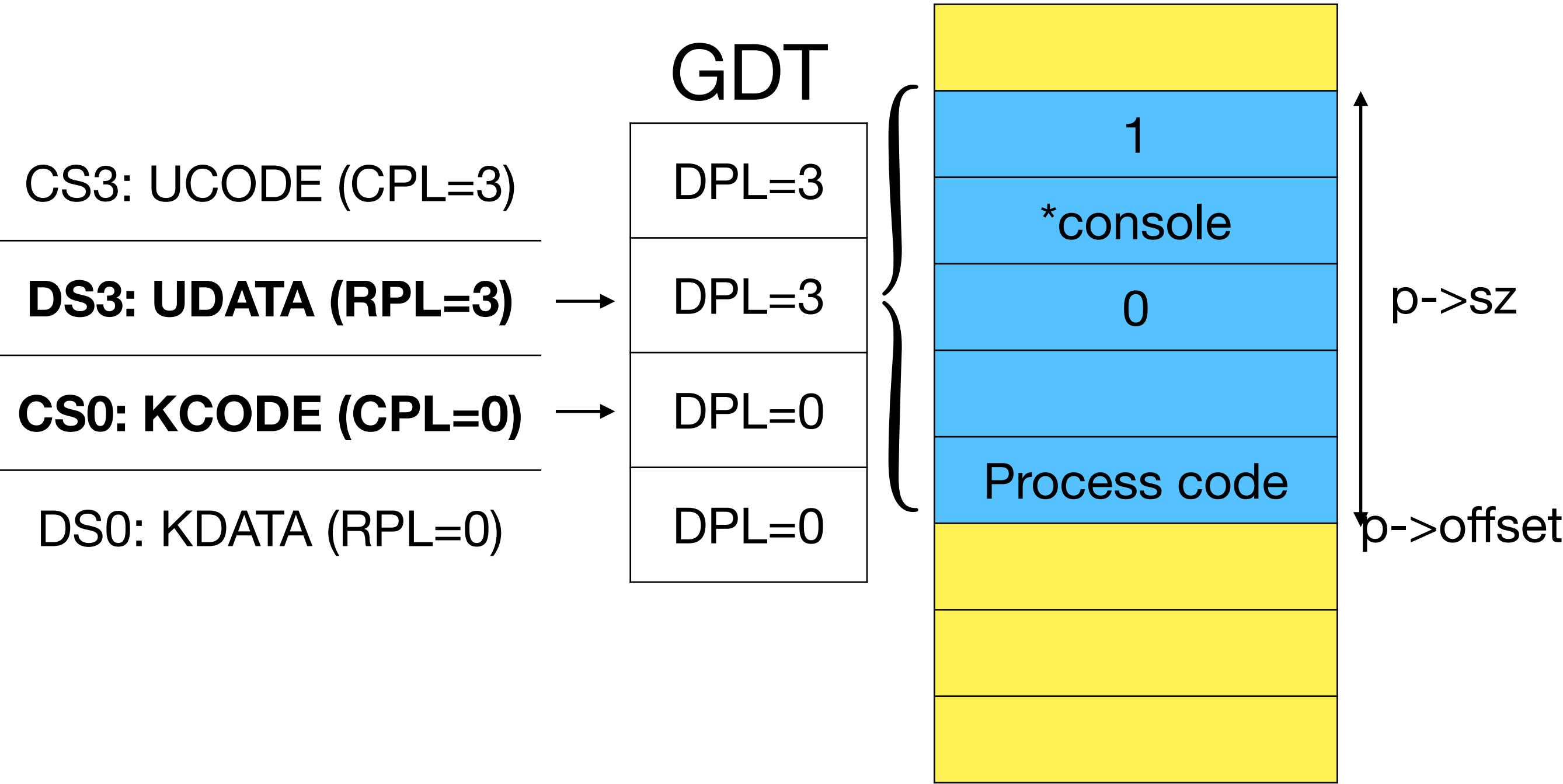
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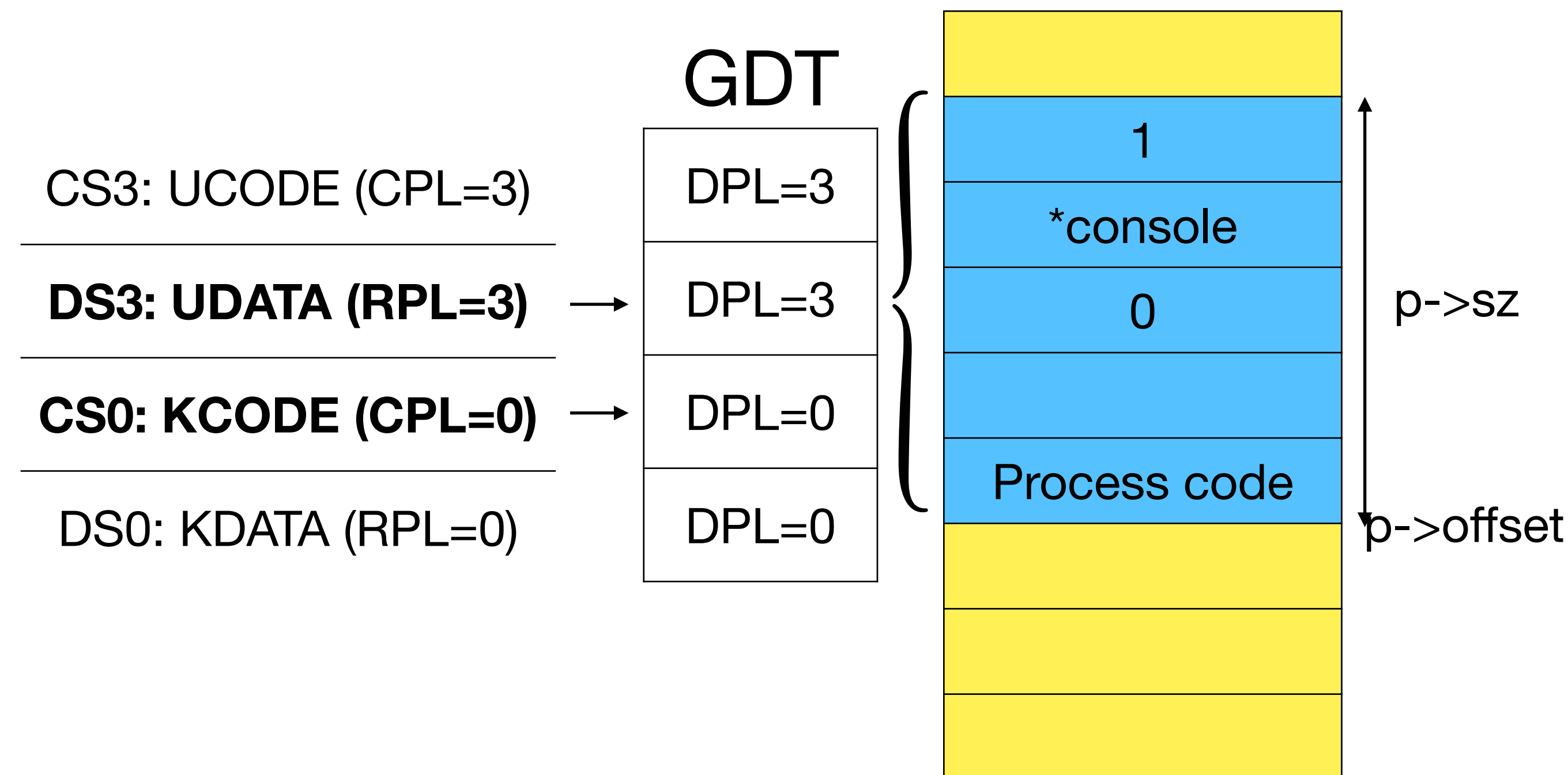
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 - $CPL \leq DPL$ and $RPL \leq DPL$



Syscall parameter checking and translation

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    if(addr >= p->sz || addr+4 > p->sz)  
        return -1;  
    *ip = *(int*)(addr + p->offset);  
}
```

- Syscall parameters must be in process' address space:
 - Check virtual address (VA) is within limit
 - Add base to VA to get physical address (PA)
- Alternative design: use process' DS to read syscall parameters
 - $CPL \leq DPL$ and $RPL \leq DPL$
- Linux has `copy\_from\_user`. Initial Linux versions had security vulnerabilities due to missing checks

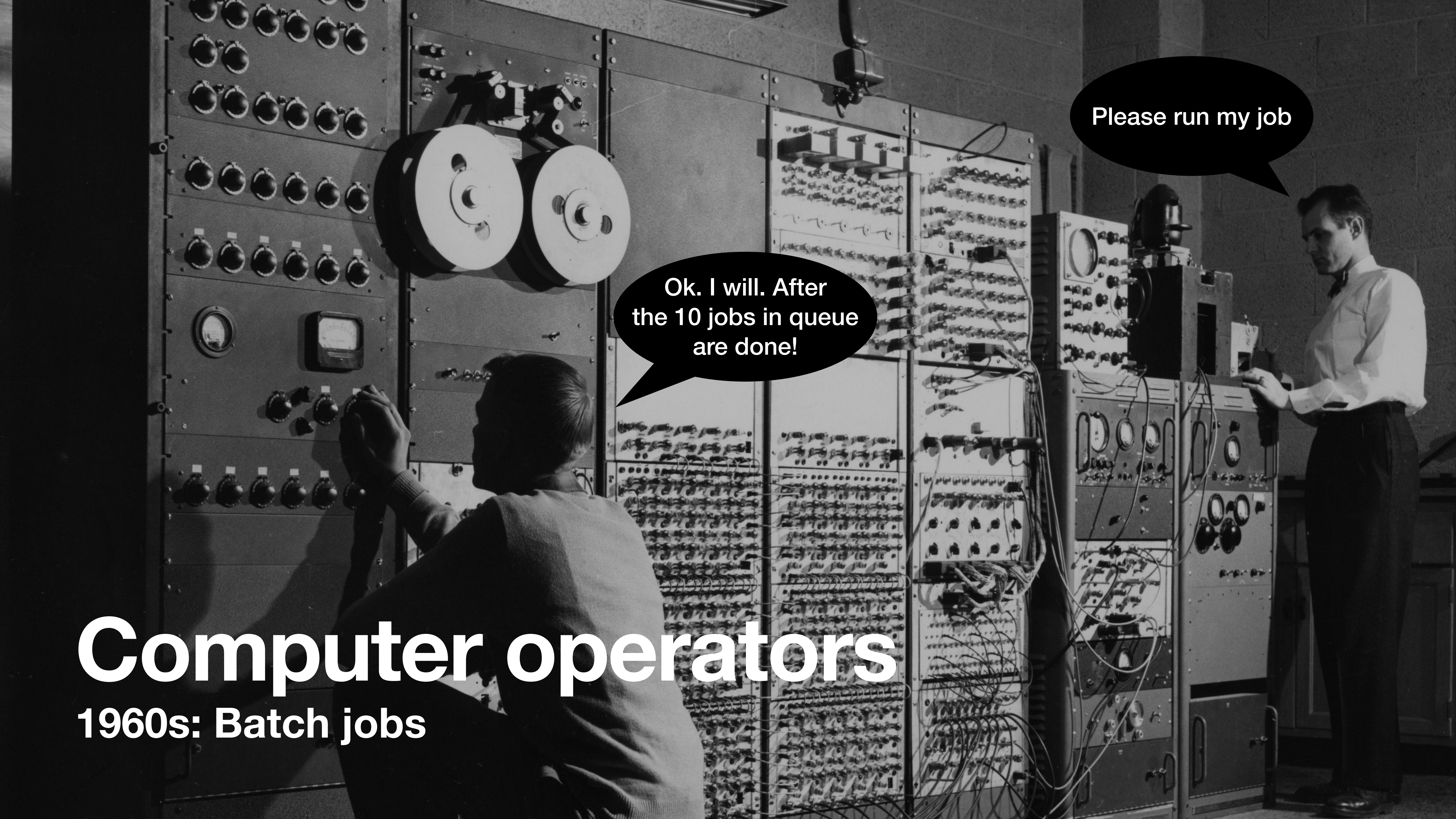


Scheduling policies

OSTEP Ch. 7-9

Scheduling problem

- A number of processes can be runnable.
 - Which process to run next?
 - How long to run each process for?
- What should scheduler try to optimise?
 - What do we know about the behaviour of each process?



Please run my job

Ok. I will. After
the 10 jobs in queue
are done!

Computer operators

1960s: Batch jobs

Batch job characteristics

- Once started, each job runs to completion
- Run-time of each job is known
- Main metric: turnaround time = (completion time - arrival time)

FIFO

- Avg turnaround time = $(10+20+30)/3 = 20$

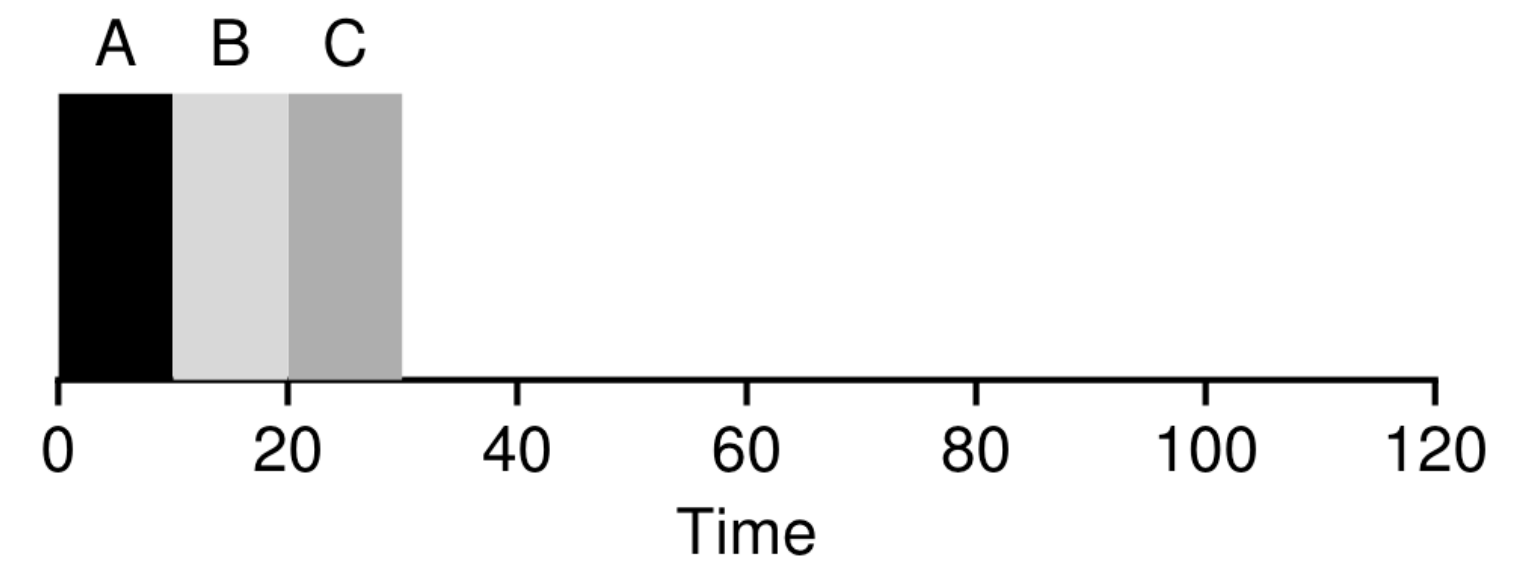


Figure 7.1: FIFO Simple Example

FIFO

- Avg turnaround time = $(10+20+30)/3 = 20$
- Avg turnaround time = $(100+110+120)/3 = 110$

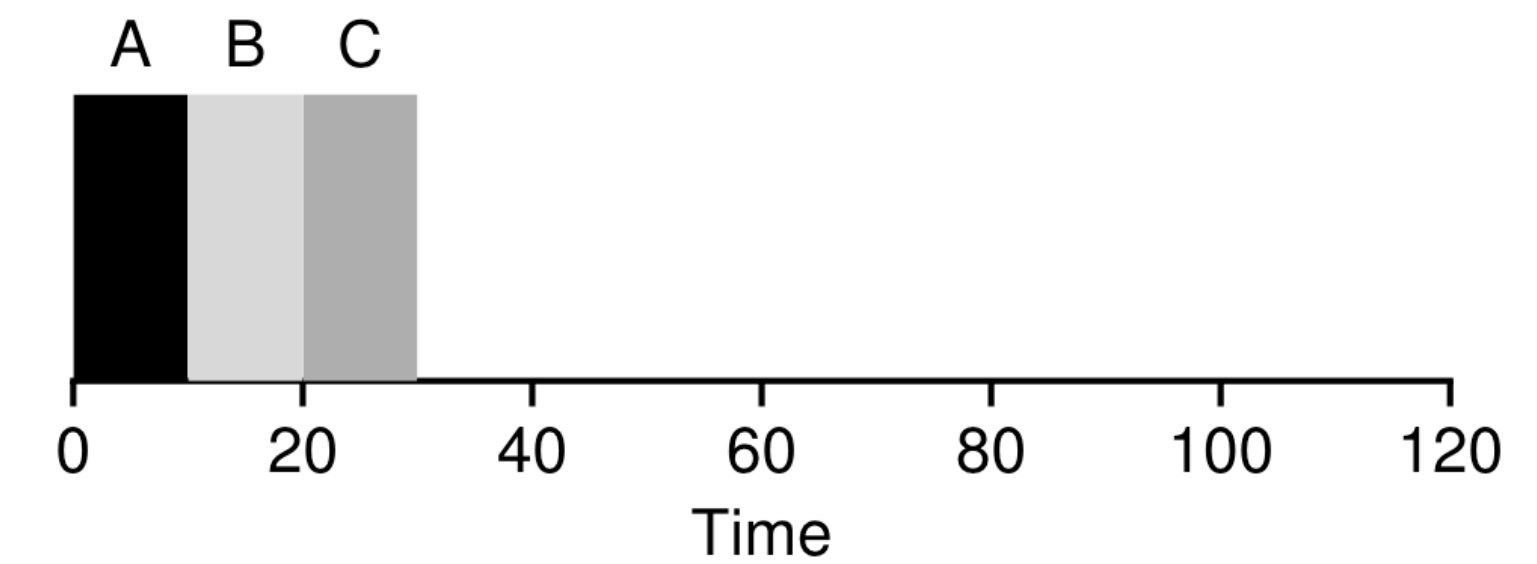


Figure 7.1: FIFO Simple Example

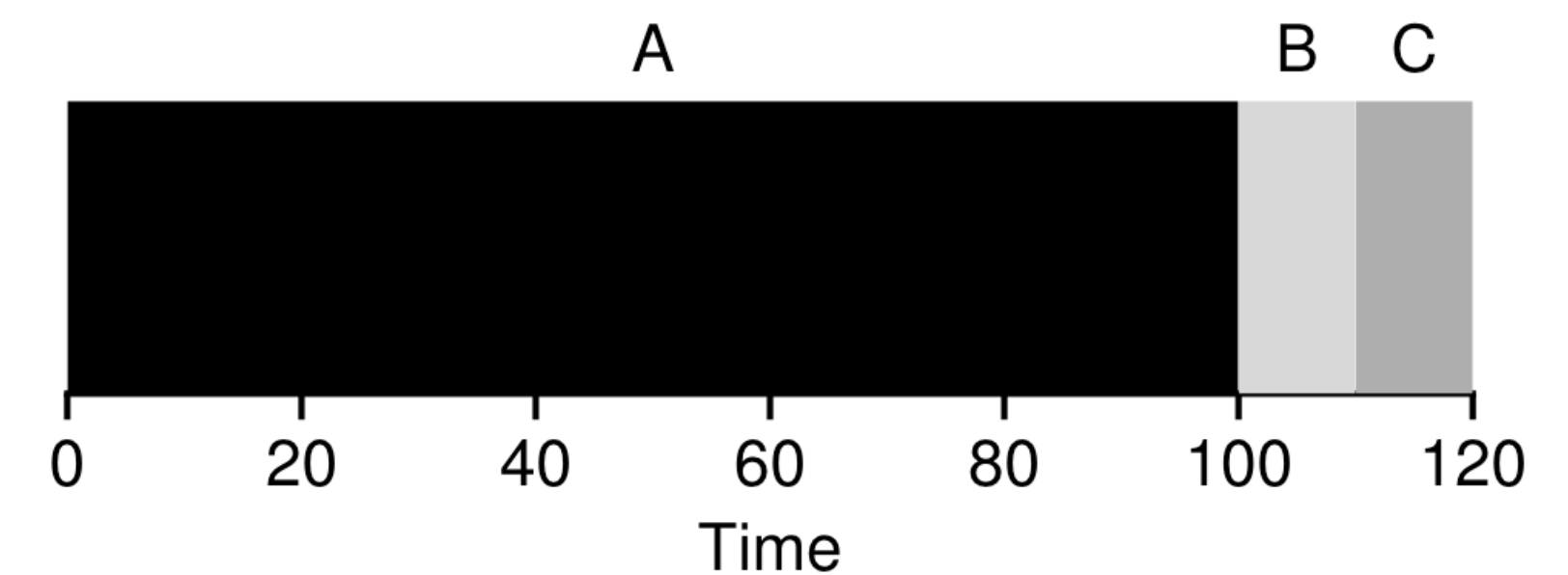


Figure 7.2: Why FIFO Is Not That Great

FIFO

- Avg turnaround time = $(10+20+30)/3 = 20$
- Avg turnaround time = $(100+110+120)/3 = 110$
- Convoy effect: fast jobs are stuck behind slow jobs

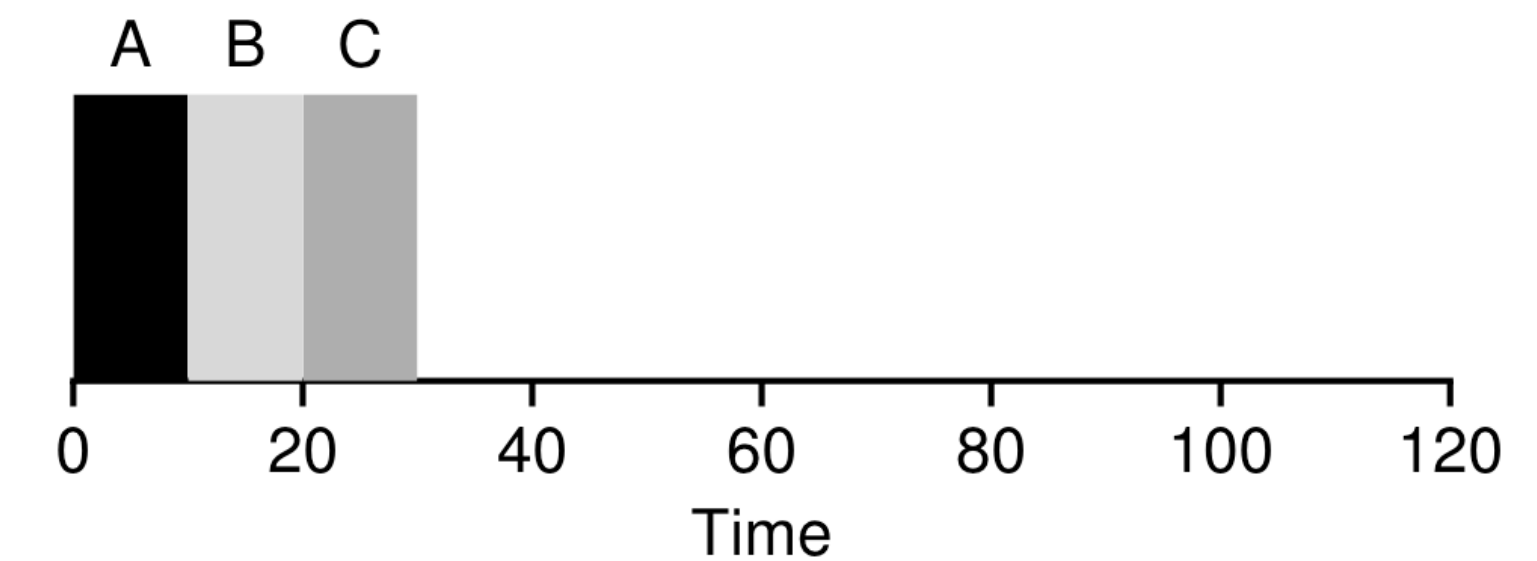
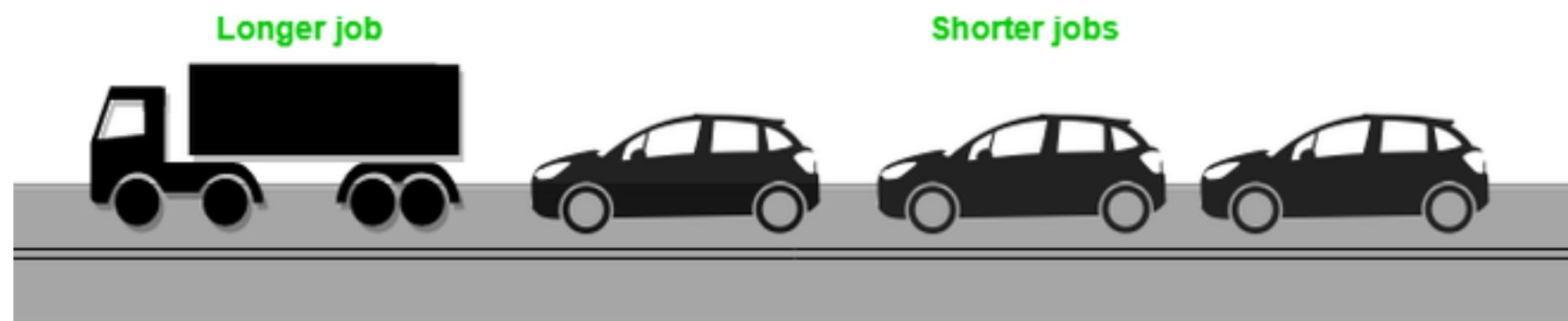


Figure 7.1: FIFO Simple Example

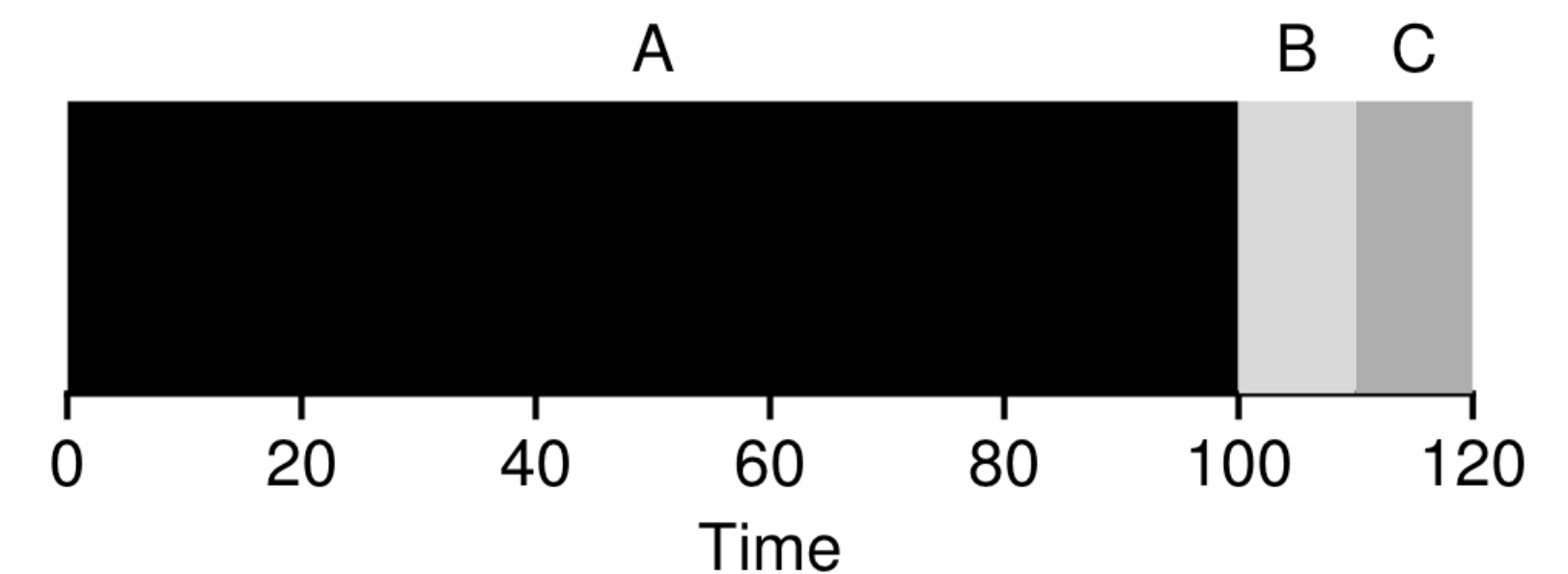


Figure 7.2: Why FIFO Is Not That Great

Shortest job first

- Avg turnaround time = $(10+20+120)/3 = 50$
- Optimal scheduling algorithm

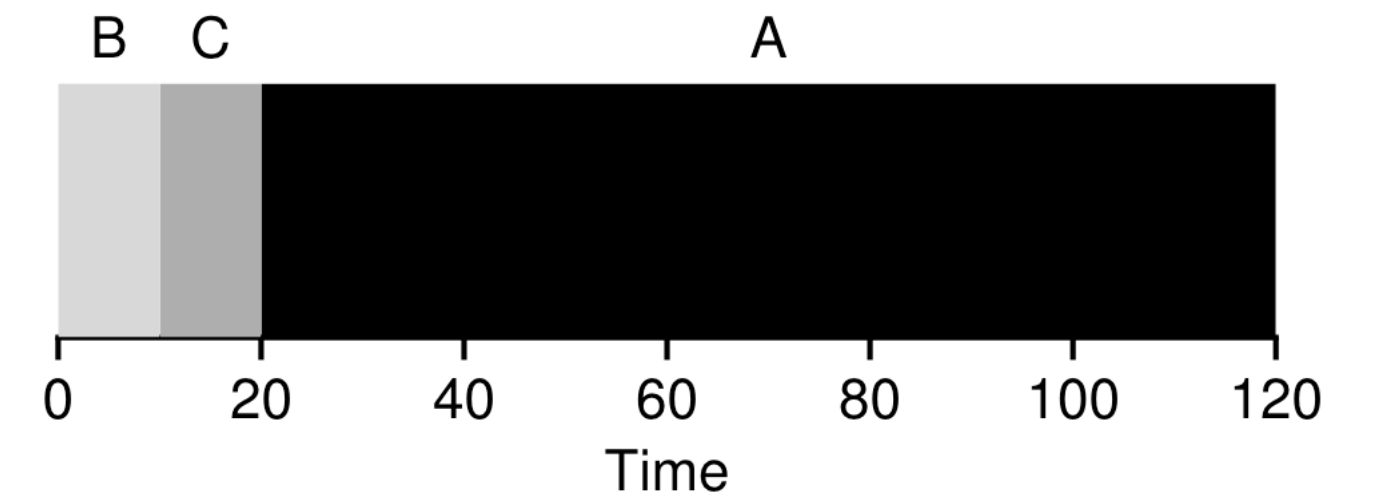


Figure 7.3: SJF Simple Example

Shortest job first

- Avg turnaround time = $(10+20+120)/3 = 50$
- Optimal scheduling algorithm
- Jobs may not arrive at the same time.

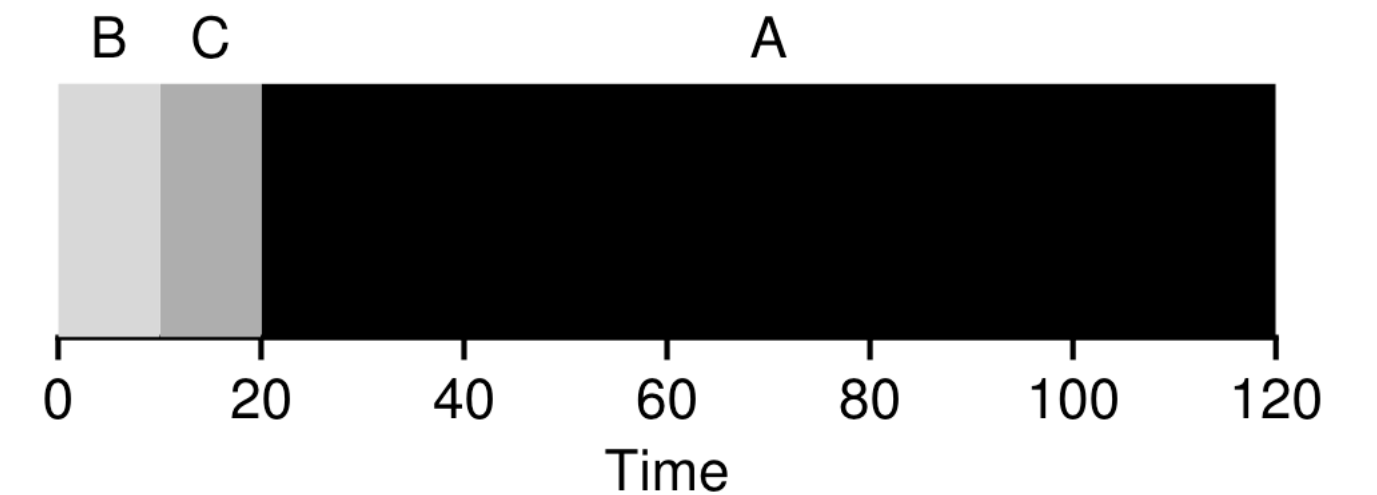


Figure 7.3: SJF Simple Example

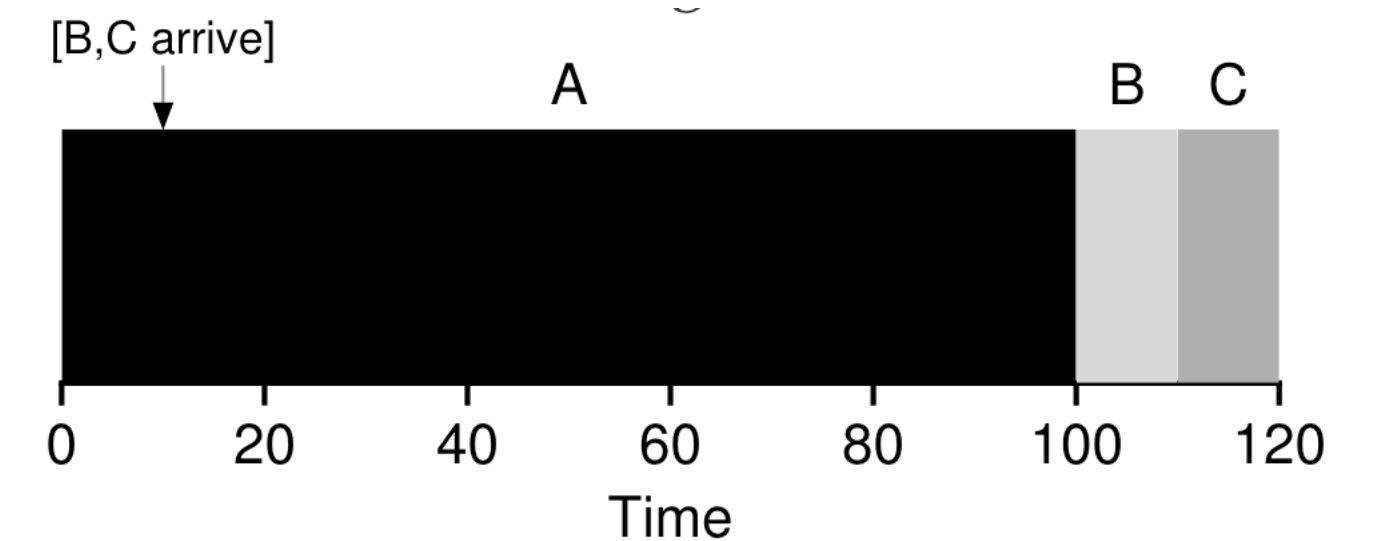


Figure 7.4: SJF With Late Arrivals From B and C

Shortest job first

- Avg turnaround time = $(10+20+120)/3 = 50$
- Optimal scheduling algorithm
- Jobs may not arrive at the same time.
- Avg turnaround time = $(100+100+110)/3 = 103.33$

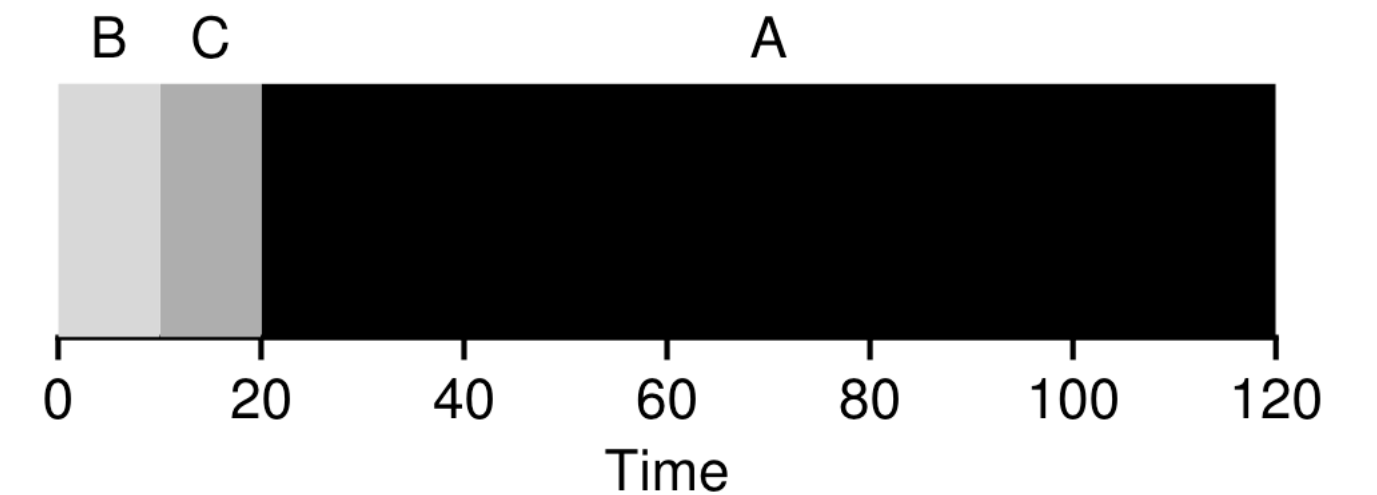


Figure 7.3: SJF Simple Example

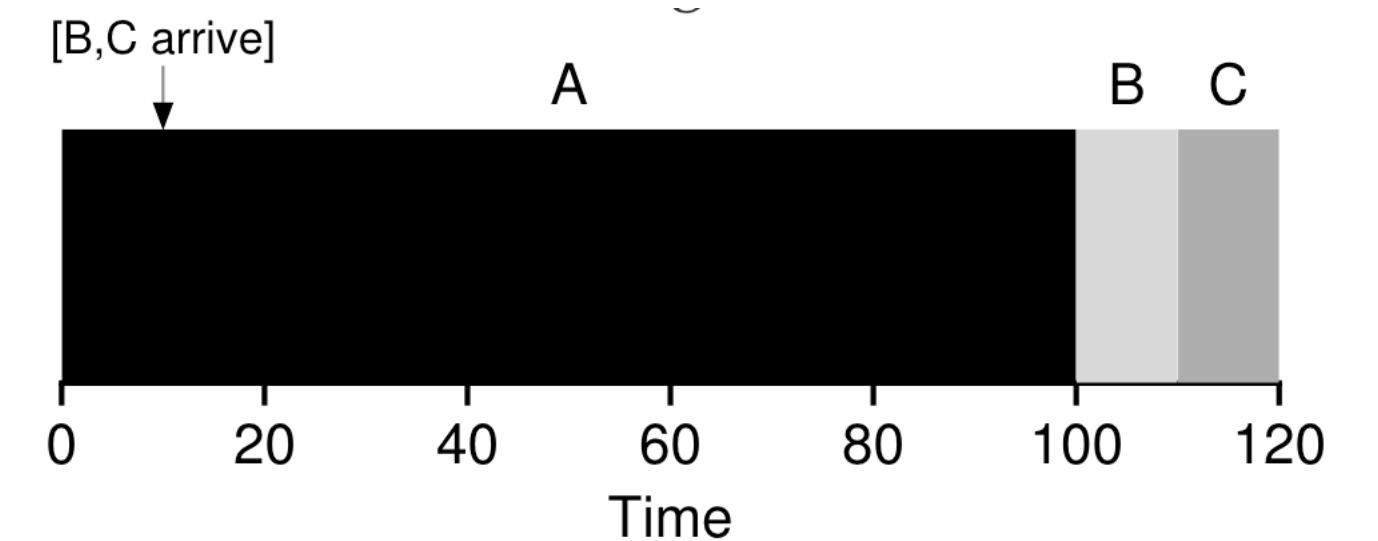


Figure 7.4: SJF With Late Arrivals From B and C

Shortest job first

- Avg turnaround time = $(10+20+120)/3 = 50$
- Optimal scheduling algorithm
- Jobs may not arrive at the same time.
- Avg turnaround time = $(100+100+110)/3 = 103.33$
- Again leads to convoy effect: fast jobs are stuck behind slow jobs

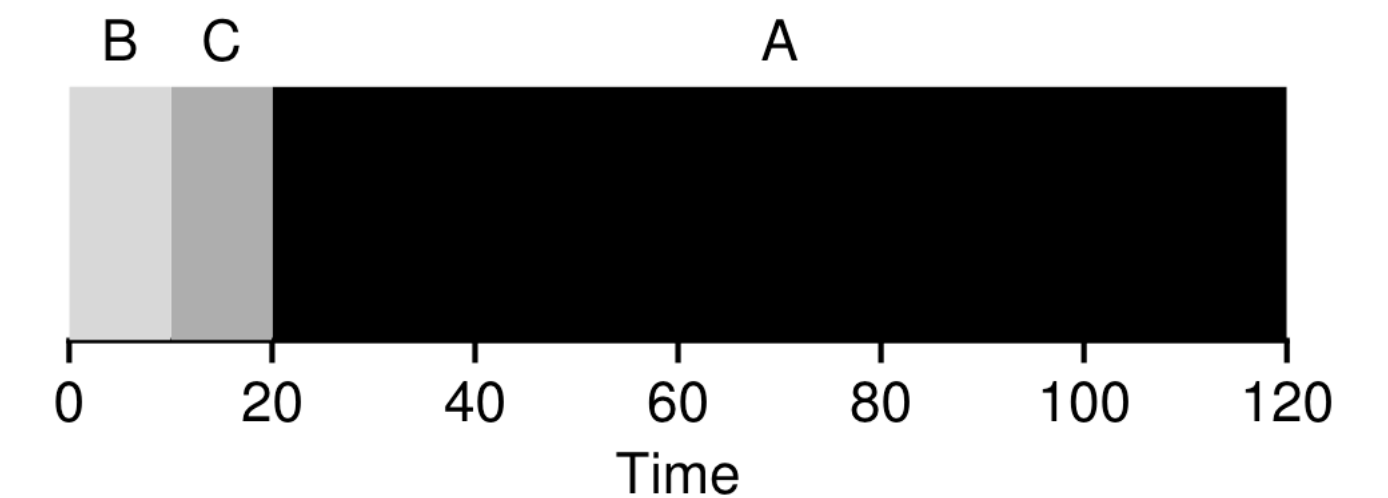
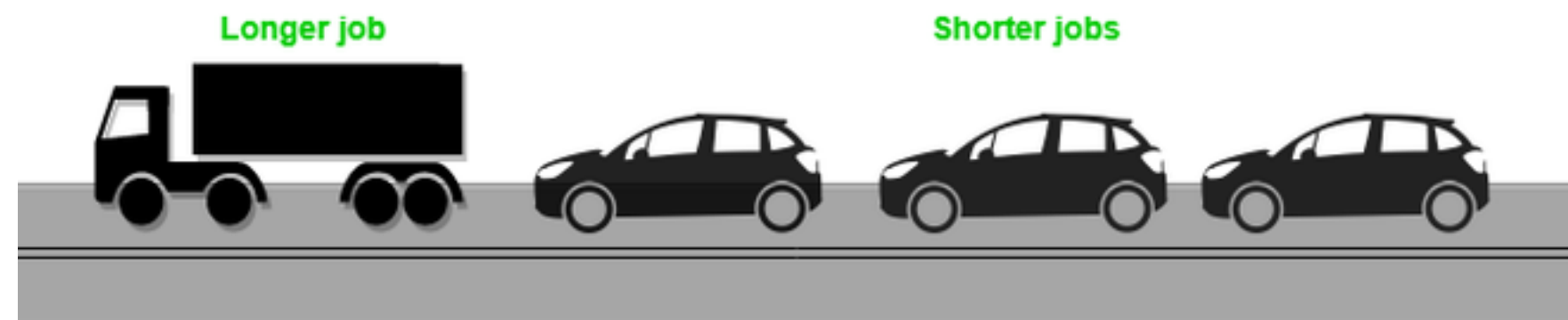


Figure 7.3: SJF Simple Example

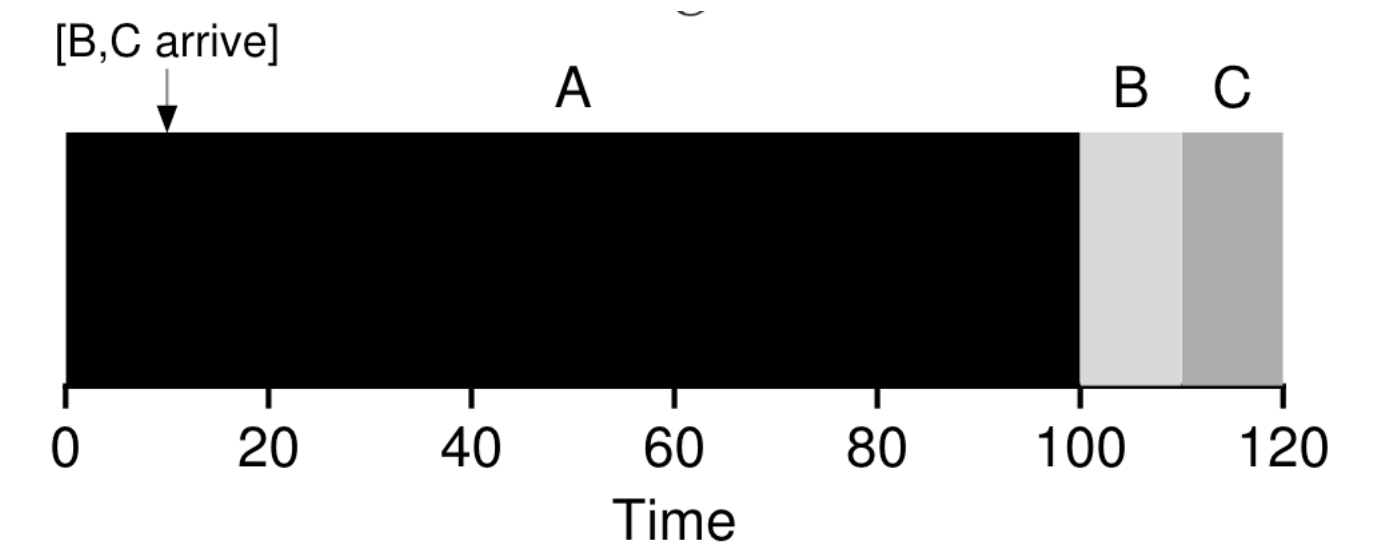


Figure 7.4: SJF With Late Arrivals From B and C

Batch job characteristics

- ~~Once started, each job runs to completion~~
- Preemptive schedulers switch to other processes before a job finishes
- Run-time of each job is known
- Main metric: turnaround time = (completion time - arrival time)

Shortest time to completion first

- Avg turnaround time = $(10+20+120)/3 = 50$
- Optimal scheduling algorithm

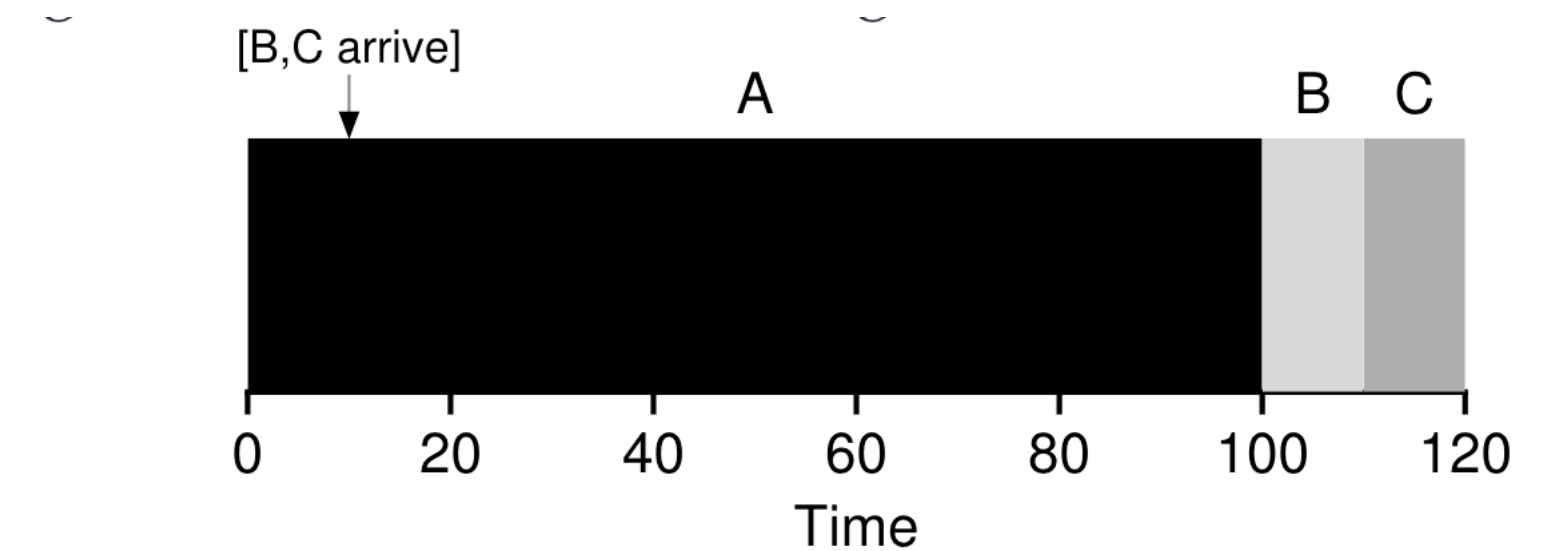


Figure 7.4: SJF With Late Arrivals From B and C

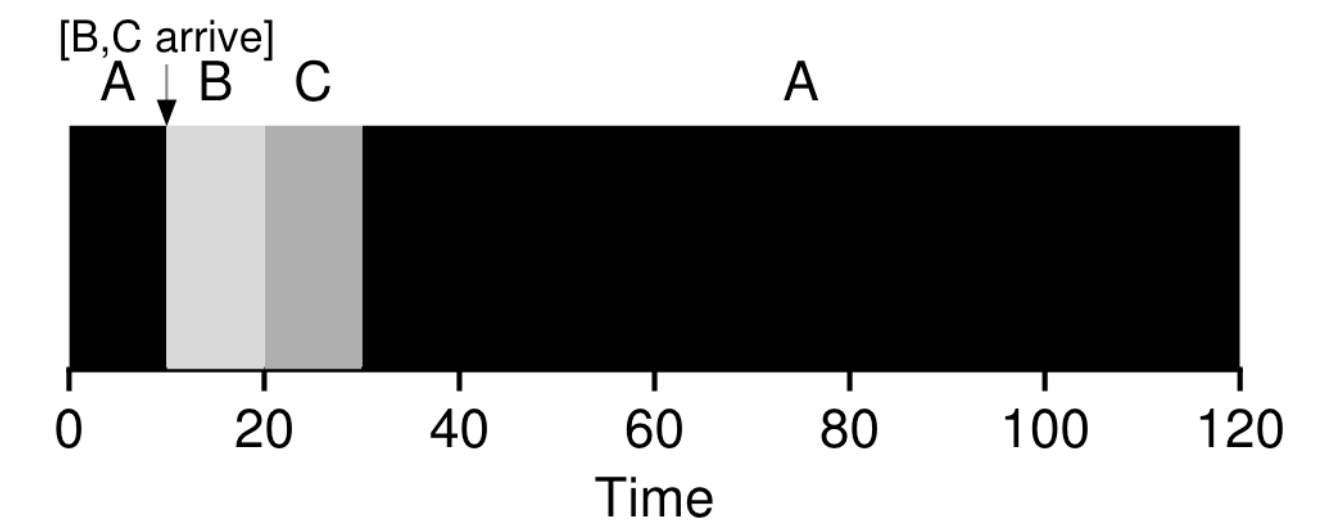
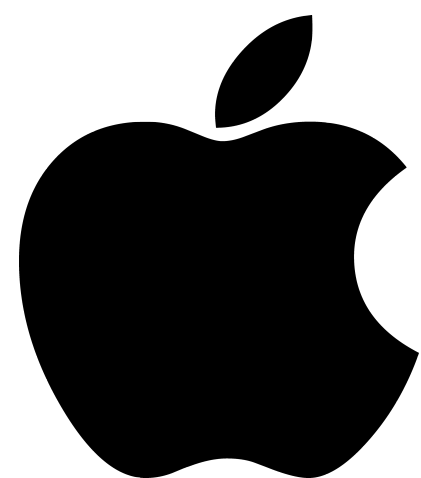
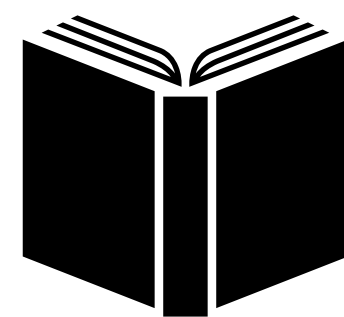
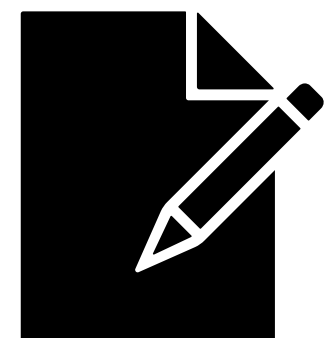
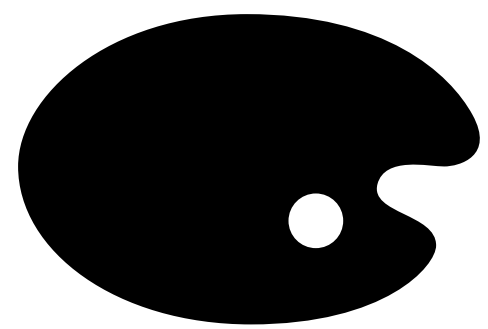


Figure 7.5: STCF Simple Example

Personal computers

1980s: Interactive jobs!



UNIX®



Interactive job characteristics

- Preemptive schedulers switch to other processes before a job finishes
- ~~Run-time of each job is known~~
- ~~Main metric: turnaround time = (completion time - arrival time)~~
- Main metric: response time = (first run - arrival time)

Interactive job characteristics

- Preemptive schedulers switch to other processes before a job finishes
- ~~Run-time of each job is known~~
- ~~Main metric: turnaround time = (completion time - arrival time)~~
- Main metric: response time = (first run - arrival time)
- Avg response time in SJF = $(0+5+10)/3 = 5$

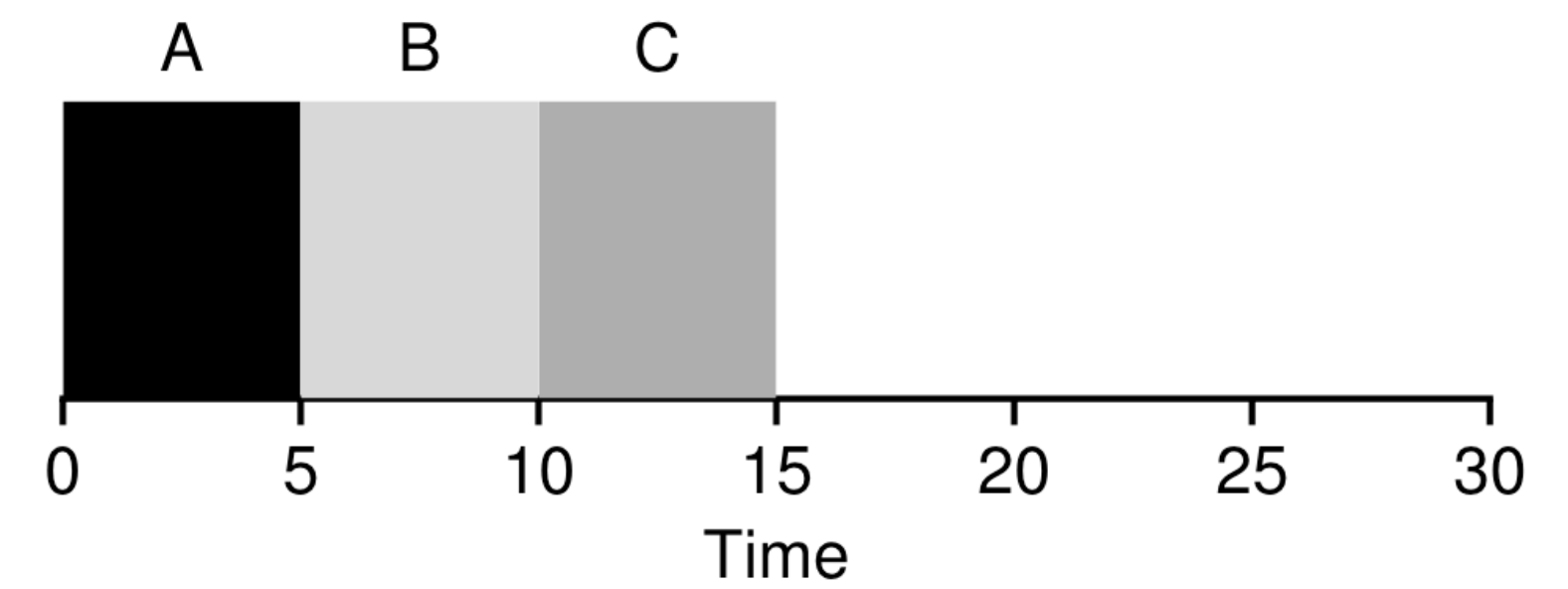


Figure 7.6: SJF Again (Bad for Response Time)

Round robin

- Run for a time slice T (scheduling quantum)

Round robin

- Run for a time slice T (scheduling quantum)
- Avg response time = $(0+1+2)/3 = 1$

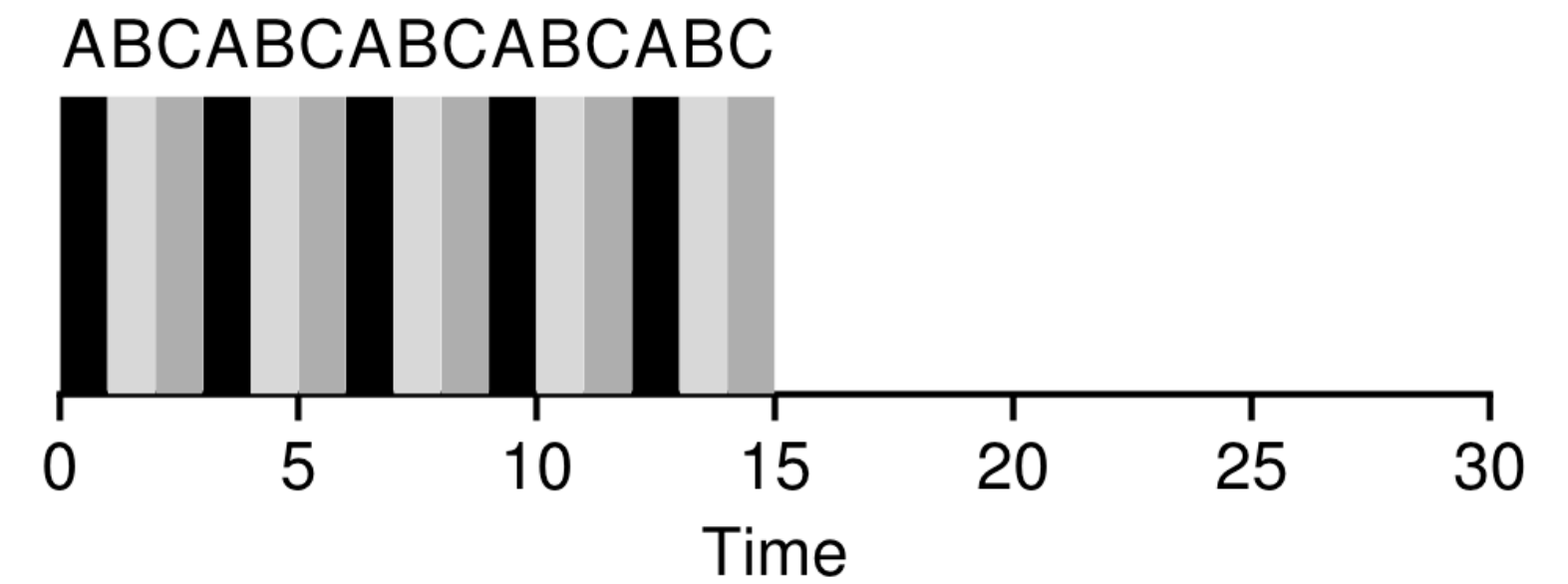


Figure 7.7: Round Robin (Good For Response Time)

Round robin

- Run for a time slice T (scheduling quantum)
- Avg response time = $(0+1+2)/3 = 1$
- How long should the time slice T be?

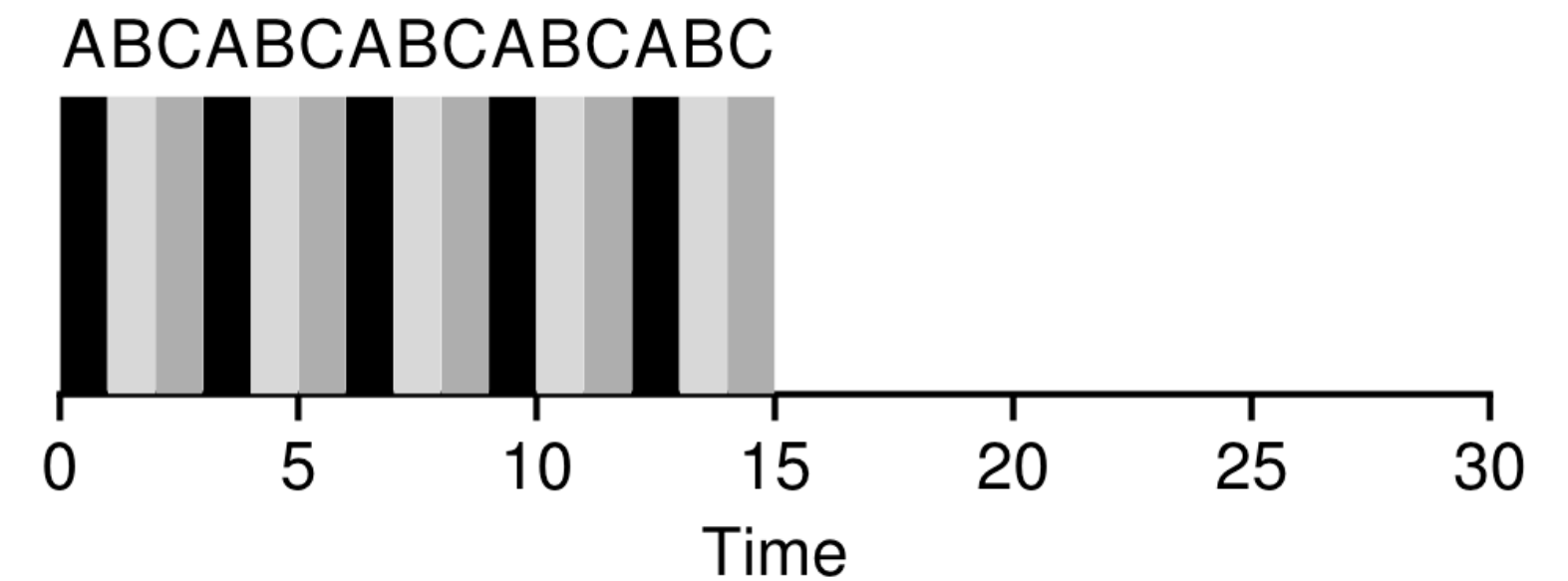


Figure 7.7: Round Robin (Good For Response Time)

Round robin

- Run for a time slice T (scheduling quantum)
- Avg response time = $(0+1+2)/3 = 1$
- How long should the time slice T be?
 - $T \gg$ context switch time (~ 1 us)

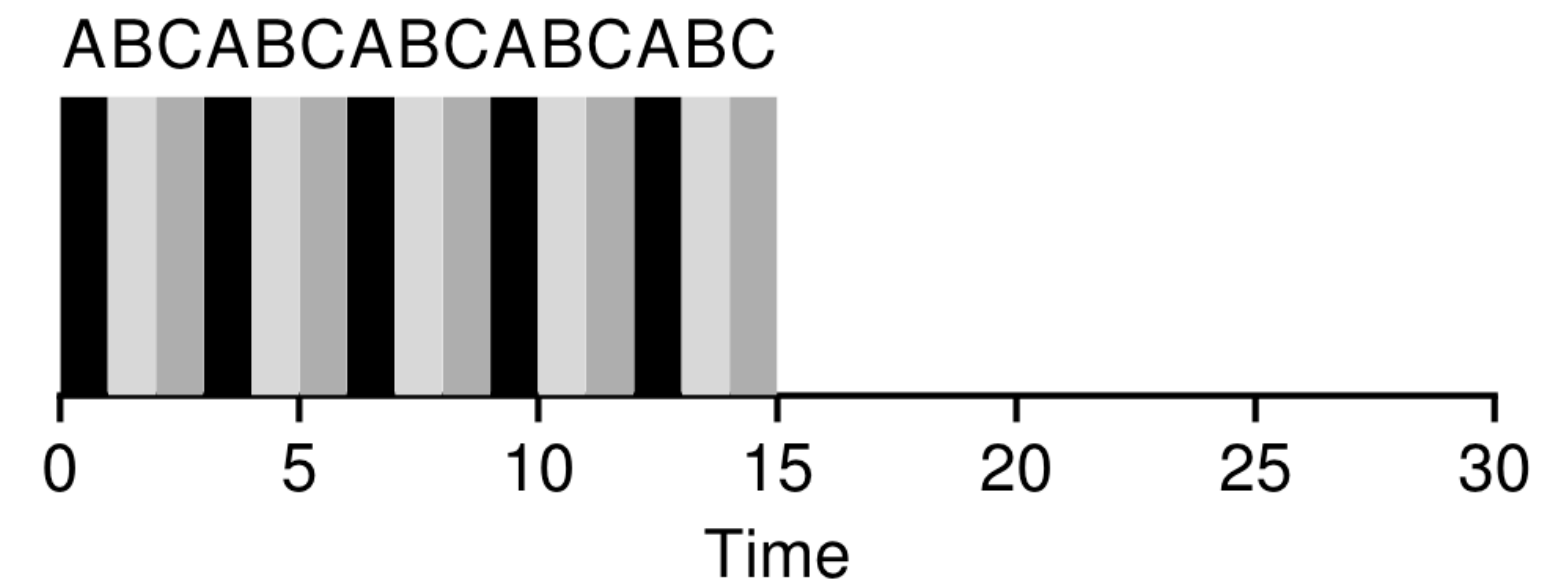


Figure 7.7: Round Robin (Good For Response Time)

Round robin

- Run for a time slice T (scheduling quantum)
- Avg response time = $(0+1+2)/3 = 1$
- How long should the time slice T be?
 - $T \gg$ context switch time (~ 1 us)
 - $T \sim$ human response time (few ms)

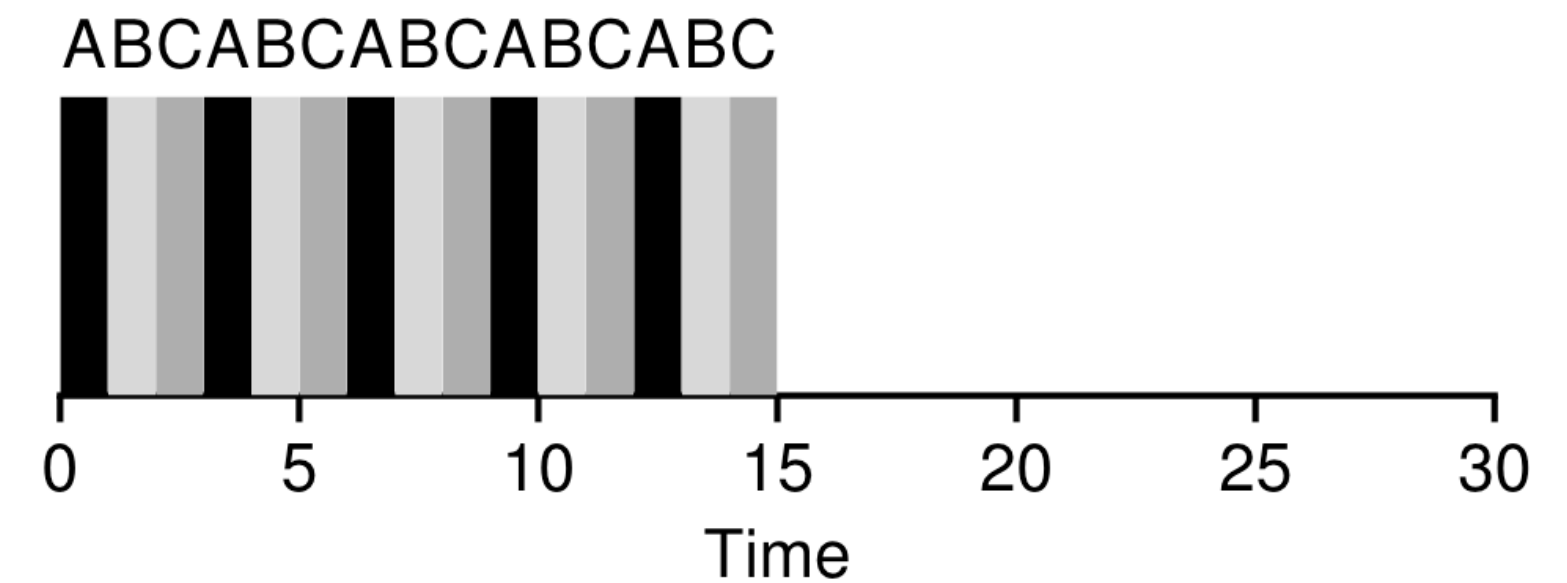


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 - $T \sim$ Used to be ~ 10 ms in 1970s. Still ~ 10 ms. CPU speeds: 60MHz $\rightarrow$ 3GHz.

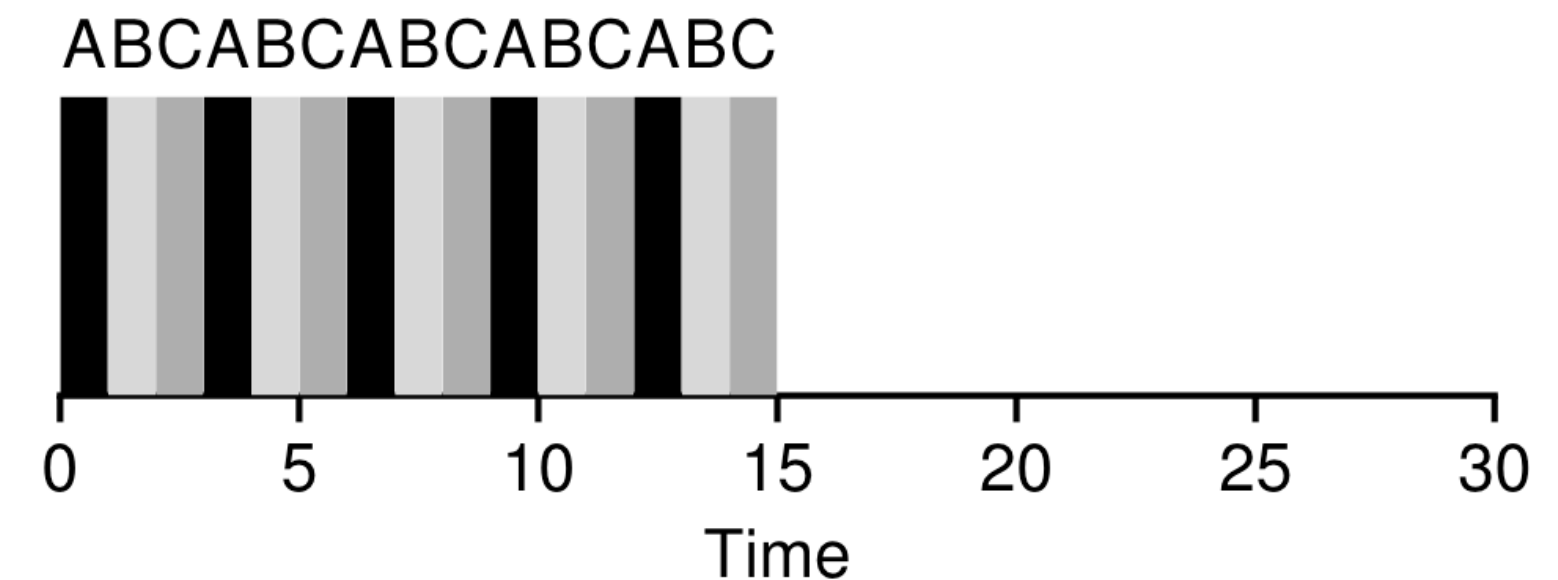


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- Worst policy for turnaround time: elongates completion times of all jobs

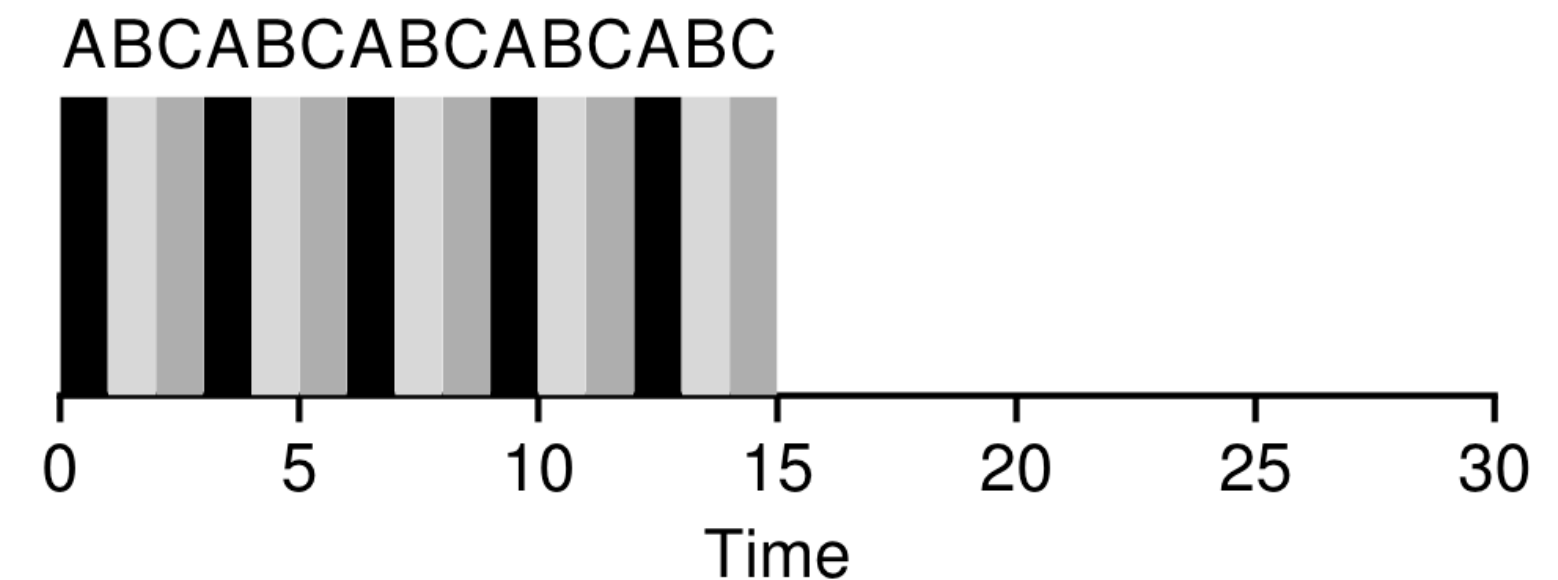


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- Worst policy for turnaround time: elongates completion times of all jobs
- Gives “fair share” to all jobs within small amount of time

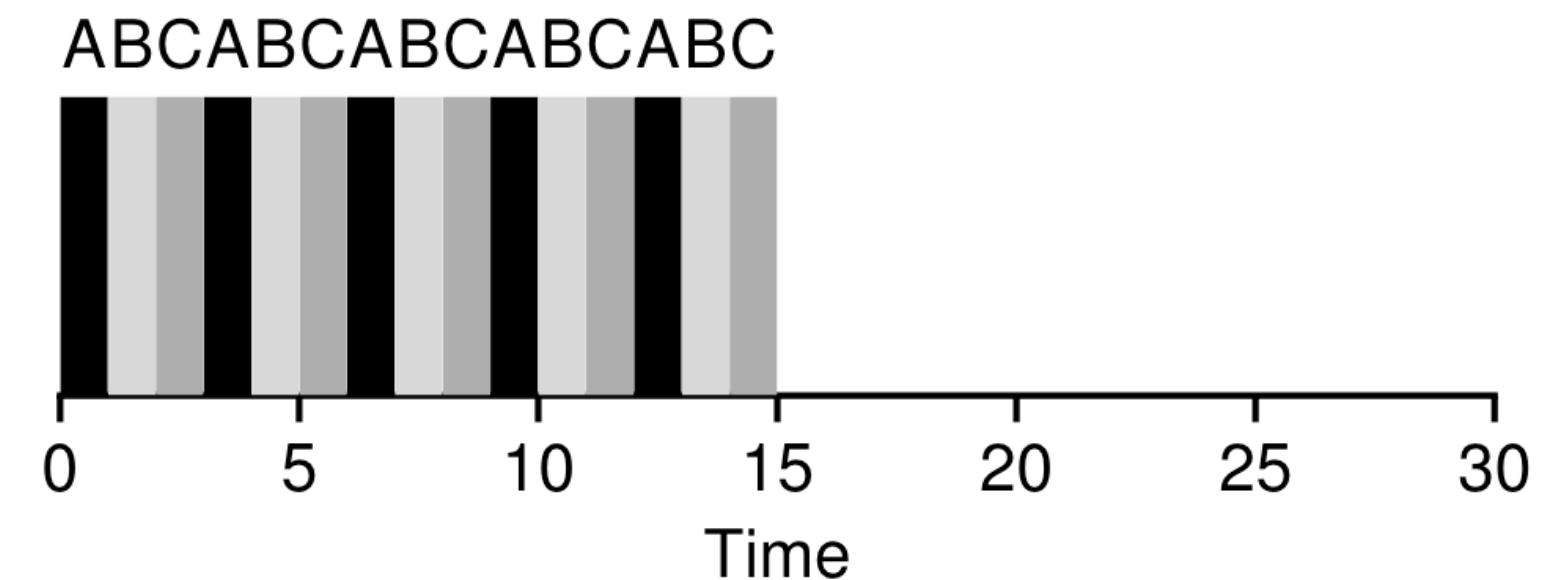


Figure 7.7: Round Robin (Good For Response Time)

Proportional share

- All jobs are not equally important

Proportional share

- All jobs are not equally important
 - Dropbox syncing should get lesser CPU time than music playback and gameplay

Proportional share

- All jobs are not equally important
 - Dropbox syncing should get lesser CPU time than music playback and gameplay
 - Give “proportional share” to jobs based on their importance

Stride scheduling

```
current = remove_min(queue); // pick client with minimum pass
schedule(current);           // use resource for quantum
current->pass += current->stride; // compute next pass using stride
insert(queue, current);      // put back into the queue
```

- Higher priority processes have low strides

Stride scheduling

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current = remove_min(queue); // pick client with minimum pass
schedule(current);           // use resource for quantum
current->pass += current->stride; // compute next pass using stride
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- Higher priority processes have low strides
- Example: A:B:C = 2:1:5

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```

- Higher priority processes have low strides
- Example: A:B:C = 2:1:5

Pass(A) (stride=100)	Pass(B) (stride=200)	Pass(C) (stride=40)	Who Runs?
0	0	0	A
100	0	0	B
100	200	0	C
100	200	40	C
100	200	80	C
100	200	120	A
200	200	120	C
200	200	160	C
200	200	200	...

Figure 9.3: Stride Scheduling: A Trace

Stride scheduling

```
current = remove_min(queue); // pick client with minimum pass
schedule(current);           // use resource for quantum
current->pass += current->stride; // compute next pass using stride
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```

- Higher priority processes have low strides
- Example: A:B:C = 2:1:5
- Deterministic scheduler

Pass(A) (stride=100)	Pass(B) (stride=200)	Pass(C) (stride=40)	Who Runs?
0	0	0	A
100	0	0	B
100	200	0	C
100	200	40	C
100	200	80	C
100	200	120	A
200	200	120	C
200	200	160	C
200	200	200	...

Figure 9.3: Stride Scheduling: A Trace

Lottery scheduling

- Probabilistic scheduler
 - $A:B \sim 3:1$
 - Generate a random number 0-99 in each time quantum
 - 0-74: run A, 75-99: run B

Lottery scheduling

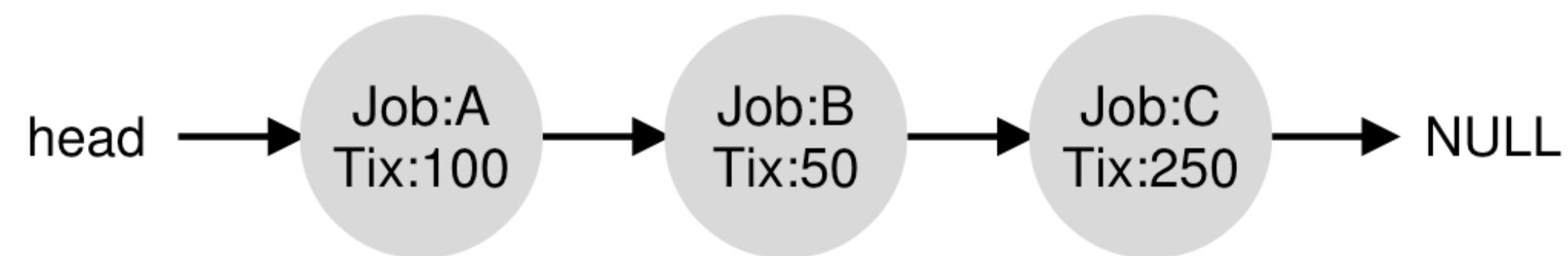
- Probabilistic scheduler
 - A:B ~ 3:1
 - Generate a random number 0-99 in each time quantum
 - 0-74: run A, 75-99: run B

63 85 70 39 76 17 29 41 36 39 10 99 68 83 63 62 43 0 49 49

A A A A A A A A A A A A A A A A
 B B B B

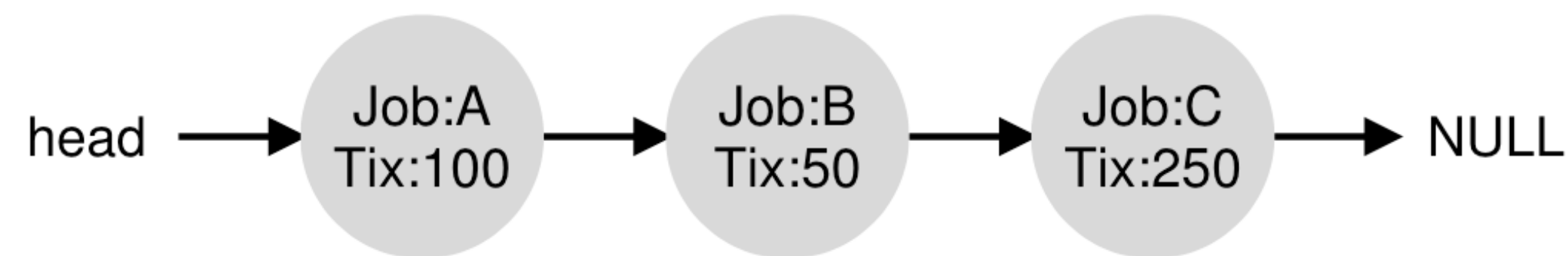
Lottery scheduling implementation

- Each process holds its own number of tickets



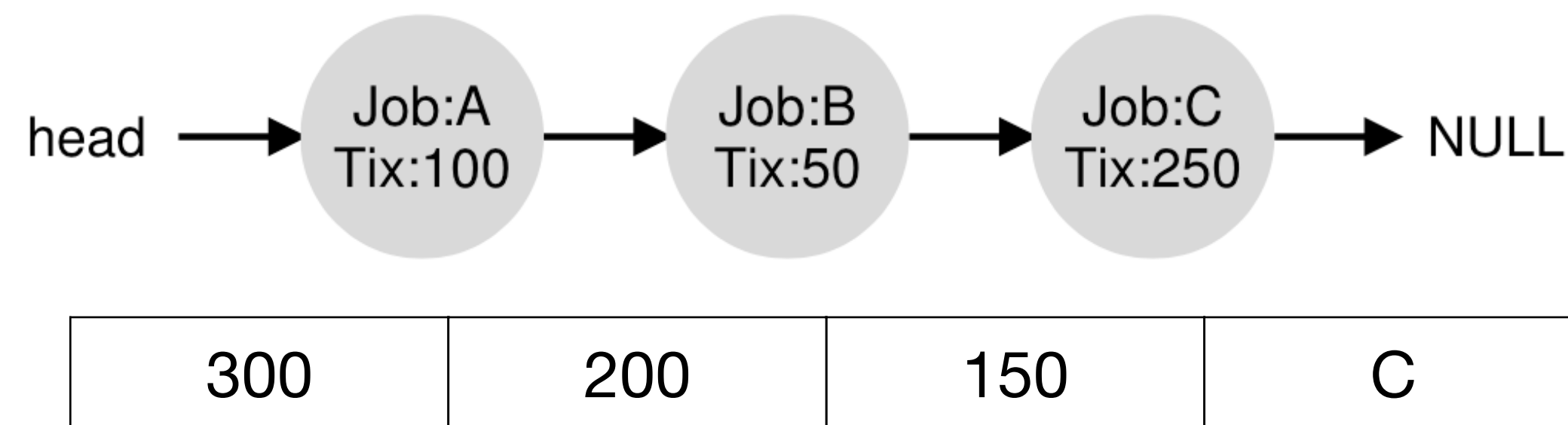
Lottery scheduling implementation

- Each process holds its own number of tickets
- Generate a random number (e.g. 300) between 0 and total number of tickets (e.g. 400) in each time quantum
- Traverse list while reducing by tickets to find the process



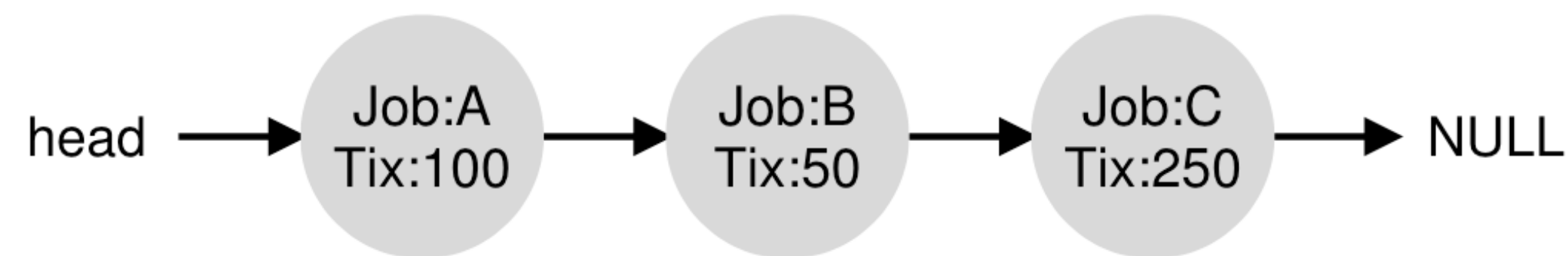
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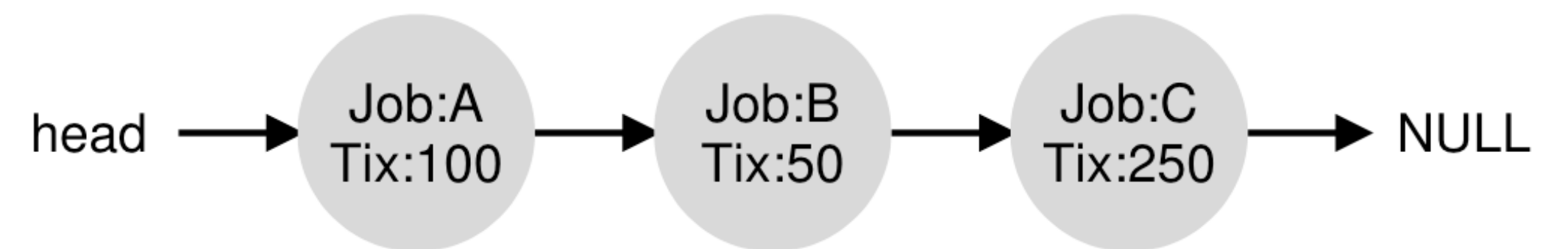
300	200	150	C
125	25		B

Setting priority

- When processes are trusted, they can raise or lower their priority
 - Example: music player lowers its priority when user pauses the song

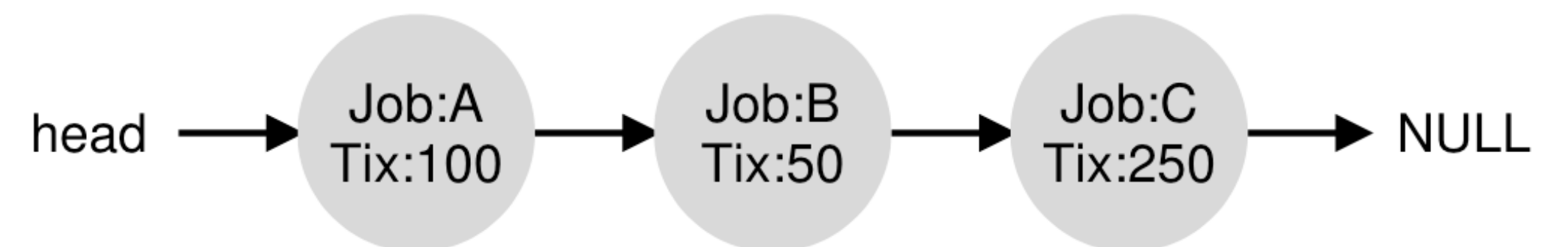
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Setting priority

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 - e.g, B changes its tickets to 250
- Otherwise, users can manually lower priority:



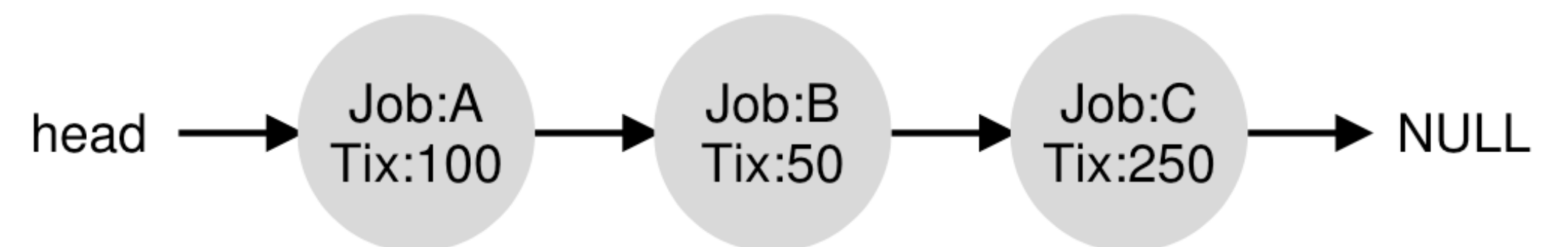
Setting priority

- When processes are trusted, they can raise or lower their priority
 - Example: music player lowers its priority when user pauses the song
 - e.g, B changes its tickets to 250

- Otherwise, users can manually lower priority:

```
time taskset -c 1 ./nicedemo
```

```
time nice -n 19 taskset -c 1 ./nicedemo
```



Process states

- Processes can be blocked upon doing IO

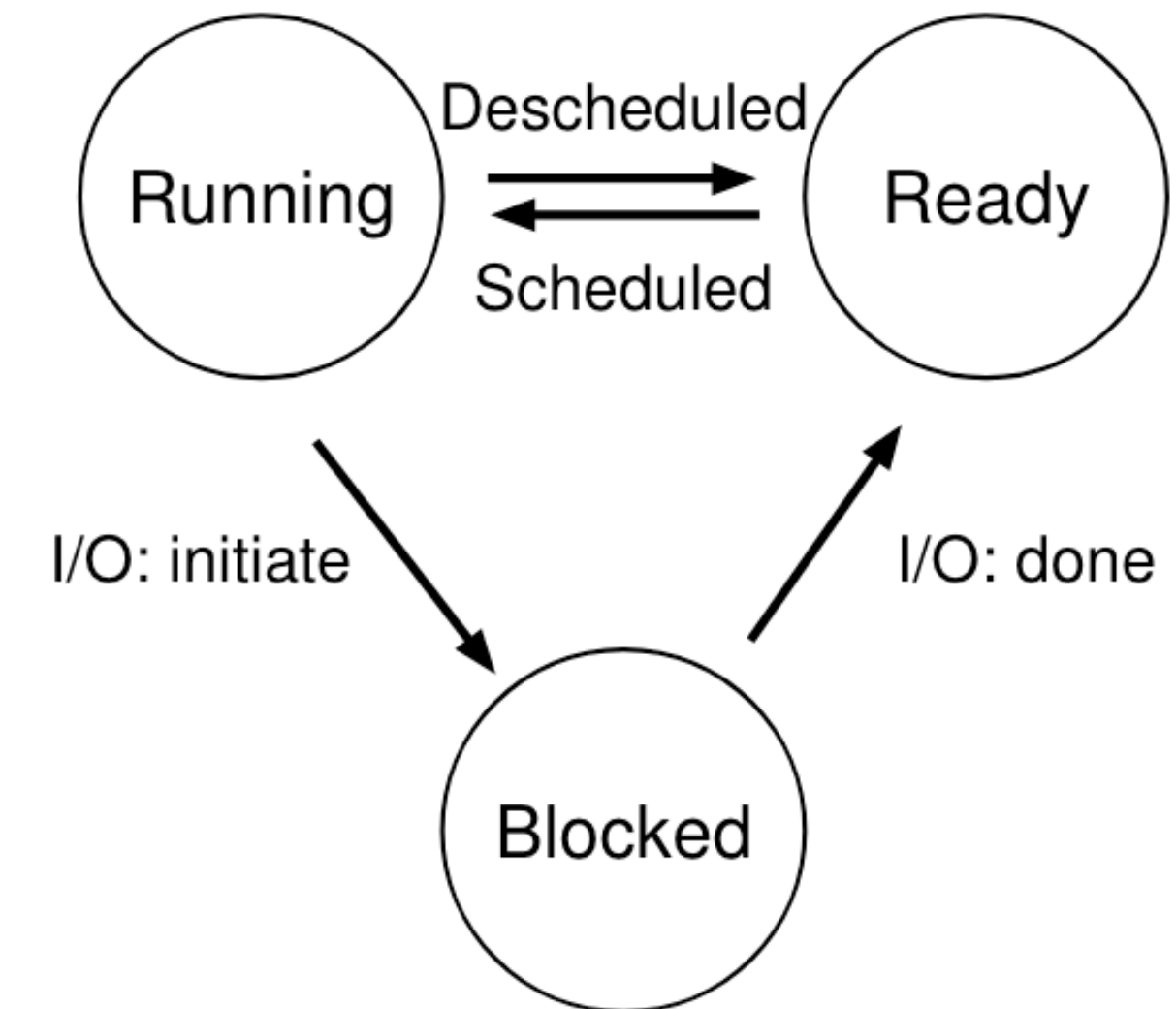


Figure 4.2: Process: State Transitions

Process states

- Processes can be blocked upon doing IO

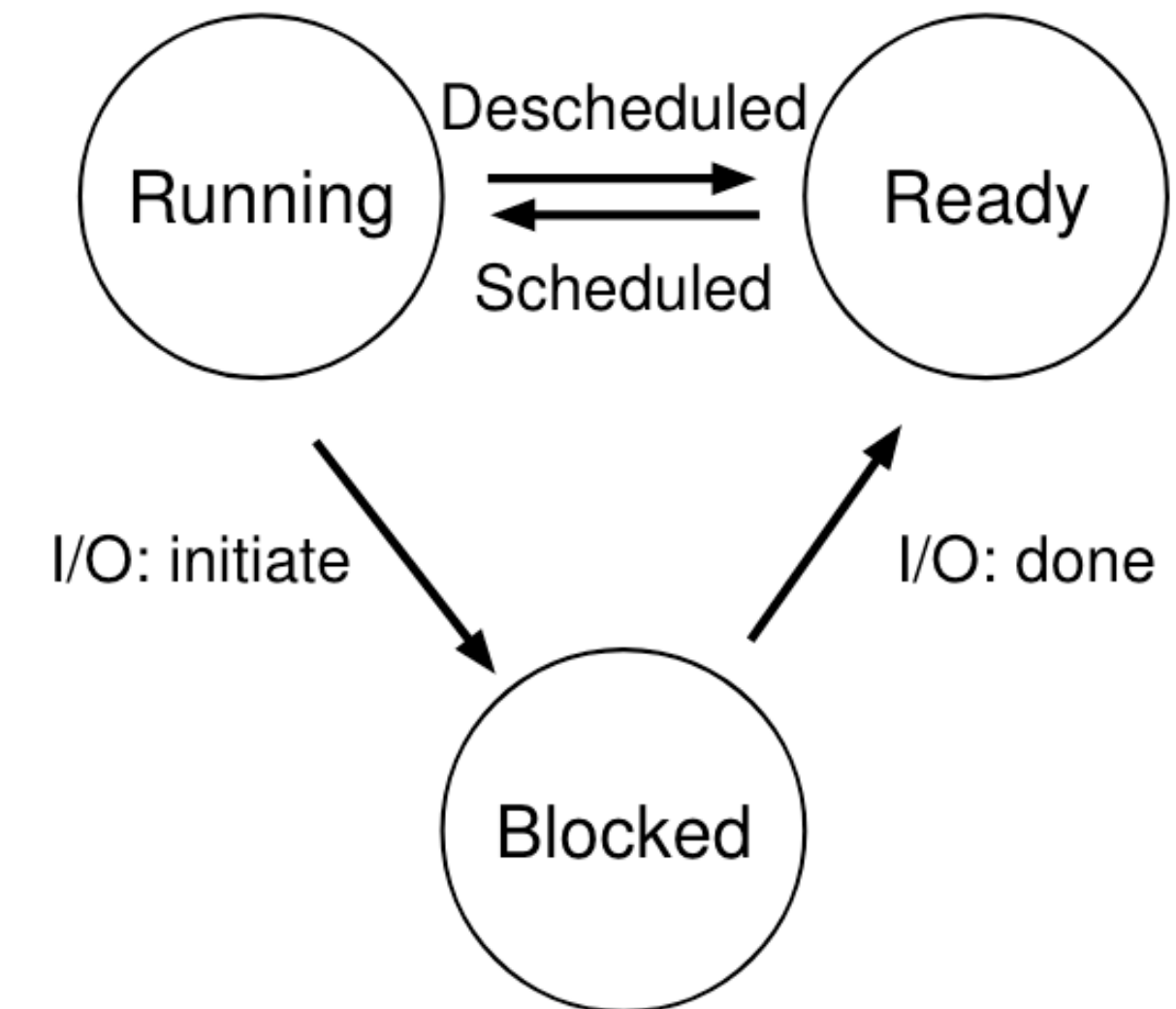


Figure 4.2: Process: State Transitions

Sleep in xv6 disk driver

```
void sleep(void *chan) {                                proc.c
    struct proc *p = myproc();
    // Go to sleep.
    p->chan = chan;
    p->state = SLEEPING;
    sched();
    // Tidy up.
    p->chan = 0;
}

void wakeup(void *chan) {
    struct proc *p;
    for(p = ptable.proc; p < &ptable.proc[NPROC]; p++)
        if(p->state == SLEEPING && p->chan == chan)
            p->state = RUNNABLE;
}
```

```
// Per-process state                                proc.h
struct proc {
    ...
    void *chan;      // If non-zero, sleeping on chan
}
```

```
void ideintr(void) {                                    ide.c
    struct buf *b = idequeue;
    if(!(b->flags & B_DIRTY) && idewait(1) >= 0)
        insl(0x1f0, b->data, BSIZE/4);
    b->flags |= B_VALID;
    b->flags &= ~B_DIRTY;
    wakeup(b);
}

void iderw(struct buf *b){
    if(idequeue == b)
        idestart(b);
    while((b->flags & (B_VALID|B_DIRTY)) != B_VALID){
        noop();
        sleep(b);
    }
}
```

IO aware scheduling

- Mixture of CPU intensive and IO bound processes

IO aware scheduling

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 - Turnaround time is important for CPU intensive processes, e.g., training an ML model

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- Challenge: OS does not know which process is CPU intensive / IO bound

IO aware scheduling

- Mixture of CPU intensive and IO bound processes
 - Turnaround time is important for CPU intensive processes, e.g., training an ML model
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- Challenge: OS does not know which process is CPU intensive / IO bound
- Round robin (fair share) and proportional share algorithms assume all processes are CPU intensive

IO aware scheduling

- Mixture of CPU intensive and IO bound processes
 - Turnaround time is important for CPU intensive processes, e.g., training an ML model
 - Response time is important for interactive IO bound processes, e.g., editor
- Challenge: OS does not know which process is CPU intensive / IO bound
- Round robin (fair share) and proportional share algorithms assume all processes are CPU intensive
- Example: Round robin scheduler (xv6) gives poor response time to interactive processes

CPU	1	1	1	2	2	2	3	3	3	1	1	1	2	2	2	3	3	3	1
Disk																			

CPU	1	1	2	3	3	3	2	2	2	3	3	3	1	1	2	3	3	3	2
Disk			1	1	1	1	1								1	1	1	1	1

Multi-level feedback queue (MLFQ)

- Strict priority using multiple queues
 - $(A > C) \Rightarrow A$ runs
 - $(A = B) \Rightarrow$ Round robin

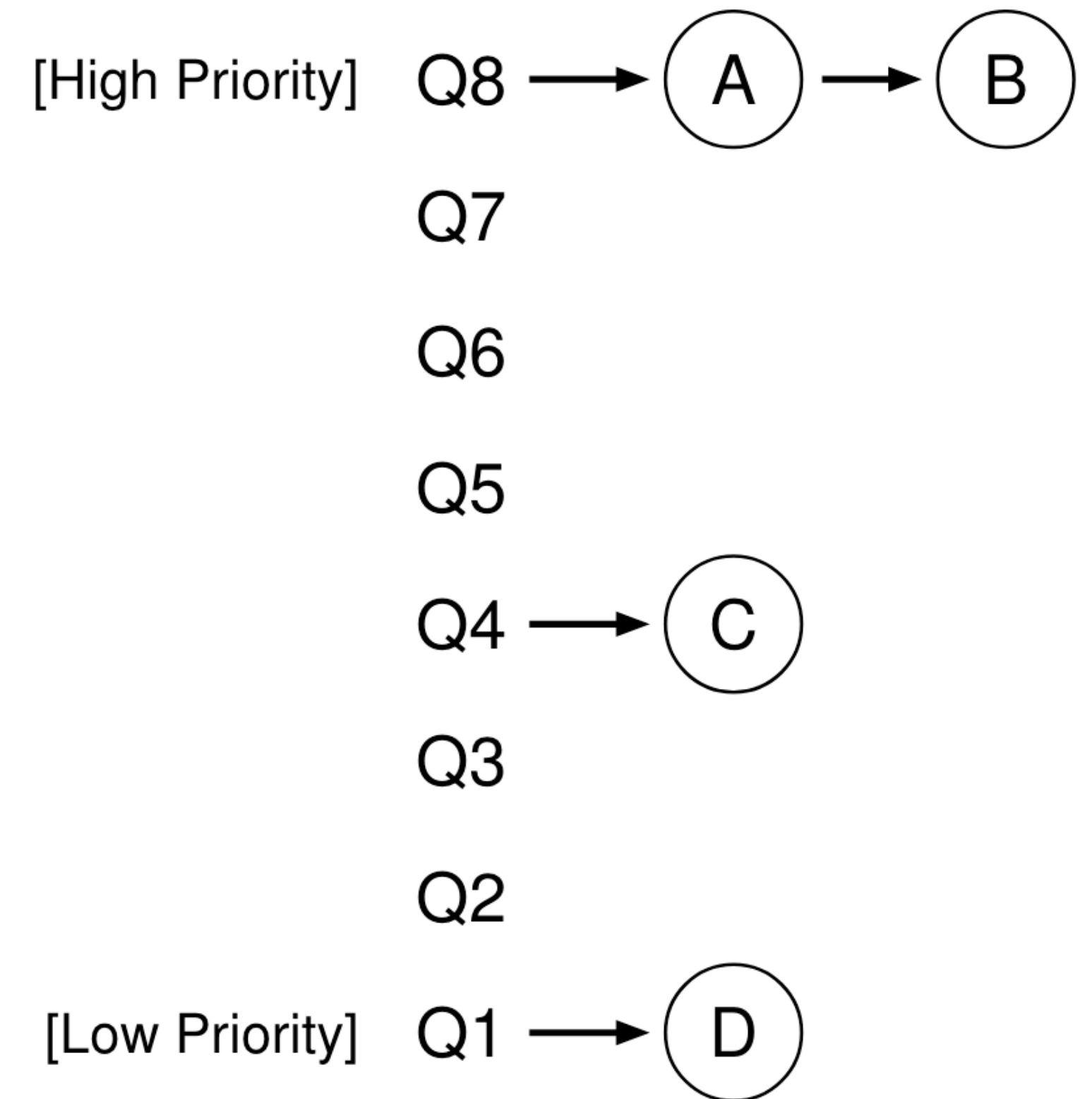


Figure 8.1: MLFQ Example

Multi-level feedback queue (MLFQ)

- Strict priority using multiple queues
 - $(A > C) \Rightarrow A$ runs
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- Goal: IO bound (interactive) processes shall take higher priority for CPU over CPU-intensive processes

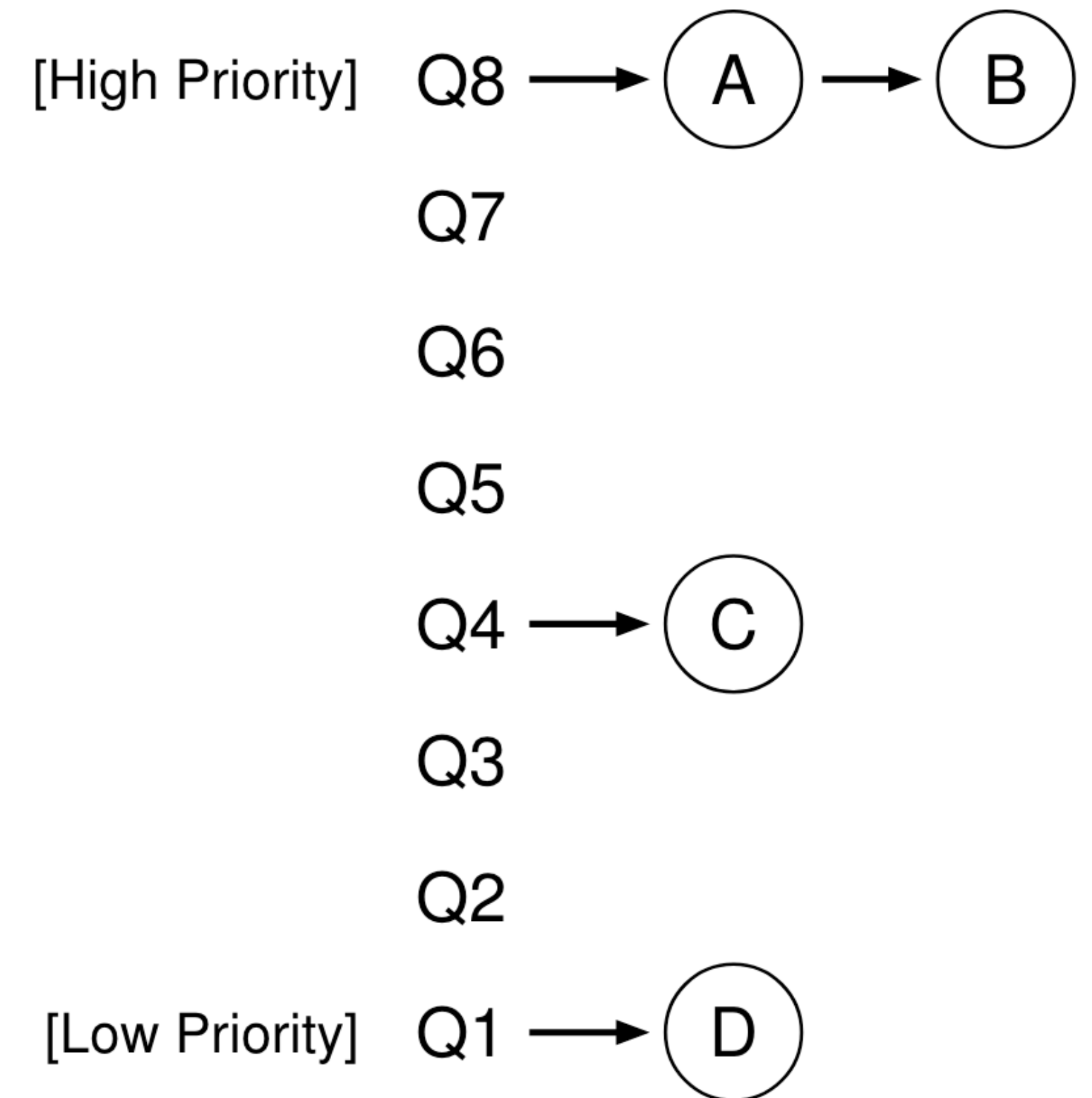


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Multi-level feedback queue (MLFQ)

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- Goal: IO bound (interactive) processes shall take higher priority for CPU over CPU-intensive processes
- Challenge: OS does not know nature of the process

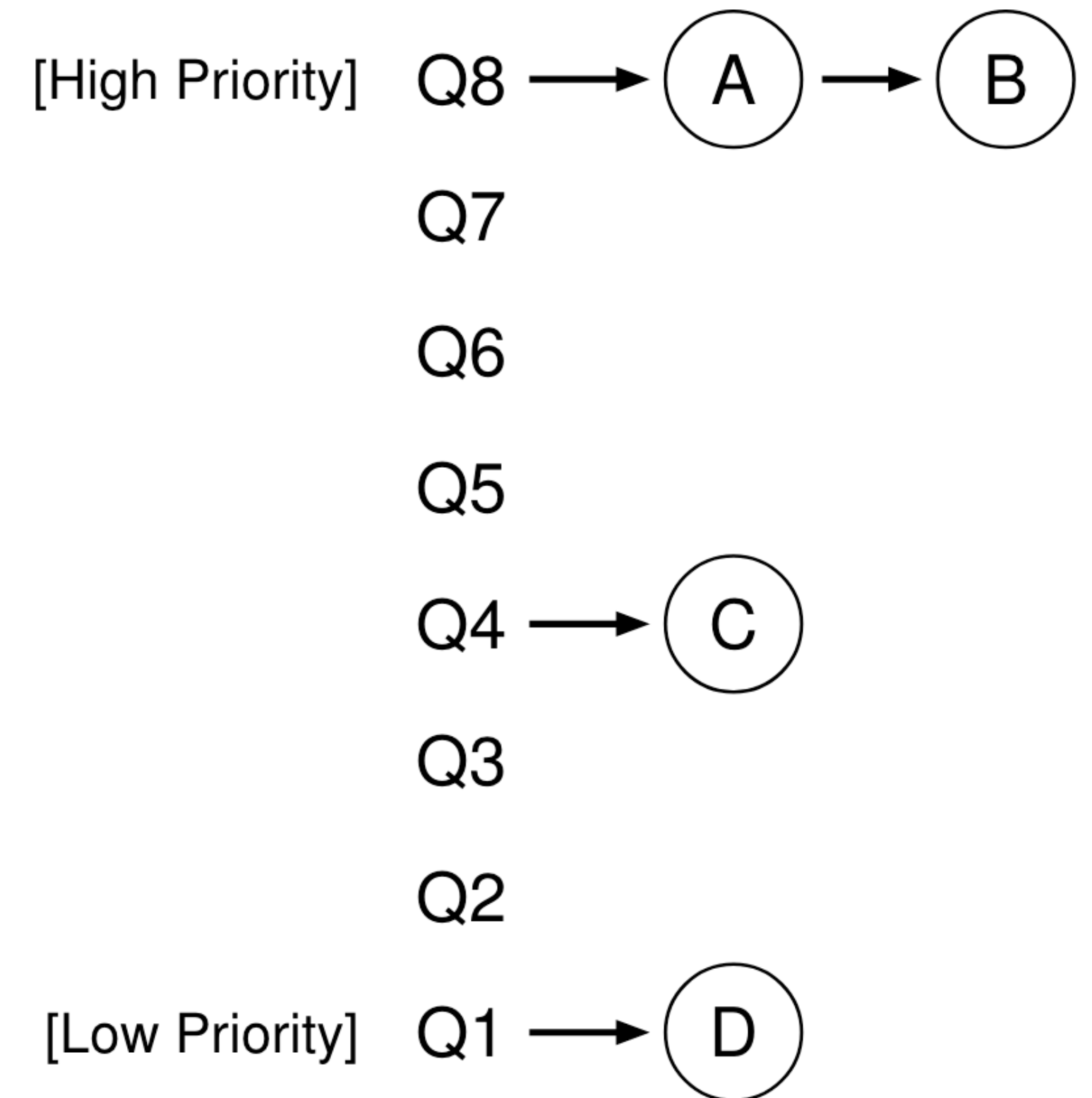


Figure 8.1: MLFQ Example

MLFQ level management

- Observation: A CPU intensive process completely uses up its time slice.
e.g, process 2

CPU	1	1	2	3	3	3	2	2	2	3	3	3	1	1	2	3	3	3	2
Disk			1	1	1	1	1								1	1	1	1	1

MLFQ level management

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CPU	1	1	2	3	3	3	2	2	2	3	3	3	1	1	2	3	3	3	2
Disk			1	1	1	1	1								1	1	1	1	1

- Goal: IO bound (interactive) processes shall take higher priority for CPU over CPU-intensive processes
 - If a process completely uses up its time slice, lower its priority

MLFQ level management

- Observation: A CPU intensive process completely uses up its time slice.
e.g, process 2

CPU	1	1	2	3	3	3	2	2	2	3	3	3	1	1	2	3	3	3	2
Disk			1	1	1	1	1								1	1	1	1	1

- Goal: IO bound (interactive) processes shall take higher priority for CPU over CPU-intensive processes
 - If a process completely uses up its time slice, lower its priority

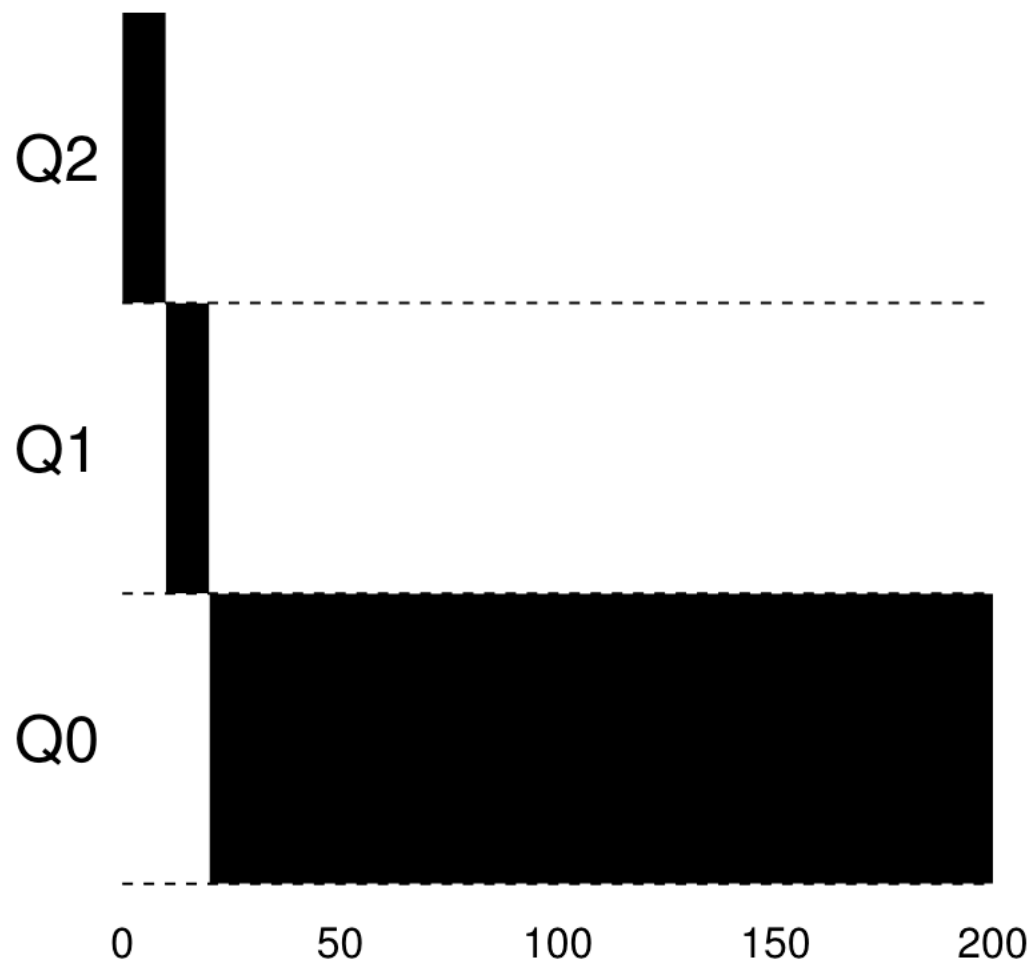


Figure 8.2: Long-running Job Over Time

MLFQ approximates STCF

- Short jobs get priority over long-running jobs

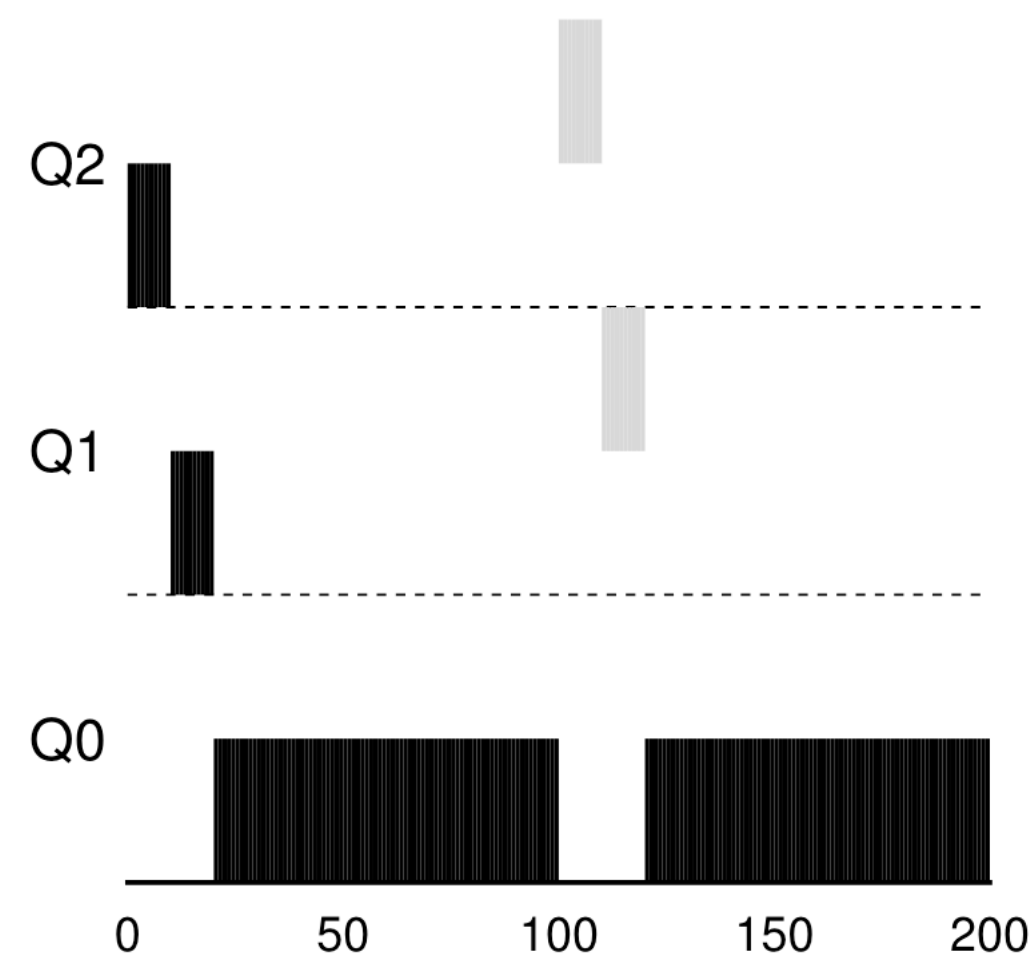


Figure 8.3: Along Came An Interactive Job

MLFQ approximates STCF

- Short jobs get priority over long-running jobs
- IO intensive jobs are prioritised over CPU intensive processes => better response time for interactive jobs

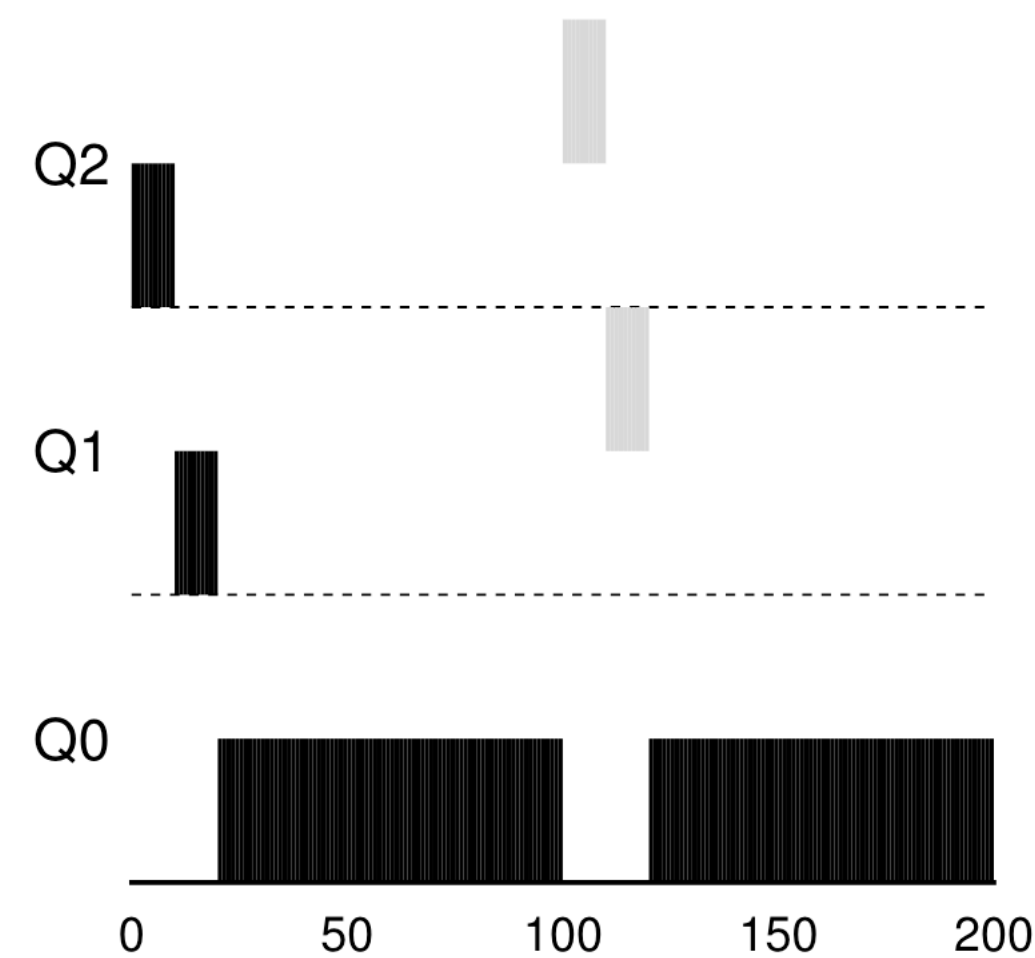


Figure 8.3: Along Came An Interactive Job

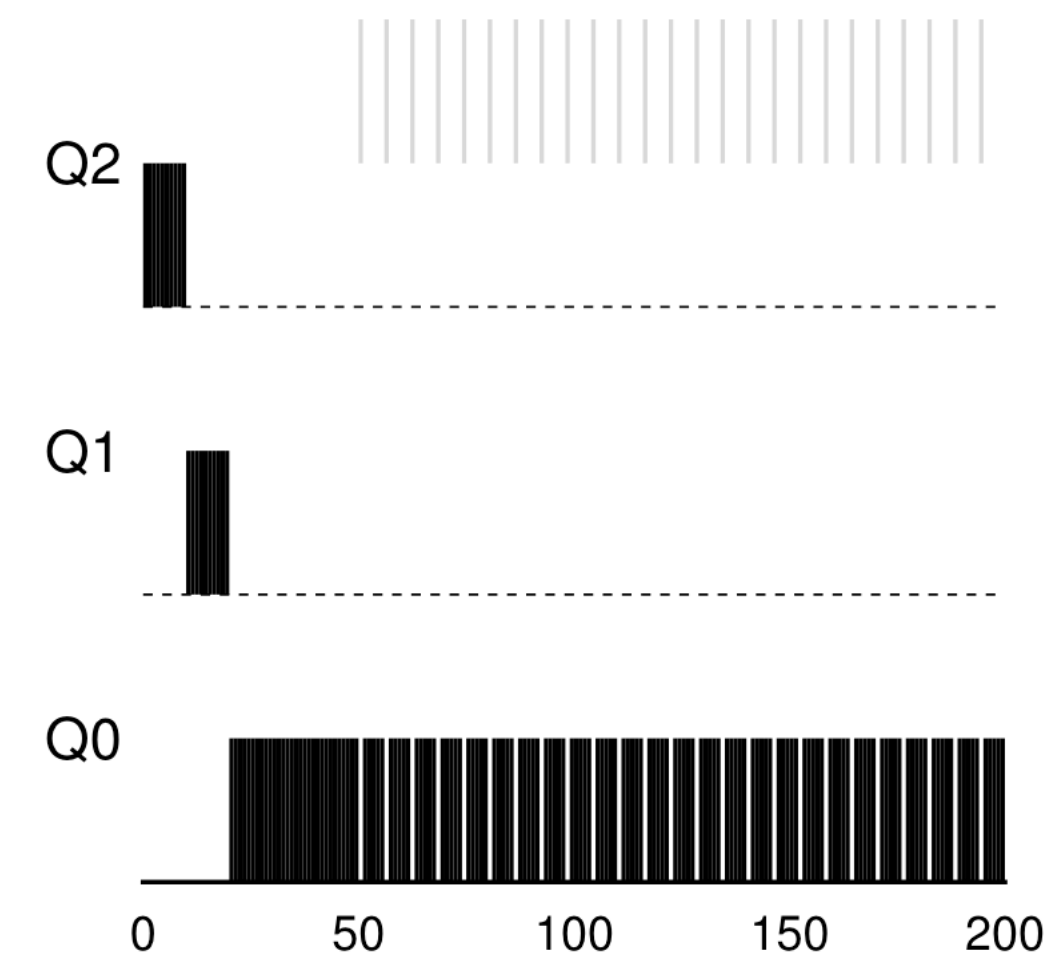
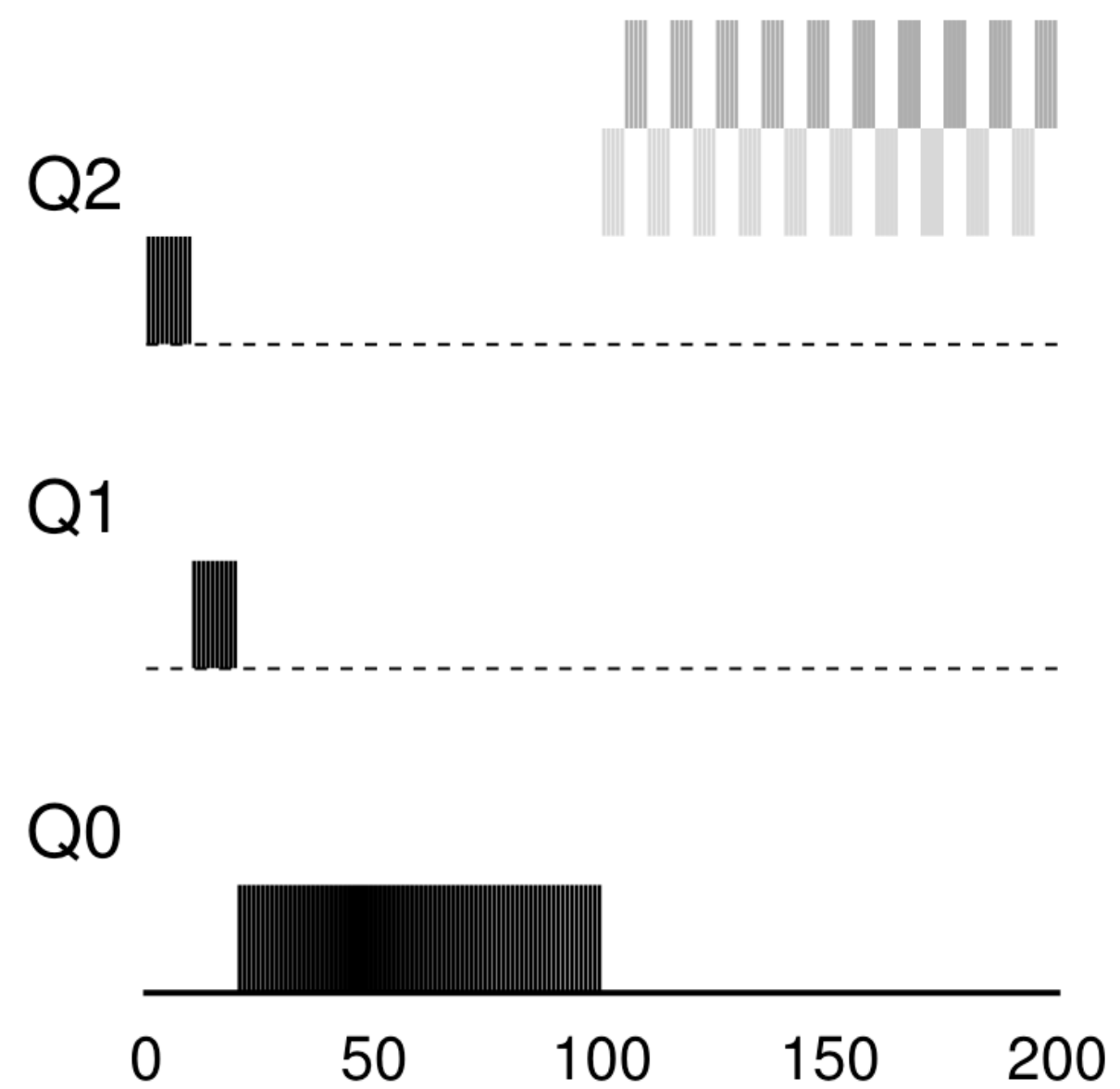


Figure 8.4: A Mixed I/O-intensive and CPU-intensive Workload

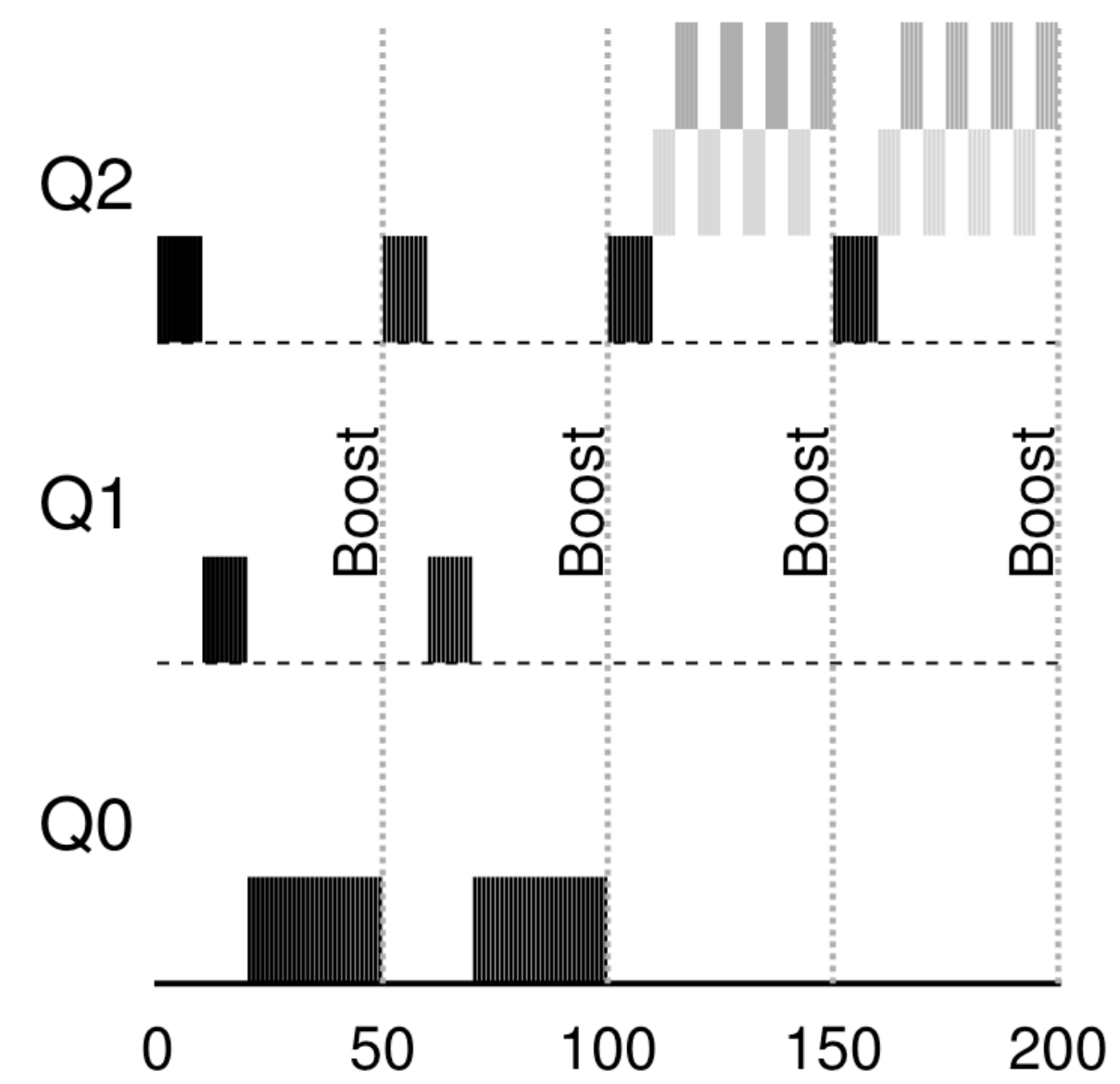
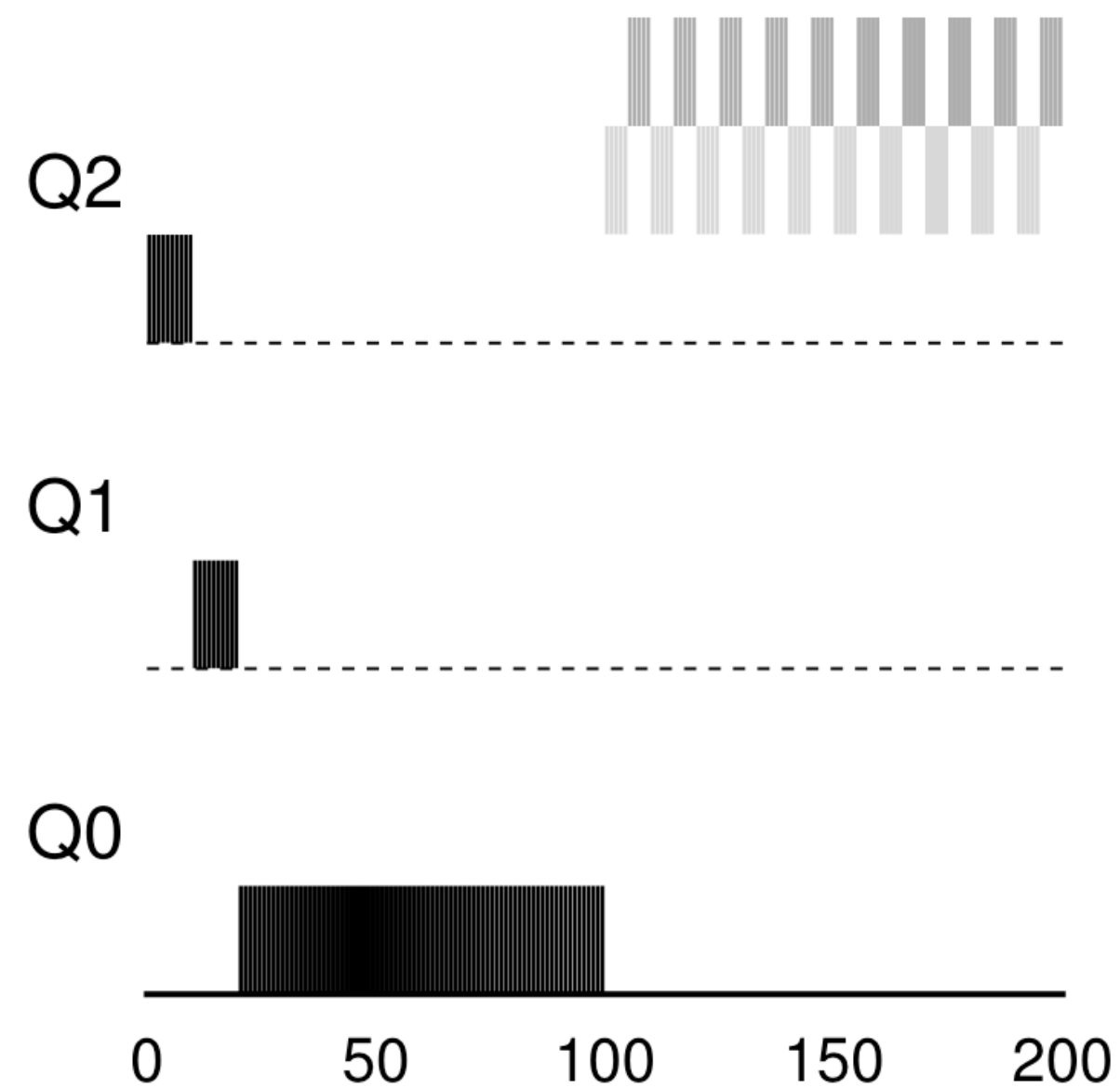
Starvation freedom

- Many IO bound jobs can starve CPU intensive jobs



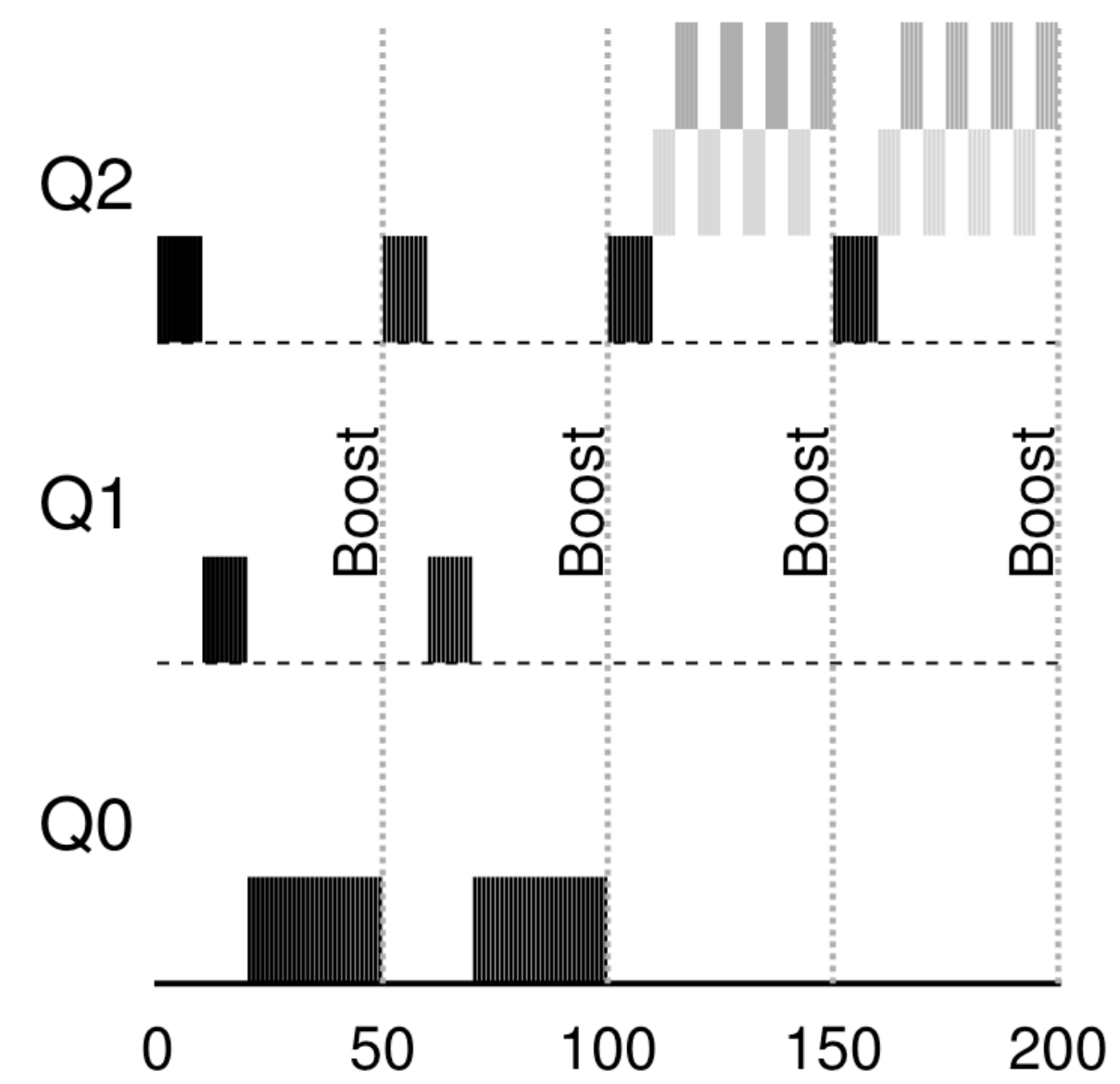
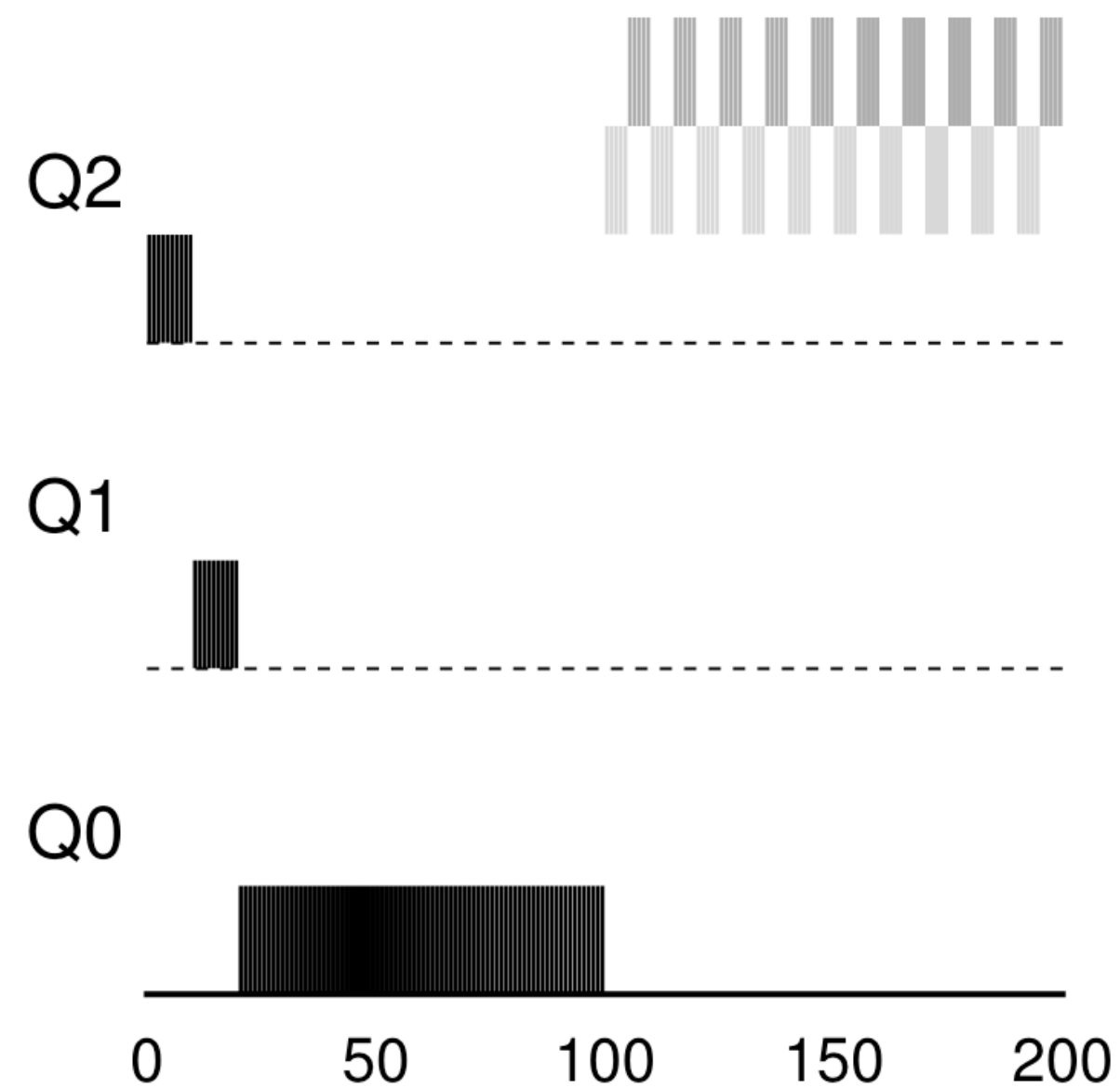
Starvation freedom

- Many IO bound jobs can starve CPU intensive jobs
- Boost all jobs to highest priority queue after some time quantum



Starvation freedom

- Many IO bound jobs can starve CPU intensive jobs
- Boost all jobs to highest priority queue after some time quantum
- Also helps if CPU intensive job has become IO intensive job



Reduce scheduling overhead for CPU intensive jobs

- If there are no IO bound jobs, CPU intensive jobs run without preemption for a longer time slice

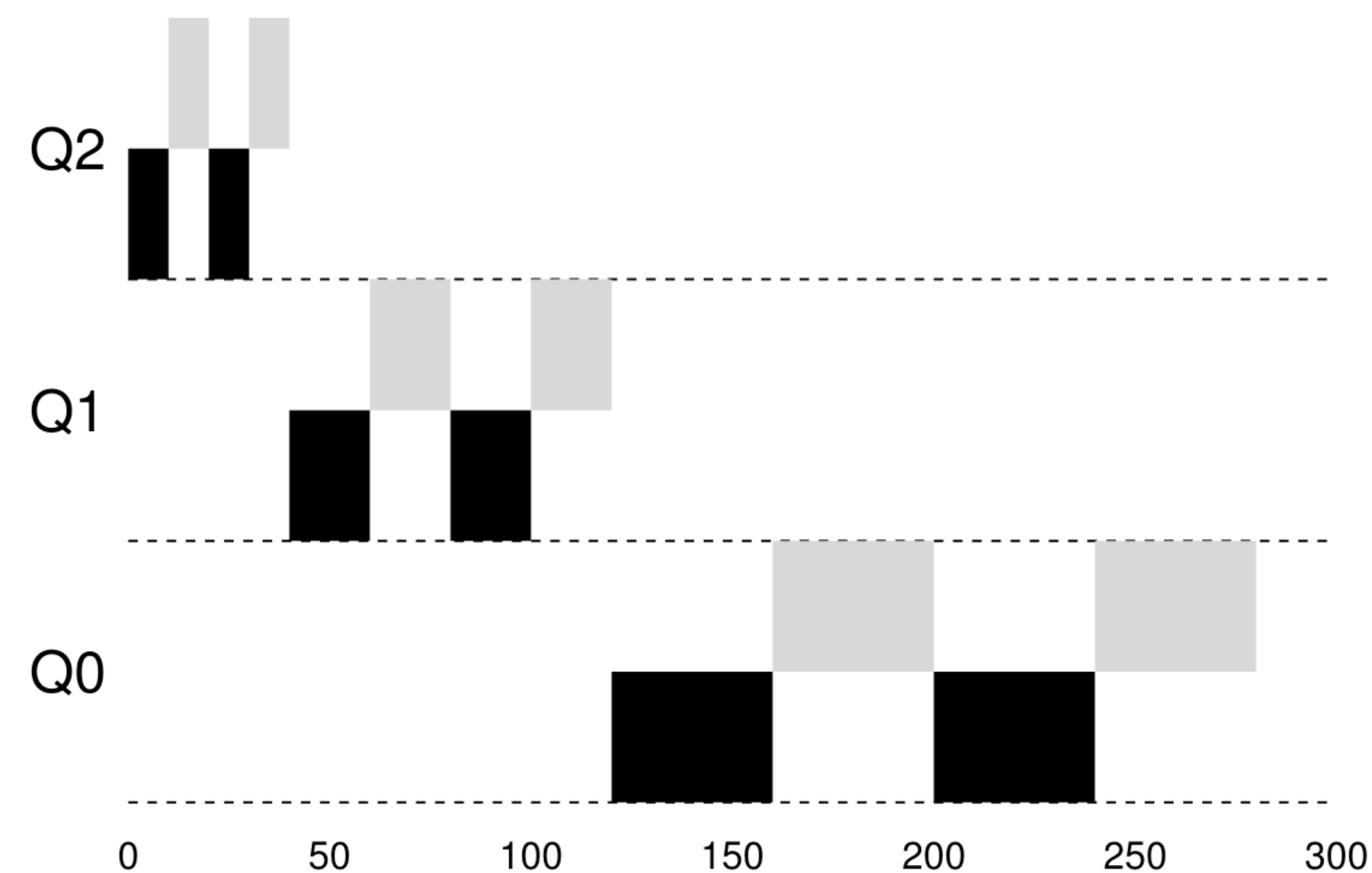
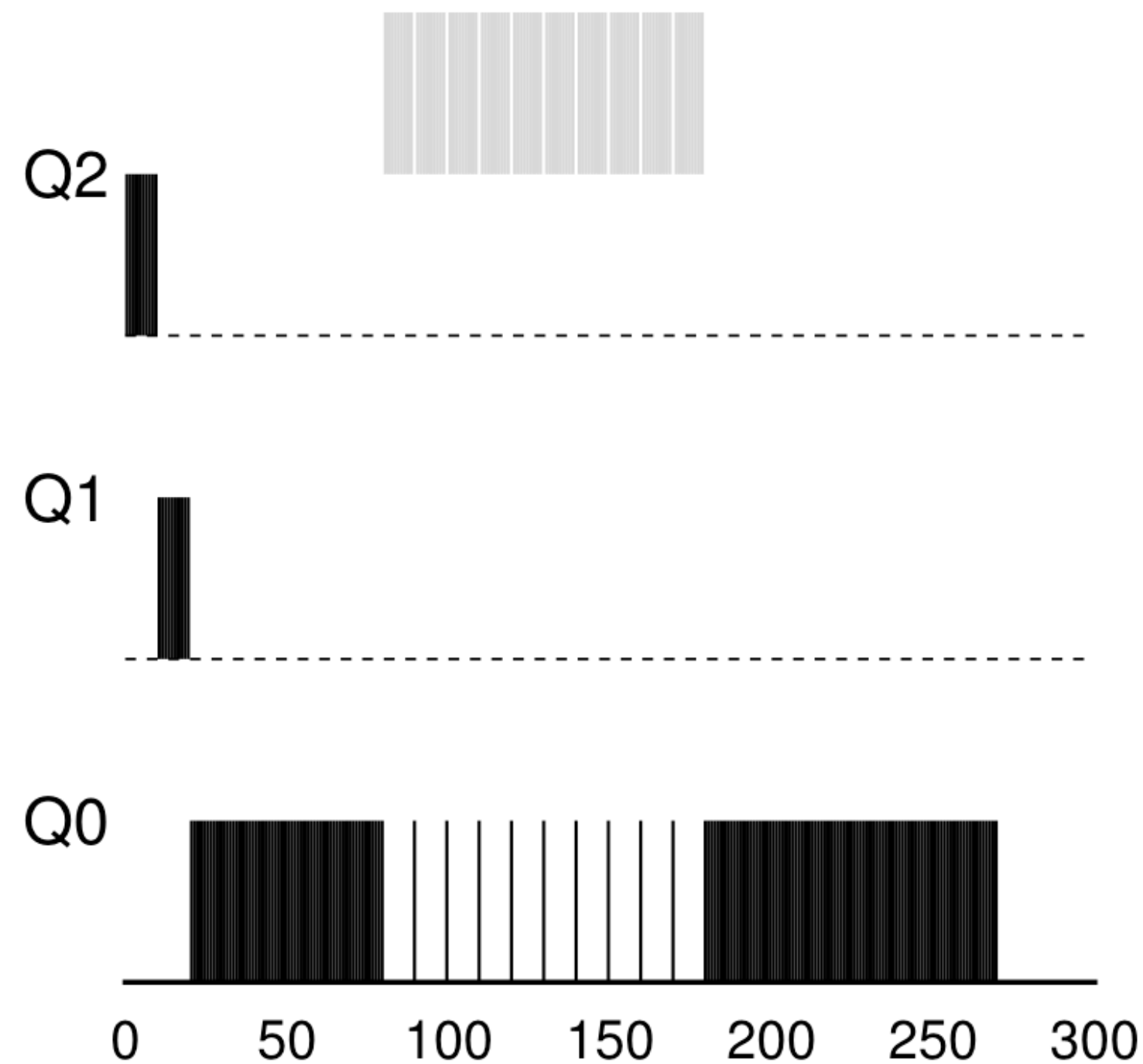


Figure 8.7: Lower Priority, Longer Quanta

Scheduler attacks

- Example: yield CPU right before time quanta expires to stay in highest priority
- In Linux 2.4 ~2003. Freezes up desktops!



Linux Completely Fair Scheduler (CFS)

- Priority scheduler:

```
static const int prio_to_weight[40] = {  
    /* -20 */      88761,      71755,      56483,      46273,      36291,  
    /* -15 */      29154,      23254,      18705,      14949,      11916,  
    /* -10 */       9548,       7620,       6100,       4904,       3906,  
    /*  -5 */       3121,       2501,       1991,       1586,       1277,  
    /*   0 */       1024,        820,        655,        526,        423,  
    /*   5 */        335,        272,        215,        172,        137,  
    /*  10 */        110,         87,         70,         56,         45,  
    /*  15 */         36,         29,         23,         18,         15,  
};
```

Linux Completely Fair Scheduler (CFS)

- Priority scheduler:
 - High priority processes have low nice values. They get longer time slices
 - $\text{weight}[\text{nice}] \approx 1024 * 0.8^{\text{nice}}$

$$\text{time_slice}_k = \frac{\text{weight}_k}{\sum_{n=0}^{n-1} \text{weight}_i} \cdot \text{sched_latency}$$

```
static const int prio_to_weight[40] = {
    /* -20 */      88761,      71755,      56483,      46273,      36291,
    /* -15 */      29154,      23254,      18705,      14949,      11916,
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    /*  10 */        110,         87,         70,         56,         45,
    /*  15 */         36,         29,         23,         18,         15,
};
```

Completely Fair Scheduler (CFS)

- Each process has vruntime. Update vruntime after each time process is run.
- Maintain runnable processes in a red-black tree (sorted by vruntime). Run the process with lowest vruntime.
- When a new process is run:

$$vruntime_i = vruntime_i + \frac{weight_0}{weight_i} \cdot runtime_i$$

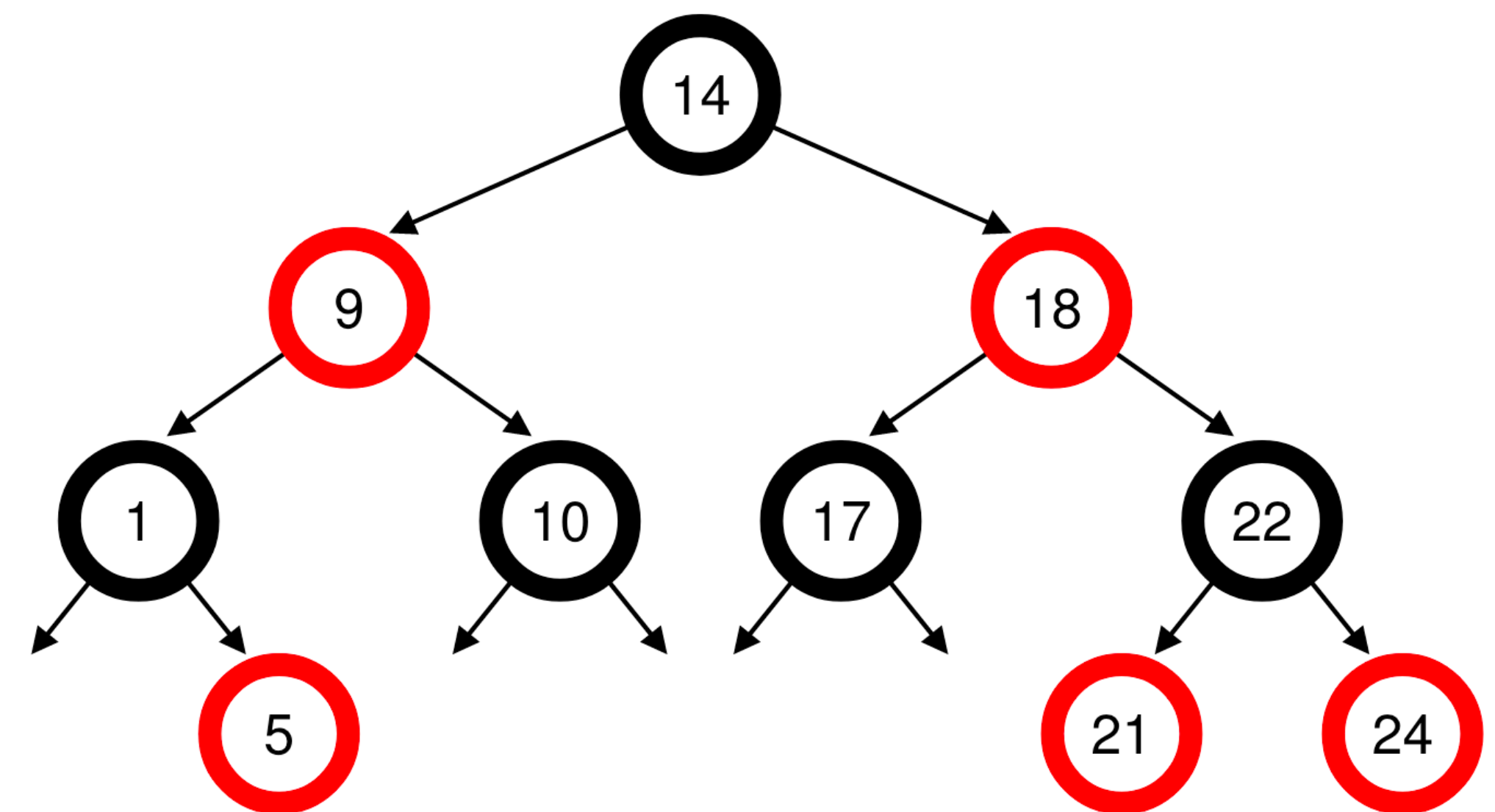


Figure 9.5: CFS Red-Black Tree

Completely Fair Scheduler (CFS)

- Each process has vruntime. Update vruntime after each time process is run.
- Maintain runnable processes in a red-black tree (sorted by vruntime). Run the process with lowest vruntime.
- When a new process is run:
 - Assign it lowest vruntime (instead of assigning it zero). Otherwise the new process monopolises the CPU

$$vruntime_i = vruntime_i + \frac{weight_0}{weight_i} \cdot runtime_i$$

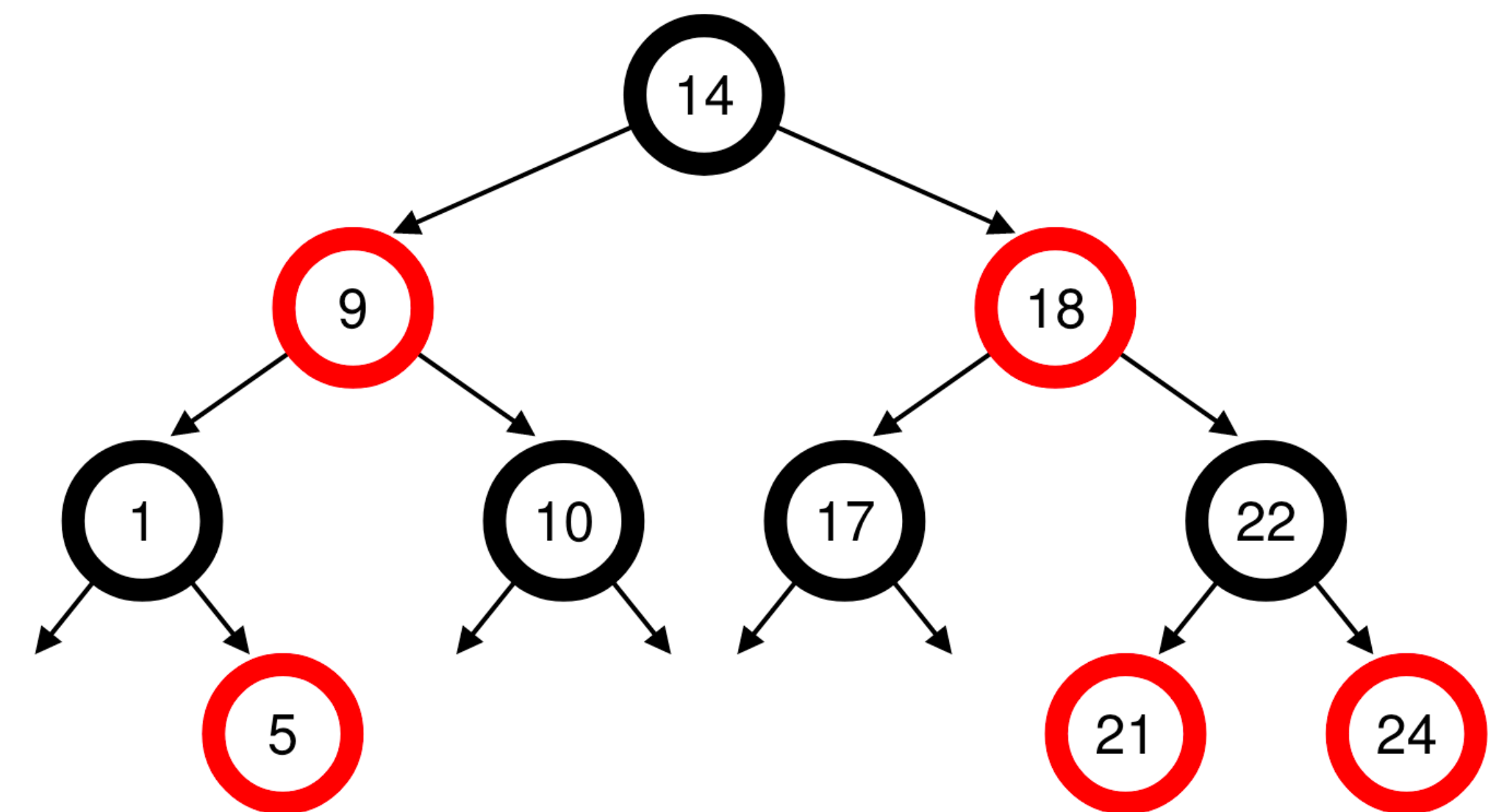
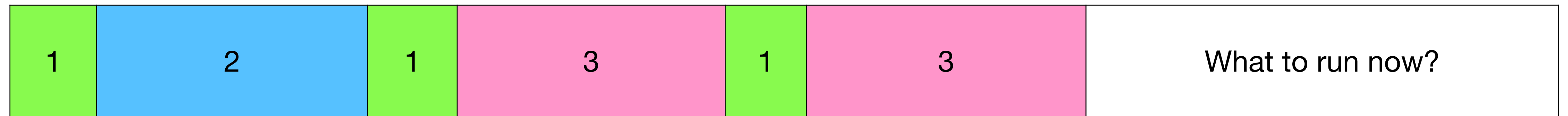


Figure 9.5: CFS Red-Black Tree

CFS: fair but not responsive

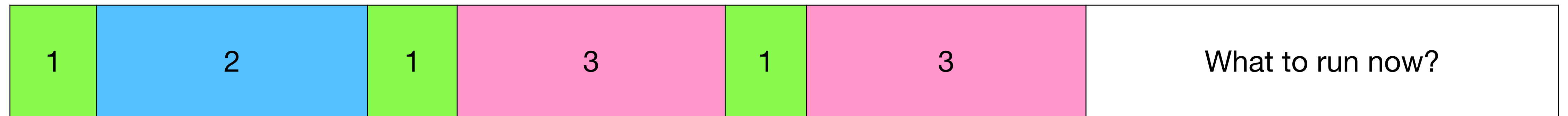
Eligible Earliest Virtual Deadline First (EEVDF) from Linux 6.6 (2024)



- | |
|------------------------------|
| 1. Latency-sensitive process |
| 2. Batch process |
| 3. Batch process |

CFS: fair but not responsive

Eligible Earliest Virtual Deadline First (EEVDF) from Linux 6.6 (2024)

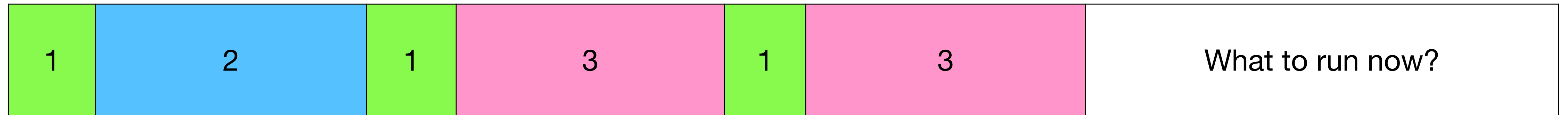


- $\text{vruntime}(1) = \text{vruntime}(2)$

1. Latency-sensitive process
2. Batch process
3. Batch process

CFS: fair but not responsive

Eligible Earliest Virtual Deadline First (EEVDF) from Linux 6.6 (2024)

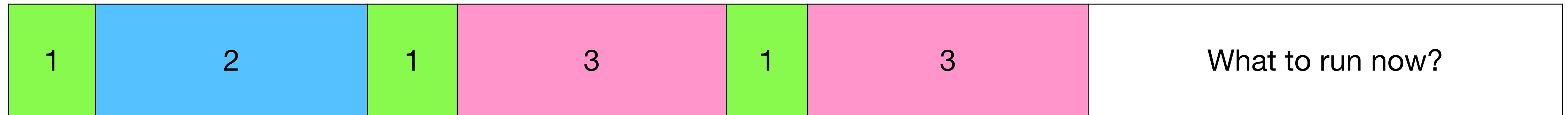


- $\text{vruntime}(1) = \text{vruntime}(2)$
- CFS: schedule either. Latency-sensitive process 1 misses deadlines!

1. Latency-sensitive process
2. Batch process
3. Batch process

CFS: fair but not responsive

Eligible Earliest Virtual Deadline First (EEVDF) from Linux 6.6 (2024)



- $\text{vruntime}(1) = \text{vruntime}(2)$
- CFS: schedule either. Latency-sensitive process 1 misses deadlines!
- EEVDF: since latency-sensitive process 1 is asking for “*smaller time slices*”, it has “*earlier virtual deadline*”. Run 1. Improves responsiveness.

1. Latency-sensitive process
2. Batch process
3. Batch process

Scheduler attacks

Before Linux 2.6 (2003)

- Kernel code was *non-preemptible*.

Scheduler attacks

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- Syscalls also disable interrupts like hardware interrupts

Scheduler attacks

Before Linux 2.6 (2003)

- Kernel code was *non-preemptible*.
- Syscalls also disable interrupts like hardware interrupts
- If process makes large system calls (e.g., large writes), it could exceed its time slice

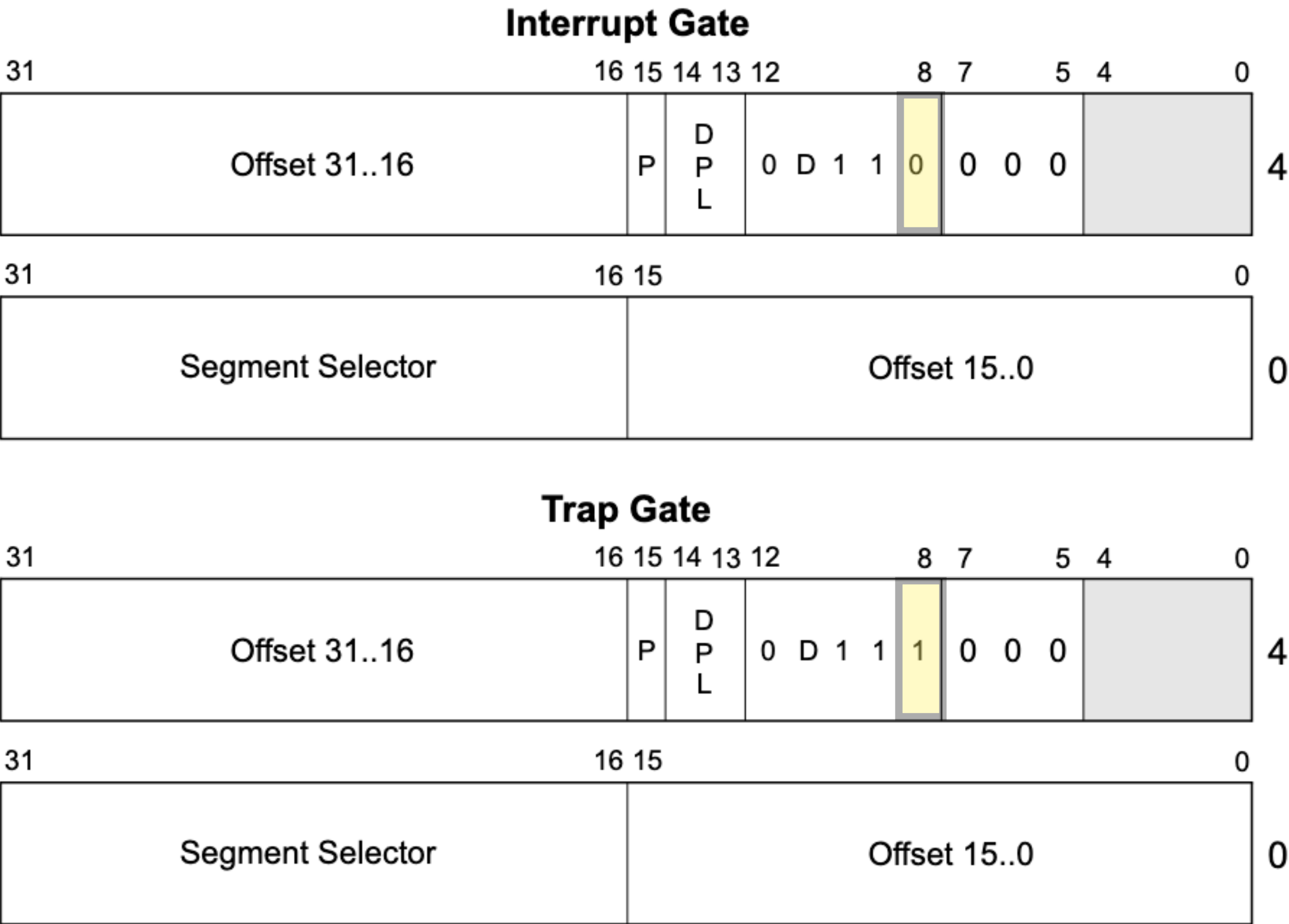
Scheduler attacks

Before Linux 2.6 (2003)

- Kernel code was *non-preemptible*.
- Syscalls also disable interrupts like hardware interrupts
- If process makes large system calls (e.g., large writes), it could exceed its time slice
- From Linux 2.6: kernel code is preemptible, e.g., keep interrupts enabled during syscalls

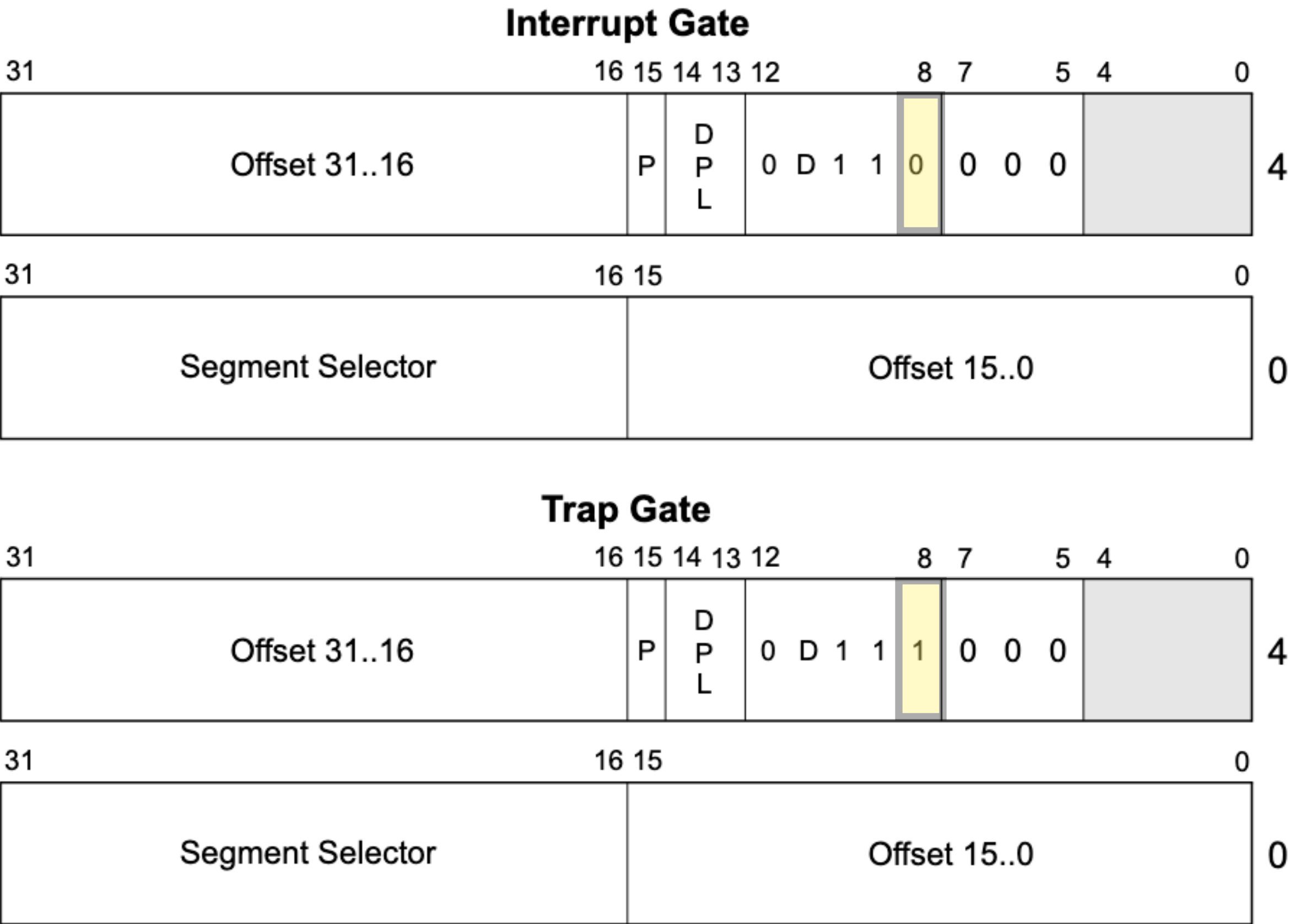
Keeping interrupts enabled during syscalls

- Set “trap gates” in Interrupt Descriptor Table



Keeping interrupts enabled during syscalls

- Set “trap gates” in Interrupt Descriptor Table
- Hardware keeps interrupts enabled during traps

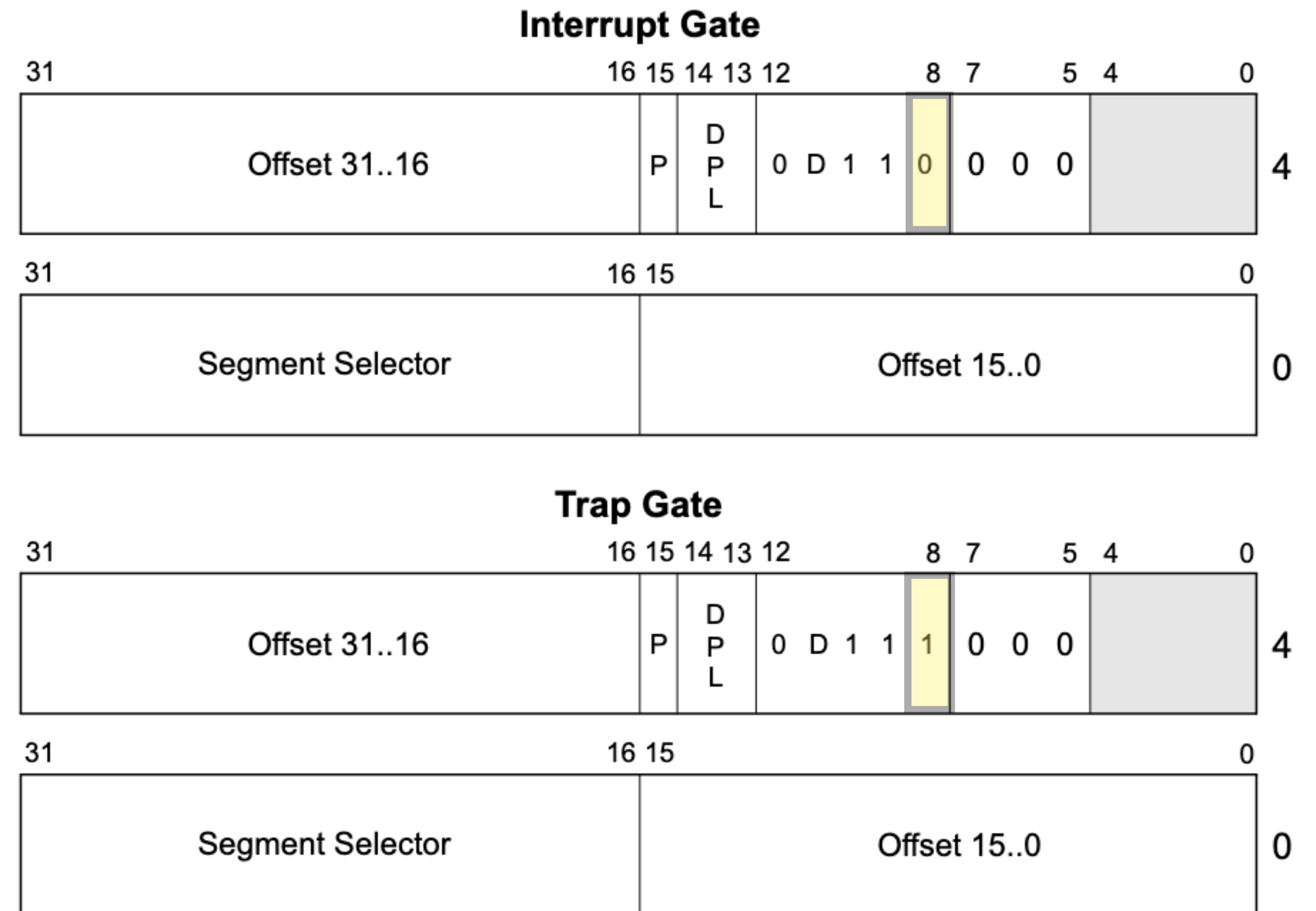


Keeping interrupts enabled during syscalls

- Set “trap gates” in Interrupt Descriptor Table
- Hardware keeps interrupts enabled during traps

```
void tvinit(void) {
    int i;
    for(i = 0; i < 256; i++)
        SETGATE(idt[i], 0, SEG_KCODE<<3, vectors[i], 0);
    SETGATE(idt[T_SYSCALL], 1, SEG_KCODE<<3, vectors[T_SYSCALL], DPL_USER);
}
```

trap.c



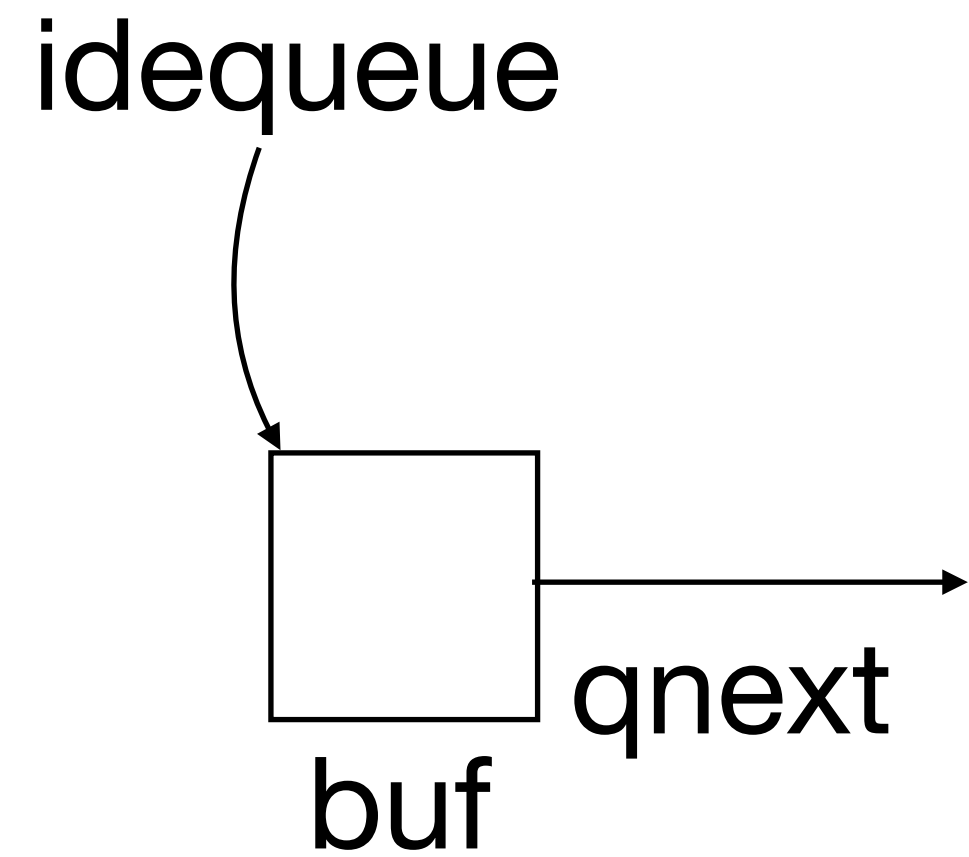
Concurrency (single processor)

Uniprocessor concurrency

- Software interrupts like system calls do not disable interrupts.
- If the kernel was updating its data structures, it needs to be careful to not run into races

Race conditions

Example: Processes reading a block in ide.c

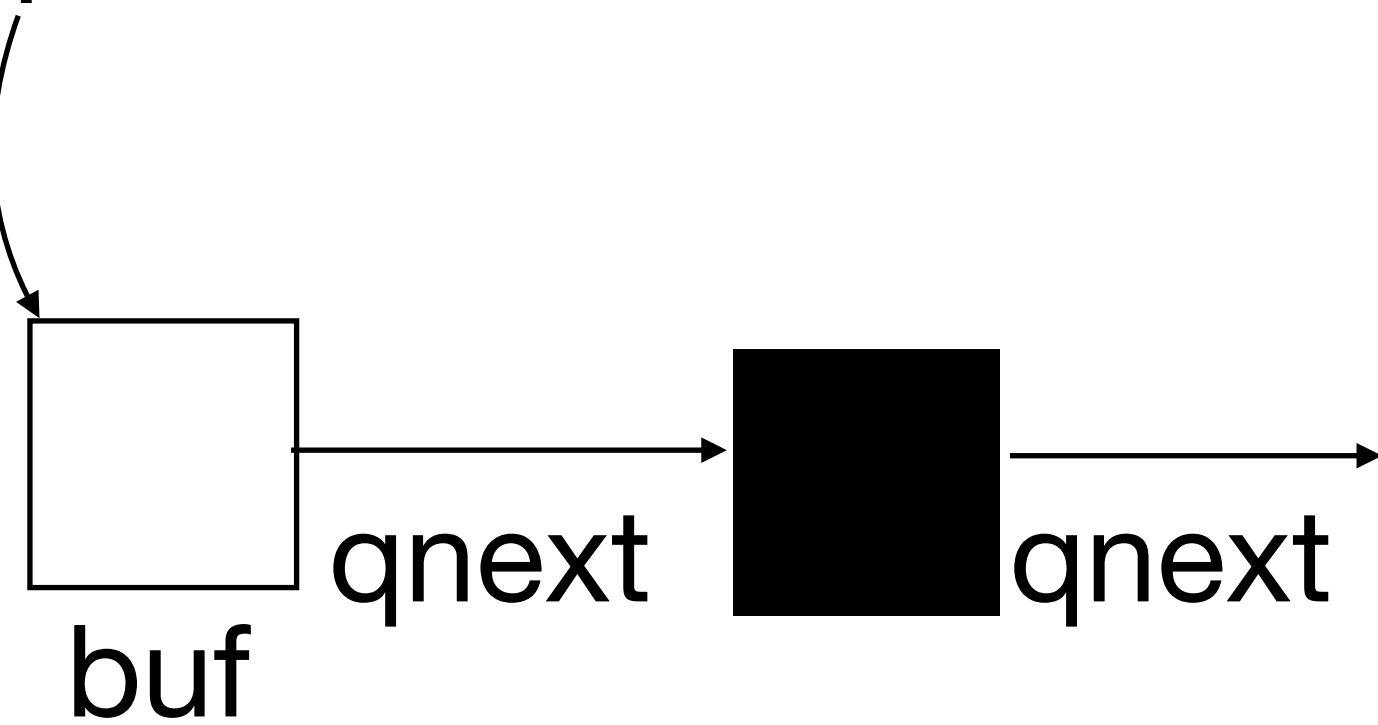


```
struct buf {  
    ..  
    struct buf *qnext; // disk queue  
    uchar data[BSIZE];  
};  
  
static struct buf *idequeue;  
  
void iderw(struct buf *b) {  
    struct buf **pp;  
    b->qnext = 0;  
    for(pp=&idequeue; *pp; pp=&(*pp)->qnext);  
    *pp = b;  
    ..  
}
```

Race conditions

Example: Processes reading a block in ide.c

idequeue



```
struct buf {
```

```
    ..
```

```
    struct buf *qnext; // disk queue
```

```
    uchar data[BSIZE];
```

```
};
```

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static struct buf *idequeue;
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```
void iderw(struct buf *b) {
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```

```
    b->qnext = 0;
```

```
    for(pp=&idequeue; *pp; pp=&(*pp)->qnext);
```

```
    *pp = b;
```

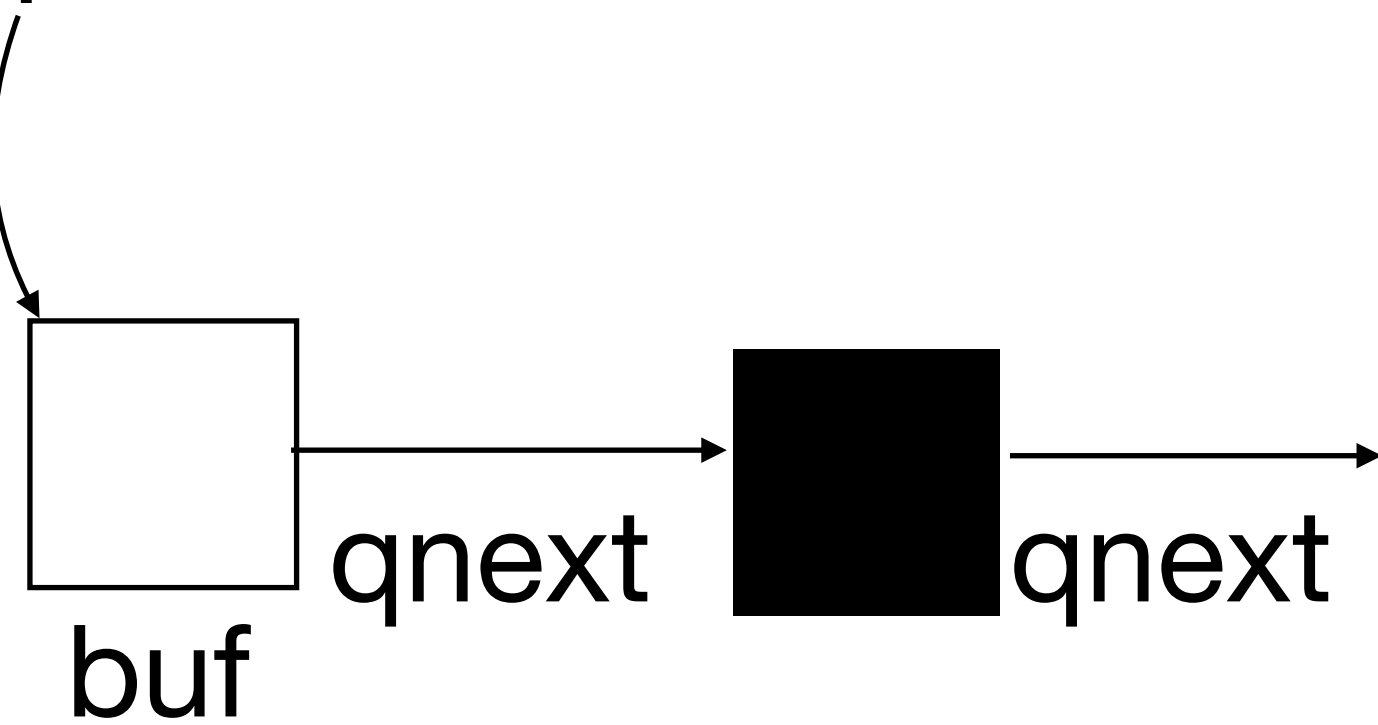
```
    ..
```

```
}
```


Race conditions

Example: Processes reading a block in ide.c

idequeue



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struct buf {
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..
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```
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```

```
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```

```
    b->qnext = 0;
```

```
    for(pp=&idequeue; *pp; pp=&(*pp)->qnext);
```

```
    *pp = b;
```

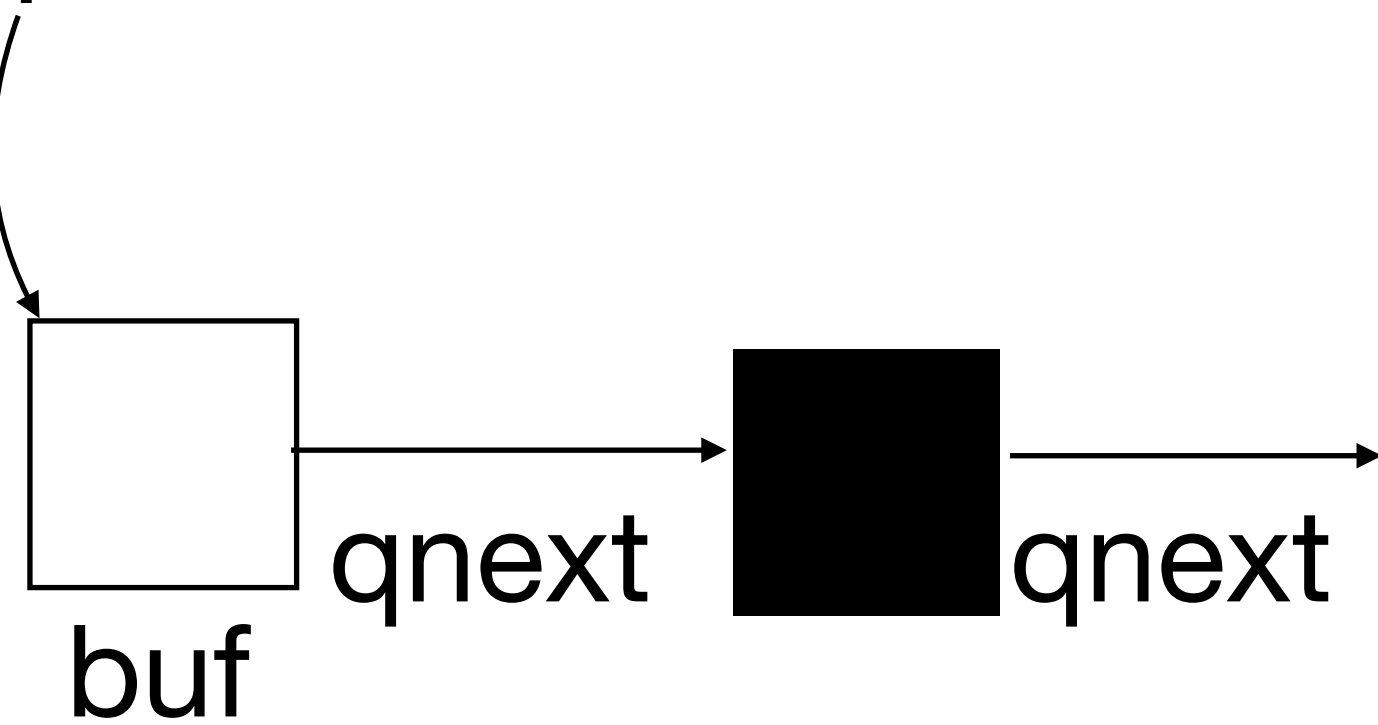
```
..
```

```
}
```

Race conditions

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idequeue



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struct buf {
```

```
..
```

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    struct buf *qnext; // disk queue
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void iderw(struct buf *b) {
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```
    struct buf **pp;
```

```
    b->qnext = 0;
```

```
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```

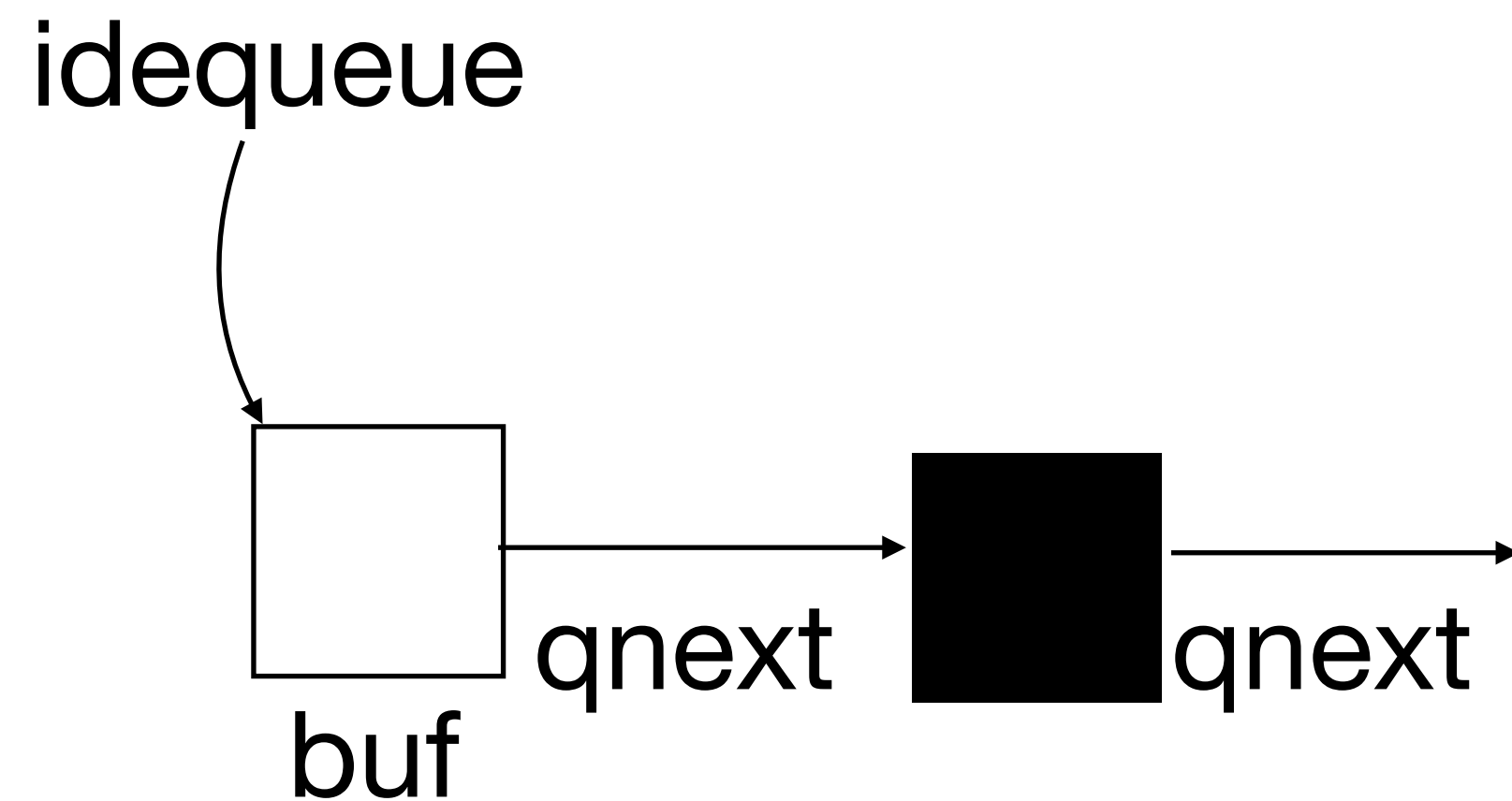
```
    *pp = b;
```

```
..
```

```
}
```


Race conditions

Example: Processes reading a block in ide.c

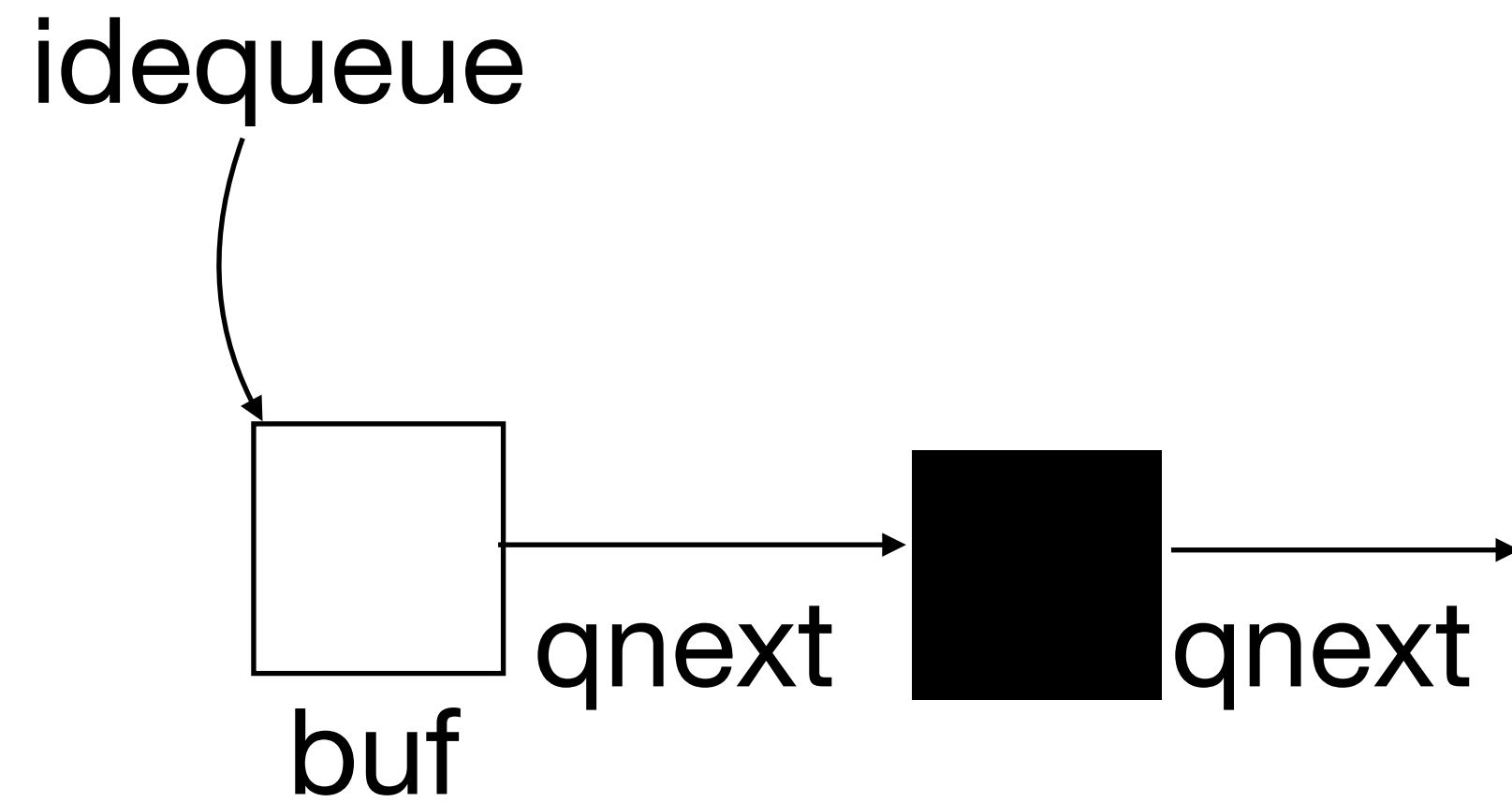


Context switch

```
struct buf {  
    ..  
    struct buf *qnext; // disk queue  
    uchar data[BSIZE];  
};  
  
static struct buf *idequeue;  
  
void iderw(struct buf *b) {  
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    b->qnext = 0;  
    for(pp=&idequeue; *pp; pp=&(*pp)->qnext);  
    *pp = b;  
    ..  
}
```

Race conditions

Example: Processes reading a block in ide.c



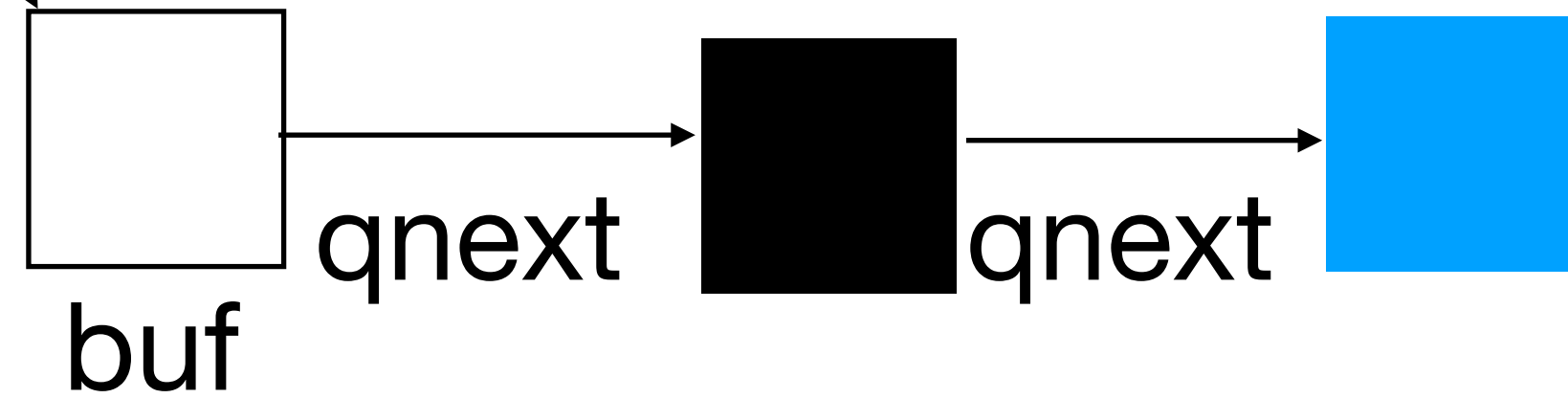
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Race conditions

Example: Processes reading a block in ide.c

idequeue



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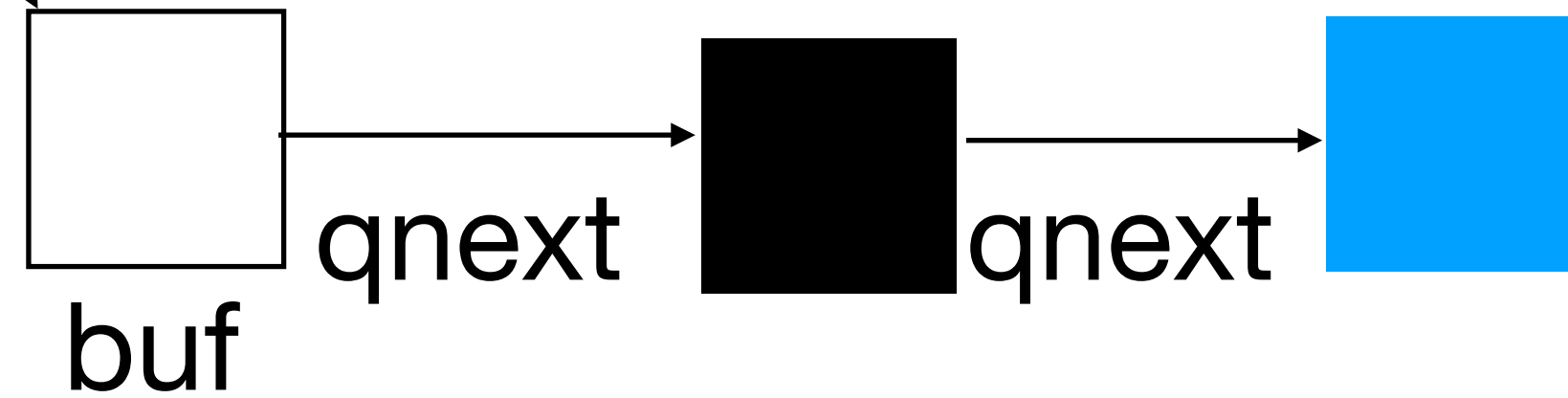
```
..
```

```
}
```


Race conditions

Example: Processes reading a block in ide.c

idequeue



Context switch

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```

```
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```

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    struct buf **pp;
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    b->qnext = 0;
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    for(pp=&idequeue; *pp; pp=&(*pp)->qnext);
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```
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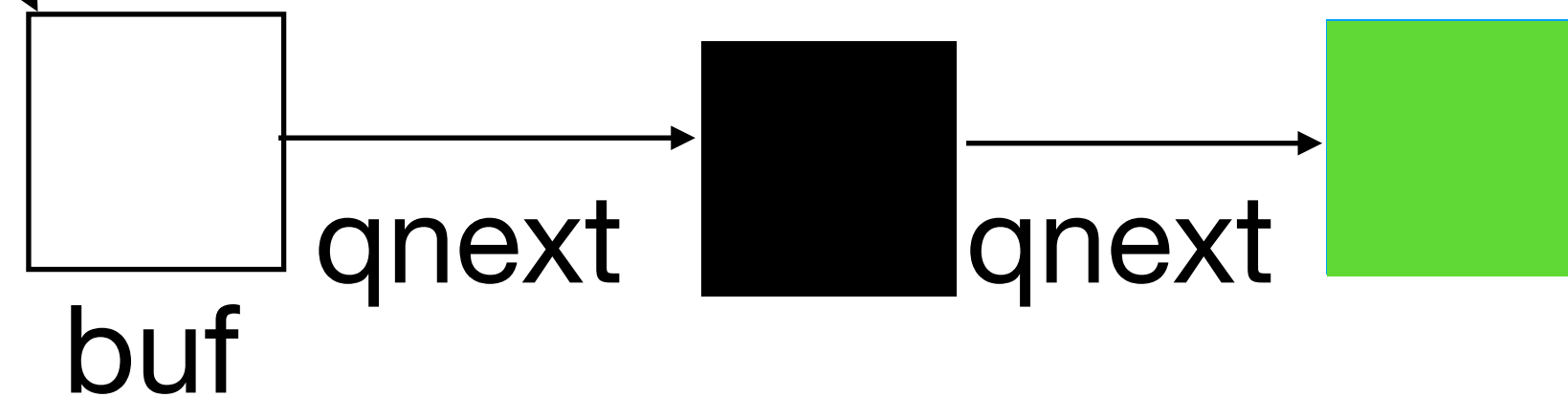
```
..
```

```
}
```

Race conditions

Example: Processes reading a block in ide.c

idequeue



Context switch

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```

```
..
```

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```
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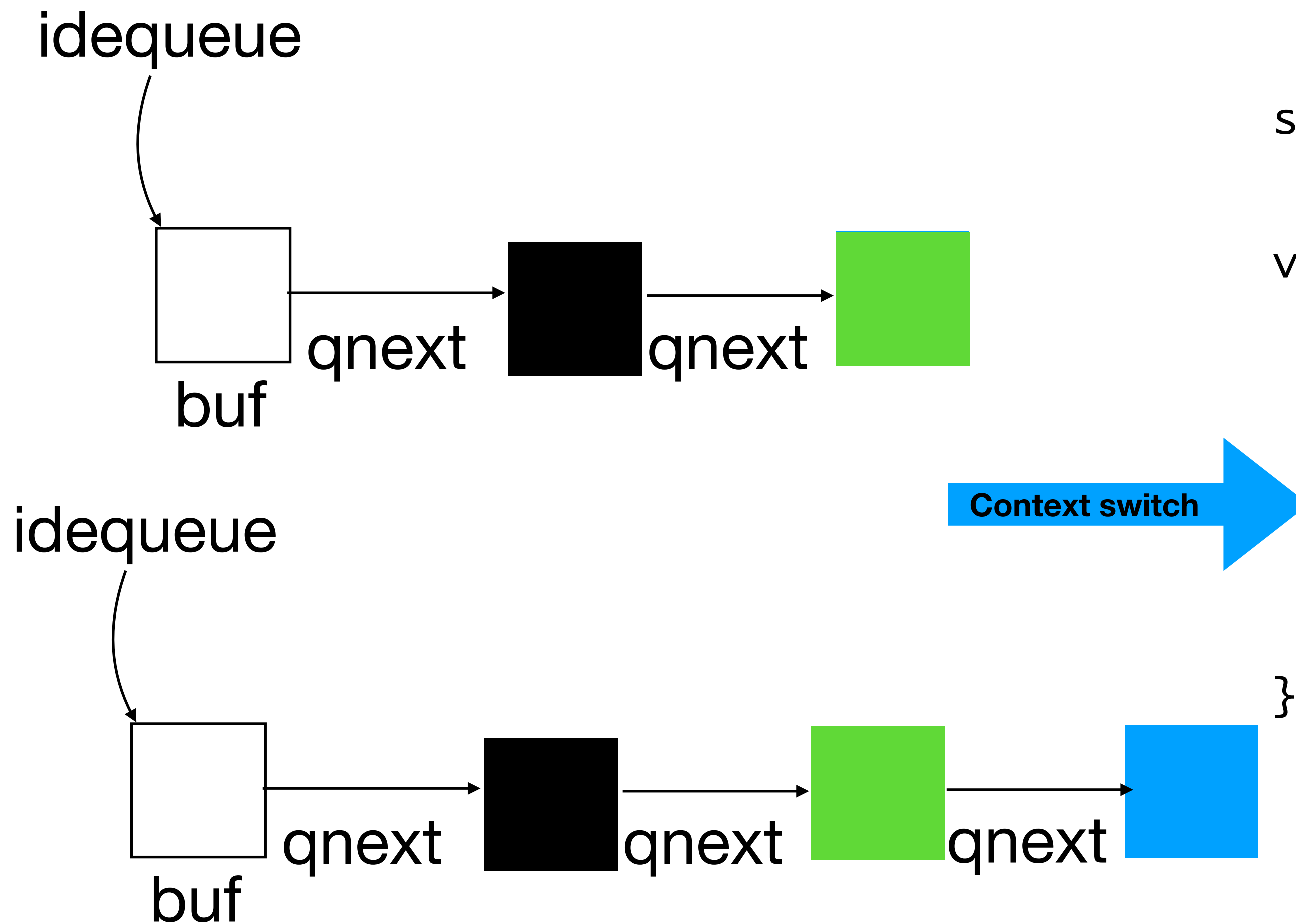
```
    *pp = b;
```

```
..
```

```
}
```

Race conditions

Example: Processes reading a block in ide.c



```
struct buf {
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```
..
```

```
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```
    uchar data[BSIZE];
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void iderw(struct buf *b) {
```

```
    struct buf **pp;
```

```
    b->qnext = 0;
```

```
    for(pp=&idequeue; *pp; pp=&(*pp)->qnext);
```

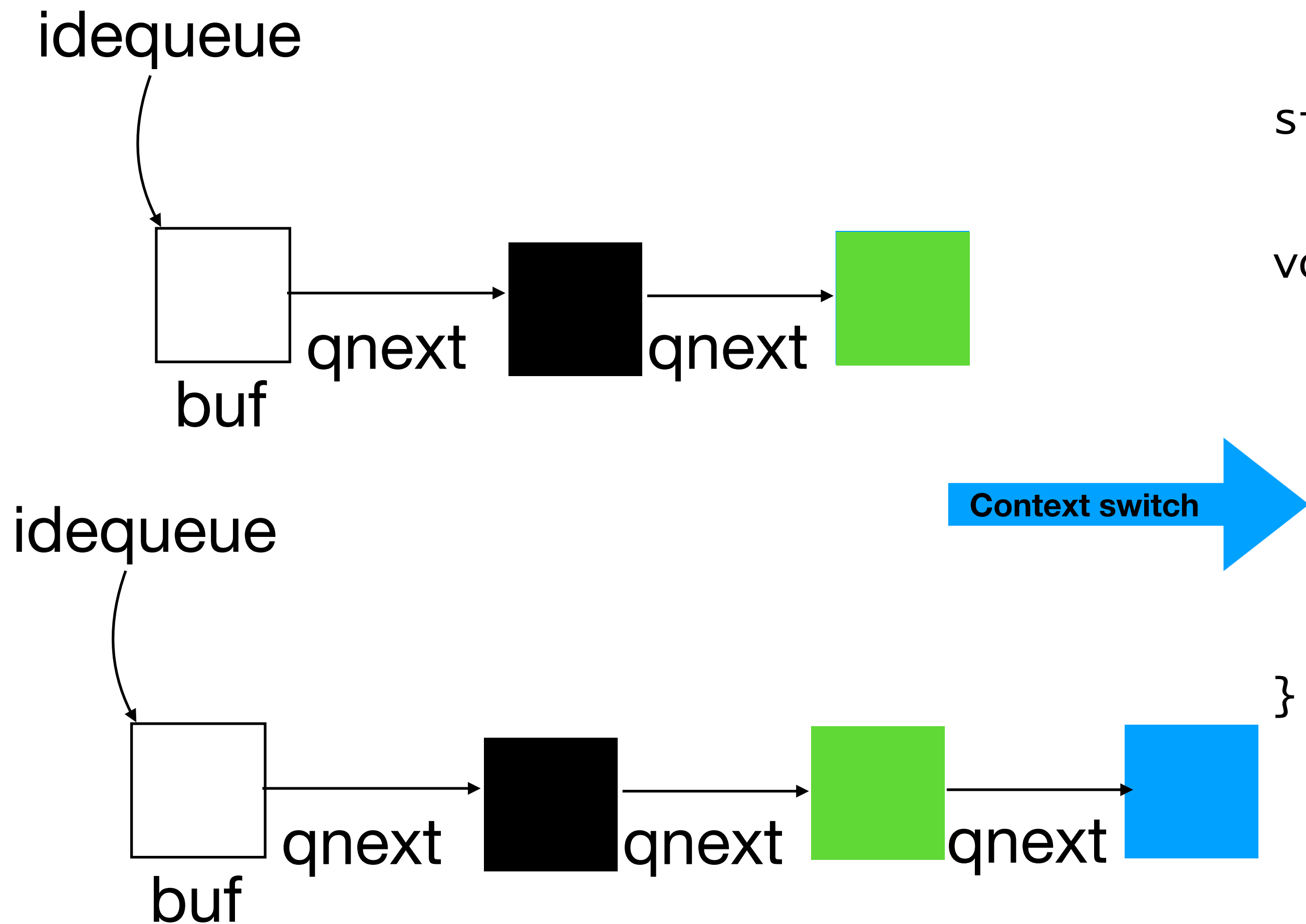
```
    *pp = b;
```

```
..
```

```
}
```


Race conditions

Example: Processes reading a block in ide.c



```
struct buf {
```

```
..
```

```
    struct buf *qnext; // disk queue
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```
    for(pp=&idequeue; *pp; pp=&(*pp)->qnext);
```

```
    *pp = b;
```

```
..
```

```
}
```

Protect critical sections!

```
void iderw(struct buf *b) {  
    struct buf **pp;  
    pushcli();  
    for(pp=&idequeue; *pp; pp=&(*pp)->qnext);  
    *pp = b;  
    ..  
    popcli();  
}
```


Protect critical sections!

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    *pp = b;  
    ..  
    popcli();  
}
```

```
void pushcli(void) {  
    int eflags = readeflags();  
    cli();  
    if(cpu->ncli == 0)  
        cpu->intena = eflags & FL_IF;  
    cpu->ncli += 1;  
}
```

Protect critical sections!

```
void iderw(struct buf *b) {  
    struct buf **pp;  
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        cpu->intena = eflags & FL_IF;  
    cpu->ncli += 1;  
}  
  
void popcli(void) {  
    cpu->ncli--;  
    if(cpu->ncli == 0 && cpu->intena)  
        sti();  
}
```


Protect critical sections!

```
void iderw(struct buf *b) {
    struct buf **pp;
    pushcli();
    for(pp=&idequeue; *pp; pp=&(*pp)->qnext);
    *pp = b;
    ..
    popcli();
}
```

- Timer interrupt and hence context switch cannot happen

```
void pushcli(void) {
    int eflags = readeflags();
    cli();
    if(cpu->ncli == 0)
        cpu->intena = eflags & FL_IF;
    cpu->ncli += 1;
}

void popcli(void) {
    cpu->ncli--;
    if(cpu->ncli == 0 && cpu->intena)
        sti();
}
```

Summary

- Memory isolation: How to isolate memory into address spaces?
- Processes in action: Process control block, user stack<>kernel stack, sys call handling, protection
- Scheduling: response time, turnaround time, fairness, scheduler attacks
- Uniprocessor concurrency: disable interrupts around critical sections